CBSE 2026-27

EduSpark

SCIENCE

ASTRA SERIES

CLASS 10 • CBSE • BOARD EXAM 2027

$E=mc^2$

60 DAYS MEIN PURA SYLLABUS **KHATAM!**

INSIDE THIS BOOK

- **Complete Notes**
  With Real Images & Diagrams
- **Timelines & Flowcharts**
  For Better Understanding
- **MCQs, Assertion-Reason**
  Case-Based Questions
- **All 3 Parts Covered**
  Physics | Chemistry | Biology
- **English + Hindi Explanation**
  Easy to Understand
- **60-Day Study Plan**
  With Mock Tests & Revision Days

By Nitesh Bhaiya

- **Concepts** That Stick
- **PYQ Analysis** Eduspark board trends
- **Memes** Concepts that stick

Chemistry | Physics | Biology
13 chapters, 60 levels, NCERT based

ASTRA SERIES
Free e-book by Nitesh Prakash

IMPRINT

PUBLISHER'S INFORMATION

Book details, copyright and the promise behind this free e-book.

ASTRA SERIES

SCIENCE CLASS 10

CBSE | 60 Days Complete Plan | Session 2026-27

by Nitesh Prakash

BOOK DETAILS

TITLE	ASTRA SERIES, Science, Class 10 (CBSE), 60 Days Complete Plan
SESSION	2026-27
SYLLABUS	As per the latest NCERT textbook and CBSE curriculum, 2026-27
EDITION	First Edition, 2026
PUBLISHER	Eduspark
AUTHOR	Nitesh Prakash
COPYRIGHT	© 2026 Nitesh Prakash and Eduspark. All rights reserved.
PRICE	FREE e-book (not for sale)
FORMAT	Digital e-book (PDF), 262 pages
ADDRESS	S/O: Rudra Prakash, Bawan Bigha, Road No-1, Mithanpura, PO: Ramna, Dist: Muzaffarpur, Bihar 842002

CONTACT AND FEEDBACK

EMAIL	contact@eduspark.in
WEBSITE	www.eduspark.in
PLACE	Muzaffarpur, Bihar
ERRATA	Found a mistake? Write to us and we will fix it in the next edition.

GET THE EDUSPARK APP

Scan to download the Eduspark app

Video lessons, mock tests and updates for CBSE 2027.

Search "Eduspark" on Play Store

FOLLOW US FOR LATEST BOOKS

YouTube | Instagram | Telegram | Facebook

Search **Eduspark** on every platform

ANTI-PIRACY WARNING

This e-book is **FREE for every student**. Share it with your friends, because the more students it helps, the happier we are. But selling it, printing it for sale, editing it, or removing the name of the author or Eduspark is strictly not allowed. Nobody may put this book out under another name. Reprinting or selling any part of it without written permission from Eduspark is an offence under the Copyright Act, 1957.

Free to share. Never for sale.

DISCLAIMER: A PROMISE TO YOU

Dear student, this book was made with one wish: that you walk into the exam hall without fear. Every chapter and question has been checked against the latest NCERT textbook and the CBSE syllabus for 2026-27. Question tags marked Eduspark PYQ Analysis are based on our own analysis, not an official CBSE record. If you spot a mistake, tell us and we will fix it in the next edition.

Nitesh Prakash

ALL RIGHTS RESERVED

No part of this book may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying or otherwise, for sale, without the prior written permission of the author and Eduspark.

NOTE

All disputes are subject to the jurisdiction of courts in Muzaffarpur, Bihar only. Science (Class X) is a prescribed NCERT textbook; this book is an independent study guide and is not affiliated with NCERT or CBSE.

FIRST EDITION 2026 | PUBLISHED BY EDUSPARK | MUZAFFARPUR, BIHAR

FROM THE AUTHOR'S DESK

NOTE FROM NITESH BHAIYA

A few words before you start Level 1.

Paheli and Bujho say: Level 1 is waiting!

NITESH PRAKASH

Author and Educator, Eduspark

"Paheli samjhayegi, Bujho poochega, aur tum seekhoge."

CBSE CLASS 10 | SCIENCE | 60 DAYS

Dear student,

Main Nitesh hoon, and this book is my promise to you: one clear day at a time, no topic left half-done.

Every year I meet students who work hard for months and still walk into the Science paper nervous. Not because they are weak, but because their notes are scattered, their guide is too thick and nobody showed them what the board actually asks. Chemistry ke equations, Physics ke ray diagrams, Biology ke labelled diagrams: sab ek saath feels like too much.

So I built ASTRA differently. The full NCERT syllabus for 2026-27 is split into 60 days. Each day is one level: read the concept, laugh a little with Paheli and Bujho, practise the exact kind of questions the board loves, and collect your XP. Boss Levels test a full chapter, Swahaa pages help you revise it, and the Grand Mock on Day 60 shows you how ready you are.

Keep your NCERT textbook beside this book, solve every question honestly before you check the answer, and be truthful in the self-assessment tracker, because it is only for you. Every R and RA you mark is a mark you are about to win back.

Some days will feel easy and some will feel heavy. That is normal. Bas rukna nahi hai. This e-book is free for every student, so share it with a friend who needs it, and if you spot a mistake, tell us. I believe in you, and I am proud of you already.

With love and best wishes,

Nitesh Prakash

AUTHOR, ASTRA SERIES
EDUSPARK

> *Roz ek level. 60 din. Phir board exam bas ek aur level hai, aur woh bhi tum clear karoge.*
>
> NITESH BHAIYA

1 ONE DAY, ONE LEVEL
Finish today's level fully before you open tomorrow's.

2 HONEST PRACTICE
Attempt first, check later. Your mistakes are your map.

3 REVISE ON TIME
Use every Swahaa page. Revision turns effort into marks.

BEFORE YOU START

HOW TO USE THIS BOOK

One complete CBSE Class 10 Science plan. One day = one level = 4 pages.

60	13
LEVELS	CHAPTERS
8000	4
TOTAL XP	PAGES A DAY

CHEMISTRY Days 1–20 | Pages 9–92

PHYSICS Days 21–40 | Pages 93–176

BIOLOGY Days 41–60 | Pages 177–261

1 YOUR PATH THROUGH THE BOOK

STEP 1 — Read the concept
Theory in NCERT order with diagrams, tables, formulas and Class Tricks.
EVERY DAY

STEP 2 — Paheli-Bujho and memes
Paheli explains, Bujho asks your doubts. A few memes lock the idea in.
EVERY DAY

STEP 3 — Practice
MCQ, A-R, case-based, short and long answers. Then check with the answer key.
EVERY DAY

STEP 4 — Boss Level
A full chapter test on the last day of the chapter. No notes, timer on.
CHAPTER END

STEP 5 — Level Complete and XP
Tick the 5-item checklist and claim the XP you earned on the Level Complete card.
EVERY DAY

STEP 6 — Swahaa revision page
One-page chapter recap, the chapter XP you earned and your book progress.
CHAPTER END

STEP 7 — Grand Mock
A full 80-mark CBSE board-pattern paper on all 13 chapters, with answer key.
DAY 60

2 BOXES YOU WILL MEET

+125 XP
XP chip on each Day banner: the XP that Level is worth.

CLASS TRICK
Mnemonics and shortcuts to recall formulas and lists fast.

PAHELI **BUJHO**
The dialogue box: concept explained through real student doubts.

EDUSPARK PYQ ANALYSIS
How often a topic has come in board papers. Prioritise these.

EXTRA KNOWLEDGE
Beyond the syllabus, for curiosity and olympiads. Optional.

BOSS
Chapter test day. Clear it before you start the next chapter.

SWAHAA
Chapter celebration and recap page. Revise from it later.

LEVEL COMPLETE
End-of-day card: checklist, score and XP collected.

3 THE XP SYSTEM

1 CBSE mark = **100 XP**

1	60	13	8000
Day = 1 Level	Levels in all	Swahaa pages	XP in the book

Chemistry	2500
Physics	2500
Biology	3000
BOOK TOTAL	**8000 XP**

Each chapter's XP is listed in the Index (page 5). Collect all 8000 XP and you have covered the full board syllabus.

4 SUGGESTED DAILY ROUTINE

- **5 min** — **Warm up.** Revise yesterday's tricks and mistakes.
- **20 min** — **Theory.** Read the notes, diagrams and the Paheli and Bujho dialogue.
- **5 min** — **Tricks and memes.** Say every mnemonic aloud once.
- **20 min** — **Practice.** Solve all questions without looking at answers.
- **10 min** — **Check and claim.** Use the answer key, tick the checklist, note your score on the Self Assessment page (page 8) and collect your XP.

About 60 minutes a day. 60 days. Board ready.

GOLDEN RULES

1. Never skip a day. If you miss one, do two the next day.
2. Attempt first, check later. Mark every mistake honestly.
3. Clear the Boss Level before you open the next chapter.

ASTRA SERIES | SCIENCE CLASS 10

CONTENTS & BOOK MAP

60 Days, 13 chapters, 3 parts, **8000 XP**
Every day = 4 pages: theory, tricks, practice, answer key
Boss Level ends each chapter, Swahaa page celebrates it

Cover 1 | Publisher's Info 2 | Author's Note 3 | How to Use 4 | Contents 5 | OMR Guide 6 | Admit Card 7 | Self Assessment 8

CHEMISTRY
Days 1–20 • Pages 9–92 • 2500 XP

CH 1 Chemical Reactions and Equations +500 XP
Day 1 Chemical Reactions and Equations 9
Day 2 Combination and Decomposition 13
Day 3 Displacement and Redox Reactions 17
Day 4 Energy, Corrosion and Rancidity 21
★ Chapter 1 Swahaa 25

CH 2 Acids, Bases and Salts +625 XP
Day 5 Acids, Bases and Salts 26
Day 6 Ions, Strength and the pH Scale 30
Day 7 pH in Daily Life and Salts 34
Day 8 Chemicals from Common Salt 38
Day 9 Chapter 2 Boss Level BOSS 42
★ Chapter 2 Swahaa 46

CH 3 Metals and Non-metals +625 XP
Day 10 Metals and their Reactions 47
Day 11 Ionic Bonds and Compounds 51
Day 12 Extraction of Metals 55
Day 13 Corrosion and Alloys 59
Day 14 Chapter 3 Boss Level BOSS 63
★ Chapter 3 Swahaa 67

CH 4 Carbon and its Compounds +750 XP
Day 15 Carbon and its Compounds 68
Day 16 Hydrocarbons and Homologous Series 72
Day 17 Functional Groups and IUPAC Names 76
Day 18 Properties of Carbon Compounds 80
Day 19 Soaps and Detergents 84
Day 20 Chapter 4 Boss + Chemistry Mock BOSS + MOCK 88
★ Chapter 4 Swahaa 92

PHYSICS
Days 21–40 • Pages 93–176 • 2500 XP

CH 5 Light: Reflection and Refraction +900 XP
Day 21 Light and Reflection 93
Day 22 Ray Diagrams and Mirror Images 97
Day 23 Mirror Formula and Numericals 101
Day 24 Refraction and Refractive Index 105
Day 25 Spherical Lenses and Images 109
Day 26 Lens Formula and Power 113
Day 27 Chapter 5 Boss Level BOSS 117
★ Chapter 5 Swahaa 121

CH 6 The Human Eye and the Colourful World +300 XP
Day 28 The Human Eye 122
Day 29 Colourful World + Boss BOSS 126
★ Chapter 6 Swahaa 130

CH 7 Electricity +700 XP
Day 30 Current and Potential Difference 131
Day 31 Ohm's Law and Resistivity 135
Day 32 Resistors in Series 139
Day 33 Parallel and Mixed Circuits 143
Day 34 Heating Effect and Electric Power 147
Day 35 Chapter 7 Boss Level BOSS 151
★ Chapter 7 Swahaa 155

CH 8 Magnetic Effects of Electric Current +600 XP
Day 36 Magnets and Magnetic Field Lines 156
Day 37 Field of a Straight Wire and a Loop 160
Day 38 Solenoid and Electromagnet 164
Day 39 Fleming's Rule and Home Circuits 168
Day 40 Chapter 8 Boss + Physics Mock BOSS + MOCK 172
★ Chapter 8 Swahaa 176

BIOLOGY
Days 41–60 • Pages 177–261 • 3000 XP

CH 9 Life Processes +900 XP
Day 41 Life Processes and Photosynthesis 177
Day 42 Heterotrophic Nutrition and Digestion 181
Day 43 Respiration and Breathing 185
Day 44 Transportation in Humans 189
Day 45 Plant Transport and Excretion 193
Day 46 Chapter 9 Boss Level BOSS 197
★ Chapter 9 Swahaa 201

CH 10 Control and Coordination +575 XP
Day 47 Nervous System and the Human Brain 202
Day 48 Coordination in Plants 206
Day 49 Hormones in Animals and Feedback 210
Day 50 Chapter 10 Boss Level BOSS 214
★ Chapter 10 Swahaa 218

CH 11 How do Organisms Reproduce? +575 XP
Day 51 Asexual Reproduction and Variation 219
Day 52 Plant Reproduction and the Flower 223
Day 53 Reproduction in Human Beings 227
Day 54 Chapter 11 Boss Level BOSS 231
★ Chapter 11 Swahaa 235

CH 12 Heredity +450 XP
Day 55 Mendel and the Monohybrid Cross 236
Day 56 Dihybrid Cross and Sex Determination 240
Day 57 Chapter 12 Boss Level BOSS 244
★ Chapter 12 Swahaa 248

CH 13 Our Environment +500 XP
Day 58 Ecosystem and Food Chains 249
Day 59 Our Environment II + Boss BOSS 253
Day 60 Science Grand Mock, 80 marks GRAND MOCK 257
★ Ch 13 Swahaa + Book Complete 261

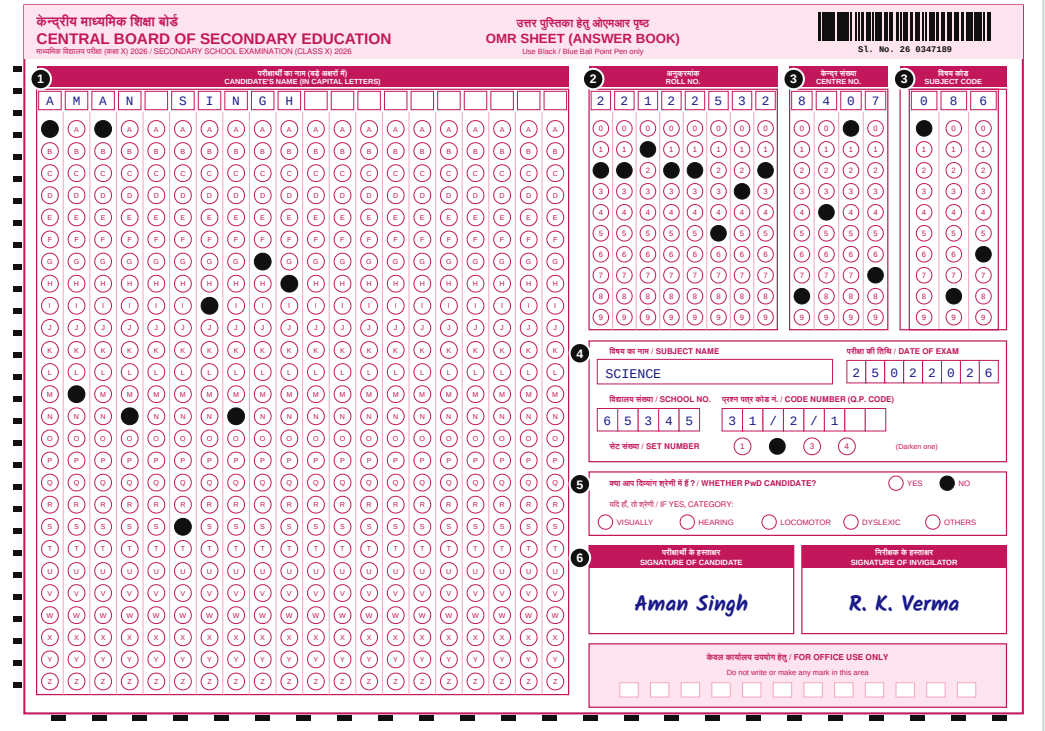
BOSS Full chapter test | GRAND MOCK 80-mark paper | ★ Swahaa: recap + XP | Total **8000 XP** | **Page 262**: review QR + ASTRA books

BEFORE THE BOARD EXAM

HOW TO FILL THE OMR SHEET

One wrong bubble can hold back your result. Fill it right, first time.

CBSE CLASS X BOARD EXAM



Sample OMR sheet, filled correctly. Black circled numbers match the steps below. *Details shown are imaginary.*

BEFORE YOU START

Pen
Blue or black ballpoint only. No gel, pencil or sketch pen.

Admit card
Keep it beside the sheet. Copy roll no., centre no. and subject code from it.

Darken fully
Fill the whole bubble, dark and inside the circle. One bubble per column.

Check twice
Match every digit before you sign. A wrong roll no. can hold your result.

No damage
Do not fold, staple or use whitener on the sheet.

1 FILL IT STEP BY STEP

1 Name
Write it in CAPITAL letters, one letter per box, exactly as on your admit card. Leave one box blank between words and darken the matching letter bubble below each box.

2 Roll No.
Copy all 8 digits from the admit card into the boxes, then darken one bubble per column (0 to 9).

3 Centre No. and Subject Code
Fill them the same way, digit by digit. Science is **086**.

4 Subject, date, school no.
Copy from the admit card. Take the **Set Number** and **Q.P. Code** from the question paper cover, every "/" included.

5 PwD
Darken YES and your category only if it applies to you; otherwise darken NO.

6 Signatures
Sign only in your own box. The invigilator signs in theirs after checking your entries.

2 CHECK BEFORE YOU HAND IT IN

WHERE EACH ENTRY COMES FROM

Admit card	Name, Roll No., Centre No., School No., Subject, Subject Code, Date of Exam
Q.P. cover	Set Number, Code Number (Q.P. Code)
Yourself	PwD status, your signature
Invigilator	Invigilator's signature, after checking the sheet

MISTAKES THAT GET AN OMR REJECTED

1. Roll No. written but bubbles left empty
2. Two bubbles darkened in one column
3. Q.P. Code copied without the "/" marks
4. Pencil, gel pen, whitener or overwriting used
5. Sheet folded, torn or written on in the office area

DARKENING A BUBBLE ● Correct ✓ Wrong ⊗ Wrong ⊙ Wrong ◑ Wrong — Blue or black ballpoint pen only

IMPORTANT: READ BEFORE THE EXAM

YOUR ADMIT CARD

Know every entry on it. Circled numbers show what you copy onto the OMR sheet.

CBSE CLASS X EXAM 2026

केन्द्रीय माध्यमिक शिक्षा बोर्ड, दिल्ली

CENTRAL BOARD OF SECONDARY EDUCATION, DELHI

ADMIT CARD FOR SECONDARY EXAMINATION 2026

Digital CBSE

6 ROLL NO.	22122532	DOB	14/08/2010	9 SCHOOL NO.	65345	8 CENTRE NO.	8407

Roll No. in Words	7 TWO CRORE TWENTY ONE LAKH TWENTY TWO THOUSAND FIVE HUNDRED THIRTY TWO
Candidate's Name	4 AMAN SINGH
Mother's Name	SUNITA DEVI
Father's / Guardian's Name	5 RAJESH KUMAR SINGH
Examination Centre	8407 : SAMPLE PUBLIC SCHOOL, SECTOR 12, NEW TOWN
PwD Category	NO
Admit Card ID	AC26X0347189

PHOTOGRAPH

2 SUB CODE	1 SUBJECT NAME	MEDIUM	3 DATE	INVIGILATOR SIGN
041	MATHEMATICS STANDARD	ENGLISH	17/02/2026	
184	ENGLISH LNG & LIT.	ENGLISH	21/02/2026	
086	**SCIENCE**	**ENGLISH**	**25/02/2026**	
002	HINDI COURSE-A	HINDI	02/03/2026	
087	SOCIAL SCIENCE	ENGLISH	07/03/2026	

ON THE QUESTION PAPER — **Series:** ZXYW4 | 10 **Set No.** 2 | 11 **Q.P. Code** 31/2/1

Aman Singh — Signature of Candidate

Signature of Principal with Seal

S. Bhardwaj — Controller of Examinations

Circles 1 to 11: entries you copy onto the OMR sheet (page 3). 10 and 11 come from the question paper.

Details shown are imaginary.

! BEFORE AND ON EXAM DAY

WHAT TO CHECK

1. Spelling of your name, mother's and father's name.
2. Photo is yours; date of birth is correct.
3. Every subject and code is right (Science = 086).
4. Centre name and address; visit it a day before.
5. Principal's signature and school seal are present.
6. Any mistake: report to your school at once, never on exam day.

EXAM DAY REPORTING

- 9:30 — Reach the centre
- 10:00 — Entry closes
- 10:15 — Question paper given, 15 min reading
- 10:30 — Start writing
- 1:30 — Exam ends (3 hours)

Always follow the times printed on your own admit card.

DOS AND DON'TS

- ✓ Carry the admit card and school ID card.
- ✓ Blue or black ballpoint pens in a transparent pouch.
- ✓ School uniform; transparent water bottle.
- ✗ No phone, smartwatch, calculator or any device.
- ✗ No chits, and nothing written on the admit card.
- ✗ Never write your name or roll no. inside the answer book.

KNOW WHERE YOU STAND

SELF ASSESSMENT

Check yourself after every level. Your score today shows you what to revise tomorrow.

WHY ASSESS YOURSELF?

1. You find your weak topics early, while there is still time to fix them.
2. Marks stop being a surprise: you know exactly where you stand before the board exam.
3. Revision becomes targeted. You re-read only what you actually got wrong.
4. Seeing your scores improve day by day builds real exam confidence.
5. It builds the habit of checking your own answers like an examiner.

HOW TO SELF-EVALUATE

1. Finish the day's practice questions (or the Boss Level) without looking at the answers.
2. Check every answer with the answer key and the marking points.
3. Write your score in the tracker below (marks scored / total marks).
4. Put a remark: E, R or RA (see the key). Be honest, nobody else is grading.
5. Revise R and RA days within a week and write the date in Revised on.

E — EASY
Concept clear, almost all correct. Move ahead.

R — REVISE
Small mistakes. Revise the tricks and wrong questions once more.

RA — RE-ATTEMPT
Many mistakes. Study the topic again and redo all questions.

MY 60-DAY TRACKER

CHEMISTRY — Days 1–20

DAY	SCORE	E / R / RA	REVISED ON
1			
2			
3			
4			
5			
6			
7			
8			
9 B			
10			
11			
12			
13			
14 B			
15			
16			
17			
18			
19			
20			

PHYSICS — Days 21–40

DAY	SCORE	E / R / RA	REVISED ON
21			
22			
23			
24			
25			
26			
27 B			
28			
29			
30			
31			
32			
33			
34			
35 B			
36			
37			
38			
39			
40			

BIOLOGY — Days 41–60

DAY	SCORE	E / R / RA	REVISED ON
41			
42			
43			
44			
45			
46 B			
47			
48			
49			
50 B			
51			
52			
53			
54 B			
55			
56			
57 B			
58			
59 B			
60	GRAND MOCK / 80		

MY XP PROGRESS TRACKER

Write the XP you earned per chapter (1 CBSE mark = 100 XP). Chemistry 2500, Physics 2500, Biology 3000.

Ch 1 Chemical Reactions _____ / 500	Ch 2 Acids, Bases, Salts _____ / 625	Ch 3 Metals and Non-metals _____ / 625	Ch 4 Carbon Compounds _____ / 750	Ch 5 Light _____ / 900	Ch 6 Human Eye _____ / 300	Ch 7 Electricity _____ / 700
Ch 8 Magnetic Effects _____ / 600	Ch 9 Life Processes _____ / 900	Ch 10 Control and Coordination _____ / 575	Ch 11 How do Organisms Reproduce _____ / 575	Ch 12 Heredity _____ / 450	Ch 13 Our Environment _____ / 500	TOTAL Science, 60 Levels _____ / 8000

B *Boss Level day (chapter test). Shaded rows: Day 20 and 40 (Boss + Mock), Day 60 Science Grand Mock.*

DAY 01 · LEVEL 1

CHEMISTRY | CHAPTER 1 | PART 1 · ASTRA SERIES

CHEMICAL REACTIONS & EQUATIONS

Goal: identify reactions, write and balance equations.

by Nitesh Prakash

01 Reactions | 02 5 Signs | 03 Equations | 04 Balancing | 05 States & Conditions | +100 XP today | Level 1 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 PHYSICAL VS CHEMICAL CHANGE

PHYSICAL	CHEMICAL
× No new substance; only shape, size or state changes	✔ A **new substance** with new properties forms
Usually reversible	Usually irreversible
Melting ice, dissolving sugar, breaking glass	Milk → curd, rusting, cooking, burning paper

2 WHAT IS A CHEMICAL REACTION?

DEFINITION

- Substances change into **new substances** with **new properties**
- Old bonds break and **new bonds form**
- Atoms are only **rearranged**, never created or destroyed

REACTANTS (what you start with (left side)) → bonds break and re-form → **PRODUCTS** (what is formed (right side))

$2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$

Reactants: magnesium, oxygen | Product: magnesium oxide

MEME BREAK

*Sodium (catches fire in water) + chlorine (poisonous gas) = the salt in your dal! New properties = **chemical reaction**.*

BUJHO *Doodh garam kiya, reaction hua?*

PAHELI *Nahi! Wo physical change hai. Doodh khatta hona chemical hai!*

EXOTHERMIC ★★★	ENDOTHERMIC
Heat **given out**; write + **Heat on the product side**	Heat **absorbed**; + **Heat on the reactant side** (or Δ on arrow)
Burning Mg, coal, natural gas; respiration; CaO + water	Heating $CaCO_3$; $Ba(OH)_2$ + NH_4Cl; AgCl in sunlight

★★★ *Nitesh Bhaiya ki guarantee hai — Exo vs Endo dimaag mein set kar lo!*

3 5 SIGNS OF A REACTION

Change in state	$NH_{3(g)} + HCl_{(g)} \rightarrow NH_4Cl_{(s)}$ two gases → white solid (fumes)
Change in colour ★★★	$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$ blue solution → light green; brown Cu coats the iron nail
Gas evolved ★☆☆	$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}\uparrow$ H_2 burns with a "pop" sound
Temperature change ★★☆	$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)}$ + **Heat** quicklime + water: beaker gets hot
Precipitate ★★★	$BaCl_{2(aq)} + Na_2SO_{4(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$ **white** ppt of $BaSO_4$; NaCl stays dissolved

BUJHO *Gas nikli, matlab reaction pakka?*

PAHELI *Sign hai, proof nahi! Asli test: naya substance bana ya nahi?*

★★★ **PYQ FAVOURITE** $BaCl_2 + Na_2SO_4 \rightarrow$ white **$BaSO_4$** (SQP 2023, SQP 2025, CBSE 2019). Also asked: $Pb(NO_3)_2 + 2KI \rightarrow$ yellow **PbI_2**. *Ye pakka aayega, Nitesh Bhaiya ki guarantee!*

Activity: Burning magnesium ribbon ★★★

1. Clean Mg ribbon with **sandpaper** (removes MgO).
2. Burn it held in **tongs**; collect the ash on a **watch glass**.

$2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$ + **Heat** + **Light**

Bright white flame, white ash; exothermic. Don't stare!

★★★ *Board ka favourite activity — Nitesh Bhaiya ki guarantee, isko yaad rakhna hi hai!*

4 CHEMICAL EQUATIONS

WORD (names only): Magnesium + Oxygen → Magnesium oxide

↓ replace names with formulae

SKELETAL (unbalanced): $Mg + O_2 \rightarrow MgO$ O: 2 vs 1 ×

↓ equalise atoms with coefficients

BALANCED (atoms equal): $2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$ ✔

LAW OF CONSERVATION OF MASS

- Mass is **neither created nor destroyed** → atoms of each element: **LHS = RHS**
- So every equation **must be balanced**

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$
H: 4 = 4, O: 2 = 2 ✔

4 g + 32 g (reactants) = 36 g (product)

Sentence → Symbols decoder

If the question says…	Write
reacts with, combines with, is added to, burns in, "and" between products	+
to form, to produce, to give, yields	→

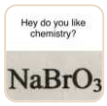
Solved Example 1: Sentence → equation

Q. "Hydrogen gas combines with nitrogen to form ammonia."
Skeletal: $H_2 + N_2 \rightarrow NH_3$ H 2 vs 3, N 2 vs 1 ×

$3H_{2(g)} + N_{2(g)} \rightarrow 2NH_{3(g)}$
Balanced: H 6 = 6, N 2 = 2 ✔

Formula toolkit

Diatomic gases are always written with "2" when free: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2. *"HON" + saare halogens jodi me!*



Criss-cross valencies for the right formula: $Ca^{2+} + OH^- \rightarrow Ca(OH)_2$

5 BALANCING ★★★ MOST ASKED

- **Hit-and-trial:** put **coefficients** in front of formulae
- Stop when every element has equal atoms on both sides

★★★ *Balancing pakka aayega — Nitesh Bhaiya ki guarantee, dimaag mein fit kar lo!*

Golden rules

1. Change only **coefficients**, never **subscripts**.
2. Balance a group (SO_4, NO_3, OH) that stays the same on both sides as **one unit**.
3. Order: metals → non-metals → H → O last. Smallest whole numbers.

× WRONG	✔ RIGHT
$H_2O \rightarrow H_2O_2$ to "fix" O	$2H_2O$ (H_2O_2 = new substance!)
$Mg + O \rightarrow MgO$	$2Mg + O_2 \rightarrow 2MgO$
CaOH	$Ca(OH)_2$

Solved Example 3: Burning of methane ★★☆

$CH_4 + O_2 \rightarrow CO_2 + H_2O$
H: 4 vs 2 × → write $2H_2O$; then O on right = 4 → write $2O_2$

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + \text{Heat}$
C 1 = 1, H 4 = 4, O 4 = 4 ✔

Solved Example 2: Iron + steam

1. Skeletal equation. Treat each formula as a sealed box.
   $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

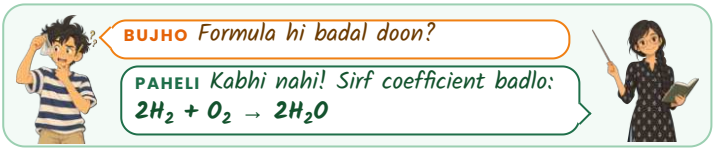
Atom	LHS	RHS
Fe	1	3 ×
H	2	2 ✔
O	1	4 ×

Start with the biggest formula, Fe_3O_4, and its biggest number: O = 4.

2. O: put 4 before H_2O. Now H = 8 on left, so 4 before H_2.
   $Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
3. Fe: 1 vs 3, so put 3 before Fe. Check: Fe 3 = 3, H 8 = 8, O 4 = 4 ✔
4. Add states (steam is a gas).
   $3Fe_{(s)} + 4H_2O_{(g)} \rightarrow Fe_3O_{4(s)} + 4H_{2(g)}$

MEME BREAK

LEFT: BALANCED (SWOLE)
$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
RIGHT: SKELETAL (CHEEMS)
$Fe + H_2O \rightarrow Fe_3O_4 + H_2$
Coefficients lagao, equation ko gym bhejo!



6 INFORMATIVE EQUATIONS

(a) Physical states ★★☆

Symbol	Meaning	Symbol	Meaning
(s)	solid	(g)	gas; ↑ = gas given off
(l)	liquid	(aq)	dissolved in water
↓	precipitate (insoluble solid separates out)		

Water: $H_2O(l)$ as liquid, $H_2O(g)$ as steam.

(b) Heat & conditions

- **+ Heat** on the product side means exothermic; on the reactant side, it means endothermic
- Δ, sunlight, electricity, pressure, catalyst → **on the arrow**

$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2(g)}$
limestone breaks down on heating

$2AgCl_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Cl_{2(g)}$
white silver chloride turns grey in sunlight

$N_{2(g)} + 3H_{2(g)} \xrightarrow[200\text{ atm, }450\text{–}500\,^\circ C]{\text{Fe (catalyst)}} 2NH_{3(g)}$
Haber process: making ammonia

A **catalyst** changes (usually speeds up) the rate and is **not used up**.

Solved Example 4: Statement → balanced equation

(a) Hydrogen sulphide burns in air to give water and sulphur dioxide. NCERT

$2H_2S(g) + 3O_2(g) \rightarrow 2H_2O(l) + 2SO_2(g)$

H 4 = 4, S 2 = 2, O 6 = 6 ✔

(b) Barium chloride reacts with aluminium sulphate to give aluminium chloride and a precipitate of barium sulphate. CBSE 2019 ★★★

$3BaCl_2(aq) + Al_2(SO_4)_3(aq) \rightarrow 2AlCl_3(aq) + 3BaSO_4(s)\downarrow$

Ba 3 = 3, Cl 6 = 6, Al 2 = 2, SO_4 3 = 3 ✔

Exam tips

- ✔ "Balanced equation" asked: an unbalanced one loses marks even with correct formulae.
- ✔ "With physical states": add (s), (l), (g) or (aq) to every formula.
- ✔ Use an arrow (→), never "=". Recount every atom at the end.
- ✔ Colours: PbI_2 yellow, $BaSO_4$ white, MgO white, $CuSO_4$ blue, $FeSO_4$ light green.

TRY YOURSELF: BALANCE THESE

Pehle khud solve karo, phir answer key check karo!

(i) $H_2 + Cl_2 \rightarrow HCl$

(ii) $Na + H_2O \rightarrow NaOH + H_2$

(iii) $KClO_3 \rightarrow KCl + O_2$

(iv) $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$

(v) $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

Day 1 mind map: revise in 2 minutes

REACTION
Reactants → products; new substance; atoms rearranged

5 SIGNS
Change in state, colour or temperature; gas; precipitate

CHEMICAL REACTIONS & EQUATIONS

3 EQUATIONS
Word → skeletal → balanced

BALANCING ★★★
Conservation of mass; coefficients only

INFORMATIVE
(s) (l) (g) (aq) ↑ ↓; + Heat (exo: product side, endo: reactant side); Δ, sunlight, catalyst on the arrow

Short answer questions

1. Dilute HCl is added to zinc granules. How will you prove that a chemical change has taken place? Give two arguments. CBSE SQP 2023
2. Write balanced equations for: (A) steam passed over red-hot iron, (B) natural gas burnt in air, (C) glucose reacting with oxygen in our body cells. CBSE 2025

7 PRACTICE MCQS

Copy-paste nahi chalega! Khud solve karo, answer key page 4 par hai.

1. Barium chloride solution is mixed with sodium sulphate solution. Which product is formed in the solid state? CBSE SQP 2023 ★★★

(a) Barium chloride (b) Barium sulphate
(c) Sodium chloride (d) Sodium sulphate

2. In the same reaction, which substance stays dissolved in the solution? CBSE SQP 2025 ★★★

(a) $BaCl_2$ (b) $BaSO_4$
(c) Na_2SO_4 (d) NaCl

3. Find *p*, *q*, *r* in the balanced equation: $p\ Pb(NO_3)_2(s) \rightarrow q\ PbO(s) + r\ NO_2(g) + O_2(g)$ CBSE SQP 2024 ★★☆

(a) 2, 2, 4 (b) 2, 4, 2
(c) 2, 4, 4 (d) 4, 2, 2

4. A magnesium ribbon is burnt over a watch glass. Which is the correct observation? CBSE SQP 2022 ★★★

(a) A brown powder of magnesium oxide is formed.
(b) A colourless gas that turns lime water milky is evolved.
(c) The ribbon burns with a brilliant white light.
(d) A reddish-brown gas smelling of burning sulphur is evolved.

5. Which statement is NOT true about the burning of magnesium ribbon in air? CBSE 2025 ★★★

(a) It burns with a dazzling white flame.
(b) A white powder is formed on burning.
(c) It is an endothermic reaction.
(d) It is an example of a combination reaction.

6. Dilute hydrochloric acid is added to granulated zinc in a test tube. The correct observation is: ★☆☆

(a) The surface of the metal becomes shiny.
(b) The reaction mixture turns milky.
(c) A pungent-smelling gas is given off.
(d) A colourless, odourless gas is evolved.

7. We balance a chemical equation because of the: ★★★

(a) law of constant proportions (b) law of conservation of mass
(c) law of multiple proportions (d) Avogadro's law

8. The balanced form of $Al + O_2 \rightarrow Al_2O_3$ is: ★★★

(a) $2Al + 3O_2 \rightarrow Al_2O_3$ (b) $4Al + 3O_2 \rightarrow 2Al_2O_3$
(c) $4Al + 2O_2 \rightarrow 2Al_2O_3$ (d) $2Al + O_2 \rightarrow Al_2O_3$

9. "+ Heat" written on the product side of an equation shows that the reaction is: ★★★

(a) endothermic (b) exothermic
(c) a precipitation reaction (d) not balanced

10. **Assertion (A):** $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$ is a balanced equation for the action of steam on iron.
Reason (R): The law of conservation of mass holds good for a chemical equation. CBSE 2020 ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	$BaSO_4$ is the insoluble white precipitate.
2	(d)	NaCl stays dissolved in water.
3	(a)	$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
4	(c)	Dazzling white light; white MgO powder.
5	(c)	It gives out heat and light: exothermic.
6	(d)	Hydrogen gas: colourless, odourless.
7	(b)	Atoms can't be created or destroyed.
8	(b)	Al 4 = 4, O 6 = 6.
9	(b)	Heat given out = exothermic.
10	(a)	Balanced because mass is conserved.

Try-yourself answers

(i) $H_2 + Cl_2 \rightarrow 2HCl$
(ii) $2Na + 2H_2O \rightarrow 2NaOH + H_2$
(iii) $2KClO_3 \rightarrow 2KCl + 3O_2$
(iv) $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$
(v) $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

Model answers: short answer

1 Zinc + dilute HCl: two proofs of a chemical change

- **Gas evolved:** colourless H_2 bubbles; burns with a "pop"
- **Temperature change:** test tube becomes warm
- Zinc dissolves; new substance $ZnCl_2$ forms

$$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$$

2 Balanced equations

$$3Fe_{(s)} + 4H_2O_{(g)} \rightarrow Fe_3O_{4(s)} + 4H_{2(g)}$$
(A) steam over red-hot iron

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + \text{Heat}$$
(B) natural gas (methane) burning

$$C_6H_{12}O_{6(aq)} + 6O_{2(aq)} \rightarrow 6CO_{2(aq)} + 6H_2O_{(l)} + \text{Energy}$$
(C) respiration in our cells

Chapter 1 hit-list: Eduspark PYQ analysis

Topic / reaction	Day	Rating
Balancing equations	1	★★★
Oxidation & reduction (redox)	3	★★★
Exothermic vs endothermic	1	★★★
$BaCl_2 + Na_2SO_4$ (white ppt)	1	★★★
$Fe + CuSO_4$ (colour change)	1–2	★★★
Burning Mg ribbon	1	★★★
AgCl / AgBr in sunlight	2	★★☆
Corrosion & rancidity	3	★★☆
Heating $Pb(NO_3)_2$ (brown NO_2)	2	★★☆

★★★ Ye sab pakka aayenge — Nitesh Bhaiya ki guarantee hai! Dimaag mein set kar lo.

Day 1 checklist: tick before you sleep

- ☐ I can spot a chemical change and name its 5 signs
- ☐ I can turn a word statement into a skeletal equation
- ☐ I can balance by hit-and-trial without changing any formula
- ☐ I can add (s), (l), (g), (aq), heat and conditions
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 1 COMPLETE +100 XP earned. Next: Level 2

BUJHO Samajh gaya! Naya substance bana toh reaction, aur balancing mein sirf coefficient badalte hain!

PAHELI Shabaash! Kal Level 2: types of reactions.

Real life: Diwali crackers

- Fuse lit → **fuel + oxidiser** react instantly
- Heat, light & gas released; fast gas = **bang**
- Metal salts glow: strontium **red**, barium **green**, sodium **yellow**

DAILY-LIFE REACTIONS

Milk → curd, rusting, cooking, digestion, respiration, cut apple turning brown

Coming up: Day 2

Types of reactions: combination, decomposition, and the famous displacement drama where iron kicks copper out of $CuSO_4$. Poor Cu!

Valency chart: keep handy

Val.	Positive ions	Negative ions
1	Na^+, K^+, Ag^+, H^+, NH_4^+	Cl^-, OH^-, NO_3^-, HCO_3^-
2	Mg^{2+}, Ca^{2+}, Zn^{2+}, Cu^{2+}, Fe^{2+}, Ba^{2+}, Pb^{2+}	O^{2-}, S^{2-}, SO_4^{2-}, CO_3^{2-}
3	Al^{3+}, Fe^{3+}	PO_4^{3-}

NITESH SIR SAYS

Kal milte hai Day 2 pe.
Consistency hi jalwa hai!

★ How we starred (Eduspark PYQ analysis): we counted how often each reaction or topic has come in Chapter 1 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 02 · LEVEL 2

CHEMISTRY | CHAPTER 1 | PART 2 · ASTRA SERIES

COMBINATION & DECOMPOSITION

Goal: classify reactions and master the five NCERT activities.

by Nitesh Prakash

01 Combination | 02 Thermal | 03 Electrolytic | 04 Photolytic | 05 Comparison | +100 XP today | Level 2 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 COMBINATION REACTION ★★★ MOST ASKED

DEFINITION

- Two or more reactants combine to form a **single product**
- General form: **A + B → AB**
- Most combination reactions **release heat** (exothermic)

REACTANTS		PRODUCT
two or more (A + B)	→ join together	only one product (AB)

Activity: Quicklime + water ★★★

1. Take a small amount of **calcium oxide** (quicklime) in a beaker.
2. Add water slowly. It reacts **vigorously** with a hissing sound; the beaker becomes **hot**.

$$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \text{Heat}$$

Quicklime + water → **slaked lime**; an exothermic reaction

★★★ *Quicklime + paani = slaked lime + garmi. Ye pakka aayega, Nitesh Bhaiya ki guarantee!*

Whitewashing: why walls shine

A solution of slaked lime is applied on walls. It reacts slowly with CO_2 in air and forms a thin layer of calcium carbonate, which gives a **shiny finish** after 2–3 days.

$$Ca(OH)_{2(aq)} + CO_{2(g)} \rightarrow CaCO_{3(s)} + H_2O_{(l)}$$

Ca 1 = 1, C 1 = 1, O 4 = 4, H 2 = 2 ✔

Note: this step gives two products, so it is not itself a combination reaction. Marble and limestone are also $CaCO_3$.

Limewater vs limewash: limewater is the clear, filtered solution of $Ca(OH)_2$ (turns milky with CO_2); limewash (milk of lime) is a suspension of $Ca(OH)_2$ used for whitewashing.

More combination examples

Type	Balanced equation
Element + element	$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \text{Heat}$ burning of coal
Element + element	$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$ formation of water
Compound + element	$2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$
Compound + compound	$NH_{3(g)} + HCl_{(g)} \rightarrow NH_4Cl_{(s)}$

2 DECOMPOSITION REACTION ★★★ MOST ASKED

DEFINITION

- A **single reactant** breaks down into **two or more simpler products**
- General form: **AB → A + B** (the reverse of combination)
- Needs energy to break bonds, so most are **endothermic**

Three ways to supply the energy

Type	Energy	Example
Thermal	heat (Δ)	$CaCO_3$, $FeSO_4$, $Pb(NO_3)_2$
Electrolytic	electricity	water → $H_2 + O_2$
Photolytic	light	AgCl, AgBr in sunlight

3 THERMAL DECOMPOSITION

Activity: Heating ferrous sulphate crystals ★★★

1. Heat a few **green** ferrous sulphate crystals in a dry boiling tube.
2. The crystals first lose water and change colour, then decompose: **smell of burning sulphur**.

$$2FeSO_{4(s)} \xrightarrow{\Delta} Fe_2O_{3(s)} + SO_{2(g)} + SO_{3(g)}$$

Fe 2 = 2, S 2 = 2, O 8 = 8 ✔

Green crystals → **reddish-brown** ferric oxide; SO_2 gives the sulphur smell.

Precautions: hold the tube with a test-tube holder, keep its mouth away from yourself and others, heat gently, and smell the gas only by wafting.

Activity: Heating lead nitrate ★★★

Heat white lead nitrate powder in a boiling tube: a crackling sound and **brown fumes** of nitrogen dioxide.

$$2Pb(NO_3)_{2(s)} \xrightarrow{\Delta} 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

Pb 2 = 2, N 4 = 4, O 12 = 12 ✔; residue PbO is yellow

Limestone: heart of the cement industry ★★☆

$$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2(g)}$$

limestone → quicklime + carbon dioxide

Quicklime is used in the manufacture of cement.

BUJHO *Lead nitrate garam kiya, brown dhuaan nikla. Kuch jal gaya kya?*

PAHELI *Kuch nahi jala! Brown fumes* NO_2 *gas ke hain, aur peela residue PbO hai.*

4 ELECTROLYTIC DECOMPOSITION ★★★ MOST ASKED

Activity: Electrolysis of water

1. Fix two **carbon electrodes** in a plastic mug of water; add a few drops of **dilute sulphuric acid**.
2. Invert two water-filled test tubes over the electrodes and pass current from a 6 V battery.
3. Bubbles form at both electrodes and push the water out of the tubes.

$$2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2H_{2(g)} + O_{2(g)}$$

H 4 = 4, O 2 = 2 ✔

Electrode	Gas	Volume	Test
Cathode (−)	H_2	2	burns with a "pop"
Anode (+)	O_2	1	candle burns brighter

BUJHO *Dono test tubes mein gas barabar honi chahiye na?*

PAHELI *Nahi!* H_2O *mein H ke 2 atom, O ka 1. Isliye* H_2 *ka volume* ***double*** *hota hai.*

Why add acid? Other examples

- Pure water is a **poor conductor**; a little dilute acid makes it conduct
- Electricity is the energy that breaks the compound

$$2NaCl_{(l)} \xrightarrow{\text{Electricity}} 2Na_{(s)} + Cl_{2(g)}$$

molten sodium chloride (used to get sodium metal)

$$2Al_2O_{3(l)} \xrightarrow{\text{Electricity}} 4Al_{(s)} + 3O_{2(g)}$$

molten alumina gives aluminium; Al 4 = 4, O 6 = 6 ✔

MEME BREAK

Bijli ka aashirwad mila, paani tootkar H_2 *aur* O_2 *ban gaya! Hydrogen double, oxygen single.*

5 PHOTOLYTIC DECOMPOSITION ★★☆

Activity: Silver chloride in sunlight ★★☆

1. Take about 2 g of **white** silver chloride in a china dish.
2. Keep it in sunlight for some time: it turns **grey**.

$$2AgCl_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Cl_{2(g)}$$

white → grey (silver metal); Ag 2 = 2, Cl 2 = 2 ✔

$$2AgBr_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Br_{2(g)}$$

pale yellow silver bromide behaves the same way

Where is photolysis used?

- ✔ **Black-and-white photography:** AgBr on film darkens where light falls.
- ✔ AgCl and AgBr are stored in **dark coloured bottles** so light cannot decompose them.

MEME BREAK

Safed AgCl dhoop mein rakha, thodi der mein grey! Silver ban gaya, Cl_2 *ud gayi.*

6 COMBINATION VS DECOMPOSITION

COMBINATION	DECOMPOSITION
Two or more reactants → **one product**	One reactant → **two or more products**
A + B → AB	AB → A + B
Usually **exothermic** (heat released)	Usually **endothermic** (heat, light or electricity absorbed)
$CaO + H_2O \rightarrow Ca(OH)_2$	$CaCO_3 \rightarrow CaO + CO_2$

★★★ *Combination ulta = decomposition! Electrolysis ka 2 : 1 pakka aayega, Nitesh Bhaiya ki guarantee.*

× WRONG	✔ RIGHT
Energy: heat only	Heat, electricity **or** light
O_2 collects at the cathode	H_2 at cathode, O_2 at anode
$FeSO_4$ residue is green	Reddish-brown Fe_2O_3
Brown fumes = smoke	Brown fumes = NO_2 gas

Exam tips

- ✔ Name the type **and** the energy: "thermal decomposition (heat)".
- ✔ Write Δ, Sunlight or Electricity over the arrow.
- ✔ Learn the colours: $FeSO_4$ green, Fe_2O_3 reddish-brown, NO_2 brown, AgCl white → grey.

Day 2 mind map: revise in 2 minutes

COMBINATION
A + B → AB; one product; usually exothermic ($CaO + H_2O$)

THERMAL ★★★
Heat: $CaCO_3$, $FeSO_4$ ($SO_2 + SO_3$), $Pb(NO_3)_2$ (brown NO_2)

TYPES OF REACTIONS I

ELECTROLYTIC ★★★
Electricity: $2H_2O \rightarrow 2H_2 + O_2$; volume 2 : 1

PHOTOLYTIC
Sunlight: AgCl, AgBr turn grey (black-and-white photography)

7 PRACTICE MCQS

Main? Haan tum! Pehle khud solve karo, answer key page 4 par hai.

1. Quicklime reacts vigorously with water and releases heat. The product formed and the type of reaction are: ★★★

(a) $Ca(OH)_2$; decomposition
(b) $Ca(OH)_2$; combination
(c) $CaCO_3$; combination
(d) CaO_2; displacement

2. On heating ferrous sulphate crystals, the gases evolved are: ★★★

(a) SO_2 only
(b) SO_2 and SO_3
(c) SO_3 and O_2
(d) H_2S and SO_2

3. The colour of the residue left after strongly heating ferrous sulphate is: ★★☆

(a) green
(b) white
(c) reddish-brown
(d) black

4. On heating lead nitrate, the products formed are: CBSE 2024 ★★★

(a) NO, PbO and O_2
(b) NO_2, Pb and O_2
(c) NO_2, PbO and N_2
(d) NO_2, PbO and O_2

5. In the electrolysis of water, the gas collected at the cathode and the volume ratio of hydrogen to oxygen are: ★★★

(a) oxygen; 2 : 1
(b) hydrogen; 2 : 1
(c) hydrogen; 1 : 2
(d) oxygen; 1 : 2

6. Silver chloride is stored in dark coloured bottles because it: ★★☆

(a) reacts with glass
(b) decomposes in light
(c) absorbs moisture
(d) is volatile

7. Which of these is a decomposition reaction? ★★★

(a) $2H_2 + O_2 \rightarrow 2H_2O$
(b) $CaCO_3 \rightarrow CaO + CO_2$
(c) $C + O_2 \rightarrow CO_2$
(d) $CaO + H_2O \rightarrow Ca(OH)_2$

8. A few drops of dilute sulphuric acid are added to water before its electrolysis to: CBSE 2020 ★★☆

(a) increase its conductivity
(b) produce more oxygen
(c) act as a reactant
(d) keep the water cool

9. **Assertion (A):** Decomposition reactions are generally endothermic.
Reason (R): Energy in the form of heat, light or electricity is needed to break down the reactant. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Whitewashed walls develop a shiny finish after two to three days.
Reason (R): Calcium hydroxide reacts slowly with carbon dioxide in air to form calcium carbonate. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1 A student heats white lead nitrate powder in a boiling tube. (a) State two observations. (b) Write the balanced equation with states. (c) Name the type of reaction.

2 In the electrolysis of water, why is the volume of gas in one test tube double that in the other? Name both gases and give a test for each. CBSE 2020

3 What is slaked lime? Write the equation for its formation. How does it give walls a shiny finish after whitewashing?

Case-based question (4 marks) ★★★

A white solid **X**, found as limestone and marble, is heated strongly. It gives a white solid **Y** and a gas **Z** that turns lime water milky. **Y** reacts vigorously with water to give **W**, which is used for whitewashing.

1 Identify X, Y, Z and W.

2 Write the balanced equation for the reaction of Y with water and name its type.

3 Is the heating of X exothermic or endothermic? Give a reason.

4 Why does a whitewashed wall shine after two to three days? Write the equation.

TRY YOURSELF: NAME THE TYPE

Combination ya decomposition? Aur energy kaunsi?

(i) $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

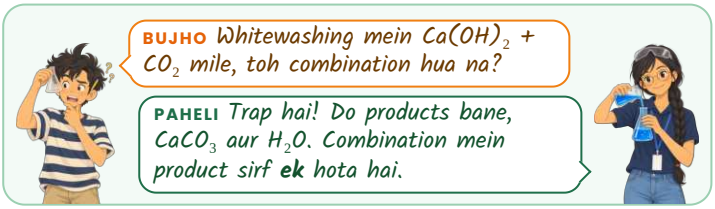
(ii) $N_2 + 3H_2 \rightarrow 2NH_3$

(iii) $2H_2O_2 \xrightarrow{Light} 2H_2O + O_2$

(iv) $2Mg + O_2 \rightarrow 2MgO$

(v) $2HgO \xrightarrow{\Delta} 2Hg + O_2$

(vi) $CaO + CO_2 \rightarrow CaCO_3$



Revision sheet: colours, gases, smells

Reaction	What you see
CaO + water	hissing; beaker gets hot
Heating $FeSO_4$	green → reddish-brown; burning sulphur smell
Heating $Pb(NO_3)_2$	brown NO_2 fumes; yellow PbO residue
Heating $CaCO_3$	CO_2 turns lime water milky
Water electrolysis	bubbles; $H_2 : O_2 = 2 : 1$
AgCl in sunlight	white → grey

Answer key: MCQs

Q	Ans	Why
1	(b)	$CaO + H_2O \rightarrow Ca(OH)_2$; one product.
2	(b)	$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$
3	(c)	Ferric oxide, Fe_2O_3, is reddish-brown.
4	(d)	$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
5	(b)	H_2 at the cathode; twice the volume of O_2.
6	(b)	Light decomposes AgCl to Ag + Cl_2.
7	(b)	One reactant gives two products.
8	(a)	Pure water conducts poorly.
9	(a)	Energy is needed to break bonds.
10	(a)	The $CaCO_3$ layer gives the shine.

Try-yourself answers

(i) Thermal decomposition
(ii) Combination
(iii) Photolytic decomposition
(iv) Combination
(v) Thermal decomposition
(vi) Combination

Case-based answers

1. X = $CaCO_3$ (limestone); Y = CaO (quicklime); Z = CO_2; W = $Ca(OH)_2$ (slaked lime).
2. $CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)}$ + **Heat**. Type: combination.
3. Endothermic: heat must be supplied to decompose $CaCO_3$.
4. $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$; the thin $CaCO_3$ layer shines.

Model answers: short and long

1 Heating lead nitrate

- **Observations:** brown fumes of NO_2; a yellow residue of PbO (crackling sound)
- **Type:** thermal decomposition

$$2Pb(NO_3)_{2(s)} \xrightarrow{\Delta} 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

2 Electrolysis of water: why 2 : 1

- Each H_2O molecule gives 2 H atoms for 1 O atom, so 2 volumes of H_2 form for 1 volume of O_2
- **H_2** (cathode): burns with a pop sound; **O_2** (anode): a candle burns brighter in it

3 Slaked lime and whitewashing

- Slaked lime is **calcium hydroxide**, $Ca(OH)_2$
- On walls it reacts with CO_2 to form a thin, shiny $CaCO_3$ layer

$$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \text{Heat}$$

Real life: decomposition around you

- **Cement:** limestone is heated in kilns to make quicklime
- **Digestion:** starch in food breaks down into simple sugars
- **Metals:** sodium and aluminium are extracted by electrolysis
- **Car airbags:** sodium azide decomposes in a flash to fill the bag with N_2

Day 2 checklist: tick before you sleep

- [ ] I can define combination and decomposition with one example each
- [ ] I can write the CaO + water and whitewashing equations
- [ ] I can write the heating equations of ferrous sulphate, lead nitrate and limestone
- [ ] I can explain electrolysis of water (2 : 1) and AgCl in sunlight
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 2 COMPLETE +100 XP earned. Next: Level 3

BUJHO *Samajh gaya! Jodna matlab combination, aur todna matlab decomposition!*

PAHELI *Shabaash! Kal Level 3: displacement aur redox ki baari.*

Name → formula bank

Common name	Chemical name	Formula
Quicklime	calcium oxide	CaO
Slaked lime	calcium hydroxide	$Ca(OH)_2$
Limestone, marble	calcium carbonate	$CaCO_3$
Green vitriol	ferrous sulphate	$FeSO_4$
Ferric oxide	iron(III) oxide	Fe_2O_3
Litharge	lead(II) oxide	PbO

EXTRA FROM CLASS

Gentle heating: $CuSO_4.5H_2O$ (blue) → $CuSO_4$ (white) + $5H_2O$
Strong heating (above about 650 °C, beyond NCERT): $2CuSO_4 \rightarrow 2CuO + 2SO_2 + O_2$; black CuO is left ($SO_3$ formed first breaks into $SO_2 + O_2$).
Cu 2 = 2, S 2 = 2, O 8 = 8 ✔

Coming up: Day 3

Displacement drama: iron kicks copper out of $CuSO_4$, plus double displacement, precipitates, and oxidation and reduction.

NITESH SIR SAYS
Har reaction ko roz ek baar likho aur colours yaad rakho. ***Kal milte hai Day 3 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each reaction or topic has come in Chapter 1 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 03 · LEVEL 3

CHEMISTRY | CHAPTER 1 | PART 3 · ASTRA SERIES

DISPLACEMENT & REDOX REACTIONS

Goal: displacement, precipitation, oxidation and reduction.

by Nitesh Prakash

01 Displacement | 02 Reactivity | 03 Precipitation | 04 Redox | 05 Agents

+100 XP today | Level 3 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 DISPLACEMENT REACTION

DEFINITION

- A **more reactive** element pushes out a **less reactive** element from its compound
- General form: **A + BC → AC + B** (A is more reactive than B)

NCERT Activity 1.9: Iron nail in $CuSO_4$ ★★★

$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$

copper sulphate (blue) → iron sulphate (light green)

- Blue colour of the solution **fades to light green**
- A **reddish-brown coat** of copper forms on the nail
- Iron is **more reactive** than copper, so it displaces Cu

More examples (both NCERT):

$$Zn_{(s)} + CuSO_{4(aq)} \rightarrow ZnSO_{4(aq)} + Cu_{(s)}$$

zinc displaces copper; blue solution turns colourless

$$Pb_{(s)} + CuCl_{2(aq)} \rightarrow PbCl_{2(aq)} + Cu_{(s)}$$

lead is above copper, so it displaces copper

MEME BREAK

*Iron nail $CuSO_4$ mein ghusa aur copper ko bahar nikal diya. Copper: "Kya tapleek hai aapko?" Iron: "Main **zyada reactive** hoon!"*

BUJHO *Toh copper bhi iron ko $FeSO_4$ se nikal dega na?*

PAHELI *Bilkul nahi! Cu kam reactive hai. Cu + $FeSO_4$ → **koi reaction nahi.***

2 REACTIVITY SERIES: WHO WINS?

MOST REACTIVE → LEAST REACTIVE

K | Na | Ca | Mg | Al | Zn | Fe | Pb | [H] | Cu | Hg | Ag | Au

- ✔ A metal **higher** in the series displaces a metal **below** it
- ✔ A metal **lower** in the series cannot displace one above it
- ✔ Metals above hydrogen displace H_2 from dilute acids

CLASS TRICK

Katrina Ne Car Mangi Alto Zen Ferrari Saturn Phir Bhi Hoyee Kyu Mili Silver Audi = K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, H, Cu, Hg, Ag, Au. Class version includes Tin (Sn) between Fe and Pb; NCERT's list skips Sn, both are correct.

REACTIVITY CHECK

*Zn + $CuSO_4$: reaction ho gaya, sab khush. Cu + $ZnSO_4$: kuch nahi hua. **Series yaad karo!***

3 DOUBLE DISPLACEMENT REACTION

DEFINITION

- Two compounds **exchange their ions** to form two new compounds
- General form: **AB + CD → AD + CB**
- **Three types:** (1) **precipitation**: forms an **insoluble solid** (precipitate, ppt), shown with ↓ or (s), e.g. $BaSO_4$ below; (2) **neutralisation**: HCl + NaOH → NaCl + H_2O; (3) **gas-forming** (vinegar + baking soda): $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2\uparrow$

NCERT Activity 1.10: $BaCl_2$ + Na_2SO_4 ★★★

$$Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$$

Ba^{2+} and Na^+ swap partners

- A **white**, insoluble solid of $BaSO_4$ settles down
- NaCl stays dissolved in water

$$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)}\downarrow + 2KNO_{3(aq)}$$

yellow ppt of lead iodide

Precipitate colours: learn by heart

Precipitate	Formed from	Colour
$BaSO_4$	$BaCl_2 + Na_2SO_4$	**White**
PbI_2	$Pb(NO_3)_2 + KI$	**Yellow**
AgCl	$AgNO_3 + NaCl$	**White** (curdy)
$CaCO_3$	$Ca(OH)_2 + CO_2$ (limewater)	**White** (milky)
CuS	$CuSO_4 + H_2S$	**Black**

BUJHO *Ppt matlab? Sab kuch paani mein ghul jaata hai na?*

PAHELI *Nahi! $BaSO_4$ ghulta nahi, **white solid** neeche baith jaata hai.*

★★★ *$BaCl_2$ + Na_2SO_4 ka white ppt aur PbI_2 ka yellow ppt: **Nitesh Bhaiya ki guarantee, board mein aayega!***

4 OXIDATION AND REDUCTION

OXIDATION	REDUCTION
✔ Gain of oxygen	✔ Loss of oxygen
✔ Loss of hydrogen	✔ Gain of hydrogen
$2Cu + O_2 \rightarrow 2CuO$	$CuO + H_2 \rightarrow Cu + H_2O$

NCERT Activity 1.11: Heating copper powder ★★★

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

brown copper powder → **black** copper(II) oxide

Now pass **hydrogen gas** over the hot black CuO:

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

black coating turns **brown** again: copper is back

DEFINITION

- **Redox reaction**: one substance is **oxidised** while another is **reduced** in the same reaction
- Oxidation and reduction always happen **together**

OXIDATION: H_2 gains oxygen → H_2O

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

REDUCTION: CuO loses oxygen → Cu

MEME BREAK

*CuO ne apna oxygen H_2 ko de diya. CuO: "Ouch, mera oxygen!" Isi ko kehte hain **reduction**.*

BUJHO *Ek hi reaction mein oxidation aur reduction dono kaise?*

PAHELI *Jo O deta hai, wo reduce hota hai; jo O leta hai, wo oxidise. **Dono saath-saath!***

Solved Example: Displacement and redox together ★★☆

$$Fe_2O_{3(s)} + 2Al_{(s)} \rightarrow 2Fe_{(l)} + Al_2O_{3(s)} + \text{Heat}$$

Fe 2 = 2, Al 2 = 2, O 3 = 3 ✔

- Al is more reactive than Fe, so it **displaces** iron
- Fe_2O_3 loses O: **reduced**; Al gains O: **oxidised**
- This thermite reaction is used to **join railway tracks**

EXTRA FROM CLASS

Non-metals displace too: a more reactive halogen pushes out a less reactive one (**F_2 > Cl_2 > Br_2 > I_2**). **$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$**; the solution turns orange-brown.

Real life: redox all around you

- **Respiration:** glucose is oxidised to CO_2 and water, giving energy
- **Burning fuel:** carbon is oxidised to CO_2
- **Extracting zinc:** ZnO is reduced by carbon (coke)

More redox examples: spot O and H ★★★

$$ZnO_{(s)} + C_{(s)} \rightarrow Zn_{(s)} + CO_{(g)}$$

ZnO loses O: **reduced**; C gains O: **oxidised**

$$MnO_{2(s)} + 4HCl_{(aq)} \rightarrow MnCl_{2(aq)} + 2H_2O_{(l)} + Cl_{2(g)}\uparrow$$

MnO_2 loses O: **reduced**; HCl loses H: **oxidised**

$$H_2S_{(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)} + S_{(s)}$$

H_2S loses H: **oxidised**; Cl_2 gains H: **reduced**

$$4Na_{(s)} + O_{2(g)} \rightarrow 2Na_2O_{(s)}$$

Na gains O: **oxidised**; O_2 is reduced

5 OXIDISING AND REDUCING AGENTS

OXIDISING AGENT — gives oxygen or removes hydrogen; **itself gets reduced**

→ always a pair

REDUCING AGENT — removes oxygen or gives hydrogen; **itself gets oxidised**

Identify: substance oxidised, reduced and the agents ★★★

Reaction	Oxidised (RA)	Reduced (OA)
$CuO + H_2 \rightarrow Cu + H_2O$	H_2	CuO
$ZnO + C \rightarrow Zn + CO$	C	ZnO
$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$	HCl	MnO_2
$H_2S + Cl_2 \rightarrow 2HCl + S$	H_2S	Cl_2

RA = reducing agent, OA = oxidising agent. The substance oxidised is the reducing agent; the substance reduced is the oxidising agent.

BUJHO *Oxidising agent toh khud oxidise hota hoga na?*

PAHELI *Ulta! Wo **dusron ko oxidise** karta hai aur khud reduce hota hai.*

× WRONG	✔ RIGHT
Oxidising agent gets oxidised	Oxidising agent gets **reduced**
In $CuO + H_2$, Cu is oxidised	CuO is **reduced**; H_2 is oxidised
With MnO_2, HCl is the oxidising agent	MnO_2 is the oxidising agent

★★★ *Oxidised ya reduced pehchano, agent batao: Nitesh Bhaiya ki guarantee, har saal ka sawaal!*

BEYOND NCERT: THE ELECTRON VIEW

- **Oxidation** = loss of electrons; **reduction** = gain of electrons
- In $Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$, iron loses $2e^-$ and Cu^{2+} gains $2e^-$
- So the iron nail reaction is a displacement **and** a redox reaction

Trick: oxidising agent oxygen deta hai, khud reduce. Reducing agent oxygen leta hai, khud oxidise!

6 EXAM TIPS AND QUICK RECAP

Exam tips

- ✔ Displacement: always name the **colour change** and the coating (brown Cu).
- ✔ Write ↓ or (s) for every precipitate and name its colour.
- ✔ For redox, first mark who **gains or loses O/H**, then name the agents.
- ✔ Check balancing: MnO_2 + **4**HCl gives **2**H_2O.

TRY YOURSELF: OXIDISED OR REDUCED?

Har reaction mein oxidised, reduced aur dono agents likho. Answer page 4 par!

(i) $PbO + C \rightarrow Pb + CO$ (ii) $2Mg + CO_2 \rightarrow 2MgO + C$

(iii) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

(iv) $SO_2 + 2H_2S \rightarrow 3S + 2H_2O$

7 PRACTICE MCQS

Answer dekh ke solve kiya toh horse bhi hasega! Khud karo, key page 4 par hai.

1. An iron nail is put into each of two test tubes: P with copper(II) sulphate solution and Q with silver nitrate solution. Which observation is correct? CBSE SQP 2025 ★★★

(a) P: blue coating on the nail; Q: no reaction
(b) P: brown coating on the nail; Q: silver coating on the nail
(c) No reaction in either test tube
(d) P: no reaction; Q: silver coating on the nail

2. Which reaction is different from the other three? CBSE 2024 ★★★

(a) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(b) $CaO + H_2O \rightarrow Ca(OH)_2$
(c) $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$
(d) $ZnCl_2 + H_2S \rightarrow ZnS + 2HCl$

3. Pick the correct statement about the reaction $CuO + H_2 \rightarrow Cu + H_2O$. CBSE Compartment 2024 ★★★

(a) CuO is oxidised and H_2 is reduced.
(b) H_2 is oxidised and CuO is reduced.
(c) CuO is the reducing agent.
(d) H_2 is the oxidising agent.

4. In $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$, the oxidising agent is: ★★★

(a) MnO_2 (b) HCl
(c) $MnCl_2$ (d) Cl_2

5. Lead powder is added to blue copper sulphate solution. After a few hours the solution becomes colourless. This is an example of a: CBSE SQP 2024 ★★☆

(a) combination reaction (b) decomposition reaction
(c) displacement reaction (d) double displacement reaction

6. Lead nitrate solution is mixed with potassium iodide solution. The yellow precipitate formed is of: ★★★

(a) potassium nitrate (b) lead iodide
(c) lead nitrate (d) potassium iodide

7. Copper powder heated in a china dish turns black on the surface. The black substance is: ★★☆

(a) copper(II) oxide (b) copper sulphate
(c) copper carbonate (d) copper hydroxide

8. In the reaction $ZnO + C \rightarrow Zn + CO$, the reducing agent is: ★★☆

(a) ZnO (b) C
(c) Zn (d) CO

9. **Assertion (A):** When an iron nail is dipped in copper sulphate solution, the blue colour of the solution fades.
Reason (R): Iron is more reactive than copper and displaces copper from copper sulphate solution. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, hydrogen acts as the oxidising agent.
Reason (R): Hydrogen removes oxygen from CuO. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer practice

1. Why does the colour of copper sulphate solution change when an iron nail is dipped in it? NCERT
2. H_2S is passed through blue $CuSO_4$ solution: the colour fades and a black ppt forms. Name the reaction type and the ppt, explain the fading, and write the equation. CBSE 2020
3. A shiny brown element X turns black when heated in air. Name X and the black compound. Write the equation, and show how the change can be reversed. NCERT ★★★

★★★ Iron nail + $CuSO_4$ ka colour change: Nitesh Bhaiya ki guarantee, ye pakka aayega!

Day 3 mind map: revise in 2 minutes

DISPLACEMENT
A + BC → AC + B; more reactive metal wins

DOUBLE DISPLACEMENT
AB + CD → AD + CB; ions swap; ppt often forms

TYPES OF REACTIONS II & REDOX

OXIDATION
Gain of O or loss of H

REDUCTION
Loss of O or gain of H

AGENTS ★★★
Oxidising agent is reduced itself; reducing agent is oxidised itself. Oxidation and reduction together = redox.

Did you know? Copper wire in $AgNO_3$

$Cu_{(s)} + 2AgNO_{3(aq)} \rightarrow Cu(NO_3)_{2(aq)} + 2Ag_{(s)}$
shiny silver crystals form; solution turns blue

Answer key: MCQs

Q	Ans	Why
1	(b)	Fe displaces Cu (brown) and Ag (silver).
2	(b)	Combination; the other three swap ions.
3	(b)	H_2 gains O; CuO loses O.
4	(a)	MnO_2 loses oxygen, so it is reduced.
5	(c)	Pb is above Cu, so it displaces Cu.
6	(b)	PbI_2 is the yellow precipitate.
7	(a)	$2Cu + O_2 \rightarrow 2CuO$ (black).
8	(b)	C takes oxygen from ZnO.
9	(a)	R correctly explains the colour change.
10	(d)	H_2 is the reducing agent; R is true.

Try-yourself answers

No.	Oxidised (RA)	Reduced (OA)
(i)	C	PbO
(ii)	Mg	CO_2
(iii)	CO	Fe_2O_3
(iv)	H_2S	SO_2

Model answers: short and long answer

1 Iron nail in $CuSO_4$

- Iron is **more reactive** than copper, so it displaces copper
- Blue $CuSO_4$ changes to light green $FeSO_4$; brown Cu deposits on the nail

$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$

2 H_2S and $CuSO_4$

- **Double displacement** (precipitation) reaction; black ppt = **copper sulphide** (CuS)
- Colour fades because blue $CuSO_4$ is used up as Cu^{2+} ions leave as CuS

$$CuSO_{4(aq)} + H_2S_{(g)} \rightarrow CuS_{(s)}\downarrow + H_2SO_{4(aq)}$$

3 X = copper (Cu); black compound = copper(II) oxide (CuO)

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

oxidation of copper

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

reversed by passing H_2 over hot CuO (reduction)

BOARD ANSWER FRAME: REDOX

Balanced equation with states → mark who gains or loses O/H → name the agents (OA is reduced, RA is oxidised).

NITESH SIR SAYS

Jo oxygen le, wo oxidise; jo oxygen de, wo reduce. Bas itna yaad rakho!

Day 3 checklist: tick before you sleep

- ☐ I can explain displacement using the reactivity series
- ☐ I can write the $BaSO_4$ and PbI_2 precipitation equations
- ☐ I can define oxidation and reduction using oxygen and hydrogen
- ☐ I can name the oxidising and reducing agents in a redox reaction
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 3 COMPLETE +100 XP earned. Next: Level 4

BUJHO *Samajh gaya! Zyada reactive metal kam reactive ko nikalta hai, aur oxidising agent khud reduce hota hai!*

PAHELI *Shabaash! Kal Level 4: exothermic, corrosion aur rancidity.*

Formula sheet: must-write equations of Day 3

$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$

$$Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$$

$$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)}\downarrow + 2KNO_{3(aq)}$$

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

$$ZnO_{(s)} + C_{(s)} \rightarrow Zn_{(s)} + CO_{(g)}$$

$$MnO_{2(s)} + 4HCl_{(aq)} \rightarrow MnCl_{2(aq)} + 2H_2O_{(l)} + Cl_{2(g)}\uparrow$$

Lab file: the three NCERT activities

Activity	Observation	Type
1.9: Fe + $CuSO_4$	Blue → light green; brown coat	Displacement
1.10: $BaCl_2$ + Na_2SO_4	White ppt of $BaSO_4$	Double displacement
1.11: heating Cu	Brown → black (CuO)	Oxidation

Coming up: Day 4

Exothermic and endothermic reactions, rusting, rancidity and the **Chapter 1 Boss Level** test!

★ How we starred (Eduspark PYQ analysis): we counted how often each reaction or topic has come in Chapter 1 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 04 LEVEL 4

CHEMISTRY | CHAPTER 1 | PART 4 ASTRA SERIES

ENERGY, CORROSION & RANCIDITY

Goal: heat changes, rusting, rancidity and the Boss Level.

by Nitesh Prakash

01 Exo vs Endo | 02 Respiration | 03 Corrosion | 04 Rancidity | 05 Boss Level | +200 XP today | Level 4 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 EXOTHERMIC VS ENDOTHERMIC

DEFINITION

- **Exothermic:** a reaction in which **heat is given out**
- **Endothermic:** a reaction in which **heat is absorbed**

EXOTHERMIC ★★★	ENDOTHERMIC ★★☆
Heat **released**; gets **warm**	Heat **absorbed**; gets **cool**
+ **Heat** on the product side	+ **Heat** with reactants (or Δ)
Burning, respiration, compost	Photosynthesis, $CaCO_3$

Exothermic: heat given out ★★★

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \text{Heat}$
burning of coal

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)} + \text{Heat}$
burning of natural gas (methane)

$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \text{Heat}$
quicklime + water: slaked lime, beaker becomes hot

Endothermic: heat absorbed ★★☆

$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2(g)}$
limestone needs strong heating

$2AgCl_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Cl_{2(g)}$
energy from light; white → grey

$6CO_{2(g)} + 6H_2O_{(l)} \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_{6(aq)} + 6O_{2(g)}$
photosynthesis: plants absorb sunlight energy

EXTRA FROM CLASS

$N_{2(g)} + O_{2(g)} \xrightarrow{\text{very high temp.}} 2NO_{(g)}$
endothermic; happens in lightning and hot car engines

MEME BREAK
*Coal jal raha, stove on: "This is fine." Heat bahar = **exothermic**!*

2 RESPIRATION AND COMPOST

Respiration: an exothermic reaction ★★★

Rice, potatoes and bread contain **carbohydrates**. Digestion breaks them down into **glucose**. In our cells, glucose combines with oxygen and releases energy.

$C_6H_{12}O_{6(aq)} + 6O_{2(aq)} \rightarrow 6CO_{2(aq)} + 6H_2O_{(l)} + \text{Energy}$
glucose + oxygen → carbon dioxide + water + energy

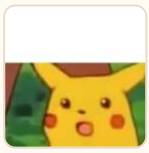
Balance check: C 6 = 6, H 12 = 12, O 18 = 18 ✔

BUJHO *Saans lena hi respiration hai na? Toh heat kahan se aayi?*

PAHELI *Saans lena sirf breathing hai. Respiration cells ke andar glucose ka oxidation hai, jisme energy nikalti hai. Isliye ye exothermic hai!*

Decomposition of vegetable matter into compost ★★★

- Kitchen waste and leaves are broken down by microbes into **compost**
- Heat is **released**, so a compost pit feels **warm** inside: **exothermic**
- Most decomposition reactions need energy (endothermic); compost making is the famous **exception**



MEME BREAK
*Teacher: "Decomposition endothermic hoti hai." Compost pit andar se garam nikla. Bujho: **exothermic decomposition?!***

★★★ *Respiration aur compost dono exothermic: Nitesh Bhaiya ki guarantee, MCQ mein pakka aayega!*

How to spot the type from an equation ★★☆

You see	Type
+ Heat or + Energy on the **right**	**Exothermic**
+ Heat on the **left**; Δ, sunlight or electricity on the arrow	**Endothermic**
Test tube becomes hot	**Exothermic**
Test tube becomes cold	**Endothermic**

3 CORROSION AND RUSTING

DEFINITION

- **Corrosion:** a metal is slowly **eaten away** when attacked by air, moisture, acids or other substances around it
- **Rusting:** corrosion of **iron**; a reddish-brown, flaky coat forms

Rusting of iron ★★★

$$4Fe_{(s)} + 3O_{2(g)} + 2xH_2O_{(l)} \rightarrow 2Fe_2O_3.xH_2O_{(s)}$$

rust = hydrated iron(III) oxide, reddish-brown

Balance check: Fe 4 = 4, O (6 + 2x) = (6 + 2x), H 4x = 4x ✔

- Rusting needs **both air (oxygen) and water (moisture)**

Activity: which nail rusts? ★★★

Tube	Condition	Result
A	Nail in ordinary tap water (air + water)	**Rusts**
B	Boiled water + oil layer on top (no air)	No rust
C	Dry air with anhydrous $CaCl_2$ (no water)	No rust

MEME BREAK
*Iron Man ko bhi **hawa + paani** mil jaaye toh jung lag jaayegi. Suit pe paint ki parat zaroori hai!*

Corrosion of other metals ★★☆

Metal	Coating formed	Due to
Silver	Black: silver sulphide, Ag_2S	H_2S in air
Copper	Green: basic copper carbonate, $CuCO_3.Cu(OH)_2$	Moist CO_2 in air

Preventing rusting ★★★

1. **Painting** gates, grills and bridges
2. **Oiling or greasing** machine parts and cycle chains
3. **Galvanising:** coating iron with a thin layer of **zinc**
4. **Chrome plating** (taps, cycle handles) and tin coating
5. **Alloying:** stainless steel (iron + chromium + nickel) does not rust

MEME BREAK *Paint ya oil ki parat **hawa aur nami** ko iron se door rakhti hai. Parat tooti, rusting shuru!*

BUJHO *Ye iron clips baarish mein bhool gaya, kal tak kuch hoga?*

PAHELI *Hawa + nami = jung! Clips sookhe rakho, ya oil laga do. Paani ke bina rust nahi lagta.*

★★★ ***Rusting ke liye hawa + paani dono chahiye: Nitesh Bhaiya ki guarantee, activity wala question pakka!***

4 RANCIDITY

DEFINITION

- **Rancidity:** fats and oils in food get **oxidised** when left exposed to air for long
- Food develops a **bad smell and taste** and is unsafe to eat

Preventing rancidity ★★★

- ✔ **Antioxidants** added to food containing fats and oils
- ✔ Chips packets **flushed with nitrogen**, an unreactive gas
- ✔ Storing food in **airtight containers**
- ✔ **Refrigeration:** low temperature slows oxidation
- ✔ Keeping food away from light; vacuum packing

BUJHO *Chips packet mein aadhi hawa bhar ke loot rahe hain!*

PAHELI *Hawa nahi, **nitrogen** hai! Oxygen chips ko rancid kar deta, nitrogen unhe crispy aur fresh rakhta hai.*

MEME BREAK
*Mummy ka chhupaya namkeen, 3 mahine baad khola. Smell aur taste dono kharab: **rancidity**!*

Corrosion vs rancidity: both are oxidation

	Corrosion	Rancidity
Happens to	Metals	Fats and oils in food
Caused by	Air, moisture, acids	Oxygen of air
Result	Metal gets weak	Bad smell and taste
Stopped by	Paint, oil, zinc coat	Antioxidants, N_2, airtight

× WRONG	✔ RIGHT
Respiration is endothermic	It releases energy, so it is exothermic
Rust is Fe_2O_3	Rust is hydrated: $Fe_2O_3.xH_2O$
Iron rusts in dry air	It needs air **and** water
Chips packets contain oxygen	They are flushed with nitrogen

5 PRACTICE MCQS

Pehle khud solve karo, answer key page 3 ke end mein hai!

1. Which of the following is an endothermic reaction? ★★★
(a) Burning of coal (b) Respiration
(c) Heating calcium carbonate (d) Quicklime added to water

2. Making compost from vegetable matter is an example of: ★★★
(a) an endothermic reaction (b) an exothermic reaction
(c) a photolytic decomposition (d) a displacement reaction

3. The chemical formula of rust is: ★★★
(a) Fe_2O_3 (b) Fe_3O_4
(c) $Fe_2O_3.xH_2O$ (d) FeO

4. An iron nail in boiled water under a layer of oil does not rust because: ★★☆
(a) oil reacts with iron (b) no air reaches it
(c) the water is hot (d) there is no light

5. The black coating formed on silver articles is: ★★☆
(a) Ag_2O (b) AgCl
(c) Ag_2S (d) $AgNO_3$

6. Packets of potato chips are flushed with: ★★★
(a) oxygen (b) hydrogen
(c) nitrogen (d) chlorine

7. Galvanisation is the process of coating iron with: ★★☆
(a) tin (b) zinc
(c) chromium (d) copper

8. **Assertion (A):** Respiration is an exothermic reaction.
Reason (R): During respiration, glucose is oxidised in our cells and energy is released. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

★★★ *Rust hamesha hydrated iron(III) oxide hota hai! Nitesh Bhaiya ki guarantee, formula wala MCQ yahin se.*

Answer key: practice MCQs

Q	Ans	Why
1	(c)	$CaCO_3$ needs heat to break down.
2	(b)	Compost pit becomes warm: heat released.
3	(c)	Rust is hydrated iron(III) oxide.
4	(b)	Boiling removes air; oil stops fresh air.
5	(c)	Silver reacts with H_2S in air.
6	(c)	Nitrogen is unreactive; no oxidation.
7	(b)	A zinc layer protects the iron.
8	(a)	Energy released, because glucose is oxidised.

Short answer questions

1. Why do we apply paint on iron articles? NCERT
2. Oil and fat containing food items are flushed with nitrogen. Why? NCERT
3. Explain corrosion and rancidity with one example each. NCERT

Model answers: short answer

1 Paint on iron
- Paint keeps **air and moisture** away from iron, so it does not rust

2 Nitrogen in food packets
- Nitrogen is **unreactive**; it pushes out oxygen, so fats are not oxidised and food does not turn **rancid**

3 Corrosion and rancidity
- **Corrosion:** metal eaten away by air, moisture or acids; e.g. rusting of iron
- **Rancidity:** oxidation of fats and oils in food; e.g. stale chips smell bad

6 BOSS LEVEL: CHAPTER 1 TEST

BOSS LEVEL: CHAPTER 1

Days 1–4 | 21 marks | 30 minutes

PAHELI *Poora chapter ek saath! Timer lagao, book band karo, phir answers se milao.*

BUJHO *Har equation balance karke likhunga, boss ko harana hai!*

SECTION A: MCQS (1 MARK EACH)

1. Which of the following is a chemical change? ★★☆
(a) Melting of wax (b) Dissolving salt
(c) Souring of milk (d) Boiling water

2. To balance $Na + H_2O \rightarrow NaOH + H_2$, the coefficients are: ★★★
(a) 1, 1, 1, 1 (b) 2, 2, 2, 1
(c) 2, 1, 2, 1 (d) 1, 2, 1, 2

3. On heating ferrous sulphate crystals, the residue left is: ★★★
(a) green (b) white
(c) reddish-brown (d) blue

4. In electrolysis of water, the volume ratio of gas at the cathode to gas at the anode is: ★★★
(a) 1 : 2 (b) 2 : 1
(c) 1 : 1 (d) 8 : 1

5. $Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$ is a: ★★★
(a) combination reaction (b) decomposition reaction
(c) displacement reaction (d) double displacement

6. The reaction between sodium sulphate and barium chloride solutions is a: ★★★
(a) displacement (b) double displacement
(c) combination (d) decomposition

7. In $CuO + H_2 \xrightarrow{\Delta} Cu + H_2O$, the substance oxidised is: ★★★

(a) CuO (b) H_2
(c) Cu (d) H_2O

8. In $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$, the oxidising agent is: ★★☆

(a) MnO_2 (b) HCl
(c) $MnCl_2$ (d) Cl_2

9. **Assertion (A):** Silver chloride turns grey when kept in sunlight.
Reason (R): Sunlight decomposes silver chloride into silver and chlorine. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Rusting of iron is a slow oxidation reaction.
Reason (R): Iron rusts quickly in dry air. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Section B and C: written questions

1. What happens when water is added to quicklime? Write the balanced equation and name the type of reaction. **(2)**
2. Lead nitrate is heated in a dry test tube. State one observation, write the balanced equation and name the reaction. **(2)**
3. In $ZnO + C \rightarrow Zn + CO$, identify the substance oxidised and the substance reduced. **(2)**
4. A shiny brown metal X turns black (Y) when heated in air. When hydrogen is passed over hot Y, X is obtained again. (a) Identify X and Y. (b) Write both balanced equations and name each reaction type. (c) What colour change is seen when an iron nail is dipped in the blue solution of a salt of X? **(5)**

★★★ *Boss Level ke ye sawaal = poore chapter ka revision. Nitesh Bhaiya ki guarantee, board mein yahi type aayega!*

Boss Level answers: MCQs

Q	Ans	Why
1	(c)	A new substance (curd) forms.
2	(b)	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
3	(b)	Fe_2O_3 is reddish-brown.
4	(b)	H_2 at cathode is double the O_2.
5	(c)	Fe is more reactive; it displaces Cu.
6	(b)	White $BaSO_4$ ppt forms.
7	(b)	H_2 gains oxygen and becomes H_2O.
8	(a)	MnO_2 gives oxygen; it is reduced.
9	(a)	Photolytic decomposition explains it.
10	(c)	Iron needs air and moisture to rust.

Boss Level answers: written questions

1 Quicklime + water (2)

$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \textbf{Heat}$
combination reaction; also exothermic

2 Heating lead nitrate (2)

> **Brown fumes** of NO_2 are seen

$2Pb(NO_3)_{2(s)} \xrightarrow{\Delta} 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$
thermal decomposition

3 ZnO + C (2)

> **Oxidised:** carbon (C gains oxygen → CO)
> **Reduced:** zinc oxide (ZnO loses oxygen → Zn)

4 Metal X and black Y (5)

> **(a)** X = copper (Cu); Y = copper(II) oxide (CuO)

$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$
(b) combination; copper is oxidised

$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(g)}$
redox: CuO is reduced, H_2 is oxidised

> **(c)** Blue $CuSO_4$ solution turns **light green** ($FeSO_4$); a **brown** coat of Cu forms on the nail

Day 4 checklist: tick before you sleep

- ☐ I can tell exothermic from endothermic and give two examples of each
- ☐ I can write the respiration equation and explain why compost pits are warm
- ☐ I can explain rusting, its equation and four ways to prevent it
- ☐ I can explain rancidity and why chips packets are flushed with nitrogen
- ☐ I cleared the Boss Level and checked every answer with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 4 COMPLETE

+200 XP earned. Chapter 1 cleared. Next: Level 5

BUJHO *Samajh gaya! Respiration exothermic, rust ko hawa + paani chahiye, aur chips ko nitrogen bachata hai!*

PAHELI *Shabaash! Chapter 1 ka boss haar gaya. Kal se acids, bases and salts.*

NITESH SIR SAYS
Boss Level mein jo galat hua, us topic ka Day phir se padho. ***Revision hi asli jalwa hai!***

CHEMISTRY | CHAPTER 1 COMPLETE

CHAPTER 1 SWAHAA!

Chemical Reactions and Equations | Days 1–4

BUJHO *Reaction pehchanna, equation balance karna, redox, corrosion... sab samajh gaya! Chapter 1 Swahaa!*

PAHELI *Shabaash Bujho! Ab Acids, Bases aur Salts ki baari. Kal milte hain Day 5 pe!*

CHAPTER XP EARNED
+500 XP

LEVELS CLEARED
4 levels (Days 1–4)

BOOK PROGRESS
Level 4 / 60 500 XP total

WHAT YOU HAVE MASTERED

- ✔ Chemical change and its 5 signs
- ✔ Combination and decomposition
- ✔ Oxidation, reduction and redox
- ✔ Writing and balancing equations
- ✔ Displacement and double displacement
- ✔ Exo/endothermic, corrosion, rancidity

NEXT CHAPTER Chapter 2: Acids, Bases and Salts (Day 5) *Jalwa jaari rahe!*

DAY 05 · LEVEL 5

CHEMISTRY | CHAPTER 2 | PART 1 · ASTRA SERIES

ACIDS, BASES & SALTS

Goal: test acids and bases, and predict how they react.

by Nitesh Prakash

01 Basics · 02 Indicators · 03 Metals · 04 Carbonates · 05 Neutralisation

+100 XP today | Level 5 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 ACIDS VS BASES

ACIDS	BASES
Sour in taste (lemon, curd, vinegar)	Bitter; soapy to touch; corrosive
Turn **blue litmus red**	Turn **red litmus blue**
HCl, H_2SO_4, HNO_3, acetic acid	NaOH (caustic soda), KOH (caustic potash), $Ca(OH)_2$

2 INDICATORS ★★★ MOST ASKED

DEFINITION

- An **indicator** is a substance that shows, by a change in **colour** or **smell**, whether a solution is **acidic** or **basic** (never taste lab chemicals!)
- Three types: **natural**, **synthetic** and **olfactory**

Colour chart: learn it by heart ★★★

Indicator	In acid	In base
Litmus (natural)	red	blue
Turmeric (natural)	yellow	reddish-brown
Red cabbage (natural)	red	green
China rose (natural)	dark pink	green
Phenolphthalein (synthetic)	colourless	pink
Methyl orange (synthetic)	red	yellow

China rose (hibiscus) petals soaked in warm water give the indicator: dark pink (magenta) in acid, green in base. **Hydrangea**, **petunia** and **geranium** petals also work; litmus comes from **lichen**.

MEME BREAK

*Haldi ka daag sabun se dhoya aur daag **laal-bhoora** ho gaya! Soap is basic, and turmeric turns **reddish-brown** in a base.*

BUJHO *Phenolphthalein acid mein laal hota hai na?*

PAHELI *Nahi! Acid mein **colourless**, base mein **pink**. Acid mein laal hota hai **methyl orange**.*

Olfactory indicators: test by smell ★★☆

- **Olfactory** = related to the sense of smell; the odour changes in an acid or a base
- **Onion**, **vanilla** extract and **clove** oil: the smell **disappears in a base** (NaOH) but stays in an acid (HCl)
- Useful for a visually impaired student to tell an acid from a base

Solved Example 1: Only red litmus given

Q. Three test tubes hold distilled water, an acid and a base. You have only red litmus paper. Identify each. NCERT

1. Dip red litmus in all three. The one that turns it **blue** is the **base**.
2. Dip this blue litmus in the other two. The one that turns it **red** is the **acid**.
3. The tube that changes neither paper is **distilled water**.

★★★ *Indicator colour chart har saal aata hai: Nitesh Bhaiya ki guarantee, isse ratt lo!*

3 ACIDS AND BASES WITH METALS ★★★

GENERAL RULE

- Acid + Metal → **Salt + Hydrogen gas**
- Hydrogen of the acid is **displaced** by the metal

$$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}\uparrow$$

Zinc granules + dilute sulphuric acid: bubbles of H_2

Activity: The "pop" test for hydrogen ★★★

1. Add dilute H_2SO_4 to zinc granules; pass the gas into **soap solution**.
2. Soap bubbles filled with gas rise up.
3. Bring a **burning candle** near a bubble: it burns with a **pop sound**. The gas is **H_2**.

Same result with dilute HCl, HNO_3 (very dilute) and other reactive metals.

Base + metal: zinc and sodium hydroxide ★★☆

$$2NaOH_{(aq)} + Zn_{(s)} \rightarrow Na_2ZnO_{2(aq)} + H_{2(g)}\uparrow$$

Na_2ZnO_2 = sodium zincate; H_2 again gives the pop test

Not every metal reacts with a base; zinc and aluminium do.

4 ACIDS WITH CARBONATES ★★★ MOST ASKED

GENERAL RULE

- Metal carbonate or hydrogencarbonate + Acid → **Salt + Carbon dioxide + Water**
- Brisk **effervescence** (fizzing) shows CO_2 is given off

$Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)\uparrow$
Sodium carbonate + hydrochloric acid

$NaHCO_3(s) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)\uparrow$
Sodium hydrogencarbonate (baking soda) + hydrochloric acid

Activity: Lime water test ★★★

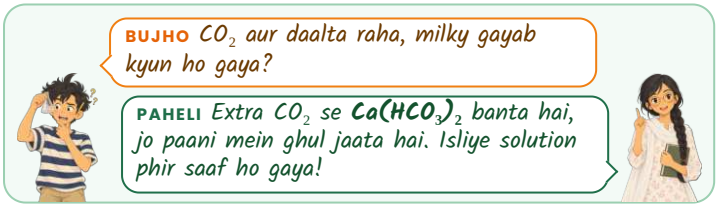
1. Pass the gas through **lime water** [$Ca(OH)_2$ solution]: it turns **milky** due to white $CaCO_3$.
2. Keep passing **excess CO_2**: the milkiness **disappears**, as soluble calcium hydrogencarbonate forms.

$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s)\downarrow + H_2O(l)$
lime water turns milky

$CaCO_3(s) + H_2O(l) + CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$
excess CO_2: milkiness disappears

Limestone, chalk and marble are all forms of $CaCO_3$, so each fizzes with dilute acid and gives CO_2.

MEME BREAK
*Baking soda + nimbu ka ras = **fizz hi fizz**! Wahi CO_2, wahi acid + hydrogencarbonate reaction.*



Reactions at a glance: 6 patterns ★★★

Reaction	Products
Acid + metal	Salt + $H_2\uparrow$
Base (NaOH) + metal (Zn, Al)	Salt + $H_2\uparrow$
Acid + carbonate or hydrogencarbonate	Salt + H_2O + $CO_2\uparrow$
Acid + base (neutralisation)	Salt + H_2O
Acid + metallic oxide (basic)	Salt + H_2O
Base + non-metallic oxide (acidic)	Salt + H_2O

Exam tips

- ✔ Name the gas **and** its test: H_2 → pop sound; CO_2 → lime water milky.
- ✔ Write state symbols and ↑ for gases in every final equation.
- ✔ For indicators, always give the colour in **both** acid and base.
- ✔ "Effervescence" usually means CO_2 from a carbonate.

5 NEUTRALISATION ★★★

DEFINITION

- The reaction of an **acid** with a **base** to give **salt and water** is called **neutralisation**
- Acid + Base → Salt + Water (heat is given out)

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$
Sodium hydroxide + hydrochloric acid → sodium chloride + water

Activity: Watch neutralisation happen ★★☆

1. Take 2 mL dilute NaOH and add 2 drops of **phenolphthalein**: it turns **pink**.
2. Add dilute HCl drop by drop: the pink colour **disappears** (base neutralised).
3. Add a few drops of NaOH again: the **pink returns**.

MEME BREAK
LEFT: NaOH + PHENOLPHTHALEIN
Solution is **pink**, all happy
RIGHT: HCl ADDED DROP BY DROP
Pink fades, solution turns **colourless**
Acid ne base ko neutralise kar diya, pink gayab!

6 OXIDES WITH ACIDS AND BASES

Metallic oxides are basic ★★★

$CuO(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + H_2O(l)$
black CuO dissolves and the solution turns **blue-green**
Metal oxide + Acid → Salt + Water, just like a base.

Non-metallic oxides are acidic ★★☆

$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s)\downarrow + H_2O(l)$
base + non-metal oxide → salt + water
Non-metal oxide + Base → Salt + Water, just like an acid.

MEME BREAK
Acid + base = salt + paani? Toh salt sirf namak nahi: NaCl, KNO_3, $CaCl_2$, $CuSO_4$, sab salts hain!

× WRONG	✔ RIGHT
Phenolphthalein: pink in acid	**Pink only in a base**; colourless in acid
$Na_2CO_3 + HCl \rightarrow NaCl + CO_2$	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
Gas from acid + metal is CO_2	It is **H_2**, shown by the pop test

REAL LIFE: WHY NOT IN COPPER OR BRASS?
Curd, pickles and other sour foods contain acids. In **copper or brass** vessels these acids react with the metal and form harmful salts. Store them in glass or steel.

Solved Example 2: Find compound A ★★★

Q. Metal compound A reacts with dilute HCl and gives effervescence. The gas puts out a burning candle. One product is calcium chloride. Write the balanced equation. NCERT

- Gas puts out a flame and fizzes out of an acid → **CO_2**, so A is a carbonate
- Product is $CaCl_2$ → A is **calcium carbonate, $CaCO_3$**

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)\uparrow$$

Ca 1 = 1, C 1 = 1, O 3 = 3, H 2 = 2, Cl 2 = 2 ✔

Day 5 mind map: revise in 2 minutes

INDICATORS ★★★
Litmus, turmeric, red cabbage; phenolphthalein, methyl orange; onion, vanilla, clove

METALS
Acid + metal → salt + H_2; NaOH + Zn → $Na_2ZnO_2 + H_2$

ACIDS & BASES: REACTIONS

CARBONATES
Acid + carbonate → salt + H_2O + CO_2; lime water milky

NEUTRALISATION
Acid + base → salt + water

OXIDES
Metallic oxides are basic (react with acids); non-metallic oxides are acidic (react with bases)

★★★ **PYQ FAVOURITE** Zinc + dilute H_2SO_4 gives **H_2** (CBSE Term-1 SQP 2021); methyl orange is **red** in acid and **yellow** in base (CBSE SQP 2025).

7 PRACTICE MCQS

Pehle khud solve karo, phir answer key dekho. Cheating se XP nahi milta!

1. Dilute sulphuric acid is added to zinc granules and the gas formed is passed through soap solution. The gas filling the soap bubbles is: CBSE Term-1 SQP 2021 ★★★

(a) nitrogen (b) hydrogen
(c) oxygen (d) carbon dioxide

2. Methyl orange is added to dilute HCl and to aqueous NaOH. Its colours are, respectively: CBSE SQP 2025 ★★★

(a) orange, red (b) red, yellow
(c) red, orange (d) yellow, red

3. A turmeric stain on a white shirt turns reddish-brown when soap is scrubbed on it. Why? (I) Soap is acidic. (II) Soap is basic. (III) Turmeric gives a reddish tinge in bases. (IV) Turmeric gives a reddish tinge in acids. CBSE Term-1 SQP 2021 ★★★

(a) (I) and (III) (b) (II) and (III)
(c) (I) and (IV) (d) (II) and (IV)

4. Excess CO_2 is passed through lime water. The milkiness disappears because: CBSE SQP 2025 ★★★

(a) $CaCO_3$ changes into CaO
(b) insoluble $CaCO_3$ changes into soluble $Ca(HCO_3)_2$
(c) $Ca(OH)_2$ is formed again
(d) insoluble $CaCO_3$ changes into insoluble $Ca(HCO_3)_2$

5. Which of these is an olfactory indicator? ★★☆

(a) Litmus (b) Turmeric
(c) Vanilla extract (d) Methyl orange

6. Two drops of phenolphthalein are added to dilute NaOH solution. The solution becomes: ★★★

(a) colourless (b) pink
(c) red (d) yellow

7. Zinc is heated with sodium hydroxide solution. What does it give? ★★☆

(a) ZnO and H_2O (b) Na_2ZnO_2 and H_2
(c) $Zn(OH)_2$ and Na (d) no reaction takes place

8. Black copper(II) oxide is stirred with dilute HCl. What colour does the solution turn? ★★☆

(a) colourless (b) blue-green
(c) pink (d) yellow

9. **Assertion (A):** Carbon dioxide is an acidic oxide.
Reason (R): CO_2 reacts with lime water, which is a base, to give salt and water. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Phenolphthalein is a natural indicator.
Reason (R): Phenolphthalein turns pink in a basic solution. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short answer questions

1. Which gas is usually given off when an acid reacts with a metal? Give an example with an equation, and state how you will test for this gas. NCERT
2. Write the balanced equations, with states, for: (a) dilute sulphuric acid with zinc granules, (b) dilute HCl with magnesium ribbon, (c) dilute sulphuric acid with sodium hydroxide. NCERT

Case-based question: Sara's titration

Sara adds two drops of phenolphthalein to 2 mL of dilute NaOH: it turns **pink**. She adds dilute H_2SO_4 drop by drop; after **40 drops** it turns colourless. One more drop of NaOH makes it pink again.
Based on CBSE SQP 2025 ★★★

1. Why did the solution first turn pink, and why did it later become colourless?
2. How many drops of the same acid will 4 mL of the same NaOH need?
3. Write the balanced equation and name the type of reaction.

TRY YOURSELF (ANSWERS ON PAGE 4)

Complete and balance:

(i) Mg + HCl →
(ii) KOH + HNO_3 →
(iii) ZnO + H_2SO_4 →
(iv) $CaCO_3$ + HCl →

Answer key: MCQs

Q	Ans	Why
1	(b)	Zn + dil. H_2SO_4 gives H_2; it burns with a pop.
2	(b)	Methyl orange: red in acid, yellow in base.
3	(b)	Soap is basic; turmeric turns reddish-brown in bases.
4	(b)	$Ca(HCO_3)_2$ dissolves in water.
5	(c)	Its smell disappears in a base.
6	(b)	Phenolphthalein is pink in a base.
7	(b)	Sodium zincate and hydrogen gas form.
8	(b)	Copper(II) chloride solution is blue-green.
9	(a)	Non-metal oxide + base → salt + water.
10	(d)	It is synthetic; R is true.

Model answers: short answer

1 Gas from acid + metal: hydrogen

- Hydrogen gas is given off, e.g. zinc with dilute sulphuric acid
- **Test:** bring a burning candle near the gas; it burns with a **pop sound**

$$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}\uparrow$$

2 Balanced equations

$$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}\uparrow$$
(a) zinc granules + dilute sulphuric acid

$$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}\uparrow$$
(b) magnesium ribbon + dilute hydrochloric acid

$$2NaOH_{(aq)} + H_2SO_{4(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$
(c) neutralisation: Na 2 = 2, S 1 = 1, O 6 = 6, H 4 = 4 ✔

Answers: Sara's case

- Phenolphthalein is pink in a base (NaOH). The acid neutralises the base, so the solution becomes colourless.
- **80 drops**: double the base needs double the acid.
- It is **neutralisation** (and double displacement); no precipitate forms, as Na_2SO_4 stays dissolved.

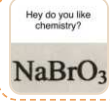
$$2NaOH_{(aq)} + H_2SO_{4(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$

Answers: Try yourself

(i) $Mg + 2HCl \rightarrow MgCl_2 + H_2$
(ii) $KOH + HNO_3 \rightarrow KNO_3 + H_2O$
(iii) $ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O$
(iv) $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

BUJHO *Toh har metal acid se H_2 deta hai?*

PAHELI *Nahi! **Copper** aur **silver** jaise kam reactive metals dilute acid se H_2 nahi dete. Chapter 3 mein detail!*



★★★ *Zn + acid, carbonate + acid, lime water, neutralisation: ye chaar pakka aayenge, skip mat karna! Nitesh Bhaiya ki guarantee!*

Day 5 checklist: tick before you sleep

- ☐ I can give the colours of litmus, turmeric, red cabbage, phenolphthalein and methyl orange in acid and base
- ☐ I can name the olfactory indicators and say how they work
- ☐ I can write acid + metal and NaOH + Zn equations, with the pop test
- ☐ I can write acid + carbonate equations and explain the lime water test
- ☐ I solved all 10 MCQs, including neutralisation and oxide questions

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Dimaag nahi chal raha? Page 1–2 phir se padho.

LEVEL 5 COMPLETE +100 XP earned. Next: Level 6

BUJHO *Samajh gaya! Metal se H_2, carbonate se CO_2, aur acid + base = salt + paani!*

PAHELI *High five! Kal Level 6: pH scale.*

Day 5 hit-list: Eduspark PYQ analysis

Topic / reaction	Rating
Indicator colours (litmus, phenolphthalein, methyl orange, turmeric)	★★★
Zn + dilute H_2SO_4: H_2 and the pop test	★★★
Lime water test; excess CO_2 clears milkiness	★★★
Acid + carbonate or hydrogencarbonate	★★★
Neutralisation with phenolphthalein	★★☆
Olfactory indicators; NaOH + Zn	★★☆

CLASS QUESTION

Q. Are all acids sour and all bases bitter?
Ans. Generally acids taste sour and bases bitter, but taste is not a reliable or safe test. **Coffee** tastes bitter yet is slightly **acidic** (pH about 5), so bitter does not always mean basic. Always use an indicator; never taste chemicals in the lab.

Which acid gives which salt? HCl → chlorides (NaCl, $CaCl_2$); H_2SO_4 → sulphates ($ZnSO_4$, Na_2SO_4); HNO_3 → nitrates (KNO_3)

NITESH SIR SAYS
*Indicator ka colour table aaj hi yaad karo. **Kal pH ka jalwa!***

★ Stars (Eduspark PYQ analysis): how often a topic came in Chapter 2 board and sample papers; year tags only where the year is known.

DAY 06 · LEVEL 6

CHEMISTRY | CHAPTER 2 | PART 2 · ASTRA SERIES

IONS, STRENGTH & THE pH SCALE

Goal: ions in water, dilution, strength and the pH scale.

by Nitesh Prakash

01 Ions in Water | 02 Conduction | 03 Dilution | 04 Strength | 05 pH Scale

+100 XP today | Level 6 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 ACIDS & BASES IN WATER ★★★

DEFINITION

- **Acid:** gives **$H^+(aq)$ ions** when dissolved in water
- **Base:** gives **$OH^-(aq)$ ions** when dissolved in water
- H^+ cannot exist alone; it joins water to form **hydronium ion, H_3O^+**

$$HCl(aq) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$$

HCl gives H^+, which is carried as H_3O^+

$$H^+ + H_2O(l) \rightarrow H_3O^+(aq)$$

so $H^+(aq)$ and $H_3O^+(aq)$ mean the same thing in answers

ACID IN WATER	→ together they make water	BASE IN WATER
H^+ / H_3O^+ ions blue litmus → red		OH^- ions red litmus → blue

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

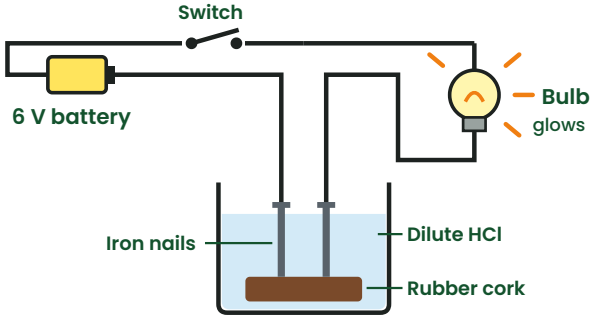
neutralisation, in one line (Day 5)

MEME BREAK

*Glucose ($C_6H_{12}O_6$) and alcohol (C_2H_5OH): "Hum mein bhi H hai, hum acid kyun nahi?" Kyunki paani mein **hydrogen ions nahi dete**.*

2 CONDUCTION EXPERIMENT ★★☆

NCERT activity: do acids conduct?



- Fix two iron nails on a rubber cork in a beaker; connect them to a 6 V battery through a bulb and a switch
- Pour **dilute HCl**, switch on: the **bulb glows**
- Repeat with dilute H_2SO_4 (glows), then glucose and alcohol solutions (bulb **does not glow**)

Result and conclusion

Solution	Bulb	Reason
Dilute HCl, H_2SO_4	✔ glows	free H^+ ions carry current
Glucose, alcohol	× no glow	no ions formed in water

Conclusion: acids give $H^+(aq)$ ions in water and these ions carry the current. Bases conduct due to OH^- ions.

BUJHO *Glucose mein bhi toh H hai, phir bulb kyun nahi jala?*

PAHELI *Kyunki glucose paani mein H^+ ions nahi banata. Ions nahi, toh current nahi!*

3 ACIDS NEED WATER ★★★

NCERT activity: acids need water to act

$$NaCl(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HCl(g)\uparrow$$

salt + conc. H_2SO_4; $CaCl_2$ guard tube keeps the gas dry

DRY LITMUS	MOIST LITMUS
× Dry blue litmus: **no change**	✔ Moist blue litmus turns **red**
No water, so HCl does not ionise; no H^+ ions	HCl dissolves in the water and gives H_3O^+ ions

MEME BREAK

*Dry HCl gas litmus ko red kar dega? **Bilkul nahi!** Acidic behaviour sirf **aqueous solution** mein dikhta hai, kyunki paani ke bina hydrogen ions bante hi nahi.*

4 BASES AND ALKALIS ★★☆

DEFINITION

- A base that **dissolves in water** is called an **alkali**: NaOH, KOH, $Ca(OH)_2$, NH_4OH
- All alkalis are bases, but **all bases are not alkalis**: $Cu(OH)_2$, $Fe(OH)_3$ do not dissolve

$$NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-(aq)$$

alkalis feel soapy, taste bitter and are corrosive

5 DILUTION OF ACIDS & BASES ★★★ MOST ASKED

DEFINITION

- **Dilution:** mixing an acid or base with water to **decrease the concentration** of ions per unit volume
- Mixing a concentrated acid or base with water is **highly exothermic** (a lot of heat is released)

Safety rule: acid into water, never water into acid

- Add the **acid slowly to water** with **constant stirring**; the large amount of water absorbs the heat
- Water added to concentrated acid: heat is released suddenly, the mixture **splashes out** and causes burns
- Local overheating can even **crack the glass** container

× WRONG	✓ RIGHT
Water poured into conc. acid	Acid added slowly to water
Fast mixing, no stirring	Slow mixing with stirring

MEME BREAK
*Conc. acid mein paani daal diya? Garam mixture uchhal ke haath par: **ouch!** Yaad rakho, **acid ko paani mein**, dheere se, hilate hue.*

What dilution does to the ions

Diluting...	Ion per unit volume	pH moves
an acid	$[H_3O^+]$ **decreases**	up, towards 7
a base	$[OH^-]$ **decreases**	down, towards 7

Dilution never crosses 7: a diluted acid stays acidic, a diluted base stays basic.

BUJHO *Jaldi hai, conc. acid mein hi paani daal doon?*

PAHELI *Bilkul nahi! Heat se mixture uchhal ke bahar aayega. Hamesha acid ko paani mein, hilate hue.*

★★★ *"Acid to water, why?" 3-mark sawaal har saal ka hai: Nitesh Bhaiya ki guarantee!*

Exam tips

- ✔ For acids in water write "H^+(aq) or H_3O^+(aq)", not a bare H^+.
- ✔ Bulb glows = ions present. Glucose and alcohol have H but give no ions.
- ✔ Dilution answer keywords: "highly exothermic", "acid to water", "slowly with stirring".

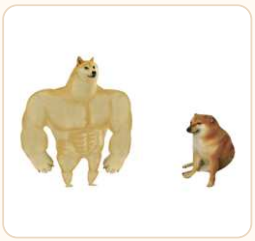
6 STRONG & WEAK ACIDS, BASES ★★★ MOST ASKED

STRONG	WEAK
Ionises **completely** in water; gives **many** H^+ (or OH^-) ions	Ionises **partially**; gives **fewer** H^+ (or OH^-) ions
Acids: HCl, H_2SO_4, HNO_3	**Acids:** CH_3COOH, H_2CO_3, citric acid
Bases: NaOH, KOH	**Bases:** NH_4OH, $Mg(OH)_2$

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$
strong: every HCl molecule breaks into ions

$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$
weak: only a few molecules ionise (⇌ = partial, reversible)

MEME BREAK
*Acetic acid thoda sa ionise; HCl ka har molecule toot jaata hai. **100% ionisation = strong acid.***



MEME BREAK
LEFT: STRONG (HYDROCHLORIC ACID)
lower pH, bright bulb, fast fizz with Mg
RIGHT: WEAK (ACETIC ACID)
same concentration: higher pH, dim bulb
Same concentration, alag taaqat: H^+ ions ka khel hai!

Strength vs concentration: don't mix them up

	Strength	Concentration
Meaning	degree of ionisation	amount of acid in a volume
Words	strong / weak	concentrated / dilute
On dilution	does not change	decreases

So: dilute HCl is still a **strong** acid; concentrated acetic acid is still a **weak** acid.

CLASS QUESTION

Q. Why is HF (hydrofluoric acid) not stored in glass bottles?
Ans. HF reacts with silica (SiO_2) in glass and eats it away, so it is kept in **plastic (polythene)** or wax-lined bottles.

$SiO_2(s) + 4HF(aq) \rightarrow SiF_4(g) + 2H_2O(l)$

7 THE pH SCALE ★★★ MOST ASKED

DEFINITION

- **pH scale:** measures the **H^+ ion concentration** of a solution, from **0** (very acidic) to **14** (very basic)
- "p" comes from **potenz**, German for power
- **Higher the H^+ concentration, lower the pH**

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

0: STRONGLY ACIDIC — 7: NEUTRAL — 14: STRONGLY BASIC

← acidic: H^+ increases, pH falls | basic: OH^- increases, pH rises →

Read the scale ★★★

pH	Nature	Ions in excess
< 7	acidic (lower pH, more acidic)	H^+ more than OH^-
= 7	neutral (pure water)	$H^+ = OH^-$
> 7	basic (higher pH, more basic)	OH^- more than H^+

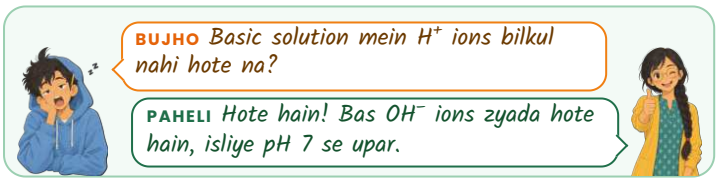
pH of common substances (approx.)

Substance	pH	Substance	pH
Gastric juice	1.2	Pure water	7
Lemon juice	2.2	Blood	7.4
Vinegar	about 3	Milk of magnesia	10
Milk	about 6.5	NaOH solution	14

Universal indicator

- A **mixture of several indicators**; shows a different colour at each H^+ concentration, so it gives the **pH value**
- Litmus only says acid or base; universal indicator also says **how acidic or basic**

pH	Colour	pH	Colour
0–2	red	7	green
3–4	orange	8–10	blue
5–6	yellow	11–14	violet



Solved Example: hydronium ions CBSE SQP 2024 ★★★

Q. Solutions A, B, C have pH 6, 8 and 9.5. Which has the most hydronium ions? Arrange them in increasing order of H^+ ion concentration.

- Lowest pH = most H^+, so **A (pH 6)** has the maximum hydronium ions
- Increasing $[H^+]$: **C (9.5) < B (8) < A (6)**

8 PRACTICE MCQS

Kyun nahi ho rahi padhai? Phone side mein, pehle khud solve karo! Answer key page 4 par hai.

1. Which of the following substances, when dissolved in an equal volume of water, will have the highest pH value? CBSE SQP 2025 ★★★

(a) Sulphuric acid (b) Acetic acid
(c) Magnesium hydroxide (d) Sodium hydroxide

2. Two nails dipped in a solution are connected to a battery and a bulb. The bulb does not glow. The solution is likely to be: CBSE SQP 2024 ★★★

(a) HNO_3 (b) C_2H_5OH
(c) H_2SO_4 (d) CO_2 in water

3. HCl gas from NaCl and conc. H_2SO_4 changes the colour of moist blue litmus paper. This is due to the presence of: CBSE SQP 2022 ★★☆

(a) an acid (b) a base
(c) H^+(aq) in the solution (d) litmus

4. Which option correctly matches the solution, its pH paper colour, pH and nature? CBSE 2023 ★★★

(a) Lemon juice: orange, 3, basic
(b) Milk of magnesia: blue, 10, basic
(c) Gastric juice: red, 6, acidic
(d) Pure water: yellow, 7, neutral

5. Solution P has pH 6 and Q has pH 8. So: ★★★

(a) P has more H^+ ions and is acidic
(b) Q has more H^+ ions and is acidic
(c) P is basic and Q is acidic
(d) Both are neutral

6. The safe way to dilute concentrated sulphuric acid is to: ★★★

(a) add water to the acid quickly
(b) add the acid to water slowly with stirring
(c) add both together at once
(d) heat the acid, then add water

7. Which of the following is a weak acid? ★★☆

(a) HCl (b) HNO_3
(c) CH_3COOH (d) H_2SO_4

8. Adding an acid to water involves: (i) ionisation (ii) neutralisation (iii) dilution (iv) salt formation ★★☆

(a) (i) and (ii) (b) (i) and (iii)
(c) (ii) and (iii) (d) (ii) and (iv)

9. A few drops of universal indicator turn a solution green. Its pH and nature are: ★★☆

(a) pH 1, strongly acidic
(b) pH 7, neutral
(c) pH 10, basic
(d) pH 14, strongly basic

10. **Assertion (A):** Dry HCl gas does not change the colour of dry blue litmus paper.
Reason (R): HCl produces H^+ ions only in the presence of water. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

11. **Assertion (A):** A concentrated acid is diluted by adding it slowly to water with constant stirring.
Reason (R): Mixing a concentrated acid with water is a highly exothermic process. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(d)	NaOH: strong base, completely ionised; most OH^-.
2	(b)	Ethanol gives no ions, so no current.
3	(c)	HCl dissolves in the moisture and gives H^+(aq).
4	(b)	Milk of magnesia: blue, pH about 10, basic.
5	(a)	Lower pH means more H^+; pH 6 is acidic.
6	(b)	Heat is absorbed by the large amount of water.
7	(c)	Acetic acid ionises only partially.
8	(b)	The acid ionises and its ions get diluted.
9	(b)	Green means neutral, pH 7.
10	(a)	No water, no H^+ ions; R explains A.
11	(a)	Highly exothermic, so add acid slowly.

Short and long answer questions

1. Why does an aqueous solution of an acid conduct electricity, while a glucose solution does not? (2 marks) NCERT
2. While diluting an acid, why is it recommended that the acid be added to water and not water to the acid? (2 marks) NCERT
3. How is the concentration of H_3O^+ ions affected when an acid solution is diluted? (1 mark) NCERT
4. (a) Describe an activity to show that HCl is acidic only in water. (b) Differentiate strong and weak acids, with one example each. (5 marks)

Model answers

1 Acid conducts, glucose does not

- Acid in water gives **H^+(aq) ions**; moving ions carry current
- Glucose does not ionise in water: **no ions, no current**

2 Acid to water

- Dilution is **highly exothermic**; water into acid makes the mixture **splash** and burn, and glass may crack
- Acid added slowly with stirring: the water absorbs the heat

3 On dilution

H_3O^+ ions per unit volume **decrease**, so pH rises towards 7.

4 (a) Dry vs moist litmus

- Add conc. H_2SO_4 to common salt; pass the gas through a $CaCl_2$ guard tube
- Dry blue litmus: no change; moist blue litmus turns red, so H^+ ions form only in water

$$NaCl_{(s)} + H_2SO_{4(l)} \rightarrow NaHSO_{4(s)} + HCl_{(g)}\uparrow$$

(b) Strong acid ionises completely, e.g. HCl; weak acid ionises partially, e.g. CH_3COOH.

× COMMON MISTAKE	✓ CORRECT IDEA
Dilute HCl is a weak acid	Dilute HCl is still strong
Bases have no H^+ ions	They have fewer H^+ than OH^-
pH 9 is more acidic than pH 4	Lower pH = more acidic

Day 6 checklist: tick before you sleep

- [ ] I can explain hydrogen, hydronium and hydroxide ions with equations
- [ ] I can describe the conduction and dry HCl activities
- [ ] I know why acid is always added to water
- [ ] I can tell strength from concentration and read the pH scale
- [ ] I solved all 11 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai, ab chain ki neend lo! 4 se kam? Page 1–3 phir se padho.

LEVEL 6 COMPLETE

+100 XP earned. Next: Level 7

BUJHO *Samajh gaya! H^+ zyada toh pH kam, aur acid hamesha paani mein!*

PAHELI *Shabaash! Kal Level 7: pH in everyday life.*

Day 6 mind map: revise in 2 minutes

IONS
Acid gives H^+(aq) or H_3O^+; base gives OH^-(aq) ions

ONLY IN WATER
Bulb glows with acids; dry HCl does not affect dry litmus

ACIDS & BASES IN WATER

DILUTION ★★★
Exothermic; acid into water, slowly, stirring

STRENGTH
Complete vs partial ionisation; not the same as concentration

pH SCALE ★★★
0–14; 7 neutral; below 7 acidic, above 7 basic; more H^+ = lower pH; universal indicator gives the value

Coming up: Day 7

pH in everyday life: your stomach, your teeth, the soil, a bee sting, acid rain. Plus the family of salts and their pH.

NITESH SIR SAYS
*pH ka khel samajh gaye toh Chapter 2 aadha jeet liya. **Kal milte hai Day 7 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 2 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 07 · LEVEL 7

CHEMISTRY | CHAPTER 2 | PART 3 · ASTRA SERIES

pH IN DAILY LIFE & SALTS

Goal: apply pH to daily life and predict the pH of salts.

by Nitesh Prakash

01 Body & Rain | 02 Digestion | 03 Soil & Teeth | 04 Natural Acids | 05 Salts & pH | +100 XP today | Level 7 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 pH IN OUR BODY AND RAIN

KEY IDEA

- Living things survive only in a **narrow pH range**
- Our body works well between **pH 7.0 and 7.8**; a change upsets body processes
- Rain water with **pH below 5.6** is **acid rain**

Quick pH numbers to remember

Where	pH	What it means
Gastric juice (stomach)	about 2	HCl helps digest food
Mouth, tooth decay	< 5.5	enamel starts to corrode
Acid rain	< 5.6	harms fish and plants
Fresh milk	about 6	slightly acidic
Human body	7.0-7.8	blood is about 7.4
Toothpaste	about 8	mild base, fights decay

Acid rain: why rivers suffer ★★☆

- SO_2 and NO_2 from burning fuels dissolve in rain and make acids
- Acid rain flows into rivers and **lowers their pH**; aquatic life finds it hard to survive
- It also corrodes marble ($CaCO_3$) monuments

2 pH IN THE DIGESTIVE SYSTEM

- Stomach makes **hydrochloric acid (HCl)**; it helps digest food without harming the stomach
- **Indigestion:** too much acid causes pain and irritation
- **Antacids** are **mild bases** that neutralise the extra acid

$$Mg(OH)_{2(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$$

Milk of magnesia: Mg 1 = 1, O 2 = 2, H 4 = 4, Cl 2 = 2 ✔

$$NaHCO_{3(s)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

Baking soda is also used as an antacid

MEME BREAK

*Pet mein jalan? "Kya tapleek hai aapko?" Tapleek = extra HCl. Ilaaj = antacid, jo karta hai **neutralisation**!*

BUJHO *Pet mein jalan hai. NaOH pee loon? Base hi to hai!*

PAHELI *Bilkul nahi! NaOH strong aur corrosive hai. **Mild base** lo: milk of magnesia ya baking soda.*

3 pH OF SOIL

- Most plants grow best when soil pH is **close to 7**
- **Too acidic soil:** add a base: quicklime (CaO), slaked lime ($Ca(OH)_2$) or chalk ($CaCO_3$)
- **Too basic soil:** add decaying organic matter (manure, compost), which is acidic

BUJHO *Mitti ka pH 4.5 aaya. Thoda nimbu daal doon?*

PAHELI *Ulta ho jayega! Acidic mitti ko **base** chahiye, jaise quicklime, slaked lime ya chalk.*

4 TOOTH DECAY ★★★ MOST ASKED

1. Bacteria in the mouth break down **sugar and food particles** left after eating and produce **acids**.
2. The pH of the mouth falls; decay starts when it goes **below 5.5**.
3. Enamel is **calcium hydroxyapatite** (a form of calcium phosphate), the hardest substance in the body. It does not dissolve in water but is corroded by acid.
4. **Prevention:** rinse the mouth after eating; brush with toothpaste, which is basic and neutralises the acid.

★★★ *Tooth decay = pH 5.5 se neeche. Ye number pakka aayega, Nitesh Bhaiya ki guarantee!*

5 SELF-DEFENCE BY ANIMALS AND PLANTS

STING	RELIEF
Bee or ant sting: injects an acid (ant: methanoic acid)	Rub a mild base: **baking soda**
Nettle leaf: stinging hairs inject **methanoic acid**	Rub a **dock leaf** (it contains a base)
Wasp sting: injects a basic liquid	Rub a mild acid: **vinegar**

BUJHO *Jungle mein ek paudha chhua. Haath mein jalan, aankh mein aansu!*

PAHELI *Nettle hai! Uske baal methanoic acid daalte hain. Paas wale* ***dock ke patte*** *ragdo, unme base hota hai.*

6 NATURALLY OCCURRING ACIDS ★★★

Natural source	Acid present
Vinegar	**Acetic acid (ethanoic acid)**
Orange, lemon	**Citric acid**
Tamarind	**Tartaric acid**
Tomato	**Oxalic acid**
Sour milk (curd)	**Lactic acid**
Ant sting, nettle sting	**Methanoic acid (formic acid)**

These **organic acids** are weak acids, safe in food. Mineral acids (HCl, H_2SO_4, HNO_3) are strong.

7 SALTS AND FAMILIES OF SALTS

DEFINITION

- A **salt** forms when an acid neutralises a base: Acid + Base → Salt + Water
- Its **positive ion** comes from the base, its **negative ion** from the acid
- Salts with the **same positive or same negative ion** form a **family**

NCERT activity: salts, parent acids and bases ★★☆

Salt	Formula	Acid	Base
Potassium sulphate	**K_2SO_4**	H_2SO_4	KOH
Sodium sulphate	**Na_2SO_4**	H_2SO_4	NaOH
Calcium sulphate	**$CaSO_4$**	H_2SO_4	$Ca(OH)_2$
Magnesium sulphate	**$MgSO_4$**	H_2SO_4	$Mg(OH)_2$
Copper sulphate	**$CuSO_4$**	H_2SO_4	$Cu(OH)_2$
Sodium chloride	**NaCl**	HCl	NaOH
Sodium nitrate	**$NaNO_3$**	HNO_3	NaOH
Sodium carbonate	**Na_2CO_3**	H_2CO_3	NaOH
Ammonium chloride	**NH_4Cl**	HCl	NH_4OH

Sulphate family: K_2SO_4, Na_2SO_4, $CaSO_4$, $MgSO_4$, $CuSO_4$. **Sodium family:** Na_2SO_4, NaCl, $NaNO_3$, Na_2CO_3. **Chloride family:** NaCl, NH_4Cl.

MEME BREAK

SO_4^{2-} to Na, K, Ca, Mg, Cu: "Humare liye kaam karoge?" Sab haan bole, ban gayi ***sulphate family****!*

× WRONG	✓ RIGHT
Every salt is neutral	pH depends on the parent acid and base
"Salt" means only NaCl	NaCl is common salt; $CuSO_4$, Na_2CO_3 are salts too
$NaHCO_3$ has H, so it is acidic	Weak acid + strong base: basic

8 pH OF SALTS ★★★ MOST ASKED

NEUTRAL pH = 7	ACIDIC pH < 7	BASIC pH > 7
Strong acid + strong base	Strong acid + weak base	Weak acid + strong base
NaCl, KNO_3, Na_2SO_4, K_2SO_4	NH_4Cl, $CuSO_4$, $ZnSO_4$, $AlCl_3$	Na_2CO_3, CH_3COONa, $NaHCO_3$

KNOW THE PARENTS

Strong acids: HCl, H_2SO_4, HNO_3. **Weak acids:** CH_3COOH, H_2CO_3. **Strong bases:** NaOH, KOH. **Weak bases:** NH_4OH and insoluble hydroxides such as $Cu(OH)_2$, $Zn(OH)_2$.

★★★ *Salt ka swabhav = jo parent strong, uska. Dono strong? Neutral! Nitesh Bhaiya ki guarantee, ye rule har saal aata hai.*

Solved example: predict the pH

Q. Sodium acetate, CH_3COONa: acidic, basic or neutral?

$$CH_3COOH(aq) + NaOH(aq) \rightarrow CH_3COONa(aq) + H_2O(l)$$

Weak acid + strong base → **basic salt**, pH more than 7 ✓

NCERT activity: test the pH of these salts ★★★

Salt	Acid	Base	Nature
NaCl	HCl (S)	NaOH (S)	Neutral
KNO_3	HNO_3 (S)	KOH (S)	Neutral
$AlCl_3$	HCl (S)	$Al(OH)_3$ (W)	Acidic
$ZnSO_4$	H_2SO_4 (S)	$Zn(OH)_2$ (W)	Acidic
$CuSO_4$	H_2SO_4 (S)	$Cu(OH)_2$ (W)	Acidic
CH_3COONa	CH_3COOH (W)	NaOH (S)	Basic
Na_2CO_3	H_2CO_3 (W)	NaOH (S)	Basic
$NaHCO_3$	H_2CO_3 (W)	NaOH (S)	Basic

S = strong, W = weak. Dissolve each salt in distilled water and test with pH paper.

Real life: the milkman and baking soda ★★☆

- Fresh milk has pH about 6; as it sours, **lactic acid** forms and the pH **falls**
- A milkman adds a little **baking soda**: pH shifts to slightly alkaline
- So the milk takes **longer to set into curd**; the base must first neutralise the acid formed

Exam tips

- ✓ Always write the **number**: 7.0–7.8 (body), 5.5 (teeth), 5.6 (acid rain).
- ✓ Antacid answers need the word **mild base** and the neutralisation equation.
- ✓ For a salt, name **both parents** and say which is strong before giving the pH.
- ✓ Tamarind = tartaric, tomato = oxalic: the most confused pair in MCQs.

Day 7 mind map: revise in 2 minutes

BODY & RAIN
Body pH 7.0–7.8; acid rain below 5.6 harms aquatic life

DIGESTION
Extra HCl → antacid (mild base): $Mg(OH)_2$, $NaHCO_3$

pH IN DAILY LIFE & SALTS

SOIL & TEETH
Acidic soil + lime; decay below pH 5.5

STINGS
Methanoic acid → baking soda, dock leaf

SALTS ★★★
Family = same positive or negative ion; pH follows the stronger parent: SA + SB neutral, SA + WB acidic, WA + SB basic

TRY YOURSELF: ACIDIC, BASIC OR NEUTRAL?

(i) K_2SO_4 (ii) NH_4NO_3 (iii) CH_3COOK
(iv) $(NH_4)_2SO_4$ (v) KCl (vi) K_2CO_3
Hint: name the parent acid and base first, then mark which one is strong.

9 PRACTICE MCQS

Pehle khud solve karo, answer key page 4 par hai. Cheating = apne aap se dhokha!

1. Tooth decay starts when the pH of the mouth falls below: ★★★
(a) 7.0 (b) 6.5
(c) 5.5 (d) 8.0

2. Tamarind juice turns blue litmus red because it contains: CBSE 2025 ★★★
(a) acetic acid (b) methanoic acid
(c) oxalic acid (d) tartaric acid

3. Aqueous solutions: A = potassium nitrate, B = ammonium chloride, C = sodium carbonate. The ascending order of their pH is: CBSE 2025 ★★★
(a) A < B < C (b) B < C < A
(c) C < A < B (d) B < A < C

4. Sonia tests solutions of sodium acetate, sodium chloride and ammonium chloride and gets pH values 4.6, 7.0 and 8.9. Which order matches pH 4.6, 7.0, 8.9? CBSE Practice Set 2023 ★★☆
(a) sodium acetate, sodium chloride, ammonium chloride
(b) sodium chloride, ammonium chloride, sodium acetate
(c) ammonium chloride, sodium acetate, sodium chloride
(d) ammonium chloride, sodium chloride, sodium acetate

5. The pH of a soil sample is 5. Which of these should be added to make it suitable for crops? (I) calcium chloride (II) calcium hydroxide (III) calcium sulphate (IV) calcium oxide ★★☆
(a) (I) and (II) (b) (II) and (IV)
(c) (III) and (IV) (d) only (III)

6. Which pairs belong to the same family of salts? (I) NaCl and CH_3COONa (II) $CaSO_4$ and $MgSO_4$ (III) Na_2CO_3 and $NaHCO_3$ (IV) NaCl and $MgSO_4$ ★★☆
(a) (I) and (II) only (b) (II) and (III) only
(c) (I), (II) and (III) (d) (I), (II) and (IV)

7. The aqueous solution of which salt turns red litmus blue? ★★☆
(a) NH_4Cl (b) NaCl
(c) Na_2CO_3 (d) $CuSO_4$

8. "Milk of magnesia", a common antacid, is: ★★★
(a) magnesium hydroxide (b) sodium hydroxide
(c) calcium oxide (d) magnesium sulphate

9. **Assertion (A):** Aquatic animals find it difficult to survive in a river into which acid rain flows.
Reason (R): Acid rain lowers the pH of the river water. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** An aqueous solution of sodium carbonate is basic in nature.
Reason (R): Sodium carbonate is formed from a strong acid and a weak base. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short answer questions (PYQ)

1. Comment on: (A) a bee sting is treated with baking soda paste, but a wasp sting with dilute vinegar; (B) farmers treat soil with quicklime when tilling; (C) marble sculptures are protected by treating them with certain chemicals. CBSE SQP 2024
2. (A) How does pH help us when (i) the stomach makes too much acid during indigestion, (ii) we are stung by a honey bee or nettle leaves? (B) Fresh milk has pH 6. When it turns into curd, will its pH rise or fall? Why? CBSE 2022
3. Write the name and formula of (i) an acidic salt and (ii) a basic salt. Name the parent acid and parent base of each. CBSE 2024

Case-based question (4 marks)

A farmer finds his wheat growing poorly; his soil has **pH 4.5**. In a pond nearby, fed by rain water, fish are dying. The rain water tests **pH 4.8**.

1. Is this rain acid rain? Give a reason. (1)
2. Name two substances the farmer can add to his soil, and the type of reaction that takes place. (2)
3. Calcium carbonate is added to the pond. What happens to the pH of its water? (1)

Long answer question (5 marks)

1. What is a family of salts? Give one example each of a sodium family and a sulphate family. (2)
2. Explain why an aqueous solution of ammonium chloride is acidic, while that of sodium acetate is basic. Write the equation for the formation of each salt. (2)
3. Why is a strong base like sodium hydroxide never used as an antacid? (1)

Answer key: MCQs

Q	Ans	Why
1	(c)	Decay starts below pH 5.5.
2	(d)	Tamarind has tartaric acid.
3	(d)	NH_4Cl acidic < KNO_3 neutral < Na_2CO_3 basic.
4	(d)	NH_4Cl 4.6, NaCl 7.0, CH_3COONa 8.9.
5	(b)	$Ca(OH)_2$ and CaO are bases; the others are neutral salts.
6	(c)	(I), (III) sodium family; (II) sulphate family.
7	(c)	Na_2CO_3 is a basic salt.
8	(a)	Milk of magnesia = $Mg(OH)_2$.
9	(a)	Lower pH is the reason fish suffer.
10	(c)	Na_2CO_3 comes from a weak acid (H_2CO_3) and a strong base (NaOH).

Model answers

1 Comment on the statements

- **(A)** Bee sting is acidic, so a mild base (baking soda) neutralises it; wasp sting is basic, so a mild acid (vinegar) neutralises it.
- **(B)** Soil is often too acidic; quicklime is a base and neutralises the extra acid.
- **(C)** Acid rain corrodes marble ($CaCO_3$); a protective treatment keeps the acid away.

$$CaCO_{3(s)} + H_2SO_{4(aq)} \rightarrow CaSO_{4(s)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

Ca 1 = 1, S 1 = 1, C 1 = 1, H 2 = 2, O 7 = 7 ✔

2 pH and indigestion, stings, milk

- **(i)** An antacid (mild base) such as milk of magnesia neutralises the extra HCl.
- **(ii)** Rub baking soda (or a dock leaf for nettle) to neutralise the acid.
- **(B)** pH **falls**: bacteria turn lactose into lactic acid, so curd is more acidic.

3 Acidic and basic salt

- **Acidic:** ammonium chloride, NH_4Cl, from HCl (strong) and NH_4OH (weak)
- **Basic:** sodium carbonate, Na_2CO_3, from H_2CO_3 (weak) and NaOH (strong)

4 Case-based

- **(a)** Yes: its pH (4.8) is below 5.6.
- **(b)** Quicklime (CaO) and slaked lime ($Ca(OH)_2$), or chalk; neutralisation.
- **(c)** The pH rises towards 7 as $CaCO_3$ neutralises the acid.

5 Long answer

- **Family:** salts with the same positive or negative ion. Sodium family: NaCl, Na_2SO_4. Sulphate family: K_2SO_4, $CuSO_4$.
- NH_4Cl: strong acid + weak base, so acidic. CH_3COONa: weak acid + strong base, so basic.

$$NH_4OH_{(aq)} + HCl_{(aq)} \rightarrow NH_4Cl_{(aq)} + H_2O_{(l)}$$

$$CH_3COOH_{(aq)} + NaOH_{(aq)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)}$$

- NaOH is strong and corrosive; it would damage the mouth, food pipe and stomach lining, so only mild bases are used.

Try-yourself answers

(i) neutral, (ii) acidic, (iii) basic, (iv) acidic, (v) neutral, (vi) basic.
Rule: the stronger parent decides.

Day 7 hit-list: Eduspark PYQ analysis

Topic	Rating
Tooth decay below pH 5.5	★★★
Antacids: $Mg(OH)_2$, $NaHCO_3$	★★★
pH of salts (parent acid and base)	★★★
Natural acids: tamarind, tomato, curd	★★★
Acid rain and aquatic life	★★☆
Soil treated with lime	★★☆
Bee, wasp and nettle stings	★★☆
Families of salts	★☆☆

Day 7 checklist: tick before you sleep

- ☐ I can give the key pH numbers: body 7.0–7.8, teeth 5.5, acid rain 5.6
- ☐ I can explain antacids and write the $Mg(OH)_2$ + HCl equation
- ☐ I can match natural sources with their acids
- ☐ I can name a salt's family and its parent acid and base
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 7 COMPLETE +100 XP earned. Next: Level 8

BUJHO *Samajh gaya! Salt ka swabhav strong parent se aata hai, aur daant 5.5 se neeche pH pe kharab hote hain!*

PAHELI *Shabaash! Kal Level 8: common salt se bane chemicals.*

pH FIXERS AROUND YOU

Bases: antacid, toothpaste, baking soda, quicklime, chalk, dock leaf.
Acids: vinegar (wasp sting), manure or compost (basic soil).

Coming up: Day 8

Chemicals from common salt: NaOH from the chlor-alkali process, bleaching powder, baking soda, washing soda and Plaster of Paris.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 2 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 08 · LEVEL 8

CHEMISTRY | CHAPTER 2 | PART 4 · ASTRA SERIES

CHEMICALS FROM COMMON SALT

Goal: prepare, write and use five chemicals made from NaCl.

by Nitesh Prakash

01 Chlor-alkali | 02 Bleaching Powder | 03 Sodas | 04 Hydrates | 05 PoP

+125 XP today | Level 8 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

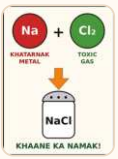
1 COMMON SALT: THE RAW MATERIAL

COMMON SALT (NaCl)

- A **neutral salt** of strong acid HCl and strong base NaOH
- **Sea water:** salt is obtained by **evaporation**; also from salt lakes: Sambhar (Rajasthan), Great Salt Lake (USA)
- **Rock salt** (*sendha namak*): beds of large crystals, often **brown** due to impurities; mined like coal at Khewra (Pakistan) and Mandi (Himachal Pradesh)

$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

neutralisation: Na 1 = 1, Cl 1 = 1, H 2 = 2, O 1 = 1 ✔



MEME BREAK

*Ek **khatarnak metal** (Na) + ek **toxic gas** (Cl_2) = khaane ka namak! Aur isi se 5 naye chemicals banenge.*

Family tree of common salt

From NaCl, by…	We get
Electrolysis of brine	NaOH, Cl_2, H_2
Cl_2 + dry slaked lime	Bleaching powder
Brine + NH_3 + CO_2	Baking soda, $NaHCO_3$
Heat $NaHCO_3$, then recrystallise	Washing soda

PoP is made from gypsum, not NaCl; NCERT teaches it in this section.

Real life: salt and freedom

- Used in food, to preserve pickles and meat, and in industry
- 1930: Gandhiji's **Dandi March** made salt from sea water

2 CHLOR-ALKALI PROCESS ★★★ MOST ASKED

DEFINITION

- **Electricity** is passed through **brine** (aqueous NaCl); it decomposes to give NaOH, Cl_2 and H_2
- Name: **chlor** for chlorine + **alkali** for sodium hydroxide

$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$

Na 2 = 2, Cl 2 = 2, H 4 = 4, O 2 = 2 ✔

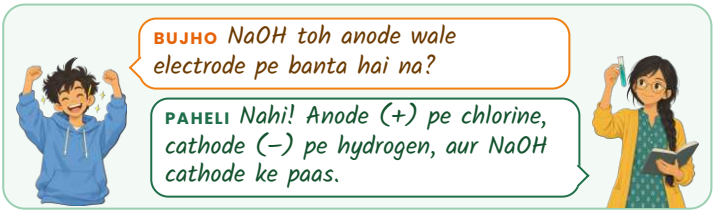
Which product, where?

Where?	What forms
Anode (+)	Cl_2 gas
Cathode (−)	H_2 gas
Near the cathode	NaOH solution

Brine = a concentrated solution of NaCl in water.

Uses of the products ★★☆

Product	Important uses
NaOH	de-greasing metals, soaps and detergents, paper making, artificial fibres
Cl_2	water treatment, swimming pools, PVC, disinfectants, CFCs, pesticides
H_2	fuels, margarine (hydrogenation of oils), ammonia for fertilisers
HCl	from H_2 + Cl_2: cleaning steel, ammonium chloride, medicines, cosmetics
Bleach	from NaOH + Cl_2: household bleaches, bleaching fabrics



★★★ *Chlor-alkali ki equation aur teeno products ke uses pakka aayenge, Nitesh Bhaiya ki guarantee!*

3 BLEACHING POWDER ★★★ MOST ASKED

PREPARATION

- **Chlorine gas** is passed over **dry slaked lime**, $Ca(OH)_2$
- Cl_2 itself comes from the chlor-alkali process
- NCERT writes bleaching powder as **$CaOCl_2$** (calcium oxychloride); its real composition is quite complex

$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$

Ca 1 = 1, O 2 = 2, H 2 = 2, Cl 2 = 2 ✔

Uses of bleaching powder

1. **Bleaching** cotton and linen (textile industry), wood pulp (paper factories) and washed clothes (laundry)
2. **Oxidising agent** in many chemical industries
3. **Disinfecting** drinking water, making it free from germs

FUN FACT

Bleaching powder ki tez smell = chlorine! Bujho ko chheenk aa gayi, aur paani ke germs ka khel khatam.

4 BAKING SODA ★★★ MOST ASKED

PREPARATION

- Chemical name: **sodium hydrogencarbonate**; made from brine with **NH_3 and CO_2**
- A **mild, non-corrosive base**

$$NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} + NH_{3(g)} \rightarrow NH_4Cl_{(aq)} + NaHCO_{3(s)}$$

Na 1 = 1, Cl 1 = 1, H 5 = 5, O 3 = 3, C 1 = 1, N 1 = 1 ✔

On heating (while cooking) ★★★

$$2NaHCO_{3(s)} \xrightarrow{\text{Heat}} Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

Na 2 = 2, H 2 = 2, C 2 = 2, O 6 = 6 ✔

The CO_2 released makes food soft; the Na_2CO_3 left behind tastes bitter.

Baking powder: soda + edible acid ★★★

- Acid used: **tartaric acid**
- With water or heat: $NaHCO_3 + H^+ \rightarrow CO_2 + H_2O$ + sodium salt of the acid
- CO_2 bubbles make bread and cake **soft and spongy**
- The acid **neutralises** the bitter Na_2CO_3, so no bitter taste

MEME BREAK

*Baking soda + tartaric acid: **perfectly balanced**, as all things should be. Cake bhi phoola, swaad bhi kadwa nahi!*

Uses of baking soda

1. To make **baking powder** (cakes, bread, dhokla)
2. In **antacids**: being basic, it neutralises extra acid in the stomach
3. In **soda-acid fire extinguishers**

BUJHO *Cake mein baking soda daala, phir bhi kadwa kyun?*

PAHELI *Garam hokar soda, sodium carbonate chhodta hai, wahi kadwa hai. Baking powder lo: tartaric acid use neutralise kar deta hai!*

5 WASHING SODA ★★☆

Preparation in three steps

1. Make baking soda from brine (equation above).
2. Heat it to get **soda ash**, anhydrous Na_2CO_3.
3. Recrystallise soda ash from water:

$$Na_2CO_{3(s)} + 10H_2O_{(l)} \rightarrow Na_2CO_3{\cdot}10H_2O_{(s)}$$

sodium carbonate decahydrate: H 20 = 20, O 13 = 13 ✔

Uses of washing soda ★★★

1. **Glass, soap and paper** industries
2. Making sodium compounds such as **borax**
3. **Cleaning agent** for domestic purposes
4. Removing **permanent hardness** of water

Baking soda vs washing soda ★★★

	Baking soda	Washing soda
Chemical name	sodium hydrogencarbonate	sodium carbonate
Formula	$NaHCO_3$	$Na_2CO_3{\cdot}10H_2O$
Nature	mild base	stronger base; harsh on skin
Main use	baking powder, antacid	cleaning, hard water, glass

6 WATER OF CRYSTALLISATION ★★★ MOST ASKED

DEFINITION

- The **fixed number of water molecules** present in **one formula unit** of a salt
- Crystals look dry, but this water is part of the crystal

Salt	Formula	H_2O
Copper sulphate (blue vitriol)	$CuSO_4{\cdot}5H_2O$	5
Washing soda	$Na_2CO_3{\cdot}10H_2O$	10
Ferrous sulphate (green vitriol)	$FeSO_4{\cdot}7H_2O$	7
Gypsum	$CaSO_4{\cdot}2H_2O$	2

Activity: heating copper sulphate crystals ★★★

- Heat blue $CuSO_4$ crystals in a dry boiling tube: **water droplets** appear and the salt turns **white**

$$CuSO_4{\cdot}5H_2O_{(s)} \xrightarrow{\text{Heat}} CuSO_{4(s)} + 5H_2O_{(g)}$$

blue → white (anhydrous)

- Add 2–3 drops of water to the white powder: the **blue colour returns**

$$CuSO_{4(s)} + 5H_2O_{(l)} \rightarrow CuSO_4{\cdot}5H_2O_{(s)}$$

white → blue: water of crystallisation regained

MEME BREAK

*$CuSO_4$ garam hone ke baad: paani gaya, neela rang gaya, poora **look hi badal gaya**. Do boond paani daalo, wapas neela hero!*

BUJHO *Crystals bilkul sukhe the, toh tube mein paani kahan se aaya?*

PAHELI *Wahi water of crystallisation! Garam karte hi nikal jaata hai, isliye neela $CuSO_4$ safed ho jaata hai.*

7 PLASTER OF PARIS (PoP) ★★★ MOST ASKED

PREPARATION

- Heat **gypsum** at **373 K**: it loses water and becomes calcium sulphate **hemihydrate**
- Formula **$CaSO_4 \cdot \frac{1}{2}H_2O$**: two formula units of $CaSO_4$ share one water molecule
- **Heat with care:** above 373 K all water is lost, giving anhydrous $CaSO_4$, called **dead-burnt plaster**, which does not set with water

$$CaSO_4 \cdot 2H_2O(s) \xrightarrow{373\ K} CaSO_4 \cdot \tfrac{1}{2}H_2O(s) + 1\tfrac{1}{2}H_2O(g)$$

gypsum → Plaster of Paris

Setting of PoP ★★★

- Mixed with water, PoP changes back into **gypsum**, a hard solid mass

$$CaSO_4 \cdot \tfrac{1}{2}H_2O(s) + 1\tfrac{1}{2}H_2O(l) \rightarrow CaSO_4 \cdot 2H_2O(s)$$

Ca 1 = 1, S 1 = 1, O 6 = 6, H 4 = 4 ✓

So PoP is stored in a **moisture-proof** container: moisture would make it set.

Uses of Plaster of Paris

1. Plaster for supporting **fractured bones** in the right position
2. Making **toys** and materials for **decoration**
3. Making surfaces **smooth** (walls, ceilings)

MEME BREAK

*PoP + paani = gypsum. Statue ki tarah **set**! Isliye doctor tooti haddi pe yahi lagate hai: ab hilna mana hai.*

Formula table: learn by heart ★★★

Common name	Formula	Made from
Bleaching powder	$CaOCl_2$	Cl_2 + dry $Ca(OH)_2$
Baking soda	$NaHCO_3$	brine + NH_3 + CO_2
Washing soda	$Na_2CO_3 \cdot 10H_2O$	soda ash + $10H_2O$
Plaster of Paris	$CaSO_4 \cdot \frac{1}{2}H_2O$	gypsum at 373 K
Caustic soda	NaOH	electrolysis of brine

★★★ *PoP, bleaching powder aur copper sulphate ki equations har saal aati hai, Nitesh Bhaiya ki guarantee!*

× WRONG	✓ RIGHT
PoP = $CaSO_4 \cdot H_2O$	$CaSO_4 \cdot \frac{1}{2}H_2O$ (hemihydrate)
Washing soda = $NaHCO_3$	$Na_2CO_3 \cdot 10H_2O$
Cl_2 over lime water	Cl_2 over **dry** slaked lime
H_2 at the anode	H_2 at the cathode (−)

Exam tips

- ✔ Gypsum is heated at exactly **373 K**; write it over the arrow
- ✔ Hydrates: write the dot and the water count, e.g. $CuSO_4 \cdot 5H_2O$
- ✔ "Uses" questions: give the use with the reason (basic, oxidising, sets hard)
- ✔ Colours: $CuSO_4 \cdot 5H_2O$ blue, anhydrous $CuSO_4$ white

8 PRACTICE MCQS

Pehle khud solve karo, phir answer key se milao. Bujho ki tarah guess mat karna!

1. A mild, non-corrosive basic salt is: CBSE SQP 2023 ★★★
(a) $Ca(OH)_2$ (b) NaCl
(c) NaOH (d) $NaHCO_3$

2. A chemical compound used in glass, soap and paper industries is: CBSE 2024 ★★★
(a) washing soda (b) baking soda
(c) bleaching powder (d) common salt

3. Which compound has the maximum number of water molecules of crystallisation in one formula unit? CBSE 2024 ★★★
(a) $FeSO_4$ (b) $CuSO_4$
(c) $CaSO_4$ (d) Na_2CO_3

4. In the chlor-alkali process, the gas formed at the anode is: ★★★
(a) hydrogen (b) chlorine
(c) oxygen (d) carbon dioxide

5. Bleaching powder is made by passing chlorine over: ★★☆
(a) quicklime (b) dry slaked lime
(c) limestone (d) gypsum

6. The chemical formula of Plaster of Paris is: ★★★
(a) $CaSO_4 \cdot 2H_2O$ (b) $CaSO_4 \cdot \frac{1}{2}H_2O$
(c) $CaSO_4 \cdot H_2O$ (d) $CaSO_4$

7. On strong heating, blue copper sulphate crystals turn: ★★★
(a) green (b) black
(c) white (d) red

8. Baking powder is a mixture of baking soda and: ★★☆
(a) tartaric acid (b) hydrochloric acid
(c) washing soda (d) common salt

9. **Assertion (A):** Fresh milk to which baking soda is added takes a longer time to set as curd.
Reason (R): Baking soda decreases the pH value of fresh milk to below 6. CBSE Term-1 SQP 2021 ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Plaster of Paris is stored in a moisture-proof container.
Reason (R): Plaster of Paris reacts with water and sets into a hard mass of gypsum. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9 SHORT ANSWER QUESTIONS

1. A compound X prepared from gypsum hardens when mixed with the proper quantity of water. Identify X and write its formula. State the difference in chemical composition between baking soda and baking powder. CBSE 2023 (3 marks)

2. Chlorine gas is prepared by the electrolysis of brine. Write the chemical equation. Name the other products and give one use of each. CBSE SQP 2023 (3 marks)

3. Write balanced equations for: (a) heating blue copper sulphate crystals; (b) heating sodium hydrogencarbonate during cooking. CBSE 2023 (2 marks)

4. Write the equation for the preparation of baking soda. Why is it also an ingredient of antacids? CBSE Compartment 2025 (2 marks)

Answer key: MCQs

Q	Ans	Why
1	(d)	$NaHCO_3$ is a mild, non-corrosive base.
2	(a)	Washing soda: glass, soap, paper.
3	(d)	$Na_2CO_3 \cdot 10H_2O$ has 10 water molecules.
4	(b)	Cl_2 at the anode; H_2 at the cathode.
5	(b)	Cl_2 + dry $Ca(OH)_2 \rightarrow CaOCl_2 + H_2O$
6	(b)	Hemihydrate: $CaSO_4 \cdot \frac{1}{2}H_2O$.
7	(c)	It loses water: anhydrous $CuSO_4$ is white.
8	(a)	A mild edible acid: tartaric acid.
9	(c)	Baking soda is basic; it raises the pH.
10	(a)	PoP sets with moisture, so keep it dry.

Coming up: Day 9

The Chapter 2 **Boss Level**: one full test on indicators, pH, salts and today's chemicals. Revise the formula table tonight!

NITESH SIR SAYS
*Formula table roz ek baar likho. **Kal Boss Level, taiyaar raho!***

Model answers

1 X = Plaster of Paris
- Formula $CaSO_4 \cdot \frac{1}{2}H_2O$; made by heating gypsum at 373 K
- **Baking soda:** only $NaHCO_3$. **Baking powder:** $NaHCO_3$ + a mild edible acid such as tartaric acid

2 Electrolysis of brine

$$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)} + H_{2(g)}$$

- **NaOH:** soaps and detergents
- **H_2:** fuel; ammonia for fertilisers
- **Cl_2:** water treatment; bleaching powder

3 Two heating reactions

$$CuSO_4 \cdot 5H_2O_{(s)} \xrightarrow{\text{Heat}} CuSO_{4(s)} + 5H_2O_{(g)}$$

(a) blue crystals → white powder

$$2NaHCO_{3(s)} \xrightarrow{\text{Heat}} Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)} \uparrow$$

(b) CO_2 makes the food soft

4 Baking soda and antacids

$$NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} + NH_{3(g)} \rightarrow NH_4Cl_{(aq)} + NaHCO_{3(s)}$$

- It is a mild base, so it **neutralises excess acid** in the stomach and gives relief

Day 8 checklist: tick before you sleep

- ☐ I can write the chlor-alkali equation and name the product at each electrode
- ☐ I can prepare bleaching powder, baking soda and washing soda with equations
- ☐ I can tell baking soda, baking powder and washing soda apart
- ☐ I can explain water of crystallisation and the PoP equations
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2–3 phir se padho.

LEVEL 8 COMPLETE +125 XP earned. Next: Level 9

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 2 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 09 LEVEL 9

CHEMISTRY | CHAPTER 2 | BOSS LEVEL

ACIDS, BASES & SALTS

Goal: revise Chapter 2, then clear the Boss test.

by Nitesh Prakash

01 Revision Map | 02 14 MCQs | 03 Case Study | 04 Written Answers | 05 Answer Key | +200 XP Boss Level | Level 9 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 2 IN ONE MAP

Quick revision map: the whole chapter

INDICATORS
Natural: litmus, turmeric, red cabbage; synthetic: phenolphthalein, methyl orange; olfactory: onion, vanilla, clove

REACTIONS ★★★
Acid + metal → salt + H_2; acid + carbonate → salt + H_2O + CO_2; acid + base → salt + water

ACIDS, BASES & SALTS

IONS IN WATER
In water, acids give hydrogen ions, H^+ (as H_3O^+); bases give OH^-; soluble bases are alkalis

pH SCALE ★★★
0–14; below 7 acidic, 7 neutral, above 7 basic; lower pH means more H^+ ions

STRENGTH AND DILUTION
Strong acids (HCl, H_2SO_4, HNO_3) ionise fully; weak (CH_3COOH, H_2CO_3) partly. Dilution is exothermic: add acid to water slowly, never water to acid.

2 INDICATOR COLOURS

Indicator	In acid	In base
Litmus	Blue → red	Red → blue
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow
Turmeric	Stays yellow	Reddish-brown
Red cabbage	Red/pink	Green
Onion, vanilla	Smell stays	Smell is lost

3 REACTIONS OF ACIDS AND BASES

$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}\uparrow$

$2NaOH_{(aq)} + Zn_{(s)} \rightarrow Na_2ZnO_{2(aq)} + H_{2(g)}\uparrow$
Base + metal (sodium zincate)

$Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$

$Ca(OH)_{2(aq)} + CO_{2(g)} \rightarrow CaCO_{3(s)}\downarrow + H_2O_{(l)}$
Lime water turns milky

$CaCO_{3(s)} + H_2O_{(l)} + CO_{2(g)} \rightarrow Ca(HCO_3)_{2(aq)}$
Excess CO_2: milkiness disappears

$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

$CuO_{(s)} + 2HCl_{(aq)} \rightarrow CuCl_{2(aq)} + H_2O_{(l)}$
Metal oxide is basic: blue-green solution

4 CHEMICALS FROM COMMON SALT

$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$
Chlor-alkali: Cl_2 at anode, H_2 at cathode

$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$
Bleaching powder (dry slaked lime)

$NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} + NH_{3(g)} \rightarrow NH_4Cl_{(aq)} + NaHCO_{3(s)}$
Baking soda

$2NaHCO_{3(s)} \xrightarrow{\text{Heat}} Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)}\uparrow$
Heating baking soda

$Na_2CO_{3(s)} + 10H_2O_{(l)} \rightarrow Na_2CO_3{\cdot}10H_2O_{(s)}$
Washing soda (recrystallisation)

$CuSO_4{\cdot}5H_2O_{(s)} \xrightarrow{\text{Heat}} CuSO_{4(s)} + 5H_2O_{(g)}$
Blue crystals turn white

$CaSO_4{\cdot}2H_2O_{(s)} \xrightarrow{373\ K} CaSO_4{\cdot}\frac{1}{2}H_2O_{(s)} + 1\frac{1}{2}H_2O_{(g)}$
Gypsum → Plaster of Paris

$CaSO_4{\cdot}\frac{1}{2}H_2O_{(s)} + 1\frac{1}{2}H_2O_{(l)} \rightarrow CaSO_4{\cdot}2H_2O_{(s)}$
PoP sets into hard gypsum

5 pH SCALE AND SALTS

Lemon juice 2.2, water 7, blood 7.4, milk of magnesia 10, NaOH 14

- Human body works in pH **7.0–7.8**; rain below **5.6** is acid rain
- Tooth enamel corrodes below pH **5.5**; brush after meals
- Ant sting, nettle = **methanoic acid**; bee sting: rub baking soda
- Excess stomach acid: antacid like milk of magnesia, $Mg(OH)_2$

Made from	Salt pH	Examples
Strong acid + strong base	7 (neutral)	NaCl, Na_2SO_4, KNO_3
Strong acid + weak base	below 7 (acidic)	NH_4Cl, $(NH_4)_2SO_4$
Weak acid + strong base	above 7 (basic)	Na_2CO_3, CH_3COONa

★★★ *Chlor-alkali, bleaching powder, baking soda aur PoP ke equations baar baar poochhe jaate hain: Nitesh Bhaiya ki guarantee!*

BUJHO *Boss Level kal hai aur poora chapter abhi bacha hai!*

PAHELI *Tension nahi! Pehle yeh map padho, phir test khud solve karo. Practice ki galti exam mein nahi hogi!*

6 BOSS LEVEL TEST: CHAPTER 2

INSTRUCTIONS

Time: 60 minutes. **Maximum marks:** 40. Section A: Q1–14, MCQs, 1 mark each. Section B: Q15, case-based, 4 marks. Section C: Q16–20, short answers, 2 or 3 marks. Section D: Q21–22, long answers, 5 marks each.

Boss Level mode ON! Phone side mein rakho, timer lagao aur key tab tak mat dekho.

SECTION A: MULTIPLE CHOICE — 14 × 1 = 14

1. Which of the following is a mild, non-corrosive basic salt?
CBSE SQP 2023 ★★★
(a) $Ca(OH)_2$ (b) NaCl
(c) NaOH (d) $NaHCO_3$

2. Methyl orange is added to dilute hydrochloric acid and to aqueous sodium hydroxide. Its colours in the two solutions are:
CBSE SQP 2025 ★★★
(a) orange; red (b) red; yellow
(c) red; orange (d) yellow; red

3. A compound used in the glass, soap and paper industries is:
CBSE 2024 ★★☆
(a) washing soda (b) baking soda
(c) bleaching powder (d) common salt

4. Solution A turns phenolphthalein pink. When solution B is added to A, the pink colour disappears. Which statement is true?
CBSE SQP 2024 ★★★
(a) A is strongly basic and B is a weak base.
(b) A is strongly acidic and B is a weak acid.
(c) A has pH greater than 7 and B has pH less than 7.
(d) A has pH less than 7 and B has pH greater than 7.

5. When excess carbon dioxide is passed through lime water, the milkiness disappears because: CBSE SQP 2025 ★★☆
(a) soluble $CaCO_3$ changes to soluble $Ca(HCO_3)_2$
(b) insoluble $CaCO_3$ changes to soluble $Ca(HCO_3)_2$
(c) soluble $CaCO_3$ changes to insoluble $Ca(HCO_3)_2$
(d) insoluble $CaCO_3$ changes to insoluble $Ca(HCO_3)_2$

6. Identify the compound A in the reaction $NaCl + NH_3 + H_2O + CO_2 \rightarrow A + NH_4Cl$ CBSE Compartment 2024 ★★★
(a) Na_2CO_3 (b) $(NH_4)_2CO_3$
(c) $NaHCO_3$ (d) NH_4HCO_3

7. Aqueous solutions: A = potassium nitrate, B = ammonium chloride, C = sodium carbonate. Their pH in ascending order is:
CBSE 2025 ★★☆
(a) A < B < C (b) B < C < A
(c) C < A < B (d) B < A < C

8. Tamarind juice turns blue litmus red. The acid present in it is:
CBSE 2025 ★★☆
(a) acetic acid (b) methanoic acid
(c) oxalic acid (d) tartaric acid

9. A curry stain on a white shirt turns reddish-brown when soap is scrubbed on it and yellow again after washing with plenty of water. The reasons are: (I) soap is acidic; (II) soap is basic; (III) turmeric turns reddish in bases; (IV) turmeric turns reddish in acids.
CBSE Term-1 SQP 2021 ★★☆
(a) (I) and (II) (b) (II) and (III)
(c) (I) and (IV) (d) (II) and (IV)

10. In a conductivity test, the bulb does not glow at all. The liquid in the beaker is most likely: ★★☆
(a) dilute HNO_3 (b) C_2H_5OH (alcohol)
(c) dilute H_2SO_4 (d) dilute HCl

11. The parent acid and parent base of calcium carbonate are:
CBSE Term-1 SQP 2021 ★★☆
(a) HCl and NaOH (b) H_2CO_3 and $Ca(OH)_2$
(c) H_2SO_4 and $Ca(OH)_2$ (d) HNO_3 and $Ca(OH)_2$

12. To dilute a concentrated acid safely, you should add: ★★★
(a) water to the acid quickly
(b) the acid to water slowly, with stirring
(c) water to the acid slowly
(d) both together at once

13. **Assertion (A):** Fresh milk to which baking soda is added takes a longer time to set as curd.
Reason (R): Baking soda decreases the pH of fresh milk to below 6.
CBSE Term-1 SQP 2021 ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

14. **Assertion (A):** Metal oxides are acidic in nature.
Reason (R): Calcium hydroxide reacts with carbon dioxide to form a salt and water. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B: CASE-BASED — 1 × 4 = 4

READ THE PASSAGE

Electricity passed through **brine** (concentrated aqueous NaCl) gives sodium hydroxide, so the process is called the **chlor-alkali process**. Chlorine forms at the anode, hydrogen at the cathode, and NaOH solution near the cathode. Sodium hydroxide formed here is a strong alkali. All three products are useful; hydrogen, for example, is a fuel and is used to make ammonia for fertilisers.

15. Answer these, based on the passage: ★★★
(a) Write the balanced equation for this process. [1]
(b) Name the gas formed at the anode; give one use. [1]
(c) Chlorine reacts with dry slaked lime to form a compound Z. Name Z, write the equation for its formation and give one use of Z. [2]

OR

(c) Hydrogen and chlorine from this process combine to form a compound. Name it and give one use. Also give two uses of sodium hydroxide. [2]

SECTION C: SHORT ANSWERS — 12 marks

16. Give reasons: CBSE 2020 ★★★ [2]
(a) Sodium hydrogencarbonate is used as an antacid.
(b) Blue copper sulphate crystals turn white on strong heating.

17. Answer both parts: CBSE 2023 ★★★ [2]
(a) Compound X, made from gypsum, hardens when mixed with the right amount of water. Identify X and write down its chemical formula.
(b) How does baking powder differ from baking soda in chemical composition?

18. The pH of rain water collected from cities A and B is 6.1 and 5.3. Which city has more acidic rain water? Justify. State, with reason, what would happen to aquatic life in a pond into which the rain water of city B flows. CBSE Compartment 2024 ★★☆ [3]

19. Answer both parts: CBSE 2022 ★★★ [3]
(a) What would you use to counteract the change in pH in: (i) too much acid in the stomach during indigestion; (ii) a honey-bee or nettle sting?
(b) Fresh milk has a pH of 6. When it turns into curd, will its pH increase or decrease? Why?

20. Three solutions A, B and C have pH values 6, 8 and 9.5. CBSE SQP 2024 ★★☆ [2]
(a) In which solution are the most hydronium ions present?
(b) Arrange A, B and C in increasing order of H^+ ions.

Short answer mein equation aur reason dono likho, tabhi poore marks milenge!

Time plan: 60 minutes

1. Section A (14 MCQs): **15 min**
2. Section B (case study): **8 min**
3. Section C (short answers): **17 min**
4. Section D (long answers): **15 min**
5. Final check of equations and states: **5 min**

Common mistakes to avoid

× WRONG	✓ RIGHT
Higher pH = more acidic	Lower pH = more acidic
Water poured into acid	Acid added to water, slowly
PoP = $CaSO_4 \cdot 2H_2O$	PoP = $CaSO_4 \cdot \frac{1}{2}H_2O$
Cl_2 + lime water	Cl_2 + dry slaked lime
Every salt is neutral	NH_4Cl acidic, Na_2CO_3 basic
Baking soda = Na_2CO_3	Baking soda = $NaHCO_3$

SECTION D: LONG ANSWERS — 2 × 5 = 10

21. Acids and their salts: ★★★
(a) Write balanced chemical equations for the reaction of an acid with (i) a metal, (ii) a base and (iii) a metal carbonate. Name the main product in each case. CBSE 2025 [3]
(b) Name the acid and the base from which (i) sodium chloride and (ii) ammonium sulphate are obtained. State whether each salt is acidic, basic or neutral, with reason. CBSE Compartment 2024 [2]

22. Salts with water of crystallisation: ★★★
(a) What is water of crystallisation? Write the formulae of crystalline copper sulphate and washing soda. CBSE Compartment 2025 [1]
(b) How is Plaster of Paris obtained from gypsum? Write the equation. Why is it stored in a moisture-proof container? [2]
(c) Write the two equations for making washing soda from baking soda. Give one use of washing soda. [2]

7 ANSWER KEY AND MODEL ANSWERS

Section A: MCQ answers

Q	Ans	Why
1	(d)	$NaHCO_3$ is a mild base, so it is non-corrosive.
2	(b)	Red in acid, yellow in base.
3	(a)	Washing soda, $Na_2CO_3 \cdot 10H_2O$.
4	(c)	Pink means basic; B neutralises A, so B is acidic.
5	(b)	$Ca(HCO_3)_2$ is soluble in water.
6	(c)	$NaHCO_3$ forms; NH_4Cl is the by-product.
7	(d)	NH_4Cl acidic, KNO_3 neutral, Na_2CO_3 basic.
8	(d)	Tamarind contains tartaric acid.
9	(b)	Soap is basic; turmeric turns reddish-brown in bases.
10	(b)	Alcohol gives no ions in water, so no current flows.
11	(b)	$H_2CO_3 + Ca(OH)_2 \rightarrow CaCO_3 + 2H_2O$
12	(b)	Dilution is highly exothermic; acid may splash.
13	(c)	A is true; baking soda raises the pH of milk. R is false.
14	(d)	Metal oxides are basic, so A is false; R is true.

15 Case: chlor-alkali process

$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$
(a) Chlor-alkali process

- (b) **Chlorine** at the anode; used in water treatment, PVC, disinfectants
- (c) Z = **bleaching powder**, $CaOCl_2$; used to bleach cotton and linen and to disinfect drinking water

$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$

- OR (c): $H_2 + Cl_2$ give **HCl** (cleaning steel, medicines). NaOH: soaps and detergents, paper making, degreasing metals

BOARD MARKING TIPS

- Balanced equation with state symbols = full marks
- Write the product's name along with its formula

16 Antacid; copper sulphate turns white

- (a) $NaHCO_3$ is a mild, non-corrosive base; it neutralises extra HCl in the stomach
- (b) Heating removes the water of crystallisation; anhydrous $CuSO_4$ is white

$$CuSO_4{\cdot}5H_2O_{(s)} \xrightarrow{\text{Heat}} CuSO_{4(s)} + 5H_2O_{(g)}$$

17 Plaster of Paris; baking powder

- (a) X = **Plaster of Paris**, $CaSO_4{\cdot}\frac{1}{2}H_2O$
- (b) Baking soda is only $NaHCO_3$; baking powder is $NaHCO_3$ + a mild edible acid such as tartaric acid

18 Acid rain and aquatic life

- City **B** (pH 5.3): lower pH means more H^+ ions, so more acidic
- Rain below pH 5.6 is acid rain; it lowers the pond's pH
- Aquatic animals survive only in a narrow pH range, so many may die

19 Counteracting pH changes

- (a)(i) An antacid such as milk of magnesia, $Mg(OH)_2$, or baking soda
- (a)(ii) Bee sting: baking soda paste; nettle (methanoic acid): rub a dock leaf
- (b) pH **decreases**: lactic acid forms as milk sets into curd

20 Hydronium ions and pH

- (a) **A** (pH 6): lowest pH, most H_3O^+ ions
- (b) C (9.5) < B (8) < A (6)

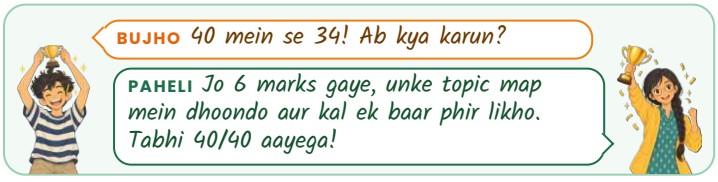
21 Reactions of acids; parent acid and base

$$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}\uparrow$$
Main product: zinc chloride (salt) and hydrogen gas

$$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$$
Main product: sodium chloride (salt) and water

$$Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$
Main product: sodium chloride, water and carbon dioxide

- (b)(i) NaCl: HCl + NaOH; **neutral** (strong acid + strong base)
- (b)(ii) $(NH_4)_2SO_4$: H_2SO_4 + NH_4OH; **acidic** (strong acid + weak base)



MEME BREAK
Score dekh ke khush ho gaye, par galat answers dobara nahi likhe? Boss abhi zinda hai!

NITESH SIR SAYS
Galat answers ek "mistake notebook" mein likho aur exam se pehle wahi padho. ***Chapter 2 cleared, jalwa hai!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 2 board and sample-paper questions. Year tags appear only where the paper year is known.

22 Hydrates, Plaster of Paris, washing soda

- (a) A fixed number of water molecules in one formula unit of a salt crystal: $CuSO_4{\cdot}5H_2O$, $Na_2CO_3{\cdot}10H_2O$
- (b) Heat gypsum at 373 K:

$$CaSO_4{\cdot}2H_2O_{(s)} \xrightarrow{373\ K} CaSO_4{\cdot}\tfrac{1}{2}H_2O_{(s)} + 1\tfrac{1}{2}H_2O_{(g)}$$

- Moisture turns it back into hard gypsum, so keep it dry:

$$CaSO_4{\cdot}\tfrac{1}{2}H_2O_{(s)} + 1\tfrac{1}{2}H_2O_{(l)} \rightarrow CaSO_4{\cdot}2H_2O_{(s)}$$

- (c) Heat baking soda, then recrystallise sodium carbonate:

$$2NaHCO_{3(s)} \xrightarrow{\text{Heat}} Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

$$Na_2CO_{3(s)} + 10H_2O_{(l)} \rightarrow Na_2CO_3{\cdot}10H_2O_{(s)}$$

- Use: glass, soap and paper industries; removing permanent hardness of water

Day 9 checklist: tick before you sleep

- [ ] I revised the one-page map and every key equation
- [ ] I attempted all 14 MCQs without looking at the key
- [ ] I solved the case study and all five short answers
- [ ] I wrote both long answers with balanced equations
- [ ] I checked my answers with the key and marked my score

My score: ____ / 5 | Test score: ____ / 40

5/5 = Jalwa hai! 4 se kam? Page 1 ka map phir se padho.

Revise these five first

1. Chlor-alkali equation, electrode products and their uses
2. Making of bleaching powder, baking soda, washing soda and PoP
3. pH of salts: parent acid and base decide the nature
4. Indicator colours: litmus, phenolphthalein, methyl orange
5. Acid + metal, carbonate, base and metal oxide equations

LEVEL 9 COMPLETE

+200 XP earned. Chapter 2 cleared. Next: Level 10

BUJHO *Boss Level clear! Chapter 2 poora samajh gaya!*

PAHELI *Shabaash! Kal milte hain Level 10 pe.*

CHEMISTRY | CHAPTER 2 COMPLETE

CHAPTER 2 SWAHAA!

Acids, Bases and Salts | Days 5–9

BUJHO *Litmus se lekar Plaster of Paris tak, pH scale bhi ab dost hai. Chapter 2 Swahaa!*

PAHELI *High five, Bujho! Ab chalte hain Metals aur Non-metals ki duniya mein.*

CHAPTER XP EARNED
+625 XP

LEVELS CLEARED
5 levels (Days 5–9)

BOOK PROGRESS
Level 9 / 60 1125 XP total

WHAT YOU HAVE MASTERED

- ✔ Indicators and reactions of acids and bases
- ✔ pH scale and universal indicator
- ✔ Family of salts and pH of salts
- ✔ H^+ and OH^- ions, strength, dilution
- ✔ pH in everyday life
- ✔ Chemicals from common salt, Plaster of Paris

NEXT CHAPTER Chapter 3: Metals and Non-metals (Day 10) *Aadhi jung jeet li!*

DAY 10 · LEVEL 10

CHEMISTRY | CHAPTER 3 | PART 1 · ASTRA SERIES

METALS & THEIR REACTIONS

Goal: properties, reactions and the reactivity series.

by Nitesh Prakash

01 Properties | 02 Exceptions | 03 Oxygen & Water | 04 Acids & Salts | 05 Reactivity Series | +100 XP today | Level 10 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 METALS VS NON-METALS: PHYSICAL PROPERTIES

METALS	NON-METALS
✔ **Lustrous** (shiny surface)	× Dull, no shine
Malleable and **ductile**	**Brittle** when solid
Good conductors of heat and electricity	Poor conductors (insulators)
Sonorous: ring when struck	Not sonorous
Hard; high melting and boiling points	Soft solids or gases; low melting points
Solids at room temperature	Solid, liquid (bromine) or gas

METALLOIDS: properties of both metals and non-metals
B, Si, Ge, As, Sb, Te | Si and Ge are semiconductors

DEFINITION

- **Malleability:** can be beaten into **thin sheets**. Gold and silver are the most malleable.
- **Ductility:** can be drawn into **thin wires**. Gold is the most ductile metal (not platinum): 1 g gives about 2 km of wire.
- **Sonorous:** rings when struck, so school bells are metal.

2 THE FAMOUS EXCEPTIONS ★★★ MOST ASKED

Exceptions board examiners love

- ✔ **Mercury** is the only liquid metal at room temperature.
- ✔ **Gallium** and **caesium** melt in your palm (low melting points).
- ✔ **Sodium** and **potassium** are so soft they cut with a knife; low density too.
- ✔ **Lead** and **mercury** are poor conductors of heat.
- ✔ **Silver** is the best conductor, then **copper**.
- ✔ **Iodine** is a shiny non-metal, and **bromine** is a liquid non-metal at room temperature.
- ✔ **Graphite** (carbon) conducts electricity; **diamond** is the hardest natural substance with a very high melting point.

★★★ *Exceptions se 1 mark pakka aata hai: Nitesh Bhaiya ki guarantee, list ratt lo!*

MEME BREAK

*Iron Man ka suit metal ka kyun? Strong, malleable aur tough. Asli hero **Fe** hai, bhai!*

BUJHO *Non-metal current nahi chalate, toh pencil ki nib bhi safe hai na?*

PAHELI *Nahi Bujho! Nib graphite ki hai: non-metal, phir bhi current chalata hai. Yahi exceptions exam mein sabse zyada aate hain!*

3 METALS + OXYGEN

METAL + OXYGEN (burn or heat in air) → combination → **METAL OXIDE** (mostly basic in nature)

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

copper (reddish-brown) → copper(II) oxide (black)

$$4Al_{(s)} + 3O_{2(g)} \rightarrow 2Al_2O_{3(s)}$$

Al 4 = 4, O 6 = 6 ✔

$$4Na_{(s)} + O_{2(g)} \rightarrow 2Na_2O_{(s)}$$

Na 4 = 4, O 2 = 2 ✔

Metal	With oxygen
K, Na	React so fast they catch fire in open air, so they are **kept in kerosene**
Mg, Al, Zn, Pb	A thin **oxide layer** forms and stops further oxidation
Mg	Burns with a dazzling white flame
Fe	Does not burn as a block; iron filings burn brightly in a flame
Cu	Does not burn; a black layer of CuO forms on heating
Ag, Au	No reaction even at high temperature

Did you know?

Phosphorus, a non-metal, catches fire in air, so it is stored **under water**. Sodium and potassium are stored under **kerosene** instead, because they react with water.

Amphoteric oxides: both acid and base nature ★★★

- Oxides that react with **both acids and bases** to give salt and water: **Al_2O_3**, **ZnO**

$$Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$$
acts as a base: Al 2 = 2, O 3 = 3, H 6 = 6, Cl 6 = 6 ✔

$$Al_2O_3(s) + 2NaOH(aq) \rightarrow 2NaAlO_2(aq) + H_2O(l)$$
acts as an acid; $NaAlO_2$ = sodium aluminate

Most metal oxides do not dissolve in water. Those that do give **alkalis**: $Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$

★★★ **PYQ FAVOURITE** The metal used in the thermit process burns in air to give an **amphoteric oxide**: the metal is **Al** and the oxide is **Al_2O_3** (CBSE 2023).

4 METALS + WATER ★★★

- Metal + water → **metal oxide + H_2**; a soluble oxide then gives **metal hydroxide**

Metal	Reacts with	What you see
K, Na	cold water	Violent; so much heat that the H_2 catches fire
Ca	cold water	Less violent; Ca **floats** as H_2 bubbles stick to it
Mg	hot water	Mg also **floats** on H_2 bubbles
Al, Zn, Fe	steam only	Metal oxide + H_2
Pb, Cu, Ag, Au	none	No reaction with water at all

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)\uparrow + \textbf{Heat}$$
Na 2 = 2, H 4 = 4, O 2 = 2 ✔

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)\uparrow$$

$$Mg(s) + 2H_2O(l) \xrightarrow{\text{hot water}} Mg(OH)_2(aq) + H_2(g)\uparrow$$

$$2Al(s) + 3H_2O(g) \rightarrow Al_2O_3(s) + 3H_2(g)\uparrow$$
Al 2 = 2, O 3 = 3, H 6 = 6 ✔

$$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)\uparrow$$
Fe 3 = 3, O 4 = 4, H 8 = 8 ✔ (Fe_3O_4, not Fe_2O_3!)

BUJHO *Calcium toh metal hai, phir paani pe tairta kyun hai?*

PAHELI *H_2 ke bubbles uski surface pe chipak jaate hain, wahi upar uthaate hain. Garam paani mein Mg bhi aise hi tairta hai!*

5 METALS + DILUTE ACIDS

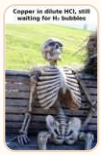
- Metal + dilute acid → **salt + hydrogen gas** (only metals **above H** in the series)

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)\uparrow$$

$$Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)\uparrow$$

$$2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)\uparrow$$
Al 2 = 2, H 6 = 6, Cl 6 = 6 ✔

Rate of bubbling: **Mg > Al > Zn > Fe**. Copper shows **no reaction**.



MEME BREAK
Copper, dilute HCl ko dekh ke: "Main H se neeche hoon, mera kuch nahi bigad sakte!"

Nitric acid exception and aqua regia ★★★

- **HNO_3** is a strong **oxidising agent**: it oxidises the H_2 formed to water and is itself reduced to N_2O, NO or NO_2. So **H_2 is not evolved**.
- Exception: **Mg** and **Mn** give H_2 with **very dilute** HNO_3.
- **Aqua regia** ("royal water"): freshly mixed conc. HCl and conc. HNO_3 in a **3 : 1** ratio. It dissolves **gold** and **platinum**.

EXTRA FROM CLASS NOCl + nascent Cl attack Au and Pt

$$HNO_3 + 3HCl \rightarrow NOCl + Cl_2 + 2H_2O$$
H 4 = 4, N 1 = 1, O 3 = 3, Cl 3 = 3 ✔ (NOCl = nitrosyl chloride)

★★★ *HNO_3 exception aur aqua regia ka 3 : 1: Nitesh Bhaiya ki guarantee, board mein aata hai!*

6 METALS + SALT SOLUTIONS ★★★

DISPLACEMENT

- A **more reactive** metal pushes a **less reactive** metal out of its salt solution
- Metal A + salt of B → salt of A + B (only if A is above B)

$$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$$
blue solution turns light green; reddish-brown Cu coats the iron nail

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$
blue colour fades; reddish-brown Cu deposits on zinc

$$Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$
solution turns blue; shiny silver deposits on copper

No reaction: $Cu(s) + FeSO_4(aq) \rightarrow$ no reaction since Cu is below Fe.

MEME BREAK
Zyada reactive metal aaya, copper ko solution se bahar! Mg aur Fe dono yahi karte hain.

7 REACTIVITY SERIES ★★★ MOST ASKED

Metals arranged in **decreasing order of reactivity**. Hydrogen is the reference point.

K *Kal*	**Na** *Naani*	**Ca** *Cake*	**Mg** *Mangwayi*	**Al** *Aaj*	**Zn** *Zara*	**Fe** *Feeka*
Pb *Pada*	**H** *Hum*	**Cu** *Chupke*	**Hg** *Hi*	**Ag** *Aadha*	**Au** *Aur*	

MOST REACTIVE — H: reference — LEAST REACTIVE

Trick yaad karo: "Kal Naani Cake Mangwayi, Aaj Zara Feeka Pada; Hum Chupke Hi Aadha Aur kha gaye!"

MEME BREAK

LEFT: POTASSIUM (TOP OF THE SERIES)
Cold water mein daalo, aag laga deta hai!
RIGHT: GOLD (BOTTOM OF THE SERIES)
Paani, hawa, acid: kuch farak nahi padta.
Upar wala jitna active, neeche wala utna noble!

Use the series to predict

- **Above H**: displaces H_2 from dilute acids. **Below H** (Cu, Hg, Ag, Au): no H_2.
- **Higher metal** displaces a lower metal from its salt solution.
- Bottom metals (Ag, Au) are found **free in nature** and stay shiny for years.

TRY YOURSELF: REACTION OR NO REACTION?

Series dekho, phir likho: reaction hoga ya nahi? Answer page 4 par.

(i) $Zn + FeSO_4$
(ii) $Ag + CuSO_4$
(iii) $Fe + ZnSO_4$
(iv) Mg + dil. HCl
(v) Cu + dil. H_2SO_4
(vi) $Pb + CuCl_2$

Solved Example: Identify the metal

Q. Metal X reacts with cold water, gives hydroxide Y, and floats on the water. Name X and Y and give the equation.

- Reacts with **cold** water but not violently, and **floats**, so X is **calcium**
- The H_2 bubbles stick to Ca and lift it; Y = **calcium hydroxide**

$$Ca_{(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + H_{2(g)}\uparrow$$

Ca 1 = 1, O 2 = 2, H 4 = 4 ✔

8 EXAM TIPS: COMMON MISTAKES

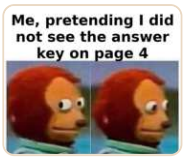
× WRONG	✔ RIGHT
$Cu + 2HCl \rightarrow CuCl_2 + H_2$	No reaction: Cu is below H
Zn + dil. HNO_3 gives H_2	HNO_3 oxidises H_2; nitrogen oxides form
$3Fe + 4H_2O \rightarrow Fe_2O_3 + \ldots$	Iron + steam gives Fe_3O_4
Al_2O_3 is a basic oxide	Al_2O_3 and ZnO are amphoteric
Sodium is stored in water	Stored in kerosene

Short answer questions

1. Give reasons: (A) Copper, not steel, is used to make hot water tanks. (B) Sodium is kept immersed in kerosene. (C) Aluminium is highly reactive, yet it is used to make cooking utensils. NCERT
2. Complete and balance: (A) $Al_2O_3 + HCl \rightarrow$ (B) $K_2O + H_2O \rightarrow$ (C) $Fe + H_2O$ (steam) $\rightarrow$ CBSE 2020
3. A man offered to polish dull gold bangles with a special solution. They sparkled like new, but their weight dropped sharply. Which solution did he use and why? NCERT

9 PRACTICE MCQS

Copy-paste nahi chalega! Khud solve karo, answer key page 4 par hai.



1. Metal X is used in the thermit process. When X is burnt in air, it gives an amphoteric oxide Y. X and Y are: CBSE 2023 ★★★
(a) Fe and Fe_2O_3 (b) Al and Al_2O_3
(c) Fe and Fe_3O_4 (d) Zn and ZnO

2. The arrangement of copper, tin, lead and mercury according to the reactivity series is: Delhi Gov. SQP 2024 ★★★
(a) Sn > Pb > Cu > Hg (b) Pb > Cu > Hg > Sn
(c) Cu > Hg > Sn > Pb (d) Hg > Sn > Pb > Cu

3. Which statements are correct? (I) Al_2O_3 reacts with both dil. HCl and NaOH(aq). (II) CaO reacts with both dil. HCl and NaOH(aq). (III) ZnO reacts with both dil. HCl and NaOH(aq). (IV) SO_2 reacts with neither. CBSE SQP 2025 ★★★
(a) (I) and (II) (b) (I) and (III)
(c) (II) and (IV) (d) (III) and (IV)

4. Dilute HCl is added to Al, Cu, Fe, Mg and Zn separately. The correct observation is: CBSE Compartment 2025 ★★☆
(a) most vigorous with Al; no reaction with Fe
(b) most vigorous with Mg; no reaction with Fe
(c) most vigorous with Mg; no reaction with Cu
(d) most vigorous with Zn; no reaction with Cu

5. Which metal is stored under kerosene? ★★☆
(a) Magnesium (b) Sodium
(c) Aluminium (d) Zinc

6. Which metal reacts with steam but not with cold or hot water? ★★☆
(a) Calcium (b) Magnesium
(c) Iron (d) Potassium

7. The non-metal that is a liquid at room temperature is: ★★★
(a) mercury (b) bromine
(c) iodine (d) gallium

8. Aqua regia is a freshly prepared mixture of: ★★☆
(a) conc. HCl and conc. HNO_3 in 1 : 3
(b) conc. HCl and conc. HNO_3 in 3 : 1
(c) conc. H_2SO_4 and conc. HCl in 3 : 1
(d) conc. HNO_3 and conc. H_2SO_4 in 1 : 1

Q9 and Q10: choose (a), (b), (c) or (d) as given below.

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

9. **Assertion (A):** A piece of zinc gets a reddish-brown coating when kept in copper sulphate solution for some time. **Reason (R):** Copper is a more reactive metal than zinc. CBSE 2024 ★★★

10. **Assertion (A):** Hydrogen gas is not evolved when zinc reacts with dilute nitric acid. **Reason (R):** Nitric acid is a strong oxidising agent; it oxidises the hydrogen formed to water. ★★☆

Answer key: MCQs

Q	Ans	Why
1	(b)	Al burns to Al_2O_3, an amphoteric oxide.
2	(a)	Sn and Pb are above H; Cu, then Hg below.
3	(b)	CaO is basic; SO_2 is acidic and reacts with NaOH.
4	(c)	Mg is most reactive; Cu is below H.
5	(b)	Na catches fire in air.
6	(c)	Fe reacts only with steam: gives Fe_3O_4.
7	(b)	Mercury is a metal; bromine is the liquid non-metal.
8	(b)	3 parts conc. HCl : 1 part conc. HNO_3.
9	(c)	Zn is more reactive and displaces Cu.
10	(a)	HNO_3 oxidises H_2 to H_2O.

Model answers: short answer

1 Reasons

- (A) Copper does not react with hot water or steam; iron in steel slowly reacts and rusts.
- (B) Sodium reacts so vigorously with air and moisture that it catches fire; kerosene keeps them away.
- (C) A thin, tough layer of Al_2O_3 protects it from further reaction; it is also light and a good conductor of heat.

2 Balanced equations

$$Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$$

$$K_2O(s) + H_2O(l) \rightarrow 2KOH(aq)$$

$$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)\uparrow$$

3 The bangle mystery

- He used **aqua regia** (conc. HCl : conc. HNO_3 = 3 : 1).
- It dissolved the dull outer layer of gold, so the bangles shone, but the dissolved gold was lost and the weight fell.

Try-yourself answers

(i) Yes: $ZnSO_4$ + Fe
(ii) No: Ag is below Cu
(iii) No: Fe is below Zn
(iv) Yes: $MgCl_2$ + H_2
(v) No: Cu is below H
(vi) Yes: $PbCl_2$ + Cu

Coming up: Day 11

How metals and non-metals react: why sodium gives away an electron and chlorine grabs it, and how **NaCl** and **$MgCl_2$** are formed.

Real life: Metals around you

- **Aluminium foil** wraps food: malleable, light, does not rust.
- **Copper wires** at home: ductile, and the best conductor after silver.
- **Gold and silver** jewellery stays shiny: they sit at the bottom of the series.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 3 board and sample-paper questions. Year tags appear only where the paper year is known.

Day 10 mind map: revise in 2 minutes

PROPERTIES
Lustrous, malleable, ductile, sonorous and good conductors (Hg is a liquid)

OXYGEN
Metal oxides are basic; Al_2O_3 and ZnO amphoteric

METALS & THEIR REACTIONS

WATER
K, Na, Ca cold; Mg hot; Al, Zn, Fe steam; Cu, Ag, Au none

ACIDS
Above H gives salt + H_2; no H_2 with HNO_3

SERIES ★★★
K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au; a higher metal displaces a lower one from its salt solution

Day 10 hit-list: Eduspark PYQ analysis

Day 10 topic	Rating
Reactivity series and displacement	★★★
Amphoteric oxides (Al_2O_3, ZnO)	★★★
Metals with water (Na, Ca, Mg, Fe)	★★★
Exceptions in physical properties	★★☆
HNO_3 exception and aqua regia	★★☆

Day 10 checklist: tick before you sleep

- ☐ I can compare metals and non-metals and list the exceptions
- ☐ I can write metal + oxygen and metal + water equations
- ☐ I can explain amphoteric oxides with two equations
- ☐ I can use the reactivity series to predict displacement
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 10 COMPLETE +100 XP earned. Next: Level 11

BUJHO *Samajh gaya! Upar wala metal neeche wale ko bahar nikaalta hai, aur H se neeche wale metal acid se H_2 nahi dete!*

PAHELI *Shabaash! Kal Level 11 mein hum ionic bonds aur electron-dot banana seekhenge.*

NITESH SIR SAYS
Reactivity series roz ek baar bolo, 7 din mein dimaag mein chhap jayegi. ***Kal milte hai Day 11 pe!***

DAY 11 · LEVEL 11

CHEMISTRY | CHAPTER 3 | PART 2 · ASTRA SERIES

IONIC BONDS & COMPOUNDS

Goal: show electron transfer and explain ionic properties.

by Nitesh Prakash

01 Octet Rule | 02 Ions | 03 Electron-dot | 04 Ionic Properties | 05 Non-metals | +100 XP today | Level 11 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHY DO ATOMS REACT?

OCTET RULE

- Noble gases (He 2; Ne 2, 8; Ar 2, 8, 8) have a **full outer shell**, so they hardly react
- Other atoms react to get **8 electrons** in the outermost shell (an octet; 2 for the first shell, a duplet)
- Metals **lose** their few valence electrons; non-metals **gain** electrons

Valence electrons decide the ion

Atom	Z	Shells	Does what	Ion
Na	11	2, 8, 1	loses 1 e^-	Na^+ (2, 8)
Mg	12	2, 8, 2	loses 2 e^-	Mg^{2+} (2, 8)
Al	13	2, 8, 3	loses 3 e^-	Al^{3+} (2, 8)
O	8	2, 6	gains 2 e^-	O^{2-} (2, 8)
Cl	17	2, 8, 7	gains 1 e^-	Cl^- (2, 8, 8)

MEME BREAK

*Sodium ne apna **ikloota valence electron** chlorine ko de diya. Sodium: "Ouch, mera dil!" Par ab sodium ion ka octet complete hai.*

2 CATION AND ANION

CATION (+)	ANION (-)
Formed when a **metal** atom **loses** electrons	Formed when a **non-metal** atom **gains** electrons
Electrons fewer than protons: positive charge	Electrons more than protons: negative charge
$Na \rightarrow Na^+ + e^-$ $Mg \rightarrow Mg^{2+} + 2e^-$	$Cl + e^- \rightarrow Cl^-$ $O + 2e^- \rightarrow O^{2-}$

BUJHO *Sodium ne electron de diya, toh sodium ion mein ek proton bhi kam ho gaya?*

PAHELI *Nahi Bujho! Nucleus ke 11 protons wahi rehte hain. Electrons 10 bache, isliye charge +1 hai.*

3 ELECTRON-DOT: NaCl ★★★

IONIC (ELECTROVALENT) BOND

- Formed when **electrons are transferred** from a metal atom to a non-metal atom
- Opposite ions are held by **strong electrostatic attraction**
- Such compounds are **ionic** or **electrovalent** compounds

× metal electron | ● non-metal electron | curved arrow: electron transferred

$Na \rightarrow Na^+ + e^-$ $\quad Cl + e^- \rightarrow Cl^-$

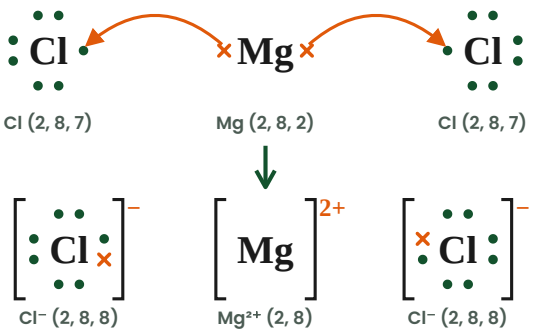
Na^+ like neon (2, 8); Cl^- like argon (2, 8, 8)

$2Na_{(s)} + Cl_{2(g)} \rightarrow 2NaCl_{(s)}$

Overall: Na 2 = 2, Cl 2 = 2 ✓

★★★ *Cl ko 8wa electron mila, full sukoon! Ye diagram har saal aata hai: Nitesh Bhaiya ki guarantee!*

4 ELECTRON-DOT: $MgCl_2$ ★★★



$Mg \rightarrow Mg^{2+} + 2e^-$ $\quad 2Cl + 2e^- \rightarrow 2Cl^-$

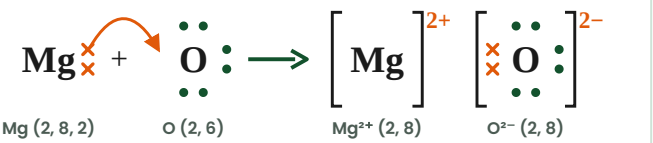
Mg gives two electrons, each Cl takes one: $Mg^{2+}(Cl^-)_2 = MgCl_2$

LINK TO VALENCY

Valency = electrons lost or gained: Na 1, Mg 2, Al 3, Cl 1, O 2. Total charge is zero: in Al_2O_3, $2 \times (+3) + 3 \times (-2) = 0$.

Solved Example 1: magnesium oxide ★★☆

Q. Show the formation of magnesium oxide by the transfer of electrons. NCERT



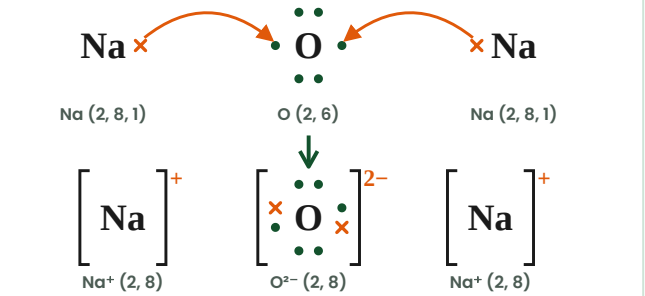
Mg gives **both** valence electrons to one O atom. Ions: Mg^{2+} and O^{2-}.

$$2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$$

Mg 2 = 2, O 2 = 2 ✔

Solved Example 2: sodium oxide ★★☆

Q. Show the formation of sodium oxide by the transfer of electrons. NCERT



Each Na gives **one** electron; O needs two, so **two** Na atoms are used. Ions: $2Na^{+}$ and O^{2-}.

$$4Na_{(s)} + O_{2(g)} \rightarrow 2Na_2O_{(s)}$$

Na 4 = 4, O 2 = 2 ✔

5 COMMON MISTAKES

× WRONG	✔ RIGHT
Both Mg electrons go to one Cl	One electron to each of two Cl atoms
A shared pair drawn between Na and Cl	A curved arrow shows the transfer
Ions drawn with no brackets or charge	Square brackets and the charge on each ion
Mg^{+} or MgCl in the answer	Mg^{2+} and the formula $MgCl_2$

MEME BREAK
*Electron-dot ekdum perfect banaya... phir **brackets aur charge** likhna bhool gaye. Marks gaye!*

TRY YOURSELF: SHOW THE TRANSFER

1. Potassium (19) and chlorine (17): draw the electron-dot formation of potassium chloride and write both ions with their shells.
2. Calcium (20) and oxygen (8): show how calcium oxide forms. How many electrons move, and which ions form?

6 PROPERTIES OF IONIC COMPOUNDS ★★★

Four properties with reasons

Property	Ionic compounds	Reason
Physical nature	Solids; hard but **brittle**	Strong attraction between + and − ions; break into pieces under pressure
Melting and boiling points	**High**	A lot of energy is needed to overcome the strong forces between ions
Solubility	Soluble in **water**; insoluble in kerosene, petrol	Water is a polar solvent and pulls the ions apart
Conduction	Only when **molten** or **in water**	Ions must be free to move

Compound	Melting point (K)	Boiling point (K)
NaCl	1074	1686
$MgCl_2$	981	1685
CaO	2850	3120

Solved Example 3: spot the ionic compound ★★☆

Q. Which of these are ionic compounds?

Substance	Melting point	Solid conducts?	Molten conducts?
A	1074 K	No	Yes
B	1235 K	Yes	Yes
C	387 K	No	No
D	2850 K	No	Yes

Ans. **A and D**: high melting point, and they conduct only when molten. B conducts as a solid (a **metal**); C melts low and never conducts.

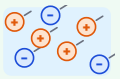
Do they conduct electricity? ★★★

SOLID

× no current
ions fixed in a rigid lattice

MOLTEN

✔ conducts
heat frees the ions

IN WATER

✔ conducts
ions move in solution

MEME BREAK
*Solid NaCl: "Main conduct karu? Mere ions toh **fixed** hain!" Pighlao ya paani mein ghol do, tab karega.*

BUJHO *Toh NaCl ek molecule hai na, ek Na aur ek Cl wala?*

PAHELI *Nahi! Ionic compound mein alag molecule nahi hote; sodium aur chloride ions ka giant crystal hota hai. NaCl sirf 1 : 1 ratio batata hai.*

7 PHYSICAL PROPERTIES OF NON-METALS

Rule and exception ★★☆

Property	Non-metals (general)	Exception
State	Solids or gases	**Bromine** is a liquid
Lustre	Dull	**Iodine** is lustrous
Hardness	Soft, brittle solids	**Diamond** is the hardest natural substance
Conduction	Poor conductors of heat and electricity	**Graphite** conducts electricity
Melting point	Generally low	**Diamond** has a very high melting point

Metal exceptions: mercury is a liquid; gallium and caesium melt on the palm; sodium and potassium are soft enough to cut with a knife; lead and mercury are poor conductors of heat.

Day 11 mind map: revise in 2 minutes

OCTET
8 electrons in the outer shell is stable (noble gas)

IONS
Metal loses e^-: cation; non-metal gains e^-: anion

IONIC BONDS & COMPOUNDS

ELECTRON-DOT ★★★
NaCl, $MgCl_2$, MgO, Na_2O with arrows, brackets, charges

PROPERTIES ★★★
Hard, brittle, high melting point, water-soluble, conduct when molten or dissolved

NON-METALS
Dull, brittle, poor conductors; exceptions: bromine, iodine, graphite, diamond

NUDGE
*Kerosene ne NaCl se kaha "ghul jao". NaCl ne seedha mana kar diya! Ionic compounds kerosene, petrol mein **insoluble** hote hain.*

Short answer questions

1. Write the electronic configurations of X (atomic number 12) and Y (atomic number 17). Show the formation of the compound formed when X and Y combine, and write its name and formula. What happens when an electric current is passed through its aqueous solution? Name the ion that moves to (i) the cathode, (ii) the anode.
CBSE Compartment 2024 (5 marks)

2. A farmer dissolved a copper sulphate fertiliser in water and sprayed it on his copper-deficient field; it worked. Later he mixed the same fertiliser in kerosene and it did not work. Give the reason. (2 marks)

3. (a) Why do ionic compounds have high melting points?
NCERT (b) Why does solid sodium chloride not conduct electricity, while its solution in water does? (2 marks)

8 PRACTICE MCQS

Sab milke solve karenge? Nahi! Pehle khud solve karo, answer key page 4 par hai.

1. In which form do electrovalent compounds conduct electricity?
CBSE Competency 2025 ★★★
(a) Only in the solid form
(b) In the solid form and in aqueous solution
(c) In aqueous solution and in the molten form
(d) In the solid, molten and aqueous forms

2. Which ions are present in sodium oxide? CBSE SQP 2023 ★★★
(a) Na^+ and O^- (b) $2Na^+$ and O^{2-}
(c) Na^{2+} and O^{2-} (d) Na^+ and O_2^{2-}

3. The electronic configurations of X, Y and Z are X: 2, 8, 7; Y: 2, 8, 2; Z: 2, 8. Which statement is correct? CBSE SQP 2024 ★★☆
(a) Y and Z are metals.
(b) Y and X are non-metals.
(c) X is a non-metal and Y is a metal.
(d) Y is a non-metal and Z is a metal.

4. The cation and anion present in aluminium oxide are: ★★☆
(a) Al^{2+} and O^{2-} (b) Al^{3+} and O^{2-}
(c) Al^{3+} and O^{3-} (d) Al^+ and O^-

5. How many electrons does one magnesium atom transfer when $MgCl_2$ is formed? ★★★
(a) one (b) two
(c) three (d) four

6. The electronic configuration of the Mg^{2+} ion is: ★★☆
(a) 2, 8, 2 (b) 2, 8
(c) 2, 8, 8 (d) 2, 6

7. Which of these is NOT a property of ionic compounds? ★★★
(a) High melting point (b) Soluble in water
(c) Conduct electricity in the solid state (d) Hard but brittle

8. Which non-metal is lustrous? ★★☆
(a) Sulphur (b) Iodine
(c) Phosphorus (d) Bromine

9. **Assertion (A):** The compound $MgCl_2$ does not conduct electricity in the solid state.
Reason (R): Movement of ions is not possible in the solid state of an ionic compound. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Ionic compounds conduct electricity in the solid state.
Reason (R): Ionic compounds are made up of oppositely charged ions. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(c)	Ions move only when molten or dissolved.
2	(b)	Two Na each lose 1 e^-; O gains 2.
3	(c)	7 valence e^-: non-metal; 2: metal.
4	(b)	Al loses 3 e^-; O gains 2 e^-.
5	(b)	Mg (2, 8, 2) gives one e^- to each Cl.
6	(b)	Mg^{2+} has 10 electrons: 2, 8.
7	(c)	Solid: ions fixed, so no conduction.
8	(b)	Iodine is a lustrous non-metal.
9	(a)	R correctly explains A.
10	(d)	A is false (no conduction as solid); R is true.

Model answers: short answer

1 X = magnesium, Y = chlorine

- X (12): 2, 8, 2 = Mg; Y (17): 2, 8, 7 = Cl
- Mg gives one electron to each of two Cl atoms (diagram in Section 4)
- Compound: **magnesium chloride, $MgCl_2$**
- In water its ions are free: the solution conducts and **electrolysis** takes place
- (i) Cathode: **Mg^{2+}** (cation); (ii) anode: **Cl^-** (anion)

$$Mg_{(s)} + Cl_{2(g)} \rightarrow MgCl_{2(s)}$$

Mg 1 = 1, Cl 2 = 2 ✔

2 Copper sulphate in water vs kerosene

- $CuSO_4$ is an **ionic compound**: soluble in water, so Cu^{2+} ions reach the plants
- Ionic compounds are **insoluble in kerosene**, so no ions were released and the spray failed

3 Melting point and conduction

- (a) Strong electrostatic attraction between oppositely charged ions needs a lot of energy to break
- (b) In solid NaCl the ions are held in a rigid lattice; in water they are free to move and carry charge

Try-yourself answers

- **(i)** K (2, 8, 8, 1) gives one electron to Cl (2, 8, 7): $[K]^+$ (2, 8, 8) and $[Cl]^-$ (2, 8, 8); formula KCl.
- **(ii)** Ca (2, 8, 8, 2) gives two electrons to one O (2, 6): $[Ca]^{2+}$ (2, 8, 8) and $[O]^{2-}$ (2, 8); formula CaO.

Day 11 hit-list: Eduspark PYQ analysis

Topic	Rating
Electron-dot formation of NaCl, $MgCl_2$	★★★
Conduction: molten or aqueous, not solid	★★★
Why ionic compounds have high melting points	★★★
Ions in Na_2O, MgO, Al_2O_3	★★☆
Solubility: water vs kerosene	★★☆
Non-metal exceptions (iodine, graphite, diamond)	★★☆

Exam tips: electron-dot answers

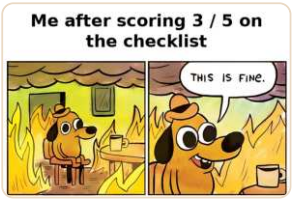
- ✔ Show only **valence** electrons: dots for one atom, crosses for the other.
- ✔ Draw a **curved arrow** from each transferred electron to the atom that receives it.
- ✔ Write ions in **square brackets** with the charge; add shell configurations below.
- ✔ Conduction answer: "molten or aqueous, not solid" **plus the reason** (ions free to move).
- ✔ Balance charges for the formula: Mg^{2+} needs two Cl^-; O^{2-} needs two Na^+.

Day 11 checklist: tick before you sleep

- ☐ I can explain the octet rule and why metals form cations
- ☐ I can draw the electron-dot formation of NaCl and $MgCl_2$
- ☐ I can show the formation of MgO and Na_2O and name the ions
- ☐ I can state four properties of ionic compounds with reasons
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 11 COMPLETE +100 XP earned. Next: Level 12

BUJHO *Samajh gaya! Metal electron deta hai, non-metal leta hai, aur ulte charge wale ions chipak jaate hain. Yahi ionic bond hai!*

PAHELI *Shabaash! Kal Level 12: ores se metal nikaalna.*

Coming up: Day 12

Occurrence and extraction of metals: ores and gangue, roasting vs calcination, the thermit reaction that welds railway tracks, and electrolytic refining of copper.

IONIC COMPOUNDS AT HOME

Common salt (NaCl), quicklime (CaO) and baking soda ($NaHCO_3$).

NITESH SIR SAYS

Electron-dot diagram teen baar haath se banao, exam mein haath khud chalega. ***Kal milte hai Day 12 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 3 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 12 · LEVEL 12

CHEMISTRY | CHAPTER 3 | PART 3 · ASTRA SERIES

EXTRACTION OF METALS

Goal: trace a metal from its ore to the pure metal.

by Nitesh Prakash

01 Ores & Gangue | 02 Extraction Flow | 03 Calcination & Roasting | 04 Reduction | 05 Refining | +100 XP today | Level 12 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 MINERALS, ORES & GANGUE

DEFINITIONS

- **Mineral:** an element or compound that **occurs naturally** in the Earth's crust
- **Ore:** a mineral from which a metal can be extracted **profitably**. All ores are minerals, but all minerals are not ores.
- **Gangue:** unwanted impurities in the ore, such as soil, sand and rock
- **Metallurgy:** the steps of extracting a metal from its ore and refining it

Common ores to know ★★☆

Metal	Ores (formula)
Mercury	cinnabar HgS
Copper	copper glance Cu_2S, copper pyrites $CuFeS_2$, cuprite Cu_2O
Zinc	zinc blende ZnS, calamine $ZnCO_3$, zincite ZnO
Iron	haematite Fe_2O_3, magnetite Fe_3O_4, siderite $FeCO_3$, iron pyrites FeS_2
Lead	galena PbS
Aluminium	bauxite $Al_2O_3 \cdot 2H_2O$ (hydrated alumina)
Sodium	rock salt NaCl

Where metals occur: it depends on reactivity

- **Low reactivity** (Au, Ag, Pt, Cu): found **free** in nature; Cu and Ag also as sulphide or oxide ores
- **Middle** (Zn, Fe, Pb): mostly as **oxides, sulphides or carbonates**
- **High reactivity** (K, Na, Ca, Mg, Al): **never free**, always as compounds

Fact file: oxygen is the most abundant element in the earth's crust (about 46%), silicon is second, and **aluminium** is third and the **most abundant metal** (about 8%). **Astatine** is the rarest naturally occurring element. Many ores are oxides because oxygen is so reactive and abundant.

MEME BREAK

Gold ka swag: "Main free state mein milta hoon!" Least reactive metal, isliye kisi se react hi nahi karta.

2 EXTRACTION FLOW CHART

ORE → Enrichment: gangue removed

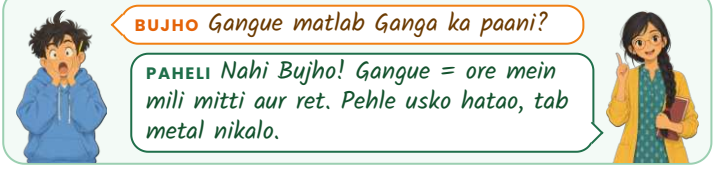
HIGH REACTIVITY (K, Na, Ca, Mg, Al)	MIDDLE (Zn, Fe, Pb)	LOW REACTIVITY (Hg, Cu)
Electrolysis of the molten ore	**Calcination** (carbonate) or **roasting** (sulphide)	Sulphide ore of the metal
Pure metal at the cathode	Oxide of the metal	**Roasting:** heat in air; the oxide is reduced by heat alone
	Reduction with carbon or a more reactive metal	**Metal, then refining**
	Metal, then refining	

EXAM FAVOURITE

Learn the three branches by reactivity and draw neat arrows.

Step 1: Enrichment of ore

- Removing gangue from the ore **before** extraction
- Uses the difference in **physical or chemical properties** of ore and gangue
- **Hydraulic washing** (gravity separation): ore powder is washed in a stream of water; heavy ore particles settle, light gangue is washed away
- Others: magnetic separation, froth flotation



Why convert ores into oxides first?

Sulphide and carbonate ores are converted to **metal oxides** because it is much easier to obtain a metal from its oxide than from its sulphide or carbonate. NCERT Exemplar

3 LOW REACTIVITY: HEAT ALONE WORKS

Mercury from cinnabar (HgS) ★★★

$$2HgS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2HgO_{(s)} + 2SO_{2(g)}$$
Step 1, roasting: Hg 2 = 2, S 2 = 2, O 6 = 6 ✔

$$2HgO_{(s)} \xrightarrow{\Delta} 2Hg_{(l)} + O_{2(g)}$$
Step 2, further heating reduces the oxide to liquid mercury

Copper from copper glance (Cu_2S) ★★☆

$$2Cu_2S_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2Cu_2O_{(s)} + 2SO_{2(g)}$$
Cu 4 = 4, S 2 = 2, O 6 = 6 ✔

$$2Cu_2O_{(s)} + Cu_2S_{(s)} \xrightarrow{\Delta} 6Cu_{(s)} + SO_{2(g)}$$
Cu 6 = 6, O 2 = 2, S 1 = 1 ✔

SPIDER-SENSE
*Question mein "liquid metal from its sulphide ore" dikha? Turant socho: **mercury aur cinnabar**!*

4 CALCINATION VS ROASTING

CALCINATION	ROASTING
Carbonate ores	**Sulphide** ores
Strong heating in **limited air** (or no air)	Strong heating in **excess air**
Gives metal oxide + **CO_2**	Gives metal oxide + **SO_2**
Example: calamine, $ZnCO_3$	Example: zinc blende, ZnS

$$ZnCO_{3(s)} \xrightarrow[\text{limited air}]{\Delta} ZnO_{(s)} + CO_{2(g)}$$
Calcination: Zn 1 = 1, C 1 = 1, O 3 = 3 ✔

$$2ZnS_{(s)} + 3O_{2(g)} \xrightarrow[\text{excess air}]{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}$$
Roasting: Zn 2 = 2, S 2 = 2, O 6 = 6 ✔

★★★ *Trick: Carbonate → Calcination, Sulphide → roaSting. Nitesh Bhaiya ki guarantee, ye table pakka aayegi!*

× WRONG	✔ RIGHT
Calcination and roasting are reduction steps	Both only change the ore into an oxide; reduction comes next
Roasting releases CO_2	Roasting releases SO_2; calcination releases CO_2

Real life: a crack in the railway track

A train stops because of a crack in the track. Engineers fill the gap with **thermit**: a mixture of iron(III) oxide and aluminium powder is ignited, molten iron flows into the crack and joins the rails on cooling.

5 REDUCTION: OXIDE TO METAL

(a) Reduction with carbon (coke) ★★★

$$ZnO_{(s)} + C_{(s)} \xrightarrow{\Delta} Zn_{(s)} + CO_{(g)}$$
Carbon removes oxygen: it is the reducing agent

Coke is the cheap carbon used in industry. ZnO is **reduced** (loses oxygen) and C is **oxidised** (gains oxygen).

(b) Thermit reaction: aluminium as reducer ★★★

$$Fe_2O_{3(s)} + 2Al_{(s)} \rightarrow 2Fe_{(l)} + Al_2O_{3(s)} + \textbf{Heat}$$
Fe 2 = 2, Al 2 = 2, O 3 = 3 ✔ (iron comes out molten)

$$3MnO_{2(s)} + 4Al_{(s)} \rightarrow 3Mn_{(l)} + 2Al_2O_{3(s)} + \textbf{Heat}$$
Mn 3 = 3, Al 4 = 4, O 6 = 6 ✔

- So much heat is released that the iron formed is **molten**
- Used to **weld railway tracks** and cracked machine parts
- Al is the reducing agent; Fe_2O_3 is the oxidising agent

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Blast furnace: haematite + coke + limestone give **pig iron**.

$C + O_2 \rightarrow CO_2$ $\quad$ $CO_2 + C \rightarrow 2CO$

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ (CO is the reducer)

$CaCO_3 \rightarrow CaO + CO_2$ $\quad$ $CaO + SiO_2 \rightarrow CaSiO_3$ (slag)

MEME BREAK
Thermit mein Al ne Fe_2O_3 se oxygen chheena, aur itni garmi nikli ki iron pighal gaya. Railway track jud gaya!

BUJHO *Carbon se toh sab reduce ho jayega na? Sodium bhi?*

PAHELI *Nahi! Na, Mg, Al ko oxygen carbon se zyada pyaara hai. Inhe electrolysis se nikalte hain.*

6 ELECTROLYTIC REDUCTION

Highly reactive metals: Na, Mg, Ca, Al ★★★

- These metals have **more affinity for oxygen than carbon**, so carbon cannot reduce their oxides
- They are obtained by **electrolysis of their molten chlorides** (Al from molten aluminium oxide)
- Metal is deposited at the **cathode** (−); chlorine is released at the **anode** (+)

$$2NaCl_{(l)} \xrightarrow{\text{Electric current}} 2Na_{(l)} + Cl_{2(g)}$$
Electrolysis of molten sodium chloride: Na 2 = 2, Cl 2 = 2 ✔

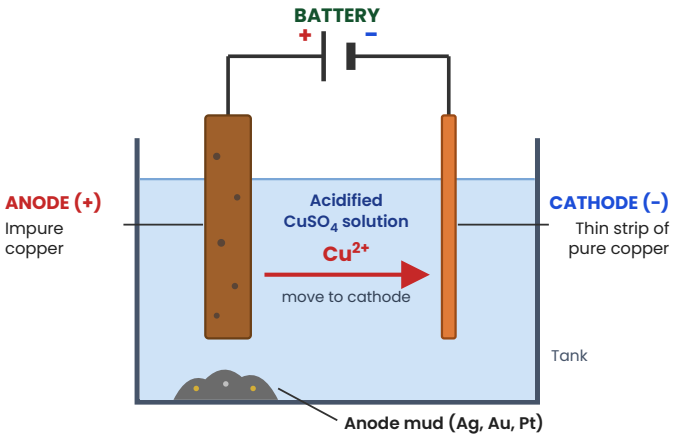
CATHODE $Na^+ + e^- \rightarrow Na$

ANODE $2Cl^- \rightarrow Cl_2 + 2e^-$

OFFICIAL NOTICE
*K, Na, Ca, Mg, Al ke oxides par **carbon ki entry band**! Inke liye sirf electrolysis chalega.*

7 ELECTROLYTIC REFINING OF COPPER

Labelled diagram: draw this in the exam ★★★



How it works

ANODE $Cu_{(s)} \rightarrow Cu^{2+}_{(aq)} + 2e^-$ (impure Cu dissolves)

CATHODE $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$ (pure Cu deposits)

- Anode: thick block of **impure** metal, joined to the **positive** terminal
- Cathode: thin strip of **pure** metal, joined to the **negative** terminal
- Electrolyte: **acidified copper sulphate** solution (a soluble salt of the same metal)
- Soluble impurities stay in the solution; **insoluble** ones settle below the anode as **anode mud**
- Also refined this way: Zn, Sn, Ni, Ag, Au

DID YOU KNOW?

The anode mud from copper refining is not waste: it contains **silver and gold**, which are recovered from it.

Which method for which metal? ★★☆

Metal	Position	Method of extraction
K, Na, Ca, Mg, Al	top	electrolysis of molten compound
Zn, Fe, Pb	middle	roasting or calcination, then reduction with carbon
Hg, Cu	low	roasting; oxide reduced by heating alone
Au, Ag, Pt	bottom	found free; only purified

Exam tips: common mistakes

- ✔ Method depends on **reactivity**: high = electrolysis, middle = carbon reduction, low = heating alone.
- ✔ Write **Δ** over the arrow for roasting, calcination and HgO decomposition.
- ✔ In refining, **impure metal is always the anode**; never swap the electrodes in the diagram.
- ✔ Label all five parts of the refining diagram: anode, cathode, electrolyte, anode mud, battery.

8 PRACTICE MCQS

Solve all 10 first, then check with the answer key on page 4.

1. Cinnabar is an ore of: ★★★

(a) zinc (b) mercury
(c) copper (d) iron

2. A metal X is used in the thermit process. When X is burnt in air, it gives an amphoteric oxide Y. X and Y are: CBSE 2023 ★★★

(a) Fe and Fe_2O_3 (b) Al and Al_2O_3
(c) Fe and Fe_3O_4 (d) Cu and CuO

3. Heating a carbonate ore strongly in limited air is called: ★★★

(a) roasting (b) calcination
(c) smelting (d) refining

4. Which gas is given off when zinc blende is roasted? ★★☆

(a) CO_2 (b) H_2
(c) SO_2 (d) O_2

5. Electrolysis of its molten chloride gives which metal? ★★★

(a) Iron (b) Copper
(c) Zinc (d) Sodium

6. In the electrolytic refining of copper, the anode is made of: ★★★

(a) a thin strip of pure copper (b) a block of impure copper
(c) graphite (d) zinc

7. The electrolyte used in refining copper is: ★★☆

(a) acidified copper sulphate solution
(b) sodium chloride solution
(c) dilute nitric acid
(d) zinc sulphate solution

8. Which metal is usually found in the free state in nature? ★☆☆

(a) Sodium (b) Gold
(c) Zinc (d) Aluminium

9. ★★★ **Assertion (A):** Zinc oxide can be reduced to zinc metal by heating with carbon.
Reason (R): Carbon is less reactive than zinc.
CBSE Practice Set 2023

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

10. ★★☆ **Assertion (A):** Sodium is obtained by electrolysis of molten sodium chloride and not by heating its oxide with carbon.
Reason (R): Sodium has a greater affinity for oxygen than carbon.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

QUICK CHECK

Calcination gives off _____; roasting gives off _____.
Answers: CO_2, SO_2

Answer key: MCQs

Q	Ans	Why
1	(b)	Cinnabar = HgS, the ore of mercury.
2	(b)	Al reduces Fe_2O_3; Al_2O_3 is amphoteric.
3	(b)	Carbonate + limited air = calcination.
4	(c)	Sulphide ore gives SO_2 on roasting.
5	(d)	Na is too reactive for carbon reduction.
6	(b)	Impure metal is always the anode.
7	(a)	A soluble salt of the same metal.
8	(b)	Least reactive, so it stays free.
9	(c)	A true; R false: carbon removes O from ZnO.
10	(a)	Na holds O more strongly than C can.

Short and long answer questions

1. A reddish-brown metal X, when heated in air, gives a black compound Y, which on heating in hydrogen gives X back. X is refined by electrolysis. Identify X and Y and draw a labelled diagram of the refining of X. **CBSE SQP 2023**
2. A metal that is a liquid at room temperature is obtained by heating its sulphide in air. Name the metal and its ore, and write the reactions. **Delhi SQP 2025**
3. An element X is stored in kerosene and cannot be extracted from its ore by a reducing agent. (A) Can X be stored in water? (B) Identify X and write how it is extracted. **CBSE SQP 2025**
4. Differentiate between calcination and roasting with one balanced equation each.

Model answers

1 X = copper (Cu), Y = copper(II) oxide (CuO)

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

Cu 1 = 1, O 1 = 1, H 2 = 2 ✔

Diagram: draw the one in Section 7.

2 Mercury (Hg); ore: cinnabar (HgS)

$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$, then $2HgO \rightarrow 2Hg + O_2$ (both on heating; see Section 3).

3 X = sodium (potassium is also accepted)

- (A) No. It reacts violently with water and catches fire.
- (B) Electrolysis of molten NaCl: $Na^+ + e^- \rightarrow Na$ at the cathode; $2Cl^- \rightarrow Cl_2 + 2e^-$ at the anode.

4 Calcination vs roasting

- **Calcination:** carbonate ore heated strongly in limited air; CO_2 released
- **Roasting:** sulphide ore heated strongly in excess air; SO_2 released

$$ZnCO_{3(s)} \xrightarrow{\Delta} ZnO_{(s)} + CO_{2(g)}$$

$$2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}$$

Day 12 mind map: revise in 2 minutes

ORE
Mineral that gives metal profitably; gangue removed first

TO OXIDE
Carbonate: calcination; sulphide: roasting

EXTRACTION OF METALS

REDUCTION
Heat alone (Hg, Cu); carbon (Zn); Al (thermit)

ELECTROLYSIS ★★★
Na, Mg, Ca, Al from molten compounds

REFINING
Anode impure Cu, cathode pure Cu, acidified $CuSO_4$; anode mud below the anode

Day 12 checklist: tick before you sleep

- ☐ I can define mineral, ore and gangue with an example
- ☐ I can draw the extraction flow chart for low, middle and high reactivity metals
- ☐ I can compare calcination and roasting with balanced equations
- ☐ I can write the thermit reaction and explain electrolytic reduction
- ☐ I can draw and label the electrolytic refining of copper

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–3 phir se padho.

LEVEL 12 COMPLETE +100 XP earned. Next: Level 13

BUJHO *Samajh gaya! Ore saaf karo, oxide banao, reduce karo, phir refine karo!*

PAHELI *Shabaash! Kal Level 13: corrosion aur alloys.*

Day 12 hit-list: Eduspark PYQ analysis

Topic	Rating
Refining of copper (labelled diagram)	★★★
Calcination vs roasting with equations	★★★
Thermit reaction and its use	★★★
Hg from cinnabar; Cu from Cu_2S	★★☆
Why Na, Mg, Al need electrolysis	★★☆

NITESH SIR SAYS
*Diagram ek baar khud banao, exam mein haath apne aap chalega. **Practice hi jalwa hai!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 3 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 13 · LEVEL 13

CHEMISTRY | CHAPTER 3 | PART 4 · ASTRA SERIES

CORROSION & ALLOYS

Goal: explain corrosion, prevent rusting, know key alloys.

by Nitesh Prakash

01 Corrosion · 02 Rusting Test · 03 Prevention · 04 Alloys · 05 Gold Carats

+125 XP today | Level 13 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHAT IS CORROSION?

DEFINITION

- A metal is **slowly eaten up** by air, moisture, acids or other substances around it
- The metal gets **oxidised** and goes back to a stable compound, like its ore
- Bridges, ships, railings and car bodies get weak; huge loss every year

Three famous examples ★★★

Metal	Attacked by	Coating formed
Iron	moist air	**Rust**: reddish-brown, flaky; hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$
Silver	sulphur compounds (H_2S) in air	**Black** coat of silver sulphide, Ag_2S (tarnishing)
Copper	moist CO_2 in air	**Green** coat of basic copper carbonate, $CuCO_3 \cdot Cu(OH)_2$

Equations behind the coats ★★☆

$4Ag(s) + 2H_2S(g) + O_2(g) \rightarrow 2Ag_2S(s) + 2H_2O(l)$

Silver tarnish: Ag 4 = 4, H 4 = 4, S 2 = 2, O 2 = 2 ✓

$2Cu(s) + H_2O(l) + CO_2(g) + O_2(g) \rightarrow CuCO_3 \cdot Cu(OH)_2(s)$

Green coat on copper: Cu 2 = 2, C 1 = 1, H 2 = 2, O 5 = 5 ✓

MEME BREAK

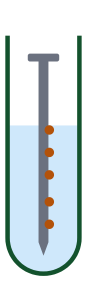
*Mummy ki chandi ki payal kaali ho gayi! Hawa ki H_2S ne silver ko **black Ag_2S** bana diya.*

BUJHO *Aluminium bhi oxygen se react karta hai, toh usme rust kyun nahi lagta?*

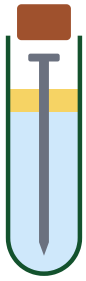
PAHELI *Rust sirf iron ka hota hai! Aluminium par Al_2O_3 ki patli layer ban jaati hai jo aage corrosion rok deti hai.*

2 RUSTING: AIR + WATER ★★★ MOST ASKED

Activity: Conditions for rusting (NCERT)



A: nails in tap water, tube open
air ✓ water ✓
Rust forms



B: boiled distilled water, oil layer on top, corked
air × water ✓
No rust

C: anhydrous $CaCl_2$, corked
air ✓ water ×
No rust

- **Boiled** water: dissolved air is driven out; **oil layer** stops air dissolving again
- **Anhydrous $CaCl_2$** absorbs moisture, so the air in tube C stays dry
- Rust only in A: iron rusts only when **both air (oxygen) and water** are present

★★★ *Air ya paani? Dono chahiye! Teen test tube wala experiment pakka aayega, Nitesh Bhaiya ki guarantee!*

$4Fe(s) + 3O_2(g) + 2xH_2O(l) \rightarrow 2Fe_2O_3 \cdot xH_2O(s)$

Rust: hydrated iron(III) oxide. Fe 4 = 4, O 6 + 2x = 6 + 2x, H 4x = 4x ✓

RUSTING SPEEDS UP WHEN

- ✓ Air is humid, as in the rainy season and near the sea
- ✓ Water has **salt** in it (sea water): ships and coastal railings rust fast
- ✓ Air carries acidic gases (pollution) or the iron is scratched

Rust is **porous and flaky**: it falls off and exposes fresh iron, so rusting goes on and on.

MEME BREAK

Baarish mein cycle bahar chhod di? Rust: "Mere liye toh party hai!" Paani + hawa = rust ka buffet.

3 PREVENTION OF CORROSION

Six ways to stop rusting ★★★

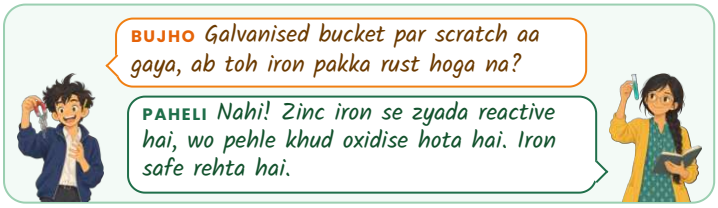
Method	How it protects	Used on
Painting	coat of paint keeps out air and moisture	railings, bridges, car bodies
Oiling, greasing	thin film of oil or grease blocks air and water	machine parts, tools, cycle chains
Galvanising ★★★	thin coat of **zinc** on iron or steel	buckets, roof sheets, water pipes
Chrome plating	chromium layer by electroplating; shiny and rust-proof	taps, cycle handlebars
Anodising ★★☆	thicker protective Al_2O_3 layer on aluminium	cookware, window frames
Alloying	mix iron with other metals, e.g. stainless steel	utensils, surgical tools

Galvanising: protected even when scratched ★★★

- Zinc first acts as a **barrier** between iron and moist air
- If the zinc coat breaks, iron still does not rust: zinc is **more reactive than iron** and gets oxidised first
- **Sacrificial protection:** iron is joined to or coated with a **more reactive metal (Zn or Mg)** that corrodes first, e.g. galvanising, Mg blocks on ship hulls and buried pipelines

Anodising of aluminium

1. Clean aluminium article is made the **anode**; dilute sulphuric acid is the electrolyte.
2. Oxygen released at the anode reacts with aluminium and **thickens the Al_2O_3 layer**.
3. The thick oxide layer resists corrosion and can be dyed in attractive colours.



× WRONG	✓ RIGHT
Rusting is caused by water alone	Rusting needs both air and water
Aluminium and copper "rust"	Only iron rusts; other metals corrode
Galvanising coats iron with tin	Galvanising coats iron with zinc

MEME BREAK
Cycle ki chain ko kabhi oil nahi diya, ab rust aur awaaz dono free mein! Oiling karo, chain bachao.

4 ALLOYS ★★★ MOST ASKED

DEFINITION

- An alloy is a **homogeneous mixture** of two or more metals, or a metal and a non-metal
- Made by melting the main metal and dissolving the others in it, then cooling
- **Amalgam**: an alloy in which one of the metals is **mercury**

Common alloys: composition and uses

Alloy	Made of	Special property and use
Steel	Fe + C (small amount)	hard, strong: buildings, bridges, rails
Stainless steel ★★★	Fe + Ni + Cr (+ a little C)	does not rust: utensils, surgical tools
Brass ★★★	Cu + Zn	harder than copper: utensils, taps, musical instruments
Bronze ★★★	Cu + Sn	strong, resists corrosion: statues, medals, bells
Solder ★★☆	Pb + Sn	**low melting point**: welding electrical wires
Amalgam ★☆☆	Hg + another metal	soft paste that hardens: dental fillings

Alloy vs pure metal ★★☆

- ✔ **Harder and stronger**: pure iron is soft and stretches when hot; a little carbon makes it hard steel
- ✔ **More resistant to corrosion** than the pure metal
- ✔ **Lower electrical conductivity** than the pure metal
- ✔ **Lower melting point** than its constituents in some cases, e.g. solder

COMMON MISTAKE

Steel is **not** a metal-only alloy: it contains carbon, a non-metal. Brass, bronze and amalgam contain only metals. CBSE 2020

5 GOLD: 24 VS 22 CARAT ★★☆

24 CARAT	22 CARAT
Pure gold (24 parts out of 24)	22 parts gold + **2 parts copper or silver**
Very soft; bends easily, not good for ornaments	Harder and more durable: used for jewellery

Solved example: Purity of gold

Q. What percentage of gold is present in 22 carat and in 18 carat gold?

- 22 carat: 22/24 × 100 = **91.67%** gold
- 18 carat: 18/24 × 100 = **75%** gold

MEME BREAK

TOP: 24 CARAT, PURE GOLD
Bahut soft, payal turant mud jaayegi
BOTTOM: 22 CARAT
22 gold + 2 copper or silver: strong aur classy
Thoda copper milao, jewellery ko tuxedo pehnao!

Exam tips

- ✔ Draw and label all three test tubes; write what each one proves.
- ✔ Write rust as **hydrated iron(III) oxide**; the formula alone can lose marks.
- ✔ "Why is a galvanised article safe when scratched?" Answer with reactivity: zinc is above iron.
- ✔ Name the constituents in the right order: brass Cu + Zn, bronze Cu + Sn, solder Pb + Sn.
- ✔ Silver turns **black**, copper turns **green**, iron turns **reddish-brown**: never mix them up.

Day 13 mind map: revise in 2 minutes

CORROSION
Iron: rust; silver: black Ag_2S; copper: green coat

RUSTING ★★★
Needs air + water; rust only in tube A

CORROSION & ALLOYS

PREVENTION
Paint, oil, zinc, chromium, anodising, alloying

ALLOYS ★★★
Steel, stainless steel, brass, bronze, solder, amalgam

GOLD
24 carat = pure and soft; 22 carat = 22 gold + 2 Cu or Ag

TRY YOURSELF: NAME THE BEST METHOD

(i) bicycle chain
(ii) iron roof sheets
(iii) car body
(iv) shiny water tap
(v) aluminium pressure cooker
(vi) surgical scissors

Short and long answer questions

1 Copper domes of old buildings slowly turn green. (A) Why do the domes appear green though copper is reddish-brown? (B) Should the copper domes be replaced by iron domes? Justify. (C) Why did ancient architects use copper sheets for domes? CBSE SQP 2025

2 Iron nails coated with zinc do not rust, even when the coat is damaged. (A) Name this process and give two other methods to prevent rusting. (B) Where will rusting be the most: an iron bucket electroplated with zinc, iron hinges on a gate, or a painted iron fence? Give a reason. CBSE SQP 2022

3 Describe an activity with a labelled diagram to show that both air and water are needed for the rusting of iron. Why is boiled water and a layer of oil used in one test tube? (5 marks)

6 PRACTICE MCQS

Padhai tabhi hogi jab MCQs khud solve karoge! Answer key page 4 par hai.

1. Rust is chemically: ★★★
(a) Fe_3O_4 (b) hydrated iron(III) oxide
(c) iron(II) sulphate (d) iron(II) oxide

2. In the three test tube experiment, rust appears only in the tube with: ★★★
(a) boiled water + oil (b) anhydrous $CaCl_2$
(c) tap water, open (d) all three tubes

3. Oil is poured over the boiled water in test tube B to: ★★☆
(a) add moisture to the air
(b) stop air from dissolving in the water
(c) dry the air in the tube
(d) speed up rusting

4. Silver articles turn black in air because of a coating of: ★★★
(a) Ag_2O (b) Ag_2S
(c) AgCl (d) $AgNO_3$

5. The green coating formed on copper in moist air is: ★★☆
(a) copper(II) oxide (b) copper sulphate
(c) basic copper carbonate (d) copper nitrate

6. Galvanisation is the process of coating iron with a thin layer of: ★★★
(a) tin (b) chromium
(c) zinc (d) aluminium

7. Which alloy contains a non-metal as one of its constituents? CBSE 2020 ★★★
(a) Brass (b) Bronze
(c) Amalgam (d) Steel

8. In stainless steel, iron is mixed with: NCERT Exemplar ★★★
(a) Ni and Cr (b) Cu and Cr
(c) Ni and Cu (d) Cu and Au

9. **Assertion (A):** A galvanised iron article stays protected from rusting even if the zinc coating is broken.
Reason (R): Zinc is more reactive than iron. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Aluminium does not corrode easily in air.
Reason (R): Aluminium is less reactive than iron. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

QUICK RECALL: ONE-MARK QUESTIONS

(a) Gas in air that blackens silver?
(b) Metal used for galvanising?
(c) Constituents of bronze?
(d) Why is solder used for wires?

Answer key: MCQs

Q	Ans	Why
1	(b)	Rust = $Fe_2O_3 \cdot xH_2O$, hydrated iron(III) oxide.
2	(c)	Only tube A has both air and water.
3	(b)	Oil seals the water from air.
4	(b)	Silver reacts with H_2S to give black Ag_2S.
5	(c)	Green coat = basic copper carbonate.
6	(c)	Galvanising = zinc coating.
7	(d)	Steel contains carbon, a non-metal.
8	(a)	Stainless steel = Fe + Ni + Cr.
9	(a)	Zinc corrodes first and protects iron.
10	(c)	A true; Al is more reactive than Fe, but its Al_2O_3 layer protects it.

Try-yourself and quick recall answers

(i) oiling or greasing
(ii) galvanising
(iii) painting
(iv) chrome plating
(v) anodising
(vi) alloying (stainless steel)
(a) hydrogen sulphide, H_2S
(b) zinc
(c) copper and tin
(d) its melting point is low

Model answers: short and long answer

1 Copper domes (CBSE SQP 2025)

- (A) Copper reacts with moist CO_2 in air to form a **green coat of basic copper carbonate**
- (B) No. Iron would rust; flaky rust falls off, exposes fresh iron and the dome weakens. The copper coat protects the metal below
- (C) Copper is highly **malleable**: thin sheets can be shaped into a dome, and it does not rust

2 Zinc-coated nails (CBSE SQP 2022)

- (A) **Galvanisation**. Other methods: painting, oiling or greasing, chrome plating, alloying
- (B) **Iron hinges on a gate**: bare iron in contact with both air and moisture; the bucket and fence are coated

3 Activity: air and water both needed

- Take three test tubes with clean iron nails: A with tap water (open), B with boiled distilled water + oil layer (corked), C with anhydrous $CaCl_2$ (corked); draw the diagram from page 1
- After a few days, rust only in A. B has no air (boiling removes air, oil stops it re-entering); C has no moisture ($CaCl_2$ absorbs it)
- Conclusion: rusting needs **both air and water**

Day 13 hit-list: Eduspark PYQ analysis

Topic	Rating
Rusting experiment (air + water)	★★★
Galvanisation and why it works	★★★
Alloys: constituents and uses	★★★
Silver black, copper green	★★★
Methods of preventing rusting	★★☆
22 carat gold, anodising	★★☆

Day 13 checklist: tick before you sleep

- [ ] I can define corrosion and name the coats on iron, silver and copper
- [ ] I can draw the three test tube experiment and explain each tube
- [ ] I can list six ways to prevent rusting and explain galvanisation
- [ ] I know the constituents and uses of six alloys, and 22 carat gold
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 13 COMPLETE +125 XP earned. Next: Level 14

BUJHO *Samajh gaya! Rust ke liye hawa aur paani dono chahiye, aur zinc khud kurbaan hokar iron ko bachata hai!*

PAHELI *Shabaash! Kal Level 14: Chapter 3 Boss Level.*

Real life: why kitchens love these

- Pressure cookers and kadhais: **anodised aluminium**, hard and non-rusting
- Steel plates and spoons: **stainless steel** never rusts
- Pooja thali and temple bells: **brass and bronze** stay bright for years

Did you know? The iron pillar of Delhi

The iron pillar near the Qutub Minar was made by Indian iron workers more than **1600 years ago** and has hardly rusted. Ancient Indians had a special way of working iron that resists rusting.

Coming up: Day 14

Chapter 3 **Boss Level**: one test on the whole chapter, from reactivity series to alloys. Revise Days 10–13 tonight!

NITESH SIR SAYS

*Iron ko paint chahiye, tumhe revision. **Roz thoda padho, rust nahi lagega!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 3 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 14 · LEVEL 14

CHEMISTRY | CHAPTER 3 | BOSS LEVEL

CHAPTER 3 BOSS LEVEL

Goal: test all of Metals and Non-metals in 75 minutes.

by Nitesh Prakash

01 Revision Map · 02 MCQs · 03 Short Answers · 04 Long Answers · 05 Answer Key · +200 XP today | Level 14 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 3 REVISION MAP

Chapter 3 mind map: revise in 5 minutes

PROPERTIES
Metals: lustrous, malleable, ductile, good conductors. Exceptions: Hg is a liquid; Na, K are soft; iodine is lustrous; graphite conducts

REACTIONS
Metal + O_2 → basic oxide (Al_2O_3, ZnO amphoteric); + water → hydroxide or oxide + H_2; + dilute acid → salt + H_2 (not with HNO_3)

REACTIVITY SERIES ★★★
K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au; a more reactive metal displaces a less reactive one from its salt solution

METALS & NON-METALS

IONIC BONDS
Metal loses e^-, non-metal gains; hard, high m.p., soluble in water; conduct electricity only in molten or aqueous form

EXTRACTION
Low: heat alone (HgS, Cu_2S); middle: roast or calcine, then reduce (C, Al); high: electrolysis of molten compound

REFINING, CORROSION & ALLOYS ★★★
Refining: anode impure, cathode pure strip, anode mud below. Rusting needs air + water; silver turns black (Ag_2S); copper turns green. Brass Cu + Zn, bronze Cu + Sn, solder Pb + Sn

Must-know equations: write them daily

$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)}\uparrow + \textbf{Heat}$
Na, K react violently with cold water: stored in kerosene

$Al_2O_{3(s)} + 2NaOH_{(aq)} \rightarrow 2NaAlO_{2(aq)} + H_2O_{(l)}$
amphoteric oxide reacts with a base too (sodium aluminate)

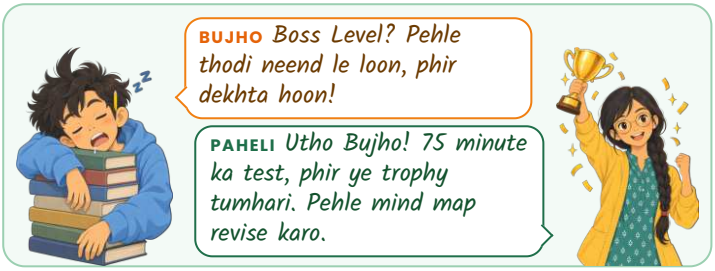
$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$
displacement: Fe is above Cu in the series

$2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}\uparrow$
roasting: sulphide ore heated in excess air

$ZnCO_{3(s)} \xrightarrow{\Delta} ZnO_{(s)} + CO_{2(g)}\uparrow$
calcination: carbonate ore heated in limited air

$ZnO_{(s)} + C_{(s)} \xrightarrow{\Delta} Zn_{(s)} + CO_{(g)}$
reduction of the oxide with carbon (coke)

$Fe_2O_{3(s)} + 2Al_{(s)} \rightarrow 2Fe_{(l)} + Al_2O_{3(s)} + \textbf{Heat}$
thermit reaction: molten iron joins railway tracks



BOSS LEVEL RULES

- ✔ **Time:** 75 minutes. **Maximum marks:** 40. Close the book and set a timer.
- ✔ **A:** 14 MCQs × 1 = 14. **B:** 5 short answers = 12. **C:** 2 long answers × 5 = 10. **D:** 1 case-based question = 4.
- ✔ Write every equation balanced, with state symbols. Draw diagrams in pencil and label them.

Phone side mein rakho, timer lagao. Boss se ladna hai, scroll nahi karna!

2 SECTION A: MCQS

1. Which equations form an amphoteric oxide, and what are A and B? (1) 2Pb + O_2 → A PbO (2) B Zn + O_2 → 2ZnO ★★☆

(a) only 1; A = 1, B = 1 (b) both; A = 2, B = 2
(c) only 2; A = 2, B = 1 (d) both; A = 1, B = 2

2. Match (A) forms an amphoteric oxide, (B) melts on the palm, (C) gives H_2 with very dilute HNO_3, (D) cannot displace H_2 from acids, with (i) Ga (ii) Au (iii) Al (iv) Mn. Code for A, B, C, D: ★★☆

(a) iii, i, iv, ii (b) i, iii, ii, iv
(c) iii, iv, i, ii (d) iv, i, iii, ii

3. Metal X reacts with cold water, Y only with hot water and Z only with steam. Their increasing order of reactivity is: ★★★

(a) X < Y < Z (b) Z < Y < X
(c) Y < Z < X (d) Z < X < Y

4. An element with atomic number ____ will form a basic oxide. CBSE SQP 2023 ★★☆

(a) 7 (2, 5) (b) 17 (2, 8, 7)
(c) 14 (2, 8, 4) (d) 11 (2, 8, 1)

5. In the reaction $3MnO_2 + 4Al \rightarrow 3Mn + 2Al_2O_3$ + heat, the substance oxidised is: CBSE SQP 2023 ★★★

(a) MnO_2 (b) Al
(c) Mn (d) Al_2O_3

6. When magnesium chloride is formed, a magnesium atom: ★★★

(a) gains two electrons (b) loses two electrons
(c) shares two electrons (d) loses one electron

7. Mercury is obtained from its ore cinnabar (HgS) by: ★★☆

(a) heating it in air to form HgO, then heating the HgO further
(b) reducing it with carbon in a blast furnace
(c) electrolysis of molten cinnabar
(d) heating it strongly with aluminium powder

8. Zinc carbonate is changed into zinc oxide by heating it strongly in limited air. This process is: ★★★

(a) roasting (b) calcination
(c) refining (d) electrolysis

9. In the electrolytic refining of copper, the cathode is a: ★★★

(a) block of impure copper (b) thin strip of pure copper
(c) graphite rod (d) zinc plate

10. In stainless steel, iron is mixed with: NCERT Exemplar ★★☆

(a) nickel and chromium (b) copper and chromium
(c) nickel and copper (d) copper and gold

11. On hammering, substance M flattens into a sheet and rings; substance N breaks into powder. So: CBSE SQP 2024 ★★☆

(a) both are metals (b) both are non-metals
(c) M metal, N non-metal (d) M non-metal, N metal

12. Silver articles turn black in air. The black coating is of: ★★☆

(a) silver oxide (b) silver sulphide
(c) silver nitrate (d) silver carbonate

A–R dekh ke dimaag ghoom gaya? Pehle A check karo, phir R, phir dono ka link!

13. **Assertion (A):** Ionic compounds have high melting and boiling points.
Reason (R): A large amount of energy is needed to break the strong electrostatic forces between oppositely charged ions. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

14. **Assertion (A):** Gold is found in the free state in nature.
Reason (R): Gold is a highly reactive metal. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

How to attack Sections B, C and D

- ✔ 2 marks = two clear points; 3 marks = three points or an equation with a reason.
- ✔ Every equation: balanced, with (s), (l), (g) or (aq), and Δ on the arrow where heat is needed.
- ✔ Time plan: A in 15 min, B in 25 min, C and D in 30 min, 5 min to recheck.

3 SECTION B: SHORT ANSWERS

15. State reasons: (A) Zinc oxide is an amphoteric oxide. (B) Calcium starts floating when it is put in water. CBSE 2024 [2]

16. During the electrolytic refining of silver: (A) What are the cathode and the anode made of? (B) What is anode mud? CBSE Compartment 2025 [2]

17. How does galvanisation protect iron from rusting? Why does a galvanised article stay protected even if its zinc coating is scratched? [2]

18. Name the reaction between iron(III) oxide and aluminium. Write its balanced equation. Why is it called a displacement reaction? Give one use of this reaction. CBSE 2024 [3]

19. A student is given copper(I) sulphide (Cu_2S) and copper(I) oxide (Cu_2O). Show with balanced equations how copper can be obtained from them. Why is no reducing agent like carbon needed here? CBSE SQP 2024 [3]

4 SECTION C: LONG ANSWERS

20. (A) Write the electronic configurations of sodium (Z = 11), magnesium (Z = 12) and chlorine (Z = 17). (B) Show the formation of NaCl and $MgCl_2$ by the transfer of electrons. (C) Why do ionic compounds have high melting points? Why do they conduct electricity in the molten state but not as solids? [5]

21. A reddish-brown metal X, when heated in air, gives a black compound Y. When Y is heated in hydrogen gas, X is obtained back. X is refined by electrolysis and the refined X is used in electrical wiring. (A) Identify X and Y and write both equations. (B) Draw a labelled diagram of the electrolytic refining of X. (C) Write the reactions at the anode and the cathode. Based on CBSE SQP 2023 [5]

5 SECTION D: CASE-BASED

CASE: THE GREEN DOMES

Many old buildings have domes made of thin copper sheets. Freshly fitted copper is reddish-brown and shiny, but over the years the domes turn green. Iron does not turn green; it slowly rusts and crumbles. CBSE SQP 2025

1. Why does a copper dome turn green? Name the green substance. [1]
2. Should the copper domes be replaced by iron domes to avoid the change of colour? Give a reason. [1]
3. Why did old builders choose copper for domes? Give two properties of copper that help. [2]

★★★ *NaCl aur magnesium chloride ka electron-dot, thermit equation aur refining diagram: ye pakka aayenge, Nitesh Bhaiya ki guarantee!*

6 ANSWER KEY

Answer key: Section A (MCQs)

Q	Ans	Why
1	(b)	PbO and ZnO are amphoteric; A = B = 2.
2	(a)	Al, Ga, Mn, Au in that order.
3	(b)	Na (cold) > Mg (hot) > Fe (steam).
4	(d)	2, 8, 1 is sodium, a metal: basic Na_2O.
5	(b)	Al gains oxygen, so it is oxidised; MnO_2 is reduced.
6	(b)	Mg (2, 8, 2) gives 2 e^- → Mg^{2+}.
7	(a)	2HgS → 2HgO → 2Hg; low-reactivity metal.
8	(b)	Carbonate + limited air = calcination.
9	(b)	Pure Cu deposits on the cathode strip.
10	(a)	Fe + Ni + Cr (with a little C).
11	(c)	Malleable and sonorous: metal. Brittle: non-metal.
12	(b)	Ag reacts with sulphur in air: black Ag_2S.
13	(a)	Strong ionic forces need much energy; R explains A.
14	(c)	Gold is very unreactive, so R is false.

Displacement wala sawaal galat hua? Series dekho: Zn aur Fe dono Cu ko uske salt se hata dete hain, Cu kisi ko nahi!

Class Question: one non-metal, two oxides

Q. Non-metal A forms two oxides: B is neutral and poisonous (incomplete burning); C is acidic and causes global warming. Identify A, B and C.

- A = **carbon**; B = **carbon monoxide, CO** (neutral oxide); C = **carbon dioxide, CO_2** (acidic oxide, greenhouse gas)
- $2C + O_2 \rightarrow 2CO$ (limited air); $C + O_2 \rightarrow CO_2$ (plenty of air)

Score card: how did you fight the boss?

Marks	Result
36–40	BOSS DEFEATED: chapter mastered
28–35	STRONG WIN: fix the small slips
20–27	ALMOST THERE: revise weak days
Below 20	RETRY: re-read Days 10–13

- **Where to revise:** Q1–4, 11, 15 → Day 10; Q6, 13, 20 → Day 11; Q5, 7–9, 14, 16, 18, 19, 21 → Day 12; Q10, 12, 17, 22 → Day 13.

36 se upar? Bujho ki tarah naacho, boss khatam!

Boss Level tracker: write your percentage

Chapter 1 Boss (Day 4) ____ %
Chapter 2 Boss (Day 9) ____ %
Chapter 3 Boss (Day 14) ____ %

Model answers: Section B

15 Reasons: ZnO and calcium

- (A) ZnO reacts with both acids and bases to give salt and water.

$$ZnO(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2O(l)$$

$$ZnO(s) + 2NaOH(aq) \rightarrow Na_2ZnO_2(aq) + H_2O(l)$$

sodium zincate

- (B) Hydrogen bubbles formed stick to the calcium, so it floats.

16 Refining of silver

- (A) Cathode: thin strip of pure silver. Anode: block of impure silver.
- (B) Insoluble impurities that settle at the bottom, below the anode, are called anode mud.

17 Galvanisation

- A thin coat of zinc keeps air and moisture away from the iron.
- Zinc is more reactive than iron, so even if the coat is scratched, zinc reacts first and the iron stays protected.

Model answers: Section B (3 marks)

18 Thermit reaction

- Name: thermit reaction (highly exothermic).

$$Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + \text{Heat}$$

Fe 2 = 2, Al 2 = 2, O 3 = 3 ✔

- Al is more reactive than Fe, so it displaces Fe from its oxide.
- Use: the molten iron joins railway tracks and cracked machine parts.

19 Copper from Cu_2S

- Roasting: part of the sulphide ore is heated in air.

$$2Cu_2S(s) + 3O_2(g) \xrightarrow{\Delta} 2Cu_2O(s) + 2SO_2(g)\uparrow$$

Cu 4 = 4, S 2 = 2, O 6 = 6 ✔

$$2Cu_2O(s) + Cu_2S(s) \xrightarrow{\Delta} 6Cu(s) + SO_2(g)\uparrow$$

Cu 6 = 6, O 2 = 2, S 1 = 1 ✔

- Cu is low in the series, so heating alone reduces it.

Model answers: Section C

20 Ionic bonding (5 marks)

- (A) Na: 2, 8, 1; Mg: 2, 8, 2; Cl: 2, 8, 7.
- (B) Na gives 1 e^- to Cl; Mg gives 1 e^- to each of two Cl atoms. Each ion gets a stable octet.

FORMATION OF SODIUM CHLORIDE

FORMATION OF MAGNESIUM CHLORIDE

- (C) A lot of energy is needed to break the strong attraction between oppositely charged ions, so the melting point is high.
- In a solid the ions are fixed in place. When molten, they move freely and carry the current.

Model answers: Sections C and D

21 Copper and its refining (5 marks)

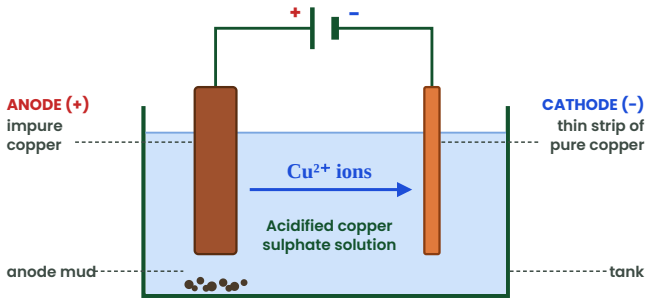
- (A) X = copper (Cu); Y = copper(II) oxide (CuO), black.

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

CuO is reduced; H_2 is oxidised

(B) ELECTROLYTIC REFINING OF COPPER



- (C) Anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ (impure copper dissolves).
- Cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ (pure copper deposits).

22 Case: the green domes

- (1) Copper reacts with moist carbon dioxide in air and gets a green coat of basic copper carbonate.
- (2) No. Iron rusts, and rust flakes off, so the dome would get weaker; the green coat on copper protects the metal below it.
- (3) Copper is highly malleable, so thin sheets can be shaped into domes; it also resists corrosion and lasts long.

Most asked in Chapter 3: Eduspark PYQ analysis

Topic	Day	Rating
Reactivity series and displacement	10	★★★
Amphoteric oxides (Al_2O_3, ZnO)	10	★★★
Thermit reaction	12	★★★
Corrosion and its prevention	13	★★★
Electron-dot formation of NaCl, $MgCl_2$	11	★★☆
Electrolytic refining, anode mud	12	★★☆
Roasting vs calcination	12	★★☆
Alloys and carats of gold	13	★★☆

Class Question: golden flame and Z = 17

Q. Element A gives a golden-yellow flame; B has atomic number 17. Identify A and B, their compound and its type of bonding.

- A = **sodium** (Na, 2, 8, 1); B = **chlorine** (Cl, 2, 8, 7)
- Na gives one electron to Cl: $Na \rightarrow Na^+ + e^-$, $Cl + e^- \rightarrow Cl^-$
- Compound **NaCl**, **ionic (electrovalent) bond**: high melting point, conducts electricity when molten or in water

Bonus rapid-fire round: quick answers

Best conductor of heat and electricity: **silver**. Non-metal liquid at room temperature: **bromine**. Metals that melt on your palm: **gallium, caesium**. Non-metal that conducts: **graphite**. Mixture that dissolves gold: **aqua regia (3 : 1)**. Coating of zinc on iron: **galvanisation**.

Common mistakes that cost marks

× WRONG	✓ RIGHT
Metal + HNO_3 gives H_2	HNO_3 oxidises the H_2 formed to water
Al_2O_3 is a basic oxide	Al_2O_3, ZnO: amphoteric
Cathode = impure metal	Anode = impure; cathode = thin strip of pure metal
Solid NaCl conducts	Only molten or dissolved NaCl conducts
Roasting uses limited air	Roasting: excess air (sulphides); calcination: limited air (carbonates)

Day 14 checklist: tick before you sleep

- ☐ I revised the mind map and all 7 must-know equations
- ☐ I attempted all 14 MCQs without looking at the answer key
- ☐ I wrote every short answer with balanced equations and state symbols
- ☐ I drew the electron-dot and refining diagrams neatly from memory
- ☐ I scored myself honestly and revised the days of my wrong answers

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Revision skip kiya toh Boss Level pe cycle gir jaayegi!

LEVEL 14 COMPLETE **CHAPTER 3 COMPLETE**

+200 XP earned. Next: Level 15

BUJHO *Samajh gaya! Reactivity series yaad hai, toh extraction aur displacement dono yaad hain!*

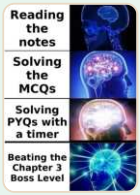
PAHELI *Boss defeated! Kal se Chapter 4: carbon ki duniya.*

Coming up: Day 15

Carbon and its compounds: covalent bonds, why carbon is so versatile, and electron-dot structures of methane.

NITESH SIR SAYS

Chapter 3 khatam! Galtiyan note karo aur Sunday ko dobara test do. ***Consistency hi jalwa hai!***



CHEMISTRY | CHAPTER 3 COMPLETE

CHAPTER 3 SWAHAA!

Metals and Non-metals | Days 10–14

BUJHO *Sodium kerosene mein, reactivity series zubaan pe, rust se bachna bhi yaad. Chapter 3 Swahaa!*

PAHELI *Kamaal kar diya! Ab carbon ka jaadu seekhte hain, Day 15 se.*

CHAPTER XP EARNED
+625 XP

LEVELS CLEARED
5 levels (Days 10–14)

BOOK PROGRESS
Level 14 / 60 1750 XP total

WHAT YOU HAVE MASTERED

- ✔ Physical properties of metals and non-metals
- ✔ Reactivity series and amphoteric oxides
- ✔ Extraction, roasting, calcination, refining
- ✔ Reactions with oxygen, water and acids
- ✔ Ionic bonds and electron-dot structures
- ✔ Corrosion, its prevention and alloys

NEXT CHAPTER Chapter 4: Carbon and its Compounds (Day 15) *Bas ek chapter aur!*

DAY 15 · LEVEL 15

CHEMISTRY | CHAPTER 4 | PART 1 · ASTRA SERIES

CARBON & ITS COMPOUNDS

Goal: covalent bonds, electron-dot structures, allotropes.

by Nitesh Prakash

01 Covalent Bond | 02 Electron-dot | 03 Bond Types | 04 Allotropes | 05 Versatile Carbon | +100 XP today | Level 15 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CARBON: THE BASICS

Carbon at a glance

Symbol, atomic number	**C, 6**
Electronic configuration	**K 2, L 4**
Valence electrons	**4** (valency 4); a non-metal
Earth's crust	Only 0.02% (carbonates, hydrogencarbonates, coal, petroleum)
Atmosphere	0.03% as CO_2
Found in	Food, clothes, medicines, fuels and our own body

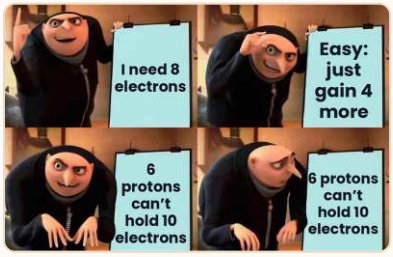
Where carbon occurs

Free state (as element)
- Diamond
- Graphite
- Fullerenes

Combined state (as compounds)
- CO_2 in air
- Carbonates: marble, limestone
- Fossil fuels; all living things

2 WHY CARBON SHARES ELECTRONS ★★★

GAIN 4 ELECTRONS	LOSE 4 ELECTRONS
Would form C^{4-} anion	Would form C^{4+} cation
× Nucleus with 6 protons cannot hold 10 electrons	× Removing 4 electrons needs a very large amount of energy



MEME BREAK
*Plan fail! Isliye carbon ne **sharing** chuni: 4 electrons share karo, octet complete.*

COVALENT BOND

- A bond formed by the **sharing of an electron pair** between two atoms
- Both atoms reach a **noble gas configuration** (octet; H gets a duplet)
- Carbon shares its 4 valence electrons: it is **tetravalent**

3 ELECTRON-DOT STRUCTURES ★★★ MOST ASKED

1. Write valence electrons: H 1, C 4, N 5, O 6, Cl 7.
2. Share pairs until every atom has 8 electrons (H has 2).
3. One shared pair = one bond (–). Two pairs (=), three pairs (≡).

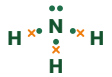
Hydrogen, H_2
H–H: single bond

Chlorine, Cl_2
Cl–Cl: single bond

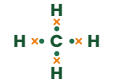
Oxygen, O_2
O=O: double bond

Nitrogen, N_2
N≡N: triple bond

Water, H_2O
2 single bonds, 2 lone pairs



Ammonia, NH_3
3 single bonds, 1 lone pair



Methane, CH_4
4 single C–H bonds

Carbon dioxide, CO_2
O=C=O: 2 double bonds

• electrons of one atom × electrons of the other atom. In each molecule, every atom ends with a full outer shell.

Solved Example: Drawing carbon dioxide

1. Valence electrons: C has 4, each O has 6.
2. C shares 2 electrons with each O, and each O shares 2 with C: two shared pairs per side.
3. Check: C has 4 pairs = 8 ✔; each O has 2 shared + 2 lone pairs = 8 ✔. Result: **O=C=O**.

★★★ *Methane aur carbon dioxide ka electron-dot structure har saal aata hai: Nitesh Bhaiya ki guarantee, roz ek baar bana ke dekho!*

BUJHO N_2 *mein bhi* O_2 *jaisa double bond hota hai na?*

PAHELI *Nahi! N ke 5 valence electrons, octet ke liye 3 share karta hai: N≡N, triple bond.*

Properties of covalent compounds

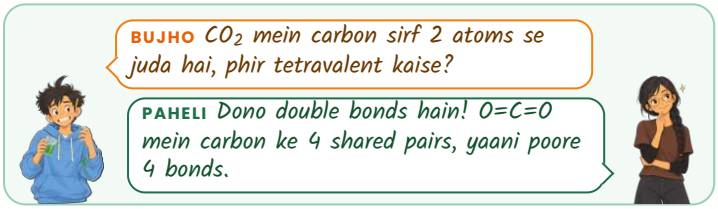
Compound	m.p. (K)	b.p. (K)
Acetic acid (CH_3COOH)	290	391
Chloroform ($CHCl_3$)	209	334
Ethanol (CH_3CH_2OH)	156	351
Methane (CH_4)	90	111

- **Low melting and boiling points:** forces between molecules are weak
- **Poor conductors of electricity:** no ions are formed

4 SINGLE, DOUBLE AND TRIPLE BONDS ★★☆

Bond	Shared pairs	Shared e^-	Examples
Single (–)	1	2	H–H, Cl–Cl, C–H in CH_4
Double (=)	2	4	O=O, O=C=O
Triple (≡)	3	6	N≡N

Between the same two atoms, strength grows with shared pairs: **triple** > **double** > **single**.



Solved Example: Counting covalent bonds

Q. How many covalent bonds are there in a molecule of pentane, C_5H_{12}?

- C–C bonds in a chain of 5 carbons = **4**
- C–H bonds = number of H atoms = **12**
- Total = 4 + 12 = **16 covalent bonds**

Shortcut for an alkane C_nH_{2n+2}: bonds = **3n + 1**. CH_4 4, C_2H_6 7, C_3H_8 10, C_5H_{12} 16 ✔

MEME BREAK
*Nahi Bujho! CH_4 mein koi ion nahi bante, electrons **share** hote hain. Yahi covalent compound hai.*

Exam tips

- ✔ Show only the **outer-shell (valence)** electrons in a dot structure.
- ✔ Show every shared pair **between** the two atoms; lone pairs stay on their own atom.
- ✔ Count after drawing: each atom has 8 electrons around it (H has 2).
- ✔ Write the line formula beside the dot structure: H–H, O=O, N≡N.

5 ALLOTROPES OF CARBON ★★★ MOST ASKED

ALLOTROPES

- Different **physical forms** of the same element
- Physical properties differ; **chemical properties are the same**
- Crystalline: **diamond, graphite, fullerene (C-60)**; amorphous: **coal, coke, charcoal, lampblack (soot)**

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \textbf{Heat}$

Diamond, graphite or fullerene: each burns to give only CO_2. Same element, same chemistry.

Property	Diamond	Graphite
Bonding	Each C bonded to **4** other C atoms	Each C bonded to **3** other C atoms
Structure	Rigid 3-D network of carbon atoms	Hexagonal layers held loosely, one above another
Hardness	**Hardest** natural substance known	Soft and slippery
Electricity	Non-conductor	**Good conductor**
Uses	Cutting and drilling tools, jewellery	Pencil lead, dry lubricant, electrodes

DIAMOND
Har carbon ke 4 dost, 3-D lock: isliye sabse hard!

GRAPHITE
Sirf 3 dost, layers phisal jaati hain: pencil mein ghis jaata hoon.

Fullerene (C-60): 60 carbon atoms in a **football** shape made of **20 hexagons + 12 pentagons**; named after the architect Buckminster Fuller.

Why graphite conducts but diamond does not

- Graphite: each C uses only 3 of its 4 valence electrons in bonds; the 4th is free to move, so it conducts
- Diamond: all 4 electrons are locked in bonds, no free electrons

MEME BREAK
*Jab pata chale ki chamakta diamond aur kaala pencil lead, dono **pure carbon** hain!*

Organic compounds: a historic surprise

- Carbon compounds were once thought to need a **"vital force"** found only in living things
- In 1828, **Friedrich Wöhler** made urea, an organic compound, from ammonium cyanate in the lab

$NH_4CNO \xrightarrow{\Delta} NH_2CONH_2$

ammonium cyanate → urea: same atoms (N 2, H 4, C 1, O 1), rearranged ✔

6 VERSATILE NATURE OF CARBON ★★★ MOST ASKED

CATENATION

- Carbon's unique ability to **form bonds with other carbon atoms**, giving large molecules
- The C–C bond is short and **very strong**, so long chains, branches and rings are stable
- **Class Question:** why less catenation in silicon? Si atoms are bigger, so the Si–Si bond is long and **weak**: Si–H chains stop at about **7–8 Si atoms** and are very reactive

LONG CHAIN	BRANCHED	RING
C–C–C–C–C	C–C–C–C | C	

TETRAVALENCY

- Valency 4: one carbon bonds with **4 other atoms**: C, H, O, N, S, Cl and more
- Small atom: the nucleus holds shared pairs strongly, so the bonds are **strong and stable**
- Result: about **three million** known carbon compounds

MEME BREAK

*Partner dhoondhne ka tension hi nahi: carbon **khud se hi** bond bana leta hai, isi ko kehte hain catenation!*

Common mistakes

× WRONG	✓ RIGHT
CO_2 drawn as O–C–O	O=C=O, two double bonds
N=N in nitrogen	N≡N, a triple bond
Diamond conducts too	Only graphite conducts
C forms C^{4+}/C^{4-} ions	C shares electrons (covalent)

TRY YOURSELF: DRAW THE ELECTRON-DOT STRUCTURES

(i) HCl
(ii) Br_2
(iii) Ethane, C_2H_6
(iv) Chloromethane, CH_3Cl

Short answer questions

1. Draw the electron-dot structures of (a) CH_4, (b) CO_2 and (c) N_2. Name the type of bond in each. NCERT
2. Why does carbon form covalent bonds? Give two reasons why covalent compounds have low melting points and do not conduct electricity. NCERT
3. Diamond and graphite are both pure carbon. Why is diamond hard and a non-conductor while graphite is soft and a good conductor?
4. Draw the electron-dot structure of a sulphur molecule made up of eight sulphur atoms. NCERT

7 PRACTICE MCQS

Neend aa rahi hai? Pehle ye 10 MCQs, answer key page 4 par hai!

1. The number of valence electrons in a carbon atom (atomic number 6) is:
(a) 2 (b) 4
(c) 6 (d) 8

2. The bond in a nitrogen molecule is: ★★★
(a) single covalent (b) double covalent
(c) triple covalent (d) ionic

3. Shared electron pairs in a CO_2 molecule: ★★★
(a) 2 (b) 3
(c) 4 (d) 6

4. Which of these is a good conductor of electricity? ★★☆
(a) Diamond (b) Graphite
(c) Sulphur (d) Phosphorus

5. The number of covalent bonds in pentane (C_5H_{12}) is: ★★☆
(a) 5 (b) 12
(c) 16 (d) 17

6. Carbon forms covalent bonds because it: ★★★
(a) can neither gain nor lose 4 electrons easily
(b) is a metal
(c) has a very large atom
(d) forms ions easily

7. The property of carbon atoms to link with one another to form long chains is called:
(a) tetravalency (b) catenation
(c) isomerism (d) allotropy

8. In diamond, each carbon atom is bonded to how many other carbon atoms?
(a) 2 (b) 3
(c) 4 (d) 6

9. **Assertion (A):** Diamond is the hardest natural substance.
Reason (R): In diamond, each carbon atom is bonded to four other carbon atoms, forming a rigid three-dimensional structure. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Covalent compounds are generally poor conductors of electricity.
Reason (R): Covalent compounds have strong forces of attraction between their molecules. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	Configuration 2, 4: four outer electrons.
2	(c)	N shares 3 pairs: N≡N.
3	(c)	Two double bonds = 4 shared pairs.
4	(b)	Graphite has free electrons.
5	(c)	4 C–C + 12 C–H = 16.
6	(a)	C^{4-} and C^{4+} are both unstable.
7	(b)	Self-linking of carbon = catenation.
8	(c)	4 bonds each: rigid 3-D network.
9	(a)	The rigid network explains hardness.
10	(c)	Forces between molecules are weak.

Model answers: short answer

2 Why covalent, and why low m.p. and non-conducting

- C cannot gain 4 e^- (6 protons cannot hold 10 electrons) or lose 4 e^- (needs too much energy), so it **shares**
- Weak forces between molecules → **low m.p. and b.p.**
- No ions formed → **poor conductors**

3 Diamond vs graphite

- Diamond: each C bonded to 4 C in a rigid 3-D network, no free electrons → hard, non-conductor
- Graphite: each C bonded to 3 C in layers that slide; one free electron per C → soft, good conductor

4 S_8 molecule

- Eight S atoms form a **crown-shaped ring**
- Each S shares one pair with each neighbour (2 single bonds) and keeps **2 lone pairs**: 8 electrons around every S

Q1: see the electron-dot structures of CH_4 (4 single), CO_2 (2 double) and N_2 (triple) on page 1.

Chapter 4 hit-list: Eduspark PYQ analysis

Topic	Day	Rating
Electron-dot structures (CH_4, CO_2, N_2)	15	★★★
Diamond vs graphite	15	★★★
Catenation and tetravalency	15	★★★
Homologous series, isomers	16	★★★
Functional groups, naming	17	★★★
Ethanol, ethanoic acid, esters	18	★★★
Soaps, micelles, hard water	19	★★☆

Real life: carbon at work

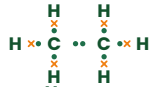
- Diamond-tipped saws and drills cut stone and glass
- Graphite rods are electrodes in dry cells; graphite powder lubricates locks
- Your pencil "lead" is graphite mixed with clay, with no lead at all

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 4 board and sample-paper questions. Year tags appear only where the paper year is known.

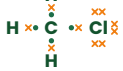
Try-yourself answers

(i) H–Cl: 1 shared pair, 3 lone pairs on Cl

(ii) Br–Br: like Cl_2, 1 shared pair and 3 lone pairs on each Br



(iii) 7 single bonds: 1 C–C + 6 C–H



(iv) 3 C–H + 1 C–Cl; 3 lone pairs on Cl

Day 15 checklist: tick before you sleep

- ☐ I can explain why carbon forms covalent bonds
- ☐ I can draw electron-dot structures of H_2, O_2, N_2, CH_4 and CO_2
- ☐ I can tell single, double and triple bonds by shared pairs
- ☐ I can compare diamond, graphite and fullerene
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 15 COMPLETE +100 XP earned. Next: Level 16

BUJHO *Samajh gaya! Carbon sharing karta hai, aur diamond-graphite dono carbon hi hain!*

PAHELI *Shabaash! Kal Level 16 mein milte hain hydrocarbons se.*

Day 15 mind map: revise in 2 minutes

COVALENT BOND
Sharing of electron pairs; octet (H: duplet)

ELECTRON-DOT ★★★
Shared pairs shown as dots; CH_4, CO_2 and N_2 asked most

CARBON: PART 1

BOND TYPES
Shared electrons: single 2, double 4, triple 6

ALLOTROPES ★★★
Diamond (4 bonds), graphite (3 bonds), C-60

VERSATILE CARBON
Catenation + tetravalency = about three million compounds

Coming up: Day 16

Hydrocarbons: saturated and unsaturated, chains, branches and rings, isomers and the homologous series. Carbon's big family!

NITESH SIR SAYS
*Electron-dot structure roz ek baar khud banao. **Practice hi jalwa hai!***

DAY 16 | LEVEL 16

CHEMISTRY | CHAPTER 4 | PART 2 | ASTRA SERIES

HYDROCARBONS & HOMOLOGOUS SERIES

Goal: hydrocarbon types, isomers, homologous series.

by Nitesh Prakash

01 Saturated | 02 Unsaturated | 03 Chains & Rings | 04 Isomers | 05 Homologous Series | +100 XP today | Level 16 / 60

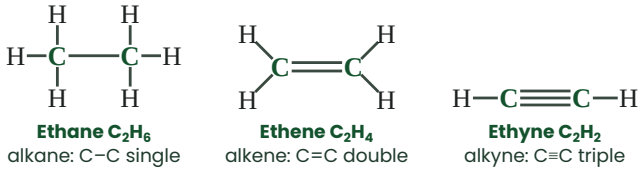
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 HYDROCARBONS: SATURATED VS UNSATURATED

DEFINITION

- **Hydrocarbons** are compounds made of **carbon and hydrogen only**
- **Saturated:** carbon atoms joined by **single bonds only**
- **Unsaturated:** at least one carbon–carbon **double or triple bond**

The C_2 family: one, two or three bonds



Ethane C_2H_6 — alkane: C–C single | **Ethene C_2H_4** — alkene: C=C double | **Ethyne C_2H_2** — alkyne: C≡C triple

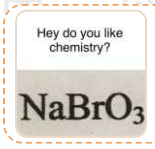
Every carbon still makes **4 bonds**. More bonds between the two carbons means **fewer hydrogens**.

Electron-dot link (Day 15): single bond = 1 shared pair (2 electrons), double = 2 pairs (4), triple = 3 pairs (6).

CLASS TRICK: COUNT THE BONDS

Double bond = 2, triple = 3. **Alkane = C + H − 1, Alkene = C + H, Alkyne = C + H + 1**. Check: C_2H_6 = 7, C_2H_4 = 6 (4 C–H + 2 in C=C), C_2H_2 = 5 (2 C–H + 3 in C≡C).

SATURATED ★★★	UNSATURATED
Only **single** C–C bonds	At least one **C=C** or **C≡C** bond
Alkanes: C_nH_{2n+2}	Alkenes C_nH_{2n}; alkynes C_nH_{2n-2}
Less reactive; substitution reactions	More reactive; addition reactions
Burn with a **clean blue** flame	Burn with a **sooty yellow** flame
Methane, ethane, propane	Ethene, propene, ethyne



★★★ *Alkane, alkene, alkyne ke general formula pakka aayenge, Nitesh Bhaiya ki guarantee hai! Yaad nahi kiya toh jawab hoga "Nah bro"!*

2 ALKANES, ALKENES AND ALKYNES

First five members: names and formulae

C atoms	Alkane	Alkene	Alkyne
Formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}
1 (meth)	Methane CH_4	none	none
2 (eth)	Ethane C_2H_6	Ethene C_2H_4	Ethyne C_2H_2
3 (prop)	Propane C_3H_8	Propene C_3H_6	Propyne C_3H_4
4 (but)	Butane C_4H_{10}	Butene C_4H_8	Butyne C_4H_6
5 (pent)	Pentane C_5H_{12}	Pentene C_5H_{10}	Pentyne C_5H_8

Names: prefix for C count + ane / ene / yne.

Class Question: Why no methene or methyne? C=C or C≡C needs **two carbon atoms**; methane has only one. So the first alkene is **ethene (C_2H_4)**, the first alkyne **ethyne (C_2H_2)**.

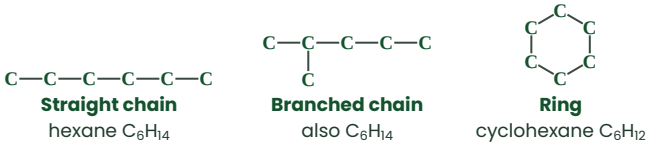
Solved Example 1: Formula from the general formula

(a) Alkyne with 5 C: n = 5, so H = 2(5) − 2 = 8. Formula: **C_5H_8** (pentyne).
(b) Which is saturated, C_3H_6 or C_4H_{10}? For n = 4, 2n + 2 = 10, so C_4H_{10} is a **saturated** alkane. C_3H_6 fits C_nH_{2n}: an alkene.
(c) Covalent bonds in an alkane = **3n + 1**. Ethane: 1 C–C + 6 C–H = **7**; propane 10; butane 13. ★★☆

3 CHAINS, BRANCHES AND RINGS

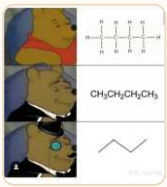
- Catenation lets carbon atoms join into **straight chains**, **branched chains** and **rings** (closed chains)
- Chains and rings can be saturated or unsaturated

Three ways six carbons can link



Straight chain hexane C_6H_{14} | **Branched chain** also C_6H_{14} | **Ring** cyclohexane C_6H_{12}

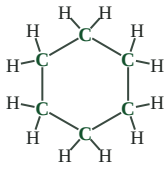
Carbon skeletons only; each C gets enough H to make 4 bonds.



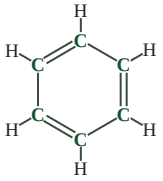
MEME BREAK

Butane ke teen roop: full structure, condensed $CH_3CH_2CH_2CH_3$ aur zig-zag. Molecule ek hi hai, bas likhne ka style alag!

Rings: cyclohexane vs benzene



Cyclohexane C_6H_{12}
all single bonds: saturated



Benzene C_6H_6
3 alternate C=C: unsaturated

- Cyclohexane: **2 H on every C**; saturated rings are **cycloalkanes**, formula C_nH_{2n}
- Benzene: **1 H on every C**; unsaturated rings like benzene are called **aromatic** compounds
- Cyclopentane C_5H_{10}: a five-carbon ring with only single bonds
- **Cycloalkenes C_nH_{2n-2}** (cyclohexene C_6H_{10}); **cycloalkynes C_nH_{2n-4}**

BUJHO *Cyclohexane mein ring hai, toh ye unsaturated hoga na?*

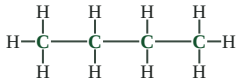
PAHELI *Nahi Bujho! Ring ka matlab unsaturated nahi. Cyclohexane mein sab single bonds hain, toh saturated. Benzene mein double bonds hain, isliye unsaturated.*

4 STRUCTURAL ISOMERS

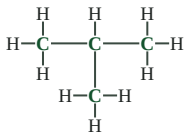
DEFINITION

- Compounds with the **same molecular formula** but **different structures** are **structural isomers**
- They have different physical properties, such as boiling points

Butane C_4H_{10}: two isomers



$CH_3–CH_2–CH_2–CH_3$
n-Butane (straight chain)

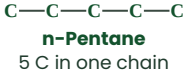


$CH_3–CH(CH_3)–CH_3$
Isobutane (2-methylpropane)

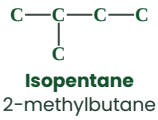
Both have 4 C and 10 H, but the carbons are **arranged differently**.

MEME BREAK
Isobutane, n-butane se: "Kaun? Main? Haan bhai, main bhi C_4H_{10} hoon, bas branch wala!"

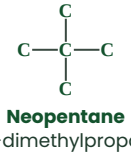
Pentane C_5H_{12}: three isomers ★★★



n-Pentane
5 C in one chain



Isopentane
2-methylbutane



Neopentane
2,2-dimethylpropane

Alkane	CH_4, C_2H_6, C_3H_8	C_4H_{10}	C_5H_{12}	C_6H_{14}
Isomers	1 each (none)	2	3	5

Isomerism in alkanes starts at butane (4 carbon atoms).

× WRONG	✔ RIGHT
Bending the chain into an L or zig-zag makes a new isomer	A bent drawing is the **same** compound; check the longest chain and the branches

BUJHO *Maine toh pentane ke 5 isomers bana diye, ab toh topper pakka?*

PAHELI *Do toh same nikle, bas chain mod di thi. Longest chain gino: 5 C, 4 C, 3 C. Pentane ke sirf 3 isomers!*

5 HOMOLOGOUS SERIES

DEFINITION

- A **homologous series** is a family of compounds with the **same functional group** (or same general formula) and similar chemical properties
- Successive members differ by a **$–CH_2–$** unit

Characteristics of a homologous series ★★★

- ✔ Same **general formula** for all members
- ✔ Successive members differ by **CH_2** and by **14 u** in mass
- ✔ **Similar chemical properties**: decided by the same functional group
- ✔ **Gradation in physical properties**: melting and boiling points rise as mass rises; solubility in a solvent changes in order
- ✔ Members can be made by similar general methods

Mass ladder: watch the 14 u steps

Alkane	Mass (u)	Alcohol	Mass (u)
CH_4	16	CH_3OH	32
C_2H_6	30	C_2H_5OH	46
C_3H_8	44	C_3H_7OH	60
C_4H_{10}	58	C_4H_9OH	74

Each step adds $–CH_2–$: 12 + 2 × 1 = **14 u**. (C = 12 u, H = 1 u, O = 16 u)

ISOMERS	HOMOLOGUES
Same molecular formula	Formulae differ by **CH_2**
Same molecular mass	Masses differ by **14 u**
n-butane and isobutane	Methane, ethane, propane

Solved Example 2: Series and next member

(a) Group these: CH_4, C_2H_4, C_3H_8, C_4H_8, C_2H_6.
- C_nH_{2n+2}: CH_4, C_2H_6, C_3H_8 (alkanes)
- C_nH_{2n}: C_2H_4, C_4H_8 (alkenes)

(b) Next homologue of C_3H_6 → add CH_2 → **C_4H_8**; of C_2H_5OH → **C_3H_7OH**.

(c) Trap: the **4th alkene** is pentene **C_5H_{10}**, not butene, because the alkene series starts at ethene. ★★☆

Formula finder: name the type in 3 steps

1. Count carbons: that number is **n**
2. Compare H with **2n + 2** (alkane), **2n** (alkene or cycloalkane), **2n − 2** (alkyne)
3. Example: C_4H_6 has n = 4 and 2n − 2 = 6, so it fits the **alkyne** formula (butyne)

TRY YOURSELF

Pehle khud karo, answer key page 4 par hai!

1. Write the formula of the alkene with 6 carbon atoms.
2. Pick the saturated ones: C_5H_{12}, C_5H_{10}, C_2H_2, C_6H_{12} (cyclohexane).
3. How many covalent bonds are there in pentane, C_5H_{12}?
4. Write the next homologue of CH_3OH and its molecular mass.

Common mistakes: avoid these

- "Ring means unsaturated": wrong; cyclohexane is saturated
- C_nH_{2n} fits **alkenes and cycloalkanes**; read the structure, not only the formula
- There is no "methene" or "methyne": these series start at 2 C
- Isomers have the **same** formula; homologues differ by CH_2

Ye galtiyan ki toh marks katenge, aur phir yahi face banega!

Exam tips

- ✔ Draw **every bond and every H**; check that each carbon shows exactly 4 bonds
- ✔ For isomers, give the **structure and the name** of each one
- ✔ Show the working for 14 u: $CH_2 = 12 + 2 \times 1$
- ✔ Write "C≡C" with three lines and "C=C" with two; a missing line changes the compound

Day 16 mind map: revise in 2 minutes

SATURATED
Alkanes C_nH_{2n+2} with single bonds only

UNSATURATED
Alkenes C_nH_{2n} and alkynes C_nH_{2n-2} have C=C or C≡C bonds

HYDROCARBONS

CHAINS & RINGS
Straight, branched, ring; C_6H_{12} and C_6H_6

ISOMERS ★★★
Same formula, new structure; butane 2, pentane 3

HOMOLOGOUS SERIES
Differ by CH_2 = 14 u; similar chemical, graded physical properties

6 PRACTICE MCQS

Padhai nahi ho rahi? 10 MCQs solve karo, dimaag apne aap start ho jayega!

1. The general formula of alkynes is: ★★★
(a) C_nH_{2n} (b) C_nH_{2n+2}
(c) C_nH_{2n-2} (d) C_nH_n

2. Which of these is an unsaturated hydrocarbon? ★★★
(a) C_3H_8 (b) C_2H_6
(c) C_3H_4 (d) C_4H_{10}

3. How many structural isomers does pentane have? NCERT ★★★
(a) 2 (b) 3
(c) 4 (d) 5

4. Formulae of cyclohexane and benzene, respectively: ★★☆
(a) C_6H_{12}, C_6H_6 (b) C_6H_6, C_6H_{12}
(c) C_6H_{14}, C_6H_6 (d) C_6H_{12}, C_6H_{12}

5. Which pair belongs to the same homologous series? ★★★
(a) CH_4, C_2H_4 (b) C_2H_6, C_3H_8
(c) C_2H_2, C_2H_4 (d) C_3H_6, C_3H_8

6. Which is NOT a feature of a homologous series? ★★☆
(a) Same general formula (b) Similar chemical properties
(c) Same physical properties (d) Members differ by CH_2

7. Number of single covalent bonds in propane, C_3H_8: ★★☆
(a) 8 (b) 10
(c) 11 (d) 12

8. n-Butane and isobutane are: ★★★
(a) homologues (b) structural isomers
(c) allotropes (d) identical compounds

9. **Assertion (A):** Benzene is an unsaturated hydrocarbon.
Reason (R): Cyclohexane has only single bonds in its ring. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Successive members of a homologous series differ in mass by 14 u.
Reason (R): They differ from each other by a $–CH_2–$ unit. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. How many structural isomers can you draw for pentane? Draw them and name each. (3 marks) NCERT
2. Give three differences between saturated and unsaturated hydrocarbons, with one example of each. (3 marks)
3. What is a homologous series? Write two of its characteristics. Give the formula of the fourth member of the alkene series. (3 marks)

Answer key: MCQs

Q	Ans	Why
1	(c)	C_nH_{2n-2}: C_2H_2, C_3H_4 and so on.
2	(c)	C_3H_4 fits C_nH_{2n-2}: propyne, a triple bond.
3	(b)	n-pentane, isopentane, neopentane.
4	(a)	Cyclohexane C_6H_{12}; benzene C_6H_6.
5	(b)	Both alkanes; they differ by one CH_2.
6	(c)	Physical properties change gradually; they are not the same.
7	(b)	2 C–C + 8 C–H = 10 (3n + 1 with n = 3).
8	(b)	Same formula C_4H_{10}, different structures.
9	(b)	Both true; benzene is unsaturated due to its C=C bonds, not because of cyclohexane.
10	(a)	CH_2 = 12 + 2 = 14 u explains the mass gap.

Model answers: short and long

1 Pentane, C_5H_{12}: three isomers

- **n-Pentane:** $CH_3–CH_2–CH_2–CH_2–CH_3$
- **Isopentane (2-methylbutane):** $CH_3–CH(CH_3)–CH_2–CH_3$
- **Neopentane (2,2-dimethylpropane):** $C(CH_3)_4$, four CH_3 groups on one central carbon

2 Saturated vs unsaturated

- **Bonds:** single C–C bonds only vs at least one C=C or C≡C
- **Formula:** C_nH_{2n+2} vs C_nH_{2n} or C_nH_{2n-2}
- **Flame and reactivity:** clean blue flame, less reactive vs sooty yellow flame, more reactive (addition reactions)
- **Examples:** ethane C_2H_6 vs ethene C_2H_4

3 Homologous series

- Compounds with the same functional group or general formula, where successive members differ by $–CH_2–$
- Characteristics: masses differ by 14 u; similar chemical properties, graded physical properties
- Alkenes: ethene, propene, butene, **pentene C_5H_{10}** is the 4th member

Day 16 hit-list: Eduspark PYQ analysis

Topic	Rating
Isomers of butane and pentane	★★★
Homologous series: definition, CH_2, 14 u	★★★
General formulae of alkanes, alkenes, alkynes	★★★
Saturated vs unsaturated (flame test)	★★☆
Cyclohexane and benzene structures	★★☆
Counting covalent bonds	★☆☆

Coming up: Day 17

Functional groups (–OH, –CHO, –COOH and more) and IUPAC naming: how propanol, propanal, propanone and propanoic acid get their names.

Day 16 checklist: tick before you sleep

- [ ] I can tell saturated from unsaturated and draw ethane, ethene and ethyne
- [ ] I can write alkane, alkene and alkyne formulae from the general formulae
- [ ] I can draw straight, branched and ring structures, including benzene
- [ ] I can draw the 2 isomers of butane and the 3 isomers of pentane
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 16 COMPLETE +100 XP earned. Next: Level 17

BUJHO *Samajh gaya! Ring ho ya chain, sirf single bonds hain toh saturated. Same formula, alag structure: isomers!*

PAHELI *Shabaash! Kal Level 17: functional groups aur IUPAC naming ki baari.*

Try-yourself answers

1. Hexene, C_6H_{12} (n = 6 in C_nH_{2n})
2. Saturated: C_5H_{12} (alkane) and cyclohexane C_6H_{12} (single bonds in a ring)
3. Pentane: 3(5) + 1 = **16** bonds (4 C–C + 12 C–H)
4. C_2H_5OH (ethanol): 24 + 5 + 16 + 1 = **46 u**

Real life: hydrocarbons around you

- **CNG** is mainly methane; **LPG** is mainly butane (with propane)
- **Petrol**, kerosene and **candle wax** are mixtures of alkanes
- **Ethyne** (acetylene) burns in oxygen in welding torches
- Kekulé proposed the benzene ring structure in 1865

EXTRA KNOWLEDGE (HIGH LEVEL)

Not in current CBSE syllabus. Types of isomers:

- **Chain:** butane / 2-methylpropane (C_4H_{10})
- **Position:** propan-1-ol / propan-2-ol (C_3H_8O)
- **Functional:** ethanol C_2H_5OH / dimethyl ether CH_3OCH_3
- **Stereo:** same bonds, different 3-D arrangement

NITESH SIR SAYS

*Roz 5 minute structures haath se draw karo. **Jo draw karega, wahi yaad rakhega!** Kal milte hai Day 17 pe.*

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 4 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 17 · LEVEL 17

CHEMISTRY | CHAPTER 4 | PART 3 · ASTRA SERIES

FUNCTIONAL GROUPS & IUPAC NAMES

Goal: spot the group, then name the compound.

by Nitesh Prakash

01 Heteroatoms | 02 Group Table | 03 Word Roots | 04 Naming Rules | 05 Examples | +100 XP today | Level 17 / 60

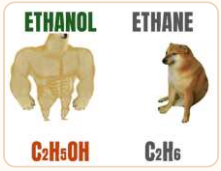
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 HETEROATOMS & FUNCTIONAL GROUPS

DEFINITION

- **Heteroatom:** any atom other than C and H (**O, N, S, Cl, Br**) that replaces H in a carbon chain
- **Functional group:** the heteroatom or group of atoms that gives the compound its **characteristic chemical properties**
- It joins the chain through its **free valency** (the line in –OH), in place of one H

HYDROCARBON $CH_3–CH_3$ ethane (a gas) → replace one H by –OH → **ALCOHOL** $CH_3–CH_2–OH$ ethanol (a liquid)



MEME BREAK

*Ek H hatao, –OH lagao: gas wala ethane ban gaya liquid ethanol! Naya group matlab **naye properties**.*

Key idea: what the group decides

- ✔ **Same functional group → similar chemical properties.** Methanol, ethanol and propanol all react with sodium.
- ✔ **Chain length** changes mainly physical properties, such as melting and boiling points.

BUJHO *Ethanol mein –OH hai, toh ye NaOH jaisa base hoga?*

PAHELI *Nahi! Yahan –OH carbon se covalent bond se juda hai, hydroxide ion nahi deta. Ethanol litmus ka rang nahi badalta.*

Common names you will hear

IUPAC name	Common name	Met in
methanal	formaldehyde	formalin
ethanol	ethyl alcohol	sanitisers
ethanoic acid	acetic acid	vinegar (5–8% solution)
propanone	acetone	nail-polish remover
methanoic acid	formic acid	ant stings

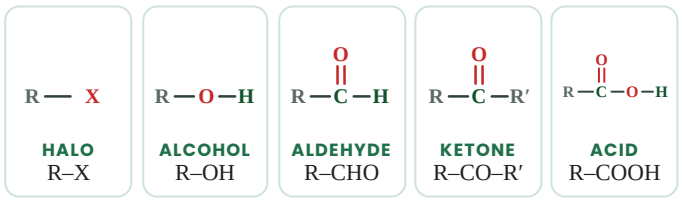
2 FUNCTIONAL GROUPS TABLE ★★★ MOST ASKED

The five groups you must know

Class	Group	Ending	Example
Halo	**–Cl, –Br**	**chloro-, bromo-**	CH_3Cl, chloromethane
Alcohol	**–OH**	**-ol**	CH_3OH, methanol
Aldehyde	**–CHO**	**-al**	HCHO, methanal
Ketone	**>C=O**	**-one**	CH_3COCH_3, propanone
Carboxylic acid	**–COOH**	**-oic acid**	CH_3COOH, ethanoic acid
Alkene, alkyne	**C=C, C≡C**	**-ene, -yne**	$CH_2=CH_2$, ethene

Halo is written as a prefix; all the others are suffixes. C=C and C≡C are not heteroatoms, but are named with an ending in the same way.

How each group is drawn



R, R′ = rest of the carbon chain; X = Cl or Br. Aldehyde: C=O at the **end** of the chain (bonded to H). Ketone: C=O in the **middle** (bonded to two carbons).

★★★ *Group, ending aur example: ye table exam ka favourite hai, Nitesh Bhaiya ki guarantee!*

SHORTCUT: THE ENDING TELLS THE FAMILY

-ol alcohol, -al aldehyde, -one ketone, -oic acid carboxylic acid.

3 WORD ROOTS: COUNT THE CARBONS

C atoms	1	2	3	4	5
Root	Meth	Eth	Prop	But	Pent
C atoms	6	7	8	9	10
Root	Hex	Hept	Oct	Non	Dec

Memory trick: "**M**onkeys **E**at **P**eeled **B**ananas" for 1 to 4 carbons. From 5 onwards the roots are number words you already know: pentagon, hexagon, octopus, decade.

4 IUPAC NAMING RULES ★★★ MOST ASKED

IUPAC (International Union of Pure and Applied Chemistry) names have four parts:

PREFIX	+	WORD ROOT	+	BOND	+	SUFFIX
chloro-, bromo- for halogens		meth, eth, prop… (no. of C)		-ane, -ene, -yne (single, =, ≡)		-ol, -al, -one, -oic acid

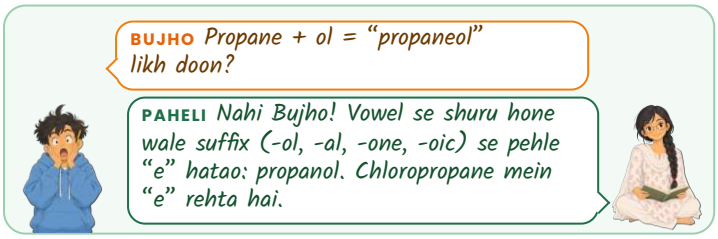
chloro + **prop** + **ane** = chloropropane | **prop** + **an** + **ol** = propanol

Read every name backwards: suffix = group, middle = bond type, root = carbon count

Four steps to name any compound

1. Count the carbons in the **longest chain**, including the C of –CHO or –COOH. 3 C → root **prop**.
2. Check the bonds: all single → **-ane**; one C=C → **-ene**; one C≡C → **-yne**.
3. Add the group. A suffix starting with a vowel **drops the final "e"**: propan**e** + ol = **propanol**. A halogen goes in front as a prefix: **chloro**propane.
4. Number the chain from the end **nearer the group**, e.g. propan-1-ol, 1-chloropropane. For Class 10 the plain name is enough unless asked.

× WRONG	✔ RIGHT
propaneol, propaneal	propanol, propanal (drop the "e")
CH_3CHO = methanal	ethanal: the CHO carbon counts, so 2 C
methanone, ethanone	propanone is the simplest ketone (3 C)
propanechloro	chloropropane (halogen = prefix)



Exam tips

- ✔ The carbon of –CHO and –COOH is **part of the chain**: HCOOH is methanoic acid, CH_3COOH is ethanoic acid.
- ✔ A ketone needs **at least 3 carbons**: its C=O carbon must sit between two carbons.
- ✔ Halogens are always a **prefix**; all other groups here are a **suffix**.
- ✔ When a structure is asked, write the condensed formula ($CH_3–CH_2–CHO$), not just the molecular formula (C_3H_6O).

REMEMBER: THE SIMPLEST MEMBER

Methanol, methanal, methanoic acid and chloromethane have 1 C; ethene and ethyne 2 C; **propanone needs 3 C**.

5 WORKED NAMES: THE PROPANE FAMILY

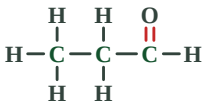
Solved Example 1: Formula → name ★★★

Formula	Group	Build-up	IUPAC name
$CH_3–CH_2–CH_2–Cl$	Halo	chloro + propane	**chloropropane**
$CH_3–CH_2–CH_2–OH$	Alcohol	propan(e) + ol	**propanol**
$CH_3–CH_2–CHO$	Aldehyde	propan(e) + al	**propanal**
$CH_3–CO–CH_3$	Ketone	propan(e) + one	**propanone**
$CH_3–CH_2–COOH$	Carboxylic acid	propan(e) + oic acid	**propanoic acid**
$CH_3–CH=CH_2$	Alkene	prop + ene	**propene**
$CH_3–C≡CH$	Alkyne	prop + yne	**propyne**
$CH_3–CH_2–Br$	Halo	bromo + ethane	**bromoethane**

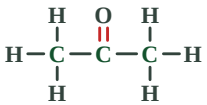
MEME BREAK

*Propanal (CH_3CH_2CHO) aur propanone (CH_3COCH_3): dono ka formula **C_3H_6O**! Group alag, isliye naam aur properties alag.*

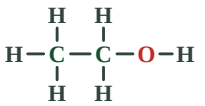
Solved Example 2: Full structural formulae ★★★



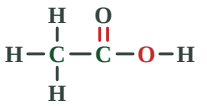
PROPANAL
$CH_3–CH_2–CHO$



PROPANONE
$CH_3–CO–CH_3$



ETHANOL
$CH_3–CH_2–OH$



ETHANOIC ACID
$CH_3–COOH$

Every carbon shows 4 bonds and every H one bond. Propanal and propanone have the same formula, C_3H_6O, but different groups.

Solved Example 3: Name → structure ★★★

Q. Draw the structures of ethanoic acid, bromopentane, butanone and hexanal. Are structural isomers possible for bromopentane? NCERT

Name	Condensed formula	Molecular
Ethanoic acid	$CH_3–COOH$	$C_2H_4O_2$
Bromopentane	$CH_3–CH_2–CH_2–CH_2–CH_2–Br$	$C_5H_{11}Br$
Butanone	$CH_3–CO–CH_2–CH_3$	C_4H_8O
Hexanal	$CH_3–CH_2–CH_2–CH_2–CH_2–CHO$	$C_6H_{12}O$

Isomers: yes: 1-, 2- and 3-bromopentane (Br position changes), plus branched chains (how to name them: Extra Knowledge box, page 4).

MEME BREAK

*–CHO aur –COOH ka carbon bhi chain mein gina jaata hai! Isliye CH_3COOH = 2 C = **ethanoic acid**.*

Day 17 mind map: revise in 2 minutes

HETEROATOM
O, N, S, Cl or Br in place of H; the group sets the reactions

WORD ROOT
Meth 1, Eth 2, Prop 3, But 4, Pent 5 … Dec 10

FUNCTIONAL GROUPS & NAMES

ENDINGS ★★★
-ol (OH), -al (CHO), -one (C=O), -oic acid (COOH)

PREFIX
chloro-, bromo- in front: chloropropane, bromoethane

BONDS
-ane (single), -ene (C=C), -yne (C≡C); drop the final "e" before a vowel suffix: propanol, propanal, propanone

Families: one group, many members ★★☆

C	Alcohols (-ol)	Aldehydes (-al)	Acids (-oic acid)
1	methanol	methanal	methanoic acid
2	ethanol	ethanal	ethanoic acid
3	propanol	propanal	propanoic acid
4	butanol	butanal	butanoic acid
5	pentanol	pentanal	pentanoic acid

Each member differs from the next by **CH_2** (14 u): a homologous series. The 2 C members: C_2H_5OH, CH_3CHO, CH_3COOH. Ketones start at 3 C: propanone CH_3COCH_3, butanone $CH_3COC_2H_5$.

Solved Example 4: Name these ★★☆

Q. How would you name: (i) $CH_3–CH_2–Br$ (ii) HCHO (iii) $HC≡C–CH_2–CH_2–CH_2–CH_3$? NCERT

- (i) 2 C, halogen Br → **bromoethane**
- (ii) 1 C, aldehyde group → **methanal**
- (iii) 6 C, one triple bond → **hexyne** (hex-1-yne), C_6H_{10}

Also: $CH_3CH_2CH_2COOH$ (4 C, –COOH) is **butanoic acid**, and $CH_3COCH_2CH_3$ (4 C, C=O in the middle) is **butanone**.

TRY YOURSELF: NAME OR DRAW

(i) $CH_3CH_2CH_2CH_2OH$ (ii) $CH_3CH_2CH_2CHO$
(iii) $CH_3CH_2CH_2CH_2COOH$ (iv) $CH_3CH_2CH_2CH_2Cl$
(v) $CH_3CH_2CH_2Br$ (vi) Draw ethanal
(vii) Draw pentanal (viii) Draw methanoic acid

Answers are on page 4, below the model answers.

Short and long answer questions

1. What is a functional group? Name the group present in, and give the IUPAC name of: (i) $CH_3CH_2CH_2OH$ (ii) CH_3CH_2COOH (iii) $CH_3COCH_2CH_3$ (iv) CH_3CH_2CHO **(3 marks)**
2. Write the condensed structural formulae of: (i) propene (ii) propyne (iii) chloroethane (iv) pentanoic acid. **(2 marks)**
3. Why is CH_3COCH_3 named propanone and not propanal? Explain why the simplest ketone must have three carbon atoms. **(3 marks)**

6 PRACTICE MCQS

Kyun nahi ho rahi padhai? Phone side mein rakho, 10 MCQs khud solve karo!

1. The functional group present in propanal is: ★★★
(a) –OH (b) –CHO
(c) >C=O (d) –COOH

2. The IUPAC name of CH_3COCH_3 is: ★★★
(a) propanal (b) propanone
(c) propanol (d) ethanone

3. The name ending used for a carboxylic acid is: ★★☆
(a) -ol (b) -al
(c) -one (d) -oic acid

4. The IUPAC name of $CH_3–CH_2–Br$ is: NCERT ★★★
(a) bromomethane (b) bromoethane
(c) bromopropane (d) ethanebromo

5. Which of the following compounds is a ketone? ★★☆
(a) CH_3CHO (b) CH_3COOH
(c) $CH_3COCH_2CH_3$ (d) CH_3CH_2OH

6. How many carbon atoms does one hexanal molecule have? ★☆☆
(a) 4 (b) 5
(c) 6 (d) 7

7. Propanal and propanone have the same: ★★☆
(a) functional group (b) molecular formula, C_3H_6O
(c) name ending (d) chemical properties

8. The correct formula of ethanoic acid is: ★★★
(a) HCOOH (b) CH_3COOH
(c) CH_3CH_2COOH (d) CH_3CHO

9. **Assertion (A):** Methanol and ethanol show similar chemical properties.
Reason (R): Both contain the same functional group, –OH. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The IUPAC name of CH_3CHO is methanal.
Reason (R): The carbon atom of the –CHO group is counted in the parent chain. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Case-based question (4 marks)

A teacher wrote four compounds on the board: **A** CH_3CH_2OH, **B** CH_3CHO, **C** CH_3COCH_3 and **D** CH_3COOH.

1. Name the functional group present in B and in C. **(1)**
2. Write the IUPAC names of A and D. **(1)**
3. Which compound, as a 5–8% solution in water, is sold as vinegar? Write the formula and IUPAC name of its next higher homologue. **(2)**

Answer key: MCQs

Q	Ans	Why
1	(b)	–CHO is the aldehyde group; propan**al**.
2	(b)	C=O in the middle: ketone, propan**one**.
3	(d)	Acids end in -oic acid, e.g. ethanoic acid.
4	(b)	2 C + Br as prefix: bromoethane.
5	(c)	C=O between two carbons: butanone.
6	(c)	Hex = 6; the CHO carbon is included.
7	(b)	Both are C_3H_6O; groups differ.
8	(b)	2 C including the COOH carbon.
9	(a)	Same group → similar properties.
10	(d)	CH_3CHO is ethanal; R is true.

Model answers: short and long answer

1 Functional group and names

The atom or group of atoms that decides the chemical properties of an organic compound.

Compound	Group	IUPAC name
$CH_3CH_2CH_2OH$	alcohol, –OH	**propanol**
CH_3CH_2COOH	carboxylic acid, –COOH	**propanoic acid**
$CH_3COCH_2CH_3$	ketone, >C=O	**butanone**
CH_3CH_2CHO	aldehyde, –CHO	**propanal**

2 Condensed formulae

(i) $CH_3–CH=CH_2$ (ii) $CH_3–C\equiv CH$
(iii) $CH_3–CH_2–Cl$
(iv) $CH_3–CH_2–CH_2–CH_2–COOH$, pentanoic acid

3 Propanone, not propanal

- In $CH_3–CO–CH_3$, the C=O carbon is **in the middle**, bonded to two carbons: that makes it a **ketone** (-one)
- In propanal, $CH_3–CH_2–CHO$, the C=O is at the **end** and bonded to H: an **aldehyde** (-al)
- A ketone's C=O carbon needs a carbon on **each side**, so at least 3 C: propanone is the simplest

Try-yourself answers

(i) butanol (ii) butanal
(iii) pentanoic acid (iv) chlorobutane
(v) bromopropane (vi) $CH_3–CHO$
(vii) $CH_3–CH_2–CH_2–CH_2–CHO$ (viii) H–COOH

Case-based answers

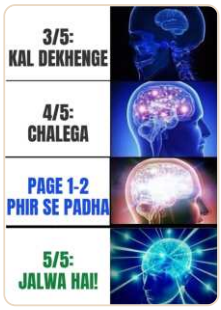
- (i) B: aldehyde group, –CHO; C: ketone group, >C=O
- (ii) A: ethanol; D: ethanoic acid
- (iii) D, ethanoic acid. Next homologue: CH_3CH_2COOH, **propanoic acid**

Coming up: Day 18

Chemical properties: combustion, oxidation, addition and substitution, plus the two stars of the chapter, ethanol and ethanoic acid.

Day 17 checklist: tick before you sleep

- ☐ I can define a heteroatom and a functional group
- ☐ I can write the five groups with their formula and name ending
- ☐ I know the word roots meth to dec for 1 to 10 carbons
- ☐ I can name a compound from its formula and draw it from its name
- ☐ I solved all 10 MCQs and checked them with the key



My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 17 COMPLETE +100 XP earned. Next: Level 18

BUJHO *Samajh gaya! Root se carbon gino, ending se group pehchano: propanal, propanone, propanoic acid, sab clear!*

PAHELI *Shabaash! Kal Level 18: chemical properties of carbon compounds.*

Day 17 hit-list: Eduspark PYQ analysis

Topic	Rating
Functional group table (group, ending, example)	★★★
Naming propanal, propanone, propanol, propanoic acid	★★★
Drawing a structure from its name	★★★
Halo prefix: chloro-, bromo-	★★☆
Word roots and -ane, -ene, -yne	★★☆
Common names: acetone, acetic acid, formalin	★☆☆
Simplest ketone needs 3 C (propanone)	★★☆

Extra Knowledge (High Level) not in current CBSE syllabus

Naming branched alkanes: (1) longest carbon chain = parent; (2) number from the end nearest a branch (lowest locants); (3) branch name with its position, di-/tri- for repeats, different branches in alphabetical order.

$CH_3–CH(CH_3)–CH_3$	**2-methylpropane**
$CH_3–CH(CH_3)–CH(CH_3)–CH_3$	**2,3-dimethylbutane**
$CH_3–CH(CH_3)–CH(C_2H_5)–CH_2–CH_3$	**3-ethyl-2-methylpentane**

NITESH SIR SAYS
*Naam ratna nahi, banana seekho: root + bond + ending. **Kal milte hai Day 18 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 4 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 18 · LEVEL 18

CHEMISTRY | CHAPTER 4 | PART 4 · ASTRA SERIES

PROPERTIES OF CARBON COMPOUNDS

Goal: four key reactions, ethanol and ethanoic acid.

by Nitesh Prakash

01 Combustion | 02 Oxidation | 03 Addition | 04 Substitution | 05 Ethanol & Ethanoic Acid | +125 XP today | Level 18 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 COMBUSTION ★★★ MOST ASKED

DEFINITION

- Carbon and its compounds **burn in oxygen** to give **$CO_2 + H_2O$**
- A lot of **heat and light** is released, so most carbon compounds are used as **fuels**

$$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \text{Heat and light}$$
C 1 = 1, O 2 = 2 ✔

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + \text{Heat and light}$$
C 1 = 1, H 4 = 4, O 4 = 4 ✔

$$CH_3CH_2OH_{(l)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(l)} + \text{Heat}$$
ethanol burns with a clean flame, giving heat and light
C 2 = 2, H 6 = 6, O 7 = 7 ✔

SATURATED (e.g. CH_4)	UNSATURATED (e.g. C_2H_2)
✔ Clean **blue flame**	× **Yellow, sooty** flame with black smoke
Burn completely in air	Burn incompletely: higher carbon share
Gas stove, LPG, CNG	Saturated ones also go sooty when air is limited

Flame facts: why vessels turn black

- Black base of a cooking vessel = **air holes blocked**, fuel wasted; clean them for a blue flame
- Yellow glow = **unburnt carbon particles** heated till they glow
- A **flame forms only when gases burn**: wax and LPG give a flame, coal and charcoal just glow red
- Incomplete burning also gives **poisonous CO**; sulphur and nitrogen in coal give acidic oxides

TRY YOURSELF: BALANCE THESE COMBUSTIONS

(i) $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$ (ii) $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

Balance C, then H, then O last. Answers on page 4.

★★★ *Thanos Baba ka gyaan: combustion ki equation bhi perfectly balanced honi chahiye. Ye pakka aayega, Nitesh Bhaiya ki guarantee!*

2 OXIDATION ★★★

DEFINITION

- Controlled addition of oxygen: **alcohol → carboxylic acid**
- **Oxidising agents** are substances that **give oxygen** to others: alkaline $KMnO_4$, acidified $K_2Cr_2O_7$

$$CH_3CH_2OH + 2[O] \xrightarrow[\text{or acidified } K_2Cr_2O_7 + \text{Heat}]{\text{Alkaline } KMnO_4 + \text{Heat}} CH_3COOH + H_2O$$
ethanol → ethanoic acid; [O] = oxygen from the oxidising agent
C 2 = 2, H 6 = 6, O 3 = 3 ✔

Activity: warm ethanol + alkaline permanganate

- Add 5% alkaline $KMnO_4$ drop by drop to warm ethanol
- **Purple colour disappears** at first: $KMnO_4$ is used up in oxidising ethanol
- Excess added: colour **stays**, no ethanol left to oxidise
- Acidified $K_2Cr_2O_7$: **orange → green**

3 ADDITION (HYDROGENATION) ★★★

DEFINITION

- **Unsaturated** compounds (C=C or C≡C) **add hydrogen** to become saturated
- Needs a **catalyst: nickel (Ni) or palladium (Pd)**

$$CH_2{=}CH_2 + H_2 \xrightarrow[\text{Heat}]{\text{Ni catalyst}} CH_3{-}CH_3$$
ethene → ethane: C 2 = 2, H 6 = 6 ✔

Use: vegetable oil → vanaspati ghee

- Vegetable oils (long unsaturated chains) + H_2, Ni catalyst → **solid vanaspati ghee**
- **Catalyst:** changes the rate of a reaction without being used up itself
- Animal fats are **saturated** and bad for the heart; vegetable oils (unsaturated) are the healthier choice for cooking

MEME BREAK

*Iron Man = Fe-male. Toh hydrogenation ka superhero? **Ni-Man!** Nickel catalyst ke bina tel kabhi ghee nahi banta.*

4 SUBSTITUTION ★★☆

DEFINITION

- Saturated hydrocarbons are **fairly unreactive**
- In **sunlight**, chlorine **replaces hydrogen atoms one by one**; excess Cl_2 gives CH_2Cl_2, $CHCl_3$, CCl_4

$$CH_4 + Cl_2 \xrightarrow{\text{Sunlight}} CH_3Cl + HCl$$

methane → chloromethane: C 1 = 1, H 4 = 4, Cl 2 = 2 ✔

MEME BREAK
Substitution = displacement ka carbon version: Cl andar aaya, H bahar. Bechara H bhi ab Cu ki tarah side mein baitha hai!

Four reactions at a glance

Reaction	Who reacts	Condition	Product
Combustion	almost all	O_2, burn	$CO_2 + H_2O$ + heat
Oxidation	alcohols	alk. $KMnO_4$ or acid. $K_2Cr_2O_7$, heat	carboxylic acid
Addition	unsaturated	H_2, Ni or Pd	saturated
Substitution	saturated	Cl_2, sunlight	$CH_3Cl + HCl$

BUJHO *Combustion aur oxidation alag kaise? Dono mein oxygen hi toh hai!*

PAHELI *Combustion = jalna, $CO_2 + H_2O$ + heat. Oxidation = $KMnO_4$ ki madad se ethanol → ethanoic acid, bina jale!*

5 ETHANOL ★★★

Properties and uses of ethanol

- Colourless **liquid** at room temperature; **soluble in water** in all proportions
- Active ingredient of alcoholic drinks; good **solvent** in tincture of iodine, cough syrups and tonics
- Neutral to litmus; burns with a clean flame, so it is **added to petrol** as a cleaner fuel

Reaction 1: ethanol + sodium

$$2Na + 2CH_3CH_2OH \rightarrow 2CH_3CH_2ONa + H_2\uparrow$$

sodium ethoxide + hydrogen: Na 2 = 2, C 4 = 4, H 12 = 12, O 2 = 2 ✔

Test: the gas burns with a **"pop" sound** (hydrogen). Ethanol gives **no** fizz with $NaHCO_3$.

Reaction 2: dehydration (removal of water)

$$CH_3CH_2OH \xrightarrow[443\ K]{\text{Hot conc. } H_2SO_4} CH_2{=}CH_2 + H_2O$$

ethanol → ethene: C 2 = 2, H 6 = 6, O 1 = 1 ✔

Conc. H_2SO_4 is a **dehydrating agent**: it pulls water out of ethanol.

Harmful effects and denatured alcohol

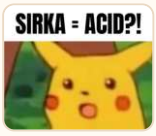
- Small amounts cause drunkenness; regular drinking leads to **serious health problems**
- **Methanol** (CH_3OH) even in small amounts can kill: in the liver it is oxidised to **methanal**, which coagulates cell protoplasm and damages the optic nerve (**blindness**)
- **Denatured alcohol** = industrial ethanol + methanol (poison) + **blue dye**, so that it cannot be misused for drinking

MEME BREAK
*Methanol ka ek ghoont: aankhein aur jaan dono khatre mein. Ethanol ki aadat: liver aur dil dono bolein **ouch!***

6 ETHANOIC ACID (ACETIC ACID) ★★★

Key facts

- Common name **acetic acid**; a **weak acid** (ionises only partly), turns blue litmus **red**
- **Vinegar = 5–8% ethanoic acid in water**: preservative in pickles and a flavouring
- **Other uses:** making cellulose acetate (rayon, photographic film), dyes, plastics and esters for perfumes; coagulating rubber latex
- Pure acid: melting point **290 K**, so it freezes in cold weather: **glacial acetic acid**



MEME BREAK
*Sirke ka khatta taste asal mein 5–8% ethanoic acid hai? **Acha!?** Aur pura acid sardi mein barf ki tarah jam jaata hai!*

Esterification: acid + alcohol → ester

$$CH_3COOH + CH_3CH_2OH \xrightarrow[\text{warm}]{\text{Conc. } H_2SO_4} CH_3COOCH_2CH_3 + H_2O$$

ethyl ethanoate (ester) + water: C 4 = 4, H 10 = 10, O 3 = 3 ✔

- Esters are **sweet-smelling**: used in **perfumes** and as **flavouring agents**
- The reaction is **slow and reversible**; conc. H_2SO_4 speeds it up

Saponification: ester + base → back to alcohol

$$CH_3COOCH_2CH_3 + NaOH \rightarrow CH_3CH_2OH + CH_3COONa$$

ethanol + sodium ethanoate: C 4 = 4, H 9 = 9, O 3 = 3, Na 1 = 1 ✔

Saponification = the reaction used to **make soap** (Day 19).

Ethanoic acid with bases and carbonates

$$CH_3COOH_{(aq)} + NaOH_{(aq)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)}$$

neutralisation: sodium ethanoate (sodium acetate) + water ✔

$$2CH_3COOH_{(aq)} + Na_2CO_{3(s)} \rightarrow 2CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

C 5 = 5, H 8 = 8, O 7 = 7, Na 2 = 2 ✔

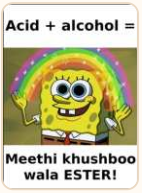
$$CH_3COOH_{(aq)} + NaHCO_{3(s)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

C 3 = 3, H 5 = 5, O 5 = 5, Na 1 = 1 ✔

Brisk effervescence; the CO_2 turns lime water milky. This is how you tell **ethanoic acid from ethanol**.

BUJHO *Acid + alcohol se ester bana, toh ye neutralisation hua na?*

PAHELI *Nahi! Neutralisation = acid + **base**. Acid + alcohol = **esterification**, conc. H_2SO_4 catalyst, meethi khushboo!*



MEME BREAK

*Chapter 1: toxic gas + toxic solid = namak. Aaj ka version: khatta acid + alcohol = **meethi khushboo wala ester!** ★★★ Esterification aur saponification pakke, Nitesh Bhaiya ki guarantee!*

7 EXAM TIPS AND COMMON MISTAKES

× WRONG	✓ RIGHT
Saturated → sooty flame	**Unsaturated** → sooty flame
Ethanol + Na → ethanoate	Sodium **ethoxide** + H_2
Hydrogenation catalyst: Fe	**Ni or Pd**
Ester + NaOH = esterification	**Saponification**
$CH_4 + Cl_2$ in the dark	Needs **sunlight**

Exam tips

- ✔ Always write the **condition over the arrow**: Ni, sunlight, conc. H_2SO_4 + 443 K, alkaline $KMnO_4$ + heat.
- ✔ Organic conversions (oxidation, addition, substitution, esterification) are written NCERT style without state symbols; the condition and product names carry the marks.
- ✔ "Distinguish ethanol and ethanoic acid": $NaHCO_3$ test (fizz with acid only) or blue litmus (red with acid only).
- ✔ Ester names: ethanoic acid + ethanol → **ethyl ethanoate**.

8 MIND MAP: REVISE IN 2 MINUTES

Day 18 mind map

COMBUSTION ★★★
$CO_2 + H_2O$ + heat; saturated blue flame, unsaturated sooty

OXIDATION
Ethanol → ethanoic acid; alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$

CHEMICAL PROPERTIES OF CARBON COMPOUNDS

ADDITION
Unsaturated + H_2, Ni or Pd; oil → vanaspati ghee

SUBSTITUTION
Saturated + Cl_2 in sunlight; CH_3Cl + HCl

ETHANOL AND ETHANOIC ACID
Ethanol + Na → ethoxide + H_2; conc. H_2SO_4, 443 K → ethene. Acid + alcohol → ester (esterification); ester + NaOH → alcohol + salt (saponification); acid + $NaHCO_3$ → CO_2 fizz

Quick quiz: name the reaction

(i) Oil → ghee (ii) $CH_4 \rightarrow CH_3Cl$ (iii) Ethanol → acid
(iv) Ester + NaOH (v) Ethanol → ethene (vi) LPG burning

Answers: (i) addition (ii) substitution (iii) oxidation (iv) saponification (v) dehydration (vi) combustion

Extra from Class: ethanol as a fuel

Molasses (from sugarcane juice) is fermented by **yeast**: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (enzyme zymase). Ethanol burns with a clean blue flame ($C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$), so it is blended with petrol as a cleaner fuel (India's **E20**; Brazil).

9 PRACTICE MCQS

Pehle khud socho! Answer key page 4 par hai.

1. Which of these burns with a yellow, sooty flame? ★★★
(a) Methane (b) Ethane
(c) Ethyne (d) Propane

2. The conversion of ethanol into ethanoic acid by alkaline $KMnO_4$ is an example of: ★★★
(a) addition (b) substitution
(c) oxidation (d) reduction

3. Which catalyst is used in the hydrogenation of vegetable oils to make vanaspati ghee? ★★★
(a) Iron (b) Nickel
(c) Conc. H_2SO_4 (d) Platinum oxide

4. $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ in sunlight is a: ★★☆
(a) addition reaction (b) substitution reaction
(c) combustion reaction (d) neutralisation

5. Sodium metal reacts with ethanol to give: ★★★
(a) sodium ethoxide + H_2 (b) sodium ethanoate + H_2
(c) sodium ethoxide + O_2 (d) sodium ethanoate + CO_2

6. Ethanol is heated at 443 K with excess hot conc. H_2SO_4. The product is: ★★☆
(a) ethane (b) ethene
(c) ethyne (d) ethanal

7. Vinegar is a solution of: ★★☆
(a) 50–60% ethanoic acid (b) 5–8% ethanoic acid
(c) 5–8% ethanol (d) pure ethanoic acid

8. Ethyl ethanoate is heated with NaOH solution to give ethanol and sodium ethanoate. This reaction is called: ★★★
(a) esterification (b) saponification
(c) hydrogenation (d) dehydration

9. **Assertion (A):** Unsaturated hydrocarbons burn in air with a yellow, sooty flame.
Reason (R): Unsaturated hydrocarbons have a higher share of carbon, so they do not burn completely in air. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Methane reacts readily with chlorine in the dark to form chloromethane.
Reason (R): Substitution of hydrogen by chlorine in saturated hydrocarbons takes place in the presence of sunlight. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. How would you distinguish experimentally between an alcohol and a carboxylic acid? (2 marks) NCERT
2. Why is the conversion of ethanol to ethanoic acid an oxidation reaction? Name two oxidising agents. (2 marks) NCERT
3. What is esterification? Describe an activity to show it, with the equation. What happens when the ester formed is heated with NaOH solution? Name this reaction. (5 marks)

Answer key: MCQs and try-yourself

Q	Ans	Why
1	(c)	Ethyne C_2H_2 is unsaturated (triple bond).
2	(c)	Oxygen is added: alcohol → acid.
3	(b)	Ni (or Pd) catalyses hydrogenation.
4	(b)	Cl replaces one H atom.
5	(a)	$2Na + 2C_2H_5OH \rightarrow 2C_2H_5ONa + H_2$
6	(b)	Dehydration gives ethene, $CH_2{=}CH_2$.
7	(b)	Vinegar = 5–8% ethanoic acid.
8	(b)	Ester + base → alcohol + salt.
9	(a)	More carbon, incomplete burning: soot.
10	(d)	No reaction in the dark; sunlight needed.
Try	(i) $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$ (ii) $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$	

Scored 8 or more? Move on. Below 6? Revise sections 1 to 6 once more, then retry.

Model answers: short and long answer

1 Alcohol vs carboxylic acid

- Add a pinch of **$NaHCO_3$** to each: brisk **effervescence** of CO_2 (turns lime water milky) only with the **acid**
- Blue litmus turns red only with the acid; the alcohol shows no change

$$CH_3COOH_{(aq)} + NaHCO_{3(s)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

2 Ethanol → ethanoic acid is oxidation

- Oxygen is **added** (and hydrogen removed): $CH_3CH_2OH \rightarrow CH_3COOH$
- Oxidising agents: **alkaline $KMnO_4$** and **acidified $K_2Cr_2O_7$**, on heating

3 Esterification and saponification

- **Esterification:** a carboxylic acid reacts with an alcohol in the presence of an acid catalyst to form a sweet-smelling **ester**
- **Activity:** 1 mL ethanol + 1 mL glacial acetic acid + a few drops of conc. H_2SO_4; warm in a water bath for 5 min; pour into 20–50 mL water: **fruity smell**

$$CH_3COOH + CH_3CH_2OH \xrightarrow[\text{warm}]{\text{Conc. } H_2SO_4} CH_3COOCH_2CH_3 + H_2O$$

- Heated with NaOH, the ester gives back the alcohol and the sodium salt of the acid: **saponification** (used to make soap)

$$CH_3COOCH_2CH_3 + NaOH \rightarrow CH_3CH_2OH + CH_3COONa$$

Day 18 hit-list: Eduspark PYQ analysis

Topic / reaction	Rating
Esterification + saponification equations	★★★
Ethanol → ethanoic acid (oxidising agents)	★★★
Hydrogenation of oils (Ni catalyst)	★★★
Clean vs sooty flame	★★★
Ethanol + Na; acid + $NaHCO_3$	★★☆
Substitution of methane in sunlight	★★☆

Day 18 checklist: tick before you sleep

- [ ] I can write combustion equations and explain clean vs sooty flames
- [ ] I can name the oxidising agents for ethanol → ethanoic acid
- [ ] I can explain hydrogenation (Ni or Pd) and substitution (sunlight)
- [ ] I can write ethanol + Na, dehydration, esterification and saponification
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho, phir hi sona.

LEVEL 18 COMPLETE +125 XP earned. Next: Level 19

BUJHO *Samajh gaya! Acid + alcohol se meetha ester, aur ester + NaOH se wapas alcohol!*

PAHELI *Shabaash! Kal Level 19: soaps and detergents.*

Coming up: Day 19

Soaps and detergents: how micelles trap oily dirt and why soap fails in hard water.

USES AT A GLANCE

Ethanol: medicines and fuel additive; ethanoic acid: vinegar and pickles; esters: perfumes and flavours; hydrogenation: vanaspati ghee; saponification: soap.

NITESH SIR SAYS

Har reaction ki condition arrow ke upar likho, wahi marks dilati hai. ***Kal milte hai Day 19 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each reaction or topic has come in Chapter 4 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 19 · LEVEL 19

CHEMISTRY | CHAPTER 4 | PART 5 · ASTRA SERIES

SOAPS & DETERGENTS

Goal: explain micelles, scum and soap vs detergent.

by Nitesh Prakash

01 Soap Structure | 02 Micelles | 03 Hard Water | 04 Soap vs Detergent | 05 Chapter Recap | +125 XP today | Level 19 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

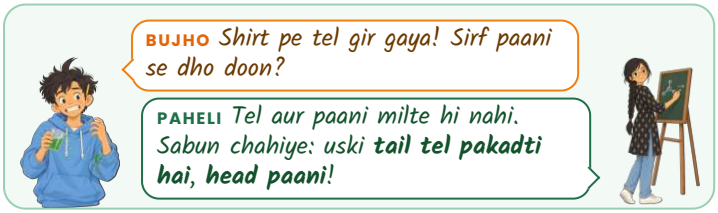
1 SOAPS AND DETERGENTS

DEFINITION

- **Soap:** the **sodium or potassium salt** of a **long-chain carboxylic acid** (fatty acid)
- Examples: sodium stearate $C_{17}H_{35}COO^-Na^+$, sodium palmitate $C_{15}H_{31}COO^-Na^+$
- **Detergent:** generally the **sodium salt of a sulphonic acid** (or an ammonium salt with chloride or bromide ions), also with a long hydrocarbon chain

Structure of a soap molecule ★★★

- **Tail** (long hydrocarbon chain): **hydrophobic**, water-hating, dissolves in oil and grease
- **Head** (ionic $-COO^-Na^+$ end): **hydrophilic**, water-loving, dissolves in water



Recap: how soap is made (saponification) ★★☆

$$(C_{17}H_{35}COO)_3C_3H_5(s) + 3NaOH(aq) \xrightarrow{\Delta} 3C_{17}H_{35}COONa(s) + C_3H_5(OH)_3(aq)$$

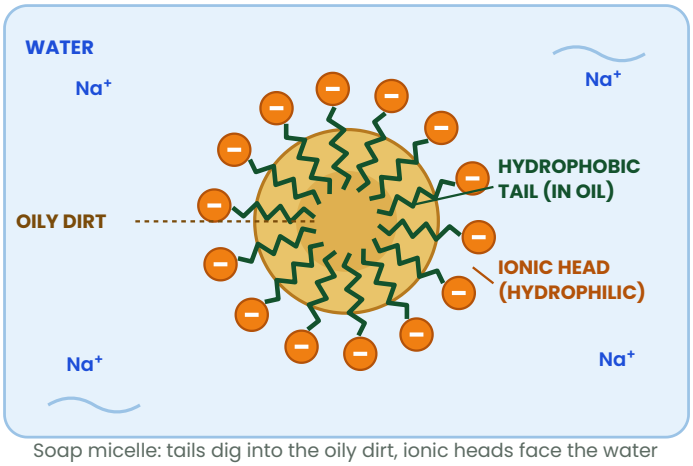
fat + caustic soda → soap + glycerol; C 57, H 113, O 9, Na 3 on each side ✔

Alkaline hydrolysis of an ester (fat or oil) with NaOH gives soap; KOH gives soft (potassium) soap.

MEME BREAK

Soap molecule to oil and water both: "Humare liye kaam karoge?" Tail bole tel, head bole paani. ***Dono se dosti!***

2 MICELLES AND CLEANSING ACTION ★★★ MOST ASKED



Soap micelle: tails dig into the oily dirt, ionic heads face the water

DEFINITION

- **Micelle:** a **spherical cluster** of soap molecules in water, with **tails inside** and **ionic heads outside**

How soap cleans: 4 steps

1. In water, soap molecules arrange so the **tails stay away from water** and the heads stay in it.
2. The tails dissolve in the **oily dirt**; the dirt is trapped at the **centre of the micelle**.
3. The ionic heads on the outside **repel each other**, so micelles stay suspended (an emulsion) and do not come together.
4. Rinsing, scrubbing or beating the cloth **washes the micelles away** with the dirt.

★★★ *Micelle ka labelled diagram + cleansing action pakka aayega: Nitesh Bhaiya ki guarantee!*

Why does soap solution look cloudy? ★★☆

- Micelles are **large enough to scatter light**, so a soap solution looks cloudy (it is a colloid)
- In **ethanol**, soap dissolves completely (the tail is soluble in ethanol), so **no micelles form**

3 HARD WATER AND SCUM ★★★ MOST ASKED

SOFT WATER	HARD WATER
Few or no Ca^{2+}/Mg^{2+} salts	Contains **calcium and magnesium salts** (hydrogencarbonates, chlorides, sulphates)
Rain water, distilled water	Well water, hand-pump water
✔ Soap gives plenty of **lather**	× Little lather; white curdy **scum** forms

Scum: the soap-wasting precipitate ★★★

- **Scum** = insoluble **calcium or magnesium salts of fatty acids**, formed when soap reacts with Ca^{2+} and Mg^{2+}

$$2C_{17}H_{35}COONa_{(aq)} + CaCl_{2(aq)} \rightarrow (C_{17}H_{35}COO)_2Ca_{(s)}\downarrow + 2NaCl_{(aq)}$$

scum (calcium stearate). C 36 = 36, H 70 = 70, O 4 = 4, Na 2 = 2, Ca 1 = 1, Cl 2 = 2 ✔

$$2C_{17}H_{35}COONa_{(aq)} + MgSO_{4(aq)} \rightarrow (C_{17}H_{35}COO)_2Mg_{(s)}\downarrow + Na_2SO_{4(aq)}$$

magnesium salts of hard water give scum the same way ✔

- Soap is **wasted** (it reacts first, cleans later); scum **sticks to clothes**, buckets and tiles

MEME BREAK
Soap to the Ca^{2+} and Mg^{2+} of hard water: "Kya tapleek hai aapko? Mera saara jhaag kha gaye!"

Activity: Test for hard water ★★☆

1. Take 10 mL distilled (soft) water in test tube A and 10 mL well or hand-pump (hard) water in test tube B.
2. Add 5 drops of **soap solution** to each and shake for the same time.
3. A: **lots of lather**. B: little lather, **curdy white scum**. Soap needed for lather in B is much more.
4. Repeat with **detergent**: lather forms in **both**. So a detergent cannot tell hard water from soft.

Result: to check hardness, always use **soap**, not detergent.

Exam tips: score full marks ★★★

1. Micelle diagram: draw **tails inside, heads outside** and write all **4 labels**.
2. Write "**insoluble** calcium or magnesium salts" for scum; the word insoluble carries the mark.
3. Name the ions: Ca^{2+} and Mg^{2+}, not "salts" alone.
4. In soap vs detergent, give **ionic group** and **hard water** points first.
5. Soap solution is **slightly basic** (salt of a weak acid and a strong base): red litmus turns blue.

4 DETERGENTS: WHY THEY WIN IN HARD WATER

- A detergent also has a **long hydrocarbon tail** and an **ionic head**, such as $-SO_3^-Na^+$
- Its charged end does **not form insoluble salts** with Ca^{2+} and Mg^{2+}, so **no scum** forms
- So detergents **lather well even in hard water**; they clean by the same **micelle** action
- Uses: washing powders, **shampoos**, products for cleaning clothes and dishes

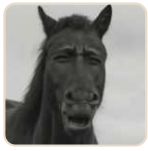
BUJHO *Aachoo! Itna detergent kyun? Sabun kaafi nahi?*

PAHELI *Hand-pump ka paani hard hai: sabun scum banata hai. **Detergent ko calcium-magnesium se farak nahi padta!***

5 SOAP VS DETERGENT ★★★

Point	Soap	Detergent
Chemical nature	Na/K salt of a long-chain **carboxylic** (fatty) acid	Na salt of a long-chain **sulphonic** acid (or ammonium salt)
Ionic group	$-COO^-Na^+$	$-SO_3^-Na^+$
Hard water	× Forms scum, little lather	✔ No scum, lathers well
Made from	Vegetable oils, animal fats	Petroleum hydrocarbons
Cleansing action	Relatively weaker	Stronger
Biodegradable?	✔ Yes	Some are **not**; foam pollutes rivers and ponds

Detergent, jab sabun hard water mein scum banata hai: Ha ha ha! Mera to jhaag ban raha hai!



Spot the type: soap or detergent?

Formula	Type
$C_{17}H_{35}COO^-Na^+$ (sodium stearate)	**Soap**
$C_{15}H_{31}COO^-K^+$ (potassium palmitate)	**Soap**
$C_{12}H_{25}C_6H_4SO_3^-Na^+$ (sodium dodecylbenzene sulphonate)	**Detergent**

Rule: $-COO^-$ head means soap; $-SO_3^-$ head means detergent.

Link to Chapter 2: softening hard water

- **Washing soda** (sodium carbonate) removes **permanent hardness**: it precipitates Ca^{2+} and Mg^{2+} as insoluble carbonates
- After softening, soap lathers easily again

Real life: where you meet them

- Bathing soap and laundry bar: **soap**; shampoo, washing powder, dishwash liquid: **detergents**

6 CHAPTER 4 QUICK RECAP

Chapter 4 recap mind map: revise in 3 minutes

CARBON AND ITS COMPOUNDS

BONDING (DAY 15)
C = 6 (2, 4): tetravalent; forms **covalent bonds** by sharing; H–H, O=O, N≡N; CH_4 four single bonds; O=C=O

ALLOTROPES
Diamond: each C to 4 C, hardest, non-conductor. Graphite: 3 C in layers, conductor. Fullerene C-60

VERSATILE CARBON
Catenation (long chains, branches, rings) + **tetravalency** → millions of compounds

HYDROCARBONS (DAY 16)
Alkane C_nH_{2n+2}, alkene C_nH_{2n}, alkyne C_nH_{2n-2}; benzene C_6H_6, cyclohexane C_6H_{12}

ISOMERS AND SERIES
Butane 2 isomers, pentane 3. Homologous series: next member + CH_2 = 14 u

FUNCTIONAL GROUPS (DAY 17)
–Cl/–Br halo; –OH -ol; –CHO -al; >C=O -one; –COOH -oic acid; C=C -ene; C≡C -yne

REACTIONS (DAY 18)
Combustion; oxidation (alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$); addition (Ni or Pd); substitution (Cl_2, sunlight)

ETHANOL AND ETHANOIC ACID
Na gives H_2; hot concentrated H_2SO_4 (443 K) gives ethene; vinegar is 5–8% ethanoic acid; esterification and saponification

SOAPS AND DETERGENTS (DAY 19)
Ionic head + hydrocarbon tail; **micelle** traps oily dirt; soap + Ca^{2+}/Mg^{2+} → scum; detergents work in hard water

Must-know equations of Chapter 4

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) + \text{Heat}$$
combustion: C 1 = 1, H 4 = 4, O 4 = 4 ✔

$$2Na(s) + 2CH_3CH_2OH(l) \rightarrow 2CH_3CH_2ONa(aq) + H_2(g)\uparrow$$
ethanol + sodium: Na 2 = 2, C 4 = 4, H 12 = 12, O 2 = 2 ✔

$$CH_3COOH(l) + C_2H_5OH(l) \xrightarrow{\text{acid catalyst}} CH_3COOC_2H_5(l) + H_2O(l)$$
esterification

$$CH_3CH_2OH(l) + 2[O] \xrightarrow{\text{alkaline } KMnO_4 + \text{heat}} CH_3COOH(l) + H_2O(l)$$
oxidation: C 2 = 2, H 6 = 6, O 3 = 3 ✔

$$CH_3COOH(aq) + NaHCO_3(s) \rightarrow CH_3COONa(aq) + H_2O(l) + CO_2(g)\uparrow$$
C 3 = 3, H 5 = 5, O 5 = 5, Na 1 = 1 ✔

7 COMMON MISTAKES

× WRONG	✔ RIGHT
Soap = salt of a sulphonic acid	Soap is a carboxylate salt
The ionic head dissolves in oil	Tail in oil, ionic head in water
Use detergent to test hard water	Use soap, not detergent
All detergents are eco-friendly	Some are non-biodegradable
Micelles clump and settle	They repel, stay suspended

8 PRACTICE MCQS

Mr Bean jaisa focus! Khud solve karo, answer key page 4 par hai.

1. Soaps are chemically: ★★★
(a) fatty acid salts (b) sulphonic acid salts
(c) glycerol esters (d) long-chain alcohols

2. In a soap micelle, the hydrocarbon tails point: ★★★
(a) towards the water outside
(b) towards the oily dirt at the centre
(c) in random directions
(d) towards the air above

3. Soap does not lather well in hard water because it contains: ★★★
(a) Na^+ and K^+ ions
(b) Ca^{2+} and Mg^{2+} ions
(c) Cl^- ions only
(d) dissolved oxygen

4. The scum formed with hard water is: ★★☆
(a) sodium chloride
(b) glycerol
(c) Ca or Mg salt of a fatty acid
(d) a detergent

5. Which substance would you use to check whether a water sample is hard? ★★☆
(a) detergent (b) soap
(c) ethanol (d) common salt

6. A soap solution looks cloudy because: ★☆☆
(a) soap is insoluble in water (b) micelles scatter light
(c) it contains scum (d) it is acidic in nature

7. The hydrophilic end of a soap molecule is: ★★☆
(a) $–COO^-Na^+$ (b) $–SO_3^-Na^+$
(c) $–CH_3$ (d) $–C_{17}H_{35}$

8. Soap forms no micelles in ethanol because: ★★☆
(a) soap reacts with ethanol to form an ester
(b) soap dissolves completely in ethanol
(c) ethanol contains Ca^{2+} ions
(d) ethanol is an ionic liquid

9. **Assertion (A):** Detergents are better than soaps for washing clothes in hard water.
Reason (R): The calcium and magnesium salts of detergents are soluble in water. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Soap removes oily dirt from clothes.
Reason (R): The ionic end of soap dissolves in oil, and the hydrocarbon end dissolves in water. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Explain the formation of scum when hard water is treated with soap. Write a balanced equation. (2 marks) NCERT
2. Would you be able to check if water is hard by using a detergent? Give a reason. (2 marks) NCERT
3. Why does micelle formation take place when soap is added to water? Will a micelle be formed in ethanol also? (3 marks) NCERT
4. Draw a labelled diagram of a soap micelle and explain the cleansing action of soap. Why are detergents preferred in hard water? (5 marks) ★★★

Answer key: MCQs

Q	Ans	Why
1	(a)	Soap = Na/K salt of a fatty acid.
2	(b)	Tails are hydrophobic and dissolve in oil.
3	(b)	Ca^{2+}, Mg^{2+} make hard water.
4	(c)	Insoluble Ca or Mg stearate, for example.
5	(b)	Detergent lathers in both waters.
6	(b)	Micelles are big enough to scatter light.
7	(a)	Carboxylate head; (b) is a detergent head.
8	(b)	Tail dissolves in ethanol, so no clustering.
9	(a)	Soluble Ca/Mg salts, so no scum.
10	(c)	R is reversed: tail in oil, head in water.

Model answers

1 Scum formation

- Hard water has Ca^{2+} and Mg^{2+} ions; soap reacts with them to give an **insoluble, curdy precipitate** (scum)
- Lather forms only after these ions are used up, so soap is wasted

$$2C_{17}H_{35}COONa_{(aq)} + CaCl_{2(aq)} \rightarrow (C_{17}H_{35}COO)_2Ca_{(s)}\downarrow + 2NaCl_{(aq)}$$

2 Detergent and hard water

- **No.** A detergent lathers well in both hard and soft water, because its Ca and Mg salts are soluble; only soap shows the difference

3 Why micelles form

- The hydrocarbon tail is **insoluble in water**, so the molecules cluster with tails inside and ionic heads facing the water
- **Not in ethanol:** the tail is soluble in ethanol, so soap simply dissolves

4 Cleansing action (5 marks)

- Draw the micelle from page 1 with all 4 labels: oily dirt, hydrophobic tail, ionic head, water
- Tails trap oily dirt at the centre, heads face water; rinsing removes the micelles with the dirt
- Detergents form no scum with Ca^{2+}/Mg^{2+}, so they clean well in hard water

SUGGESTED MARKING FOR Q4

Diagram with 4 labels: 2 marks; micelle formation: 1; removal of trapped dirt: 1; detergents in hard water: 1.

Day 19 checklist: tick before you sleep

- ☐ I can draw a soap molecule and label its head and tail
- ☐ I can draw and label a micelle and explain how it removes oily dirt
- ☐ I can write the scum equation and explain why soap fails in hard water
- ☐ I can give four differences between soaps and detergents
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Kyun nahi ho rahi padhai? Page 1–2 phir se padho.

LEVEL 19 COMPLETE +125 XP earned. Next: Level 20

BUJHO *Samajh gaya! Tail tel pakadti hai, head paani; aur hard water mein detergent hi chalega!*

PAHELI *Shabaash! Kal Level 20 hai: Boss Level aur poora Chemistry mock test.*

Chapter 4 hit-list: Eduspark PYQ analysis

Topic	Day	Rating
Electron-dot structures (CH_4, CO_2, N_2)	15	★★★
Homologous series and isomers	16	★★☆
Functional groups and IUPAC names	17	★★★
Ethanol and ethanoic acid reactions	18	★★★
Micelles and cleansing action	19	★★★
Soap vs detergent; hard water	19	★★★
Allotropes of carbon	15	★★☆

Coming up: Day 20

Boss Level for Chapter 4 plus a full Chemistry mock test covering Days 1–20, with complete answers. Revise today's mind map and the must-know equations first!

NITESH SIR SAYS

*Micelle ka diagram teen baar banao, exam mein haath khud chalega. **Consistency hi jalwa hai!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 4 board and sample-paper questions. NCERT tags mark NCERT in-text questions.

DAY 20
LEVEL 20

CHEMISTRY | CHAPTER 4 | BOSS LEVEL + MOCK TEST

BOSS LEVEL + CHEMISTRY MOCK TEST

Goal: clear Chapter 4, then test all four chemistry chapters.

by Nitesh Prakash

01 Boss Level | 02 Mock Test | 03 Answer Key | 04 Marking Points | 05 Checklist | +200 XP today | Level 20 / 60

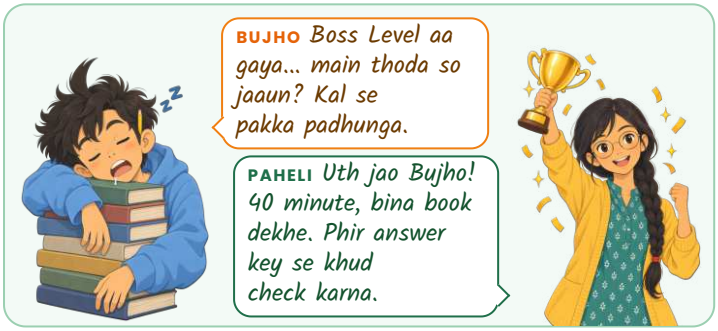
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 4 BOSS LEVEL

Time: 40 min | **Max. marks:** 24 | **Questions:** 14
Q1–10: 1 mark each; Q11–13: 3 marks each; Q14: 5 marks

TIME PLAN FOR 40 MINUTES

Q1–10 in 10 min, Q11–13 in 15 min, Q14 in 10 min, then 5 min to re-check equations and state symbols.



SECTION A: OBJECTIVE — 10 × 1 = 10 marks

1. The total number of covalent bonds in a molecule of propane, C_3H_8, is: ★★★
(a) 8 (b) 10 (c) 11 (d) 13

2. Which belongs to the same homologous series as C_3H_6? ★★★
(a) C_4H_8 (b) C_3H_8 (c) C_2H_2 (d) C_4H_{10}

3. The functional group present in propanal is: ★★☆
(a) –OH (b) –CHO (c) –COOH (d) >C=O

4. The number of electron pairs shared between the two atoms in a nitrogen molecule, N_2, is: ★★☆
(a) 1 (b) 2 (c) 3 (d) 6

5. Which of these is an unsaturated hydrocarbon? ★★☆
(a) C_2H_6 (b) C_3H_8 (c) C_3H_4 (d) C_4H_{10}

6. Ethanol reacts with sodium metal to form: ★★★
(a) sodium ethanoate + H_2 (b) sodium ethoxide + H_2
(c) sodium ethoxide + O_2 (d) sodium ethanoate + O_2

7. Vegetable oils are converted into vanaspati ghee by: ★★★
(a) oxidation (b) substitution
(c) hydrogenation (Ni catalyst) (d) saponification

8. In a soap micelle, the hydrocarbon tails point: ★★☆
(a) outwards, into the water (b) inwards, into the oily dirt
(c) in random directions (d) along the water surface

ASSERTION–REASON OPTIONS: Q9, Q10

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

9. **Assertion (A):** Diamond is a non-conductor of electricity, whereas graphite is a good conductor.
Reason (R): In diamond each carbon atom is bonded to four other carbon atoms, while in graphite each carbon atom is bonded to only three, leaving electrons free to move. ★★☆

10. **Assertion (A):** Detergents can be used for washing clothes even in hard water.
Reason (R): Detergents are sodium salts of long-chain carboxylic acids. ★★☆

SECTION B: SHORT ANSWER — 3 × 3 = 9 marks

11. (a) Why can carbon form neither C^{4+} nor C^{4-} ions? (b) Name the two properties of carbon that explain the huge number of carbon compounds. [3] ★★★

12. Write the IUPAC name and the functional group of:
(i) CH_3CH_2Br (ii) CH_3COCH_3 (iii) CH_3CH_2CHO [3] ★★★

13. How will you distinguish between ethanol and ethanoic acid by a chemical test? Write the balanced equation of the reaction. [3] ★★☆

SECTION C: LONG ANSWER — 1 × 5 = 5 marks

14. An organic compound **A** (C_2H_6O) is used as a solvent and in cough syrups. Heated with excess hot concentrated H_2SO_4 at 443 K, it gives a gas **B** that decolourises bromine water. Oxidised with alkaline $KMnO_4$, **A** gives **C**, which reacts with **A** in the presence of an acid to give a sweet-smelling compound **D**. Identify A, B, C and D and write the three equations. Name the reaction that forms D, and state what D gives when heated with sodium hydroxide. [5] ★★★

EXAM RULE

Mr Bean wali copy-paste nahi chalegi! Pehle khud likho, answer key **page 3** *par hai.*

2 CHEMISTRY MOCK TEST: DAYS 1-20

Time: 75 min | **Max. marks:** 40 | **Questions:** 22, all compulsory
Marks per question: A (MCQ) 1, B (very short) 2, C (short) 3, D (long) 5, E (case-based) 4

TIME PLAN FOR 75 MINUTES

Section A 15 min, B 12 min, C 15 min, D 10 min, E 8 min; keep the last 15 min for checking.

Timer lagao, phone door rakho, aur real board exam ki tarah ek baar mein poora paper likho!

SECTION A: MULTIPLE CHOICE — 12 × 1 = 12 marks

1. In the reaction $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$, the substance oxidised is: ★★★

(a) Fe_2O_3 (b) Al (c) Al_2O_3 (d) Fe

2. The brown fumes given off when lead nitrate is heated are of: ★★★

(a) O_2 (b) NO (c) NO_2 (d) N_2O

3. Which of the following is an endothermic process? ★★☆

(a) Respiration (b) Burning natural gas
(c) Heating $CaCO_3$ (d) Quicklime + water

4. A solution turns red litmus blue. Its pH is most likely: ★★☆

(a) 1 (b) 4 (c) 5 (d) 10

5. The chemical formula of Plaster of Paris is: ★★★

(a) $CaSO_4 \cdot 2H_2O$ (b) $CaSO_4 \cdot \frac{1}{2}H_2O$
(c) $CaSO_4$ (d) $CaCO_3$

6. Tooth decay starts when the pH of the mouth falls below: ★★☆

(a) 7.4 (b) 6.5 (c) 5.5 (d) 8.5

7. Which of these metals is the best conductor of heat? ★☆☆

(a) Lead (b) Mercury (c) Silver (d) Iron

8. Which element forms an amphoteric oxide? ★★☆

(a) Sodium (b) Magnesium (c) Aluminium (d) Sulphur

9. The next higher homologue of propanol, C_3H_7OH, is: ★★☆

(a) C_2H_5OH (b) C_4H_9OH (c) C_3H_7CHO (d) C_4H_{10}

10. Which of these burns with a yellow, sooty flame? ★★☆

(a) Methane (b) Ethane
(c) Naphthalene (d) Ethanol

ASSERTION–REASON OPTIONS: Q11, Q12

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

11. **Assertion (A):** Dry HCl gas does not change the colour of dry blue litmus paper.
Reason (R): HCl gas produces H^+ ions only in the presence of water. ★★★

12. **Assertion (A):** Copper displaces zinc from zinc sulphate solution.
Reason (R): Zinc is more reactive than copper. ★★★

SECTION B: VERY SHORT ANSWER — 5 × 2 = 10 marks

13. Balance: (a) $Al + CuCl_2 \rightarrow AlCl_3 + Cu$ (b) $BaCl_2 + Al_2(SO_4)_3 \rightarrow BaSO_4 + AlCl_3$ [2] ★★★

14. Write the chemical name, formula and one use of: (a) bleaching powder (b) baking soda. [2] ★★★

15. (a) Why is sodium kept immersed in kerosene? (b) Why do ionic compounds conduct electricity in the molten state but not in the solid state? [2] ★★☆

16. What is a homologous series? Write the general formula of alkenes and name the second member. [2] ★★☆

17. Chips packets are flushed with nitrogen gas. Why? Name the process that this prevents. [2] ★★☆

SECTION C: SHORT ANSWER — 3 × 3 = 9 marks

18. Write balanced equations for: (a) the chlor-alkali process (b) the preparation of bleaching powder (c) the preparation of Plaster of Paris from gypsum. [3] ★★★

19. Zinc is extracted from zinc blende (ZnS) and from zinc carbonate. Name the process used for each ore to get zinc oxide and write all equations, including the reduction of zinc oxide with carbon. [3] ★★★

20. Explain the cleansing action of soap with the help of micelles. Why does soap not form lather with hard water? [3] ★★★

SECTION D: LONG ANSWER — 1 × 5 = 5 marks

21. Write a balanced equation with state symbols and name the type of reaction for: (a) water added to quicklime (b) heating ferrous sulphate crystals (c) an iron nail dipped in copper sulphate solution (d) sodium sulphate solution mixed with barium chloride solution (e) hydrogen passed over heated copper(II) oxide; also name the substance oxidised and the substance reduced in (e). [5] ★★★

SECTION E: CASE-BASED — 1 × 4 = 4 marks

22. Riya added equal-sized pieces of four metals P, Q, R and S separately to dilute hydrochloric acid and recorded her observations. [4] ★★★

Metal	Observation with dilute HCl
P	Brisk effervescence; test tube becomes warm
Q	A few bubbles form slowly
R	No change at all
S	Moderate, steady effervescence

- (i) Arrange P, Q, R and S in decreasing order of reactivity. [1]
- (ii) Which metal could be copper? Give a reason. [1]
- (iii) P is magnesium. Write the balanced equation for its reaction with dilute HCl and give the test for the gas evolved. [2]

END OF QUESTION PAPER

Stop the timer. Check every answer with the answer key and marking points on pages 3 and 4, then fill in the score tracker.

3 ANSWER KEY AND MARKING POINTS

Boss Level: MCQ answers

Q	Ans	Why
1	(b)	2 C–C bonds + 8 C–H bonds = 10
2	(a)	Both are alkenes, C_nH_{2n}; they differ by CH_2
3	(b)	Aldehyde group –CHO; suffix "-al"
4	(c)	N≡N triple bond = 3 shared pairs
5	(c)	C_3H_4 is propyne, C_nH_{2n-2} (triple bond)
6	(b)	$2Na + 2C_2H_5OH \rightarrow 2C_2H_5ONa + H_2$
7	(c)	Addition of hydrogen with Ni or Pd catalyst
8	(b)	Tails dissolve in oil; ionic heads face the water
9	(a)	Free electrons in graphite carry the current
10	(c)	R is false: detergents are sulphonate or ammonium salts; the sodium salts of carboxylic acids are soaps

Boss Level: model answers, Q11–12

11 Why carbon shares electrons (1 + 1 + 1)

- C^{4-}: a nucleus with 6 protons cannot hold 10 electrons
- C^{4+}: removing 4 electrons needs a very large amount of energy; so carbon **shares** electrons and forms covalent bonds
- **(b) Catenation** (long chains, branches, rings) and **tetravalency** (bonds with O, H, N, S, Cl and others)

12 IUPAC names (1 each)

Formula	IUPAC name	Functional group
CH_3CH_2Br	Bromoethane	Halo (–Br)
CH_3COCH_3	Propanone	Ketone (>C=O)
CH_3CH_2CHO	Propanal	Aldehyde (–CHO)

Boss Level: model answer, Q13

13 Test: 1, observation: 1, equation: 1

- Add sodium hydrogencarbonate to both: **ethanoic acid** gives brisk effervescence; the gas turns lime water milky (CO_2)
- **Ethanol:** no effervescence. (Ethanoic acid also turns blue litmus red.)

$CH_3COOH_{(aq)} + NaHCO_{3(s)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$

Boss Level: model answer, Q14 (2 + 2 + 1)

- **A** = ethanol, CH_3CH_2OH; **B** = ethene, $CH_2=CH_2$
- **C** = ethanoic acid, CH_3COOH; **D** = ethyl ethanoate, $CH_3COOC_2H_5$

$CH_3CH_2OH_{(l)} \xrightarrow[443\ K]{\text{Hot conc. } H_2SO_4} CH_2=CH_{2(g)} + H_2O_{(l)}$

Dehydration; B decolourises bromine water

$CH_3CH_2OH_{(aq)} + 2[O] \xrightarrow[\text{Heat}]{\text{Alkaline } KMnO_4} CH_3COOH_{(aq)} + H_2O_{(l)}$

Oxidation

$CH_3COOH_{(l)} + CH_3CH_2OH_{(l)} \xrightarrow{\text{Acid}} CH_3COOC_2H_{5(l)} + H_2O_{(l)}$

Esterification: D is the sweet-smelling ester

$CH_3COOC_2H_{5(l)} + NaOH_{(aq)} \rightarrow C_2H_5OH_{(aq)} + CH_3COONa_{(aq)}$

Saponification: D + NaOH gives ethanol and sodium ethanoate

Mock Test: Section A answers

Q	Ans	Why
1	(b)	Al gains oxygen, so Al is oxidised; Fe_2O_3 is reduced
2	(c)	$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
3	(c)	$CaCO_3 \rightarrow CaO + CO_2$ needs heat
4	(d)	Bases turn red litmus blue; pH above 7
5	(b)	$CaSO_4 \cdot ½H_2O$
6	(c)	Below pH 5.5 the tooth enamel corrodes
7	(c)	Silver and copper are the best; lead and mercury are poor
8	(c)	Al_2O_3 reacts with both acids and bases
9	(b)	Add CH_2: C_4H_9OH, butanol
10	(c)	Unsaturated and aromatic compounds give a sooty flame
11	(a)	Without water, HCl gives no H^+ ions
12	(d)	A is false: Cu is less reactive than Zn; R is true

Mock Test: Section B, Q13–14

13 Balanced equations (1 + 1)

$2Al_{(s)} + 3CuCl_{2(aq)} \rightarrow 2AlCl_{3(aq)} + 3Cu_{(s)}$

Al 2 = 2, Cu 3 = 3, Cl 6 = 6 ✔

$3BaCl_{2(aq)} + Al_2(SO_4)_{3(aq)} \rightarrow 3BaSO_{4(s)}\downarrow + 2AlCl_{3(aq)}$

Ba 3 = 3, Cl 6 = 6, Al 2 = 2, S 3 = 3, O 12 = 12 ✔

14 Bleaching powder and baking soda (½ each)

- **(a)** Calcium oxychloride, $CaOCl_2$: disinfecting drinking water, bleaching cotton and linen
- **(b)** Sodium hydrogencarbonate, $NaHCO_3$: antacid, baking powder, soda-acid fire extinguisher

Mock Test: Section B, Q15–17

15 Sodium and ionic compounds (1 + 1)

- **(a)** Sodium reacts so vigorously with oxygen and moisture that it catches fire in air; kerosene keeps air away
- **(b)** In the solid, ions are held in a rigid lattice and cannot move; in the melt, ions move freely and carry charge

16 Homologous series (1 + 1)

- Compounds with the **same functional group** and similar chemical properties; successive members differ by CH_2 (14 u)
- Alkenes: C_nH_{2n}; second member: **propene**, $CH_3CH=CH_2$

17 Nitrogen flushing (1 + 1)

- Nitrogen is unreactive and replaces oxygen, so the fats and oils in chips are not oxidised
- It prevents **rancidity** (bad smell and taste)

Mock Test: Section C, Q18

18 Chemicals from common salt (1 each)

$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$

(a) Chlor-alkali: Cl_2 at the anode, H_2 at the cathode

$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$

(b) Bleaching powder

$CaSO_4 \cdot 2H_2O_{(s)} \xrightarrow{373\ K} CaSO_4 \cdot ½H_2O_{(s)} + 1½H_2O_{(g)}$

(c) Plaster of Paris from gypsum

Mock Test: Section C, Q19

19 Extraction of zinc (1 each)

$2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{Heat} 2ZnO_{(s)} + 2SO_{2(g)}$

Roasting: sulphide ore heated strongly in excess air

$ZnCO_{3(s)} \xrightarrow{Heat} ZnO_{(s)} + CO_{2(g)}$

Calcination: carbonate ore heated strongly in limited air

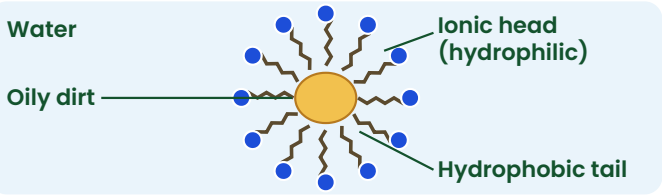
$ZnO_{(s)} + C_{(s)} \xrightarrow{Heat} Zn_{(s)} + CO_{(g)}$

Reduction of zinc oxide with carbon

Mock Test: Section C, Q20

20 Cleansing action of soap (2 + 1)

- A soap molecule has an **ionic (hydrophilic) head** that dissolves in water and a long **hydrocarbon (hydrophobic) tail** that dissolves in oil
- In water the molecules form **micelles**: tails point inwards and trap the oily dirt, heads face the water outside
- The micelles stay suspended and the dirt is rinsed away with the water
- **Hard water:** Ca^{2+} and Mg^{2+} ions react with soap to form an insoluble **scum**, so soap is wasted and little lather forms



Mock Test: Section D marking points (1 each)

$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \text{Heat}$

(a) Combination; exothermic

$2FeSO_{4(s)} \xrightarrow{Heat} Fe_2O_{3(s)} + SO_{2(g)} + SO_{3(g)}$

(b) Thermal decomposition; green crystals turn reddish-brown

$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$

(c) Displacement; blue solution turns light green

$Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$

(d) Double displacement; white precipitate

$CuO_{(s)} + H_{2(g)} \xrightarrow{Heat} Cu_{(s)} + H_2O_{(g)}$

(e) Redox: H_2 is oxidised and CuO is reduced

Mock Test: Section E marking points

- **(i)** P > S > Q > R (1)
- **(ii)** R; copper is below hydrogen in the reactivity series, so it cannot displace hydrogen from dilute acid (1)
- **(iii)** Bring a burning splinter near the mouth of the test tube: the gas burns with a pop sound, so it is hydrogen (1 + 1)

$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}\uparrow$

★★★ *Balancing, redox, pH, reactivity series, esterification: ye sab pakka aayenge, Nitesh Bhaiya ki guarantee hai!*

Find your weak chapter: score tracker

Chapter	Questions	Max.	Mine	Revise
1. Chemical reactions and equations	1, 2, 3, 13, 17, 21	12	___	Days 1–4
2. Acids, bases and salts	4, 5, 6, 11, 14, 18	9	___	Days 5–9
3. Metals and non-metals	7, 8, 12, 15, 19, 22	12	___	Days 10–14
4. Carbon and its compounds	9, 10, 16, 20	7	___	Days 15–19
Mock test total	22 questions	**40**	___	
Chapter 4 Boss Level	14 questions	**24**	___	Days 15–19

How to use the marking points

- Give yourself marks **only** for points that match the key; partial marks are fine
- A balanced equation without state symbols loses ½ mark in many papers: always add them
- Write every wrong question in a "mistake notebook" and re-solve it after 3 days

Day 20 checklist: tick before you sleep

- ☐ I finished the Chapter 4 Boss Level (24 marks) in 40 minutes
- ☐ I wrote the full 40-mark mock test in one sitting of 75 minutes
- ☐ I checked every answer against the key and marking points
- ☐ I found my weak chapter from the marks table above
- ☐ I re-read the Days of that chapter and corrected my mistakes

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Weak chapter ke Days phir se padho.

LEVEL 20 COMPLETE

+200 XP earned. Chemistry cleared. Next: Level 21

BUJHO *Samajh gaya! Chemistry ke chaaron chapter clear, Boss bhi haar gaya!*

PAHELI *Shabaash! Ab Physics ki duniya mein milte hain.*

NITESH SIR SAYS

*Mock ke marks kam aaye toh tension mat lo. **Galtiyon ki list hi asli revision list hai!** Chemistry complete, ab Physics ki taiyari.*

CHEMISTRY | CHAPTER 4 COMPLETE | CHEMISTRY COMPLETE

CHAPTER 4 SWAHAA!

Carbon and its Compounds | Days 15–20

BUJHO *Covalent bond, isomers, soap ka micelle, mock test bhi! Chapter 4 Swahaa, poori Chemistry Swahaa!*

PAHELI *Trophy tumhari hai, Bujho! 20 levels clear. Ab Physics ki taraf chalte hain.*

CHAPTER XP EARNED
+750 XP

LEVELS CLEARED
6 levels (Days 15–20)

BOOK PROGRESS
Level 20 / 60 2500 XP total

WHAT YOU HAVE MASTERED

- ✔ Covalent bonding, tetravalency, catenation
- ✔ Functional groups and IUPAC names
- ✔ Ethanol, ethanoic acid, esterification
- ✔ Hydrocarbons, isomers, homologous series
- ✔ Combustion, oxidation, addition, substitution
- ✔ Soaps, detergents and micelles

NEXT CHAPTER Day 21: Physics begins

Chemistry cleared!

DAY 21 · LEVEL 21

PHYSICS | CHAPTER 5 | PART 1 · ASTRA SERIES

LIGHT & REFLECTION

Goal: master reflection and the terms of spherical mirrors.

by Nitesh Prakash

01 Light | 02 Reflection Laws | 03 Plane Mirror | 04 Spherical Mirrors | 05 Mirror Terms | +125 XP today | Level 21 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHAT IS LIGHT?

DEFINITION

- **Light** is a form of **energy** that produces the **sensation of vision** in our eyes.
- It is an **electromagnetic wave**, so it needs **no medium**: it travels even through vacuum.
- Speed of light in vacuum (or air): $c = 3 \times 10^8$ **m/s**, the fastest speed known.

How light behaves around us

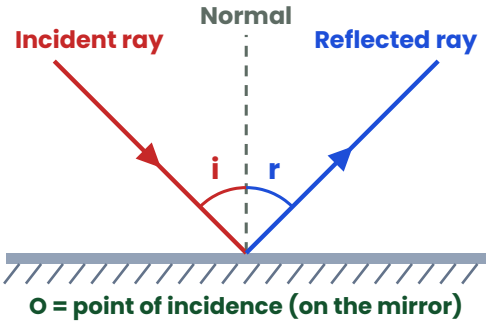
- ✔ **Travels in straight lines** (rectilinear propagation). Proof: an opaque object casts a **sharp shadow**.
- ✔ We see an object only when light from it (its own or reflected) **enters our eyes**.
- ✔ Shiny, polished surfaces such as a mirror **reflect most of the light** falling on them.
- ✔ Sunlight takes about **8 minutes** to reach the Earth.

2 REFLECTION OF LIGHT ★★★ MOST ASKED

DEFINITION

- **Reflection:** the bouncing back of light into the **same medium** when it falls on a polished surface.

The terms in one figure



- **Incident ray:** the ray that falls on the surface; **reflected ray:** the ray that bounces back
- **Normal:** the line at 90° to the surface at the point of incidence O
- **Angle of incidence (∠i):** between incident ray and normal; **angle of reflection (∠r):** between reflected ray and normal

Laws of reflection ★★★

1. The **angle of incidence is equal to the angle of reflection**: ∠i = ∠r.
2. The incident ray, the normal at the point of incidence and the reflected ray **all lie in the same plane**.

These laws hold for **every reflecting surface**, plane or spherical.

★★★ *Dono laws of reflection har saal 1 mark mein aate hai: Nitesh Bhaiya ki guarantee, ∠i = ∠r ratna nahi, samajhna hai!*

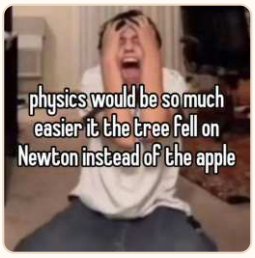
BUJHO *Ray seedhi mirror pe 90° pe giri, toh ∠i = 90° hoga na?*

PAHELI *Nahi! Angles normal se naapte hai. Normal ke along giri ray ka ∠i = 0°, aur wo usi raste wapas aati hai.*

Solved Example 1: Angle work ★★☆

Q. A ray strikes a plane mirror making an angle of 30° with the **mirror surface**. Find ∠i, ∠r and the angle between the incident and reflected rays.

- Angle with the mirror = 30°, so angle with the normal: ∠i = 90° − 30° = **60°**
- By the first law, ∠r = ∠i = **60°**
- Angle between incident and reflected rays = ∠i + ∠r = **120°**



MEME BREAK

Newton ke sir pe seb gira toh gravity mili. Hamare sir pe sirf syllabus girta hai! Chalo, Light chapter pyaar se shuru karte hai.

3 IMAGE: REAL OR VIRTUAL?

DEFINITION

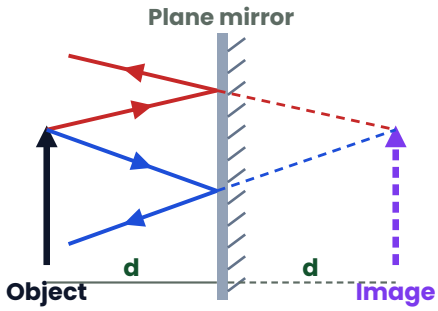
- **Image:** the point where rays from a point of the object **actually meet**, or **appear to meet**, after reflection or refraction.

Real image vs virtual image ★★★

Real image	Virtual image
Rays **actually meet**	Rays only **appear to meet** (when produced backwards)
Can be **obtained on a screen**	**Cannot** be obtained on a screen
Always **inverted**	Always **erect**
Example: picture on a cinema screen	Example: your face in a plane mirror

4 IMAGE IN A PLANE MIRROR ★★★ MOST ASKED

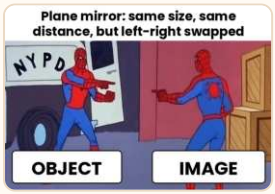
How the image forms



Reflected rays diverge; produced backwards (dashed), they meet behind the mirror at the image.

Characteristics of the image (plane mirror) ★★★

1. **Virtual and erect**; it cannot be caught on a screen.
2. **Same size** as the object (magnification m = +1).
3. As far **behind** the mirror as the object is **in front** of it.
4. **Laterally inverted**: left looks right and right looks left.



MEME BREAK

*Object aur image ek doosre ko point kar rahe hai: same size, same distance. Bas left-right ulta! Yahi hai **lateral inversion**.*

Lateral inversion in daily life

- Raise your right hand in front of a mirror: the image raises its **left** hand.
- Ambulances carry **AMBULANCE** written in mirror letters, so drivers ahead read it correctly in their rear-view mirror.

Solved Example 2: Plane mirror distances ★★☆

Q. Riya stands 2 m in front of a plane mirror. (a) How far is her image from her? (b) She walks 0.5 m towards the mirror. Now how far is the image from her?

- (a) Image is 2 m behind the mirror, so distance = 2 + 2 = **4 m**
- (b) New object distance = 1.5 m, so image distance from her = 1.5 + 1.5 = **3 m**

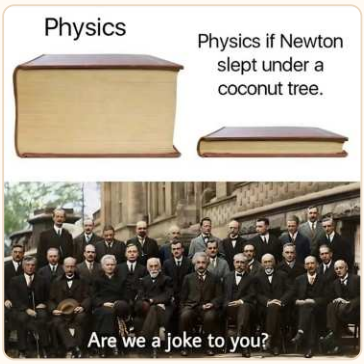
5 SPHERICAL MIRRORS ★★★ MOST ASKED

DEFINITION

- **Spherical mirror:** a mirror whose reflecting surface is a **part of a hollow sphere** of glass.

Concave vs convex

	Concave mirror	Convex mirror
Reflecting surface	curved **inwards**, facing the centre of the sphere	curved **outwards**
Polished (painted) side	outer surface	inner surface
Action on parallel rays	**converging** mirror	**diverging** mirror
Spoon test	inner side of a shiny spoon	back (outer) side of the spoon

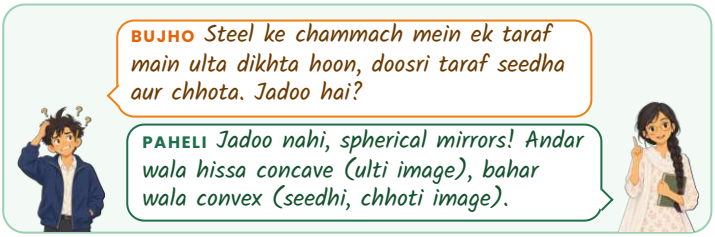


MEME BREAK

Physics ki kitaab ki motai dekh ke darr lag raha hai? Tension nahi, roz ek chhota topic: aaj sirf mirrors!

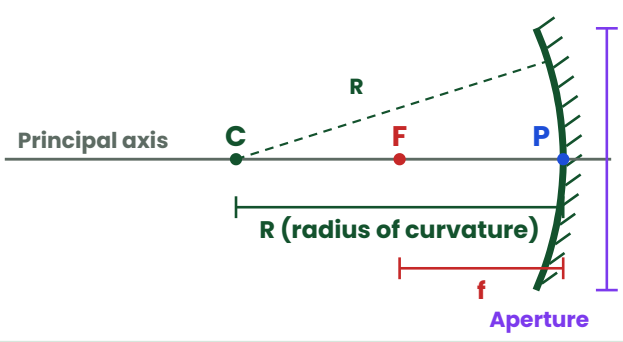
REMEMBER: CAVE GOES IN

Con**cave** is hollowed in like a **cave**; con**vex** bulges out. Concave **converges** light, convex **diverges** it.



6 MIRROR TERMS ★★★ MOST ASKED

Concave mirror with all its terms



Know each term

- **Pole (P):** the centre of the reflecting surface of the mirror
- **Centre of curvature (C):** the centre of the sphere of which the mirror is a part; it lies **in front** of a concave mirror and **behind** a convex mirror
- **Radius of curvature (R):** the radius of that sphere, i.e. the distance PC
- **Principal axis:** the straight line through P and C; it is normal to the mirror at P
- **Aperture:** the diameter of the reflecting surface

Principal focus and focal length ★★★

- **Concave:** rays parallel to the principal axis, after reflection, **actually meet** at a point F on the axis: the **principal focus** (real, in front)
- **Convex:** such rays, after reflection, **appear to diverge** from a point F behind the mirror (virtual focus)
- **Focal length (f):** the distance PF between the pole and the principal focus

R = 2f: the link ★★★

For spherical mirrors of **small aperture**, the principal focus lies midway between P and C:

$$R = 2f \text{ or } f = R / 2$$

Solved Example 3: Using R = 2f ★★☆

Q. The radius of curvature of a spherical mirror is 20 cm. What is its focal length? NCERT

- f = R / 2 = 20 cm / 2 = **10 cm**
- For a concave mirror F is 10 cm in front; for a convex mirror 10 cm behind

Q. A concave mirror has f = 15 cm. Where is C? **Ans.** R = 2 × 15 = **30 cm** in front of the pole.

ACTIVITY: FIND THE FOCUS WITH SUNLIGHT

- Hold a concave mirror facing the Sun and move a sheet of paper in front of it until a **sharp, bright spot** appears.
- Hold it there for a minute: the paper starts to smoke and may burn. The spot is the **image of the Sun**, formed at the **principal focus**.
- Distance from mirror to paper ≈ focal length. Caution: never look at the Sun or its reflection directly.

★★★ *Newton ka fruit salad ho ya mirror ka R = 2f, basics strong toh numerical easy! Nitesh Bhaiya ki guarantee: R = 2f wala 1 mark pakka.*

BUJHO *Plane mirror bhi toh mirror hai, uska focal length kitna hoga?*

PAHELI *Plane mirror ko infinite radius wala sphere samjho: R = ∞, toh f = R/2 = ∞. Isliye uski image na badi hoti hai, na chhoti!*

COMMON MISTAKES

- Measuring ∠i from the **mirror surface**: always measure it from the **normal**.
- Writing that a plane mirror image is "real": it is **virtual**, so no screen can catch it.
- Mixing up sides: a concave mirror reflects from its **inner, curved-in** side.

Exam tips

- ✔ Draw the **normal as a dashed line** in every reflection figure and label ∠i and ∠r.
- ✔ For "characteristics of a plane mirror image", write all four points.
- ✔ In the definition of focus, always say **parallel to the principal axis**.

QUICK RECALL: ONE-MARK QUESTIONS

1. Speed of light in vacuum? **3×10^8 m/s**
2. ∠i for a ray along the normal? **0°**
3. Magnification of a plane mirror? **+1**
4. Which mirror is diverging? **Convex**
5. f of a mirror with R = 24 cm? **12 cm**

7 PRACTICE MCQS

Buff doge banna hai toh 10 MCQs khud solve karo, answer key baad mein!

1. The angle of incidence for a ray falling along the normal on a plane mirror is: ★★☆

(a) 90° (b) 45°
(c) 0° (d) 180°

2. A plane mirror forms an image that is: ★★★

(a) real and inverted (b) virtual, erect, same size
(c) virtual and diminished (d) real and magnified

3. Which of these mirrors is called a converging mirror? ★★☆

(a) plane (b) convex
(c) concave (d) both convex and concave

4. The radius of curvature of a spherical mirror is 32 cm. Its focal length is: ★★★

(a) 64 cm (b) 32 cm
(c) 16 cm (d) 8 cm

5. The image formed by a convex mirror can never be obtained on a screen because it is: ★★☆

(a) real (b) virtual
(c) magnified (d) inverted

6. A ray makes an angle of 25° with the surface of a plane mirror. The angle of reflection is: ★★☆

(a) 25° (b) 65°
(c) 90° (d) 115°

7. The centre of the reflecting surface of a spherical mirror is called its: ★★☆

(a) centre of curvature (b) principal focus
(c) pole (d) aperture

8. AMBULANCE is written in mirror letters on ambulances because a plane mirror image is: ★☆☆

(a) magnified (b) laterally inverted
(c) real (d) diminished

9. **Assertion (A):** The focal length of a spherical mirror of radius of curvature 30 cm is 15 cm.
Reason (R): For a spherical mirror of small aperture, the principal focus lies midway between the pole and the centre of curvature. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The image formed by a plane mirror can be taken on a screen.
Reason (R): A plane mirror forms a virtual image. ★☆☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. List four characteristics of the images formed by plane mirrors. CBSE 2017 **(2 marks)**
2. Define: (i) pole (ii) centre of curvature (iii) principal focus of a concave mirror. Write the relation between R and f. **(3 marks)**
3. Distinguish between a real image and a virtual image (any three points). Which kind does a plane mirror form? **(3 marks)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Along the normal, the angle with the normal is 0°.
2	(b)	Virtual, erect, same size, laterally inverted.
3	(c)	Parallel rays meet at F after reflection.
4	(c)	f = R/2 = 32/2 = 16 cm.
5	(b)	Rays only appear to meet; no screen catches it.
6	(b)	∠i = 90° − 25° = 65°, so ∠r = 65°.
7	(c)	The pole is the centre of the reflecting surface.
8	(b)	Left and right swap in a mirror.
9	(a)	f = R/2 = 15 cm; R explains A.
10	(d)	A virtual image cannot be taken on a screen.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Dual nature of light (from class notes): light behaves as a **wave** and also as a stream of tiny energy packets called **photons** (particle nature). Class 10 uses only the ray model.

Model answers: short and long answer

1 Plane mirror image
- virtual and erect; same size as the object
- as far behind the mirror as the object is in front
- laterally inverted

2 Definitions
- **Pole:** centre of the reflecting surface
- **C:** centre of the sphere of which the mirror is a part
- **F (concave):** point on the principal axis where rays parallel to it meet after reflection; **R = 2f**

3 Real vs virtual
Real: rays actually meet, can be taken on a screen, inverted. Virtual: rays only appear to meet, cannot be taken on a screen, erect. A plane mirror forms a **virtual** image.

Day 21 checklist: tick before you sleep

- [ ] I can define light, reflection, ∠i and ∠r
- [ ] I can state both laws of reflection
- [ ] I can list four features of a plane mirror image
- [ ] I can tell a concave mirror from a convex one
- [ ] I can define P, C, R, F, f and use R = 2f

My score: ____ / 5
5/5 = Jalwa hai! 4 se kam? Page 1–3 phir se padho.

LEVEL 21 COMPLETE +125 XP earned. Next: Level 22

BUJHO *Samajh gaya! Angle normal se, plane mirror ki image virtual, aur R = 2f. Mirror ka chakkar clear!*

PAHELI *Shabaash! Kal Level 22: concave mirror ke chhe cases.*

Day 21 hit-list: Eduspark PYQ analysis

Topic	Rating
Laws of reflection	★★★
Characteristics of plane mirror image	★★★
R = 2f and focal length	★★★
Real vs virtual image	★★☆
Pole, C and principal focus definitions	★★☆
Concave vs convex mirror	★★☆
Lateral inversion (ambulance)	★☆☆

NITESH SIR SAYS
Terms ka diagram bana ke yaad karo, ratke nahi. ***Kal milte hai Day 22 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 5 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 22 · LEVEL 22

PHYSICS | CHAPTER 5 | PART 2 · ASTRA SERIES

RAY DIAGRAMS & MIRROR IMAGES

Goal: draw any mirror image with just two rays.

by Nitesh Prakash

01 Ray Rules | 02 Concave Cases | 03 Ray Diagrams | 04 Convex Cases | 05 Uses | +125 XP today | Level 22 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

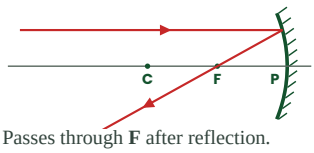
1 RULES FOR DRAWING RAY DIAGRAMS

★★★ MOST ASKED

To locate an image, take **any two** of these rays from the top of the object. Where the reflected rays **meet** (real) or **appear to meet** (virtual) is the image of that point.

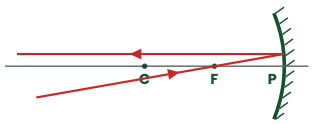
The four rays (concave mirror shown)

Rule 1: Parallel ray



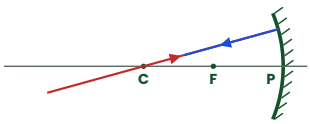
Passes through **F** after reflection.
Convex: appears to come from F behind.

Rule 2: Ray through F



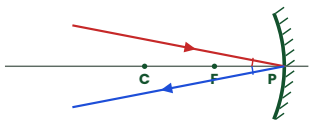
Comes out **parallel** to the axis.
Convex: ray aimed at F comes out parallel.

Rule 3: Ray through C



Hits along the normal and **retraces** its path.
Convex: ray aimed at C retraces its path.

Rule 4: Ray at the pole



Reflects **symmetrically**: ∠i = ∠r about the axis.
Convex: same rule.

BUJHO *Ray diagram mein teen-teen rays banani padegi kya?*

PAHELI *Nahi, koi bhi do rays kaafi hai. Jahan dono reflected rays milti hai, wahi image ka top hai.*

DRAWING CHECKLIST

- Arrow on every ray in the direction light travels
- Virtual rays and virtual images **dashed**
- Label P, F, C and the principal axis; object AB, image A′B′

★★★ *Ray diagram ka 2 ya 3 mark ka sawal har saal aata hai: Nitesh Bhaiya ki guarantee! Object F par toh image infinity par.*

2 IMAGES BY A CONCAVE MIRROR

★★★ MOST ASKED

All six positions at a glance ★★★

Object at	Image at	Size	Nature
Infinity	F	highly diminished, point-sized	real, inverted
Beyond C	between F and C	diminished	real, inverted
C	C	same size	real, inverted
Between C and F	beyond C	enlarged	real, inverted
F	infinity	highly enlarged	real, inverted
Between P and F	behind the mirror	enlarged	**virtual, erect**

PATTERN TO REMEMBER

Bring the object from infinity towards F: the real image moves **from F out to infinity** and grows **bigger**. Cross F (between P and F) and the image jumps **behind** the mirror: virtual, erect, enlarged.

Concave, case 1. Object at infinity

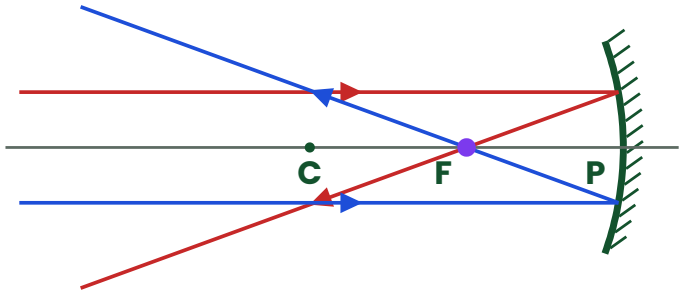


Image at F: point-sized, real, inverted

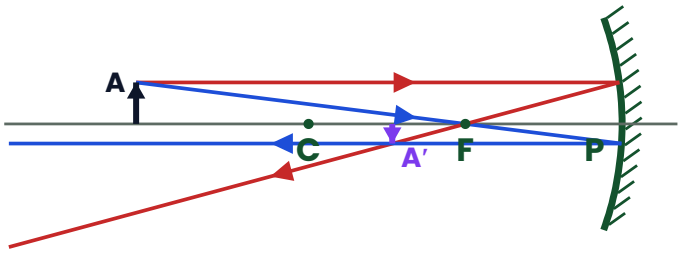
Steel ke chammach mein ulti shakal

Concave mirror: real, inverted image beyond C

MEME BREAK

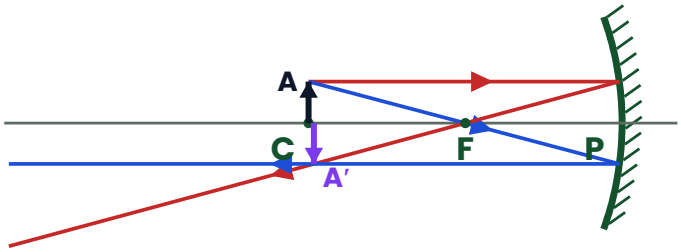
Chammach mein ulta chehra? Wo concave mirror ka case 2 hai: object beyond C, image real, inverted aur chhoti!

Concave, case 2. Object beyond C ★★★



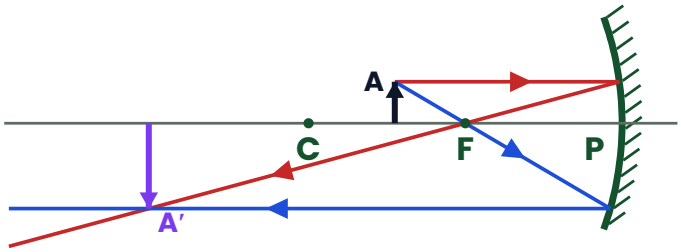
Between F and C: diminished, real, inverted

Concave, case 3. Object at C



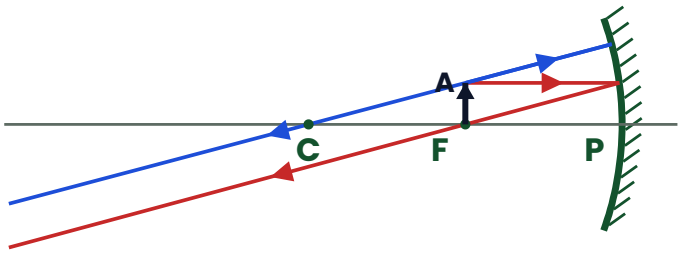
At C: same size, real, inverted

Concave, case 4. Between C and F ★★★



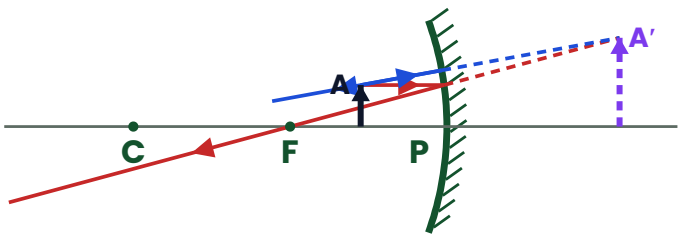
Beyond C: enlarged, real, inverted

Concave, case 5. Object at F

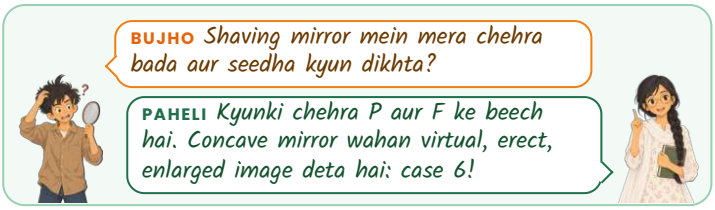


At infinity: highly enlarged, real, inverted

Concave, case 6. Between P and F ★★★



Behind mirror: enlarged, virtual, erect. This is how a shaving mirror works.



3 IMAGES BY A CONVEX MIRROR ★★☆

Only two cases, always virtual

Object at	Image at	Size	Nature
Infinity	F, behind the mirror	highly diminished, point-sized	virtual, erect
Between infinity and P	between P and F, behind	diminished	virtual, erect

Convex, case 1. Object at infinity

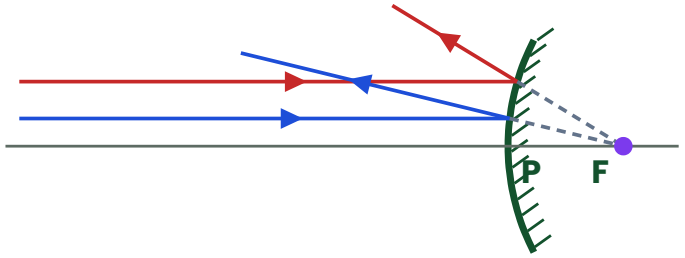


Image at F behind the mirror: point-sized, virtual, erect

Convex, case 2. Object at a finite distance ★★★

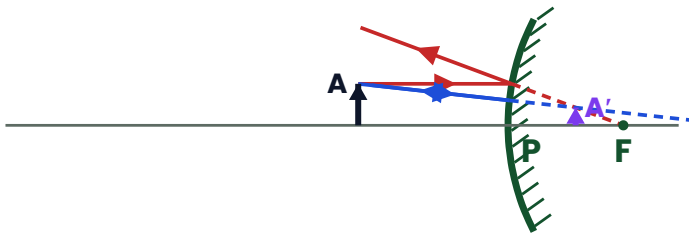


Image between P and F behind the mirror: diminished, virtual, erect

MEME BREAK

Bike mirror mein truck chhota dikha, panic! Phir yaad aaya convex mirror hamesha chhoti image deta hai. Par truck sach mein paas hai, phir panic!

4 USES OF SPHERICAL MIRRORS ★★★ MOST ASKED

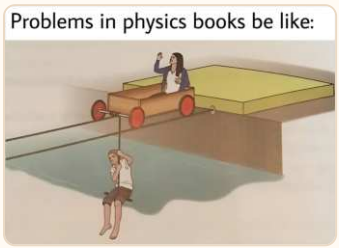
Concave mirror uses ★★★

- **Torches, vehicle headlights, searchlights:** bulb at F gives a **powerful parallel beam**
- **Shaving mirrors:** face between P and F gives an **enlarged, erect** image
- **Dentists:** see a **large image** of the teeth
- **Solar furnaces:** sunlight is **concentrated** at F to produce heat

Convex mirror uses ★★★

- **Rear-view (wing) mirrors** in vehicles
- **Big convex mirrors** in shops and at sharp road bends, for security and safety

Why convex for rear view? (1) It always gives an **erect**, diminished image. (2) It is curved outwards, so it gives a **much wider field of view**: the driver sees a large area behind.



MEME BREAK

Physics ke problems mein ladka ice pe phisalta hai, hum ray diagram mein. Practice se dono sambhal jaate hai!

Solved Example 1: Which mirror? ★★★

Q. Name the mirror that gives: (a) an erect, enlarged image; (b) an erect image of the same size; (c) an erect, diminished image of any object. **NCERT**

- (a) **Concave** mirror, object between P and F
- (b) **Plane** mirror
- (c) **Convex** mirror, for every position

ACTIVITY: FIND THE IMAGE POSITIONS

- Fix a concave mirror on a stand and place a lit candle far away; move a screen until a sharp image appears (near F).
- Bring the candle closer step by step: beyond C, at C, between C and F. Each time, find the image on the screen and note its size.
- With the candle between P and F, no screen catches the image: look into the mirror to see an enlarged, erect, virtual image.

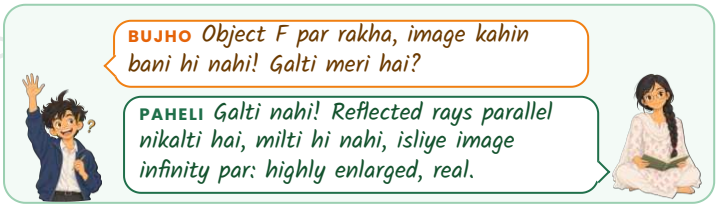
Solved Example 2: Range for a magnified image ★★★

Q. Rohit wants an erect, magnified image of an object using a converging mirror of focal length 40 cm. (a) In what range should the object be placed? (b) State one use of this kind of image formation. **Mod. CBSE SQP 2022**

- A converging mirror is concave; an erect, magnified (virtual) image forms only when the object is between P and F
- (a) Place the object **between 0 and 40 cm** from the mirror
- (b) Shaving or make-up mirror, dentist's mirror

COMMON MISTAKES

- Drawing the virtual image with a **solid** line: use dashes.
- Writing "image at F" for an object at F: the image is at **infinity**.
- Saying a convex mirror can form a real image: it **never** can.



Exam tips

- ✔ Ray diagram questions usually ask for **two rays**: choose the parallel ray and the ray through F or C.
- ✔ Write position, size and nature as three separate words: "beyond C, enlarged, real and inverted".
- ✔ For "why convex mirror in vehicles", write both reasons: erect image and wider field of view.

5 PRACTICE MCQS

Aisa haal na ho! 10 MCQs khud solve karo, phir answer key se milao.

1. For which ray is the angle of incidence on a concave mirror zero? **CBSE SQP 2024** ★★★

(a) a ray parallel to the principal axis
(b) a ray passing through the centre of curvature
(c) a ray passing through the principal focus
(d) a ray incident obliquely at the pole

2. An object is placed at C of a concave mirror. The image is: ★★★

(a) at F, diminished
(b) at C, same size, inverted
(c) beyond C, enlarged
(d) behind the mirror, erect

3. The magnification produced by a concave mirror when the object is between the pole and the focus is: **CBSE SQP 2024** ★★☆

(a) m = +1
(b) m = −1
(c) m > +1
(d) m < −1

4. A concave mirror forms a real image of the same size as the object when the object is at: ★★☆

(a) F
(b) C
(c) infinity
(d) between P and F

5. The mirror used in vehicle headlights is: ★★☆

(a) plane
(b) convex
(c) concave
(d) any spherical mirror

6. The image formed by a convex mirror is always: ★★★

(a) real and inverted
(b) virtual, erect, diminished
(c) virtual, erect, enlarged
(d) real and diminished

7. To get a highly enlarged real image from a concave mirror, the object is placed: ★★☆

(a) at C
(b) just beyond C
(c) between C and F, close to F
(d) between P and F

8. Sunlight is focused by a concave mirror. The bright spot forms at the: ★★☆

(a) pole
(b) principal focus
(c) centre of curvature
(d) infinity

9. **Assertion (A):** A convex mirror is used as a rear-view mirror in vehicles.
Reason (R): A convex mirror gives an erect, diminished image and a wider field of view. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** A concave mirror always forms a real image.
Reason (R): A concave mirror is a converging mirror. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Case-based question: the reflecting telescope

Large telescopes collect starlight with a big **concave mirror**. A small **plane mirror** turns the reflected beam sideways to an eyepiece. CBSE SQP 2024

1. Where does the concave mirror form the image of a distant star, and what is its nature?
2. Why is a concave and not a convex mirror used to collect the light?
3. What does the plane mirror do to the beam?

Short and long answer questions

1. Draw a labelled ray diagram to show the image formed by a concave mirror when the object is placed between its pole and principal focus. CBSE Compartment 2024 **(2 marks)**
2. Draw ray diagrams for a concave mirror with the object (i) beyond C (ii) at F. State the position and nature of the image in each case. **(3 marks)**
3. Why do we prefer a convex mirror as a rear-view mirror? State two uses of concave mirrors. NCERT **(3 marks)**

TRY YOURSELF

Draw the ray diagram for a concave mirror with the object at C and state where the image forms.

Answer key: MCQs

Q	Ans	Why
1	(b)	It hits along the normal and retraces its path.
2	(b)	Object at C gives image at C, same size.
3	(c)	Virtual, erect, enlarged image: m > +1.
4	(b)	Same size, real, inverted at C.
5	(c)	Bulb at F gives a parallel beam.
6	(b)	Every position: virtual, erect, diminished.
7	(c)	Close to F, the image goes far beyond C, very big.
8	(b)	Parallel rays from the Sun meet at F.
9	(a)	R explains why it is preferred.
10	(d)	Object between P and F gives a virtual image.

Case-based answers

- (i) At the principal focus F; real, inverted and point-sized (object at infinity)
- (ii) A concave mirror converges light and forms a real image; a convex mirror only spreads light out
- (iii) It reflects the beam sideways towards the eyepiece without changing its size; the image is laterally inverted

Model answers

1 Object between P and F

Ray parallel to the axis reflects through F; ray along the line from C hits normally and retraces. The reflected rays diverge; produced backwards (dashed) they meet **behind the mirror**: image virtual, erect, enlarged (case 6 figure).

2 Two cases

(i) Beyond C: image between F and C, real, inverted, diminished. (ii) At F: reflected rays are parallel, image at infinity, real, inverted, highly enlarged (cases 2 and 5).

3 Rear view and uses

Convex mirror: always erect, diminished image and a wider field of view. Concave uses: shaving mirror, headlights or torches (any two of those listed).

Day 22 checklist: tick before you sleep

- ☐ I can state the four rules for ray diagrams
- ☐ I can fill the six-case concave mirror table from memory
- ☐ I can draw any concave mirror ray diagram with two rays
- ☐ I know the two convex mirror cases
- ☐ I can give uses of concave and convex mirrors with reasons

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Table phir se likho.

LEVEL 22 COMPLETE +125 XP earned. Next: Level 23

BUJHO *Samajh gaya! Do rays, ek milan point, wahi image. Six cases bhi yaad!*

PAHELI *Shabaash! Kal Level 23: sign convention aur mirror formula.*

Day 22 hit-list: Eduspark PYQ analysis

Topic	Rating
Concave mirror image table (six cases)	★★★
Ray diagram: object between P and F	★★★
Convex mirror as rear-view mirror	★★★
Uses of concave mirrors	★★★
Rules for drawing ray diagrams	★★☆
Ray diagram: object beyond C or at C	★★☆
Convex mirror image cases	★★☆

NITESH SIR SAYS

*Diagram mein arrow aur dashed line hi marks dilate hai. Roz ek diagram banao. **Kal milte hai Day 23 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 5 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 23 · LEVEL 23

PHYSICS | CHAPTER 5 | PART 3 · ASTRA SERIES

MIRROR FORMULA & NUMERICALS

Goal: solve any mirror numerical with the right signs.

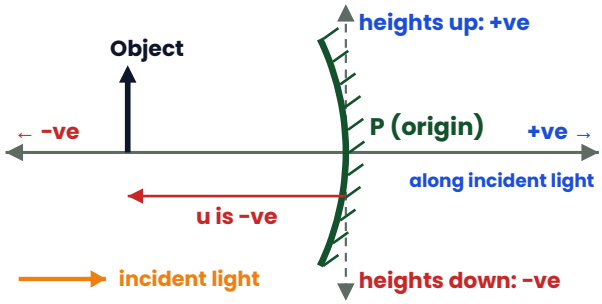
by Nitesh Prakash

01 Sign Convention | 02 Mirror Formula | 03 Magnification | 04 Solved Numericals | 05 Practice | +125 XP today | Level 23 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 NEW CARTESIAN SIGN CONVENTION ★★★ MOST ASKED

Pole = origin, principal axis = x-axis



DEFINITION

- The object is always placed to the **left** of the mirror, so light falls from left to right.
- All distances are measured from the **pole (P)**.
- Distances in the direction of **incident light** (to the right) are **+ve**; distances against it (to the left) are **−ve**.
- Heights **above** the principal axis are **+ve**; heights **below** it are **−ve**.

CLASS TRICK: GAANTH BAANDH LO

"Numericals solve karne ke liye isse gaanth baandh lo"

Mirror	u	v	f	h_o	h_i
Concave	−ve	depends	−ve	+ve	depends
Convex	−ve	+ve	+ve	+ve	+ve

"Depends": real image (in front) → v and h_i −ve; virtual image (behind) → v and h_i +ve. A convex mirror always forms a virtual, erect image, so its v and h_i are always +ve.

MEME BREAK

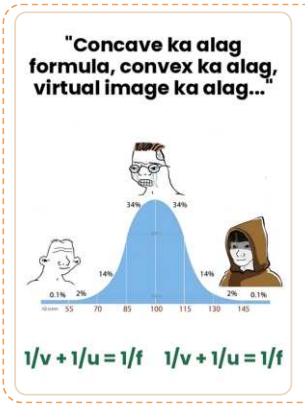
u ka sign? Seagull ki tarah yaad karo: real object hamesha mirror ke left, isliye u hamesha −ve.

2 MIRROR FORMULA ★★★ MOST ASKED

The relation between u, v and f ★★★

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

- **u** = object distance, **v** = image distance, **f** = focal length; all measured from P
- Valid for **all** spherical mirrors and all object positions, if signs are put correctly
- Also remember f = R/2. **Derivation is not required** for the board exam



★★★ *Mirror formula ka 3 mark ka numerical har paper mein: Nitesh Bhaiya ki guarantee! Formula ek, bas sign sahi daalo.*

Five steps for every numerical

1. Draw a quick sketch: mirror, P, object on the left.
2. Write the given values **with signs** (gaanth wali table).
3. Write the formula: 1/v + 1/u = 1/f.
4. Substitute with brackets, e.g. 1/(−27); solve for the unknown.
5. Read the sign of the answer: v −ve = real, in front; v +ve = virtual, behind.

ACTIVITY: FOCAL LENGTH OF A CONCAVE MIRROR

- Focus the image of a far-away window or tree on a screen using a concave mirror.
- Measure the distance from the mirror to the screen: that is the **focal length** (object at infinity, image at F).
- Example: mirror stand at the 10 cm mark, sharp image on the screen at the 40 cm mark: f = 40 − 10 = **30 cm**, so R = 2f = **60 cm**.

3 MAGNIFICATION ★★★ MOST ASKED

DEFINITION

- **Magnification (m):** the ratio of the height of the image to the height of the object.
- $m = h_i / h_o = -v / u$; it has **no unit**.

What the value of m tells you ★★★

m	Meaning
negative	image real and inverted
positive	image virtual and erect
$\|m\| > 1$	image enlarged
$\|m\| < 1$	image diminished
$m = +1$	same size, erect: plane mirror
$m = -1$	same size, real, inverted: object at C of a concave mirror

BUJHO *Question mein "real image, 3 times bada" likha hai. m = +3 lun ya −3?*

PAHELI *Real image hamesha ulti hoti hai, isliye m = −3. Virtual image hoti toh m = +3 leti.*

Signs at a glance ★★☆

Situation	v	h_i	m
Concave, real image (object beyond F)	−	−	−
Concave, object at C (image at C)	−	−	−1
Concave, virtual image (object between P and F)	+	+	+
Convex, any position	+	+	+ (< 1)
Plane mirror	+	+	+1

FORMULA CARD: REVISE IN 30 SECONDS

$f = R / 2$ **$1/v + 1/u = 1/f$**

$m = h_i / h_o = -v / u$ **$h_i = m \times h_o$**

Plane mirror: m = +1, image as far behind as object in front

4 SOLVED NUMERICALS ★★★ MOST ASKED

Solved Numerical 1: Screen position (from class notes) ★★★

Q. An object 7.0 cm in size is placed at 27 cm in front of a concave mirror of focal length 18 cm. Where should a screen be placed to get a sharp image? Find the size and nature of the image. NCERT

Given (with signs): $u = -27$ cm, $f = -18$ cm, $h_o = +7.0$ cm

Formula: $1/v + 1/u = 1/f$

- $1/v = 1/f - 1/u = 1/(-18) - 1/(-27) = -1/18 + 1/27$
- $= (-3 + 2)/54 = -1/54$, so **v = −54 cm**
- $m = -v/u = -(-54)/(-27) = -2$; $h_i = m \times h_o = -2 \times 7.0 = -14$ **cm**

Answer: Screen at **54 cm in front** of the mirror; image 14 cm tall, **real, inverted, enlarged**.

Solved Numerical 2: Convex mirror (from class notes) ★★★

Q. An object is placed at 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image. NCERT

Given (with signs): $u = -10$ cm, $f = +15$ cm

Formula: $1/v + 1/u = 1/f$

- $1/v = 1/f - 1/u = 1/15 - 1/(-10) = 1/15 + 1/10$
- $= (2 + 3)/30 = 5/30 = 1/6$, so **v = +6 cm**
- $m = -v/u = -(6)/(-10) =$ **+0.6**

Answer: Image **6 cm behind** the mirror; **virtual, erect, diminished**.

Solved Numerical 3: Concave mirror, image size ★★★

Q. An object is placed 25 cm in front of a concave mirror of focal length 15 cm. Find the position of the image. If the object is 4.0 cm tall, also find the nature and size of the image. CBSE Compartment 2025

Given (with signs): $u = -25$ cm, $f = -15$ cm, $h_o = +4.0$ cm

Formula: $1/v + 1/u = 1/f$

- $1/v = 1/(-15) - 1/(-25) = -1/15 + 1/25 = (-5 + 3)/75 = -2/75$
- **v = −37.5 cm**
- $m = -v/u = -(-37.5)/(-25) =$ **−1.5**; $h_i = -1.5 \times 4.0 =$ **−6.0 cm**

Answer: Image **37.5 cm in front**; real, inverted, enlarged, 6.0 cm tall.

Solved Numerical 4: Rear-view mirror ★★★

Q. A convex rear-view mirror has a radius of curvature of 3.00 m. A bus is 5.00 m from it. Find the position, nature and size of the image. NCERT

Given (with signs): $R = +3.00$ m, so $f = R/2 = +1.50$ m; $u = -5.00$ m

Formula: $1/v + 1/u = 1/f$

- $1/v = 1/1.50 + 1/5.00 = 0.667 + 0.200 = 0.867\ m^{-1}$
- **v = +1.15 m**
- $m = -v/u = -(1.15)/(-5.00) =$ **+0.23**

Answer: Image **1.15 m behind** the mirror; virtual, erect, 0.23 times the size of the bus.

Solved Numerical 5: Find f from magnification ★★★

Q. A concave mirror forms a real image three times the size of an object placed 10 cm in front of it. Where is the image, and what is the focal length? NCERT

Given (with signs): $u = -10$ cm, real image so $m = -3$

Formula: $m = -v/u$, then $1/v + 1/u = 1/f$

- $-3 = -v/(-10)$, so **v = −30 cm**
- $1/f = 1/v + 1/u = -1/30 - 1/10 = -4/30$
- **f = −7.5 cm**

Answer: Image **30 cm in front** of the mirror; focal length 7.5 cm (concave).

MEME BREAK

u = +30 likha? Batman ka thappad pakka! Real object ka u ***hamesha negative.***

Solved Numerical 6: Erect, magnified image ★★☆

Q. An object is placed 6 cm in front of a concave mirror of focal length 12 cm. Find the position, nature and magnification of the image.

Given (with signs): u = −6 cm, f = −12 cm
Formula: 1/v + 1/u = 1/f

- 1/v = 1/(−12) − 1/(−6) = −1/12 + 2/12 = 1/12
- **v = +12 cm** (positive: behind the mirror)
- m = −v/u = −(12)/(−6) = **+2**

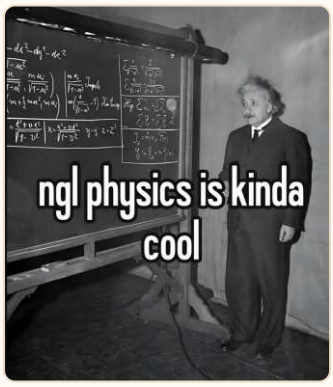
Answer: Image **12 cm behind** the mirror; virtual, erect, twice the size (object between P and F).

COMMON MISTAKES

- Writing f of a concave mirror as +ve: it is **−ve**.
- Forgetting brackets: 1/(−27), not −1/27 mixed with +1/27.
- Stopping at v: always state **nature** and **size** as well.

Exam tips

- ✔ Write units with every answer (cm or m) and the sign meaning in words.
- ✔ Use fractions with an LCM; avoid long decimals until the last step.
- ✔ If v comes out +ve for a concave mirror, the object is between P and F: the image is virtual.



MEME BREAK
Sign convention samajh aaya toh physics sach mein cool lagti hai! Ab numericals khud try karo.

TRY YOURSELF: NUMERICALS

1. Concave mirror, f = 20 cm, object at 30 cm. Find v and m.
2. Convex mirror, f = 20 cm, object at 20 cm. Find v and m.
3. Concave mirror, f = 10 cm, object at 5 cm. Find v and m; state the nature.

QUICK RECALL: ONE-MARK QUESTIONS

1. Sign of u for a real object? **Always −ve**
2. Sign of f for a convex mirror? **+ve**
3. Unit of magnification? **None**
4. m for an image at C of a concave mirror? **−1**
5. v is +ve for a concave mirror when? **Object between P and F**

Case-based question: the dentist's mirror

A dentist holds a small concave mirror of focal length 4 cm at a distance of 3 cm from a tooth.

1. Find the position of the image of the tooth.
2. Find the magnification. Describe the image.
3. Why does the dentist not use a convex mirror?

5 PRACTICE MCQS

Sign galat, answer galat, dil toota! 10 MCQs dhyan se karo.

1. In the New Cartesian sign convention, the focal length of a concave mirror is taken as: ★★★
(a) positive (b) negative
(c) zero (d) positive or negative

2. The magnification produced by a plane mirror is: ★★☆
(a) +1 (b) −1
(c) 0 (d) infinite

3. A negative value of magnification means the image is: ★★★
(a) virtual and erect (b) real and inverted
(c) always diminished (d) always enlarged

4. An object is 20 cm from a concave mirror of focal length 10 cm. The image distance is: ★★☆
(a) −10 cm (b) −20 cm
(c) +20 cm (d) −40 cm

5. The radius of curvature of a convex mirror is 32 cm. Its focal length is: ★★☆
(a) −16 cm (b) +16 cm
(c) +32 cm (d) −32 cm

6. For a convex mirror, the image distance v is always: ★★☆
(a) negative (b) positive
(c) zero (d) equal to u

7. A mirror gives m = +0.5 for every position of the object. The mirror is: ★★☆
(a) plane (b) concave
(c) convex (d) none of these

8. For a concave mirror, u = −15 cm and v = −30 cm. The magnification is: ★★☆
(a) +2 (b) −2
(c) +0.5 (d) −0.5

9. Assertion (A): The focal length of a convex mirror is taken as positive.
Reason (R): The principal focus of a convex mirror lies behind the mirror, in the direction of incident light. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** A concave mirror can produce a magnification of +2.
Reason (R): A concave mirror forms an enlarged, virtual, erect image when the object is between P and F. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. An object 5.0 cm long is placed 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position, nature and size of the image. NCERT **(3 marks)**
2. State the New Cartesian sign convention for spherical mirrors. Write the mirror formula and the formula for magnification. **(3 marks)**
3. The magnification of a spherical mirror is −1. Name the mirror and state where the object is placed. **(2 marks)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Focus is in front, against incident light.
2	(a)	Same size and erect.
3	(b)	Minus sign: real, inverted.
4	(b)	1/v = −1/10 + 1/20 = −1/20; at C.
5	(b)	f = R/2 = +16 cm, behind.
6	(b)	Image always behind the mirror.
7	(c)	Always erect, diminished: convex.
8	(b)	m = −v/u = −(−30)/(−15) = −2.
9	(a)	F lies behind, along incident light.
10	(a)	m = +2 is virtual, erect, enlarged.

Model answers

1 Convex mirror, R = 30 cm
f = +15 cm, u = −20 cm. 1/v = 1/15 + 1/20 = 7/60, v = **+8.6 cm** (behind). m = −v/u = 8.57/20 = +0.43; $h_i = 0.43 \times 5.0 =$ **+2.1 cm**. Virtual, erect, diminished.

2 Sign convention
State the four rules on page 1; mirror formula 1/v + 1/u = 1/f; m = h_i/h_o = −v/u.

3 m = -1
Concave mirror; real, inverted image of the same size, so the object is at **C** (u = −2f).

4 Try-yourself answers
(1) v = −60 cm, m = −2. (2) v = +10 cm, m = +0.5. (3) v = +10 cm, m = +2: virtual, erect, enlarged.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Where the mirror formula comes from (derivation deleted from the syllabus): draw the ray diagram for an object beyond C. Two pairs of **similar triangles** give A′B′/AB = B′P/BP and A′B′/AB = B′F/FP. Equating them and putting in the signed distances (BP = −u, B′P = −v, FP = −f) leads to **1/v + 1/u = 1/f**.

Case-based answers

- (i) u = −3 cm, f = −4 cm: 1/v = −1/4 + 1/3 = 1/12, so **v = +12 cm** (behind the mirror)
- (ii) m = −v/u = −12/(−3) = **+4**: virtual, erect, 4 times enlarged
- (iii) A convex mirror always gives a diminished image; the dentist needs an enlarged one

Day 23 checklist: tick before you sleep

- ☐ I can state the four sign convention rules
- ☐ I know the gaanth wali sign table by heart
- ☐ I can write 1/v + 1/u = 1/f and m = −v/u
- ☐ I can solve a mirror numerical with signs and units
- ☐ I can read nature and size from the signs of v and m

My score: ____ / 5
5/5 = Jalwa hai! 4 se kam? Solved numericals phir se karo.

Coming up: Day 24

Refraction of light: why a pencil looks bent in water, the laws of refraction, refractive index and the glass slab.

LEVEL 23 COMPLETE +125 XP earned. Next: Level 24

BUJHO *Samajh gaya! Pehle sign, phir formula, phir answer ka matlab. Numericals ab aasan!*

PAHELI *Shabaash! Kal Level 24: refraction of light.*

Day 23 hit-list: Eduspark PYQ analysis

Topic	Rating
Mirror formula numericals	★★★
New Cartesian sign convention	★★★
Magnification m = −v/u and its sign	★★★
Image size from magnification	★★★
Finding f from magnification	★★☆
Convex mirror numericals (rear view)	★★☆
Signs of v, h_i and m by image type	★★☆

NITESH SIR SAYS
Har numerical mein pehle signs likho, phir formula. Sign sahi toh answer sahi. ***Kal milte hai Day 24 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 5 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 24 · LEVEL 24

PHYSICS | CHAPTER 5 | PART 4 · ASTRA SERIES

REFRACTION & REFRACTIVE INDEX

Goal: know why light bends, Snell's law and refractive index.

by Nitesh Prakash

01 Refraction | 02 Snell's Law | 03 Refractive Index | 04 Glass Slab | 05 Numericals | +125 XP today | Level 24 / 60

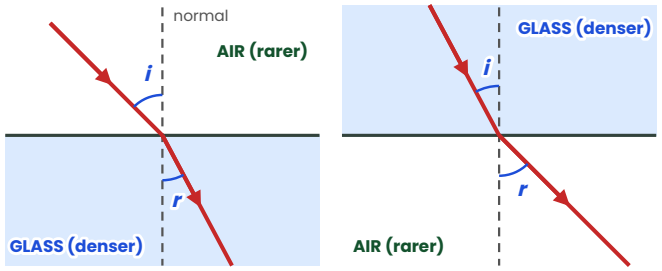
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 REFRACTION OF LIGHT

DEFINITION

- **Refraction:** the change in the **direction** of light when it passes **obliquely** from one transparent medium into another
- It happens because light has a **different speed** in different media: about 3×10^8 **m/s** in vacuum, slower in water and glass
- **Optically denser** medium: light is slower there (glass is denser than air). **Optically rarer**: light is faster

Figure 1: Which way does the ray bend? ★★★



(a) Bends TOWARDS normal (b) Bends AWAY from normal

Angles are always measured from the **normal** (the dashed line at 90° to the surface), never from the surface.

Rule: rarer and denser

- ✔ **Rarer → denser** (air → glass): speed falls, ray bends **towards** the normal, so $r < i$
- ✔ **Denser → rarer** (glass → air): speed rises, ray bends **away from** the normal, so $r > i$

★★★ *Towards ya away from normal: ye ek line har saal MCQ ya 1-mark mein aati hai, Nitesh Bhaiya ki guarantee!*

NCERT in-text questions: quick answers ★★☆

1 Air into water, obliquely: towards or away?

Towards the normal. Water is optically denser, so light slows down on entering it.

2 Kerosene 1.44, turpentine 1.47, water 1.33: where is light fastest?

In **water**, the medium with the lowest refractive index ($v = c/n$).

When does refraction NOT take place? ★★☆

1. **Normal incidence:** the ray falls along the normal ($i = 0°$). It goes straight through with $r = 0°$. The speed still changes, but the direction does not.
2. **Same optical density:** both media bend light equally (same refractive index), so the ray goes straight. Example: a glass rod dipped in glycerine almost disappears.

MEME BREAK

*Ray normal pe aayi? **i = 0°, r = 0°**. Na bend, na drama: seedha andar, seedha bahar. Newton bhi khush!*

BUJHO *Agar light normal pe aaye to speed bhi same rehti hai na?*

PAHELI *Nahi Bujho! Speed to glass mein phir bhi kam hoti hai. Sirf **direction** nahi badalti, kyunki koi tirchha angle hi nahi hai.*

2 REFRACTION AROUND US

Everyday examples

- A **pencil** partly dipped in water looks **bent** at the water surface
- A swimming pool or tank looks **shallower** than it really is (bottom appears raised)
- A **lemon** in a glass of water looks **bigger** from the side
- Letters under a thick glass slab look **raised**
- All happen because light from the object bends at the surface before it reaches our eye

MEME BREAK

*Pencil paani mein **toota hua** dikha? Scroll of truth: pencil theek hai, light ne surface pe **direction** badli hai!*

3 LAWS OF REFRACTION

The two laws ★★★

1. The incident ray, the refracted ray and the normal at the point of incidence **all lie in the same plane**.
2. **Snell's law:** for a given pair of media and a given colour of light, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is a **constant**.

$$n_{21} = \sin i / \sin r = \text{constant}$$

The constant is the **refractive index of medium 2 with respect to medium 1**. It has **no unit** (it is a ratio).

CLASS TRICK: SINE VALUES YOU NEED

Refractive Index: darna nahi hai! Kam sawaal aate hain, aur aate bhi hain to max 2 marks.

Angle	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1

Solved Example 1: Using Snell's law ★★☆

Q. A ray in air falls on glass at $i = 60°$. The refractive index of the glass is √3. Find the angle of refraction.

- **Given:** $i = 60°$, $n = \sqrt{3}$
- **Formula:** $n = \sin i / \sin r \Rightarrow \sin r = \sin i / n$
- **Substitute:** $\sin r = (\sqrt{3}/2) \div \sqrt{3} = 1/2$
- **Answer:** $r = \mathbf{30°}$ (smaller than 60°: bent towards the normal, as expected)

4 REFRACTIVE INDEX

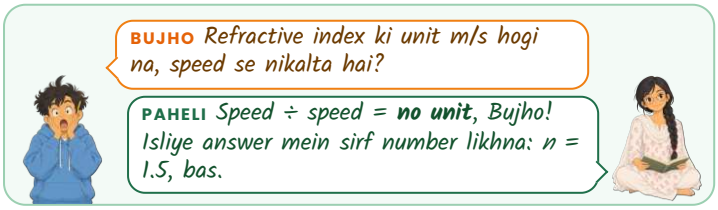
DEFINITION

- **Absolute refractive index** of a medium: speed of light in **vacuum (or air)** divided by the speed of light in that medium
- $\boldsymbol{n_m = c / v}$, where $c = 3 \times 10^8$ m/s. Medium 1 is **fixed**: always vacuum or air
- It is a pure number (no unit) and is always more than 1 for a material medium

Solved Example 5: Relative refractive index ★★☆

Q. $n_{glass} = 1.50$ and $n_{water} = 1.33$. Find (i) n of glass w.r.t. water (ii) n of water w.r.t. glass.

- **Formula:** $n_{21} = n_2 / n_1$
- (i) $n_{gw} = 1.50 / 1.33 \approx \mathbf{1.13}$
- (ii) $n_{wg} = 1.33 / 1.50 \approx \mathbf{0.89}$ (less than 1: water is rarer than glass)



Remember: n depends on the nature of the medium and the colour of light, not on the angle of incidence.

Relative refractive index ★★★

Refractive index of medium 2 **with respect to** medium 1 (here nothing is fixed; both media can be anything):

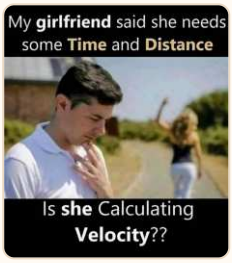
$$n_{21} = v_1 / v_2 = n_2 / n_1$$

Also $n_{12} = v_2 / v_1 = 1 / n_{21}$. Example: n of glass w.r.t. water = 1.50 / 1.33 ≈ 1.13.

CLASS TRICK: N.B. FROM SIR'S NOTES

- Jitna jyada denser medium, utni kam speed of light, isliye $\mathbf{n \propto 1/v}$.
- Jitna jyada denser medium, utna jyada refractive index; aur $\mathbf{n \propto}$ **bending**: n bada, bending zyada.
- Absolute mein fix thing: air ya vacuum (medium 1). Relative mein koi fix nahi, dono medium alag alag.

Note (NCERT): "denser" here means **optical density**, not mass density. Kerosene ($n = 1.44$) is optically denser than water (1.33) but lighter in mass density.



MEME BREAK

Speed = distance / time yaad hai? Yahan bhi wahi khel: $\mathbf{n = c / v}$. Medium mein light jitni slow, n utna bada!

Solved Examples 2 to 4: Speed and n ★★★

2 Speed becomes one-third

Light's speed in a medium is $c/3$. Then $n = c \div (c/3) = \mathbf{3}$.

3 Glass, n = 1.50 (NCERT)

$v = c/n = (3 \times 10^8) / 1.50 = \mathbf{2 \times 10^8}$ **m/s**.

4 Carbon disulphide, n = 1.63

Light travels **1.63 times slower** in CS_2 than in vacuum: $v = 3 \times 10^8 / 1.63 \approx \mathbf{1.84 \times 10^8}$ **m/s**.

NCERT table: absolute refractive index ★★☆

Medium	n	Medium	n
Air	1.0003	Canada balsam	1.53
Ice	1.31	Rock salt	1.54
Water	1.33	Carbon disulphide	1.63
Alcohol	1.36	Dense flint glass	1.65
Kerosene	1.44	Ruby	1.71
Fused quartz	1.46	Sapphire	1.77
Turpentine oil	1.47	Diamond	**2.42**
Benzene	1.50	Crown glass	1.52

Highest optical density: **diamond**. Lowest: **air**.

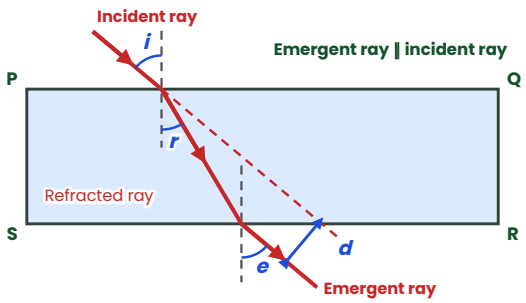
TRY YOURSELF: QUICK NUMERICALS

1. Speed of light in a medium is 2.25×10^8 m/s. Find n.
2. n of a medium is 2. Find the speed of light in it.
3. $n_A = 1.2$, $n_B = 1.5$. Find n_{BA}.

Answers: 1.33; 1.5×10^8 m/s; 1.25.

5 REFRACTION THROUGH A GLASS SLAB

Figure 2: Rectangular glass slab PQRS ★★★



What happens at each surface

1. At **PQ** (air → glass): the ray bends **towards** the normal ($r < i$).
2. At **SR** (glass → air): the ray bends **away from** the normal and comes out as the **emergent ray**.
3. The bending at the two faces is **equal and opposite**, so **angle of emergence *e* = angle of incidence *i***.
4. The emergent ray is **parallel** to the incident ray but **shifted sideways**.

DEFINITION

> **Lateral displacement (d):** the **perpendicular distance** between the incident ray (produced forward) and the emergent ray

Lateral displacement depends on ★★☆

1. **Angle of incidence** (larger *i*, larger shift)
2. **Thickness** of the slab (thicker slab, larger shift)
3. **Refractive index** of the slab (larger *n*, larger shift)

MEME BREAK
Andar ghusi, mudi, bahar nikli, phir mudi: e = i, emergent ray incident ke parallel. Perfectly balanced!

Activity (NCERT): tracing the path

- Fix a glass slab on paper and draw its outline PQRS
- Draw a slanting line on PQ as the incident ray; fix two pins on it
- Look from the other side and fix two more pins in line with the images of the first two
- Join the points: the emergent ray is parallel to the incident ray; measure *i*, *r* and *e*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

$n_{21} \times n_{12} = 1$, because $(v_1/v_2) \times (v_2/v_1) = 1$. A relative index can be less than 1; an absolute index of a material is always more than 1.

BUJHO *Slab se nikal kar ray parallel kyun aati hai? Andar to mud gayi thi!*

PAHELI *Dono faces **parallel** hain, Bujho. PQ pe jitna normal ki taraf mudi, SR pe utna hi door mudi. Isliye **e = i**, bas thoda side shift.*

★★★ *Glass slab ka labelled diagram with i, r, e aur lateral displacement: Nitesh Bhaiya ki guarantee!*

6 COMMON MISTAKES AND EXAM TIPS

× WRONG	✓ RIGHT
Angle measured from the surface	Measure *i*, *r*, *e* from the normal
Writing a unit for *n*	*n* is a ratio: no unit
Denser = heavier liquid	Denser = optically denser (higher *n*)
Lateral displacement = shift along the slab	Perpendicular distance between the two rays
Emergent ray drawn in the same line as the incident ray	Parallel to it, but shifted sideways

Exam tips

- ✔ In diagrams show arrows on every ray and dashed normals at both faces
- ✔ Light bends towards the normal when it **slows down**; away when it **speeds up**
- ✔ Higher *n* means lower speed: $v = c / n$
- ✔ For relative index write the order carefully: $n_{21} = n_2 / n_1$
- ✔ Write *n* as a plain number, with no unit

7 PRACTICE MCQS

1. A ray of light passes from air into glass obliquely. It: ★★★
(a) bends away from the normal (b) bends towards the normal
(c) goes straight (d) is reflected back

2. If $n = 1.5$ for a medium, the speed of light in it is: ★★★
(a) 1.5×10^8 m/s (b) 2×10^8 m/s
(c) 3×10^8 m/s (d) 4.5×10^8 m/s

3. A ray falls normally on a glass slab. Angle of refraction: ★★☆
(a) 0° (b) 45°
(c) 90° (d) equal to the critical angle

4. From the NCERT table, light travels fastest in: NCERT ★★☆
(a) kerosene (b) turpentine oil
(c) water (d) benzene

5. For a rectangular glass slab, the emergent ray is: ★★★
(a) perpendicular to the incident ray (b) parallel to the incident ray
(c) along the normal (d) in line with the refracted ray

6. The SI unit of refractive index is: ★☆☆
(a) m/s (b) metre
(c) dioptre (d) it has no unit

7. Light goes from medium 1 ($v = 2 \times 10^8$ m/s) to medium 2 ($v = 1.5 \times 10^8$ m/s). n_{21} is: ★★☆

(a) 0.75 (b) 1.33
(c) 1.5 (d) 2

8. Lateral displacement does NOT depend on: ★☆☆

(a) angle of incidence (b) thickness of slab
(c) refractive index of slab (d) colour of the slab's frame

9. **Assertion (A):** A ray of light bends towards the normal when it goes from water into air.
Reason (R): Light travels faster in air than in water. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The refractive index of diamond (2.42) is the highest in the NCERT table.
Reason (R): Light travels slowest in diamond among these media. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. State the laws of refraction of light. If the speed of light in vacuum is 3×10^8 m/s, find the absolute refractive index of a medium in which light travels at 1.5×10^8 m/s. **(3 marks)**
2. Draw a labelled ray diagram for refraction through a rectangular glass slab. Mark i, r, e and the lateral displacement. Why is the emergent ray parallel to the incident ray? **(3 marks)**
3. The refractive index of diamond is 2.42. What is the meaning of this statement? **(2 marks)** NCERT

LEVEL 24 COMPLETE +125 XP earned. Next: Level 25

BUJHO *Samajh gaya! Slow hui to normal ki taraf, fast hui to door. Aur n = c/v, bas itna sa!*

PAHELI *Shabaash! Kal Level 25: lens aur unki images.*

Day 24 hit-list: Eduspark PYQ analysis

Topic	Rating
Towards / away from normal with reason	★★★
Glass slab diagram, $e = i$, lateral displacement	★★★
$n = c/v$ numericals	★★★
Laws of refraction, Snell's law	★★★
Reading the NCERT n table	★★☆
Relative refractive index	★★☆
Everyday examples of refraction	★☆☆

Answer key: MCQs

Q	Ans	Why
1	(b)	Air → glass is rarer → denser: towards the normal.
2	(b)	$v = 3 \times 10^8 / 1.5 = 2 \times 10^8$ m/s.
3	(a)	Normal incidence: no bending, $r = 0°$.
4	(c)	Water has the lowest n (1.33) of the four.
5	(b)	$e = i$ as the faces are parallel.
6	(d)	A ratio of two speeds: no unit.
7	(b)	$n_{21} = v_1/v_2 = 2/1.5 \approx 1.33$.
8	(d)	Only i, thickness and n matter.
9	(d)	Water → air bends **away**; R is true.
10	(a)	Highest n means lowest speed: R explains A.

Model answers: short and long answer

1 Laws and n

- Incident ray, refracted ray and normal lie in one plane
- $\sin i / \sin r$ = constant (Snell's law)
- $n = c/v = (3 \times 10^8)/(1.5 \times 10^8) = \mathbf{2}$

2 Glass slab

- Diagram as in Figure 2, arrows on all rays, normals dashed
- Bending at PQ and SR is equal and opposite as the faces are parallel, so $e = i$ and the emergent ray is parallel, shifted by d

3 Diamond, $n = 2.42$

- Speed of light in diamond is 1/2.42 of its speed in vacuum: $v = 3 \times 10^8 / 2.42 \approx \mathbf{1.24 \times 10^8}$ **m/s**

Day 24 checklist: tick before you sleep

- ☐ I can say why light bends and which way (towards or away from the normal)
- ☐ I can state both laws of refraction and use Snell's law
- ☐ I can use $n = c/v$ and $n_{21} = v_1/v_2$
- ☐ I can draw the glass slab diagram with i, r, e and d
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

NITESH SIR SAYS
*Refraction ka ek hi funda: light ki speed badli to direction badli. Diagram mein normal aur arrows kabhi mat bhoolna. **Kal milte hai Day 25 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Light chapter board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 25 · LEVEL 25

PHYSICS | CHAPTER 5 | PART 5 · ASTRA SERIES

SPHERICAL LENSES & IMAGES

Goal: draw lens ray diagrams and know all image cases.

by Nitesh Prakash

01 Lens Basics | 02 Lens Terms | 03 Ray Rules | 04 Convex Cases | 05 Concave Cases | +125 XP today | Level 25 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 SPHERICAL LENSES

DEFINITION

- **Lens:** a piece of transparent material (glass or plastic) bound by two surfaces, of which **one or both are spherical**
- **Convex lens:** both surfaces bulge **outwards**; it is **thicker at the middle** than at the edges
- **Concave lens:** both surfaces curve **inwards**; it is **thicker at the edges** than at the middle

Convex vs concave lens ★★★

	Convex lens	Concave lens
Shape	thick middle, thin edges	thin middle, thick edges
Action on light	**converging**: bends parallel rays to meet	**diverging**: spreads parallel rays out
Focus	real (rays actually meet)	virtual (rays only seem to come from it)
Image (real object)	real or virtual	always virtual, erect, diminished
Focal length sign	positive (+)	negative (−)

2 TERMS USED FOR LENSES

Know these terms ★★★

- **Centre of curvature (C_1, C_2):** centres of the two spheres of which the lens surfaces are parts
- **Radius of curvature:** radius of the sphere of which a lens surface is a part
- **Principal axis:** the imaginary straight line through both centres of curvature
- **Optical centre (O):** the central point of the lens; a ray through it passes **without any deviation**
- **Aperture:** the effective diameter of the circular outline of the lens
- **Principal focus (F):** convex: the point where rays parallel to the axis **meet** after refraction; concave: the point from which they **appear to diverge**. A lens has two foci, F_1 and F_2
- **Focal length (f):** distance between the optical centre and the principal focus

Note: $2F_1$ and $2F_2$ mark twice the focal length from O on either side; they play the role that C plays for a mirror.

MEME BREAK

*"Converging lens" ka mask utaaro, andar se nikla **convex lens**! Teen naam, ek hi lens, focal length **positive**. Fancy naam se darna mat.*

BUJHO *Lens ka bhi mirror jaisa ek hi focus hota hai na?*

PAHELI *Nahi Bujho! Light lens ke dono taraf se aa sakti hai, isliye **do foci** hote hain, F1 aur F2, dono optical centre se barabar doori par.*

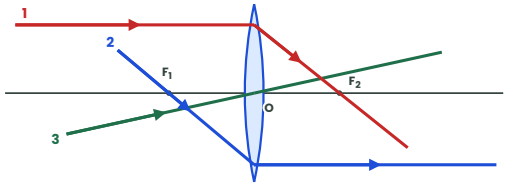
Quick test: which lens is it? ★☆☆

- ✔ Hold it close over printed letters: **bigger** letters mean convex, **smaller** letters mean concave
- ✔ Focus sunlight on paper: only a **convex** lens gives a bright spot (it can burn the paper, so never look at the Sun through it)
- ✔ Touch it: thick middle is convex, thick edges is concave

★★★ *Convex = converging, concave = diverging: ye pair har saal 1-mark mein aata hai, Nitesh Bhaiya ki guarantee!*

3 RULES FOR DRAWING RAY DIAGRAMS

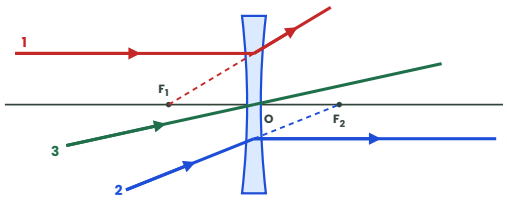
Figure 1: The three rules, convex lens



Rules in words (both lenses) ★★★

1. A ray **parallel to the principal axis**: after refraction it passes **through F_2** (convex) or **appears to come from F_1** (concave).
2. A ray **through F_1** (convex), or **directed towards F_2** (concave): after refraction it emerges **parallel** to the principal axis.
3. A ray through the **optical centre O** goes straight, **without deviation**.

Figure 2: The three rules, concave lens



MEME BREAK

Convex ho ya concave, optical centre se jaane wali ray pe dono agree: ***seedha jao, no bending!***

4 IMAGE FORMATION BY A CONVEX LENS

The six cases at a glance ★★★

Object at	Image at	Size	Nature
Infinity	at F_2	highly diminished, point-sized	real, inverted
Beyond $2F_1$	between F_2 and $2F_2$	diminished	real, inverted
At $2F_1$	at $2F_2$	same size	real, inverted
Between F_1 and $2F_1$	beyond $2F_2$	enlarged	real, inverted
At F_1	at infinity	infinitely large, highly enlarged	real, inverted
Between F_1 and O	same side as object	enlarged	**virtual, erect**

SHORTCUT: OBJECT COMES CLOSER, IMAGE RUNS AWAY

Convex lens mein object jaise jaise F_1 ki taraf aata hai, real image door jaati hai aur badi hoti jaati hai. F_1 ke andar aate hi image virtual ho jaati hai.

SHORTCUT: ANY TWO RAYS ARE ENOUGH

Two rays from the top of the object fix the top of the image. Show real rays as solid lines and back-extended (virtual) rays as dashed lines.

Case 1: Object at infinity

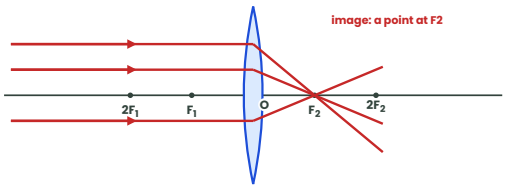


Image: real, inverted, point-sized, at F_2. Use: burning paper with sunlight.

Case 2: Object beyond $2F_1$

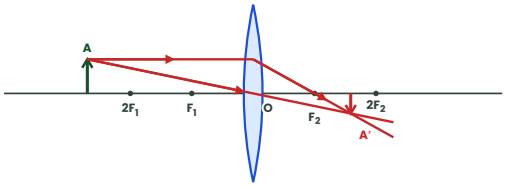


Image: real, inverted, diminished, between F_2 and $2F_2$. Use: camera.

Case 3: Object at $2F_1$

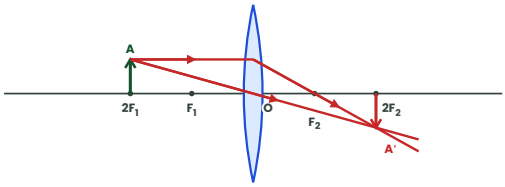


Image: real, inverted, same size, at $2F_2$.

MEME BREAK

Object F_1 pe rakha aur image balloon ki tarah ***infinity*** *ki taraf udd gayi! Rays parallel nikalti hain.*

Case 4: Object between F_1 and $2F_1$

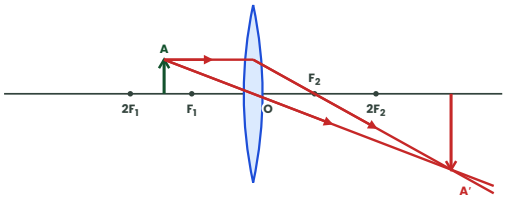


Image: real, inverted, enlarged, beyond $2F_2$. Use: projector.

Case 5: Object at F_1

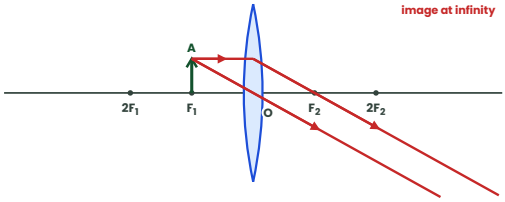


Image: real, inverted, highly enlarged, at infinity (rays emerge parallel).

Case 6: Object between F_1 and O

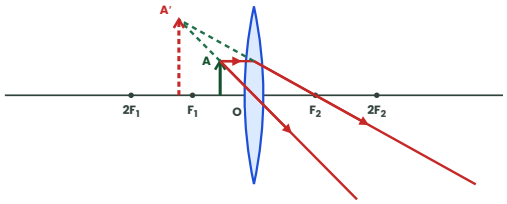


Image: virtual, erect, enlarged, on the object's side. Use: **magnifying glass**.

★★★ *Six-case table + ek ray diagram (mostly case 4 ya case 6): Nitesh Bhaiya ki guarantee!*

Solved Example: name the case ★★☆

Q. A convex lens of focal length 10 cm. Where is the image, and what is it like, for an object at (i) 25 cm (ii) 15 cm (iii) 6 cm?

- (i) 25 cm is beyond 2F (20 cm): image between F_2 and $2F_2$, real, inverted, diminished
- (ii) 15 cm is between F and 2F: image beyond $2F_2$, real, inverted, enlarged
- (iii) 6 cm is inside F: virtual, erect, enlarged, on the object's side

REMEMBER: THE MAGNIFYING GLASS

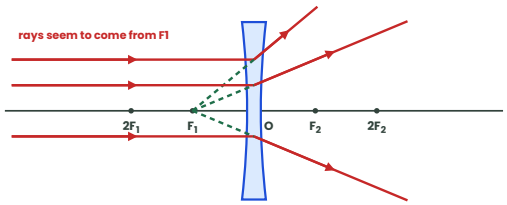
A reading lens is a **convex lens** held so that the object is **inside F_1** (case 6): the image is virtual, erect and enlarged.

5 IMAGE FORMATION BY A CONCAVE LENS

Concave lens: two cases ★★☆

Object at	Image at	Size	Nature
Infinity	at F_1	highly diminished, point-sized	virtual, erect
Between infinity and O	between F_1 and O	diminished	virtual, erect

Concave Case 1: Object at infinity



Concave Case 2: Object at a finite distance

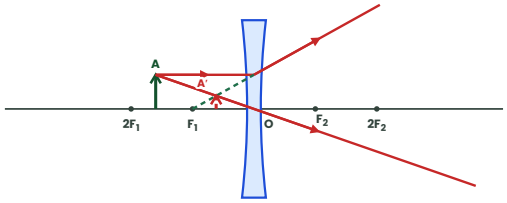


Image: virtual, erect, diminished, between F_1 and O, on the same side as the object.

BUJHO *Concave lens se screen pe real image le sakte hain?*

PAHELI *Kabhi nahi! Real object ke liye concave lens hamesha* ***virtual, erect aur chhoti*** *image banata hai, jo screen pe nahi aati.*

MEME BREAK

Concave lens ho to: ***virtual image, virtual image everywhere!*** *Real object ke liye ye* ***screen pe kabhi nahi aati.***

Uses of lenses (from class notes) ★★☆

Convex lens	Concave lens
Magnifying glass (reading small print, watch repair)	Spectacles for **myopia** (short-sightedness)
Camera, projector, microscope, telescope	Door peephole (wide view of outside)
Spectacles for **hypermetropia**, that is long-sightedness	Galilean telescope eyepiece
Focusing sunlight to burn paper	Combined with convex lenses in cameras to sharpen the image

× WRONG	✓ RIGHT
Concave lens forms a real image too	Concave lens (real object): always virtual and erect
Object at $2F_1$ gives image at F_2	Object at $2F_1$ gives image at $2F_2$, same size
Ray through O drawn bent	Ray through the optical centre goes straight
Virtual rays drawn as solid lines	Back-extended rays are always dashed

6 PRACTICE MCQS

1. A convex lens is also called a: ★★★

(a) diverging lens (b) converging lens
(c) plane lens (d) cylindrical lens

2. A ray passing through the optical centre of a lens: ★★★

(a) bends towards the axis (b) bends away from the axis
(c) goes undeviated (d) is reflected

3. An object is placed at $2F_1$ of a convex lens. The image is formed: ★★★

(a) at F_2 (b) at $2F_2$
(c) beyond $2F_2$ (d) at infinity

4. A magnifying glass uses a convex lens with the object placed: ★★★

(a) beyond $2F_1$ (b) at $2F_1$
(c) at F_1 (d) between F_1 and O

5. For a real object, a concave lens always forms an image that is: ★★★

(a) real and enlarged
(b) virtual, erect, diminished
(c) real, inverted, same size
(d) virtual and enlarged

6. For a real, inverted and enlarged image, the object must be placed: ★★☆

(a) between F_1 and $2F_1$
(b) beyond $2F_1$
(c) between F_1 and O
(d) at infinity

7. A convex lens forms the image at infinity when the object is: ★★☆

(a) at $2F_1$
(b) at F_1
(c) beyond $2F_1$
(d) at the optical centre

8. A ray parallel to the principal axis, after passing through a concave lens: ★★☆

(a) passes through F_2
(b) appears to come from F_1
(c) goes undeviated
(d) becomes parallel to the lens

9. **Assertion (A):** A convex lens can form both real and virtual images.
Reason (R): The nature of the image formed by a convex lens depends on the position of the object. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The image formed by a concave lens can be obtained on a screen.
Reason (R): A concave lens diverges the rays falling on it. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Draw a ray diagram to show the image formed by a convex lens when the object is placed between F_1 and $2F_1$. State the position, size and nature of the image. **(3 marks)**
2. Name the lens that can give (i) a real, inverted image of the same size (ii) a virtual, erect, diminished image. Where must the object be in each case? **(2 marks)**
3. Define principal focus and focal length of a concave lens. Why is its focus called virtual? **(3 marks)**

★★★ *Har ray pe arrow, virtual ray dashed: diagram ke poore marks yahi dilate hain, Nitesh Bhaiya ki guarantee!*

LEVEL 25 COMPLETE +125 XP earned. Next: Level 26

BUJHO *Samajh gaya! Do rays kaafi hain, aur concave lens hamesha chhoti virtual image deta hai.*

PAHELI *Shabaash! Kal Level 26: lens formula aur power.*

Answer key: MCQs

Q	Ans	Why
1	(b)	It bends parallel rays to meet at F_2.
2	(c)	Rule 3: no deviation at O.
3	(b)	Case 3: at $2F_2$, same size.
4	(d)	Case 6: virtual, erect, enlarged.
5	(b)	Concave lens, real object: always so.
6	(a)	Case 4: image beyond $2F_2$.
7	(b)	Case 5: rays emerge parallel.
8	(b)	Rule 1 for a concave lens.
9	(a)	R explains why both kinds occur.
10	(d)	Its image is virtual; R is true.

Model answers: short and long answer

1 Object between F_1 and $2F_1$

- Diagram as in Case 4: parallel ray through F_2, ray through O undeviated
- Image beyond $2F_2$, **enlarged, real and inverted**

2 Which lens?

- (i) Convex lens, object at $2F_1$
- (ii) Concave lens, object anywhere in front of it

3 Focus of a concave lens

- Rays parallel to the axis appear to diverge from a point F_1 on the axis after refraction: the principal focus
- Distance OF_1 is the focal length
- It is virtual because the rays do not actually pass through it; they only seem to come from it

Day 25 checklist: tick before you sleep

- ☐ I can tell a convex lens from a concave lens in three ways
- ☐ I can define optical centre, principal focus and focal length
- ☐ I can use the three ray rules for both lenses
- ☐ I can draw all six convex cases and both concave cases
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2–3 ke diagrams phir se banao.

NITESH SIR SAYS
Lens ka diagram ratna nahi, banana hai. Har din ek case pencil se banao, ek hafte mein haath khud chalega. ***Kal milte hai Day 26 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Light chapter board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 26 LEVEL 26

PHYSICS | CHAPTER 5 | PART 6 ASTRA SERIES

LENS FORMULA & POWER

Goal: solve lens numericals with correct signs and find power.

by Nitesh Prakash

01 Sign Rules | 02 Lens Formula | 03 Magnification | 04 Power | 05 Numericals | +125 XP today | Level 26 / 60

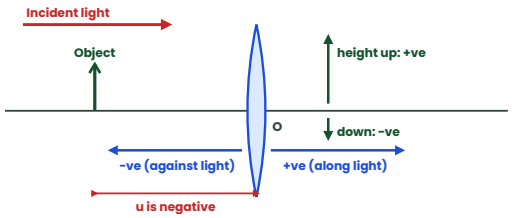
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 SIGN CONVENTION FOR LENSES

New Cartesian sign convention (5 points) ★★★

1. All distances are measured from the **optical centre O**.
2. Distances measured **in the direction of incident light** (left to right) are **positive**.
3. Distances measured **against** the incident light are **negative**, so u is always negative for a real object.
4. Heights measured **upwards** from the principal axis are **positive**.
5. Heights measured **downwards** are **negative**.

Figure 1: Signs at a glance



Sign table (from class notes) ★★★

Quantity	Convex lens	Concave lens
Object distance u	−	−
Image distance v	+ (real image)	−
Focal length f	+	−
Object height h	+	+
Image height h'	− (real image)	+

Note: the convex-lens row in the notes is for a **real** image. When a convex lens forms a **virtual** image (object inside F), v is negative and h' positive.

5 steps for every lens numerical ★★★

1. Write data with signs: u negative; f + for convex, − for concave.
2. Write the formula first: $1/v - 1/u = 1/f$.
3. Put values **in brackets**: $1/v = 1/(+10) + 1/(-15)$.
4. Take LCM, solve, keep the sign of the answer.
5. Read the sign: v + means real image, v − means virtual image.

2 LENS FORMULA AND MAGNIFICATION

DEFINITION

> **Lens formula:** relation between object distance u, image distance v and focal length f (derivation not required)

$$1/v - 1/u = 1/f$$

CLASS TRICK: FORMULA KO ULTA-SEEDHA KARO

1/v − 1/u = 1/f ko seedha 1/v = 1/f + 1/u likh lo; v nikaalna ho to yahi form sabse fast hai.

Magnification (m) ★★★

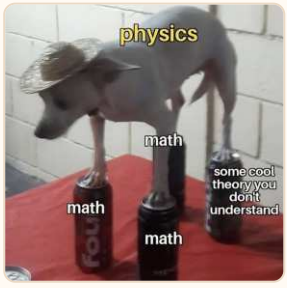
The ratio of the height of the image to the height of the object; for a lens it also equals v/u. It has **no unit**.

$$m = h'/h = v/u$$

If	It means
m negative	real and inverted image
m positive	virtual and erect image
$\|m\| > 1, = 1, < 1$	enlarged, same size, diminished

COMPARE: MIRROR VS LENS

Mirror: $1/v + 1/u = 1/f$ and $m = -v/u$. Lens: $1/v - 1/u = 1/f$ and $m = +v/u$. Do not mix them up.



MEME BREAK

Lens formula physics hai, par chalti maths pe hai: 1/v − 1/u = 1/f. Sign galat, poora answer galat!

BUJHO *Object ki doori 15 cm hai, to u = +15 likh doon?*

PAHELI *Nahi Bujho! Object light ke ulta side hai, isliye u = −15 cm. Real object ke liye u hamesha negative.*

3 SOLVED NUMERICALS: LENS FORMULA

Example 1: convex lens, real image NCERT ★★★

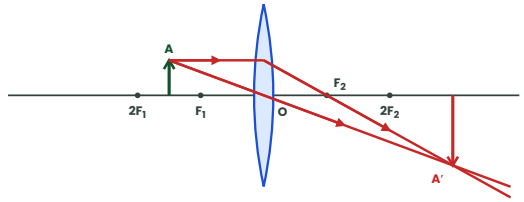
Q. A 2.0 cm tall object is placed 15 cm from a convex lens of focal length 10 cm. Find the position, size and nature of the image.

Given: h = +2.0 cm, u = −15 cm, f = +10 cm

- $1/v = 1/f + 1/u = 1/10 + 1/(−15) = (3 − 2)/30 = 1/30$, so v = +30 cm
- $m = v/u = (+30)/(−15) = −2$
- $h' = m \times h = (−2)(2.0) = −4.0$ cm

Answer: Image 30 cm on the other side of the lens; real, inverted, twice the size (4.0 cm).

Figure 2: Example 1 as a ray diagram



Object between F_1 and $2F_1$ (u = −15 cm, f = 10 cm): image beyond $2F_2$ (v = +30 cm), real, inverted, enlarged.

MEME BREAK

*Signs pehle likho, formula baad mein. Tab numerical **first attempt** mein solve hoga!*

Example 2: concave lens NCERT ★★★

Q. A concave lens has focal length 15 cm. At what distance should the object be placed so that it forms an image 10 cm from the lens? Also find the magnification.

Given: f = −15 cm, v = −10 cm (virtual image, same side)

- $1/u = 1/v − 1/f = 1/(−10) − 1/(−15) = (−3 + 2)/30 = −1/30$, so u = −30 cm
- $m = v/u = (−10)/(−30) = +1/3 \approx +0.33$

Answer: Object 30 cm in front of the lens; image virtual, erect, one-third the size.

BUJHO *v ka answer negative aaya, matlab maine galti ki?*

PAHELI *Bilkul nahi! Negative v ka matlab image **virtual** hai, object wali side. Sign hi to answer bata raha hai.*

Example 3: find size of image NCERT ★★★

Q. A 5 cm tall object is placed 25 cm from a converging lens of focal length 10 cm. Find the position, size and nature of the image.

Given: h = +5 cm, u = −25 cm, f = +10 cm

- $1/v = 1/10 + 1/(−25) = (5 − 2)/50 = 3/50$, so v = +16.7 cm
- $m = 16.7/(−25) = −0.67$, so $h' = −0.67 \times 5 = −3.3$ cm

Answer: Image 16.7 cm behind the lens; real, inverted, diminished (3.3 cm).

Example 4: the magnifying glass case ★★☆

Q. An object is placed 6 cm from a convex lens of focal length 10 cm. Find the image distance and magnification.

Given: u = −6 cm, f = +10 cm

- $1/v = 1/10 + 1/(−6) = (3 − 5)/30 = −2/30$, so v = −15 cm
- $m = (−15)/(−6) = +2.5$

Answer: v negative: image virtual, erect, 2.5 times enlarged, on the object side (case 6).

Example 5: camera lens (PYQ) CBSE SQP 2025 ★★★

Q. A flower 80 mm tall is 150 mm from a camera lens. Its image on the sensor is 20 mm tall. Find the focal length of the lens.

Given: h = +80 mm, h' = −20 mm (real, inverted image), u = −150 mm

- $m = h'/h = −20/80 = −1/4$
- $v = m \times u = (−1/4)(−150) = +37.5$ mm
- $1/f = 1/v − 1/u = 1/37.5 + 1/150 = (4 + 1)/150 = 1/30$

Answer: f = +30 mm, so it is a convex (converging) lens.

Example 6: same-size image NCERT ★★★

Q. A convex lens forms a real, inverted image of a needle 50 cm from it. Where is the needle placed if the image is the same size as the object? Find the power of the lens.

Given: v = +50 cm, m = -1 (real, inverted, same size)

- $m = v/u = −1$, so u = −50 cm (needle at $2F_1$)
- $1/f = 1/50 − 1/(−50) = 2/50$, so f = +25 cm
- $P = 100/25 = +4$ D

Answer: Needle 50 cm in front of the lens; P = +4 D.

Formula sheet: mirror vs lens ★★★

Item	Spherical mirror	Spherical lens
Main formula	$1/v + 1/u = 1/f$	$1/v − 1/u = 1/f$
Magnification	$m = −v/u$	$m = v/u$
Measured from	pole P	optical centre O
f positive for	convex mirror	convex lens
Power used?	no	yes, $P = 1/f$

UNIT ALERT

Keep u, v, f in one unit; use metre only for power.

4 POWER OF A LENS

DEFINITION

- **Power (P):** the degree of convergence or divergence of light rays achieved by a lens. It is the reciprocal of focal length.
- **SI unit:** dioptre (D). **1 dioptre** is the power of a lens whose focal length is 1 metre (1 D = 1 m^{-1}).

$$P = 1/f \text{ (in metre)} = 100/f \text{ (in cm)}$$

CLASS TRICK: N.B.

Concave lens ka power hamesha −ve, convex lens ka power +ve. Sign dekho, lens pehchano!

BUJHO *Focal length 20 cm hai, to power = 1/20 = 0.05 D?*

PAHELI *Galti! f metre mein lo: 20 cm = 0.2 m, to* ***P = 1/0.2 = +5 D****. Shortcut: 100/20 = 5 D.*

Example 7: power and focal length NCERT ★★☆

Q. (i) Find the power of a convex lens of focal length 50 cm. (ii) A lens has power −2.0 D. Find its focal length and type.

Given: (i) $f = +50$ cm (ii) $P = -2.0$ D

- (i) $P = 100/f(\text{cm}) = 100/50 = +2$ D
- (ii) $f = 1/P = 1/(-2.0) = -0.5$ m $= -50$ cm

Answer: (i) +2 D (ii) −50 cm; negative, so it is a concave lens.

Example 8: dioptre (PYQ) CBSE 2020 ★★☆

Q. Define 1 dioptre power of a lens. Find the focal length of a lens of power +0.5 D.

Given: $P = +0.5$ D

- 1 D is the power of a lens of focal length 1 m
- $f = 1/P = 1/(+0.5) = +2$ m

Answer: $f = +2$ m (200 cm); positive, so a convex lens.

Power of a combination of lenses ★★★

When lenses are placed **in contact**, the net power is the **algebraic sum** of the individual powers:

$$P = P_1 + P_2 + P_3 + \dots$$

- ✔ Opticians use this: they put test lenses together to find the power of the spectacles you need.
- ✔ Cameras and microscopes use lens combinations to get a larger, sharper image.

Example 9: combination ★★☆

Q. A convex lens of power +2.0 D is kept in contact with a concave lens of power −0.5 D. Find the power and focal length of the combination.

Given: $P_1 = +2.0$ D, $P_2 = -0.5$ D

- $P = P_1 + P_2 = +2.0 + (-0.5) = +1.5$ D
- $f = 1/P = 1/1.5 = +0.67$ m $= +66.7$ cm

Answer: +1.5 D and 66.7 cm; the combination behaves like a convex lens.

MEME BREAK
Ramanujan wali maths nahi chahiye: bas ***P = 1/f****, aur f metre mein!*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

A spectacle prescription is written in dioptres. A reading of **−1.5 D** means a concave lens of $f = 1/(-1.5) = -0.67$ m; a reading of **+2.0 D** means a convex lens of $f = +0.5$ m. You will meet these lenses again in the chapter on the human eye.

Activity: find f of a convex lens in 2 minutes ★★☆

1. Hold a convex lens facing a far-off object such as a tree or a window across the room.
2. Move a paper screen behind the lens till a sharp, inverted image appears.
3. Measure the distance from lens to screen. Rays from a far object are nearly parallel, so this distance is about the focal length.
4. Work out the power: if $f = 20$ cm, then $P = 100/20 = +5$ D.

Caution: never look at the Sun through a lens.

× WRONG	✓ RIGHT
$u = +15$ cm for a real object	$u = -15$ cm (against the light)
Lens: $m = -v/u$	Lens: $m = +v/u$; the minus sign is for mirrors
$P = 1/f$ with f in cm	$P = 1/f$ in metre, or $100/f$ in cm
Powers in contact are multiplied	Powers add: $P = P_1 + P_2$
Concave lens power is positive	Concave lens power is negative

Try it yourself (answers in brackets) ★★☆

1. Convex lens, $f = 15$ cm, object at 30 cm. Find v. (Ans: +30 cm)
2. Concave lens, $f = 10$ cm, object at 10 cm. Find v. (Ans: -5 cm)
3. Find the power of a lens of $f = -40$ cm. (Ans: -2.5 D)
4. Lenses of +5 D and −3 D in contact. Find f. (Ans: +50 cm)

★★★ *Sign convention + formula + unit: teeno likhe to numerical ke poore marks pakke, Nitesh Bhaiya ki guarantee!*

MEME BREAK
Numericals ki practice skip karke seedha kal ka ***Boss Level****? Phir paper dekh ke Tom wala yahi chehra banega!*

5 PRACTICE MCQS

1. The SI unit of power of a lens is: NCERT ★★★
(a) metre (b) dioptre
(c) watt (d) joule

2. The lens formula is: NCERT ★★★
(a) $1/v + 1/u = 1/f$ (b) $v - u = f$
(c) $1/v - 1/u = 1/f$ (d) $1/u - 1/v = 1/f$

3. A lens has power −2.5 D. It is a: ★★★
(a) concave lens, $f = -40$ cm (b) convex lens, $f = +40$ cm
(c) concave lens, $f = -25$ cm (d) convex lens, $f = +25$ cm

4. A lens gives magnification +0.5. The image is: ★★☆
(a) real, inverted, diminished (b) real, inverted, enlarged
(c) virtual, erect, enlarged (d) virtual, erect, diminished

5. For a real object placed in front of a lens, u is always: ★★☆

(a) positive
(b) negative
(c) zero
(d) positive for a convex lens only

6. Lenses of power +3 D and −1 D are kept in contact. The net power is: ★★☆

(a) +4 D
(b) −2 D
(c) +2 D
(d) +3 D

7. An object is 40 cm from a convex lens of focal length 20 cm. The image is: ★★☆

(a) 40 cm behind the lens
(b) 20 cm behind the lens
(c) 40 cm in front of the lens
(d) at infinity

8. A lens forms a virtual, erect image 3 times the size of the object. Its magnification is: ★★☆

(a) −3
(b) −1/3
(c) +1/3
(d) +3

9. The power of a convex lens of focal length 25 cm is: NCERT ★★★

(a) +0.25 D
(b) +4 D
(c) −4 D
(d) +25 D

10. **Assertion (A):** The power of a concave lens is negative.
Reason (R): The focal length of a concave lens is taken as negative in the New Cartesian sign convention. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Define 1 dioptre. Find the focal length of a lens of power −2.0 D. What type of lens is it? NCERT **(2 marks)**
2. Find the power of a concave lens of focal length 2 m. NCERT **(1 mark)**
3. A 4 cm tall object is placed 30 cm from a convex lens of focal length 20 cm. Find the position, size and nature of the image. **(3 marks)**
4. An object is placed 20 cm from a concave lens of focal length 20 cm. Find the image distance and magnification. **(3 marks)**
5. Two thin lenses of power +3.5 D and −2.5 D are placed in contact. Find the power and focal length of the combination. NCERT **(2 marks)**

LEVEL 26 COMPLETE +125 XP earned. Next: Level 27

BUJHO *Samajh gaya! Pehle signs, phir formula, aur power ke liye f metre mein.*

PAHELI *Shabaash! Lenses ka chapter ab tumhari jeb mein.*

Answer key: MCQs

Q	Ans	Why
1	(b)	1 D = 1 m^{-1}.
2	(c)	Lens: minus sign between $1/v$ and $1/u$.
3	(a)	$f = 1/(-2.5) = -0.4$ m.
4	(d)	+ means erect and virtual; 0.5 means smaller.
5	(b)	It is measured against the incident light.
6	(c)	$P = 3 + (-1) = +2$ D.
7	(a)	$1/v = 1/20 - 1/40 = 1/40$.
8	(d)	Erect means +, enlarged 3 times.
9	(b)	100/25 = +4 D.
10	(a)	$P = 1/f$ and f is negative.

Model answers: short and long answer

1 **Dioptre and −2.0 D**
- 1 D is the power of a lens of focal length 1 m
- $f = 1/(-2.0) = -0.5$ m, so it is a **concave lens**

2 **Concave lens, f = −2 m**
- $P = 1/f = 1/(-2) =$ **−0.5 D**

3 **u = −30 cm, f = +20 cm, h = +4 cm**
- $1/v = 1/20 - 1/30 = 1/60$, so $v =$ **+60 cm**
- $m = 60/(-30) = -2$, $h' =$ **−8 cm**: real, inverted, enlarged

4 **u = −20 cm, f = −20 cm**
- $1/v = 1/(-20) + 1/(-20) = -1/10$, so $v =$ **−10 cm**
- $m = (-10)/(-20) =$ **+0.5**: virtual, erect, half size

Day 26 checklist: tick before you sleep

- ☐ I can state the New Cartesian sign convention for lenses
- ☐ I can write the lens formula and the magnification formula
- ☐ I can solve a lens numerical with correct signs
- ☐ I can find power in dioptre and tell the lens type from its sign
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2 ke numericals phir se karo.

NITESH SIR SAYS
Numerical mein marks steps ke milte hain: data with signs, formula, substitution, unit. Ek bhi step mat chhodo. ***Kal milte hai Day 27, Boss Level pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Light chapter board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 27 LEVEL 27

PHYSICS | CHAPTER 5 | BOSS LEVEL

CHAPTER 5 BOSS LEVEL

Goal: the whole Light chapter in 75 minutes.

by Nitesh Prakash

01 Revision Map | 02 MCQs | 03 Short Answers | 04 Long Answers | 05 Answer Key | +150 XP today | Level 27 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 5 REVISION MAP

Chapter 5 mind map: revise in 5 minutes

REFLECTION
Angle i = angle r; incident ray, normal, reflected ray in one plane. Plane mirror: virtual, erect, same size, laterally inverted

MIRRORS
R = 2f. Concave: converging, f negative. Convex: diverging, f positive, always virtual, erect, diminished

MIRROR FORMULA ★★★
1/v + 1/u = 1/f; m = h′/h = −v/u. m negative: real and inverted; m positive: virtual and erect

LIGHT

REFRACTION
Snell: sin i / sin r = constant. n = c/v. Rarer to denser: bends towards the normal

LENSES
Convex: converging, f positive. Concave: diverging, f negative, always virtual, erect, diminished

LENS FORMULA & POWER ★★★
1/v − 1/u = 1/f; m = h′/h = v/u. P = 1/f (f in metres), unit dioptre (D). Convex P positive, concave P negative

New Cartesian sign convention: the scorecard

Measure every distance from the pole P (mirror) or optical centre O (lens). Along the incident light = positive; against it = negative.

Quantity	Sign
u (real object)	always negative
f concave mirror, concave lens	negative
f convex mirror, convex lens	positive
Height above the axis	positive
Real image (mirror)	v negative
Real image (lens)	v positive

Image cases: concave mirror

Object at	Image at	Nature, size
Infinity	F	real, inverted, point-sized
Beyond C	Between F and C	real, inverted, diminished
C	C	real, inverted, same size
Between C and F	Beyond C	real, inverted, enlarged
F	Infinity	real, inverted, highly enlarged
Between P and F	Behind the mirror	virtual, erect, enlarged

BUJHO *Boss Level? Mirror ka formula yaad hai, lens ka bhi same hi hoga na?*

PAHELI *Nahi! Mirror: 1/v + 1/u. Lens: 1/v − 1/u. Ek sign ki galti aur poora numerical gaya.*

BOSS LEVEL RULES

- ✔ **Time:** 75 minutes. **Maximum marks:** 40. Close the book and set a timer.
- ✔ **A:** 14 MCQs × 1 = 14. **B:** 5 short answers = 12. **C:** 2 long answers × 5 = 10. **D:** 1 case-based question = 4.
- ✔ Write Given (with signs), Formula, Substitution and Answer with unit in every numerical. Draw ray diagrams with a pencil and arrows.

Mirror-lens ka har doubt? Sign convention ka flex tape lagao, sab fix!

2 SECTION A: MCQS

1. The image formed by a plane mirror is: ★★★

(a) real, inverted, same size (b) virtual, erect, same size
(c) virtual, inverted, magnified (d) real, erect, diminished

2. The radius of curvature of a spherical mirror is 20 cm. Its focal length is: ★★☆

(a) 40 cm (b) 20 cm
(c) 10 cm (d) 5 cm

3. A spherical mirror and a thin spherical lens each have a focal length of −15 cm. The mirror and the lens are likely to be: NCERT ★★★

(a) both concave (b) both convex
(c) mirror concave, lens convex (d) mirror convex, lens concave

4. No matter how far you stand from a mirror, your image always appears erect. The mirror is: NCERT ★★☆

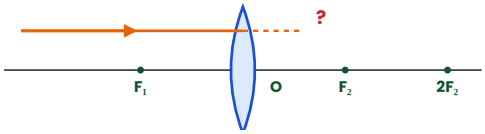
(a) only plane (b) only concave
(c) only convex (d) either plane or convex

5. When light enters glass from air, it: ★★★
(a) speeds up and bends away from the normal
(b) slows down and bends towards the normal
(c) speeds up and bends towards the normal
(d) slows down and does not bend

6. The refractive index of diamond is 2.42. This means: ★★☆
(a) light travels 2.42 times faster in diamond
(b) speed of light in diamond is 1/2.42 of its speed in vacuum
(c) diamond reflects 2.42 times more light
(d) diamond absorbs light

7. An object is at $2F_1$ of a convex lens. Where is the image? ★★★
(a) F_2, point-sized
(b) between F_2 and $2F_2$
(c) $2F_2$, same size
(d) beyond $2F_2$, enlarged

8. A ray parallel to the principal axis falls on a convex lens as shown. After refraction it will: ★★★
(a) pass through F_2
(b) pass through $2F_2$
(c) go undeviated
(d) appear to come from F_1



9. The SI unit of the power of a lens is: ★★☆
(a) metre
(b) dioptre
(c) watt
(d) candela

10. To get a virtual, erect and enlarged image from a convex lens, the object must be placed: ★★★
(a) at $2F_1$
(b) between F_1 and $2F_1$
(c) between F_1 and O
(d) at infinity

Must-know formulas: write them daily

Mirror	Lens
$1/v + 1/u = 1/f$	$1/v - 1/u = 1/f$
$m = h'/h = -v/u$	$m = h'/h = v/u$
$f = R/2$	$P = 1/f$ (f in m), 1 D = 1 m^{-1}
$n = c/v = 3 \times 10^8 / v$	$n_{21} = n_2/n_1 = v_1/v_2$

How to attack Sections B, C and D

- ✔ 2 marks = two clear points; 3 marks = three points, or a full numerical with steps.
- ✔ Numericals: Given with signs, formula, substitution, answer with unit and the nature of the image.
- ✔ Ray diagrams: two standard rays, arrows on every ray, dashed virtual rays, labels P, F, C or O, F_1, F_2.

Light ka nature samajhne gaye the, MCQs ne dimaag ka bulb jala diya!

11. Light travels fastest in which medium? (n: water 1.33, kerosene 1.44, crown glass 1.52, diamond 2.42) ★★☆
(a) water
(b) kerosene
(c) crown glass
(d) diamond

12. A mirror gives m = +0.4 for every position of the object. The mirror is: ★★☆
(a) plane
(b) concave
(c) convex
(d) either concave or plane

13. **Assertion (A):** A concave mirror can form a virtual image. **Reason (R):** When the object is between P and F, the reflected rays diverge and appear to meet behind the mirror. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

14. **Assertion (A):** The power of a concave lens is negative. **Reason (R):** A concave lens is a converging lens. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Image cases: convex lens

Object at	Image at (convex lens)
Infinity	F_2: real, inverted, point-sized
Beyond $2F_1$	Between F_2 and $2F_2$: diminished
$2F_1$	$2F_2$: real, inverted, same size
Between F_1 and $2F_1$	Beyond $2F_2$: real, enlarged
F_1	Infinity: highly enlarged
Between F_1 and O	Same side: virtual, erect, enlarged

3 SECTION B: SHORT ANSWERS

15 State the two laws of refraction. What is meant by the absolute refractive index of a medium? **(2 marks)**

16 Draw a ray diagram to show the image formed by a concave mirror when the object is placed between C and F. State the nature of the image. **(2 marks)**

17 The rear-view mirror of a bus is convex with radius of curvature 3 m. A car is 6 m behind it. Find the position, nature and magnification of the image. **(3 marks)**

18 An object is placed 10 cm from a concave lens of focal length 15 cm. Find the image distance and magnification. State the nature of the image. **(3 marks)**

19 Give two uses each of concave and convex mirrors with one reason for each. **(2 marks)**

4 SECTION C: LONG ANSWERS

20 (A) A 4 cm tall object is placed 30 cm in front of a concave mirror of focal length 20 cm. Find the position, nature and size of the image. (B) Draw the ray diagram. **(5 marks)**

21 (A) An object 3 cm high is placed 20 cm from a convex lens of focal length 15 cm. Find the position, size and nature of the image. (B) Draw the ray diagram. (C) Find the power of the lens. **(5 marks)**

5 SECTION D: CASE-BASED QUESTION

Case: the refractive index chart (4 marks)

Riya measured the absolute refractive index of some media in her school lab:

Medium	Air	Water	Kerosene	Crown glass	Diamond
n	1.0003	1.33	1.44	1.52	2.42

22 Which medium is optically the densest? **(1)**

23 Find the speed of light in crown glass ($c = 3 \times 10^8$ m/s). **(1)**

24 A ray goes from water into kerosene. Does it bend towards or away from the normal? **(1)**

25 Find the refractive index of diamond with respect to water. **(1)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Plane mirror: virtual, erect, same size, laterally inverted.
2	(c)	f = R/2 = 20/2 = 10 cm.
3	(a)	Negative f: concave mirror and concave lens.
4	(d)	Plane and convex mirrors always give erect images.
5	(b)	Glass is optically denser: speed falls, ray bends towards the normal.
6	(b)	n = c/v, so v = c/2.42.
7	(c)	Object at $2F_1$ gives a same-size image at $2F_2$.
8	(a)	A ray parallel to the axis passes through F_2.
9	(b)	1 D = 1 m^{-1}.
10	(c)	Object between F_1 and O: virtual, erect, enlarged.
11	(a)	Lowest n means highest speed: water.
12	(c)	m positive and less than 1 for every position: convex mirror.
13	(a)	R correctly explains the virtual image.
14	(c)	A is true; R is false: a concave lens is diverging.

Model answers: Section C

20 Concave mirror numerical

Given: h = +4 cm, u = −30 cm, f = −20 cm
Formula: 1/v + 1/u = 1/f, so 1/v = 1/f − 1/u
1/v = −1/20 + 1/30 = (−3 + 2)/60 = −1/60
v = −60 cm: 60 cm in front of the mirror
m = −v/u = −(−60)/(−30) = **−2**; h′ = m × h = **−8 cm**
Image: real, inverted, enlarged (twice). (B) Ray diagram: as in answer 16 (object between C and F).

Model answers: Section D

- (22.1) Diamond (highest n = 2.42).
- (22.2) v = c/n = (3 × 10⁸)/1.52 ≈ **1.97 × 10⁸ m/s**.
- (22.3) Towards the normal: kerosene (1.44) is optically denser than water (1.33).
- (22.4) n(diamond, water) = 2.42/1.33 ≈ **1.82**.

Sign convention skip kiya? Clown makeup complete! Har answer ka sign check karo.

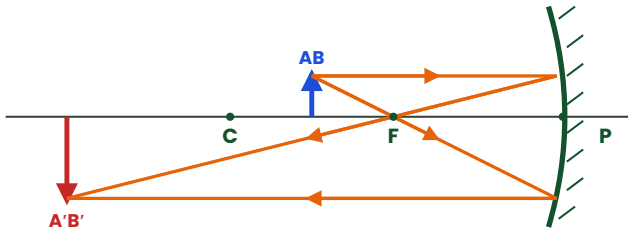
Model answers: Section B

15 Laws of refraction

- (1) The incident ray, the refracted ray and the normal at the point of incidence lie in the same plane.
- (2) Snell's law: for a given pair of media and a given colour, sin i / sin r = constant (the refractive index of the second medium with respect to the first).
- Absolute refractive index: n = speed of light in vacuum / speed of light in the medium = c/v.

16 Concave mirror, object between C and F

- Image: beyond C, real, inverted and enlarged.



Ray 1: parallel to the axis, reflects through F. Ray 2: through F, reflects parallel to the axis.

Model answers: Section B (numericals)

17 Convex rear-view mirror

Given: R = +3 m, so f = R/2 = +1.5 m; u = −6 m
Formula: 1/v + 1/u = 1/f, so 1/v = 1/f − 1/u
1/v = 1/1.5 − 1/(−6) = 4/6 + 1/6 = 5/6
v = +1.2 m (behind the mirror)
m = −v/u = −(1.2)/(−6) = **+0.2**
Image: virtual, erect, diminished (one-fifth the size).

18 Concave lens

Given: f = −15 cm, u = −10 cm
Formula: 1/v − 1/u = 1/f, so 1/v = 1/f + 1/u
1/v = −1/15 − 1/10 = (−2 − 3)/30 = −1/6
v = −6 cm (same side as the object)
m = v/u = (−6)/(−10) = **+0.6**
Image: virtual, erect, diminished.

19 Uses of spherical mirrors

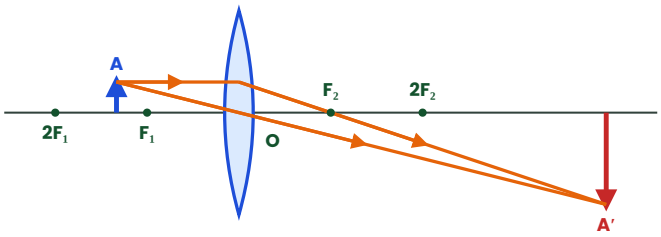
- Concave: torches and headlights (bulb at F gives a strong parallel beam); shaving and dentist's mirrors (enlarged, erect image when close); solar furnaces (concentrate sunlight).
- Convex: rear-view mirrors of vehicles: always erect image and a much wider field of view.

Light: kabhi wave, kabhi particle. Board exam mein bas sahi ray diagram banao, observer khush!

Model answers: Section C (continued)

21 Convex lens numerical

Given: h = +3 cm, u = −20 cm, f = +15 cm
Formula: 1/v − 1/u = 1/f, so 1/v = 1/f + 1/u
1/v = 1/15 − 1/20 = (4 − 3)/60 = 1/60
v = +60 cm: on the other side of the lens
m = v/u = 60/(−20) = **−3**; h′ = m × h = **−9 cm**
Image: real, inverted, enlarged (three times).
(C) P = 1/f = 1/(+0.15 m) = **+6.67 D**



(B) Object between F_1 and $2F_1$: image beyond $2F_2$. Ray 1 parallel, then through F_2; ray 2 through O, undeviated.

Most asked in Chapter 5: Eduspark PYQ analysis

Topic	Day	Rating
Mirror formula numericals	22	★★★
Lens formula and power numericals	25	★★★
Ray diagrams: concave mirror, convex lens	22, 24	★★★
Refractive index, n = c/v	23	★★★
Uses of concave and convex mirrors	21	★★☆
Laws of refraction, Snell's law	23	★★☆
Image by concave lens and convex mirror	22, 24	★★☆

Common mistakes that cost marks

× WRONG	✓ RIGHT
u written as +30 cm	u is always negative for a real object: −30 cm
Lens formula 1/v + 1/u = 1/f	Lens: 1/v − 1/u = 1/f; mirror: 1/v + 1/u = 1/f
Mirror m = v/u	Mirror m = −v/u; lens m = v/u
P = 1/f with f in cm	Convert f to metres first: P = 100/f(cm)
No arrows on rays	Every ray needs an arrow; virtual rays are dashed

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 5 board and sample-paper questions.

BUJHO *Convex mirror ki image screen pe kyun nahi aati? Main toh try karta rahunga!*

PAHELI *Kyunki wo virtual hai, rays actually milti hi nahi. Screen pe sirf real image aati hai.*

Day 27 checklist: tick before you sleep

- ☐ I revised the mind map, the sign table and all formulas
- ☐ I attempted all 14 MCQs without looking at the answer key
- ☐ I solved every numerical with Given, Formula, Substitution and Answer
- ☐ I drew both ray diagrams neatly from memory with arrows and labels
- ☐ I scored myself honestly and revised the days of my wrong answers

My score: ____ / 5

5/5 = Jalwa hai! Light chapter ka boss ab hamara hai!

LEVEL 27 COMPLETE **CHAPTER 5 COMPLETE**

+150 XP earned. Next: Level 28

BUJHO *Samajh gaya! Mirror mein −v/u, lens mein v/u. Sign convention hi asli hero hai!*

PAHELI *Boss defeated! Kal se Chapter 6: hamari apni aankh ka camera.*

Bonus from class notes: solved example

B Concave mirror: where to place the screen?

A 7 cm tall object is 27 cm in front of a concave mirror of focal length 18 cm. Where should the screen be placed for a sharp image? Find the image size.

u = −27 cm, f = −18 cm, h = +7 cm
1/v = 1/f − 1/u = −1/18 + 1/27 = (−3 + 2)/54 = −1/54
v = −54 cm: screen 54 cm in front of the mirror
m = −v/u = −(−54)/(−27) = −2; **h′ = −14 cm** (real, inverted, enlarged)

Bonus rapid-fire round: quick answers

Question	Answer
Speed of light in vacuum	3×10^8 m/s
Mirror used by dentists	Concave
Lens in a magnifying glass	Convex
Unit of refractive index	None (a ratio)
Ray through the optical centre	Goes undeviated
Power of a lens with f = −25 cm	−4 D (concave)

NITESH SIR SAYS
Chapter 5 khatam! Galat numericals dobara solve karo, sign ke saath. ***Consistency hi jalwa hai!***

PHYSICS | CHAPTER 5 COMPLETE ASTRA SERIES

CHAPTER 5 SWAHAA!

Light: Reflection and Refraction | Days 21–27

BUJHO *Mirror, lens, sign convention, numericals sab clear! Ab concave aur convex mein kabhi confuse nahi hounga.*

PAHELI *Shabaash, Bujho! Light ka chapter Swahaa. Ab chalo aankh ki duniya mein, Human Eye wait kar raha hai.*

CHAPTER XP EARNED
+900 XP

LEVELS CLEARED
7 levels (Days 21–27)

BOOK PROGRESS
Level 27 / 60 3400 XP total

WHAT YOU HAVE MASTERED

- ✔ Laws of reflection, plane mirror images
- ✔ Mirror formula, magnification, sign convention
- ✔ Convex and concave lenses, image formation
- ✔ Concave and convex mirrors, ray diagrams
- ✔ Refraction, Snell's law, refractive index
- ✔ Lens formula, magnification, power of a lens

NEXT CHAPTER Day 28: The Human Eye and the Colourful World — *Light cleared!*

DAY 28 · LEVEL 28

PHYSICS | CHAPTER 6 | PART 1 · ASTRA SERIES

THE HUMAN EYE

Goal: know every part of the eye, then fix its defects.

by Nitesh Prakash

01 Eye Structure | 02 Accommodation | 03 Near, Far Point | 04 Defects | 05 Numericals | +125 XP today | Level 28 / 60

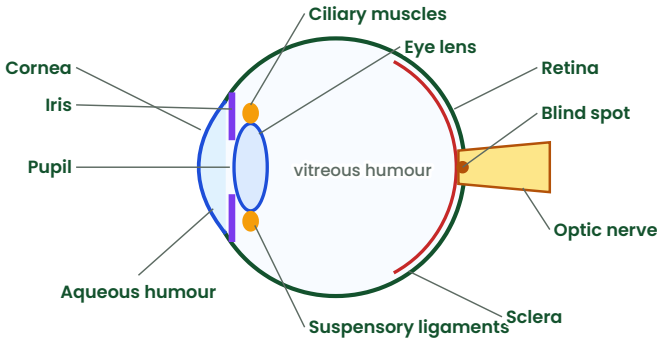
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 THE HUMAN EYE: OUR NATURAL CAMERA

DEFINITION

- **Human eye:** a natural optical instrument that lets us see objects around us. Its lens system forms a **real, inverted** image on the light-sensitive retina.
- The brain processes the signals so we see the object **erect**.

Structure of the human eye (labelled diagram)



Draw it with at least 8 labels: cornea, iris, pupil, eye lens, ciliary muscles, retina, optic nerve, blind spot.

Retina: rods and cones (from class notes)

Rod cells	Cone cells
Sensitive to **dim light**	Sensitive to **bright light**
Side (peripheral) vision	Colour and sharp central vision
No colour information	Detect colours

The retina turns the image into electrical impulses; the optic nerve takes them to the brain.

2 POWER OF ACCOMMODATION

DEFINITION

- **Power of accommodation:** the ability of the eye lens to adjust its focal length so that both near and far objects are focused on the retina.

Looking at	Ciliary muscles	Eye lens	Focal length
Distant object	relaxed	thin	increases
Nearby object	contract	thick	decreases

Range of vision of a normal eye: from 25 cm to infinity.

Parts of the eye and their functions

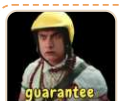
Part	Function
Sclera	Tough white outer coat; protects the eye
Cornea	Transparent front bulge of the sclera; light enters here and **most refraction** happens at its outer surface
Iris	Dark muscular ring behind the cornea; gives the eye its colour; controls the size of the pupil
Pupil	Opening in the iris; controls the light entering: **small in bright light, wide in dim light**
Eye lens	Crystalline, flexible convex lens; fine-tunes the focal length to focus the image on the retina
Ciliary muscles	Hold the lens and change its curvature, so its focal length changes
Aqueous humour	Clear fluid between the cornea and the lens
Vitreous humour	Jelly behind the lens; gives nutrients and keeps the shape of the eyeball
Retina	Light-sensitive screen with rods and cones; forms the image and makes electrical signals
Optic nerve	Carries the signals from the retina to the brain
Blind spot	Where the optic nerve leaves the retina; no rods or cones, so no vision

BUJHO *Retina pe image ulti banti hai, toh duniya ulti kyun nahi dikhti?*

PAHELI *Kyunki brain signals ko process karke image ko seedha kar deta hai. Aankh camera hai, brain editor!*

Near point and far point

- ✔ **Near point** (least distance of distinct vision): the closest point seen clearly without strain. Normal young adult: **25 cm**.
- ✔ **Far point:** the farthest point seen clearly. Normal eye: **infinity**.
- ✔ Closer than 25 cm, the lens cannot become thicker any more, so the object looks blurred.



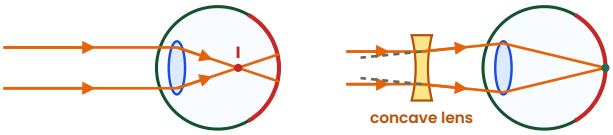
★★★ *Near point 25 cm, far point infinity, accommodation ka meaning: ye teen line har saal aati hain, Nitesh Bhaiya ki guarantee!*

3 DEFECTS OF VISION AND THEIR CORRECTION

A defect of vision (refractive defect) means the eye cannot focus the image exactly on the retina. Spectacle lenses fix it.

1. Myopia (near-sightedness) ★★★

- ✔ **Sees:** near objects clearly, but distant objects look blurred.
- ✔ **Far point** comes nearer than infinity (a few metres or less).
- ✔ **Image** of a distant object forms **in front of the retina**.
- ✔ **Causes:** (i) excessive curvature of the eye lens; (ii) elongation of the eyeball.
- ✔ **Correction:** a **concave (diverging) lens** of suitable power.



(a) Myopic eye: image in front of retina (b) Corrected: image on retina

Class notes say "excessive curvature of the cornea"; NCERT says excessive curvature of the **eye lens**. Write the NCERT line in the exam.

MEME BREAK

*Myopia mein concave, hypermetropia mein convex. Button galat dabaya toh **marks bhi blur**!*

NCERT quick Q&A

> **Q.** Why can a normal eye not see clearly an object closer than 25 cm? **A.** The ciliary muscles cannot make the lens thicker beyond a limit, so its focal length cannot decrease any further.

4 NUMERICALS ON DEFECTS OF VISION

CLASS TRICK: NUMERICAL HACK (NITESH SIR)

- ✔ **Step 1:** Find the case of defect (myopia or hypermetropia).
- ✔ **Step 2:** Find the correction lens of that defect.
- ✔ **Step 3:** If myopia: u = ∞, v = given in question. If hypermetropia: u = 25 cm, v = given in question.
- ✔ **Step 4:** Jitna bhi value aaye usme pehle negative sign laga do (such as u = −∞, u = −25 cm).
- ✔ Yaad rakho: 1/∞ = 0. Then use 1/f = 1/v − 1/u and P = 100/f (f in cm).

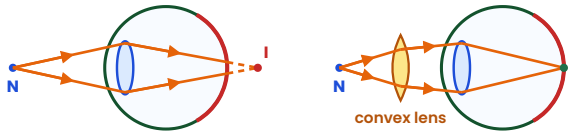
Solved Example 1 (class notes): myopia

A person cannot see objects beyond 40 cm clearly. Name the defect and find the power of the correcting lens.

Defect: myopia (far point = 40 cm), so a concave lens is needed.
Given: u = −∞, v = −40 cm (the lens must form the image of a distant object at the far point)
1/f = 1/v − 1/u = −1/40 − 0 = −1/40, so **f = −40 cm**
P = 100/f = 100/(−40) = **−2.5 D** (concave lens)

2. Hypermetropia (far-sightedness) ★★★

- ✔ **Sees:** distant objects clearly, but near objects look blurred.
- ✔ **Near point** moves farther than 25 cm.
- ✔ **Image** of a near object forms **behind the retina**.
- ✔ **Causes:** (i) focal length of the eye lens is too long; (ii) the eyeball is too small.
- ✔ **Correction:** a **convex (converging) lens** of suitable power.



(a) Hypermetropia: image behind retina (b) Corrected: image on retina

3. Presbyopia (old-age defect)

- ✔ **Sees:** near objects with difficulty; reading and writing become uncomfortable.
- ✔ **Cause:** with age the **ciliary muscles weaken** and the **eye lens loses flexibility**, so the near point recedes.
- ✔ A person may suffer from **both myopia and hypermetropia**. Class notes say near and far both blur; NCERT: mainly near vision, both only in some people.
- ✔ **Correction: bifocal lenses**: upper part concave (distant vision), lower part convex (reading).

4. Cataract

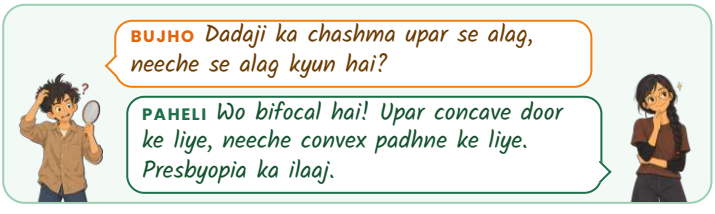
- ✔ The crystalline lens of older people often becomes **milky and cloudy**.
- ✔ Seeing through it is like looking through a frosty or fogged-up window; vision is partly or fully lost.
- ✔ **Correction:** vision is restored by **cataract surgery**.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Astigmatism (from class notes): the eye cannot focus horizontal and vertical lines at the same time, so it sees clearly in only one plane at a time. Cause: an irregularly shaped cornea or a distorted eye lens. Correction: a **cylindrical lens**.

Defects at a glance

Defect	Image	Lens
Myopia	in front of retina	concave
Hypermetropia	behind retina	convex
Presbyopia	near point recedes	bifocal
Cataract	cloudy lens	surgery



Solved Example 2 (class notes): hypermetropia

An old person can read a newspaper only when it is 80 cm away. Find the power of the lens needed to read at 25 cm.

Defect: hypermetropia (near point = 80 cm), so a convex lens is needed.
Given: u = −25 cm, v = −80 cm (image of the paper must be at his near point)
1/f = 1/v − 1/u = −1/80 + 1/25 = (−5 + 16)/400 = 11/400
f = +36.4 cm; P = 100/f = 1100/400 = **+2.75 D** (convex lens)

Check your notes: 1100/400 = 2.75, not 2.25. The correct answer is +2.75 D.

Solved Examples 3 and 4 (NCERT)

3. The far point of a myopic person is 80 cm in front of the eye. Find the nature and power of the lens needed.

u = −∞, v = −80 cm; 1/f = 1/v − 1/u = −1/80, so **f = −80 cm**
P = 100/(−80) = **−1.25 D**: a concave lens

4. The near point of a hypermetropic eye is 1 m. Find the power of the lens needed (normal near point 25 cm).

u = −25 cm, v = −100 cm
1/f = 1/v − 1/u = −1/100 + 1/25 = 3/100, so **f = +33.3 cm**
P = 100/f = 300/100 = **+3.0 D**: a convex lens

Board dhundhla, teacher dhundhli, sab dhundhla... "This is fine." Nahi bhai, eye test karao aur concave lens lagao!

Common mistakes that cost marks

× WRONG	✓ RIGHT
Myopia: image behind retina	Myopia: in front; hypermetropia: behind
Iris controls light directly	Iris controls the pupil size; the pupil lets light in
Hypermetropia: u = −∞	Hypermetropia: u = −25 cm; myopia: u = −∞
Near point of normal eye: 25 m	Near point: 25 cm; far point: infinity

EXAM TIPS: SCORE FULL MARKS

- ✓ Defect questions: write **three things**: where the image forms, two causes, and the correcting lens with a ray diagram.
- ✓ In the eye diagram, the retina is at the back and the blind spot is where the optic nerve leaves.
- ✓ Power answer: always give the sign and the lens type, e.g. −2.5 D, concave.

REMEMBER IN ONE LINE

- ✓ Myopia: image in front, concave. Hypermetropia: image behind, convex.

5 PRACTICE MCQS

1. The least distance of distinct vision for a normal young adult is about: NCERT ★★★
(a) 25 m (b) 2.5 cm
(c) 25 cm (d) 2.5 m

2. The change in focal length of the eye lens is caused by the action of the: NCERT ★★★
(a) pupil (b) retina
(c) ciliary muscles (d) iris

3. The human eye forms the image of an object at its: NCERT ★★☆
(a) cornea (b) iris
(c) pupil (d) retina

4. The eye focuses on objects at different distances by adjusting the focal length of its lens. This is due to: NCERT ★★★
(a) presbyopia (b) accommodation
(c) near-sightedness (d) far-sightedness

5. The size of the pupil is controlled by the: ★★☆
(a) cornea (b) iris
(c) retina (d) optic nerve

6. A student sees far objects clearly but not a book held close. Defect and lens: ★★★
(a) myopia, concave (b) myopia, convex
(c) hypermetropia, convex (d) hypermetropia, concave

7. The cells of the retina that respond to colour are the: ★★☆
(a) rods (b) cones
(c) ciliary cells (d) iris cells

8. A myopic person has a far point of 2 m. The power of the correcting lens is: ★★★
(a) +2 D (b) −2 D
(c) +0.5 D (d) −0.5 D

9. **Assertion (A):** A myopic person uses spectacles with a concave lens.
Reason (R): A concave lens diverges the rays from a distant object so that the image forms on the retina. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The far point of a normal eye is at infinity.
Reason (R): The eye lens is thickest when we look at distant objects. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

6 SHORT AND LONG ANSWER QUESTIONS

1 Draw a labelled diagram of the human eye. Give the functions of the iris and the pupil. **(3 marks)**

2 A student cannot read the blackboard from the last row. Name the defect, give two causes, and draw ray diagrams of the defect and its correction. **(5 marks)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Near point of a normal eye: 25 cm.
2	(c)	Ciliary muscles change the curvature of the lens.
3	(d)	The image forms on the retina.
4	(b)	Adjusting focal length = accommodation.
5	(b)	The iris controls the size of the pupil.
6	(c)	Near point has moved away: convex lens.
7	(b)	Cones detect colour; rods work in dim light.
8	(d)	f = −2 m, so P = 1/f = −0.5 D.
9	(a)	R correctly explains A.
10	(c)	A true; R false: the lens is thinnest for distant objects.

Model answers: short and long questions

11 Eye diagram, iris and pupil

- Diagram: as in Section 1, with 8 labels.
- Iris: controls the size of the pupil. Pupil: lets light in; it becomes small in bright light and wide in dim light.

12 Blackboard from the last row: myopia

- Image of a distant object forms in front of the retina.
- Causes: (i) excessive curvature of the eye lens; (ii) elongation of the eyeball.
- Correction: concave lens of suitable power. Diagrams: as in Section 3.

NCERT quick Q&A (continued)

- **Q.** What happens to the image distance in the eye when the object moves farther away? **A.** It does not change: the image stays on the retina; only the focal length of the eye lens changes.
- **Q.** Why do older people need bifocal lenses? **A.** Presbyopia: weak ciliary muscles and a stiff lens, so they may need correction for both near and far vision.

Day 28 hit-list: Eduspark PYQ analysis

Topic	Rating
Myopia: causes, correction, ray diagram	★★★
Hypermetropia: causes, correction, ray diagram	★★★
Power of accommodation, near and far point	★★★
Labelled diagram of the eye	★★☆
Presbyopia and cataract	★★☆
Power of correcting lens numericals	★★☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 6 board and sample-paper questions.

Did you know? Eye donation (NCERT)

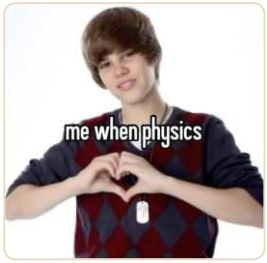
- ✔ People of any age, and even those who wear spectacles or had cataract surgery, can donate their eyes.
- ✔ Eyes must be removed within 4 to 6 hours after death.
- ✔ One pair of eyes can give vision to up to four people with corneal blindness.

Day 28 checklist: tick before you sleep

- ☐ I can draw and label the human eye with 8 parts
- ☐ I can explain accommodation, near point and far point
- ☐ I can draw the ray diagrams of myopia and hypermetropia and their correction
- ☐ I solved all four numericals with the Class Trick steps
- ☐ I know presbyopia, cataract and bifocal lenses

My score: ____ / 5

Physics ka eye chapter samajh aaya toh dil se pyaar ho gaya. 5/5 = Jalwa!



LEVEL 28 COMPLETE +125 XP earned. Next: Level 29

BUJHO *Samajh gaya! Myopia mein concave, hypermetropia mein convex, aur 1/∞ = 0!*

PAHELI *Shabaash! Kal Level 29: prism, rainbow, blue sky aur Chapter 6 ka Boss.*

Try yourself: two quick numericals

1. A myopic person's far point is 2.5 m. Find the power of the lens needed. **[Answer: −0.4 D, concave]**
2. A person's near point is 50 cm. Find the power of the lens needed to read at 25 cm. **[Answer: +2 D, convex]**

Bonus rapid-fire round: quick answers

Question	Answer
Where does most refraction happen?	Outer surface of the cornea
Part that gives the eye its colour	Iris
Screen of the eye	Retina
Point with no vision	Blind spot
Lens for presbyopia	Bifocal
Normal range of vision	25 cm to infinity

Coming up: Day 29

The Colourful World: prism, dispersion, rainbow, twinkling stars, scattering of light, and the Chapter 6 Boss Level.

NITESH SIR SAYS

Aankh ka diagram roz ek baar banao, aur numericals mein pehle defect pehchano. ***Kal milte hai Day 29 pe!***

DAY 29 · LEVEL 29

PHYSICS | CHAPTER 6 | PART 2 · ASTRA SERIES

COLOURFUL WORLD + BOSS

Goal: prism, rainbow, stars and blue sky, then the boss.

by Nitesh Prakash

01 Prism | 02 Dispersion | 03 Rainbow | 04 Atmospheric Refraction | 05 Scattering | +175 XP today | Level 29 / 60

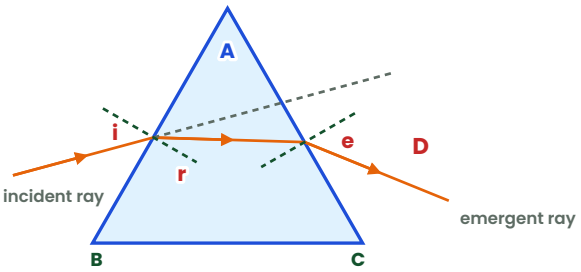
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 REFRACTION THROUGH A GLASS PRISM

DEFINITION

- **Prism:** a transparent block with two triangular bases and three rectangular side faces. The angle between its two refracting faces is the **angle of the prism (A)**.
- **Angle of deviation (D):** the angle between the incident ray produced forward and the emergent ray.

Refraction of light through a glass prism



i = angle of incidence, r = angle of refraction, e = angle of emergence, A = angle of the prism, D = angle of deviation.

KEY POINT

✔ Unlike a glass slab, the two refracting faces of a prism are not parallel, so the emergent ray is **not parallel** to the incident ray: it bends towards the **base** of the prism.

2 DISPERSION OF WHITE LIGHT

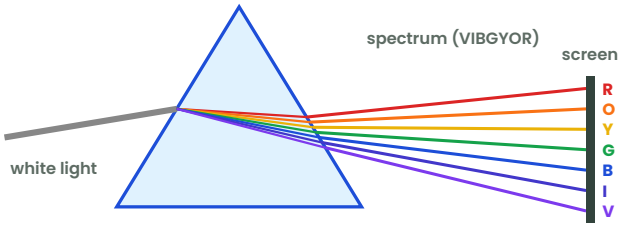
DEFINITION

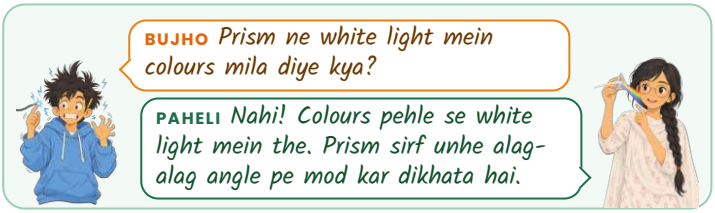
- **Dispersion:** the splitting of white light into its seven component colours.
- **Spectrum:** the band of coloured components of a light beam: **VIBGYOR** (violet, indigo, blue, green, yellow, orange, red).

Tesla aur Einstein baad mein aaye, Newton ne pehle hi prism se dikha diya: white light = 7 colours!

Why does dispersion happen? (class notes) ★★★

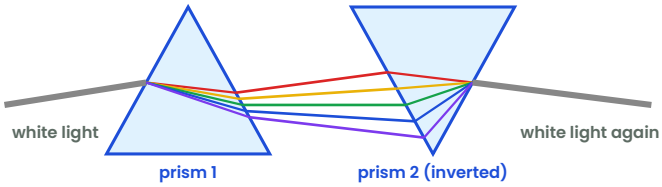
- ✔ All colours travel at the same speed in vacuum, but at **different speeds in glass**, so each colour bends through a different angle.
- ✔ **Red**: longest wavelength, travels fastest in glass, **bends the least** (top of the spectrum).
- ✔ **Violet**: shortest wavelength, travels slowest, **bends the most** (bottom of the spectrum).





Newton's experiment: recombination of the spectrum

- ✔ Newton placed a second, identical prism **upside down** in the path of the spectrum.
- ✔ All seven colours recombined and **white light** came out of the other side.
- ✔ So white light is made of seven colours (class notes: "How can we say white light has 7 colours?").



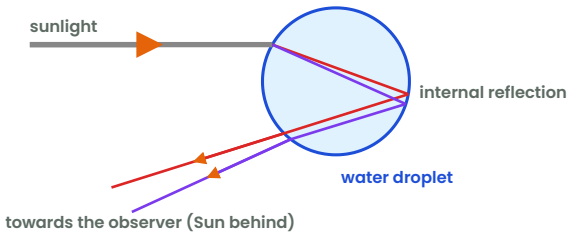
★★★ *Dispersion ka reason (red sabse kam mudta, violet sabse zyada) aur VIBGYOR: Nitesh Bhaiya ki guarantee!*

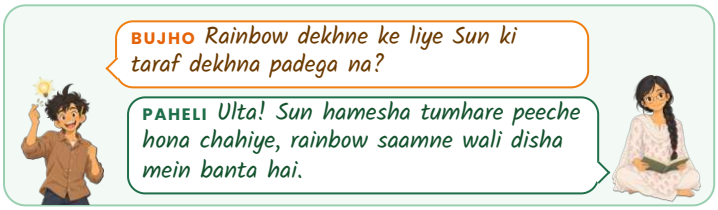
Roses are red kyunki red ki wavelength sabse lambi hai. Violet sabse zyada mudta hai, poetry mein bhi!

3 RAINBOW: A NATURAL SPECTRUM

How a rainbow forms ★★★

- ✔ A rainbow is a natural spectrum seen in the sky after a rain shower, caused by **dispersion of sunlight by tiny water droplets**.
- ✔ Each droplet acts like a small prism: it **refracts and disperses** the sunlight, **reflects it internally**, and **refracts it again** as it comes out.
- ✔ A rainbow is always seen in the direction **opposite to the Sun** (the Sun is behind you).





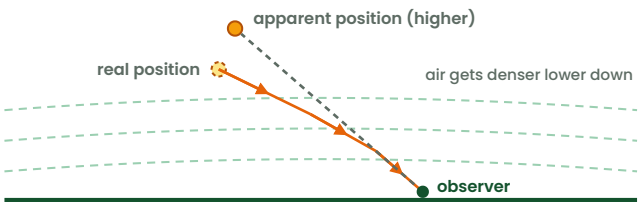
4 ATMOSPHERIC REFRACTION

DEFINITION

- > **Atmospheric refraction:** refraction of light by the Earth's atmosphere, whose air layers have different optical densities.
- > Example: objects seen through hot air above a fire seem to waver, because the hot air is optically rarer and keeps changing.

Why do stars twinkle? (class notes) ★★★

- ✔ Starlight bends towards the normal as it enters denser and denser air, so the star appears **slightly higher** than its real position.
- ✔ Stars are far away and act as **point sources**. The atmosphere keeps changing, so the apparent position keeps shifting and the light entering the eye **flickers**: brighter, then fainter. This is twinkling.



★★★ *Twinkling of stars aur planets kyun nahi twinkle karte: har saal ka 3-mark question, Nitesh Bhaiya ki guarantee!*

Why don't planets twinkle?

- ✔ Planets are much closer, so they act as **extended sources** (a collection of many point sources).
- ✔ The changes in light from all these points **average out to zero**, so there is no twinkling.

Planet ki light steady, star ki light flicker. Point source vs extended source, bas itna yaad rakho!

Advanced sunrise and delayed sunset

- ✔ Because of atmospheric refraction, we see the Sun about **2 minutes before** the actual sunrise and about **2 minutes after** the actual sunset.
- ✔ Actual sunrise: when the Sun actually crosses the horizon.
- ✔ The same effect makes the Sun's disc look **flattened** at sunrise and sunset.

5 SCATTERING OF LIGHT

DEFINITION

- > **Scattering of light:** tiny particles (air molecules, dust, droplets) take in light and send it out again in all directions.
- > **Tyndall effect:** scattering of light by colloidal particles, which makes the path of a beam of light visible.
- > Examples: sunlight entering a smoke-filled room through a small hole; sunlight through the canopy of a dense forest, where tiny water droplets in the mist scatter it.

The colour depends on particle size

- ✔ **Very fine particles** scatter mainly blue light (short wavelength).
- ✔ **Larger particles** scatter longer wavelengths too; if large enough, the scattered light looks **white** (clouds).

Why is the clear sky blue? ★★★

- ✔ Air molecules and fine particles are **smaller than the wavelength** of visible light, so they scatter **blue** (shorter wavelength) much more than **red**.
- ✔ Red light has a wavelength about **1.8 times** that of blue light. The scattered blue light enters our eyes, so the sky looks blue.
- ✔ **Sky for astronauts:** very high up there is almost no air to scatter light, so the sky looks **dark**.

Sky blue hai kyunki blue hai? Nahi, kyunki short wavelength wala blue sabse zyada scatter hota hai!

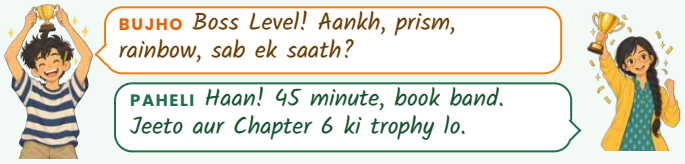
Why are danger signals red?

- ✔ Red light has the longest wavelength, so it is **scattered the least** by fog and smoke.
- ✔ So it travels farthest and is seen clearly from a distance.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Reddish Sun at sunrise and sunset (from class notes): near the horizon, sunlight crosses a thick layer of air, so most blue light is scattered away and mostly red reaches us. **Deep sea looks blue** (class notes): water absorbs red light, and the light scattered back from the water is mainly blue. **Law:** class notes write scattering $\propto 1/\lambda$; for fine particles the correct relation is $\propto 1/\lambda^4$ (Rayleigh scattering). NCERT gives no formula: just remember shorter wavelength, more scattering.

6 CHAPTER 6 BOSS LEVEL



BOSS LEVEL RULES

- ✔ **Time:** 45 minutes. **Maximum marks:** 31. Close the book and set a timer.
- ✔ **A:** 11 MCQs × 1 = 11. **B:** 4 short answers = 11. **C:** 1 long answer = 5. **D:** 1 case = 4.

1. Splitting of white light into its colours is called: ★★★
(a) reflection (b) refraction
(c) dispersion (d) scattering

2. In the spectrum formed by a glass prism, the colour that deviates the least is: ★★★
(a) red (b) yellow
(c) blue (d) violet

3. The clear sky looks blue because: ★★★
(a) air absorbs blue light
(b) fine particles scatter blue light more
(c) the sea reflects blue light
(d) blue has the longest wavelength

4. The twinkling of stars is due to: ★★★
(a) dispersion (b) atmospheric refraction
(c) scattering (d) reflection

5. A rainbow is always seen: ★★☆
(a) towards the Sun (b) opposite to the Sun
(c) only at noon (d) only at night

6. The path of a light beam through a colloid is visible due to: ★★☆
(a) dispersion (b) Tyndall effect
(c) refraction (d) reflection

Tip: in every "why" answer, name the phenomenon first, then give the reason.

7. Danger signal lights are red because red light: ★★★
(a) is scattered the least (b) is scattered the most
(c) bends the most (d) travels the slowest

8. Bifocal lenses are used to correct: ★★☆
(a) myopia (b) hypermetropia
(c) cataract (d) presbyopia

9. The far point of a myopic eye is 1 m. The power of the correcting lens is: ★★☆
(a) +1 D (b) −1 D
(c) +0.5 D (d) −2 D

10. **Assertion (A):** Planets do not twinkle.
Reason (R): Planets are close and act as extended sources, so the changes in their light average out. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

11. **Assertion (A):** The sky appears dark to astronauts at very high altitudes.
Reason (R): Scattering of light is greater at higher altitudes. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

7 SHORT ANSWERS (SECTION B)

12 Draw a labelled diagram of refraction of light through a glass prism. Mark the angles i, e, A and D. **(3 marks)**

13 Why does a star appear slightly higher than its real position? Why do planets not twinkle? **(3 marks)**

14 What is the Tyndall effect? Give two examples from daily life. **(2 marks)**

15 Explain with a labelled diagram how a rainbow is formed. **(3 marks)**

8 LONG ANSWER (SECTION C)

16 (A) What is myopia? Give two causes and show its correction with a ray diagram. (B) A person cannot see beyond 1.5 m clearly. Find the nature and power of the lens needed. **(5 marks)**

9 CASE-BASED (SECTION D)

Case 17: a morning at the hill station (4 marks)

In the morning mist, Aarav sees bright sunbeams streaming through the trees. At noon the sky is deep blue. At night the stars twinkle, but Jupiter shines steadily.

1. Name the effect that makes the sunbeams visible. **(1)**
2. Why is the noon sky blue? **(1)**
3. Why does Jupiter not twinkle? **(1)**
4. Name the phenomenon that makes stars twinkle. **(1)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Splitting of white light = dispersion.
2	(a)	Red has the longest wavelength: least deviation.
3	(b)	Fine particles scatter short-wavelength blue light more.
4	(b)	Changing air layers bend starlight: atmospheric refraction.
5	(b)	The Sun must be behind the observer.
6	(b)	Colloidal particles scatter light: Tyndall effect.
7	(a)	Red is scattered least, so it is seen from far.
8	(d)	Bifocal lenses correct presbyopia.
9	(b)	f = −1 m, so P = 1/f = −1 D.
10	(a)	R correctly explains A.
11	(c)	A true; R false: high up there is almost no air to scatter light.

Model answers: Sections B, C and D

12 Prism diagram

- Draw as in Section 1: incident ray, refracted ray, emergent ray, normals at both faces; mark i, e, A, and D between the incident ray produced and the emergent ray.

13 Star higher, planets steady

- Starlight bends towards the normal as it passes into denser air layers, so the star appears higher than it is.
- Planets are extended sources; the changes in light from their many points average out, so they do not twinkle.

14 Tyndall effect

- Scattering of light by colloidal particles, which makes the path of a light beam visible.
- Examples: a sunbeam in a smoky or dusty room; sunlight through the canopy of a misty forest.

15 Rainbow

- Water droplets act as tiny prisms: refraction and dispersion on entry, internal reflection, refraction again on exit. Seen opposite to the Sun. Diagram: as in Section 3.

16 Myopia (5 marks)

- (A) Distant objects look blurred; image forms in front of the retina. Causes: excessive curvature of the eye lens; elongation of the eyeball. Correction: concave lens (ray diagram as on Day 28).

(B) u = −∞, v = −1.5 m (far point)
1/f = 1/v − 1/u = −1/1.5, so f = −1.5 m
P = 1/f = **−0.67 D**: a concave lens

17 Case: hill station

- (1) Tyndall effect. (2) Air molecules scatter blue light (shorter wavelength) much more than red. (3) Jupiter is a planet: an extended source, so the changes average out. (4) Atmospheric refraction.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 6 board and sample-paper questions.

Score guide: 27+ Jalwa, 20 to 26 good, below 20: revise Days 28 and 29 once more.

Most asked in Chapter 6: Eduspark PYQ analysis

Topic	Day	Rating
Myopia and hypermetropia with diagrams	28	★★★
Twinkling of stars, planets do not twinkle	29	★★★
Dispersion and rainbow formation	29	★★★
Blue sky, danger signals, Tyndall effect	29	★★★
Power of accommodation, near and far point	28	★★☆
Advanced sunrise and delayed sunset	29	★★☆

Day 29 checklist: tick before you sleep

- ☐ I can draw the prism diagram and mark i, e, A and D
- ☐ I can explain dispersion, recombination and the rainbow
- ☐ I can explain twinkling of stars and advanced sunrise
- ☐ I can explain the blue sky, red danger signals and the Tyndall effect
- ☐ I attempted the Boss Level honestly and checked my score

My score: ____ / 5

5/5 = Chapter 6 ka boss khatam! Hooded Kermit ki mat suno.

LEVEL 29 COMPLETE **CHAPTER 6 COMPLETE**

+175 XP earned. Next: Level 30

BUJHO Samajh gaya! Red sabse kam mudta, blue sabse zyada scatter hota, aur stars point source hain!

PAHELI Boss defeated! Kal se Chapter 7: Electricity, current ka khel.

BUJHO Sunrise se pehle hi Sun dikh gaya, ye jaadu hai kya?

PAHELI Jaadu nahi, atmospheric refraction! Sun horizon ke neeche hota hai, phir bhi 2 minute pehle dikh jaata hai.

Coming up: Day 30

Electricity begins: electric charge, current, potential difference and how a circuit works.

NITESH SIR SAYS

Aankh aur aasmaan dono samajh aa gaye! Galat answers ka chapter dobara padho. ***Consistency hi jalwa hai!***

PHYSICS | CHAPTER 6 COMPLETE ASTRA SERIES

CHAPTER 6 SWAHAA!

The Human Eye and the Colourful World | Days 28–29

BUJHO *Myopia, hypermetropia, prism, rainbow, blue sky, sab samajh aa gaya! Ab chashma bhi physics lagta hai.*

PAHELI *Bilkul sahi, Bujho! Chapter 6 Swahaa. Next stop: Electricity, current aur circuits ki duniya.*

CHAPTER XP EARNED
+300 XP

LEVELS CLEARED
2 levels (Days 28–29)

BOOK PROGRESS
Level 29 / 60 3700 XP total

WHAT YOU HAVE MASTERED

- ✔ Structure of the human eye and its parts
- ✔ Myopia, hypermetropia, presbyopia, cataract
- ✔ Rainbow and atmospheric refraction
- ✔ Power of accommodation, near and far point
- ✔ Refraction through a prism, dispersion
- ✔ Scattering of light, Tyndall effect, blue sky

NEXT CHAPTER Day 30: Electricity

Human Eye cleared!

DAY 30 · LEVEL 30

PHYSICS | CHAPTER 7 | PART 1 · ASTRA SERIES

CURRENT & POTENTIAL DIFFERENCE

Goal: master charge, current, volt and circuit symbols today.

by Nitesh Prakash

01 Charge | 02 Current | 03 Circuit | 04 Potential Difference | 05 Symbols

+100 XP today | Level 30 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 ELECTRIC CHARGE

DEFINITION

- **Electric charge:** the property of matter that produces **electric and magnetic effects**
- Two kinds: **positive** (carried by protons) and **negative** (carried by electrons)
- Charge of one electron, **$e = 1.6 \times 10^{-19}$ C** (negative)
- SI unit: **coulomb (C)**; symbol Q or q; charge is a **scalar**

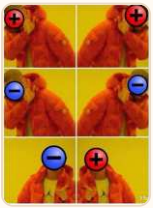
Particle	Charge	Where it sits
Electron	-1.6×10^{-19} C	outside the nucleus, free to move in metals
Proton	$+1.6 \times 10^{-19}$ C	inside the nucleus, fixed

How a body gets charged

- ✔ A body that **loses electrons** becomes **positively** charged
- ✔ A body that **gains electrons** becomes **negatively** charged
- ✔ Only electrons move; protons stay locked in the nucleus
- ✔ Like charges repel; unlike charges attract

CLASS TRICK: WHO TURNS POSITIVE?

"Dene wala hamesha bada hota hai." Jo electron **deta** hai (loses) wo **positive** ban jata hai; jo leta hai wo negative.



MEME BREAK

*Plus-plus? Nahi. Minus-minus? Nahi. **Plus-minus? Haan!** Unlike charges hi ek doosre ko attract karte hain.*

EXTRA KNOWLEDGE (HIGH LEVEL) properties of charge, from class notes

- **Additivity:** total charge = algebraic sum, e.g. +3 C and −1 C give +2 C
- **Conservation:** charge is neither created nor destroyed, only transferred
- **Invariance:** charge does not change with speed (you will see this in Class 12)
- **Quantisation:** Q = ne, where n is a whole number; no body has a charge smaller than e

2 ELECTRIC CURRENT ★★★ MOST ASKED

DEFINITION

- **Electric current:** the **rate of flow of charge** through any cross-section of a conductor
- **$I = Q / t$** (Q = net charge, t = time in seconds)
- SI unit: **ampere (A)**, named after André-Marie Ampère; 1 A = 1 C/s

One ampere and smaller units

- **1 ampere:** current when **1 coulomb** of charge flows through a cross-section in **1 second**
- 1 mA (milliampere) = 10^{-3} A; 1 μA (microampere) = 10^{-6} A
- Current is measured by an **ammeter**, always joined **in series**
- A **galvanometer** detects small currents and shows their direction

★★★ *Q = It wale sawal mein minute ko second aur mA ko A mein badalna mat bhoolna. Nitesh Bhaiya ki guarantee!*

Solved Example 1: Electrons in one coulomb ★★★

Given	Q = 1 C, $e = 1.6 \times 10^{-19}$ C
Formula	Q = ne, so n = Q / e
Solve	$n = 1 / (1.6 \times 10^{-19}) = 0.625 \times 10^{19}$
Answer	**$n = 6.25 \times 10^{18}$ electrons** make up 1 C

Solved Example 2: Charge from current ★★★

Given	I = 0.5 A, t = 10 min = 600 s (NCERT)
Formula	Q = I × t
Solve	Q = 0.5 × 600
Answer	**Q = 300 C** flows through the filament

Class question: how do metals conduct?

Metals have a large number of **free electrons** that are loosely held. When a cell is connected, these electrons drift through the metal and carry charge, so metals are good conductors.

REMEMBER: UNITS FIRST

Convert time to **seconds** and current to **amperes** before using Q = It. 2 min = 120 s; 250 mA = 0.25 A.

3 DIRECTION OF CURRENT ★★☆

	Conventional current	Electron flow
Outside the cell	**+ terminal → − terminal**	− terminal → + terminal
Fixed by	agreed before electrons were discovered	actual motion of free electrons
Used in	all circuit diagrams and answers	explaining conduction in metals

MEME BREAK
*Pata tha current + se − chalta hai, phir bhi arrow ulta bana diya? **Diagram ke marks gaye!** Conventional current hamesha + se −, electrons − se +.*

Why do electrons move at all?

- Without a cell, free electrons move **randomly** in all directions, so there is **no net flow** and no current
- A cell sets up an **electric force** across the wire; the electrons now **drift** towards the + terminal
- This steady drift of charge is the electric current

BUJHO *Electron − se + ki taraf jaata hai, toh current bhi − se + hona chahiye na?*

PAHELI *Current ki direction electrons discover hone se pehle tay ho gayi thi: **+ se −**. Electrons ulta chalte hain. Diagram mein hamesha conventional current hi dikhao.*

4 ELECTRIC CIRCUIT

DEFINITION

- **Electric circuit:** a **continuous and closed path** of an electric current
- Parts: a **source** (cell or battery), **connecting wires**, a **switch or key** and a **device** such as a bulb

Closed vs open circuit

- ✔ **Closed circuit:** key ON, path complete, current flows, bulb glows
- ✔ **Open circuit:** key OFF or wire broken, path incomplete, **no current**
- ✔ The switch only completes or breaks the path; it does not make charge

Solved Example 5: Counting electrons ★★☆

Given	I = 1.5 A, t = 2 min = 120 s, e = 1.6×10^{-19} C
Step 1	Q = It = 1.5 × 120 = 180 C
Step 2	n = Q / e = 180 / (1.6×10^{-19})
Answer	**n = 1.125 × 10^{21} electrons** pass in 2 min

5 POTENTIAL DIFFERENCE ★★★ MOST ASKED

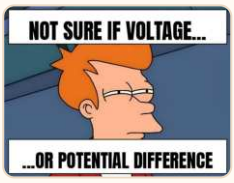
DEFINITION

- **Potential difference (V)** between two points: **work done** to move a **unit positive charge** from one point to the other
- **V = W / Q**; SI unit **volt (V)** = joule per coulomb (J/C)
- Another name for potential difference is **voltage**

One volt and the voltmeter

- **1 volt:** PD between two points when **1 joule** of work is done to move **1 coulomb** of charge between them
- Measured by a **voltmeter**, joined **in parallel** across the two points; it has a **high resistance**
- A cell keeps up the PD by the **chemical reaction** inside it; this PD drives the current
- Like water flows from a higher level to a lower level, current flows from **higher to lower potential**

CLASS TRICK: VOLTAGE = POTENTIAL DIFFERENCE
"Gangadhar hi Shaktiman hain, aur Shaktiman hi Gangadhar hain!" Voltage hi potential difference hai, aur potential difference hi voltage. Naam do, cheez ek.



MEME BREAK
*Fry confuse mat ho: **voltage aur potential difference ek hi cheez** hain.*

Solved Example 3: Work done on a charge ★★★

Given	Q = 2 C, V = 12 V (NCERT)
Formula	V = W / Q, so W = V × Q
Solve	W = 12 × 2
Answer	**W = 24 J** of work is done

Solved Example 4: Energy given to each coulomb ★★☆

Given	Battery of 6 V; Q = 1 C (NCERT)
Formula	W = V × Q
Solve	W = 6 × 1
Answer	**6 J** of energy is given to every coulomb

EXTRA KNOWLEDGE (HIGH LEVEL) beyond the Class 10 syllabus
Electric potential at a point: the work done to bring a **unit positive charge** from **infinity** to that point. Its unit is also the volt. The difference of the potentials of two points is the potential difference you use in Class 10.

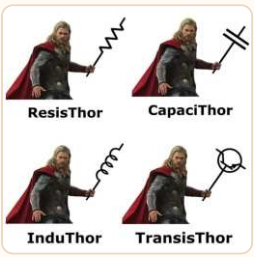
REMEMBER: THREE FORMS
V = W / Q, so W = V × Q and Q = W / V.

EXAM TIP: VOLTMETER AND AMMETER
Ammeter: **series**, very low resistance. Voltmeter: **parallel**, very high resistance.

6 CIRCUIT SYMBOLS ★★★ MOST ASKED

Symbols you must draw neatly

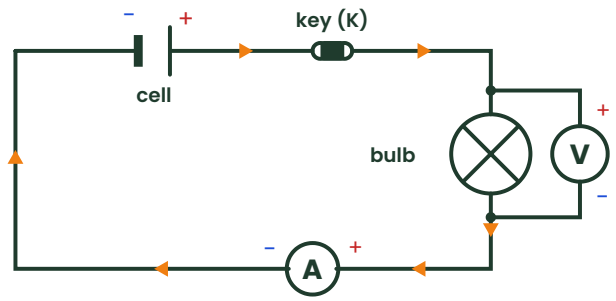
Electric cell (long plate +)	Battery: combination of cells
Plug key or switch, open (OFF)	Plug key or switch, closed (ON)
Wire joint (wires connected)	Wires crossing without joining
Electric bulb	Resistor of resistance R
Variable resistance (rheostat)	Ammeter (in series)
Voltmeter (in parallel)	Galvanometer (detects small current)



MEME BREAK

Thor ka hammer bhi symbol se pehchana jata hai! Resistor zig-zag, cell lambi-chhoti line: ***symbol galat toh diagram ke marks gaye.***

Circuit diagram: measuring I and V for a bulb ★★★



conventional current: + terminal → circuit → − terminal

Ammeter in series with the bulb; voltmeter across (in parallel with) the bulb; long plate of the cell = + terminal.

Ammeter vs voltmeter

	Ammeter	Voltmeter
Measures	current (A)	potential difference (V)
Joined	**in series**	**in parallel**
Resistance	very low	very high
+ terminal	towards the + terminal of the cell	

TRY YOURSELF: QUICK NUMERICALS

(i) 2 A flows for 5 min. Find Q. (ii) 96 C flows in 1 min. Find I.
(iii) 10 J moves 2 C. Find V. (iv) Write 3.2 mA in amperes.

Answers are on page 4, with the answer key.

× WRONG	✓ RIGHT
Ammeter in parallel with the bulb	ammeter always in series
Current flows from − to + in the circuit	conventional current: + to −
Q = It with t in minutes	convert t to seconds first
1 V = 1 C/J	1 V = 1 J/C (joule per coulomb)

BUJHO *Ammeter ko bhi bulb ke parallel laga doon, voltmeter ki tarah?*

PAHELI *Nahi! Ammeter ka resistance bahut kam hota hai. Parallel lagaya toh saara current usi mein bhaagega.* ***Ammeter series mein, voltmeter parallel mein.***

★★★ *Circuit diagram mein ammeter series, voltmeter parallel aur + terminal sahi taraf: ye 2 marks pakke hain,* ***Nitesh Bhaiya ki guarantee!***

Day 30 mind map: revise in 2 minutes

CHARGE
e = 1.6 × 10^{-19} C; lose e^- → +, gain e^- → −

CURRENT ★★★
I = Q/t; 1 A = 1 C/s; ammeter in series

CURRENT & POTENTIAL DIFFERENCE

VOLTAGE ★★★
V = W/Q; 1 V = 1 J/C; voltmeter in parallel

DIRECTION
current + to −; electrons − to +

CIRCUIT
closed path: source + wires + key + device; 6.25 × 10^{18} electrons = 1 C

BUJHO *Battery current banati hai na, isliye current khatam ho jata hai?*

PAHELI *Battery* ***potential difference*** *banati hai, chemical reaction se. Wahi PD charges ko dhakka deta hai aur current behta hai. Reaction ka saman khatam, toh cell khatam.*

Day 30 hit-list: Eduspark PYQ analysis

Topic	Rating
Define current and 1 ampere; Q = It numericals	★★★
Define potential difference and 1 volt; W = VQ	★★★
Circuit diagram with ammeter and voltmeter	★★★
Electrons in 1 C (6.25 × 10^{18})	★★☆
Symbols: cell, key, rheostat, wires crossing	★★☆
Direction of current vs electron flow	★☆☆

7 PRACTICE MCQS

1. The SI unit of electric charge is: ★★☆
(a) ampere (b) volt
(c) coulomb (d) ohm

2. One ampere is equal to: ★★★
(a) 1 J/C (b) 1 C/s
(c) 1 C × s (d) 1 J/s

3. The number of electrons that make up 1 C of charge is: ★★★
(a) 1.6×10^{19} (b) 6.25×10^{18}
(c) 1.6×10^{-19} (d) 6.25×10^{-18}

4. An ammeter is always connected in a circuit: ★★★
(a) in series (b) in parallel
(c) across the cell (d) either way

5. The unit joule per coulomb is the same as: ★★☆
(a) ampere (b) watt
(c) ohm (d) volt

6. The work done in moving 2 C of charge across a potential difference of 6 V is: ★★☆
(a) 3 J (b) 8 J
(c) 12 J (d) 4 J

7. A body has a charge of $+3.2 \times 10^{-19}$ C. It has: ★☆☆
(a) gained 2 electrons (b) lost 2 electrons
(c) lost 3 electrons (d) gained 3 protons

8. A current of 500 mA, expressed in amperes, is: ★☆☆
(a) 5 A (b) 0.05 A
(c) 0.5 A (d) 50 A

9. **Assertion (A):** A body that loses electrons becomes positively charged.
Reason (R): An electron carries a negative charge. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Electric current in a metal wire is due to the flow of electrons.
Reason (R): Conventional current flows from the negative to the positive terminal outside the cell. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Define electric current and state its SI unit. A current of 0.5 A is drawn by a bulb for 10 minutes. Find the charge that flows. **(3 marks)**
2. Define potential difference and 1 volt. How much work is done in moving a charge of 2 C across two points with a potential difference of 12 V? **(3 marks)**
3. Draw a circuit with a cell, a plug key, a bulb and an ammeter, and show a voltmeter that measures the potential difference across the bulb. **(3 marks)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Charge is measured in coulomb.
2	(b)	I = Q/t, so 1 A = 1 C/s.
3	(b)	$n = 1/(1.6 \times 10^{-19}) = 6.25 \times 10^{18}$.
4	(a)	Series, so the full current passes through it.
5	(d)	V = W/Q, so 1 V = 1 J/C.
6	(c)	W = VQ = 6 × 2 = 12 J.
7	(b)	$n = 3.2 \times 10^{-19} / 1.6 \times 10^{-19} = 2$, lost.
8	(c)	500 mA = 500×10^{-3} A = 0.5 A.
9	(a)	Losing negative electrons leaves extra positive charge.
10	(c)	A is true; conventional current flows + to −.

Model answers: short and long answer

1 Electric current
Rate of flow of charge, I = Q/t; SI unit ampere. Q = It = 0.5 × 600 s = **300 C**.

2 Potential difference
Work done to move a unit positive charge from one point to another, V = W/Q. 1 V = 1 J/C. W = VQ = 12 × 2 = **24 J**.

3 Circuit diagram
As on page 3: ammeter in series, voltmeter across the bulb, + terminals towards the cell's + terminal.

Try-yourself answers: (i) 600 C (ii) 1.6 A (iii) 5 V (iv) 3.2×10^{-3} A

Day 30 checklist: tick before you sleep

- ☐ I can define charge, current and potential difference with their SI units
- ☐ I can use Q = It and W = VQ with correct units
- ☐ I know 1 C = 6.25×10^{18} electrons and e = 1.6×10^{-19} C
- ☐ I can draw all circuit symbols and a circuit with ammeter and voltmeter
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 30 COMPLETE +100 XP earned. Next: Level 31

BUJHO *Samajh gaya! Charge behta hai toh current, aur use dhakka deta hai potential difference.*

PAHELI *Shabaash! Kal Level 31: Ohm's law aur resistance.*

NITESH SIR SAYS
Bas do formula: Q = It aur V = W/Q. Par units pe dhyan do: minute ko second, mA ko A. ***Kal milte hai Day 31 pe!***

DAY 31 · LEVEL 31

PHYSICS | CHAPTER 7 | PART 2 · ASTRA SERIES

OHM'S LAW & RESISTIVITY

Goal: use V = IR and R = ρl/A with full confidence.

by Nitesh Prakash

01 Ohm's Law | 02 V–I Graph | 03 Resistance | 04 Resistivity | 05 Materials

+100 XP today | Level 31 / 60

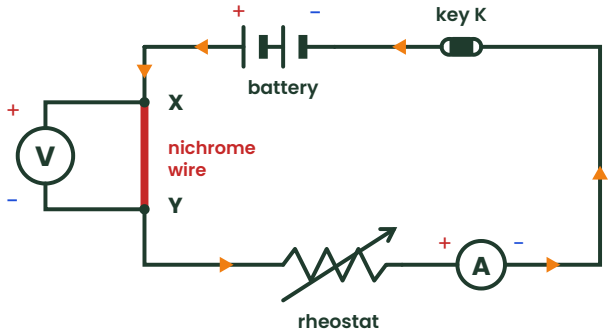
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 OHM'S LAW ★★★ MOST ASKED

DEFINITION

- **Ohm's law:** the potential difference V across a metallic conductor is **directly proportional** to the current I through it, provided its **temperature remains the same**
- V ∝ I, so **V = IR**, where R is a constant called the **resistance**
- Stated by the German physicist **Georg Simon Ohm** in 1827

Activity: verifying Ohm's law ★★☆



- Close the key; note the ammeter (I) and voltmeter (V) readings
- Change the current with the rheostat or by adding cells; take 4 to 5 readings
- V/I comes out the **same** every time: that constant is R of the wire

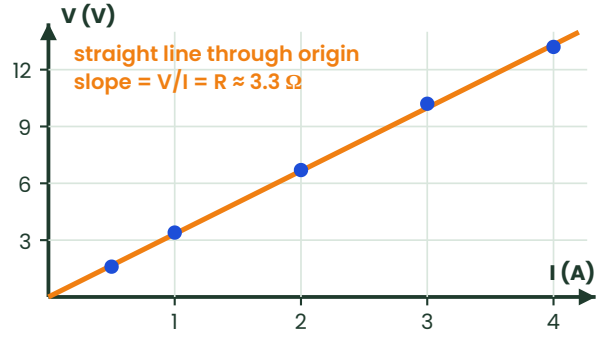
MEME BREAK

"OHMMMM..." Resistance ke monks bhi Ohm ka naam japte hain! Unit yaad rakho: ***ohm (Ω).***

BUJHO *Ohm's law toh har cheez pe lagta hoga, bulb ke filament pe bhi?*

PAHELI *Ohm's law tab tak sahi hai jab tak* ***temperature same*** *rahe. Filament bahut garam hota hai, uska resistance badh jata hai, isliye uska V–I graph poori tarah seedha nahi rehta.*

V–I graph: a straight line ★★★



I (A)	0.5	1.0	2.0	3.0	4.0
V (V)	1.6	3.4	6.7	10.2	13.2
V/I (Ω)	3.2	3.4	3.35	3.4	3.3

Data from the NCERT exercise (nichrome wire). The graph is a straight line through the origin, so V ∝ I.

2 RESISTANCE ★★★ MOST ASKED

DEFINITION

- **Resistance:** the property of a conductor to **oppose the flow of current** through it
- **R = V / I**; SI unit **ohm (Ω)**; measured by an **ohmmeter**
- **1 ohm:** resistance of a conductor when a PD of **1 V** across it drives a current of **1 A**; 1 Ω = 1 V/A

What R tells you

- ✔ For a given V, current is **inversely proportional** to R: double R, half the current
- ✔ A **rheostat** (variable resistance) changes R to control the current
- ✔ Good conductors have low R; insulators have very high R

Solved Example 1: Current through a bulb ★★★

Given	R = 1200 Ω, V = 220 V (NCERT)
Formula	I = V / R
Solve	I = 220 / 1200 = 0.183 A
Answer	**I ≈ 0.18 A** flows through the filament

3 RESISTANCE FACTORS ★★★ MOST ASKED

Factor	Relation	Example
Length l	R ∝ l	double the length → R doubles
Thickness (area A)	R ∝ 1/A	double the area → R halves
Temperature	R rises with temperature (metals)	a hot filament has more R
Nature of material	set by ρ	alloy (nichrome) > its metals

Why each factor works

- **Longer wire:** electrons travel a longer path and collide more, so R increases
- **Thicker wire:** more room for electrons to flow, so R decreases; current flows more easily through a thick wire
- **Hotter wire:** atoms vibrate more and block electrons, so R of a metal increases
- **Alloys:** have higher resistance than the pure metals they are made of, e.g. nichrome

★★★ *Ohm's law ka statement "temperature same rahe" ke saath likhna, aur V–I graph seedhi line origin se. Nitesh Bhaiya ki guarantee!*

BUJHO *Mota taar ho ya patla, current toh same hi behta hoga na?*

PAHELI *Nahi! Mote taar ka area zyada, toh resistance kam. Isliye same voltage pe* ***mote taar mein current zyada*** *behta hai. R ∝ 1/A yaad rakho.*

MEME BREAK

Resistance lambai-motai se badalta hai, par ***resistivity sirf material aur temperature*** *pe depend karti hai. Lambai se ρ badalna = Patrick logic!*

Solved Example 2: Resistance of a wire (class notes) ★★☆

Given	l = 2 m, A = 1.55 × 10^{-6} m^2, ρ = 2.8 × 10^{-8} Ω m
Formula	R = ρl / A
Solve	R = (2.8 × 10^{-8} × 2) / (1.55 × 10^{-6}) = 3.61 × 10^{-2}
Answer	**R ≈ 0.036 Ω**

EXAM SHORTCUT: STRETCHED WIRE

If a wire is stretched to **n times** its length (same volume), its resistance becomes **n^2 times**. Double the length → 4R.

4 RESISTIVITY ★★★ MOST ASKED

Derivation: R = ρl/A ★★★

- R ∝ l (length) and R ∝ 1/A (area)
- Combining: R ∝ l/A, so **R = ρ l / A**
- ρ (rho) is a constant for the material at a given temperature: the **resistivity**
- Put l = 1 m and A = 1 m^2: then **ρ = R**

DEFINITION

- **Resistivity (ρ):** the resistance of a conductor of **unit length** and **unit area of cross-section**
- ρ = RA / l; SI unit **ohm metre (Ω m)**
- **1 Ω m:** resistivity of a material when a conductor of it, 1 m long with 1 m^2 cross-section, has a resistance of 1 Ω

Resistivity: the key facts

- ✔ Depends only on the **material** and the **temperature**, not on length or thickness
- ✔ Metals and alloys: very low ρ, 10^{-8} to 10^{-6} Ω m; insulators: 10^{12} to 10^{17} Ω m
- ✔ Resistivity of an alloy is generally **higher** than that of its constituent metals

Solved Example 3: Length and radius both doubled (class notes) ★★★

Given	l′ = 2l, r′ = 2r, so A′ = π(2r)^2 = 4A
Formula	R = ρl/A
Solve	R′ = ρ(2l)/(4A) = ½ × ρl/A
Answer	**R′ = R/2**: the resistance becomes half

Solved Example 4: Wire stretched to double length (class notes) ★★★

Given	R = 4 Ω; volume stays the same, so l′ = 2l gives A′ = A/2
Formula	R = ρl/A
Solve	R′ = ρ(2l)/(A/2) = 4 × ρl/A = 4R
Answer	**R′ = 4 × 4 = 16 Ω**

Solved Example 5: Finding resistivity ★★★

Given	l = 1 m, R = 26 Ω, d = 0.3 mm, so r = 0.15 × 10^{-3} m (NCERT)
Area	A = πr^2 = 3.14 × (0.15 × 10^{-3})^2 = 7.07 × 10^{-8} m^2
Solve	ρ = RA / l = 26 × 7.07 × 10^{-8} / 1
Answer	**ρ ≈ 1.84 × 10^{-6} Ω m** (manganese, from the table)

Solved Example 6: Length needed for 10 Ω ★★☆

Given	R = 10 Ω, d = 0.01 cm, so r = 5 × 10^{-5} m; ρ = 50 × 10^{-8} Ω m (NCERT)
Area	A = πr^2 = 7.85 × 10^{-9} m^2
Solve	l = RA / ρ = 10 × 7.85 × 10^{-9} / (5 × 10^{-7})
Answer	**l ≈ 0.157 m**; if the diameter is doubled, R becomes one-fourth

5 CONDUCTORS, ALLOYS AND INSULATORS ★★☆

Resistivity of some materials (NCERT table) ★★☆

	Material	ρ (Ω m) at 20 °C
Conductors	Silver	1.60×10^{-8}
	Copper	1.62×10^{-8}
	Aluminium	2.63×10^{-8}
	Tungsten	5.20×10^{-8}
	Nickel	6.84×10^{-8}
	Iron	10.0×10^{-8}
	Chromium	12.9×10^{-8}
	Mercury	94.0×10^{-8}
	Manganese	1.84×10^{-6}
Alloys	Constantan (Cu, Ni)	49×10^{-6}
	Manganin (Cu, Mn, Ni)	44×10^{-6}
	Nichrome (Ni, Cr, Mn, Fe)	100×10^{-6}
Insulators	Glass	10^{10} to 10^{14}
	Hard rubber	10^{13} to 10^{16}
	Ebonite	10^{15} to 10^{17}
	Diamond	10^{12} to 10^{13}
	Paper (dry)	10^{12}

Choosing the right material

- **Heating elements** (iron, toaster, heater): **nichrome**, because alloys have high resistivity and do not oxidise (burn) easily at high temperature
- **Bulb filament: tungsten**, as it has a very high melting point (3380 °C)
- **Connecting and transmission wires: copper and aluminium**, because of their very low resistivity
- **Best conductor:** silver (lowest ρ), but it is too costly for wiring

MEME BREAK

*Heater coil ke liye copper bola? Seedha window ke bahar! **Nichrome** chahiye: high resistivity, aur garam hokar jalta nahi.*

Solved Example 7: Heater at two voltages ★★★

Given	I = 4 A at V = 60 V (NCERT)
Step 1	R = V / I = 60 / 4 = 15 Ω
Step 2	At 120 V: I = V / R = 120 / 15
Answer	**I = 8 A**: double the voltage, double the current

CLASS QUESTION: IRON OR MERCURY?

Iron (ρ = 10.0×10^{-8} Ω m) conducts better than mercury (94.0×10^{-8} Ω m).

× WRONG	✓ RIGHT
Resistivity changes with length	ρ depends only on material and temperature
Unit of resistivity: Ω/m	unit is Ω m (ohm metre)
Thick wire has more resistance	thick wire: larger A, less R
Ohm's law holds at any temperature	only when temperature stays same

Solved Example 8: Unknown resistor ★★☆

Given	V = 12 V, I = 2.5 mA = 2.5×10^{-3} A (NCERT)
Formula	R = V / I
Solve	R = 12 / (2.5×10^{-3})
Answer	**R = 4800 Ω = 4.8 kΩ**

Exam tips

- ✔ Draw the V–I graph through the **origin**, with V on the y-axis
- ✔ In R = ρl/A, convert mm and cm to **metres**; A = πr^2 uses the **radius**
- ✔ For a stretched wire, remember the area also changes

Day 31 mind map: revise in 2 minutes

OHM'S LAW ★★★
V ∝ I at constant temperature; V = IR

RESISTANCE
R = V/I; ohm (Ω); 1 Ω = 1 V/A

OHM'S LAW & RESISTIVITY

FACTORS ★★★
R ∝ l, R ∝ 1/A, rises with temperature, depends on material

RESISTIVITY
R = ρl/A; Ω m; only material and temperature

MATERIALS
nichrome for heaters, tungsten for filaments, copper and aluminium for wires

TRY YOURSELF: QUICK NUMERICALS

(i) V = 6 V, R = 3 Ω. Find I.
(ii) I = 0.5 A, R = 20 Ω. Find V.
(iii) Length halved: new R?
(iv) Area tripled: new R?

Answers are on page 4, with the answer key.

Short and long answer questions

1. State Ohm's law. Draw the circuit used to verify it and sketch the V–I graph you expect. **(5 marks)**
2. Name the factors on which the resistance of a conductor depends. How does the resistance change when both the length and the radius of a wire are doubled? **(3 marks)**
3. Why are the coils of electric toasters and irons made of an alloy rather than a pure metal? **(2 marks)**

REMEMBER

Silver: best conductor. Tungsten melts at 3380 °C.

6 PRACTICE MCQS

MCQ TIME
Pehle khud solve karo, phir answer key dekho. ***Seedha key dekhna mana hai!***

1. The SI unit of resistance is: ★★☆
(a) volt (b) ampere
(c) ohm (d) ohm metre

2. The V–I graph of a metallic wire at constant temperature is: ★★★
(a) a curve (b) a straight line through the origin
(c) parallel to the I-axis (d) a parabola

3. If the length of a wire is doubled, its resistance becomes: ★★★
(a) half (b) double
(c) four times (d) unchanged

4. The resistivity of a wire depends on: ★★★
(a) its length (b) its area of cross-section
(c) its material (d) the current through it

5. The SI unit of resistivity is: ★★☆
(a) Ω (b) Ω/m
(c) Ω m (d) Ω m^2

6. A wire of resistance 4 Ω is stretched to twice its length. Its new resistance is: ★★★
(a) 8 Ω (b) 2 Ω
(c) 16 Ω (d) 4 Ω

7. A current of 2.5 mA flows when a 12 V battery is connected across a resistor. Its resistance is: ★★☆
(a) 30 Ω (b) 480 Ω
(c) 4800 Ω (d) 0.2 Ω

8. Which material is used for the filament of an electric bulb? ★☆☆
(a) nichrome (b) tungsten
(c) copper (d) aluminium

9. **Assertion (A):** The heating elements of electric irons are made of an alloy.
Reason (R): Alloys have high resistivity and do not oxidise readily at high temperatures. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Resistance of a conductor increases when its area of cross-section is increased.
Reason (R): Resistance is inversely proportional to the area of cross-section. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(c)	R = V/I is measured in ohm.
2	(b)	V ∝ I, so a straight line through the origin.
3	(b)	R ∝ l.
4	(c)	ρ depends on material and temperature only.
5	(c)	ρ = RA/l gives Ω m^2/m = Ω m.
6	(c)	l doubles, A halves: R × 4 = 16 Ω.
7	(c)	R = 12/(2.5 × 10^{-3}) = 4800 Ω.
8	(b)	Tungsten has a very high melting point.
9	(a)	R is exactly why alloys are chosen.
10	(d)	A is false (R decreases); R is true.

Model answers: short and long answer

1 Ohm's law
At constant temperature, V across a metallic wire is directly proportional to I through it: V = IR. Circuit as on page 1 (battery, key, ammeter, rheostat in series with wire XY, voltmeter across XY); the V–I graph is a straight line through the origin.

2 Factors
Length, area of cross-section, temperature and material. A becomes 4A, so R′ = ρ(2l)/(4A) = **R/2**.

3 Alloys in heaters
Alloys have a high resistivity and do not oxidise (burn) readily at high temperature.

Try-yourself answers: (i) 2 A (ii) 10 V (iii) R/2 (iv) R/3

Day 31 checklist: tick before you sleep

- ☐ I can state Ohm's law and draw its circuit and V–I graph
- ☐ I can define resistance, 1 ohm, resistivity and 1 ohm metre
- ☐ I know how length, area, temperature and material change R
- ☐ I can solve stretched-wire and R = ρl/A numericals
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 31 COMPLETE +100 XP earned. Next: Level 32

BUJHO *Samajh gaya! Lamba taar zyada R, mota taar kam R, aur ρ sirf material ka.*

PAHELI *Shabaash! Kal Level 32: resistors in series.*

NITESH SIR SAYS
V = IR aur R = ρl/A, dono formula se 3 se 5 marks pakke. Stretch wale sawal mein area badalna mat bhoolna. ***Kal milte hai Day 32 pe!***

DAY 32 LEVEL 32

PHYSICS | CHAPTER 7 | PART 3 ASTRA SERIES

RESISTORS IN SERIES

Goal: same current, shared voltage, added resistance.

by Nitesh Prakash

01 Series Basics | 02 Same Current | 03 Derivation | 04 Numericals | 05 Drawbacks | +100 XP today | Level 32 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

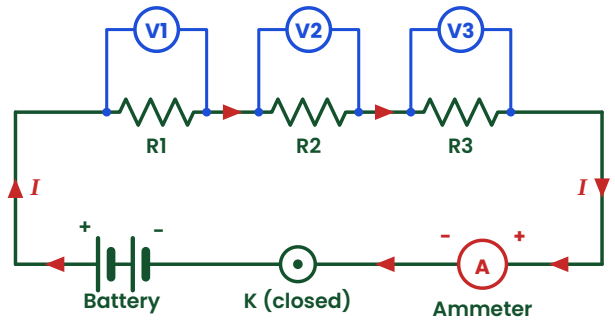
1 SERIES COMBINATION OF RESISTORS

Recap (Day 31): Ohm's law, **V = IR**. Today we join two or more resistors and find one **equivalent resistance** that can replace them all.

DEFINITION

- **Series combination:** resistors joined **end to end**, one after the other, so that charge has **only one path** through them
- The **same current** flows through each resistor when a potential difference is applied across the combination
- **Equivalent (resultant) resistance R_s:** the single resistance that draws the same current from the same battery

Circuit: three resistors in series ★★★



Ammeter in series (anywhere in the loop); each voltmeter in parallel across its resistor. Red arrows: conventional current, + to −.

Where series is used

- ✔ A **fuse** is joined in series with the appliance it protects, so all the current passes through it
- ✔ An **ammeter** is always joined in series
- ✔ **Decorative lights**: small bulbs share the mains voltage; one fused bulb darkens the string

Activity: is the current the same everywhere?

1. Join resistors of 1 Ω, 2 Ω and 3 Ω in series with a battery, a plug key and an ammeter
2. Note the ammeter reading, then shift the ammeter between resistors: **same reading**
3. Put a voltmeter across all three, then across each one: $V = V_1 + V_2 + V_3$

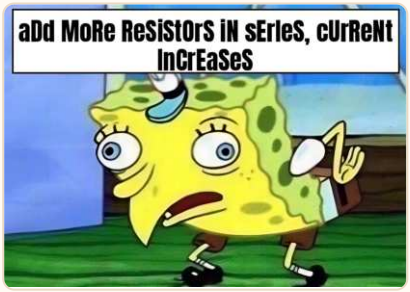
2 KEY FACTS OF A SERIES CIRCUIT ★★★ MOST ASKED

Learn these five lines

- ✔ **Current is the same** in every resistor: $I_1 = I_2 = I_3 = I$. The ammeter reads the same anywhere in the loop.
- ✔ **Potential difference divides:** $V = V_1 + V_2 + V_3$
- ✔ **Resistances add:** $R_s = R_1 + R_2 + R_3$, so R_s is **greater than the largest** single resistance
- ✔ **n equal resistors** R in series: $R_s = nR$
- ✔ **V ∝ R** (same I): $V_1 : V_2 : V_3 = R_1 : R_2 : R_3$. The bigger resistor takes the bigger share of voltage.

CLASS TRICK: K³B (CLASS KEY POINT)

"Series mein current same rehta hai, voltage divide hota hai." Notes wala derivation: $V = V_1 + V_2 + V_3 \rightarrow IR_s = IR_1 + IR_2 + IR_3 \rightarrow \mathbf{R_s = R_1 + R_2 + R_3}$. Current same hai, isliye I cancel ho jaata hai.



MEME BREAK

*"Series mein resistor badhao, current badhega"? Bilkul nahi! R_s badha toh current **ghatega**: $I = V / R_s$.*

BUJHO *Pehla resistor thoda current kha jaata hoga na? Last wale tak kam pahunchega.*

PAHELI *Bilkul nahi! Current use nahi hota. Series mein ek hi raasta hai, toh har resistor se **same current** guzarta hai. Ammeter kahin bhi lagao, reading same.*

BUJHO *Toh zyada resistor series mein lagaun toh bulb aur tez jalega?*

PAHELI *Ulta! Rs badhta hai, $I = V / R_s$ ghatta hai, isliye bulb **dimmer**. Har resistor ek naya speed breaker.*

3 DERIVATION OF SERIES RESISTANCE

★★★ MOST ASKED

Let resistors R_1, R_2, R_3 be joined in series between points A and B, with a battery of potential difference V. Let I be the current.

Step 1: $V = V_1 + V_2 + V_3$
total potential difference is shared by the three resistors ...(i)

Step 2: $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$
Ohm's law for each resistor; the current I is the same ...(ii)

Step 3: for the equivalent resistor, $V = IR_s$
one resistor R_s drawing the same current I ...(iii)

Step 4: $IR_s = IR_1 + IR_2 + IR_3$
put (ii) and (iii) in (i), then divide both sides by I

$$R_s = R_1 + R_2 + R_3$$

for n resistors: $R_s = R_1 + R_2 + \ldots + R_n$

★★★ *3 marks ka derivation: diagram banao, V = V1 + V2 + V3 likho, I cancel karo. Nitesh Bhaiya ki guarantee!*

SHORTCUT: VOLTAGE SHARE

Same I flows, so $V_1 = V \times R_1 / R_s$. Example: 5 Ω and 15 Ω on 12 V: $V_1 = 12 \times 5/20 = 3$ V, $V_2 = 9$ V.

Series at a glance

Quantity	In a series circuit
Current	Same through each resistor
Potential difference	Divides; largest across the largest R
Total resistance	$R_s = R_1 + R_2 + \ldots$, more than the largest
Add one more resistor	R_s rises, current falls
One part breaks	Circuit opens; current becomes zero

4 COMMON MISTAKES

× WRONG	✓ RIGHT
Current gets used up as it passes each resistor	The **same current** flows through every resistor
Each resistor gets the full battery voltage	Voltage **divides**: $V = V_1 + V_2 + V_3$
Adding a resistor in series raises the current	R_s rises, so **I = V / R_s falls**
Voltmeter drawn in series with a resistor	Voltmeter goes **in parallel**, ammeter in series

GRU KA PLAN

*Bulbs series mein lagaye, Rs badha, current ghata, aur bulbs **dim ho gaye**. Plan ka last step dekh ke Gru bhi hairan!*

5 SOLVED EXAMPLES

★★★ MOST ASKED

★★★ *Given, Formula, Substitution, Answer + unit: ye 4 steps likho, poore marks pakke. Nitesh Bhaiya ki guarantee!*

Solved Example 1: Lamp and conductor ★★★

Q. An electric lamp of 20 Ω and a conductor of 4 Ω are joined in series to a 6 V battery. Find the total resistance, the current and the potential difference across each. NCERT

Given: $R_1 = 20$ Ω, $R_2 = 4$ Ω, V = 6 V

- $R_s = R_1 + R_2 = 20 + 4 = \mathbf{24\ \Omega}$
- $I = V / R_s = 6 / 24 = \mathbf{0.25\ A}$
- $V_{lamp} = IR_1 = 0.25 \times 20 = \mathbf{5\ V}$; $V_{conductor} = 0.25 \times 4 = \mathbf{1\ V}$

Answer: 24 Ω, 0.25 A, 5 V and 1 V. Check: 5 + 1 = 6 V ✔

Solved Example 2: Three resistors ★★★

Q. Resistors of 5 Ω, 10 Ω and 15 Ω are connected in series across a 12 V supply. Find the current and the voltage across each.

Given: $R_1 = 5$ Ω, $R_2 = 10$ Ω, $R_3 = 15$ Ω, V = 12 V

- $R_s = 5 + 10 + 15 = \mathbf{30\ \Omega}$
- $I = V / R_s = 12 / 30 = \mathbf{0.4\ A}$
- $V_1 = 0.4 \times 5 = \mathbf{2\ V}$, $V_2 = 0.4 \times 10 = \mathbf{4\ V}$, $V_3 = 0.4 \times 15 = \mathbf{6\ V}$

Answer: 0.4 A; 2 V, 4 V, 6 V (ratio 1 : 2 : 3, like R). Total 12 V ✔

Solved Example 3: Five resistors ★★★

Q. A 9 V battery is connected in series with resistors of 0.2 Ω, 0.3 Ω, 0.4 Ω, 0.5 Ω and 12 Ω. How much current flows through the 12 Ω resistor? NCERT

Given: V = 9 V; five resistors in series

- $R_s = 0.2 + 0.3 + 0.4 + 0.5 + 12 = \mathbf{13.4\ \Omega}$
- $I = V / R_s = 9 / 13.4 = 0.6716$ A ≈ **0.67 A**
- In series the current is the same everywhere, so the 12 Ω resistor also carries 0.67 A

Answer: ≈ 0.67 A through the 12 Ω resistor

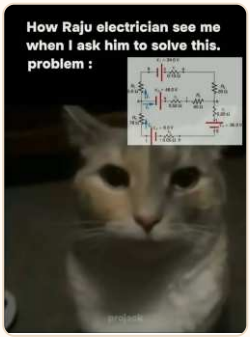
Solved Example 4: Work backwards from one voltage ★★☆

Q. A 4 Ω resistor and a resistor R are in series with a 12 V battery. The voltage across the 4 Ω resistor is 4 V. Find the current and R.

Given: $R_1 = 4$ Ω, $V_1 = 4$ V, V = 12 V

- $I = V_1 / R_1 = 4 / 4 = \mathbf{1\ A}$ (same through R)
- $V_R = V - V_1 = 12 - 4 = \mathbf{8\ V}$
- $R = V_R / I = 8 / 1 = \mathbf{8\ \Omega}$

Answer: I = 1 A, R = 8 Ω



MEME BREAK

*Circuit dekh ke Raju electrician ka bhi yahi haal! Darr mat, series mein bas **R jodo, I nikalo, V = IR lagao**. Done.*

Solved Example 5: Find the unknown resistor ★★★

Q. A 5 Ω resistor and an unknown resistor R are joined in series across 6 V. The current is 0.4 A. Find R.

Given: $R_1 = 5\ \Omega$, V = 6 V, I = 0.4 A

- $R_s = V / I = 6 / 0.4 =$ **15 Ω**
- $R_s = R_1 + R$, so $R = 15 - 5 =$ **10 Ω**

Answer: R = 10 Ω

Solved Example 6: Ammeter and voltmeter readings ★★★

Q. A battery of three cells of 2 V each, resistors of 5 Ω, 8 Ω and 12 Ω and a plug key are all in series. Find the ammeter reading and the voltmeter reading across the 12 Ω resistor. NCERT

Given: $V = 3 \times 2 = 6$ V; $R_1 = 5\ \Omega$, $R_2 = 8\ \Omega$, $R_3 = 12\ \Omega$

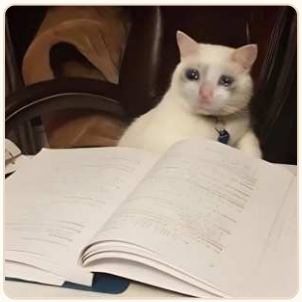
- $R_s = 5 + 8 + 12 =$ **25 Ω**
- $I = V / R_s = 6 / 25 =$ **0.24 A** (ammeter)
- $V_3 = IR_3 = 0.24 \times 12 =$ **2.88 V** (voltmeter)

Answer: Ammeter 0.24 A; voltmeter 2.88 V

6 DISADVANTAGES OF SERIES CIRCUITS ★★☆

Why homes are not wired in series

- **One fault stops all:** if one bulb fuses, the only path breaks and every appliance goes off
- **No separate control:** appliances cannot be switched on and off independently
- **Same current for all:** a bulb, fan and heater need very different currents, but series forces one current on all
- **High total resistance:** R_s is large, so the current is small and devices do not work at full power



SERIES KA DUKH

Ek bulb fuse hua aur poori series lights andhere mein. Ek gaya, sab gaye!

Day 32 mind map: revise in 2 minutes

CURRENT ★★★
Same in every resistor; ammeter reads the same anywhere

VOLTAGE ★★★
Divides: $V = V_1 + V_2 + V_3$; $V \propto R$

RESISTORS IN SERIES

RESISTANCE ★★★
$R_s = R_1 + R_2 + R_3$, more than the largest

EQUAL ONES
n resistors of R each: $R_s = nR$

DRAWBACKS
One fault stops all; no separate switching; same current forced on all; homes use parallel (Day 33)

7 PRACTICE MCQS

1. In a series combination of resistors, the quantity that is the same for every resistor is: ★★★

(a) potential difference (b) current
(c) resistance (d) heat produced

2. Resistors of 2 Ω, 3 Ω and 5 Ω are joined in series. The equivalent resistance is: ★★★

(a) 0.97 Ω (b) 5 Ω
(c) 10 Ω (d) 30 Ω

3. Equivalent resistance of n identical resistors, each R, joined in series is: ★★☆

(a) R/n (b) nR
(c) n^2R (d) R

4. 4 Ω and 8 Ω resistors in series are connected to 12 V. The potential difference across the 8 Ω resistor is: ★★★

(a) 4 V (b) 6 V
(c) 8 V (d) 12 V

5. The equivalent resistance of resistors in series is always: ★★☆

(a) less than the smallest (b) equal to the average
(c) greater than the largest (d) equal to the smallest

6. Ammeters placed at different points of a series circuit will read: ★★☆

(a) more near + terminal (b) the same current
(c) zero in the middle (d) less near − terminal

7. In a string of bulbs joined in series, one bulb fuses. The other bulbs will: ★★☆

(a) glow brighter (b) all go off
(c) glow dimmer (d) keep glowing

8. Resistors of 1 Ω, 2 Ω and 3 Ω are in series with a 6 V battery. The current is: ★★★

(a) 0.5 A (b) 1 A
(c) 2 A (d) 6 A

9. **Assertion (A):** Series arrangement is not used for domestic circuits.
Reason (R): In series, if one appliance fails the circuit breaks and all appliances stop working. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** When resistors are joined in series, the potential difference across each resistor is the same.
Reason (R): The same current flows through each resistor in a series combination. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

EXAM TIPS

- Write the unit with every answer: Ω, A or V
- Self-check: $V_1 + V_2 + V_3$ must equal the battery voltage

Day 32 hit-list: Eduspark PYQ analysis

Topic	Rating
Derivation of $R_s = R_1 + R_2 + R_3$	★★★
Numericals: R_s, current, voltage across each	★★★
Same current, divided voltage	★★★
Why series is not used at home	★★☆
n equal resistors: nR	★★☆
Voltage ratio $V \propto R$	★☆☆

Case-based question (4 marks)

Resistors of 2 Ω, 4 Ω and 6 Ω are joined in series with a 6 V battery and a key.

1. Find the equivalent resistance. **(1)**
2. Find the current in the circuit. **(1)**
3. Find the potential difference across the 6 Ω resistor. **(1)**
4. What happens to the current if the 4 Ω resistor breaks? **(1)**

Short and long answer questions

1. Derive the expression for the equivalent resistance of three resistors R_1, R_2 and R_3 joined in series. Draw the circuit diagram. **(3 marks)**
2. Why is the series arrangement not used in domestic wiring? Give two reasons. **(2 marks)**
3. A 4 Ω and a 6 Ω resistor are joined in series with a 5 V battery. Find (i) the equivalent resistance, (ii) the current and (iii) the potential difference across each resistor. **(3 marks)**

TRY YOURSELF

1. 2 Ω, 3 Ω and 7 Ω are in series across 6 V. Find the current.
2. How many 4 Ω resistors in series give 20 Ω?
3. 10 Ω and R in series across 9 V draw 0.3 A. Find R.
4. Find the ratio of the voltages across 3 Ω and 9 Ω resistors joined in series.

Answers are just below this box.

Try-yourself answers

(1) 12 Ω, so 0.5 A
(2) 5 resistors
(3) R_s = 30 Ω, R = 20 Ω
(4) 1 : 3

NITESH SIR SAYS

K³B yaad rakho: current same, voltage divide. Baaki sab V = IR se nikal jaata hai. ***Kal milte hai Day 33 pe!***

Answer key: MCQs

Q	Ans	Why
1	(b)	One path, so one current.
2	(c)	2 + 3 + 5 = 10 Ω.
3	(b)	R added n times.
4	(c)	I = 12/12 = 1 A; V = 1 × 8 = 8 V.
5	(c)	Adding only increases the total.
6	(b)	Current is the same everywhere.
7	(b)	The only path is broken.
8	(b)	I = 6/6 = 1 A.
9	(a)	R correctly explains A.
10	(d)	V divides ($V \propto R$); R is true.

Model answers: short and long answer

1 Derivation

Draw R_1, R_2, R_3 in series with battery, key and ammeter. Same current I; $V = V_1 + V_2 + V_3$. By Ohm's law $IR_s = IR_1 + IR_2 + IR_3$, so $\mathbf{R_s = R_1 + R_2 + R_3}$.

2 Not for homes

(i) If one appliance fails, the circuit breaks and all stop.
(ii) Appliances cannot be switched separately and need different currents.

3 Numerical

R_s = 4 + 6 = **10 Ω**; I = 5/10 = **0.5 A**; V_4 = 0.5 × 4 = **2 V**, V_6 = 0.5 × 6 = **3 V**.

Case-based answers

(1) 2 + 4 + 6 = 12 Ω
(2) I = 6/12 = 0.5 A
(3) V = 0.5 × 6 = 3 V
(4) Circuit opens; I = 0

Day 32 checklist: tick before you sleep

- ☐ I can define a series combination and draw its circuit
- ☐ I know: current same, voltage divides (K³B)
- ☐ I can derive $R_s = R_1 + R_2 + R_3$
- ☐ I can find R_s, I and each V in a numerical
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Ek concept chhoota toh domino ki tarah sab girega, page 1–2 phir se padho.

LEVEL 32 COMPLETE +100 XP earned. Next: Level 33

BUJHO *Samajh gaya! Current same, voltage divide, resistance add.*

PAHELI *Shabaash! Kal Level 33: parallel combination of resistors.*

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Electricity board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 33 · LEVEL 33

PHYSICS | CHAPTER 7 | PART 4 · ASTRA SERIES

PARALLEL & MIXED CIRCUITS

Goal: find R_p and solve any series-parallel circuit.

by Nitesh Prakash

01 Parallel Rules | 02 Derivation | 03 Series vs Parallel | 04 Mixed Circuits | 05 Home Wiring | +100 XP today | Level 33 / 60

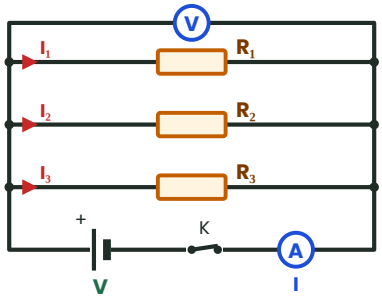
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 RESISTORS IN PARALLEL ★★★ MOST ASKED

DEFINITION

- **Parallel combination:** resistors joined **between the same two points**, so each one is connected directly across the source
- The **potential difference across every resistor is the same** (equal to V of the source)
- The **total current divides** among the branches: $I = I_1 + I_2 + I_3$

Circuit: three resistors in parallel



Ammeter A (in series) reads the total current I; voltmeter V (in parallel) reads the same p.d. across R_1, R_2 and R_3.

CLASS TRICK: SERIES VS PARALLEL IN ONE LINE

Parallel: Voltage remains same, current is divided inversely proportional to the resistance.
Series: Current remains same, voltage divides itself on all the resistors.

Activity: what the ammeters show

- ✔ Put an ammeter in each branch: I_1, I_2, I_3 come out different.
- ✔ Their sum equals the main-line reading I.
- ✔ Smallest resistor carries the **largest** current ($I = V/R$, same V).

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Junction rule (Kirchhoff's current law): the total current entering a junction equals the total current leaving it, because charge is neither created nor destroyed at the junction. That is exactly why $I = I_1 + I_2 + I_3$. Named in Class 12; not asked by name in Class 10.

2 THE PARALLEL FORMULA ★★★ MOST ASKED

Step by step (write all steps in the exam) ★★★

STEP 1 Total current = sum of branch currents: $I = I_1 + I_2 + I_3$

STEP 2 Same V across each: by Ohm's law $I_1 = V/R_1$, $I_2 = V/R_2$, $I_3 = V/R_3$

STEP 3 For the equivalent resistor R_p: $I = V/R_p$

STEP 4 So $V/R_p = V/R_1 + V/R_2 + V/R_3$; cancel V

$$\mathbf{1/R_p = 1/R_1 + 1/R_2 + 1/R_3}$$

The reciprocal of the equivalent resistance is the sum of the reciprocals

Shortcuts that save time ★★☆

Case	Equivalent resistance
Two resistors	$\mathbf{R_p = R_1R_2 / (R_1 + R_2)}$ (product over sum)
n equal resistors R	$\mathbf{R_p = R/n}$; in series it would be nR
Any parallel set	R_p is **less than the smallest** resistor

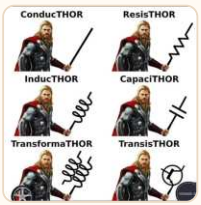
Quick check: 3 Ω ‖ 6 Ω = 18/9 = 2 Ω. For n equal resistors, $R_s : R_p = nR : R/n = \mathbf{n^2 : 1}$.

BUJHO *Resistor jodne se resistance badhna chahiye na? Parallel mein kam kaise ho gaya?*

PAHELI *Har naya branch current ko ek extra raasta deta hai. Zyada raaste matlab current ko kam rukawat, isliye R_p sabse chhote se bhi chhota.*

REMEMBER: ADDING ANY BRANCH LOWERS R_p

Even a 1 MΩ resistor added in parallel makes R_p a little smaller. Removing a branch makes R_p larger and the main current smaller.



MEME BREAK

*ResisTHOR kitne bhi lagao, **parallel** mein total resistance sabse chhote wale se bhi kam hi rahega!*

★★★ *Derivation of the parallel formula with circuit diagram: 3 marks ka favourite sawaal. Nitesh Bhaiya ki guarantee!*

3 SERIES VS PARALLEL ★★★ MOST ASKED

Compare before you calculate

Feature	Series	Parallel
Current	same in every resistor	divides; $I = I_1 + I_2 + \ldots$
Potential difference	divides; $V = V_1 + V_2 + \ldots$	same across every resistor
Equivalent R	$R_s = R_1 + R_2 + \ldots$	$1/R_p = 1/R_1 + 1/R_2 + \ldots$
Value	more than the largest	less than the smallest
One device fails	whole circuit stops	others keep working

MEME BREAK

*Yoda ki seekh: "Parallel mein jodo, **R_p sabse chhote se bhi kam** aayega, hmm." Bada aaya to galti hai!*

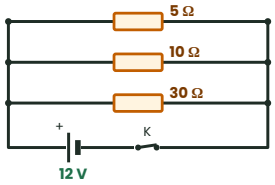
4 MIXED CIRCUITS: THE METHOD ★★★ MOST ASKED

Four steps for any series-parallel circuit

1. Spot the **parallel group** (resistors between the same two junctions) and replace it by R_p.
2. Add the series parts: $R_{total} = R_{series} + R_p$.
3. Main current $I = V / R_{total}$.
4. Work back: p.d. across each part = $I \times R$, then each branch current = V/R.

Solved Example 1: three in parallel ★★★

Q. Resistors of 5 Ω, 10 Ω and 30 Ω are connected in parallel to a 12 V battery. Find R_p, the current in each resistor and the total current.



GIVEN $R_1 = 5\ \Omega$, $R_2 = 10\ \Omega$, $R_3 = 30\ \Omega$, $V = 12$ V

FORMULA $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$; $I = V/R$

SOLVE $1/R_p = 6/30 + 3/30 + 1/30 = 10/30$, so $\mathbf{R_p = 3\ \Omega}$

CURRENTS $I_1 = 12/5 = \mathbf{2.4\ A}$, $I_2 = 12/10 = \mathbf{1.2\ A}$, $I_3 = 12/30 = \mathbf{0.4\ A}$

ANSWER $I = 12/3 = \mathbf{4\ A} = 2.4 + 1.2 + 0.4$ (check)

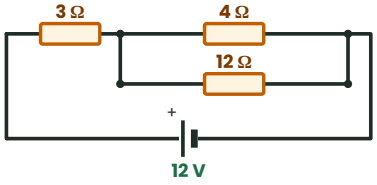
TRY YOURSELF: MIXED CIRCUITS

1. Two 4 Ω resistors in parallel, in series with a 3 Ω resistor, across 10 V. Find the main current.
2. Find R_p of 2 Ω, 4 Ω and 4 Ω in parallel.
3. What is the highest and lowest resistance from four 4 Ω resistors?

Answers are on page 4, below the model answers.

Solved Example 2: series + parallel ★★★

Q. A 3 Ω resistor is in series with a parallel pair of 4 Ω and 12 Ω, across a 12 V battery. Find the total resistance, the main current and the current in the 4 Ω resistor.



STEP 1 $R_p = (4 \times 12)/(4 + 12) = 48/16 = 3\ \Omega$

STEP 2 $R_{total} = 3 + 3 = \mathbf{6\ \Omega}$

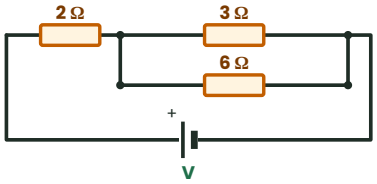
STEP 3 $I = V/R = 12/6 = \mathbf{2\ A}$

STEP 4 p.d. across 3 Ω = $2 \times 3 = 6$ V, so across the pair = $12 - 6 = 6$ V

ANSWER $I_{4\Omega} = 6/4 = \mathbf{1.5\ A}$; $I_{12\Omega} = 6/12 = 0.5$ A (sum 2 A)

Solved Example 3: get 4 Ω and 1 Ω ★★★

Q. How will you connect three resistors of 2 Ω, 3 Ω and 6 Ω to get a total resistance of (a) 4 Ω (b) 1 Ω? NCERT



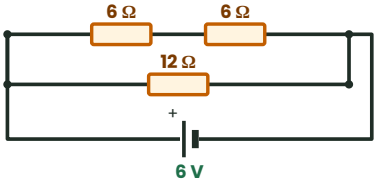
(A) 3 Ω ‖ 6 Ω = 18/9 = 2 Ω; then in series with 2 Ω: 2 + 2 = **4 Ω** (circuit above)

(B) All three in parallel: 1/R = 1/2 + 1/3 + 1/6 = 6/6, so **R = 1 Ω**

NOTE Maximum possible = 11 Ω (all series); minimum = 1 Ω (all parallel)

Solved Example 4: branch with two resistors ★★☆

Q. Two 6 Ω resistors in series form one branch; a 12 Ω resistor forms a second branch in parallel with it. The combination is connected to a 6 V battery. Find the total current and the current in each branch.



STEP 1 Branch 1 = 6 + 6 = 12 Ω; branch 2 = 12 Ω

STEP 2 $R_p = 12/2 = \mathbf{6\ \Omega}$ (two equal resistors: R/n)

STEP 3 $I = 6/6 = \mathbf{1\ A}$

ANSWER Each branch has 6 V across it, so each carries $6/12 = \mathbf{0.5\ A}$

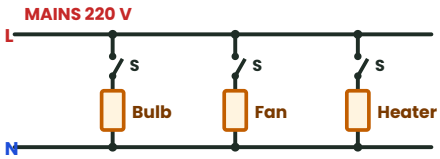
EXAM TIP: SHOW THE REDRAW

Write "3 Ω ‖ 6 Ω = 2 Ω" beside a redrawn circuit before the next step. Units on every line: Ω, A, V.

★★★ *Mixed circuit numericals 3 marks ke pakke hain: pehle parallel ko ek resistor banao, phir series jodo. Nitesh Bhaiya ki guarantee!*

5 PARALLEL WIRING AT HOME ★★★ MOST ASKED

Home circuit: every appliance in parallel



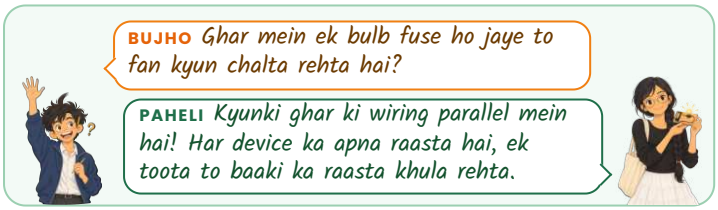
L = live wire, N = neutral wire, S = switch. Each appliance sits between L and N, so each gets the full 220 V.

Why not series for domestic circuits?

1. If one bulb fuses or one device is switched off, the circuit breaks and **every device stops**.
2. The **same current** flows through all, but a bulb and a heater need very different currents.
3. Resistances add up, so the total resistance is large and the current is too small to run the devices.

Advantages of parallel connection ★★★

- ✔ Each appliance gets the **full supply voltage** (220 V).
- ✔ If one appliance fails, the **others keep working**; each has its own switch.
- ✔ Each appliance draws the **current it needs** (I = V/R), according to its own resistance.
- ✔ Total resistance is low, so the circuit can run many devices together.



Class questions (from Nitesh Sir's notes)

Q1. Judge the equivalent resistance when 1 Ω and 10^6 Ω are connected in parallel.
Ans. R_p is less than the smallest, so it is **less than 1 Ω** (about 0.999999 Ω).
Q2. Why does the main current increase when one more bulb is switched on at home?
Ans. A new parallel branch lowers R_p; with V fixed, I = V/R_p increases.

Common mistakes that cost marks

× WRONG	✔ RIGHT
$1/R_p = 1/2 + 1/3$, so R_p = 5/6 Ω	$1/R_p = 5/6$, so R_p = **6/5 Ω**: invert at the end
Same current in all parallel branches	Same **p.d.**; currents differ
Voltmeter in series	Voltmeter in parallel, ammeter in series

6 PRACTICE MCQS

Mixed circuit dekh ke rona aa gaya? Method follow karo, 10 MCQs khud solve karo!

1. Three resistors of 6 Ω each are connected in parallel. The equivalent resistance is: ★★★
(a) 18 Ω (b) 2 Ω
(c) 3 Ω (d) 6 Ω

2. In a parallel combination, the quantity that is the same for every resistor is: ★★★
(a) current (b) potential difference
(c) resistance (d) charge

3. The equivalent resistance of 1 Ω and 10^6 Ω in parallel is: NCERT ★★☆
(a) more than 10^6 Ω (b) between 1 Ω and 10^6 Ω
(c) less than 1 Ω (d) exactly 10^6 Ω

4. 4 Ω and 12 Ω resistors in parallel give an equivalent resistance of: ★★☆
(a) 16 Ω (b) 8 Ω
(c) 3 Ω (d) 0.33 Ω

5. A wire of resistance R is cut into five equal parts, which are joined in parallel to give R′. The ratio R/R′ is: NCERT ★★★
(a) 1/25 (b) 1/5
(c) 5 (d) 25

6. Domestic appliances are connected in parallel because: ★★☆
(a) the same current flows through each (b) each gets the same voltage and works independently
(c) total resistance increases (d) it saves wire

7. How many 176 Ω resistors in parallel are needed to draw 5 A from a 220 V line? NCERT ★★☆
(a) 2 (b) 4
(c) 5 (d) 44

8. The lowest total resistance from coils of 4 Ω, 8 Ω, 12 Ω and 24 Ω is: NCERT ★★☆
(a) 48 Ω (b) 4 Ω
(c) 2 Ω (d) 1 Ω

9. **Assertion (A):** When more resistors are added in parallel, the total current drawn from the source increases.
Reason (R): The equivalent resistance of a parallel combination decreases as more resistors are added. ★★☆
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

10. **Assertion (A):** Bulbs in a house are connected in series.
Reason (R): In a series circuit, every bulb carries the same current. ★★☆
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

Short and long answer questions

1. Derive the expression for the equivalent resistance of three resistors connected in parallel. **(3 marks)**
2. Give two disadvantages of series connection and two advantages of parallel connection for domestic appliances. **(3 marks)**
3. An electric lamp of 100 Ω, a toaster of 50 Ω and a water filter of 500 Ω are connected in parallel to a 220 V source. What is the resistance of an electric iron that draws as much current as all three together, and what current flows through it? **(3 marks)** NCERT

Answer key: MCQs

Q	Ans	Why
1	(b)	R/n = 6/3 = 2 Ω.
2	(b)	Every branch is across the same two points.
3	(c)	R_p is less than the smallest resistor.
4	(c)	48/16 = 3 Ω.
5	(d)	Each part R/5; five in parallel: R/25, so R/R′ = 25.
6	(b)	Full 220 V each; one off, others work.
7	(b)	R = 220/5 = 44 Ω; n = 176/44 = 4.
8	(c)	1/R = (6 + 3 + 2 + 1)/24 = 12/24, so 2 Ω.
9	(a)	Lower R_p with fixed V means larger I.
10	(d)	Home bulbs are in parallel; R itself is true.

Model answers: short and long answer

1 Derivation of the parallel formula

$I = I_1 + I_2 + I_3$; same V across each, so $I_1 = V/R_1$, $I_2 = V/R_2$, $I_3 = V/R_3$. For R_p, $I = V/R_p$. Hence $V/R_p = V/R_1 + V/R_2 + V/R_3$, giving $\mathbf{1/R_p = 1/R_1 + 1/R_2 + 1/R_3}$. Draw the circuit with ammeter in series and voltmeter across the combination.

2 Series vs parallel at home

Series: one fault stops all; same current for devices that need different currents. Parallel: each gets 220 V; devices work independently, each has its own switch and draws the current it needs. Also, the total resistance stays low.

3 Lamp, toaster and filter

GIVEN $R_1 = 100\ \Omega$, $R_2 = 50\ \Omega$, $R_3 = 500\ \Omega$, V = 220 V

SOLVE $1/R = 1/100 + 1/50 + 1/500 = (5 + 10 + 1)/500 = 16/500$

R $R = 500/16 = \mathbf{31.25\ \Omega}$ (the iron must have this resistance)

I $I = 220/31.25 = \mathbf{7.04\ A}$

Try yourself answers: (1) 4 Ω ‖ 4 Ω = 2 Ω, total 5 Ω, I = 2 A (2) 1/R = 1/2 + 1/4 + 1/4 = 1, R_p = 1 Ω (3) highest 16 Ω, lowest 1 Ω

FORMULA RECAP

Parallel: $I = I_1 + I_2 + I_3$, V same, $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$. Mixed: $R_{total} = R_{series} + R_p$, then $I = V/R_{total}$.

Coming up: Day 34

Heating effect of current: Joule's law $H = I^2Rt$, bulbs, heaters and fuses, electric power and the kilowatt-hour.

Day 33 mind map: revise in 2 minutes

SAME V
Every branch is across the same two points

CURRENT DIVIDES
$I = I_1 + I_2 + I_3$; small R takes big I

RESISTORS IN PARALLEL

FORMULA ★★★
$1/R_p = \Sigma\ 1/R$; two: $R_1R_2/(R_1+R_2)$; n equal: R/n

HOME WIRING
Parallel: full 220 V, independent switches

MIXED CIRCUITS
Parallel group → R_p, add series parts, find I = V/R, then work back to each branch

Day 33 hit-list: Eduspark PYQ analysis

Topic	Rating
Derivation of $1/R_p$ with circuit	★★★
Mixed series-parallel numericals	★★★
Why domestic wiring is parallel	★★★
Highest / lowest resistance combinations	★★☆
Cut wire, parts in parallel (R/n^2)	★★☆

REVISION PLAN FOR TONIGHT

Redo Solved Examples 2 and 4 without looking, then attempt the Try yourself box. Time yourself: 4 minutes per numerical.

Day 33 checklist: tick before you sleep

- ☐ I can state the rules for current and p.d. in a parallel circuit
- ☐ I can derive the formula for R_p with a circuit diagram
- ☐ I can solve a mixed series-parallel numerical step by step
- ☐ I can explain why homes use parallel wiring
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

LEVEL 33 COMPLETE +100 XP earned. Next: Level 34

BUJHO Samajh gaya! Parallel mein voltage same, current bant jaata hai!

PAHELI Shabaash! Kal Level 34: heating effect, power aur bijli ka bill.

NITESH SIR SAYS
Circuit pehle redraw karo, phir parallel group ko ek resistor banao. **Diagram saaf to answer saaf!**

DAY 34 · LEVEL 34

PHYSICS | CHAPTER 7 | PART 5 · ASTRA SERIES

HEATING EFFECT & ELECTRIC POWER

Goal: use $H = I^2Rt$, $P = VI$ and the kWh with confidence.

by Nitesh Prakash

01 Joule's Law | 02 Bulb & Heater | 03 Fuse | 04 Electric Power | 05 kWh & Bill | +100 XP today | Level 34 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 THREE EFFECTS OF CURRENT ★★☆

Electric current can produce

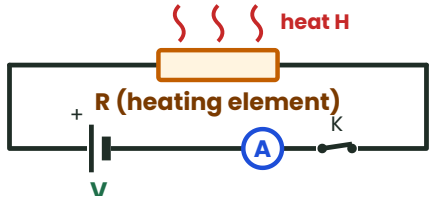
Effect	What happens	Used in
Heating	conductor gets hot	heater, iron, bulb, fuse
Magnetic	wire behaves like a magnet	electromagnet, bell (Day 36)
Chemical	solution decomposes	electroplating, refining

2 HEATING EFFECT OF CURRENT ★★★ MOST ASKED

DEFINITION

- **Heating effect:** when a current flows through a resistor, part of the electrical energy is **converted into heat**
- The source keeps doing work to push charge through the resistance; this work appears as **heat energy**
- Purely resistive circuit: **all** the energy from the source is dissipated as heat

Circuit: current through a heating element



More current, more resistance or more time means more heat. Nichrome coil gets red-hot; copper wires stay cool.

Energy chain in a heater

Cell / mains chemical or electrical energy → **Moving charges** work done = VQ → **Resistor** heat energy H

CLASS NOTE: THREE EFFECTS OF CURRENT

Current ke teen effect: Heating (nichrome), Magnetic, Chemical. Aaj ka hero: heating effect!

REMEMBER: WHY CONNECTING WIRES STAY COOL

Copper connecting wires have very low resistance, so I^2Rt is tiny there. The heat appears in the high-resistance element.

3 JOULE'S LAW OF HEATING ★★★ MOST ASKED

JOULE'S LAW

- The heat produced in a resistor is directly proportional to (i) the **square of the current**, I^2 (ii) the **resistance**, R (iii) the **time** for which current flows, t

Derivation of $H = I^2Rt$ ★★★

STEP 1 Charge Q moves through p.d. V: work done W = VQ
STEP 2 Charge in time t: Q = It, so W = VIt
STEP 3 This work is converted to heat: H = VIt
STEP 4 By Ohm's law V = IR, so H = (IR)It

$$H = I^2Rt = VIt = V^2t/R$$

H in joule (J), I in ampere, R in ohm, t in second

Read the law like a graph ★★☆

- ✔ Double I → heat becomes **4 times** ($H \propto I^2$).
- ✔ Double R (same I) → heat doubles.
- ✔ Double t → heat doubles.

BUJHO *Heater ka coil laal ho jata hai, par usse judi copper ki taar thandi kyun rehti?*

PAHELI *Same current, par coil ka resistance bahut zyada hai. $H = I^2Rt$, to garmi wahin banti jahan R bada hai.*

Solved Example 1: heat from charge ★★☆

Q. Compute the heat generated while transferring 96000 C of charge in one hour through a potential difference of 50 V. NCERT

GIVEN Q = 96000 C, V = 50 V, t = 1 h
FORMULA H = VQ (work on the charge)
ANSWER $H = 50 \times 96000 = \mathbf{4.8 \times 10^6\ J}$

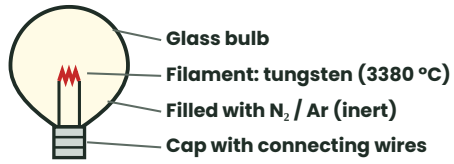
lly = I love you
lmy = I miss you
Brb = Be right back
pV = nRT
F = ma
KE = (1/2)mv^2
p1v1 = p2v2
E = mc^2

MEME BREAK

Formula list mein ek aur pyaar jod lo: $H = I^2Rt$. Current double, garmi chaar guna!

4 USES OF THE HEATING EFFECT ★★★ MOST ASKED

Electric bulb ★★★



- ✔ Filament made of **tungsten**: very high melting point (3380 °C), glows white-hot without melting.
- ✔ Bulb filled with chemically inactive **nitrogen and argon** to prolong the life of the filament.
- ✔ Most of the energy is given out as **heat**; only a small part as light.

Heater, iron, toaster, kettle

- ✔ Heating element is a coil of **nichrome** (alloy of nickel and chromium).
- ✔ The coil sits in grooves of a **china clay (porcelain)** plate, which is a heat-resistant insulator.

Class question: why only nichrome for the heating coil? ★★★

Ans. (i) Nichrome has **high resistivity**, so more heat is produced (H = I^2Rt). (ii) It has a **high melting point**. (iii) It **does not oxidise (burn) readily** at high temperature, so it can stay red-hot for long.

Electric fuse ★★★

- ✔ A short wire of an alloy with **low melting point** (aluminium, copper, iron, lead), connected **in series** with the device.
- ✔ If the current exceeds the safe value, the wire heats up, **melts and breaks the circuit**, protecting the appliance.
- ✔ Common ratings: 1 A, 2 A, 3 A, 5 A, 10 A.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Modern homes use a **miniature circuit breaker (MCB)**: it switches off automatically on overload and can be reset by a switch, so no fuse wire needs replacing.

BUJHO *Fuse ki jagah moti copper ki taar laga dein to kya hoga?*

PAHELI *Wo pighlegi hi nahi! Zyada current mein appliance jal jayega ya aag lag sakti hai. Sahi rating ka fuse hi lagao.*

Solved Example 2: resistance of a bulb ★★☆

Q. Find the resistance of the filament of a 100 W, 220 V bulb. Which has more resistance: this bulb or a 60 W, 220 V bulb?

FORMULA $P = V^2/R$, so $R = V^2/P$

100 W $R = 220^2/100 = 48400/100 = \mathbf{484\ \Omega}$

60 W $R = 48400/60 = \mathbf{806.7\ \Omega}$: the 60 W bulb has more resistance

Solved Example 3: choosing a fuse ★★★

Q. An electric iron of 1 kW is used on a 220 V supply. Which fuse should be used: 2 A, 5 A or 10 A?

GIVEN P = 1000 W, V = 220 V

FORMULA I = P/V

SOLVE I = 1000/220 = **4.54 A**

ANSWER Use the **5 A** fuse: just above the normal current

Solved Example 4: heat in an iron ★★★

Q. An electric iron of resistance 20 Ω takes a current of 5 A. Calculate the heat developed in 30 s. NCERT

GIVEN R = 20 Ω, I = 5 A, t = 30 s

FORMULA $H = I^2Rt$

SOLVE $H = 5^2 \times 20 \times 30 = 25 \times 600$

ANSWER $H = \mathbf{15000\ J = 1.5 \times 10^4\ J}$

Power ratings you will meet (typical values)

Appliance	Power	Current at 220 V
LED bulb	10 W	about 0.05 A
Ceiling fan	75 W	about 0.34 A
Electric iron	1000 W	about 4.5 A
Geyser	2000 W	about 9.1 A

High-power appliances draw large currents, so they need thicker wires and higher fuse ratings.

Undesirable heating

- ✔ Heat in wires, motors and computers wastes energy and can damage insulation.
- ✔ Thick copper wires and cooling fans keep this loss low.

Common mistakes that cost marks

× WRONG	✔ RIGHT
kWh is a unit of power	kWh is a unit of **energy**
Time in minutes put directly in kWh	Convert to **hours** first
Higher wattage bulb has higher R	$R = V^2/P$: higher P, **lower** R

FUSE SAFETY

Always replace a blown fuse with one of the **same rating**. Never use a thicker wire or a nail in its place.

MEME BREAK

Joule aur kWh ki dosti: ek hi energy, bas 3.6×10^6 *ka farak. Unit badlo, answer nahi!*

★★★ *Tungsten filament, nichrome coil, fuse in series: har saal ek na ek pakka aata hai. Nitesh Bhaiya ki guarantee!*

5 ELECTRIC POWER ★★★ MOST ASKED

DEFINITION

- **Electric power:** the rate at which electrical energy is consumed (or work is done) in a circuit, P = W/t
- SI unit: **watt (W)**. **1 W** is the power consumed by a device that carries **1 A** when operated at a p.d. of **1 V**
- 1 kilowatt (kW) = 1000 W

Derivation: three forms of P ★★★

STEP 1 P = W/t and W = VQ, so P = VQ/t
STEP 2 Q/t = I, so **P = VI**
STEP 3 Put V = IR: **P = I^2R**
STEP 4 Put I = V/R: **P = V^2/R**

$$P = VI = I^2R = V^2/R$$

Same current (series): use I^2R. Same voltage (parallel, home): use V^2/R

Solved Example 5: power of a bulb ★★★

Q. An electric bulb connected to a 220 V generator draws a current of 0.50 A. What is its power and its resistance? NCERT

GIVEN V = 220 V, I = 0.50 A
POWER P = VI = 220 × 0.50 = **110 W**
RESISTANCE R = V/I = 220/0.50 = **440 Ω**

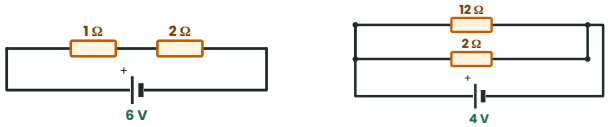
Solved Example 6: heater on the mains ★★☆

Q. An electric heater of resistance 8 Ω draws 15 A from the service mains for 2 hours. Calculate the rate at which heat is developed in the heater. NCERT

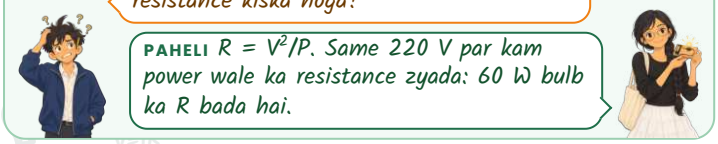
FORMULA Rate of heat = power, P = I^2R
ANSWER P = 15^2 × 8 = 225 × 8 = **1800 W = 1800 J/s**

Solved Example 7: compare the power in the 2 Ω resistor ★★★

Q. Compare the power used in the 2 Ω resistor in: (i) a 6 V battery in series with 1 Ω and 2 Ω (ii) a 4 V battery in parallel with 12 Ω and 2 Ω. NCERT



(I) I = 6/(1 + 2) = 2 A; P = I^2R = 2^2 × 2 = **8 W**
(II) V across 2 Ω = 4 V; P = V^2/R = 16/2 = **8 W**
ANSWER Power is the **same, 8 W**, in both cases



REMEMBER: RATED BULB ON LOWER VOLTAGE

R stays fixed, so P ∝ V^2. A 220 V, 100 W bulb on 110 V gives $(1/2)^2$ × 100 = **25 W**.

6 ELECTRICAL ENERGY ★★★ MOST ASKED

DEFINITION

- **Electrical energy** used by a device = power × time: **E = P × t**
- SI unit: joule (J). Practical unit: watt-hour (Wh)
- Commercial unit: **kilowatt-hour (kWh)**, called **1 unit** on the electricity bill

How big is 1 kWh? ★★★

1 kWh = 1000 W × 3600 s = **3.6×10^6 J**
Energy used by a 1000 W device running for 1 hour

Bill = units (kWh) × rate per unit. Units = (watts × hours) / 1000.

MEME BREAK
*Bill pe **1 unit** likha hai? Wo 1 joule nahi, poore **3.6 × 10^6 J** hai!*

Solved Example 8: refrigerator bill ★★★

Q. An electric refrigerator rated 400 W operates 8 hours a day. What is the cost of the energy to operate it for 30 days at ₹ 3.00 per kWh? NCERT

ENERGY E = 400 W × 8 h × 30 = 96000 Wh = **96 kWh**
COST 96 × 3.00 = **₹ 288**

Solved Example 9: which uses more energy ★★☆

Q. Which uses more energy: a 250 W TV set in 1 hour, or a 1200 W toaster in 10 minutes? NCERT

TV E = 250 W × 1 h = **250 Wh**
TOASTER E = 1200 W × (10/60) h = **200 Wh**
ANSWER The **TV** uses more energy (250 Wh > 200 Wh)

Exam tips

- ✔ Write H = I^2Rt with units: J, A, Ω, s.
- ✔ For "why tungsten / why nichrome", give two reasons each.
- ✔ Fuse: say "in series" and "low melting point" every time.

Short and long answer questions

1. State Joule's law of heating and derive H = I^2Rt. **(3 marks)**
2. An electric motor takes 5 A from a 220 V line. Find its power and the energy consumed in 2 h. **(2 marks)** NCERT
3. Two lamps, 100 W and 60 W, are connected in parallel to a 220 V supply. What current is drawn from the line? **(2 marks)** NCERT
4. Why is tungsten used for bulb filaments, and why are heater coils made of an alloy? **(2 marks)**

7 PRACTICE MCQS

Ek side: "bijli ka bill joule mein?"
Doosri side: "units = kWh", mast view.
Ab MCQs khud solve karo!

1. The heat produced in a resistor is given by: ★★★

(a) IRt (b) I^2Rt
(c) IR^2t (d) I^2R/t

2. The filament of an electric bulb is made of tungsten because it has: ★★★

(a) low resistivity (b) a very high melting point
(c) a low melting point (d) low cost

3. A fuse wire is connected: ★★★

(a) in parallel with the appliance (b) in series with the appliance
(c) between live and earth (d) only in the neutral wire

4. 1 kWh is equal to: ★★★

(a) 3.6×10^3 J (b) 3.6×10^5 J
(c) 3.6×10^6 J (d) 36×10^6 J

5. If the current through a resistor is doubled, the heat produced in the same time becomes: ★★☆

(a) double (b) half
(c) four times (d) unchanged

6. Which of the following does not represent electrical power in a circuit? NCERT ★★☆

(a) I^2R (b) IR^2
(c) VI (d) V^2/R

7. A bulb rated 220 V, 100 W is operated on 110 V. The power consumed is: NCERT ★★☆

(a) 100 W (b) 75 W
(c) 50 W (d) 25 W

8. **Assertion (A):** Nichrome is used in the heating elements of electric heaters.
Reason (R): Nichrome has high resistivity and does not oxidise readily at high temperature. ★★☆

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

9. **Assertion (A):** The commercial unit of electrical energy is the watt.
Reason (R): 1 kWh = 3.6×10^6 J. ★☆☆

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

TRY YOURSELF: BILLS AND POWER

1. Five 60 W bulbs run 6 h a day and a 1 kW iron runs 1 h a day. Find the units used in 30 days and the bill at ₹ 5 per unit.
2. Find the resistance of a 60 W, 220 V bulb.

Answers: 84 units, ₹ 420; R = V^2/P = 48400/60 = 806.7 Ω.

Answer key: MCQs

Q	Ans	Why
1	(b)	Joule's law: $H = I^2Rt$.
2	(b)	Melting point 3380 °C; glows without melting.
3	(b)	In series, so it breaks the whole current path.
4	(c)	1000 W × 3600 s.
5	(c)	$H \propto I^2$: $2^2 = 4$.
6	(b)	IR^2 has wrong units.
7	(d)	R fixed; $P \propto V^2$: half V gives P/4 = 25 W.
8	(a)	R gives the exact reason for A.
9	(d)	Unit of energy is kWh; watt is power.

Model answers: numericals

2 Motor

POWER P = VI = 220 × 5 = **1100 W**

ENERGY E = 1.1 kW × 2 h = **2.2 kWh** = $2.2 \times 3.6 \times 10^6$ = **7.92×10^6 J**

3 Two lamps in parallel

TOTAL P 100 + 60 = 160 W

CURRENT I = P/V = 160/220 = **0.73 A**

Day 34 hit-list: Eduspark PYQ analysis

Topic	Rating
Joule's law with derivation	★★★
kWh to J and electricity bill	★★★
Tungsten, nichrome, fuse reasons	★★★

Day 34 checklist: tick before you sleep

- ☐ I can state Joule's law and derive $H = I^2Rt$
- ☐ I can explain the bulb, heater and fuse using the heating effect
- ☐ I can use $P = VI = I^2R = V^2/R$
- ☐ I can convert kWh to joules and calculate a bill
- ☐ I solved all 9 MCQs and checked them with the key

My score: ____ / 5

LEVEL 34 COMPLETE +100 XP earned. Next: Level 35

BUJHO *Samajh gaya! Garmi I^2Rt, power VI, aur bill kWh mein!*

PAHELI *Shabaash! Kal Level 35: Chapter 7 ka Boss Level test.*

NITESH SIR SAYS

Har numerical mein units likho aur kWh ko joule mein badalna mat bhoolo. ***Unit sahi, marks pakke!***

DAY 35 · LEVEL 35

PHYSICS | CHAPTER 7 | BOSS LEVEL

CHAPTER 7 BOSS LEVEL

Goal: test all of Electricity in 90 minutes.

by Nitesh Prakash

01 Revision Map · 02 MCQs · 03 Short Answers · 04 Case & Long · 05 Answer Key · +200 XP today | Level 35 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 7 REVISION MAP ★★★ MOST ASKED

Chapter 7 mind map: revise in 5 minutes

CURRENT & P.D.
I = Q/t (A); V = W/Q (V); 1 C = 6.25 × 10^{18} electrons

OHM'S LAW
V = IR at constant temperature; V–I graph a straight line

ELECTRICITY

RESISTANCE ★★★
R = ρl/A; ρ depends on material and temperature

COMBINATIONS
$R_s = R_1 + R_2 + \ldots$; $1/R_p = 1/R_1 + 1/R_2 + \ldots$

HEAT, POWER, ENERGY
$H = I^2Rt$; $P = VI = I^2R = V^2/R$; 1 kWh = 3.6 × 10^6 J

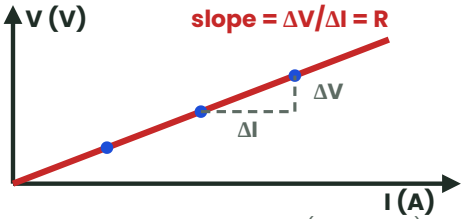
Must-know formulas: write them daily ★★★

Quantity	Formula	SI unit
Current	I = Q/t	ampere (A)
Potential difference	V = W/Q	volt (V)
Resistance	R = V/I = ρl/A	ohm (Ω)
Resistivity	ρ = RA/l	ohm metre (Ω m)
Heat	$H = I^2Rt$	joule (J)
Power	P = VI	watt (W)
Energy	E = Pt	J or kWh

CLASS TRICK: VOLTAGE = POTENTIAL DIFFERENCE

"Gangadhar hi Shaktiman hain or Shaktiman hi Gangadhar hain": potential difference aur voltage ek hi cheez ke do naam hain.

V–I graph you must draw ★★☆



A straight line through the origin shows V ∝ I (Ohm's law). A steeper line means a larger resistance.

How to attack the Boss: 90 minutes

1. Section A (20 MCQs, 1 mark each): **20 min**
2. Section B (5 × 2 marks): **12 min**
3. Section C (4 × 3 marks): **20 min**
4. Section D (2 cases × 4 marks): **14 min**
5. Section E (2 × 5 marks): **18 min**; final check of units: 6 min

Total 60 marks. Write Given, Formula, Substitution and Answer with unit in every numerical.

BOSS INTRO

*Ohm, Volta, Ampere, Joule: saare scientists dekh rahe hain. Aaj **Boss Level** mein unka naam roshan karo!*

2 SECTION A: MCQS ★★★ MOST ASKED

1. The number of electrons that make up a charge of 1 C is: ★★★
(a) 1.6 × 10^{-19} (b) 6.25 × 10^{18}
(c) 6.25 × 10^{19} (d) 1.6 × 10^{19}

2. 1 volt is equal to: ★★★
(a) 1 J/C (b) 1 C/J
(c) 1 J × 1 C (d) 1 A/s

3. An ammeter is always connected: ★★☆
(a) in series (b) in parallel
(c) across the cell (d) across the resistor

4. The resistivity of a conductor depends on: ★★★
(a) its length (b) its area of cross-section
(c) its material and temperature (d) its shape

5. A wire of resistance R is stretched to twice its length (volume unchanged). Its new resistance is: ★★★
(a) R/2 (b) 2R
(c) 4R (d) R

6. The slope of a V–I graph (V on the y-axis) of an ohmic conductor gives: ★★☆
(a) conductance (b) resistance
(c) resistivity (d) power

7. Using three 3 Ω resistors, which value can NOT be obtained? ★★☆

(a) 9 Ω (b) 1 Ω
(c) 4.5 Ω (d) 6 Ω

8. A charge of 150 C flows through a wire in 1 minute. The current is: ★★☆

(a) 150 A (b) 2.5 A
(c) 0.4 A (d) 9000 A

9. The work done in moving 2 C of charge across a p.d. of 12 V is: NCERT ★★★

(a) 6 J (b) 12 J
(c) 24 J (d) 14 J

10. A 100 W bulb glows for 10 hours. The energy used is: ★★☆

(a) 1 kWh (b) 10 kWh
(c) 0.1 kWh (d) 1000 kWh

11. If the current in a heater coil is halved, the heat produced in the same time becomes: ★★★

(a) half (b) one-fourth
(c) double (d) four times

12. Heating elements are made of alloys rather than pure metals because alloys: NCERT ★★★

(a) have low resistivity (b) melt easily
(c) have high resistivity and do not oxidise readily (d) are good conductors

13. **Assertion (A):** The resistivity of a material does not depend on the length of the wire.
Reason (R): Resistivity is a characteristic property of the material at a given temperature. ★★☆

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

14. **Assertion (A):** In a parallel circuit, the equivalent resistance is less than the smallest resistance.
Reason (R): In a parallel circuit, the potential difference across each resistor is different. ★★☆

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

15. The SI unit of resistivity is: ★★☆

(a) Ω (b) Ω m
(c) Ω/m (d) Ω m^2

16. The device used to change the resistance in a circuit is a: ★☆☆

(a) ammeter (b) voltmeter
(c) rheostat (d) galvanometer

17. The conventional direction of electric current is: ★★☆

(a) the direction of electron flow (b) opposite to the direction of electron flow
(c) from negative to positive terminal outside the cell (d) random

MCQ TIP
Read all four options before you tick.

18. Wire A has twice the length and half the area of cross-section of wire B (same material). $R_A : R_B$ is: ★★★

(a) 1 : 1 (b) 2 : 1
(c) 4 : 1 (d) 1 : 4

19. A 60 W and a 100 W bulb (both 220 V) are joined in series to 220 V. Which glows brighter? ★★☆

(a) 100 W bulb (b) 60 W bulb
(c) both equally (d) neither glows

20. A fuse wire should have: ★★☆

(a) high melting point (b) low melting point
(c) very low resistance (d) high melting point and low resistance

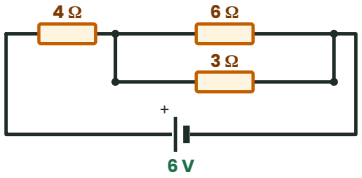
3 SECTIONS B AND C ★★★ MOST ASKED

Section B: Short Answers (2 marks each)

1. Define 1 volt. State Ohm's law.
2. Why are copper and aluminium wires used for electricity transmission?
3. Find the equivalent resistance of 10 Ω and 15 Ω (a) in series (b) in parallel.
4. Why is a series arrangement not used for domestic circuits? Give two reasons.
5. Draw a schematic diagram of a circuit with a battery of three 2 V cells, resistors of 5 Ω, 8 Ω and 12 Ω and a plug key, all in series. Find the current. NCERT

Section C: Short Answers (3 marks each)

1. The resistance of a metal wire of length 1 m is 26 Ω at 20 °C. If the diameter of the wire is 0.3 mm, what is the resistivity of the metal? NCERT
2. An electric iron consumes energy at 840 W when heating is at the maximum and 360 W when at the minimum, on 220 V. Find the current and resistance in each case. NCERT
3. A hot plate has two coils A and B of 24 Ω each. Find the current drawn from a 220 V supply when they are used (a) separately (b) in series (c) in parallel. NCERT
4. In the circuit below, find (a) the total resistance (b) the main current (c) the current in the 3 Ω resistor.



Resistivity aur electric iron wale numericals: pakka aayenge, isliye do baar practice karo!

BUJHO Section D mein units kaise nikalun? Watt ko seedha 30 se guna kar doon?

PAHELI Watt × hour = Wh; 1000 se bhaag do to kWh, phir din aur rate se guna karo!

4 SECTIONS D AND E ★★★ MOST ASKED

Section D, Case 1: Riya's home (4 marks)

Riya's home runs on a 220 V supply. Every day she uses 4 LED bulbs of 10 W for 5 h, 2 fans of 75 W for 10 h and a refrigerator of 200 W for 24 h. The electricity board charges ₹ 6 per unit.

1. How are these appliances connected, and why? **(1)**
2. Find the total energy used per day in kWh. **(1)**
3. Find the electricity bill for 30 days. **(1)**
4. What current flows in the main line when all of them run together? **(1)**

Case 2: choosing the right material (4 marks)

Resistivity (Ω m) at 20 °C: silver 1.60×10^{-8}, copper 1.62×10^{-8}, aluminium 2.63×10^{-8}, tungsten 5.20×10^{-8}, nichrome 100×10^{-6}, glass 10^{10} to 10^{14}.

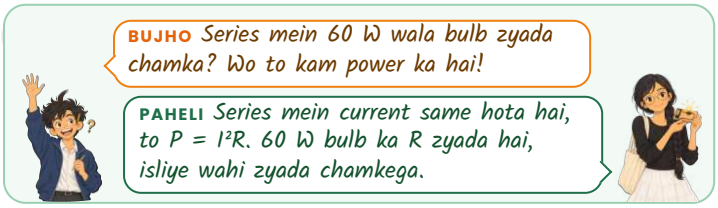
1. Which is the best conductor in the list? **(1)**
2. Which material is used in heating elements and which in bulb filaments? **(1)**
3. Why is copper used for wiring rather than silver? **(1)**
4. A wire's length and area are both doubled. What happens to its resistance and resistivity? **(1)**

Section E: Long Answers (5 marks each)

1. (a) Derive the expression for the equivalent resistance of three resistors R_1, R_2 and R_3 connected in series. **(3)** (b) Resistors of 5 Ω, 10 Ω and 15 Ω are connected in series to a 30 V battery. Find the current and the p.d. across each resistor. **(2)**
2. (a) Define electric power and derive $P = I^2R$ and $P = V^2/R$. **(2)** (b) A 1.5 kW geyser runs 2 h every day. Find the energy used in 30 days and the cost at ₹ 7 per unit. Which fuse (5 A or 10 A) suits it on 220 V? **(3)**

Common mistakes that cost marks

× WRONG	✓ RIGHT
Diameter used as radius in A = πr^2	Halve the diameter first
mm^2 left as it is	Convert to m^2: 1 mm = 10^{-3} m
Rated power used at a different voltage	R stays fixed; find R = V^2/P first
Unit missing in final answer	Every answer needs Ω, A, V, W or J



5 ANSWER KEY ★★★ MOST ASKED

Answer key: Section A (MCQs)

Q	Ans	Why
1	(b)	$1/(1.6 \times 10^{-19}) = 6.25 \times 10^{18}$.
2	(a)	V = W/Q.
3	(a)	Same current must pass through it.
4	(c)	Not on l or A.
5	(c)	l doubles, A halves: R × 4.
6	(b)	V/I = R.
7	(d)	Possible: 9, 1, 4.5, 2 Ω.
8	(b)	I = 150/60.
9	(c)	W = VQ = 12 × 2.
10	(a)	0.1 kW × 10 h.
11	(b)	$H \propto I^2$.
12	(c)	High ρ, no oxidation.
13	(a)	R explains A.
14	(c)	R is false: p.d. is the same.
15	(b)	ρ = RA/l: Ω × m^2/m.
16	(c)	Rheostat varies R.
17	(b)	Positive to negative outside the cell.
18	(c)	2 ÷ ½ = 4.
19	(b)	Same I; higher R (60 W) gets more I^2R.
20	(b)	It must melt on overload.

Score card: how did you fight the boss?

Marks	Result
54–60	BOSS DEFEATED
42–53	STRONG WIN: fix the small slips
30–41	ALMOST THERE: revise weak days
Below 30	RETRY: re-read Days 30–34

› **Where to revise:** Q1–3, 8, 9, 16, 17 → Day 30; Q4–6, 13, 15, 18 → Day 31; Q7, 14 → Days 32–33; Q10–12, 19, 20 → Day 34.

Bonus rapid-fire round: quick answers (from class notes)

Question	Answer
1 mA and 1 μA in ampere	10^{-3} A, 10^{-6} A
Charge on one electron	1.6×10^{-19} C
Instrument to detect a small current	galvanometer
Voltmeter: series or parallel?	parallel, high resistance
Ammeter resistance	very low
Resistance of a conductor depends on	l, A, material, temperature

Boss Level tracker: write your percentage

Ch 5 (Day 26)	Ch 6 (Day 29)	Ch 7 (Day 35)
____ %	____ %	____ %

Model answers: Sections B and C

B1 Volt and Ohm's law

1 V: p.d. when 1 J of work moves 1 C of charge (1 V = 1 J/C). Ohm's law: at constant temperature, V ∝ I, so V = IR. **(1 + 1)**

B2 Copper and aluminium

Very low resistivity, so little energy is lost as heat; they are also good, cheap conductors. **(2)**

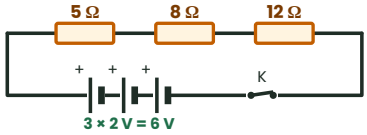
B3 10 Ω and 15 Ω

Series: 10 + 15 = **25 Ω**. Parallel: (10 × 15)/25 = **6 Ω**. **(1 + 1)**

B4 No series at home

One fault stops every device; all devices get the same current though they need different currents. **(1 + 1)**

B5 Series circuit



R = 5 + 8 + 12 = 25 Ω; I = 6/25 = **0.24 A**. **(1 diagram + 1)**

C1 Resistivity

GIVEN R = 26 Ω, l = 1 m, d = 0.3 mm, so r = 0.15×10^{-3} m

AREA $A = \pi r^2 = 3.14 \times (0.15 \times 10^{-3})^2 = 7.07 \times 10^{-8}$ m²

ANSWER $\rho = RA/l = 26 \times 7.07 \times 10^{-8} =$ **1.84×10^{-6} Ω m**

C2 Electric iron

MAX I = P/V = 840/220 = **3.82 A**; R = V^2/P = 48400/840 = **57.6 Ω**

MIN I = 360/220 = **1.64 A**; R = 48400/360 = **134.4 Ω**

C3 Hot plate coils

(A) I = 220/24 = **9.17 A** (each coil alone)

(B) R = 48 Ω; I = 220/48 = **4.58 A**

(C) R = 12 Ω; I = 220/12 = **18.33 A**

C4 Mixed circuit

(A) 6 Ω ‖ 3 Ω = 18/9 = 2 Ω; total = 4 + 2 = **6 Ω**

(B) I = 6/6 = **1 A**

(C) p.d. across pair = 1 × 2 = 2 V; $I_{3\Omega}$ = 2/3 = **0.67 A**

Class questions: Nitesh Sir's favourites

Q1. Why is a voltmeter connected in parallel and an ammeter in series?

Ans. A voltmeter measures p.d. between two points, so it goes across them (high resistance, draws little current). An ammeter must carry the full current, so it goes in series (very low resistance).

Q2. Why does the resistance of a metal wire increase when it gets hot?

Ans. Resistivity of metals increases with temperature, so R = ρl/A increases.

Q3. How many electrons flow per second when a current of 1 A flows?

Ans. n = I t / e = 1/(1.6×10^{-19}) = **6.25×10^{18}** electrons.

Most asked in Chapter 7: Eduspark PYQ analysis

Topic	Rating
Series and parallel numericals	★★★
Resistivity: R = ρl/A numericals	★★★
Joule's law, power, kWh bill	★★★
Ohm's law with V–I graph	★★☆
Fuse, tungsten, nichrome reasons	★★☆

Model answers: Sections D and E

D1 Riya's home

(I) In **parallel**: each gets 220 V and works independently

(II) 4 × 10 × 5 + 2 × 75 × 10 + 200 × 24 = 6500 Wh = **6.5 kWh**

(III) 6.5 × 30 = 195 units; 195 × 6 = **₹ 1170**

(IV) P = 40 + 150 + 200 = 390 W; I = 390/220 = **1.77 A**

D2 Materials

(I) **Silver** (lowest ρ)

(II) Heating: **nichrome**; filament: **tungsten**

(III) Copper is almost as good and much cheaper

(IV) R = ρl/A: 2l/2A, so **R unchanged**; ρ unchanged (same material)

E1 Series resistors

(a) Same I in each; V = $V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3$. With V = IR_s: **$R_s = R_1 + R_2 + R_3$** (diagram 1 mark). (b) R_s = 30 Ω, I = **1 A**; V = **5 V, 10 V, 15 V**.

E2 Power and geyser

(a) P = rate of using electrical energy = W/t = VI; V = IR gives P = I^2R; I = V/R gives P = V^2/R. (b) E = 1.5 × 2 × 30 = **90 kWh**; cost = 90 × 7 = **₹ 630**; I = 1500/220 = 6.82 A, so the **10 A** fuse.

Day 35 checklist: tick before you sleep

- ☐ I attempted all 20 MCQs in 20 minutes
- ☐ I wrote Given, Formula and units in every numerical
- ☐ I can solve series, parallel and mixed circuits
- ☐ I can do resistivity, power and kWh bill numericals
- ☐ I checked my paper with the key and noted weak days

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Weak days phir se padho.

LEVEL 35 COMPLETE +200 XP earned. Next: Level 36

BUJHO *Boss defeat ho gaya! Electricity ab apni dost!*

PAHELI *Shabaash! Kal Level 36: Magnetic Effects of Electric Current.*

Coming up: Day 36

Magnetic Effects of Electric Current: magnetic field, field lines and the compass, the start of Chapter 8.

NITESH SIR SAYS

Galtiyan ek page pe likho aur Sunday ko yahi test dobara do. ***Consistency hi jalwa hai!***

PHYSICS | CHAPTER 7 COMPLETE ASTRA SERIES

CHAPTER 7 SWAHAA!

Electricity | Days 30–35

BUJHO *Ohm ka law, series-parallel, I squared R t, kilowatt-hour, sab numericals solve! Ab bijli ka bill bhi main check karunga.*

PAHELI *Zabardast, Bujho! Electricity Swahaa. Ab current ka magnetic jaadu dekhte hain.*

CHAPTER XP EARNED
+700 XP

LEVELS CLEARED
6 levels (Days 30–35)

BOOK PROGRESS
Level 35 / 60 4400 XP total

WHAT YOU HAVE MASTERED

- ✔ Charge, current, potential difference
- ✔ Resistance, resistivity and its factors
- ✔ Heating effect of current, Joule's law
- ✔ Ohm's law and V–I graph
- ✔ Resistors in series and in parallel
- ✔ Electric power, energy and kWh

NEXT CHAPTER Day 36: Magnetic Effects of Electric Current — *Electricity cleared!*

DAY 36 · LEVEL 36

PHYSICS | CHAPTER 8 | PART 1 · ASTRA SERIES

MAGNETS & MAGNETIC FIELD LINES

Goal: see the invisible field around a magnet and draw it right.

by Nitesh Prakash

01 Oersted | 02 Bar Magnet | 03 Magnetic Field | 04 Field Lines | 05 Properties | +100 XP today | Level 36 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

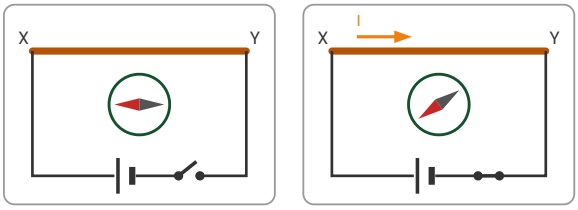
1 MAGNETS & OERSTED'S DISCOVERY

DEFINITION

- **Magnet:** an object that attracts iron, nickel and cobalt and points north–south when freely suspended
- **Natural magnet:** **lodestone** (magnetite, Fe_3O_4), a rock found in the Earth's crust
- **Man-made shapes:** bar magnet (long rectangle), horseshoe magnet (U-shaped), compass needle

Oersted's discovery (1820) ★★★

Hans Christian **Oersted** (19th century) noticed by accident that a compass needle placed near a wire **deflected** whenever current flowed. First proof that **electricity and magnetism are linked**.



Key open: needle rests along XY **Key closed: needle deflects**

- ✔ Reverse the current → the needle deflects the **opposite** way
- ✔ The unit of magnetic field strength, the **oersted**, is named after him

BUJHO *Wire mein sirf current hai, magnet toh hai hi nahi. Phir compass kyun ghooma?*

PAHELI *Kyunki **bahta hua current** khud apne chaaron taraf magnetic field banata hai. Yahi electricity aur magnetism ka rishta hai!*

MEME BREAK

*Current se compass hil gaya? Ab physics "**fysics**" nahi, full electromagnetism hai!*

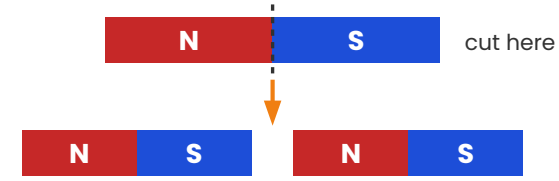
EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

The Earth is a magnet. A compass points north because the Earth behaves like a huge bar magnet whose **magnetic south pole lies near the geographic North Pole**. The north pole of the needle is attracted to it.

2 PROPERTIES OF A BAR MAGNET

Five facts to remember

1. A freely suspended bar magnet always comes to rest in the **north–south** direction. The end pointing north is the **north pole (N)**.
2. **Like poles repel**, unlike poles attract.
3. Poles always exist **in pairs** (a dipole). They **can never be separated**.
4. The attracting power is strongest at the **poles** and weakest in the middle.
5. The space around a magnet where its effect is felt is its **magnetic field**.



Each piece is a complete magnet: poles never separate

Magnetic and non-magnetic materials

Magnetic (attracted)	Non-magnetic
iron, steel, nickel, cobalt	wood, plastic, glass, copper, aluminium

BUJHO *Magnet ko beech se tod diya toh ek tukda sirf N aur doosra sirf S hoga?*

PAHELI *Nahi Bujho! Har tukda khud ek **poora magnet** ban jaata hai, naye N aur S ke saath. Akela pole kabhi nahi milta.*

How to find the N pole of an unmarked magnet

- Hang it freely by a thread; the end that settles facing **geographic north** is its N pole
- Check: that end **repels** the N pole of a compass needle

ATTRACTION IS NOT PROOF; REPULSION IS

An iron nail is attracted by both poles. Only **repulsion** proves that a body is a magnet, because it needs a like pole.

3 MAGNETIC FIELD & COMPASS NEEDLE

DEFINITION

- **Magnetic field:** the region around a magnet in which its force can be detected
- It has both **magnitude** (strength) and **direction**
- **Compass needle:** a tiny bar magnet; its north end shows the direction of the field at that point

Activity 1: iron filings show the pattern ★★☆

1. Fix a sheet of white paper on a drawing board and place a bar magnet at its centre.
2. Sprinkle iron filings evenly around it and **tap the board gently**.
3. The filings arrange themselves in a definite pattern of curves from pole to pole.

Why? Each filing becomes a tiny magnet and lines up along the field. The pattern shows the **magnetic field lines**.

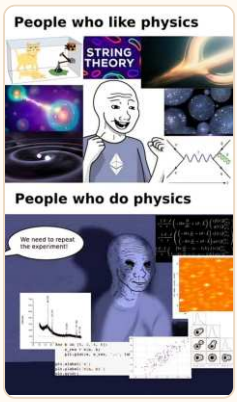
Activity 2: plotting field lines with a compass

1	Place the magnet on paper and mark its outline. Put a compass near the **north pole**.
2	Mark the positions of the two ends of the needle with dots.
3	Move the compass so that its **south end** sits on the dot where the north end was. Mark again.
4	Repeat until you reach the south pole. Join the dots with a smooth curve: one field line.
5	Repeat from other points. The deflection is **largest near the poles**.

Iron filings vs compass: what each tells you

Tool	Shows the pattern?	Shows the direction?
Iron filings	Yes, the whole pattern at once	No
Compass needle	Yes, plotted point by point	**Yes**, the N end points along the field

★★★ *Field lines kabhi cross nahi karti: crossing point par compass do directions dikhayega, jo impossible hai. Nitesh Bhaiya ki guarantee!*

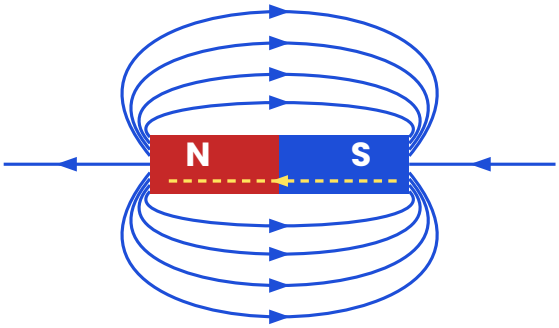


MEME BREAK
*Physics pasand karne wale: "wow, magnet!" Physics karne wale: compass ke dots **phir se plot karo**, experiment repeat!*

4 MAGNETIC FIELD LINES ★★★ MOST ASKED

DEFINITION

- **Magnetic field lines:** imaginary curves drawn to represent a magnetic field
- The **tangent** at any point gives the direction of the field there (the way a north pole would move)

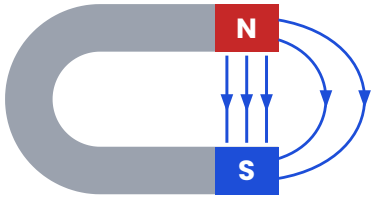


Outside: N → S | Inside the magnet: S → N (dashed)

Properties of field lines ★★★

1. Outside the magnet they **emerge from the north pole and merge at the south pole**; inside the magnet they run from S to N.
2. So they are **closed, continuous curves**.
3. **Crowded lines = stronger field.** They are closest at the poles (strongest) and far apart in the middle (weaker).
4. **No two field lines ever intersect.**

Uniform and non-uniform field



parallel, equally spaced = uniform

Horseshoe magnet: uniform field in the gap between the poles

- ✔ **Uniform field:** same strength and direction everywhere; drawn as **parallel, equally spaced** lines
- ✔ **Non-uniform field:** strength changes from point to point; lines curve and spread, as around a bar magnet

MEME BREAK
*Intezaar karte karte skeleton ban jaoge, par do field lines **kabhi nahi milengi**!*

Day 36 mind map: revise in 2 minutes

OERSTED 1820
Current in a wire deflects a compass: electricity makes magnetism

BAR MAGNET
Rests N–S; like poles repel; poles in pairs, never separate

MAGNETIC FIELD & FIELD LINES

FIELD
Region of magnetic force; has strength and direction

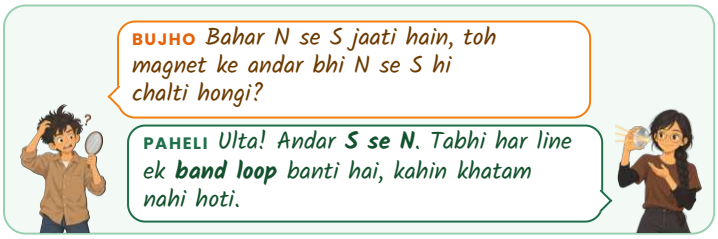
FIELD LINES ★★★
N to S outside, S to N inside; closed; never intersect

STRENGTH
Crowded lines = strong field; strongest at the poles

Why can field lines never intersect? ★★★

Class notes Q. Field lines show the **direction of the magnetic force** on a north pole. If two lines crossed, the field at that point would have **two directions**, and a compass placed there would point two ways at once. A force cannot have two directions at one point, so field lines never intersect.

× WRONG	✔ RIGHT
Field lines start at N and end at S	They are closed loops: N to S outside, S to N inside
The field is strongest in the middle	Strongest at the poles, where lines are crowded
Field lines can cross where two magnets meet	Never: the fields add to one direction
An attracted rod must be a magnet	Only repulsion proves it is a magnet



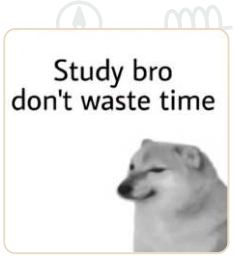
MEME BREAK
Do field lines ek dusre ko cross kar sakti hain? Bugs Bunny bole: ***No!*** *Rule yaad rakho: N se S, closed loops, no crossing.*

Exam tips

- ✔ Always put **arrows** on field lines; without arrows the diagram loses marks.
- ✔ Draw lines **closer near the poles** and wider in the middle.
- ✔ Label N and S on the magnet **before** drawing any field line.

5 PRACTICE MCQS

Study bro mode ON: phone side mein, 10 MCQs khud solve karo!



1. Oersted discovered that a current-carrying wire: ★★★
(a) attracts iron filings only at its ends
(b) deflects a nearby compass needle
(c) becomes hot and glows
(d) has no effect on a magnet

2. A freely suspended bar magnet comes to rest in the: ★★☆
(a) east–west direction
(b) north–south direction
(c) vertical direction
(d) any direction

3. Inside a bar magnet, the magnetic field lines run: ★★★
(a) from N to S
(b) from S to N
(c) in circles
(d) nowhere; there is no field inside

4. Where is the magnetic field of a bar magnet strongest? ★★☆
(a) At the middle
(b) At the poles
(c) Equal everywhere
(d) Far away from it

5. A bar magnet is broken into two equal halves at its middle. Each half: ★★★
(a) has only one pole
(b) loses all magnetism
(c) is a complete magnet with N and S
(d) has two north poles

6. The direction of the magnetic field at a point is given by: ★★☆
(a) the south end of a compass needle
(b) the north end of a compass needle
(c) iron filings only
(d) the length of the field line

7. Which property shows that two field lines cannot cross? ★★☆
(a) Field lines are closed curves
(b) Field at a point has only one direction
(c) Field lines are crowded at poles
(d) Field lines start at N

8. Lodestone is a natural magnet. It is an ore of: ★☆☆
(a) copper
(b) aluminium
(c) iron
(d) zinc

9. **Assertion (A):** Magnetic field lines never intersect each other. **Reason (R):** At the point of intersection, a compass needle would point in two directions, which is not possible. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Magnetic field lines are closed curves. **Reason (R):** Outside a magnet, field lines go from the south pole to the north pole. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	Oersted, 1820: the needle deflected.
2	(b)	It points geographic north–south.
3	(b)	S to N inside, making closed loops.
4	(b)	Lines are most crowded at the poles.
5	(c)	Poles always exist in pairs.
6	(b)	North end points along the field.
7	(b)	One point, one direction of force.
8	(c)	Magnetite, Fe_3O_4, an iron ore.
9	(a)	R is the exact reason for A.
10	(c)	A true; R false: outside it is N to S.

Short and long answer questions

1. Reema placed a compass needle near a wire in a circuit and saw it deflect when the circuit was switched ON. Name the effect and the scientist who first observed it. What happens if the battery is reversed? CBSE 2025 **(2 marks)**
2. A student says that the magnetic field at X, near a pole of a bar magnet, is stronger than at Y, near its middle. Is the claim correct? Give a reason and draw the field lines with arrows. SQP 2025 **(3 marks)**
3. List four properties of magnetic field lines. Why do two field lines never intersect? **(3 marks)**

Model answers

1 Oersted's effect

- The magnetic effect of current; first seen by **Hans Christian Oersted** in 1820
- The current in the wire produces a magnetic field that turns the needle
- Reversing the battery reverses the current, so the needle deflects the **opposite way**

2 X is stronger

- Correct: field lines are **closer together** near the poles, so the field at X is stronger
- Draw closed N-to-S curves with arrows, crowded at the poles

3 Properties

- N to S outside, S to N inside; closed curves; crowded where strong; never intersect
- If they crossed, one point would have two field directions, which is impossible

Case-based question (4 marks)

Riya sprinkled iron filings around a bar magnet on a sheet of paper and tapped it. Then she placed a compass at different points.

1. Why do the filings form a pattern? **(1)**
2. Where are the filings most crowded, and what does that show? **(1)**
3. Can the filings alone show the direction of the field? What should she use? **(2)**

Day 36 checklist: tick before you sleep

- [ ] I can state Oersted's observation and what it proved
- [ ] I know five properties of a bar magnet
- [ ] I can plot field lines with a compass and explain iron filings
- [ ] I can draw bar-magnet field lines with correct arrows
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 36 COMPLETE +100 XP earned. Next: Level 37

BUJHO Samajh gaya! Field lines N se S, band loop, aur kabhi cross nahi.

PAHELI Shabaash! Kal Level 37: current wale taar ka magnetic field.

Day 36 hit-list: Eduspark PYQ analysis

Topic	Rating
Properties of magnetic field lines	★★★
Why field lines never intersect	★★★
Drawing bar-magnet field lines with arrows	★★★
Oersted's experiment	★★☆
Plotting field lines with a compass	★★☆
Poles cannot be separated	★☆☆

NITESH SIR SAYS

Magnet ko samjho toh field lines apne aap ban jaati hain: N se S, band loop, kabhi cross nahi. ***Kal milte hai Day 37 pe!***

Case-based answers

- (i) Each filing is magnetised and lines up along the field
- (ii) At the poles, where the field is strongest
- (iii) No, filings show only the pattern; a compass shows direction

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 37 · LEVEL 37

PHYSICS | CHAPTER 8 | PART 2 · ASTRA SERIES

FIELD OF A STRAIGHT WIRE & A LOOP

Goal: find the field around a current and its direction.

by Nitesh Prakash

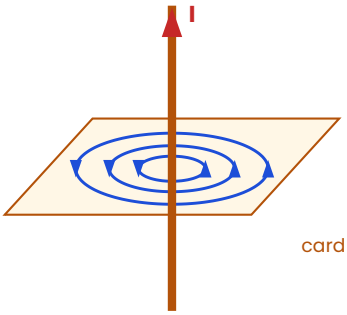
01 Straight Wire | 02 Thumb Rule | 03 B ∝ I, 1/r | 04 Circular Loop | 05 Coil Turns | +100 XP today | Level 37 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

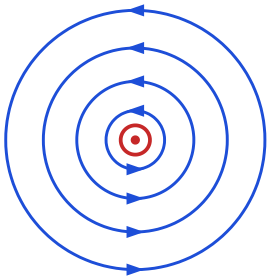
1 FIELD AROUND A STRAIGHT CONDUCTOR

Activity: wire through a cardboard ★★☆

1. Pass a thick copper wire **vertically** through the centre of a horizontal cardboard; connect it to a battery, key and rheostat.
2. Sprinkle iron filings on the cardboard, switch on the current and **tap gently**.
3. The filings form **concentric circles** centred on the wire. A compass placed on a circle shows the direction.



Current up: anticlockwise



Top view: current out of page
gaps grow outward = weaker

DEFINITION

- **Pattern:** field lines around a straight wire are **concentric circles** in a plane perpendicular to the wire, centred on the wire
- **Spacing:** the circles get larger and farther apart away from the wire, because the field **fades with distance**
- **Reverse the current** → the direction of the circles reverses

SYMBOLS YOU MUST KNOW

⊙ current coming **out of** the page (you see the arrow tip); ⊗ current going **into** the page (you see the tail feathers).

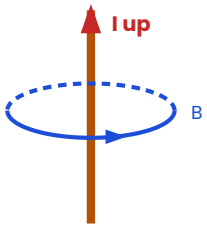
BUJHO *Straight wire ka field bhi straight line mein hoga na?*

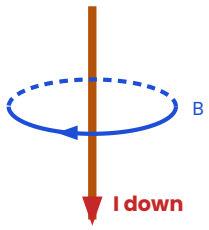
PAHELI *Nahi Bujho! Wire seedha hai, par field uske chaaron taraf* ***gol gol circles*** *mein ghoomta hai, wire ke perpendicular plane mein.*

2 RIGHT-HAND THUMB RULE ★★★ MOST ASKED

DEFINITION

- **Maxwell's right-hand thumb rule:** hold the current-carrying wire in your right hand with the **thumb along the current**; the **curled fingers** give the direction of the magnetic field lines
- It gives the **direction** of the field when the direction of the current is known





Direction at a glance ★★★

Current in a vertical wire	Field seen from above
Upwards (thumb up)	**Anticlockwise**
Downwards (thumb down)	**Clockwise**

Note on the class notes: "thumb up, fingers clockwise" is the view from **below**. Seen from **above** (the NCERT convention), current up gives an **anticlockwise** field.

Current upar ja raha hai... field clockwise hai ya anticlockwise??

MEME BREAK

Clockwise ya anticlockwise? Mr Krabs mat bano, ***right hand ka thumb*** *current ki taraf karo, ungliyan jawab de dengi!*

★★★ *Thumb = current, fingers = field. Ye rule har saal aata hai, Nitesh Bhaiya ki guarantee!*

3 WHAT DECIDES THE STRENGTH? ★★★

Two factors that decide B

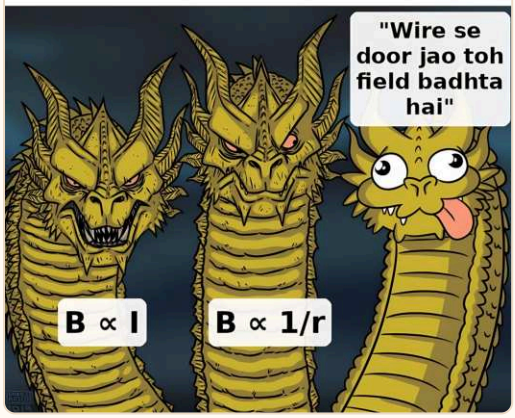
Factor	Relation	Meaning
Current, I	**B ∝ I**	double the current, double the field
Distance, r	**B ∝ 1/r**	farther from the wire, weaker field

Note: the class notes write "B = 1/r"; the correct form is **B ∝ 1/r** (proportional, not equal).

Activity: change I, then change r

- Increase the current with the **rheostat**: the compass deflection **increases**
- Keep the current fixed and move the compass **away** from the wire: the deflection **decreases**
- Conclusion: B rises with I and falls with distance r

Straight wire ke magnetic field ke rules:



MEME BREAK

*Do heads sahi: B ∝ I aur B ∝ 1/r. Teesra head galat: wire se door jao toh field **kam** hota hai, badhta nahi.*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

SI unit of magnetic field: the **tesla (T)** (from the class notes).
For a long straight wire, $B = \mu_0 I / (2\pi r)$, which shows B ∝ I and B ∝ 1/r together.
Clock-face rule for a coil: current **anticlockwise** as you look at the face → that face is a **north pole**; clockwise → south pole.

Solved Example 1: two parallel wires ★★☆

- **Q.** Two long parallel wires carry equal currents. What is the field midway between them when the currents are (a) in the same direction, (b) in opposite directions?
- (a) The two fields are equal and opposite there → **zero**
- (b) The two fields point the same way → **twice** the field of one wire

Solved Example 2: power line overhead ★★★

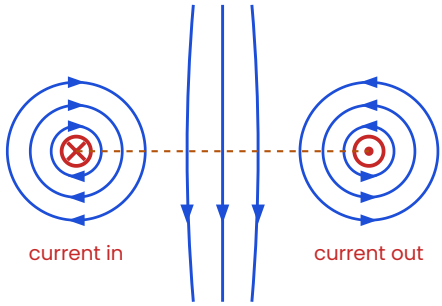
Q. A current flows through a horizontal power line from **east to west**. Find the direction of the magnetic field at a point directly below it and directly above it. NCERT

- Point the right thumb from east to west along the wire
- Fingers curl: **below** the wire the field points from **north to south**
- **Above** the wire the field points from **south to north**
- If the current flows west to east, both directions reverse

4 FIELD DUE TO A CIRCULAR LOOP

Activity: a loop through a cardboard

Bend a wire into a circular loop and pass it through a cardboard so that the cardboard cuts the loop at two points. Sprinkle iron filings and pass the current.



Section through the loop: almost straight lines at the centre

What the pattern shows ★★★

1. Near each point where the wire crosses the cardboard, the lines are **concentric circles** (each small part of the loop acts like a straight wire).
2. The circles grow **larger** as we move away from the wire.
3. At the **centre**, the arcs become so large that the lines are **almost straight** and **perpendicular to the plane** of the loop.
4. Every part of the loop makes the field at the centre point in the **same direction**, so the fields add up.

Strength at the centre of a coil

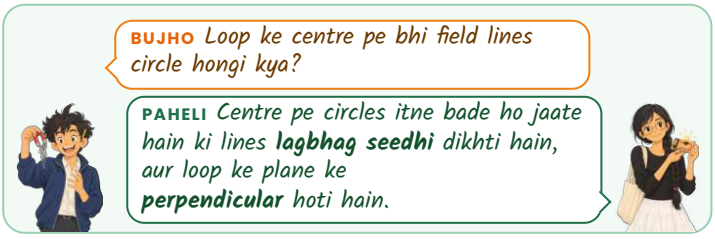
Increase	Field at centre
Current, I	increases
Number of turns, n	increases n times
Radius of the loop	decreases

Solved Example 3: change the coil ★★☆

- **Q.** A coil gives a field B at its centre. What is the field if (a) the turns are doubled, (b) the current is halved, (c) both are done together?
- (a) B ∝ n → **2B**
- (b) B ∝ I → **B/2**
- (c) 2 × 1/2 = 1 → field stays **B**

THINK OF IT THIS WAY

A loop is many tiny straight wires; inside it, all their fields point the same way.



Day 37 mind map: revise in 2 minutes

STRAIGHT WIRE
Concentric circles in a plane perpendicular to the wire

THUMB RULE ★★★
Thumb along current, curled fingers give the field

FIELD OF A CURRENT

FACTORS
B ∝ I and B ∝ 1/r for a straight wire

LOOP
Circles near the wire; almost straight at the centre

COIL
n turns give n times the field; smaller radius, stronger field

Straight wire vs circular loop

	Straight wire	Circular loop
Shape of lines	concentric circles	circles near the wire, straight at the centre
Strongest	close to the wire	near the wire; at the centre all parts add
Depends on	I, r	I, n, radius

× WRONG	✓ RIGHT
B = 1/r	B ∝ 1/r: proportional, not equal
Field lines are straight along the wire	Concentric circles around the wire
Use the left hand for the field of a wire	Right-hand thumb rule
More turns, same field at the centre	n turns give n times the field

BUJHO *Do parallel wires mein same direction mein current, toh beech mein field double ho jayega?*

PAHELI *Ulta! Beech ke point par dono wires ke fields **barabar aur opposite** hote hain, isliye total **zero**. CBSE 2023 mein yahi poocha tha.*

MEME BREAK
*Thumb rule mein left hand laga diya? Physics se nafrat ho jayegi! **Right hand** hi yaad rakho.*

Exam tips

- ✓ Always say **"viewed from above"** or **"viewed from below"** with clockwise or anticlockwise.
- ✓ Use ⊙ for current out of the page and ⊗ for current into the page.
- ✓ For a coil, mention **all three** factors: current, number of turns and radius.

5 PRACTICE MCQS

1. The magnetic field lines around a straight current-carrying wire are: ★★★

(a) straight lines parallel to the wire
(b) concentric circles centred on the wire
(c) radial lines starting from the wire
(d) ellipses

2. A vertical wire carries current upwards. Seen from above, the field lines are: ★★★

(a) clockwise
(b) anticlockwise
(c) straight up
(d) straight down

3. If the distance from a straight wire is doubled, the magnetic field becomes: ★★☆

(a) double
(b) half
(c) four times
(d) unchanged

4. The rule used to find the direction of the field around a wire is: ★★★

(a) Fleming's left-hand rule
(b) right-hand thumb rule
(c) Ohm's law
(d) Snell's law

5. At the centre of a current-carrying circular loop, the field lines are: ★★★

(a) concentric circles
(b) almost straight, perpendicular to the loop
(c) parallel to the plane of the loop
(d) absent

6. The resultant field at a point midway between two long parallel wires carrying equal currents in the same direction is: CBSE 2023 ★★☆

(a) double of one wire
(b) along the wires
(c) zero
(d) vertically upward

7. Which change does NOT increase the field at the centre of a coil? ★★☆

(a) more turns
(b) more current
(c) a larger radius
(d) a smaller radius

8. A current flows east to west in a horizontal wire. The field directly below it points: ★★★

(a) north to south
(b) south to north
(c) east to west
(d) vertically up

9. **Assertion (A):** The field at the centre of a current-carrying circular coil increases when its number of turns is increased.
Reason (R): The current in each turn has the same direction, so the fields of all turns add up. CBSE 2023 ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The magnetic field of a straight current-carrying wire is the same at all distances from it.
Reason (R): Field lines of a straight wire are concentric circles. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	Concentric circles in a perpendicular plane.
2	(b)	Thumb up: fingers curl anticlockwise from above.
3	(b)	B ∝ 1/r, so double r gives half B.
4	(b)	Maxwell's right-hand thumb rule.
5	(b)	Large arcs look straight at the centre.
6	(c)	The two fields are equal and opposite.
7	(c)	Larger radius means weaker field at the centre.
8	(a)	Thumb west, fingers curl N to S below.
9	(a)	Fields of all turns add; R explains A.
10	(d)	A false: B falls with distance. R true.

Short and long answer questions

1. Draw a labelled diagram to show the pattern of magnetic field lines produced by a current-carrying straight conductor. Mark the direction of the field lines. CBSE 2024 **(2 marks)**
2. Mona noticed that the deflection of a compass needle fell as she moved it away from a current-carrying wire. (a) Why? (b) Name one change in the circuit that would increase the deflection. SQP 2024 **(2 marks)**
3. State the right-hand thumb rule. Draw the field lines of a current-carrying circular loop and list three factors on which the field at its centre depends. **(5 marks)**

Model answers

1 Straight conductor

- Concentric circles centred on the wire, closer near it; for current upward the arrows run anticlockwise seen from above (1 mark circles, 1 mark arrows)

2 Mona's compass

- (a) B ∝ 1/r: the field weakens with distance
- (b) Increase the current, e.g. add a cell or lower the rheostat resistance

3 Thumb rule and loop

- Thumb along the current, curled fingers give the field direction
- Circles near the wire, almost straight lines at the centre, perpendicular to the loop
- Factors: current, number of turns, radius (smaller radius, stronger field)

Case-based question (4 marks)

A vertical wire passes through a horizontal card. With current flowing upwards, a student places compasses at points P (near the wire) and Q (farther away).

1. What is the shape of the field lines on the card? **(1)**
2. Seen from above, which way do the lines run? **(1)**
3. At which point is the field stronger, and why? What happens to the directions if the current is reversed? **(2)**

Day 37 checklist: tick before you sleep

- [ ] I can describe the field pattern around a straight wire
- [ ] I can state and apply the right-hand thumb rule
- [ ] I know B ∝ I and B ∝ 1/r for a straight wire
- [ ] I can draw the field of a circular loop and list its three factors
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 37 COMPLETE +100 XP earned. Next: Level 38

BUJHO *Samajh gaya! Thumb current ki taraf, ungliyan field ki taraf, aur loop ke centre pe lines seedhi.*

PAHELI *Shabaash! Kal Level 38: solenoid aur electromagnet.*

Day 37 hit-list: Eduspark PYQ analysis

Topic	Rating
Right-hand thumb rule and its use	★★★
Field pattern of a straight wire (diagram)	★★★
Factors: current and distance	★★★
Field of a circular loop (diagram)	★★☆
Effect of number of turns	★★☆
Field midway between two wires	★☆☆

NITESH SIR SAYS

*Haath sahi chuno toh direction kabhi galat nahi hogi: **right hand**, thumb current ki taraf. **Kal milte hai Day 38 pe!***

Case-based answers

- (i) Concentric circles centred on the wire
- (ii) Anticlockwise (right-hand thumb rule)
- (iii) At P, since B ∝ 1/r; reversing the current reverses every direction

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 38 LEVEL 38

PHYSICS | CHAPTER 8 | PART 3 ASTRA SERIES

SOLENOID & ELECTROMAGNET

Goal: make a magnet you can switch on and off.

by Nitesh Prakash

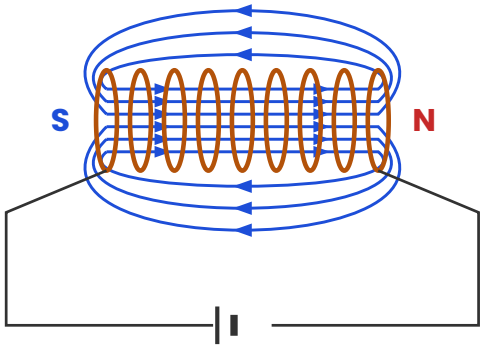
01 Solenoid | 02 Uniform Field | 03 Strength | 04 Electromagnet | 05 Comparison | +100 XP today | Level 38 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 THE SOLENOID

DEFINITION

- **Solenoid:** a coil of many circular turns of **insulated copper wire** wrapped closely in the shape of a **cylinder**
- When current flows, it produces a field just like a **bar magnet**: one end behaves as a **north pole**, the other as a **south pole**



Inside: parallel lines S → N (uniform) | Outside: N → S

Field pattern of a solenoid ★★★

1. **Outside:** field lines run from the **N end to the S end**, like a bar magnet.
2. **Inside:** lines run from **S to N** and are **parallel straight lines**.
3. Parallel, equally spaced lines mean the field inside is the **same at all points**: a **uniform field**.
4. Inside and outside lines together form **closed loops**.

Activity: make your own solenoid ★★☆

1. Wind insulated copper wire closely on a pencil or a cardboard tube and connect it to a cell and a key.
2. Close the key and bring a compass near each end.
3. One end **attracts** the north pole of the needle and the other **repels** it: the coil is now a magnet; reversing the cell swaps the ends.

★★★ *Solenoid ke andar field **uniform**: parallel, barabar doori wali lines. Board mein pakka, Nitesh Bhaiya ki guarantee!*

Solenoid vs bar magnet

	Solenoid	Bar magnet
Field pattern	same: N to S outside, S to N inside	
Freely suspended	both rest in the north–south direction	
Switch off	field disappears	stays a magnet
Reverse polarity	reverse the current	not possible

MEME BREAK

*Solenoid ne bar magnet ka pura field pattern **copy** kar liya, Mr Bean style! Bas switch off karte hi magnetism gayab.*

Which end is north?

- Curl the fingers of your right hand along the current in the turns: the **thumb points to the N end**
- Reverse the current and the N and S ends swap

2 STRENGTH OF THE SOLENOID'S FIELD ★★★

It depends on (class notes)

1. **Number of turns**: more turns, stronger field
2. **Current** through the coil: more current, stronger field
3. **Material of the core**: a soft iron core makes it much stronger
4. **Radius of the coil**: a narrower coil gives a stronger field at its centre

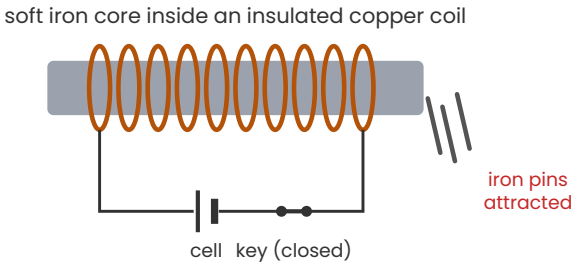
BUJHO *Solenoid ke andar field uniform hai, toh beech aur end pe same strength hogi?*

PAHELI *Andar ke lambe hisse mein haan, **uniform**. Sirf bilkul ends ke paas lines phailne lagti hain, wahan field thoda kam hota hai.*

3 ELECTROMAGNET ★★★ MOST ASKED

DEFINITION

- **Electromagnet:** a **temporary magnet** made by placing a **soft iron core** inside a current-carrying solenoid
- It is a magnet **only while current flows**; it is demagnetised when the current is switched off



Four properties (class notes)

1. **Temporary:** magnetism lasts only while the current flows.
2. **Polarity can be reversed** by reversing the direction of the current.
3. **Strength can be changed** by changing the current (or the number of turns).
4. It is much stronger than the solenoid alone because the soft iron core gets magnetised.

Note on the class notes: polarity is **reversed** (N and S swap) by reversing the current; it is the **strength** that is increased or decreased.

MEME BREAK

*Switch off karte hi magnet gayab! Yahi hai electromagnet ka **temporary** nature: current gaya, magnetism gaya.*

Activity: making an electromagnet

- Wind about 50 turns of insulated wire on a large iron nail; connect a cell and a key
- Close the key: the nail picks up pins. Open it: the pins drop
- Add more turns or one more cell: it picks up **more** pins

Why soft iron, not steel? ★★☆

- Soft iron is **magnetised strongly** by the solenoid and **loses its magnetism at once** when the current stops
- Steel keeps its magnetism after the current stops, so a steel core gives a **permanent magnet**
- A copper or wooden core does not get magnetised at all

Uses of electromagnets

- ✔ **Cranes** in scrap yards lift and drop heavy iron loads by switching the current on and off
- ✔ **Electric bells**, relays and electric door locks
- ✔ **Loudspeakers** and headphones
- ✔ Separating iron pieces from waste and from ores

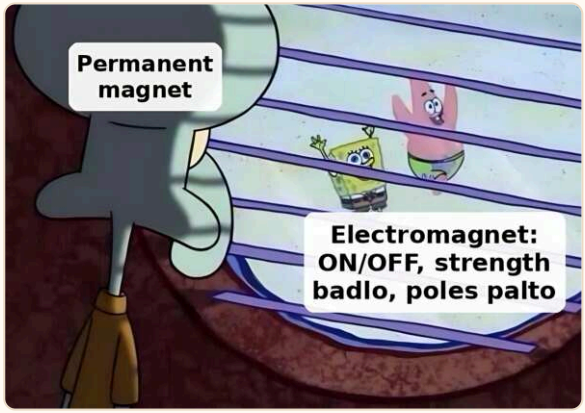
CLASS TRICK

"Permanent magnet ka properties theek ulta hain electromagnet se :)"

Electromagnet: temporary, strength badlo, polarity palto. Permanent magnet: sab ulta, sab fixed.

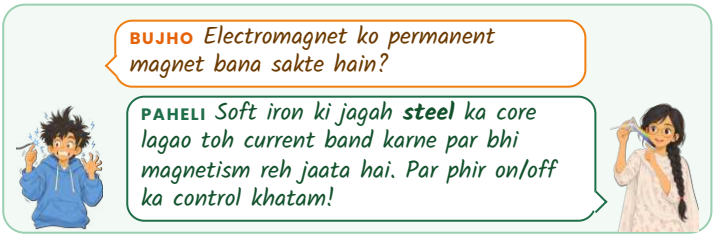
4 ELECTROMAGNET VS PERMANENT MAGNET ★★★

Property	Electromagnet	Permanent magnet
Nature	temporary	permanent
Strength	can be changed	fixed
Polarity	can be reversed	cannot be reversed
Core	soft iron	steel, alloys such as alnico
Needs current?	yes	no
Strength possible	very strong	comparatively weak



MEME BREAK

*Electromagnet: on/off, strength aur poles sab control mein. Permanent magnet window se dekhta reh gaya: "main toh bas **fixed** hoon bhai."*



Increase the strength (material fixed) ★★★

- **Increase the current** through the coil
- **Increase the number of turns** in the coil

Asked as: "List two ways of increasing the strength of an electromagnet if the material of the electromagnet is fixed." **CBSE 2020**

Day 38 mind map: revise in 2 minutes

SOLENOID
Many close turns of insulated copper wire in a cylinder

FIELD ★★★
Like a bar magnet; uniform, parallel lines inside

SOLENOID & ELECTROMAGNET

STRENGTH
Turns, current, core material, coil radius

ELECTROMAGNET
Soft iron core in a solenoid; temporary magnet

VS PERMANENT
Electromagnet: variable, reversible; permanent: fixed

Conceptual Q&A: think, then read ★★★

1 Can a freely suspended current-carrying solenoid stay in any direction?
> No. It behaves like a bar magnet and rests in the **north–south** direction; reversing the current turns it round

2 Why is the field inside a long solenoid uniform?
> The fields of all the turns point the same way inside and add up evenly, giving parallel, equally spaced lines

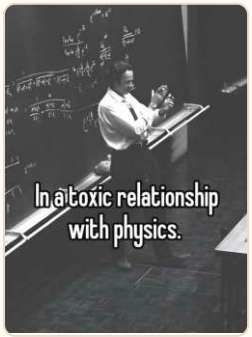
3 A copper rod is put inside the solenoid. Does it become a magnet?
> No. Copper is not a magnetic material; only a magnetic material such as soft iron gets magnetised

4 Why are field lines inside a solenoid drawn S to N?
> Field lines are closed loops: they leave the N end outside and must return through the inside

× WRONG	✓ RIGHT
Field inside a solenoid is zero	Strong and uniform: parallel lines S to N
Electromagnets use a steel core	Soft iron core: loses magnetism quickly
Reversing current increases strength	Reversing current reverses the polarity
Any metal core will do	Only magnetic materials (soft iron) work

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

For a long solenoid, $B = \mu_0 nI$, where n is the number of turns per metre; inside a long solenoid the radius hardly matters.
Electric motor, electromagnetic induction and the generator are no longer in the CBSE Class 10 syllabus.



MEME BREAK
*Physics ke saath toxic relationship? Solenoid, electromagnet, sab ek hi kahani hai: **current se magnet**. Pyaar ho jayega!*

5 PRACTICE MCQS

1. A solenoid is a coil of many turns of insulated copper wire wound in the shape of a: ★★☆
(a) sphere (b) cylinder
(c) cone (d) flat spiral

2. The magnetic field inside a long straight current-carrying solenoid is: CBSE 2025 ★★★
(a) zero (b) maximum at the centre only
(c) uniform throughout its length (d) maximum at the ends

3. The field pattern of a current-carrying solenoid resembles that of a: ★★★
(a) straight wire (b) horseshoe magnet
(c) bar magnet (d) circular loop only

4. Which core makes the best electromagnet? ★★★
(a) steel (b) copper
(c) soft iron (d) wood

5. The polarity of an electromagnet can be reversed by: ★★☆
(a) increasing the current (b) reversing the current
(c) adding more turns (d) removing the core

6. Which of these is NOT a factor for the strength of a solenoid's field? ★☆☆
(a) number of turns (b) current
(c) nature of the core (d) colour of the wire

7. Steel is used to make permanent magnets because it: ★★☆
(a) loses magnetism quickly (b) retains magnetism
(c) is a poor conductor (d) is non-magnetic

8. Electromagnets are used in all of these EXCEPT: ★★☆
(a) electric bell (b) scrap-yard crane
(c) loudspeaker (d) dry cell

9. **Assertion (A):** A soft iron core is used in an electromagnet.
Reason (R): Soft iron is magnetised strongly and loses its magnetism as soon as the current is switched off. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The magnetic field lines inside a solenoid are parallel straight lines.
Reason (R): The field inside a long solenoid is uniform. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Case-based question (4 marks)

A scrap-yard crane lifts old cars with a large electromagnet and drops them into a crusher.

1. Which core material should the electromagnet have, and why? **(1)**
2. How does the operator drop the load? **(1)**
3. Give two ways to lift heavier loads with this core. **(1)**
4. How can the poles of the electromagnet be swapped? **(1)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Turns wound closely as a cylinder.
2	(c)	Parallel, equally spaced lines inside.
3	(c)	One end N, other end S.
4	(c)	Strong, temporary magnetisation.
5	(b)	N and S swap with the current.
6	(d)	Colour has no effect.
7	(b)	Steel keeps its magnetism.
8	(d)	A dry cell is a source, not a magnet.
9	(a)	R explains why soft iron is used.
10	(a)	Uniform field is shown by parallel lines.

Short and long answer questions

1. Define the term solenoid. Draw the pattern of field lines around a current-carrying solenoid and state how it can be used to magnetise soft iron. SQP 2025 **(3 marks)**
2. Why can two field lines never cross? Draw the field lines of a long current-carrying solenoid, state the conclusion from the lines inside, and name two factors on which its field depends. CBSE 2025 **(5 marks)**
3. A current-carrying solenoid is suspended freely. Will it stay in any direction? What happens when the current is reversed? CBSE 2020 **(2 marks)**

Model answers

1 Solenoid and soft iron

- A coil of many close turns of insulated copper wire in the shape of a cylinder
- Diagram: N to S outside, parallel lines S to N inside
- Soft iron placed inside is magnetised and becomes an electromagnet

2 Field lines and factors

- If they crossed, the field there would have two directions: impossible
- Parallel lines inside: the field is uniform there
- Factors: number of turns, current (also the core material)

3 Suspended solenoid

- No: it settles north–south like a bar magnet
- Reversing the current swaps its poles, so it turns through 180°

Case-based answers

- (i) Soft iron: magnetised strongly, loses magnetism at once when switched off
- (ii) Switch off the current: the core is demagnetised
- (iii) Increase the current or the number of turns
- (iv) Reverse the current

BUJHO *Solenoid ke bahar bhi field uniform hoga na?*

PAHELI *Nahi! Bahar lines mudti hain. Uniform sirf **andar** hai.*

Day 38 checklist: tick before you sleep

- [ ] I can define a solenoid and draw its field lines
- [ ] I know why the field inside a solenoid is uniform
- [ ] I can list four factors for the strength of a solenoid's field
- [ ] I can compare an electromagnet with a permanent magnet
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 38 COMPLETE +100 XP earned. Next: Level 39

BUJHO *Samajh gaya! Solenoid = bar magnet ki copy. Soft iron daalo toh electromagnet.*

PAHELI *Shabaash! Kal Level 39: current wale taar par force aur ghar ki wiring.*

Day 38 hit-list: Eduspark PYQ analysis

Topic	Rating
Field lines of a solenoid (diagram)	★★★
Uniform field inside a solenoid	★★★
Electromagnet: definition and soft iron core	★★★
Electromagnet vs permanent magnet	★★☆
Factors for the strength of the field	★★☆
Uses of electromagnets	★☆☆

NITESH SIR SAYS

*Solenoid samajh liya toh electromagnet free mein mil gaya: coil + current + soft iron. **Kal milte hai Day 39 pe!***

Coming up: Day 39

Force on a current-carrying conductor, Fleming's left-hand rule, AC and DC, and domestic electric circuits.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 39 LEVEL 39

PHYSICS | CHAPTER 8 | PART 4 ASTRA SERIES

FLEMING'S RULE & HOME CIRCUITS

Goal: find the force direction, then wire a safe home.

by Nitesh Prakash

01 Force on Wire | 02 Fleming's LHR | 03 AC vs DC | 04 Home Wiring | 05 Fuse & Safety | +100 XP today | Level 39 / 60

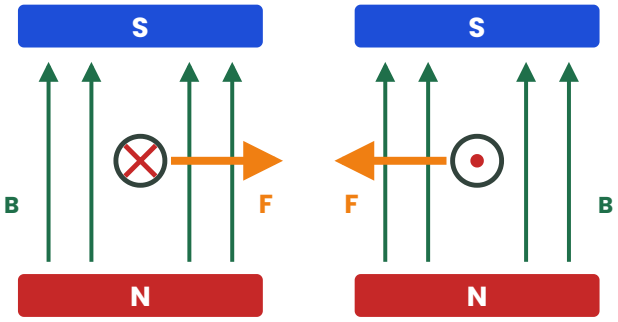
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 FORCE ON A CURRENT-CARRYING WIRE ★★★

A current-carrying wire makes its own magnetic field (Day 37). So when it is placed in the field of a magnet, the two fields push on each other: the **magnet exerts a force on the conductor**, and (as Ampère suggested) the conductor exerts an equal and opposite force on the magnet.

Activity: the jumping aluminium rod

A light aluminium rod AB hangs horizontally from two wires between the poles of a horseshoe magnet (N below, S above, so the field points **upward**). It is connected in series with a battery, a key and a rheostat.



Current into page: F right | Current reversed: F left

- Key closed: the rod is **pushed sideways** (displaced)
- Current reversed: the rod moves the **opposite way**
- Magnet flipped (field reversed): direction again reverses
- Rheostat lowered (more current): bigger displacement

KEY RESULTS

- The force is **maximum** when the current is **perpendicular** to the field
- The force is **zero** when the current is **parallel** to the field
- Its direction depends on **both** the current and the field
- Its size grows with current, field strength and length of wire in the field

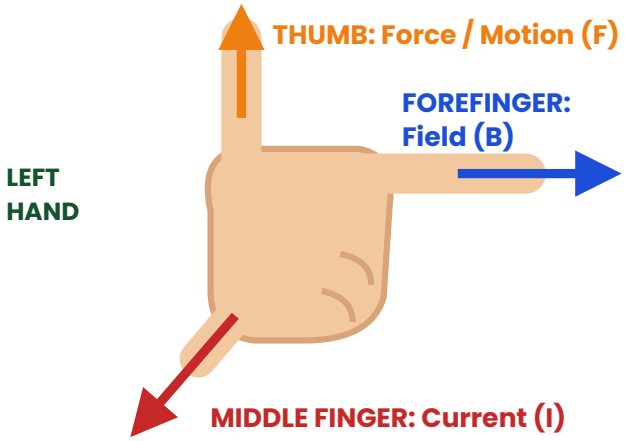
BUJHO *Current bhi ulta, magnet bhi ulta: toh rod bhi ulti jayegi na?*

PAHELI *Nahi! Do baar ulta matlab seedha. Force ki direction* ***wahi rahegi.*** *Sirf ek ko reverse karo tab direction badlegi.*

2 FLEMING'S LEFT-HAND RULE ★★★ MOST ASKED

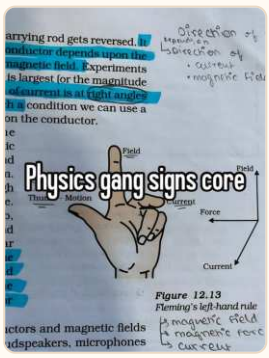
STATEMENT

Stretch the **thumb, forefinger and middle finger of the left hand** so that they are mutually perpendicular. If the **forefinger** points along the **magnetic field** and the **middle finger** along the **current**, then the **thumb** points along the **motion or force** on the conductor.



MEMORY TRICK: FBI ON THE LEFT HAND

Force = thumb, **B** (field) = forefinger, **I** (current) = middle finger. Thumb upar, F-B-I order mein ungliyaan kholo.



MEME BREAK

Exam hall mein sabke haath hawa mein! ***Left hand*** *hi use karna, right hand wala thumb rule Day 37 ka hai.*

Solved Example 1: find the field NCERT

A positively charged alpha particle moving **towards west** is deflected **towards north**. Find the direction of the magnetic field.

- Current = direction of positive charge = **west** (middle finger)
- Force = **north** (thumb)
- Forefinger then points **upward**: the field is **vertically upward**

Solved Example 2: an electron beam

An electron beam moves to the **right** through a field pointing **into the page**. Which way is the force?

- Electrons move right, so conventional current is to the **left**
- Field into the page, current left: thumb points **downward**
- Force on the beam is **towards the bottom of the page**

Devices using this force: electric motor, electric fan, loudspeakers and moving-coil meters (galvanometer, ammeter). Their working is not in the current syllabus.

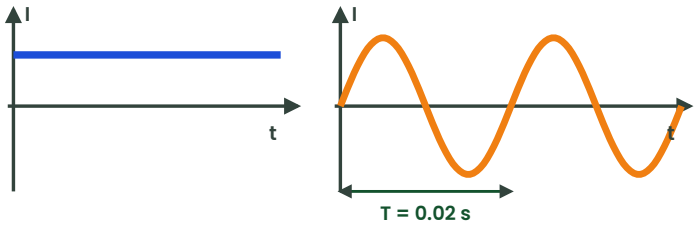
FROM CLASS NOTES: MAGNETISM INSIDE YOU

Nerve impulses are tiny currents, so the **heart and the brain** produce weak but significant magnetic fields. **MRI** (magnetic resonance imaging) uses magnetic fields to form images of the inside of the body.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Size of the force: $F = B\,I\,L \sin\theta$, where θ is the angle between the wire and the field. At $\theta = 90°$, $F = BIL$ (maximum); at $\theta = 0°$, $F = 0$. That is why the rod moves most when it is perpendicular to the field.

3 DC VS AC: KINDS OF CURRENT ★★☆

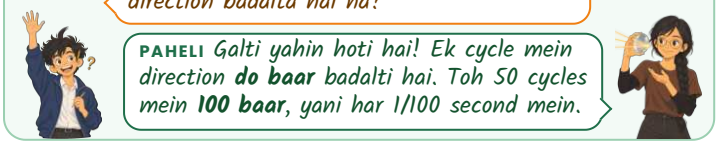


DC: one direction — AC: reverses direction

Point	DC (direct current)	AC (alternating current)
Direction	always one direction	reverses periodically
Sources	cells, batteries	power stations, AC generators
Storage	can be stored in cells	cannot be stored
Transmission	big losses over long distances	long distances with **little energy loss**

IN INDIA

- Frequency of AC = **50 Hz**: 50 complete cycles every second
- Time period $T = 1/f = 1/50 =$ **0.02 s**
- Direction reverses **twice per cycle**, i.e. after every **1/100 s**, or **100 times a second**



QUICK CHECK

Direction changes of Indian AC in one minute: 2 per cycle × 50 cycles per second × 60 s = **6000**.

4 DOMESTIC ELECTRIC CIRCUITS ★★★ MOST ASKED

STEP 1 Power reaches the house through overhead poles or underground cables

↓ ↓

STEP 2 Main fuse, then the electricity meter (kWh)

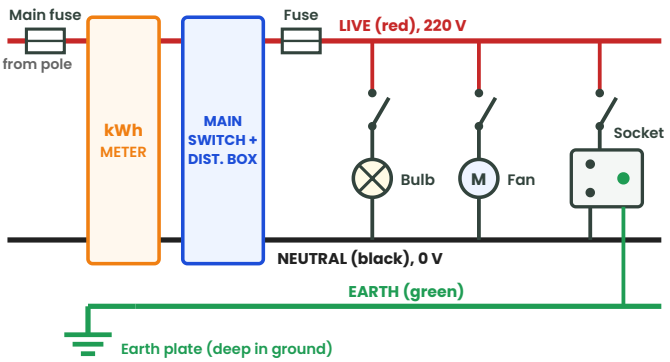
↓ ↓

STEP 3 Main switch and distribution box: separate circuits, each with its own fuse

The three wires

Wire	Insulation	Job
Live (positive)	red	brings current in; at 220 V
Neutral (negative)	black	carries current back; at 0 V
Earth	green	safety wire to a metal plate buried deep in the earth near the house

Potential difference between live and neutral in India = **220 V**. Newer wiring may use brown (live), blue (neutral) and green-yellow (earth); for the exam, write the NCERT colours.



Domestic circuit: every appliance is in **parallel**, with its switch and fuse in the **live** wire.

Why parallel, and why two circuits?

- ✔ **Parallel:** every appliance gets the full 220 V and can be switched on or off independently
- ✔ **15 A circuit** for high-power appliances: geyser, air conditioner, refrigerator, electric iron, room heater
- ✔ **5 A circuit** for low-power ones: bulbs, fans, TV, mobile chargers

Earthing: why the third wire? CBSE 2025

Appliances with a **metal body** (electric iron, toaster, refrigerator, table fan) have their body joined to the earth wire. If live wire touches the body, the leaked current flows to the earth through this **low-resistance path**. The body stays at the potential of the earth, so the user does not get a severe shock.

★★★ *Live red, neutral black, earth green, 220 V, 50 Hz: ye five facts har saal aate hain, Nitesh Bhaiya ki guarantee!*

5 SHORT CIRCUIT & OVERLOADING

★★★ MOST ASKED

SHORT CIRCUIT	OVERLOADING
Live and neutral wires come in **direct contact**, because the insulation is damaged or the appliance is faulty. Resistance becomes almost zero, so the current **rises suddenly** to a very large value.	Current drawn exceeds the safe limit because **too many appliances** run on one socket, or the **supply voltage rises** accidentally. Wires overheat slowly.
Result: sparks, heating, fire	**Result:** overheating, fire risk

MEME BREAK

*Insulation ghisi, live ne neutral ko chhua: **short circuit!** Current achanak itna ki sparks hi sparks.*

Electric fuse: the most important safety device

- A short piece of wire of **low melting point** and relatively high resistance, made of a suitable alloy
- Connected in **series**, in the **live wire**, at the start of the circuit
- Works on the **heating effect of current** ($H = I^2Rt$)
- If the current exceeds its rating, the wire **melts** and breaks the circuit, saving wiring and appliances
- Common ratings: 1 A, 2 A, 3 A, 5 A, 10 A, 15 A
- **MCB** (miniature circuit breaker): a switch that **trips automatically** on excess current; reset it instead of replacing a wire

Precautions against overloading NCERT

- ✔ Do not connect too many appliances to a single socket
- ✔ Use a fuse or MCB of the correct rating in every circuit
- ✔ Run high-power appliances on the 15 A circuit only
- ✔ Replace damaged or old wiring and loose plugs

Solved Example 3: will the fuse blow? NCERT

An electric oven of **2 kW** runs on a 220 V domestic circuit of current rating **5 A**. What happens?

1. Given: P = 2 kW = 2000 W, V = 220 V
2. Formula: I = P / V
3. I = 2000 / 220 = **9.09 A**
4. 9.09 A > 5 A, so the circuit is overloaded: the **fuse melts** and cuts off the supply. Run the oven on a 15 A circuit.

MEME BREAK

*220 V AC ki power: **unlimited power!** Isliye fuse, MCB aur earthing ke bina kabhi kaam nahi.*

MEME BREAK

*Metal body + earthing = chill. Metal body, no earthing, live wire chhua = **jhatka**. Third wire mat bhoolo!*

Solved Example 4: choose the fuse

Which fuse suits a 1 kW electric iron on 220 V: 2 A, 5 A or 15 A?

1. I = P / V = 1000 / 220 = **4.55 A**
2. Pick the rating just above the normal current: **5 A**. A 2 A fuse would melt at once; a 15 A fuse would not protect the iron.

× WRONG	✔ RIGHT
Fuse in the neutral wire	Fuse and switch always in the live wire
Fuse wire: high melting point	Low melting point, so it melts first
Earth wire carries current all the time	Only when current leaks to the body
Appliances in series	In parallel: same 220 V each

BUJHO *Fuse ko neutral wire mein laga doon toh bhi toh circuit tootega na?*

PAHELI *Circuit tootega, par appliance phir bhi **220 V live** se juda rahega. Chhuoge toh shock! Isliye fuse hamesha **live wire** mein.*

Short and long answer questions

1. State Fleming's left-hand rule. A wire carrying current from south to north is placed in a magnetic field acting vertically upward. Find the direction of the force on it. **(3 marks)**
2. Explain the function of an electric fuse in a domestic circuit. Why is it always connected in the live wire? CBSE 2024 **(3 marks)**
3. What is short-circuiting? How is it different from overloading? Name two devices that protect a circuit from both. **(3 marks)**

Exam tips

- ✔ Draw the three fingers at right angles and label F, B, I: diagrams fetch marks
- ✔ Direction answers need a word: up, down, north, south, into or out of the page
- ✔ For electrons, first flip the direction to get conventional current
- ✔ Fuse questions: always write "in series, in the live wire"

6 PRACTICE MCQS

Phone side mein, pen haath mein! 10 MCQs khud solve karo, phir key dekho.

1. The direction of force on a current-carrying conductor in a magnetic field is given by: ★★★
(a) right-hand thumb rule (b) Fleming's left-hand rule
(c) Ohm's law (d) Joule's law

2. The force on a current-carrying conductor in a magnetic field is maximum when the angle between current and field is: ★★★
(a) 0° (b) 45°
(c) 90° (d) 180°

3. In Fleming's left-hand rule, the middle finger shows the direction of: ★★☆
(a) magnetic field (b) force
(c) motion (d) current

4. A proton enters vertically downwards into a uniform magnetic field acting horizontally from west to east. The force on it acts towards the: SQP 2025 ★★★
(a) north (b) south
(c) east (d) west

5. The frequency of AC supplied in India is: ★★★
(a) 50 Hz (b) 60 Hz
(c) 100 Hz (d) 220 Hz

6. The insulation colour of the live wire as per NCERT is: ★★☆
(a) black (b) green
(c) red (d) blue

7. An electric fuse works on the: ★★★
(a) magnetic effect of current (b) heating effect of current
(c) chemical effect of current (d) induction of current

8. Domestic appliances are connected in parallel so that each: ★★☆
(a) draws the same current (b) gets the same voltage, 220 V
(c) has the same resistance (d) uses less power

9. **Assertion (A):** In domestic circuits, the earth wire is connected to a metal plate buried deep inside the earth.
Reason (R): The earth wire keeps the metallic body of an appliance at the potential of the earth, so the user does not get a severe shock. CBSE 2025 ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The current in an AC supply of 50 Hz changes its direction 50 times in one second.
Reason (R): The current in AC reverses its direction twice in every cycle. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	Fleming's LHR gives the force direction.
2	(c)	F is maximum when I is perpendicular to B.
3	(d)	Middle finger = current.
4	(b)	v down, B east: thumb points south.
5	(a)	India: 50 Hz.
6	(c)	Live red, neutral black, earth green.
7	(b)	Fuse melts by heating (I^2Rt).
8	(b)	Parallel: same 220 V for each.
9	(a)	R correctly explains A.
10	(d)	It changes 100 times a second; R is true.

Model answers: short and long answer

1 Fleming's rule and the force
Statement as in Section 2. Middle finger north (current), forefinger up (field): the thumb points **east**. Force: **towards east**.

2 Fuse
- A low-melting-point wire in series; it **melts** when current exceeds its rating and cuts the supply
- In the live wire, so the appliance is **cut off from the 220 V side** and is safe to touch

3 Short circuit vs overloading
- Short circuit: live touches neutral, so current shoots up suddenly
- Overloading: too many appliances or a voltage surge; current exceeds the rating
- Protection: **fuse** and **MCB**

Day 39 checklist: tick before you sleep

- ☐ I can explain the aluminium rod activity
- ☐ I can apply Fleming's left-hand rule
- ☐ I can compare AC and DC and explain 50 Hz
- ☐ I can draw a home circuit with all 3 wires
- ☐ I solved all 10 MCQs and checked the key

My score: ____ / 5
5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 39 COMPLETE +100 XP earned. Next: Level 40

BUJHO *Samajh gaya! Left hand se force, live wire mein fuse, metal body ke liye earthing: done!*

PAHELI *Shabaash! Kal Level 40: Boss Level aur poore Physics ka mock test.*

NITESH SIR SAYS
Left hand force ke liye, right hand field ke liye. Haath mat ulto, marks mat gawao. ***Kal milte hai Day 40 pe!***

DAY 40 LEVEL 40

PHYSICS | CHAPTER 8 | BOSS LEVEL + MOCK TEST

BOSS LEVEL + PHYSICS MOCK TEST

Goal: clear Chapter 8, then test all four physics chapters.

by Nitesh Prakash

01 Boss Level | 02 Mock Test | 03 Answer Key | 04 Marking Points | 05 Checklist | +200 XP today | Level 40 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 8 BOSS LEVEL

PAPER PATTERN

Time: 40 min | Max. marks: 24 | Questions: 14
Q1–10: 1 mark each; Q11–13: 3 marks each; Q14: 5 marks
Plan: Q1–10 in 10 min, Q11–13 in 15, Q14 in 10, re-check 5

BUJHO *Boss Level aa gaya... main thoda so jaaun? Kal se pakka padhunga.*

PAHELI *Uth jao Bujho! 40 minute, bina book dekhe. Phir answer key se khud check karna.*

SECTION A: OBJECTIVE — 10 × 1 = 10 marks

1. Inside a long current-carrying solenoid, the magnetic field lines are: ★★★

(a) circles (b) parallel straight lines
(c) absent (d) spreading outwards

2. Two magnetic field lines never intersect because, at the crossing point: ★★☆

(a) compass points two ways (b) field is zero
(c) lines cancel (d) poles merge

3. The direction of the magnetic field around a straight current-carrying wire is given by: ★★★

(a) Fleming's left-hand rule (b) the right-hand thumb rule
(c) Ohm's law (d) Snell's law

4. As we move away from a straight current-carrying wire, the magnetic field: ★★☆

(a) increases (b) decreases
(c) stays the same (d) becomes zero at once

5. The field at the centre of a circular coil grows if we increase: ★★☆

(a) the radius (b) the resistance
(c) the number of turns (d) the length of leads

6. Outside a bar magnet, field lines go from: ★★★

(a) S pole to N pole (b) N pole to S pole
(c) the middle to the ends (d) nowhere: they are closed

7. A soft iron core is placed inside a solenoid to: ★★★

(a) reduce the current (b) make it a weak magnet
(c) make a strong electromagnet (d) stop heating

8. An electron beam moving east to west enters a field acting vertically downward. The force on it is towards the: ★★☆

(a) north (b) south
(c) east (d) west

ASSERTION–REASON OPTIONS

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9. **Assertion (A):** A current-carrying solenoid behaves like a bar magnet.
Reason (R): One end of the solenoid acts as a north pole and the other end as a south pole. ★★★

10. **Assertion (A):** The fuse is always connected in the live wire.
Reason (R): A fuse wire has a very high melting point. ★★☆

SECTION B: SHORT ANSWER — 3 × 3 = 9 marks

11. Draw the pattern of magnetic field lines produced by a current-carrying straight conductor and mark their direction. Which rule gives this direction? **(3)** CBSE 2024 ★★★

12. Name two factors on which the field of a current-carrying solenoid depends. List two ways to strengthen an electromagnet whose core material is fixed. **(3)** CBSE 2025 ★★★

13. A 1.1 kW heater and a 220 W TV run together on a 220 V line with a 5 A fuse. Will the fuse blow? Show the calculation. **(3)** ★★★

SECTION C: LONG ANSWER — 1 × 5 = 5 marks

14. (a) Describe an activity to show the field pattern of a current-carrying solenoid and compare it with a bar magnet. (b) Give two differences between an electromagnet and a permanent magnet. (c) State Fleming's left-hand rule. **(5)** ★★★

Diagram scale aur arrows ke saath banao

EXAM RULE

*Shortcut mat lo! Ray diagram aur circuit diagram dono pencil aur scale se, **arrows ke saath**.*

2 PHYSICS MOCK TEST: DAYS 21-39

PAPER PATTERN

Time: 75 min | Max. marks: 40 | Questions: 22, all compulsory
Marks: A (MCQ) 1, B (very short) 2, C (short) 3, D (long) 5, E (case-based) 4
Time plan: A 15 min, B 12, C 15, D 10, E 8; last 15 min for checking

REALITY CHECK

*Formula ratta, par numerical mein unit aur sign bhool gaye? Result: **2/20**. Har step likho, marks pakke!*

SECTION A: MULTIPLE CHOICE — 12 × 1 = 12 marks

1. The focal length of a spherical mirror of radius of curvature 20 cm is: ★★★
(a) 20 cm (b) 10 cm
(c) 40 cm (d) 5 cm

2. An object at the centre of curvature of a concave mirror forms an image that is: ★★★
(a) virtual and erect
(b) at F, highly diminished
(c) at C, real, inverted, same size
(d) at infinity

3. The mirror used as a rear-view mirror in vehicles is: ★★★
(a) plane (b) convex
(c) concave (d) any curved mirror

4. Glass has refractive index 1.5. Speed of light in glass is: ★★☆
(a) 1.5×10^8 m/s (b) 2×10^8 m/s
(c) 3×10^8 m/s (d) 4.5×10^8 m/s

5. The power of a lens of focal length −25 cm is: ★★★
(a) −4 D (b) +4 D
(c) −0.25 D (d) +0.04 D

6. The near point of a normal human eye is at: ★★☆
(a) infinity (b) 2.5 cm
(c) 25 cm (d) 1 m

7. Myopia is corrected by using a: ★★★
(a) concave lens (b) convex lens
(c) bifocal lens (d) prism

8. The sky looks blue because of: ★★☆
(a) dispersion (b) scattering of light
(c) reflection (d) total internal reflection

9. The magnetic field lines around a straight current-carrying wire are: ★★☆
(a) parallel lines (b) ellipses
(c) radial lines (d) concentric circles

10. Three resistors of 6 Ω each are connected in parallel. The equivalent resistance is: ★★★
(a) 2 Ω (b) 3 Ω
(c) 6 Ω (d) 18 Ω

Q11–12: use the Assertion–Reason options on page 1.

11. **Assertion (A):** Danger signal lights are red in colour.
Reason (R): Red light is scattered the least by fog and smoke. ★★★

12. **Assertion (A):** The resistivity of a wire depends on its length.
Reason (R): The resistance of a wire is given by R = ρl/A. ★★☆

SECTION B: VERY SHORT ANSWER — 5 × 2 = 10 marks

13. Draw a ray diagram for an object placed between F and P of a concave mirror. State the nature of the image. **(2)** ★★★

14. A concave lens (f = 15 cm) forms an image 10 cm from it. Find the object distance and the magnification. **(2)** NCERT ★★★

15. What is dispersion of white light? Which colour bends the most through a prism? **(2)** ★★☆

16. Why are the coils of electric toasters and irons made of an alloy rather than a pure metal? **(2)** NCERT ★★☆

17. What is the power of accommodation of the eye? Which muscles make it possible? **(2)** ★★☆

SECTION C: SHORT ANSWER — 3 × 3 = 9 marks

18. A 5 cm tall object is placed 25 cm from a convex lens of focal length 10 cm. Find the position, size and nature of the image. **(3)** NCERT ★★★

19. Resistors of 5 Ω, 10 Ω and 30 Ω are in parallel across 12 V. Find each current, total current and total resistance. **(3)** NCERT ★★★

20. An electric iron of resistance 20 Ω draws 5 A. Find (a) the heat produced in 30 s and (b) its power. **(3)** NCERT ★★★

SECTION D: LONG ANSWER — 1 × 5 = 5 marks

21. What is myopia? Give two causes and draw ray diagrams of the defect and its correction. A person cannot see beyond 1.2 m: find the lens type and power. **(5)** ★★★

SECTION E: CASE-BASED — 1 × 4 = 4 marks

22. A student records these readings for a nichrome wire: **(4)** ★★★

V (volt)	1.0	2.0	3.0	4.0
I (ampere)	0.1	0.2	0.3	0.4

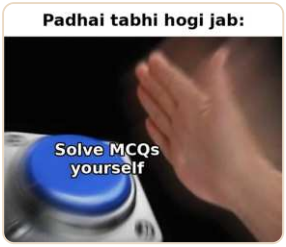
(i) Which law does the data verify? **1**
(ii) Find the resistance of the wire. **1**
(iii) A wire of the same material and thickness but twice as long replaces it. Find its resistance, with reason. **2**

EXAM TIP

Numericals: Given, Formula, Substitution, Answer with unit. **End of paper:** check with the key on pages 3–4.

3 ANSWER KEY AND MARKING POINTS

Pehle khud solve kiya? Tabhi answer key kholo, warna button dabao phir se!



Boss Level: MCQ answers

Q	Ans	Why
1	(b)	Uniform field: parallel straight lines
2	(a)	A compass cannot point two ways at once
3	(b)	Right-hand thumb rule
4	(b)	Field weakens with distance from the wire
5	(c)	More turns, stronger field
6	(b)	Outside: N to S; inside: S to N
7	(c)	Soft iron is strongly magnetised
8	(a)	Current east, B down: force north
9	(a)	R explains A
10	(c)	R is false: fuse wire has a low melting point

Boss Level: model answers, Q11–13

11 Field around a straight wire — 1 + 1 + 1

- Concentric circles centred on the wire, in a plane perpendicular to it; closer near the wire
- Arrows: current upward gives anticlockwise circles seen from above
- Rule: **right-hand thumb rule**

12 Solenoid and electromagnet — 1½ + 1½

- Field depends on the **number of turns** and the **current** (and the core material)
- To strengthen: **increase the number of turns** and **increase the current**

13 Will the fuse blow? — 1 + 1 + 1

1	Total power P = 1100 W + 220 W = 1320 W
2	I = P / V = 1320 / 220 = **6 A**
3	6 A > 5 A: the circuit is overloaded and the **fuse melts**

Boss Level: marking points, Q14

14 Solenoid, magnets, Fleming — 2 + 2 + 1

- (a) Pass a solenoid through a cardboard sheet, sprinkle iron filings, switch on and tap: the pattern matches a bar magnet; inside, lines are parallel (uniform field); one end acts as N, the other as S
- (b) Electromagnet: temporary, strength can be changed, polarity can be reversed. Permanent magnet: always magnetic, fixed strength and polarity
- (c) Thumb, forefinger, middle finger of the left hand at right angles: forefinger field, middle finger current, thumb force

Mock Test: Section A answers

Q	Ans	Why
1	(b)	f = R/2 = 10 cm
2	(c)	At C: real, inverted, same size
3	(b)	Convex: erect image, wide view
4	(b)	$v = c/n = 3\times10^8/1.5$
5	(a)	P = 1/f = 1/(−0.25 m) = −4 D
6	(c)	25 cm for a normal eye
7	(a)	Concave (diverging) lens
8	(b)	Blue is scattered more than red
9	(d)	Concentric circles around the wire
10	(a)	6/3 = 2 Ω
11	(a)	R explains A
12	(d)	A is false: resistivity is a material property; R is true

Mock Test: Section B, Q13–14

13 Concave mirror, object between F and P — 1 + 1

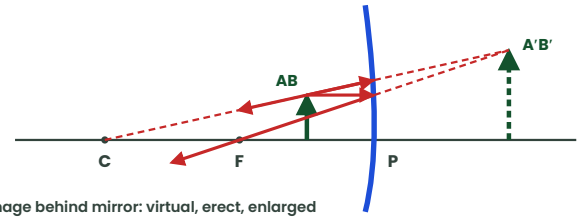


Image behind mirror: virtual, erect, enlarged

14 Concave lens — 1 + 1

1	Given: f = −15 cm, v = −10 cm (virtual image, same side)
2	1/u = 1/v − 1/f = −1/10 + 1/15 = −1/30, so **u = −30 cm**
3	m = v/u = (−10)/(−30) = **+1/3**: erect, diminished

Mock Test: Section B, Q15–17

15 Dispersion — 1 + 1

Splitting of white light into its seven colours (VIBGYOR) by a prism. **Violet** bends the most, red the least.

16 Alloy coils — 1 + 1

An alloy has a **higher resistivity** than a pure metal, and it **does not oxidise (burn) easily** at high temperature.

17 Power of accommodation — 1 + 1

The ability of the eye lens to **change its focal length** to see near and far objects clearly; done by the **ciliary muscles**.

Physics hit-list: Eduspark PYQ analysis

Topic	Rating
Ray diagrams for mirrors and lenses (all cases)	★★★
Mirror and lens formula numericals with signs	★★★
Myopia, hypermetropia and their correction	★★★
Series and parallel circuit numericals	★★★
Field lines: bar magnet, wire, loop, solenoid	★★★

Mock Test: Section C, Q18–20

18 Convex lens numerical 1 + 1 + 1

1. Given: h = +5 cm, u = −25 cm, f = +10 cm
2. 1/v = 1/f + 1/u = 1/10 − 1/25 = 3/50, so **v = +16.7 cm**
3. m = v/u = 16.7/(−25) = −0.67; h′ = m × h = **−3.3 cm**. Real, inverted, diminished, on the other side of the lens

19 Parallel resistors 1 + 1 + 1

1. Same V = 12 V across each: I_1 = 12/5 = **2.4 A**, I_2 = 12/10 = **1.2 A**, I_3 = 12/30 = **0.4 A**
2. Total I = 2.4 + 1.2 + 0.4 = **4 A**
3. $1/R_p$ = 1/5 + 1/10 + 1/30 = 10/30, so $\mathbf{R_p = 3\ \Omega}$ (check: 12/3 = 4 A)

20 Heating effect 2 + 1

1. $H = I^2Rt = 5^2 \times 20 \times 30 = \mathbf{1.5 \times 10^4\ J}$
2. $P = I^2R = 25 \times 20 = \mathbf{500\ W}$

Mock Test: Section D and E

21 Myopia 1 + 1 + 2 + 1

- Near objects clear, far objects blurred; the image forms **in front of the retina**
- Causes: **excessive curvature** of the eye lens; **elongation** of the eyeball
- Diagrams: rays from a far object meet in front of the retina; a concave lens makes them meet on the retina
- Far point 1.2 m, so f = −1.2 m; P = 1/f = **−0.83 D**, a **concave** lens

22 Case: Ohm's law 1 + 1 + 2

- (i) **Ohm's law**: V ∝ I at constant temperature
- (ii) R = V/I = 1.0/0.1 = **10 Ω**
- (iii) R ∝ l (same material and area), so double length gives **20 Ω**

Physics formula sheet: Days 21–39

Chapter	Must-know formulae and values
Light	f = R/2; mirror 1/v + 1/u = 1/f, m = −v/u; lens 1/v − 1/u = 1/f, m = v/u; n = c/v; P = 1/f (f in m), D
Human eye	near point 25 cm, far point infinity; myopia: concave lens, P = −1/far point (m); hypermetropia: convex lens
Electricity	I = Q/t; V = W/Q; V = IR; R = ρl/A; series $R = R_1 + R_2 + \ldots$; parallel $1/R = 1/R_1 + 1/R_2 + \ldots$; $H = I^2Rt$; $P = VI = I^2R = V^2/R$; 1 kWh = 3.6×10^6 J
Magnetism	right-hand thumb rule (field), Fleming's left-hand rule (force); India: 220 V, 50 Hz AC

Sign convention (New Cartesian): distances from the pole or optical centre; along the incident light +ve; real object u is -ve; f of concave mirror and concave lens -ve.

FORMULA SHEET
*Physics ke formula dekh ke darr lagta hai? Bas yaad rakho: **1/v − 1/u = 1/f**, V = IR, H = I²Rt. Simple hai boss!*

× WRONG	✔ RIGHT
u = +25 cm for a real object	u = −25 cm (against incident light)
P = 1/f with f in cm	f in metres: P = 1/(−0.25) = −4 D
Parallel $R = R_1 + R_2$	$1/R = 1/R_1 + 1/R_2$; R is smaller than each
Right hand for force on a wire	Left hand (Fleming) for force; right hand for field

Score tracker

Paper	Max.	My marks
Chapter 8 Boss Level	24	____
Physics Mock Test	40	____
Total	**64**	____

55+ Topper zone | 45–54 Strong, polish the numericals | 32–44 Revise the weak chapter | below 32 Redo Days 21–39 summaries

RESULT DAY
*Mock mein 55+ aaye? Ab toh **Jethalal wali smile** banti hai!*

Day 40 checklist: tick before you sleep

- ☐ I finished the Boss Level in 40 minutes
- ☐ I wrote the mock test in one sitting
- ☐ I checked every answer with the key
- ☐ I reworked every numerical I got wrong
- ☐ I listed my weakest chapter to revise

My score: ____ / 5
5/5 = Jalwa hai! Weak chapter ka summary phir se padho.

LEVEL 40 COMPLETE

+200 XP earned. Physics complete. Book total: 5000 XP

BUJHO *Physics khatam! Light, eye, electricity, magnet: sab boss levels clear!*

PAHELI *Shabaash! Ab Day 41 se Biology shuru. Naya chapter, naya level!*

NITESH SIR SAYS
*Physics mein formula kam, concept zyada. Diagram saaf, unit sahi, sign convention pakka. **Kal milte hai Day 41 pe!***

PHYSICS | CHAPTER 8 COMPLETE | PHYSICS COMPLETE

CHAPTER 8 SWAHAA!

Magnetic Effects of Electric Current | Days 36–40

BUJHO *Field lines, thumb rule, solenoid, Fleming ka left hand aur mock test bhi! Chapter 8 ke saath poori Physics Swahaa!*

PAHELI *Trophy phir se tumhari, Bujho! 40 levels clear. Ab Biology ki duniya mein chalte hain.*

CHAPTER XP EARNED
+600 XP

LEVELS CLEARED
5 levels (Days 36–40)

BOOK PROGRESS
Level 40 / 60 5000 XP total

WHAT YOU HAVE MASTERED

- ✔ Magnetic field and field lines
- ✔ Field of a circular loop and a solenoid
- ✔ Force on a conductor, Fleming's left-hand rule
- ✔ Right-hand thumb rule, straight wire
- ✔ Electromagnets and their uses
- ✔ AC, DC and domestic electric circuits

NEXT CHAPTER Day 41: Biology begins — *Physics cleared!*

DAY 41
LEVEL 41

BIOLOGY | CHAPTER 9 | PART 1 ASTRA SERIES

LIFE PROCESSES & PHOTOSYNTHESIS

Goal: life processes and how plants make food.

by Nitesh Prakash

01 Life Processes | 02 Nutrition Types | 03 Photosynthesis | 04 Stomata | 05 NCERT Activities | +140 XP today | Level 41 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHAT ARE LIFE PROCESSES? ★★★ MOST ASKED

DEFINITION

- **Life processes:** the basic processes that together carry out the **maintenance job** of an organism and keep it alive, even when it is sleeping or sitting still
- Molecules in living bodies keep moving all the time, so repair never stops

The four life processes ★★★

Process	What it does
Nutrition	takes in food (energy source) from outside the body
Respiration	breaks down food to release energy (mostly using oxygen)
Transportation	carries food, oxygen and wastes to every cell
Excretion	removes harmful by-products from the body

Why do we need energy all the time?

- Maintenance and repair go on even in sleep, so the body needs a **constant supply of energy**
- Energy comes from **food** taken from outside the body (carbon-based, mostly)
- Growth needs raw materials from outside too, since the body must make new molecules
- Food is broken down by **oxidising–reducing reactions** inside cells; many organisms use oxygen from outside for this

MNEMONIC: NARATE

"NaRaTE" = **N**utrition, **R**espiration, **T**ransportation, **E**xcretion. Jo NaRaTE karta hai, wahi zinda hai!

BUJHO *Main toh so raha tha, tab bhi energy kharch ho rahi thi?*

PAHELI *Haan Bujho! Sote waqt bhi heart, breathing aur cell repair chalte rehte hain. Maintenance kabhi chutti nahi leta, isliye energy roz chahiye.*

★★★ *NaRaTE ke chaar processes ka naam aur kaam: 2 marks pakke, Nitesh Bhaiya ki guarantee!*

Unicellular vs multicellular organisms

Unicellular	Multicellular
whole body surface touches the surroundings	most cells are far from the surroundings
simple **diffusion** meets the need for food, gases, waste	diffusion is too slow, so **special tissues and organs** do the job
e.g. Amoeba, Paramecium	e.g. humans, plants

2 NUTRITION AND ITS TYPES ★★★

DEFINITION

- **Nutrition:** the process of taking in food and using it to get energy and raw materials for growth and repair

Autotrophic vs heterotrophic nutrition ★★★

Autotrophic	Heterotrophic
makes its own food from simple **inorganic** substances (CO_2, water)	depends on food made by other organisms, directly or indirectly
uses sunlight and chlorophyll	breaks down complex food with **enzymes**
e.g. green plants, some bacteria	e.g. animals, fungi, most bacteria
called **autotrophs** (producers)	called **heterotrophs** (consumers)

MEME BREAK

*Biology mein pehla pyaar: **life processes**. Ye chaar samajh liye toh poora chapter apna lagega!*

★★★ *Autotrophic vs heterotrophic ka difference har saal 1–2 marks mein aata hai, Nitesh Bhaiya ki guarantee!*

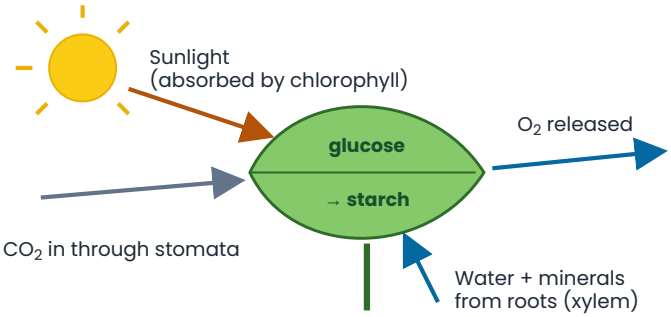
3 PHOTOSYNTHESIS ★★★ MOST ASKED

DEFINITION

- **Photosynthesis:** the process by which autotrophs take in **CO_2** and **water** and convert them into **carbohydrates** in the presence of **sunlight** and **chlorophyll**, releasing oxygen

BALANCED EQUATION

$$6CO_2 + 12H_2O \xrightarrow{\text{(sunlight, chlorophyll)}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$



Three events of photosynthesis ★★★ MOST ASKED

1. **Absorption** of light energy by chlorophyll
2. **Conversion** of light energy into chemical energy and **splitting of water** into hydrogen and oxygen
3. **Reduction** of carbon dioxide to carbohydrates

MNEMONIC: THREE EVENTS

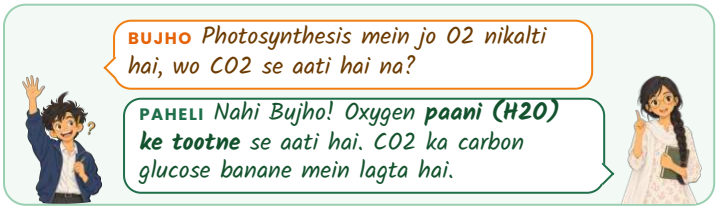
"A–C–R: Aaj Chai Ready" = **A**bsorb light, **C**onvert energy + split water, **R**educe CO_2. Chai ki tarah order fix!

Desert plants: a clever timing trick ★★☆

- Their stomata stay **closed in the day** to save water
- They take up **CO_2 at night** and store it as an intermediate
- In the day, energy absorbed by chlorophyll acts on this intermediate to make food

Chlorophyll, chloroplasts and starch

- Leaf cells contain green dots called **chloroplasts**; they hold the pigment **chlorophyll**
- Glucose not used at once is stored as **starch**, the plant's energy reserve

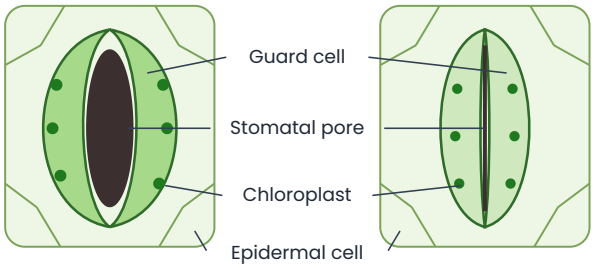


DO NOT MIX UP

- **Chloroplast** = the green organelle (the kitchen)
- **Chlorophyll** = the green pigment inside it that traps light (the stove)

4 STOMATA: DOORS OF THE LEAF ★★★ MOST ASKED

Stomata are tiny pores on the surface of the leaf. Most gas exchange for photosynthesis happens through them (gases also pass through stems and roots).



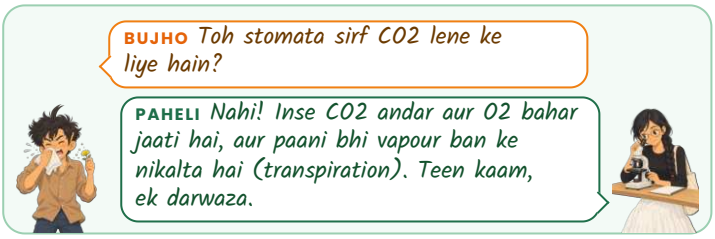
(a) Open stomatal pore (b) Closed stomatal pore

How the pore opens and closes ★★★

- Water **flows into** the guard cells → they swell → the pore **opens**
- Water **flows out** → guard cells shrink → the pore **closes**
- A lot of water is lost through open stomata, so the plant closes them when CO_2 is not needed

MNEMONIC: GUARD CELLS

"Paani aaya, darwaza khula; paani gaya, darwaza band." Swollen guard cells = open stoma.



MEME BREAK

*Plant bole: delivery app nahi chahiye, **sunlight + CO_2 + paani** se apna khana khud bana leta hoon!*

Why photosynthesis matters ★☆☆

- It is the **starting point of almost every food chain**: all heterotrophs depend on it
- It releases the **oxygen** that most organisms use for respiration
- It removes CO_2 from the air

NCERT in-text questions

Q How do autotrophs get carbon dioxide and nitrogen?

- CO_2 from air through stomata; nitrogen from soil as nitrates or nitrites

Q Name the raw materials of photosynthesis.

- CO_2, water, sunlight and chlorophyll (plus minerals for other compounds)

5 RAW MATERIALS FOR PHOTOSYNTHESIS ★★☆

Where each raw material comes from

Raw material	Source and route
Carbon dioxide	air, enters through **stomata**
Water	soil, absorbed by **roots** and carried up
Minerals (N, P, Fe, Mg)	soil, taken up by roots
Sunlight	absorbed by chlorophyll

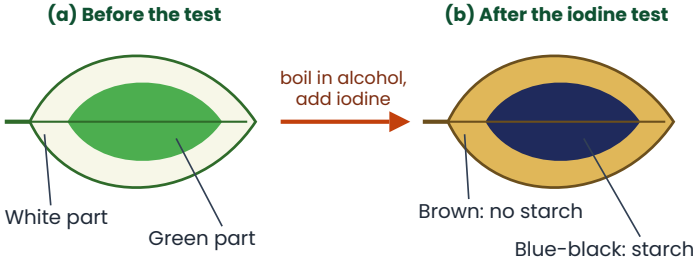
Nitrogen: why and how

- Nitrogen is essential to make **proteins** and other compounds
- Plants take it as **inorganic nitrates or nitrites**
- Or as **organic compounds** made by bacteria from atmospheric nitrogen

6 NCERT ACTIVITIES ★★★ MOST ASKED

Activity 1: is chlorophyll needed? (variegated leaf)

1. Keep a potted plant with variegated leaves (green and white parts) in the **dark for three days**, so its starch is used up (**destarching**)
2. Keep it in sunlight for about six hours; pluck a leaf and trace its green parts
3. Dip the leaf in boiling water, then boil it in **alcohol** on a water bath to remove chlorophyll
4. Dip it in dilute **iodine solution** and compare with the tracing



Result: only the parts that were green turn blue-black, so **chlorophyll is needed** for photosynthesis (starch is formed only there).

MEME BREAK
*Cat ne bhi photosynthesis try kiya! Par bina **chlorophyll** ke koi starch nahi banta, iodine test fail.*

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Using water with a heavy oxygen isotope (^{18}O), scientists showed that the oxygen released in photosynthesis comes from **water**, not from carbon dioxide.

Activity 2: is carbon dioxide needed? ★★★

1. Take two healthy potted plants of the same size; keep both in the **dark for three days**
2. Place each on a glass plate; put a watch-glass of **potassium hydroxide (KOH)** beside one plant
3. Cover each with a bell jar and seal the bottom with **vaseline** (air-tight)
4. Keep both in sunlight for two hours, then test one leaf from each for starch

Set-up	Starch test	Why
with KOH	**no** blue-black	KOH absorbs CO_2
without KOH	**blue-black**	CO_2 available

Conclusion: carbon dioxide is essential for photosynthesis.

WHY ALCOHOL, WHY IODINE?

- Alcohol dissolves **chlorophyll**, so the colour change is easy to see
- Iodine turns **starch blue-black**; boil alcohol on a **water bath**, never on a direct flame

Exam tips and common mistakes

- Write the equation with **"sunlight" and "chlorophyll"** on the arrow; both carry marks
- Photosynthesis releases **O_2** that comes from **water**, not from CO_2
- Destarching is done **first**, so that any starch found later is new
- Do not say "plants respire only at night": they respire all the time
- The plant without KOH is the **control**; vaseline keeps outside air (CO_2) from entering

MEME BREAK
*Bhookh lagne ka tension hi nahi, jab khana khud banta ho. **Autotrophs** ki smart life!*

7 PRACTICE MCQS

1. Which of these is NOT a life process? ★★☆

(a) nutrition (b) excretion
(c) photosynthesis in animals (d) respiration

2. The correct sequence of events in photosynthesis is best described as: ★★★

(a) reduction of CO_2, light absorption, splitting of water
(b) light absorption, splitting of water, reduction of CO_2
(c) splitting of water, reduction, light absorption
(d) none of these

3. Opening and closing of the stomatal pore is controlled by: ★★★

(a) epidermal cells (b) guard cells
(c) xylem (d) chloroplasts

4. In the KOH activity, KOH is used to: ★★★

(a) provide oxygen (b) absorb water
(c) absorb carbon dioxide (d) kill bacteria

5. In the iodine test, a starch-containing part of a leaf turns: ★★☆

(a) red (b) blue-black
(c) yellow (d) white

6. Desert plants take up CO_2: ★★☆

(a) only in the day (b) at night
(c) through roots (d) never

7. Plants absorb nitrogen mostly in the form of: ★★☆

(a) N_2 gas (b) nitrates or nitrites
(c) ammonia gas (d) proteins

8. Excess glucose in a plant is stored as: ★★☆

(a) glycogen (b) starch
(c) cellulose (d) sucrose

9. **Assertion (A):** The parts of a variegated leaf that were white do not turn blue-black in the iodine test.
Reason (R): White parts lack chlorophyll, so no starch is made there. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Guard cells close the stomatal pore when they lose water.
Reason (R): Plants lose a large amount of water through open stomata. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(c)	Animals do not photosynthesise.
2	(b)	Absorb, convert and split, reduce.
3	(b)	Guard cells swell or shrink.
4	(c)	KOH absorbs CO_2 inside the jar.
5	(b)	Iodine + starch gives blue-black.
6	(b)	Stomata open at night in deserts.
7	(b)	Inorganic nitrates or nitrites.
8	(b)	Starch is the plant's reserve.
9	(a)	No chlorophyll, no starch: R explains A.
10	(b)	Both true; R is why plants close them, not how.

Day 41 hit-list: Eduspark PYQ analysis

Topic	Rating
Photosynthesis equation and events	★★★
Stomata diagram and opening	★★★
Starch test activities	★★★
Autotrophic vs heterotrophic	★★☆
Desert plants and CO_2	★☆☆

Short and long answer questions

1. What are life processes? Why is energy needed even when we are asleep? **(2 marks)**
2. Write the balanced equation of photosynthesis and list its three events. NCERT **(3 marks)**
3. Draw a labelled diagram of an open and a closed stomatal pore. How do guard cells open the pore? **(3 marks)**
4. Describe an activity to show that CO_2 is essential for photosynthesis. NCERT **(5 marks)**

Model answers

1 Life processes
- Processes that maintain the body: nutrition, respiration, transport, excretion
- Repair and molecular movement never stop, so energy is needed all the time

2 Equation and events
- Equation with sunlight, chlorophyll on the arrow
- Absorption of light; conversion and splitting of water; reduction of CO_2

3 Stomata
- Diagram: two guard cells, pore, chloroplasts (2 labels each view)
- Water enters guard cells, they swell and the pore opens

4 KOH activity
- Destarch two plants, KOH beside one, seal, sunlight 2 h
- Leaf with KOH shows no starch: CO_2 is essential

Day 41 checklist: tick before you sleep

- ☐ I can name the four life processes and why energy is needed
- ☐ I can write the photosynthesis equation and its three events
- ☐ I can draw open and closed stomata
- ☐ I can explain both NCERT starch-test activities
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 41 COMPLETE +140 XP earned. Next: Level 42

BUJHO *Samajh gaya! Plant khud chef hai: CO_2, paani aur dhoop se glucose.*

PAHELI *Shabaash! Kal Level 42: hum khana kaise digest karte hain.*

NITESH SIR SAYS
Photosynthesis ka equation aur stomata ka diagram roz ek baar likho. ***Kal milte hai Day 42 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 42
LEVEL 42

BIOLOGY | CHAPTER 9 | PART 2 ASTRA SERIES

HETEROTROPHIC NUTRITION & DIGESTION

Goal: how animals take in food and digest it.

by Nitesh Prakash

01 Nutrition Types | 02 Amoeba | 03 Digestive System | 04 Enzymes | 05 Villi and Caries | +140 XP today | Level 42 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 HETEROTROPHIC NUTRITION ★★★ MOST ASKED

DEFINITION

- **Heterotrophic nutrition:** obtaining ready-made food (made by other organisms) and breaking it down with **enzymes**
- How an organism eats depends on the **form of food** available and **how it gets it**

Three types of heterotrophic nutrition ★★★ MOST ASKED

Type	How	Examples
Holozoic	takes in whole food and digests it inside the body	Amoeba, cow, humans
Saprotrophic	breaks down food **outside** the body (on dead matter) and absorbs it	fungi: bread mould, yeast, mushrooms
Parasitic	takes food from a living host **without killing it**	Cuscuta (amarbel), ticks, lice, leeches, tapeworms

MNEMONIC: THREE TYPES

"HSP": **H**olozoic = haath se poora khana; **S**aprotrophic = sadi-gali cheez pe; **P**arasitic = parayi thali, host zinda rehta hai.

Examples that confuse students

- **Cuscuta** (dodder) is a **plant** parasite: it has no chlorophyll
- **Yeast and mushrooms** are saprotrophs, not parasites
- Bacteria can be saprotrophic or parasitic, depending on the species
- Some organisms (like grass for a cow) are food; the cow is holozoic

BUJHO *Parasite apne host ko maar deta hai na?*

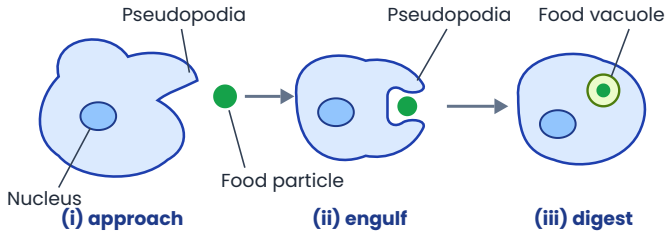
PAHELI *Nahi Bujho! Parasite host se khana leta hai par usse **turant nahi maarta**. Host mar gaya toh khana bhi khatam!*

Saprotrophs such as fungi and bacteria act as nature's **cleaners**: they break down dead plants and animals and return nutrients to the soil.

2 HOW DO ORGANISMS OBTAIN FOOD? ★★★

Single-celled organisms take food in through the **whole cell surface**; there is no separate digestive system.

Amoeba: food through pseudopodia ★★★ MOST ASKED



1. Temporary finger-like extensions of the cell, the **pseudopodia**, surround the food particle
2. They fuse to form a **food vacuole** around it
3. Inside the vacuole, complex food is broken into simpler substances that **diffuse** into the cytoplasm
4. Undigested material moves to the surface and is thrown out

Paramecium: food at a fixed spot ★★☆

- Its cell has a **definite shape**
- Food enters at a **specific spot** (oral groove)
- Hair-like **cilia** covering the cell move the food to this spot

Steps of holozoic nutrition ★★☆

Step	What happens
Ingestion	taking food into the body
Digestion	breaking complex food into simple soluble forms
Absorption	digested food passes into blood
Assimilation	cells use the food for energy, growth and repair
Egestion	throwing out undigested waste

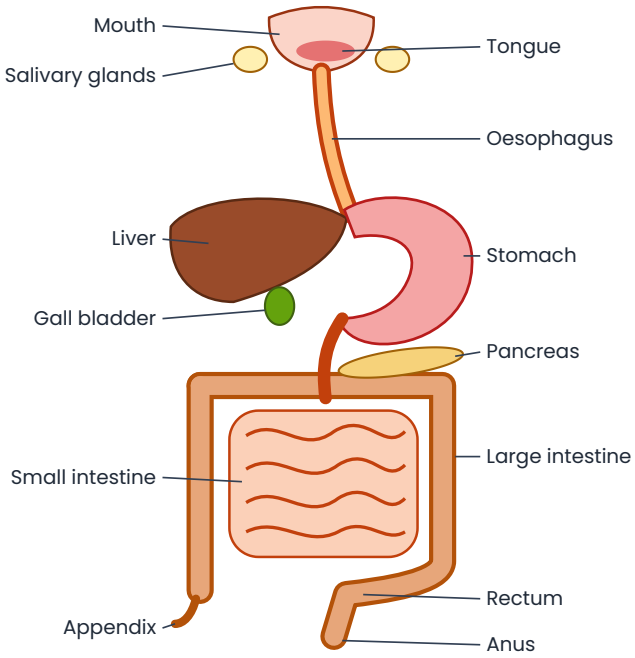
MNEMONIC: HOLOZOIC STEPS

"I Do Always Amaze Everyone" = **I**ngestion, **D**igestion, **A**bsorption, **A**ssimilation, **E**gestion.

★★★ *Amoeba ka labelled diagram (pseudopodia, food vacuole) board ka favourite hai, Nitesh Bhaiya ki guarantee!*

3 HUMAN DIGESTIVE SYSTEM ★★★ MOST ASKED

The **alimentary canal** is a long tube from the mouth to the anus. Different parts do different jobs, helped by **glands** that release digestive juices.



Human alimentary canal with associated glands (schematic)

MNEMONIC: ORDER OF THE CANAL

"Mummy Ordered Samosa, Sister Loved Aloo" = **M**outh, **O**esophagus, **S**tomach, **S**mall intestine, **L**arge intestine, **A**nus.

Glands that join the canal

- **Salivary glands** → mouth
- **Gastric glands** → stomach wall
- **Liver** and **pancreas** → small intestine
- **Intestinal glands** → wall of small intestine

Enzyme summary table ★★★ MOST ASKED

Juice	Enzyme	Acts on	Product
Saliva	Salivary amylase	Starch	Maltose (sugar)
Gastric juice	Pepsin (+ HCl)	Proteins	Peptides
Pancreatic juice	Trypsin	Proteins	Peptides
Pancreatic juice	Lipase	Emulsified fats	Fatty acids, glycerol
Pancreatic juice	Amylase	Starch	Sugars
Intestinal juice	Several enzymes	All three	Amino acids, glucose, fatty acids + glycerol

MEME BREAK

*Mouth shuru karta hai, stomach madad karta hai, par **small intestine** asli hero hai. Change my mind!*

4 MOUTH, OESOPHAGUS AND STOMACH ★★★

Mouth: the start ★★★

- **Teeth** chew and grind food; the **tongue** mixes it with saliva
- Food is made **small and soft** so enzymes can act on it
- **Salivary amylase** in saliva breaks **starch** into a simple sugar (maltose)

Oesophagus: food pipe ★★☆

- Carries food from the mouth to the stomach
- Rhythmic contraction of its muscles, called **peristalsis**, pushes food forward (it happens all along the gut)

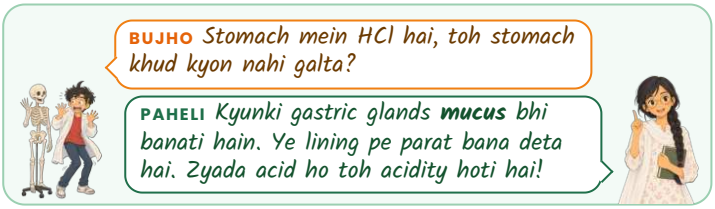
Stomach: the acid tank ★★★ MOST ASKED

Gastric juice part	Role
Hydrochloric acid (HCl)	makes the medium **acidic** for pepsin; kills germs
Pepsin	digests **proteins**
Mucus	protects the stomach lining from the acid

Muscular walls churn the food. A **sphincter muscle** lets food leave the stomach into the small intestine little by little.

MEME BREAK

*HCl sab kuch gala dena chahta hai, par **mucus** stomach wall ko bachata hai. Mucus = stomach ka bodyguard!*



MNEMONIC: WHERE DIGESTION STARTS

"S–P–F" in order of the canal: **S**tarch starts in the **mouth**, **P**rotein in the **stomach**, **F**at in the **small intestine**.

Exam tips and common mistakes

- **Pepsin** works in an **acidic** medium; **trypsin** works in an **alkaline** medium
- Bile is made by the **liver** (stored in the gall bladder), not by the gall bladder
- Write **salivary amylase** (not just "saliva") as the enzyme
- Peristalsis happens **all along the gut**, not only in the oesophagus
- Fat digestion needs **both** bile (emulsify) and lipase (digest)

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Adult humans have **32 teeth** of four kinds: incisors (cut), canines (tear), premolars and molars (grind). Enamel is the **hardest** substance in the body.

5 SMALL INTESTINE ★★★ MOST ASKED

Longest part of the canal ★★☆

- Tightly coiled; the site of **complete digestion** of carbohydrates, proteins and fats
- Its length differs with diet: **herbivores** eating grass need a **longer** small intestine to digest cellulose; **carnivores** have a shorter one

Juices that work here ★★★ MOST ASKED

Juice (source)	What it does
Bile (liver)	makes the acidic food from the stomach **alkaline**; bile salts **emulsify** fats (big globules → tiny droplets)
Pancreatic juice (pancreas)	**amylase** for starch, **trypsin** for proteins, **lipase** for emulsified fats
Intestinal juice (intestinal wall)	finishes the job: proteins → **amino acids**, complex carbohydrates → **glucose**, fats → **fatty acids + glycerol**

MNEMONIC: PANCREATIC ENZYMES

"Pancreas ka TAL" = **T**rypsin (protein), **A**mylase (starch), **L**ipase (fat). Yaad rakho: bile mein koi enzyme nahi!

MEME BREAK

*Bile ki deal: bade fat globules do, chhote droplets lo. Ise kehte hain **emulsification**, aur lipase khush!*

BUJHO *Bile fat ko digest karta hai na?*

PAHELI *Nahi! Bile mein **enzyme hi nahi** hota. Wo sirf fat ko chhote droplets mein todta hai (emulsify). Digest toh **lipase** karta hai.*

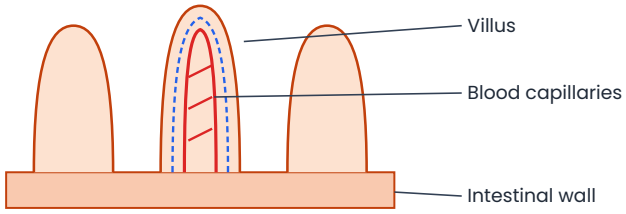
Absorption and assimilation

- **Absorption:** digested food passes through the walls of the villi into the **blood**
- **Assimilation:** body cells use the absorbed food for **energy, growth and repair**

Large intestine and anus ★★☆

- Unabsorbed food enters the **large intestine**, where more villi absorb **water**
- The rest leaves the body through the **anus**
- The exit of waste is controlled by the **anal sphincter**

Villi: built for absorption ★★★ MOST ASKED



- The inner wall has millions of finger-like folds called **villi**
- They **increase the surface area** for absorption
- Each villus is richly supplied with **blood vessels**, which carry absorbed food to every cell, where it gives energy, builds new tissue and repairs old tissue

MEME BREAK

*Muh wala amylase ho ya pancreas wala, kaam ek hi: **starch** → **sugar**. Bas jagah alag hai!*

Dental caries (tooth decay) ★★☆

- Bacteria act on **sugars** and make **acids** that soften (demineralise) the **enamel**
- Bacterial cells and food particles stick to teeth as **dental plaque**; saliva cannot reach the enamel to neutralise the acid
- Brushing after eating removes plaque before acid forms
- If untreated, germs reach the **pulp**, causing inflammation and infection

6 PRACTICE MCQS

1. Cuscuta (amarbel) shows which mode of nutrition? ★★★

(a) saprotrophic (b) parasitic
(c) holozoic (d) autotrophic

2. In Amoeba, digestion of food takes place in the: ★★☆

(a) nucleus (b) pseudopodia
(c) food vacuole (d) cell membrane

3. Which enzyme of the pancreas digests proteins? ★★★

(a) amylase (b) trypsin
(c) lipase (d) pepsin

4. The function of HCl in the stomach is to: ★★★

(a) digest fats (b) make the medium acidic for pepsin
(c) emulsify fats (d) absorb water

5. Emulsification of fats is done by: ★★★

(a) pancreatic juice (b) bile salts
(c) saliva (d) mucus

6. Water is mainly absorbed from undigested food in the: ★★☆

(a) stomach (b) small intestine
(c) large intestine (d) oesophagus

7. Villi in the small intestine help mainly by: ★★★

(a) secreting bile (b) increasing the surface area for absorption
(c) killing germs (d) churning food

8. Herbivores have a longer small intestine than carnivores because: ★★☆

(a) they eat more
(b) cellulose takes longer to digest
(c) they drink more water
(d) they have no stomach

9. **Assertion (A):** The inner lining of the stomach is not damaged by its own hydrochloric acid.
Reason (R): Mucus secreted by gastric glands protects the stomach lining. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Bile juice is called a digestive juice although it has no enzyme.
Reason (R): Bile makes the food alkaline and emulsifies fats, helping digestion. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	A plant that feeds on a living host.
2	(c)	Food vacuole holds and digests food.
3	(b)	Trypsin digests proteins.
4	(b)	Acidic medium for pepsin; also kills germs.
5	(b)	Bile salts break fat into droplets.
6	(c)	Large intestine absorbs water.
7	(b)	More surface area, faster absorption.
8	(b)	Cellulose needs a long gut.
9	(a)	Mucus explains why the lining is safe.
10	(a)	R explains why bile still counts.

Day 42 hit-list: Eduspark PYQ analysis

Topic	Rating
Human digestive system diagram	★★★
Enzymes and their juices	★★★
Role of HCl, mucus, bile	★★★
Amoeba nutrition diagram	★★☆
Villi, absorption, dental caries	★★☆

Short and long answer questions

1. Name the three types of heterotrophic nutrition with one example each. **(3 marks)**
2. Draw a labelled diagram of Amoeba taking in food. NCERT **(2 marks)**
3. Draw the human alimentary canal and label the liver, stomach, pancreas and small intestine. Give the role of each gland. **(5 marks)**
4. What is dental caries? How can it be prevented? NCERT **(2 marks)**

Model answers

1 Types
- Holozoic (human), saprotrophic (mushroom), parasitic (Cuscuta)

2 Amoeba
- Pseudopodia around the food, food vacuole formed (Section 2 diagram)

3 Digestive system
- Liver: bile; pancreas: trypsin, lipase, amylase; gastric glands: HCl, pepsin, mucus

4 Dental caries
- Acid from bacteria on sugar softens enamel; brush after meals, less sugar

Day 42 checklist: tick before you sleep

- ☐ I can name and give examples of the three heterotrophic types
- ☐ I can draw Amoeba taking in food
- ☐ I can draw and label the human digestive system
- ☐ I know every enzyme, its juice and its product
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 42 COMPLETE +140 XP earned. Next: Level 43

BUJHO *Samajh gaya! Muh se amylase, stomach mein pepsin, aur intestine mein TAL.*

PAHELI *Shabaash! Kal Level 43: respiration aur energy ka khel.*

NITESH SIR SAYS
Digestive system ka diagram 3 baar bana lo, labels khud aa jayenge. ***Kal milte hai Day 43 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 43
LEVEL 43

BIOLOGY | CHAPTER 9 | PART 3 ASTRA SERIES

RESPIRATION & BREATHING

Goal: learn how cells release energy and how we breathe.

by Nitesh Prakash

01 Glucose Paths | 02 ATP | 03 Anaerobic | 04 Lungs | 05 Breathing

+140 XP today | Level 43 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 RESPIRATION: GETTING ENERGY FROM FOOD ★★☆

DEFINITION

- **Respiration:** the process in which food (glucose) is broken down in cells to release **energy**; this energy runs every life process
- Two parts: **breathing** (taking in O_2, giving out CO_2) and **cellular respiration** (breaking glucose inside cells)
- Respiration is a **catabolic**, energy-releasing process: the opposite of photosynthesis

ATP: the energy currency of the cell ★★★

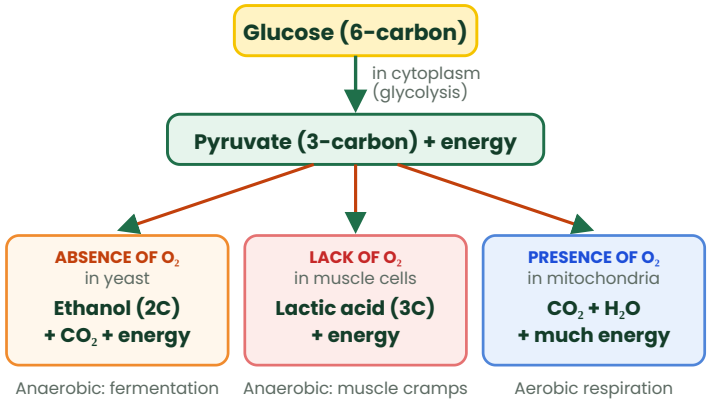
- Energy released in respiration is stored as **ATP** (adenosine triphosphate)
- When a cell needs energy, ATP breaks into **ADP + phosphate** and releases about **30.5 kJ/mol**
- ATP powers muscle contraction, protein making, nerve impulses and active transport
- Like money in your pocket: earn once, spend anywhere in the cell

MEME BREAK
*Mitochondria ATP ko aise **sprinkle** karta hai jaise salt bae namak. Isliye mitochondria = **powerhouse of the cell**, aur ATP = energy currency.*

Breathing vs respiration ★☆☆

Breathing	Cellular respiration
physical process	chemical (biochemical) process
in the lungs	in every living cell
takes in O_2, gives out CO_2	breaks glucose, releases energy
no energy released	energy released as ATP

2 BREAKDOWN OF GLUCOSE: THREE PATHWAYS ★★★ MOST ASKED



Step 1 is the same in all three: glucose splits in the cytoplasm

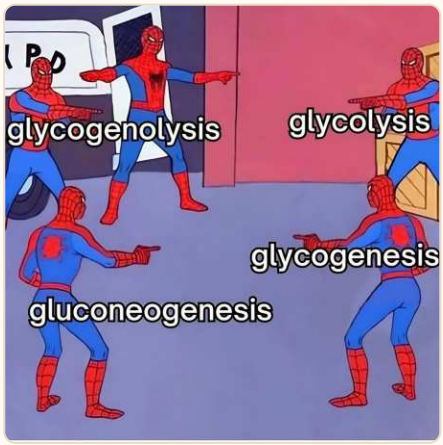
★★★ *Glucose → pyruvate hamesha cytoplasm mein; aage ka rasta O_2 decide karta hai. Ye flowchart har saal poochha jaata hai, Nitesh Bhaiya ki guarantee!*

MNEMONIC: THREE ENDS OF PYRUVATE

"Yeast ka Ethanol, Muscle ka Lactic, Mito ka CO_2–Paani." Y-E, M-L, M-CW: yeast gives ethanol + CO_2, muscles give lactic acid, mitochondria give CO_2 + water.

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

In the mitochondria, pyruvate enters the **Krebs cycle** and the electron transport chain; most of the ATP of aerobic respiration is made here. NCERT does not test these names, only the pathway flowchart.



MEME BREAK
*Glycogenolysis, glycolysis, glycogenesis, gluconeogenesis: naam sab ek jaise! Board ke liye bas ek yaad rakho: **glycolysis = glucose → pyruvate**, cytoplasm mein.*

Aerobic vs anaerobic respiration ★★★

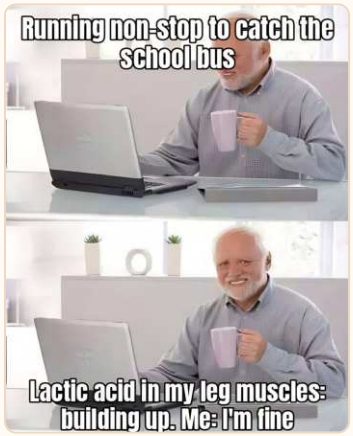
Point	Aerobic	Anaerobic
Oxygen	needed	not needed
Site	cytoplasm + mitochondria	cytoplasm only
Breakdown	complete	incomplete
Products	$CO_2 + H_2O$	ethanol + CO_2 (yeast); lactic acid (muscles)
Energy	much more (about 38 ATP per glucose)	very little (2 ATP per glucose)
Example	most plants and animals	yeast, some bacteria, our muscles in a sprint

MUSCLE CRAMPS: WHY?

During heavy exercise the muscles need energy faster than O_2 can reach them. They switch to anaerobic respiration and **lactic acid** builds up, which causes **cramps**. Rest and a massage bring blood and O_2 back.

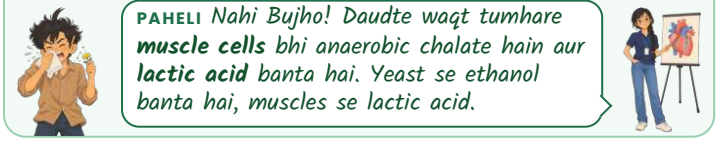
EXAM TIPS

Pathway flowchart mein har arrow pe **site** (cytoplasm / mitochondria) aur **condition** (O_2 hai ya nahi) likho. Activity questions mein hamesha **observation + conclusion** dono.



MEME BREAK

*Lactic acid jam raha hai, phir bhi "main theek hoon". Ye hi hai **anaerobic respiration** in muscles.*



3 NCERT ACTIVITIES ★★☆

Activity 1: limewater test for exhaled air ★★☆

- Pump ordinary air through limewater in one test tube; blow **exhaled air** through limewater in another
- Limewater turns **milky faster** with exhaled air
- **Conclusion:** exhaled air has more CO_2 than ordinary air; $CO_2 + Ca(OH)_2 \rightarrow CaCO_3$ (milky) + H_2O

Activity 2: yeast and sugar ★★☆

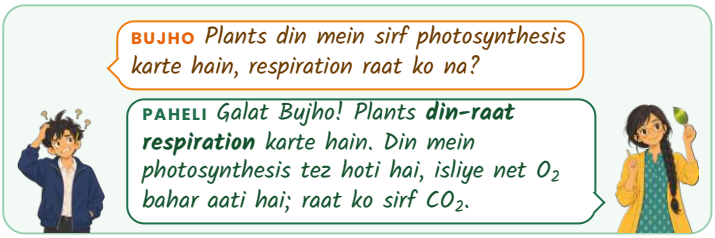
- Add yeast to a sugar solution in a stoppered test tube; pass the gas formed into limewater
- Limewater turns milky: yeast gives out **CO_2**; the solution smells of **alcohol**
- **Conclusion:** yeast respires anaerobically (fermentation)

4 EXCHANGE OF GASES IN PLANTS ★★☆

Plants breathe by diffusion ★★☆

- Gases enter and leave through **stomata** (leaves), **lenticels** (stems) and the surface of roots, by **diffusion**
- Plants have no special breathing organs; every cell is close to the air

Time	What happens	Gas out
Day	photosynthesis faster than respiration; CO_2 used up	O_2
Night	only respiration	CO_2



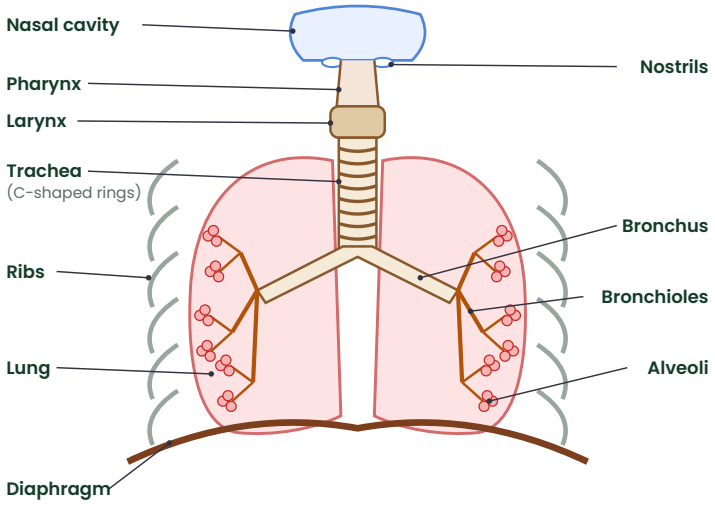
Breathing rate: water vs land ★★☆

- Water holds far **less dissolved O_2** than air
- So aquatic animals such as fish breathe **much faster** than land animals
- Fish take water in through the mouth and push it over the **gills**, where dissolved oxygen is absorbed

SMALL ORGANISMS

In **unicellular** organisms such as Amoeba, gases diffuse across the whole body surface; large animals need lungs or gills.

5 HUMAN RESPIRATORY SYSTEM ★★★ MOST ASKED



Path of air (draw it as a flow)

Nostrils → nasal cavity → pharynx → larynx → trachea → bronchi → bronchioles → alveoli

- Fine **hair** and **mucus** in the nose filter dust and germs and warm the air
- A flap, the **epiglottis**, closes the trachea while we swallow food
- **Rings of cartilage** keep the trachea open so it does not collapse when there is little air in it
- Alveoli are balloon-like sacs with thin walls and a dense network of **blood capillaries**

Mechanism of breathing ★★★

Step	Inhalation	Exhalation
Ribs	lifted up and out	move down and in
Diaphragm	contracts, flattens	relaxes, domes up
Chest cavity	becomes larger	becomes smaller
Air	rushes into the lungs	pushed out of the lungs

MNEMONIC: INHALE VS EXHALE

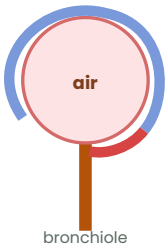
"Andar saans: Ribs Upar, Diaphragm Neeche (flat)." Inhale = chest bada, air andar. Exhale mein sab ulta: ribs neeche, diaphragm upar (dome).

Residual volume and alveoli ★★☆

- The lungs always hold some air even after breathing out: the **residual volume**
- So there is always time for O_2 to be absorbed and CO_2 to be released
- The alveoli spread out would cover about **80 m^2**: a huge surface for gas exchange

MEME BREAK

*Saans rok li? Fikar not! Lungs mein **residual volume** hamesha rehta hai, isliye O_2 absorb hona chalta rehta hai.*



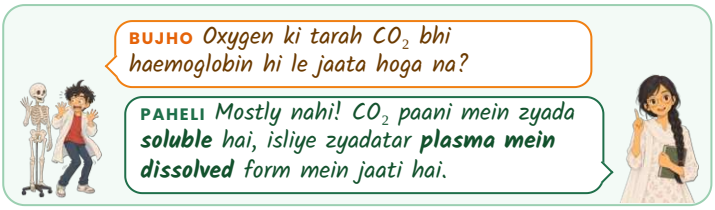
→ O_2: alveolus → blood

← CO_2: blood → alveolus

Walls one cell thick, wrapped in capillaries: gases cross by diffusion in a fraction of a second

How O_2 and CO_2 travel in blood ★★★

- **Haemoglobin**, the respiratory pigment in red blood cells, has a very high affinity for O_2 and carries it from the lungs to the tissues
- CO_2 is more **soluble in water** than O_2, so it is mostly carried **dissolved in the blood plasma**
- In large animals diffusion alone is too slow, so a pigment is needed



MNEMONIC: PATH OF AIR

"Nani Ne Pakode Laye, Tea Bina Biscuit Aayi." Nostrils, Nasal cavity, Pharynx, Larynx, Trachea, Bronchi, Bronchioles, Alveoli.

MEME BREAK

*Anaerobic respiration se bahut ATP? Nahi bhai, sirf **2 ATP** per glucose. Bahut energy chahiye toh **oxygen + mitochondria**.*

COMMON MISTAKES

- Writing that glycolysis happens in mitochondria: it happens in the **cytoplasm**
- Saying plants give out only O_2: at night they give out **CO_2**
- Calling breathing and respiration the same: breathing is physical, respiration is chemical
- Forgetting the **arrows** and site names in the pathway flowchart

Case-based question ★★☆

- Riya runs a 400 m race and feels a sharp pain in her calf muscles. (i) Which type of respiration took place in her muscles? (ii) Name the product that caused the pain. (iii) Where in the cell does the first step occur? **(1 + 1 + 1)**
- **Answers:** (i) anaerobic respiration (ii) lactic acid (iii) cytoplasm

6 PRACTICE MCQS

1. The first step of glucose breakdown, forming pyruvate, takes place in the: ★★★

(a) mitochondria (b) cytoplasm
(c) nucleus (d) chloroplast

2. In yeast, anaerobic respiration produces: ★★★

(a) lactic acid + energy (b) $CO_2 + H_2O$ + energy
(c) ethanol + CO_2 + energy (d) pyruvate only

3. Muscle cramps after a long run are due to the build-up of: ★★★

(a) ethanol (b) lactic acid
(c) pyruvic acid (d) carbon dioxide

4. The energy currency of the cell is: ★★☆

(a) ADP (b) glucose
(c) ATP (d) DNA

5. Rings of cartilage are present in the: ★★☆

(a) oesophagus (b) trachea
(c) alveoli (d) bronchioles

6. Aquatic animals breathe faster than terrestrial animals because: ★★☆

(a) water is denser
(b) water has less dissolved O_2
(c) gills are small
(d) they move more

7. Most CO_2 is carried in human blood: ★★☆

(a) by haemoglobin only
(b) in dissolved form in plasma
(c) by platelets
(d) by white blood cells

8. During inhalation the diaphragm: ★★☆

(a) relaxes and domes up
(b) contracts and flattens
(c) does not move
(d) moves into the chest

9. **Assertion (A):** The lungs always contain some air even after a forceful exhalation.
Reason (R): The residual volume of air allows time for oxygen to be absorbed and carbon dioxide to be released. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Aerobic respiration releases more energy than anaerobic respiration.
Reason (R): In aerobic respiration glucose is broken down completely into carbon dioxide and water. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Why do muscles cramp after vigorous exercise? Name the substance responsible. **(2 marks)**
2. Write the three pathways of breakdown of glucose in a flowchart and name the site of each step. **(3 marks)**
3. Draw a labelled diagram of the human respiratory system. Explain how the alveoli are designed for maximum exchange of gases. **(5 marks)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Glycolysis happens in the cytoplasm.
2	(c)	Fermentation in yeast.
3	(b)	Anaerobic respiration in muscles.
4	(c)	ATP stores and supplies energy.
5	(b)	Rings stop the trachea collapsing.
6	(b)	Water holds less oxygen than air.
7	(b)	CO_2 is more soluble in water.
8	(b)	Diaphragm flattens, chest enlarges.
9	(a)	R explains A.
10	(a)	Complete breakdown gives more energy.

YAAD RAKHNE KI TRICK: ATP COUNT

"O_2 mila toh 38 ka jackpot, O_2 nahi toh sirf 2." Aerobic: about 38 ATP per glucose; anaerobic: 2 ATP.

Model answers

1 Cramps

> Lack of O_2 in a sprint makes muscles respire anaerobically; **lactic acid** builds up and causes cramps (1 + 1)

2 Pathways

> Use the flowchart on page 1: cytoplasm step, then yeast, muscle and mitochondria branches with products (1 mark each)

3 Alveoli

> Diagram with 6 labels (2); huge surface area (about 80 m^2), very thin walls, rich blood supply, residual air (3)

Day 43 checklist: tick before you sleep

- [ ] I can draw the three pathways of glucose breakdown with sites
- [ ] I know aerobic vs anaerobic in a table and why muscles cramp
- [ ] I can explain the limewater and yeast activities
- [ ] I can draw and label the human respiratory system
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

LEVEL 43 COMPLETE +140 XP earned. Next: Level 44

BUJHO *Samajh gaya! Glucose ka pehla step cytoplasm mein, aur mitochondria mein poori energy nikalti.*

PAHELI *Shabaash! Kal Level 44: heart, blood aur double circulation.*

NITESH SIR SAYS

*Flowchart mein **site** aur **product** dono likho, poore marks wahi se aate hain.* ***Kal milte hai Day 44 pe!***

Day 43 hit-list: Eduspark PYQ analysis

Topic	Rating
Pathways of glucose breakdown (flowchart)	★★★
Aerobic vs anaerobic respiration	★★★
Respiratory system diagram	★★★
Alveoli and exchange of gases	★★☆
Gas exchange in plants, day and night	★★☆
Haemoglobin and CO_2 transport	★☆☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 44 · LEVEL 44

BIOLOGY | CHAPTER 9 | PART 4

TRANSPORTATION IN HUMANS

Goal: master blood, the heart and double circulation.

by Nitesh Prakash

01 Blood | 02 Heart | 03 Double Circulation | 04 Vessels | 05 BP and Lymph | +140 XP today | Level 44 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

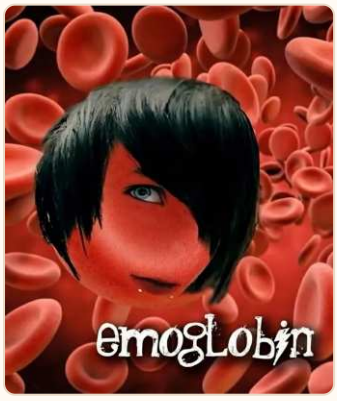
1 WHY TRANSPORT? BLOOD AS THE CARRIER ★★☆

DEFINITION

- Food, O_2, water and wastes must move to and from every cell; in big animals diffusion is far too slow
- **Blood** is a fluid **connective tissue**; the **heart** pumps it through **blood vessels**
- Humans have a closed circulatory system: blood always flows inside vessels

Components of blood ★★★

Component	Key facts
Plasma	fluid part (about 55%); carries food, CO_2, salts, hormones and nitrogenous wastes in dissolved form
RBCs	contain **haemoglobin**; carry O_2; no nucleus in humans
WBCs	fight germs (body defence); have a nucleus
Platelets	cell fragments that help in **clotting** of blood at a wound



MEME BREAK

*O_2 ko sabse zyada pyaar **haemoglobin** karta hai. Isliye RBC ke andar ye red pigment O_2 ko lungs se tissues tak le jaata hai.*

Clotting: sealing a wound ★★☆

- At a cut, **platelets** collect and start a chain of reactions
- A mesh of **fibrin** threads traps blood cells and forms a **clot**
- The clot plugs the leak, so blood loss and pressure drop are stopped

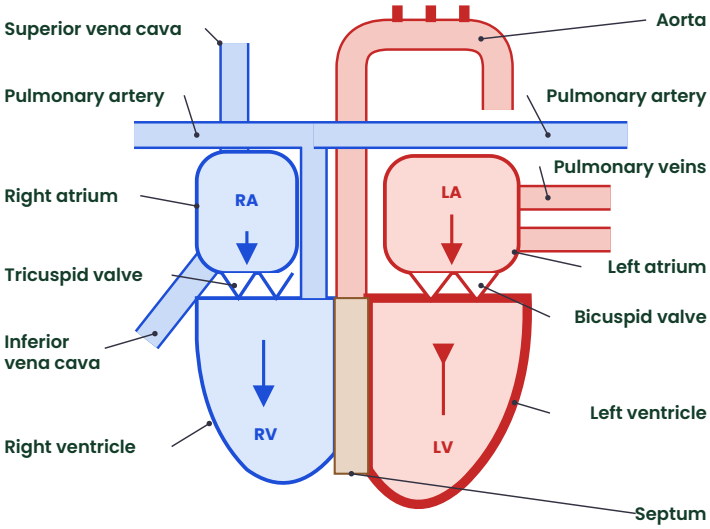
Removing the crust from your wound

Your blood platelets that have been working for days

MEME BREAK

*Wound pe jo crust banti hai na? Wo **platelets** ki mehnat hai: clot banake leak band. Usse khurachna mat!*

2 THE HUMAN HEART ★★★ MOST ASKED



Heart at a glance

- A muscular organ about the size of your **fist**, with **four chambers**: two atria on top, two ventricles below
- **Atria** receive blood (thin walls); **ventricles** pump it out (thick walls); the **left ventricle** is thickest as it pumps to the whole body
- **Valves** (tricuspid on the right, bicuspid on the left, semilunar at the arteries) stop **backflow**
- A wall called the **septum** keeps the two sides apart

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

At rest the heart beats about **72 times a minute**; each beat can be felt as the **pulse** at the wrist. The beat starts at a natural pacemaker in the wall of the right atrium (the SA node).

★★★ *Heart ka labelled diagram har board paper ka favourite hai: chambers, valves aur chaaron vessels. Nitesh Bhaiya ki guarantee!*

3 PATH OF BLOOD THROUGH THE HEART ★★★

Flow of blood (learn as a chain)

Body → vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary veins → left atrium → left ventricle → aorta → body

- Right side carries **deoxygenated** blood; left side carries **oxygenated** blood
- Both atria contract together; then both ventricles contract together

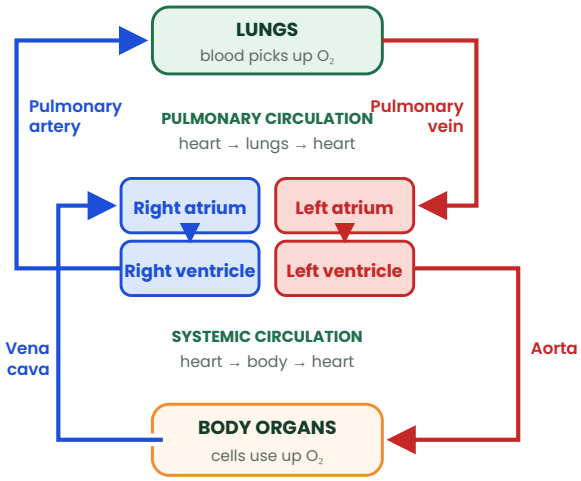
MNEMONIC: PATH OF BLOOD

"Vimal Ramu Ravi Pappu Lalu Pinku Lali Lata Aaye." Vena cava, R-atrium, R-ventricle, Pulmonary artery, Lungs, Pulmonary vein, L-atrium, L-ventricle, Aorta.

Four big vessels: quick revision ★★★

Vessel	From → To	Blood
Vena cava	body → right atrium	deoxygenated
Pulmonary artery	right ventricle → lungs	deoxygenated
Pulmonary veins	lungs → left atrium	oxygenated
Aorta	left ventricle → body	oxygenated

4 DOUBLE CIRCULATION ★★★ MOST ASKED



DEFINITION

- **Double circulation:** blood passes through the heart **twice** in one complete round of the body
- **Pulmonary circulation:** heart → lungs → heart; **systemic circulation:** heart → body → heart

EXAM TIPS

Heart diagram mein right side ko **blue** (deoxygenated) aur left side ko **red** (oxygenated) dikhao; valves aur septum label karna mat bhoolo. Viewer ka left = heart ka **right**!

Why four chambers? ★★★

- The septum keeps **oxygenated and deoxygenated** blood from mixing
- So the body gets a highly efficient supply of O_2: vital for **birds and mammals**, which need lots of energy to keep a constant body temperature

Animal	Heart	Circulation
Fish	2 chambers	single: heart → gills → body → heart
Amphibians, most reptiles	3 chambers	some mixing of blood; body temperature depends on surroundings
Birds, mammals	4 chambers	double, no mixing

BUJHO *Pulmonary artery mein toh oxygenated blood hoga, artery jo hai!*

PAHELI *Nahi Bujho!* ***Pulmonary artery*** *deoxygenated blood le jaati hai (heart → lungs), aur* ***pulmonary vein*** *oxygenated blood laati hai. Artery = heart se door, chahe blood koi bhi ho.*

5 BLOOD VESSELS ★★★

Arteries vs veins vs capillaries ★★★

Feature	Arteries	Veins	Capillaries
Carry blood	away from heart	to the heart	link arteries and veins
Blood	oxygenated (not the pulmonary artery)	deoxygenated (not the pulmonary vein)	exchange O_2, food, CO_2, wastes
Walls	thick, elastic	thin	one cell thick
Valves	absent	present	absent
Pressure	high	low	low

YAAD RAKHNE KI TRICK

"A for Away, V for Visit (heart)." Artery heart se door, Vein heart ko visit. Veins mein **valves** hote hain kyunki pressure kam hai.

Cross-sections of blood vessels

CLASS TRICK: VALVE NAMES

"Try Right, Be Left." **Tri**cuspid valve on the **right** side, **Bi**cuspid (mitral) valve on the **left**. Semilunar valves guard the aorta and pulmonary artery.

BUJHO *Capillaries itni patli kyun, bekaar nahi hai?*

PAHELI *Bilkul nahi! Capillary wall sirf* ***ek cell moti*** *hai, isliye O_2, food aur CO_2 ka exchange blood aur cells ke beech yahi hota hai.*

MEME BREAK
Cheers to the ***septum****! Isi deewar ki wajah se oxygenated aur deoxygenated blood kabhi mix nahi hote.*

6 BLOOD PRESSURE ★★★

Systolic and diastolic ★★★

- **Blood pressure:** the force that blood exerts on the walls of a vessel; much higher in arteries than in veins
- **Systolic** pressure: during ventricular contraction, normally about **120 mm Hg**
- **Diastolic** pressure: during ventricular relaxation, normally about **80 mm Hg**; normal BP is written **120/80**
- Measured with a **sphygmomanometer**
- **Hypertension** (high BP) is caused by constriction of **arterioles**, which raises resistance to blood flow; it can burst an artery and cause internal bleeding

BUJHO *120/80 mein 120 diastolic hai na?*

PAHELI *Ulta Bujho! Bada number* ***systolic*** *(ventricle contract) aur chhota number* ***diastolic*** *(ventricle relax). "****S = Squeeze = bada.****"*

7 LYMPH: THE SECOND TRANSPORT FLUID ★★☆

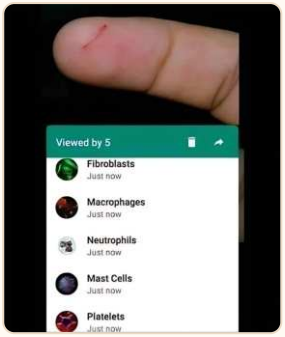
Lymph (tissue fluid) ★★☆

- Some plasma, proteins and blood cells escape through pores in capillary walls into the spaces between cells: this is **tissue fluid** or **lymph**
- Lymph is like plasma but **colourless** and has **less protein**
- It drains into **lymphatic capillaries** and then larger vessels that open into veins

Functions:

- ✔ carries **digested fat** absorbed in the intestine (lacteals)
- ✔ drains excess fluid from tissues back into the blood
- ✔ its lymphocytes help fight infection

MEME BREAK
Tissue fluid akela intezaar karta reh jaata agar ***lymph vessels*** *na hote. Yahi lymph use wapas veins tak le jaata hai.*



MEME BREAK
Ek chhota sa cut, aur "seen by 5": fibroblasts, macrophages, neutrophils, mast cells, platelets! ***Platelets*** *clot banate, WBCs germs se ladte.* ***Team blood****!*

COMMON MISTAKES

Writing that all arteries carry oxygenated blood (pulmonary artery does not). Mixing up systolic and diastolic. Calling lymph red: it is **colourless**. Forgetting valve names in the heart diagram.

NCERT in-text: quick answers

- **Components of the transport system:** heart (pumps), blood vessels (carry), blood (medium: plasma, RBCs, WBCs, platelets), lymph
- **Why separate oxygenated and deoxygenated blood in birds and mammals?** They need high energy to keep a constant body temperature, so an efficient O_2 supply is essential

MNEMONIC: FOUR PARTS OF BLOOD

"Paani, Rang, Wardi, Plaster." Plasma = paani (fluid), RBC = rang (red, O_2), WBC = wardi (soldier, defence), Platelets = plaster (seal the wound).

8 PRACTICE MCQS

1. Which blood component helps in clotting? ★★★
(a) RBCs (b) WBCs
(c) platelets (d) plasma

2. The chamber of the heart with the thickest muscular wall is the: ★★★
(a) right atrium (b) left atrium
(c) right ventricle (d) left ventricle

3. Oxygenated blood from the lungs enters the heart through the: ★★★
(a) pulmonary artery (b) pulmonary veins
(c) vena cava (d) aorta

4. Valves are present in: ★★☆
(a) arteries (b) veins
(c) capillaries (d) all vessels

5. Normal blood pressure of an adult is about: ★★★
(a) 80/120 mm Hg (b) 120/80 mm Hg
(c) 100/60 mm Hg (d) 140/100 mm Hg

6. The instrument used to measure blood pressure is: ★★☆
(a) sphygmomanometer (b) stethoscope
(c) barometer (d) thermometer

7. A fish heart has: ★★☆
(a) 2 chambers (b) 3 chambers
(c) 4 chambers (d) 1 chamber

8. Lymph differs from blood plasma as it: ★★☆

(a) is red (b) has more protein
(c) is colourless with less protein (d) has RBCs

9. **Assertion (A):** Birds and mammals have a four-chambered heart. **Reason (R):** Separation of oxygenated and deoxygenated blood gives them an efficient O_2 supply needed to maintain a constant body temperature. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** All arteries carry oxygenated blood. **Reason (R):** Arteries carry blood away from the heart to the organs. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Why is double circulation necessary in human beings? **(2 marks)**
2. Differentiate between arteries and veins (any three points). **(3 marks)**
3. Draw a labelled sectional view of the human heart and trace the path of blood from the body back to the body. **(5 marks)**

Case-based question ★★☆

> A doctor records Rohan's BP as 150/100 mm Hg. (i) Which value is systolic? (ii) Name the condition. (iii) Give one cause. **(1 + 1 + 1)**

> **Answers:** (i) 150 mm Hg (ii) hypertension (iii) constriction of arterioles raises resistance to flow

DIAGRAM PRACTICE

Aaj raat 5 minute ka timer lagao: heart ka sectional view aur double circulation flowchart bina dekhe banao, phir page 1–2 se milao.

Answer key: MCQs

Q	Ans	Why
1	(c)	Platelets start clotting.
2	(d)	LV pumps blood to the whole body.
3	(b)	Pulmonary veins bring blood from the lungs.
4	(b)	Low pressure; valves stop backflow.
5	(b)	Systolic/diastolic = 120/80.
6	(a)	Sphygmomanometer.
7	(a)	Fish: 2 chambers, single circulation.
8	(c)	Lymph is colourless, less protein.
9	(a)	R explains A.
10	(d)	Pulmonary artery carries deoxygenated blood.

Model answers

1 Double circulation

> Keeps oxygenated and deoxygenated blood separate, so tissues get a rich O_2 supply for high energy needs (2)

2 Arteries vs veins

> Use the table on page 3: direction, wall thickness, valves (1 each)

3 Heart

> Diagram with chambers, valves and four vessels (3); path chain vena cava → RA → RV → pulmonary artery → lungs → pulmonary veins → LA → LV → aorta (2)

Day 44 checklist: tick before you sleep

- ☐ I can list the four components of blood and their jobs
- ☐ I can draw and label the heart with valves and vessels
- ☐ I can trace the path of blood and explain double circulation
- ☐ I know arteries vs veins vs capillaries and BP 120/80
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

LEVEL 44 COMPLETE +140 XP earned. Next: Level 45

BUJHO *Samajh gaya! Right side blue, left side red, aur blood heart se do baar guzarta.*

PAHELI *Shabaash! Kal Level 45: plants mein transport aur excretion.*

NITESH SIR SAYS

Heart diagram roz ek baar banao, 7 din mein haath set ho jaayega. ***Kal milte hai Day 45 pe!***

Day 44 hit-list: Eduspark PYQ analysis

Topic	Rating
Heart diagram with labels	★★★
Double circulation, why 4 chambers	★★★
Arteries vs veins vs capillaries	★★★
Blood pressure 120/80, hypertension	★★☆
Components of blood, clotting	★★☆
Lymph and its functions	★☆☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 45 | LEVEL 45

BIOLOGY | CHAPTER 9 | PART 5 | ASTRA SERIES

PLANT TRANSPORT & EXCRETION

Goal: move water and food in plants and filter wastes in us.

by Nitesh Prakash

01 Xylem | 02 Transpiration | 03 Phloem | 04 Nephron | 05 Dialysis

+140 XP today | Level 45 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

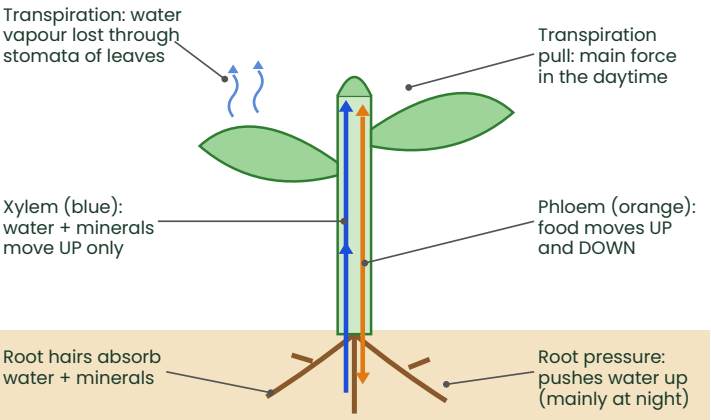
1 TRANSPORTATION IN PLANTS ★★★

Why do plants need a transport system?

1. Roots take **water and minerals** from the soil; leaves make **food**. Each must reach the other part.
2. Plants do not move and have many **dead cells** (e.g. wood), so they need **less energy** and can use a slow transport system.
3. Two conducting tissues do the job: **xylem** and **phloem** (together called vascular tissue).

DEFINITION

- **Xylem:** carries **water and minerals** from roots to leaves. Made of **vessels and tracheids** (dead, tube-like)
- **Phloem:** carries **food** (sucrose, amino acids) from leaves to other parts. Made of **sieve tubes and companion cells**



Root pressure: how water enters the xylem ★★☆

- Root cells in contact with soil **actively take up ions** (minerals)
- The ion concentration in the root becomes **higher** than in the soil
- So **water moves into the root** from the soil to remove this difference
- This makes a steady push of water up the xylem, called **root pressure**; it matters most **at night**

MNEMONIC: DAY VS NIGHT

"Din mein Dhoop, patte kheenchte hain = transpiration pull. Raat mein Root dhakelti hai = root pressure (R se R)."

2 TRANSPIRATION ★★★ MOST ASKED

DEFINITION

- **Transpiration:** the loss of water in the form of **vapour** from the aerial parts of a plant, mainly through the **stomata** of leaves

How transpiration pulls water up ★★★

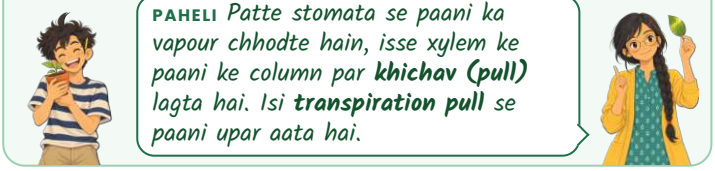
- Water evaporates from leaf cells and leaves through the stomata
- Leaf cells lose water, so they draw water from the **xylem of the leaf**
- This creates a **suction (pull)** on the unbroken water column in the xylem
- Water and minerals rise from roots to leaves: the **transpiration pull**

Two functions of transpiration ★★★

Function	How
Absorption and upward movement of water and minerals	the suction pulls the water column up the xylem
Temperature regulation	evaporation of water cools the leaf (like sweating)

BUJHO *Paani toh neeche girta hai, phir 50 metre ke ped mein upar kaise chadhta hai?*

PAHELI *Patte stomata se paani ka vapour chhodte hain, isse xylem ke paani ke column par **khichav (pull)** lagta hai. Isi **transpiration pull** se paani upar aata hai.*



MEME BREAK

*Transpiration pull sabko baantta hai: har patte tak **paani aur minerals**, xylem ke through!*

REMEMBER

Root pressure alone cannot lift water to the top of tall trees. The main force in the daytime is the **transpiration pull**.

3 TRANSLOCATION IN PHLOEM ★★★

DEFINITION

- **Translocation:** transport of soluble products of photosynthesis (and amino acids) through the **phloem**
- Food goes to storage organs (roots, fruits, seeds) and to growing parts; it can move **upward and downward**

How translocation works: uses energy ★★★

- **Sucrose** is loaded into the sieve tubes using energy from **ATP**
- This raises the **osmotic pressure** of the phloem tissue
- **Water enters** the sieve tubes, so the pressure rises further
- The pressure moves the food to tissues with **lower pressure**, where it is used or stored

EXAMPLE

In **spring**, sugar stored in the root or stem is sent **up** to the buds, which need energy to grow. Phloem moves food as the plant needs.

MEME BREAK *Xylem bas transpiration ke bharose; phloem ATP laga ke dono taraf maal bhejta hai!*

Xylem vs phloem ★★★

Point	Xylem	Phloem
Carries	water and minerals	food (sucrose, amino acids)
Direction	one way: **upward**	**both ways**, up and down
Cells	vessels, tracheids (mostly dead)	sieve tubes, companion cells (living)
Energy	no ATP of its own; transpiration pull, root pressure	**uses ATP**
Process	conduction of water	translocation

MNEMONIC: X FOR H_2O, P FOR PIZZA

"Xylem = paani (X jaise paani ka nal, sirf upar). Phloem = Pizza, yaani khana: upar bhi, neeche bhi."

THINK OF IT THIS WAY

Xylem is a one-way **water pipe** run by suction; phloem is a two-way **delivery service** that pays with ATP.

★★★ *Transpiration ke 2 functions aur transpiration pull, ye har saal aata hai, Nitesh Bhaiya ki guarantee!*

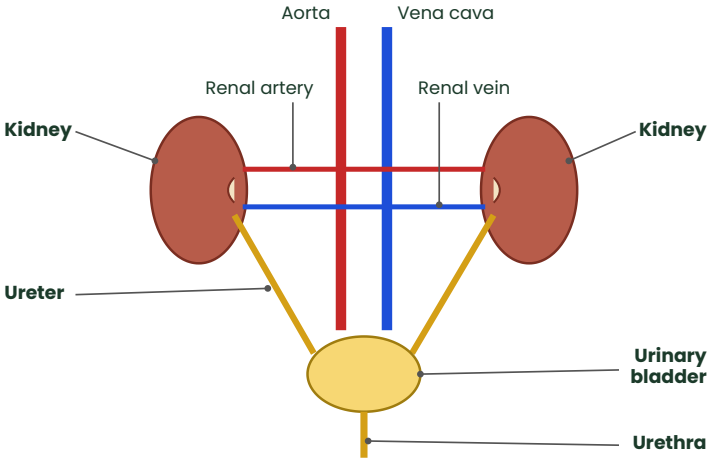
4 EXCRETION IN HUMAN BEINGS ★★★ MOST ASKED

DEFINITION

- **Excretion:** the biological process of removing **harmful nitrogenous metabolic wastes** (like urea, uric acid) from the body
- Unicellular organisms remove wastes by simple **diffusion** from the body surface into the surrounding water

Human excretory system: labelled diagram ★★★

A pair of **kidneys**, a pair of **ureters**, a **urinary bladder** and a **urethra**.



Parts and their jobs ★★★

Part	Job
Kidneys (in the abdomen, one on each side of the backbone)	filter blood and make urine
Ureters	carry urine from the kidneys to the bladder
Urinary bladder	muscular bag that stores urine; it is under nervous control
Urethra	passes urine out of the body

MNEMONIC: URINE KA ROUTE

"Kabhi Ulta Bolna Uchit nahi" = Kidney → Ureter → Bladder → Urethra.

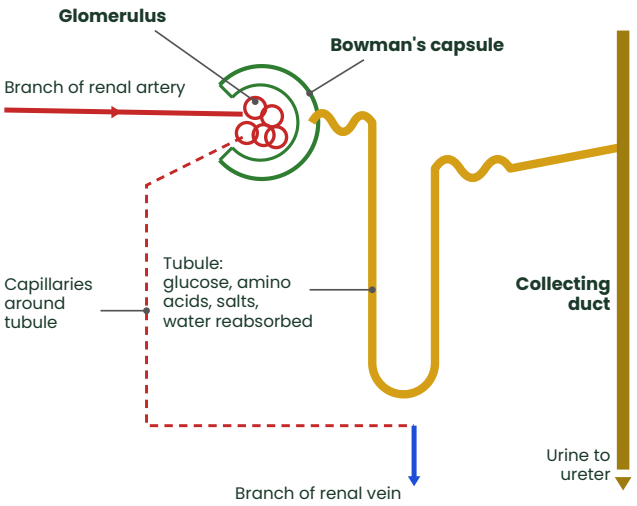
★★★ *Excretory system ka labelled diagram aur urine ka route, board ka favourite hai, Nitesh Bhaiya ki guarantee!*

× WRONG	✓ RIGHT
Phloem carries water up	Xylem carries water; phloem carries food
Urea is made in the kidney	Urea is made in the liver, filtered by the kidney
Urine is stored in the kidney	Stored in the urinary bladder
Dialysis reabsorbs glucose	No reabsorption in an artificial kidney

5 NEPHRON: THE FILTRATION UNIT ★★★ MOST ASKED

DEFINITION

- **Nephron:** the basic **filtration unit** of the kidney. Each kidney has a very large number of nephrons packed close together
- Parts: **Bowman's capsule** (a cup) with a **glomerulus** (a cluster of thin-walled capillaries) inside, a long **tubule** and a **collecting duct**



How urine is formed ★★★

1. **Filtration:** the renal artery brings blood to the glomerulus; water, glucose, amino acids, salts and urea are filtered into **Bowman's capsule**.
2. **Selective reabsorption:** in the tubule, useful substances (glucose, amino acids, salts) and a major part of the water go **back into the blood** of the capillaries.
3. The amount of water reabsorbed depends on how much **extra water** the body has and how much waste must be removed.
4. **Collecting duct:** the remaining fluid is **urine**; it goes to the ureter and is stored in the bladder.

MEME BREAK

*Roz ~180 L filtrate, par bahar sirf **1 se 2 L urine**. Baaki sab tubule mein **reabsorb**: kidney ki smart investment!*

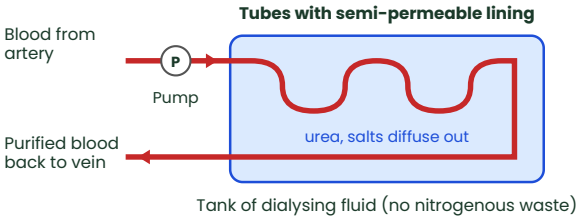
BUJHO *Glomerulus mein toh glucose bhi filter ho jaata hai, phir urine mein glucose kyun nahi aata?*

PAHELI *Kyunki **tubule** glucose, amino acids aur zyada paani ko wapas blood mein **reabsorb** kar leta hai. Healthy insaan ke urine mein glucose nahi hota.*

6 ARTIFICIAL KIDNEY (HAEMODIALYSIS) ★★☆

When the kidneys fail

- Infection, injury or less blood flow can make the kidneys stop working; poisonous wastes then **build up** in the body
- An **artificial kidney** removes nitrogenous wastes from the blood by **dialysis**



How it works ★★☆

1. Long tubes with a **semi-permeable lining** sit in a tank of **dialysing fluid**.
2. The fluid has the **same osmotic pressure as blood** but has **no nitrogenous wastes**.
3. The patient's blood is passed through the tubes; wastes **diffuse** out into the fluid.
4. The purified blood is pumped back into the patient.

KEY DIFFERENCE

A natural kidney **reabsorbs** useful substances; an artificial kidney has **no reabsorption**.

MNEMONIC: URINE FORMATION

"Filter, Reabsorb, Urine" = FRU: pehle chhaano, phir kaam ka wapas lo, bacha urine.

7 EXCRETION IN PLANTS ★★★

How plants get rid of wastes ★★★

- ✔ **Oxygen** (from photosynthesis) and CO_2 (from respiration) leave by diffusion through stomata
- ✔ **Excess water** is lost by transpiration
- ✔ Many wastes are stored in **cell vacuoles** and in **leaves that fall off**
- ✔ Some wastes are stored as **resins and gums**, mostly in **old xylem**
- ✔ Some wastes are released into the **surrounding soil**

MEME BREAK

*Purane patte waste lekar gir jaate hain: plant ka **easy excretion** trick!*

8 PRACTICE MCQS

1. Water and minerals in plants move through: ★★★
(a) phloem (b) xylem
(c) cortex (d) epidermis

2. The main force that lifts water in tall trees during the day is: ★★★
(a) root pressure (b) transpiration pull
(c) osmosis in leaves (d) gravity

3. Translocation of food in plants needs: ★★★
(a) no energy (b) energy from ATP
(c) sunlight only (d) root pressure

4. The basic filtration unit of the kidney is: ★★★
(a) neuron (b) nephron
(c) ureter (d) alveolus

5. The capillary cluster inside Bowman's capsule is the: ★★★
(a) glomerulus (b) tubule
(c) collecting duct (d) villus

6. Which is NOT normally present in the urine of a healthy person? ★★☆
(a) water (b) urea
(c) salts (d) glucose

7. Urine is stored in the: ★★☆
(a) kidney (b) ureter
(c) urinary bladder (d) urethra

8. Plants store some waste products as resins and gums in: ★★☆
(a) old xylem (b) young leaves
(c) root hairs (d) phloem sap

9. **Assertion (A):** Transpiration helps in the upward movement of water in plants.
Reason (R): Loss of water vapour from leaves creates a suction that pulls water up the xylem. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** An artificial kidney reabsorbs glucose and amino acids into the blood.
Reason (R): The dialysing fluid has the same osmotic pressure as blood but no nitrogenous wastes. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. How are water and minerals transported in plants? Give two functions of transpiration. **(3 marks)**
2. Draw a labelled diagram of the human excretory system. Which organ stores urine? **(3 marks)**
3. Draw the structure of a nephron and label Bowman's capsule, glomerulus, tubule and collecting duct. How is urine formed? **(5 marks)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Xylem conducts water and minerals.
2	(b)	Transpiration pull in the daytime.
3	(b)	Sucrose is loaded using ATP.
4	(b)	Nephron = filtration unit.
5	(a)	Glomerulus sits in the capsule.
6	(d)	Glucose is fully reabsorbed.
7	(c)	The bladder stores urine.
8	(a)	Resins and gums in old xylem.
9	(a)	R explains A.
10	(d)	A false: no reabsorption. R true.

Model answer points

- **Q1:** root ions draw water in; xylem column pulled by transpiration; functions: water and mineral uptake, temperature control
- **Q2:** kidneys, ureters, urinary bladder, urethra (plus aorta, vena cava, renal artery and vein); urine is stored in the urinary bladder
- **Q3:** filtration in glomerulus into Bowman's capsule; reabsorption in tubule; urine in collecting duct

Day 45 checklist: tick before you sleep

- ☐ I can explain root pressure and transpiration pull
- ☐ I can draw and compare xylem and phloem transport
- ☐ I can draw the human excretory system and a nephron
- ☐ I can explain urine formation and haemodialysis
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2–3 phir se padho.

LEVEL 45 COMPLETE +140 XP earned. Next: Level 46

BUJHO *Samajh gaya! Xylem paani upar, phloem khana dono taraf, aur nephron chhaan ke wapas leta hai.*

PAHELI *Shabaash! Kal Level 46: Chapter 9 ka* ***Boss Level.***

NITESH SIR SAYS
Nephron ka diagram 5 baar bana lo, labels ke saath. Exam mein ***diagram = pakke marks.*** *Kal milte hai* ***Day 46 pe!***

DAY 46
LEVEL 46

BIOLOGY | CHAPTER 9 | BOSS LEVEL

CHAPTER 9 BOSS LEVEL

Goal: test all of Life Processes in 90 minutes.

by Nitesh Prakash

01 Revision Map | 02 MCQs | 03 Short Answers | 04 Case & Long | 05 Answer Key | +200 XP Boss Level | Level 46 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 9 REVISION MAP ★★★ MOST ASKED

Chapter 9 mind map: revise in 5 minutes

NUTRITION
$6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$ (sunlight, chlorophyll); holozoic, saprotrophic, parasitic

RESPIRATION ★★★
Glucose → pyruvate (cytoplasm); aerobic in mitochondria: $CO_2 + H_2O$ + ATP; yeast: ethanol; muscle: lactic acid

LIFE PROCESSES

TRANSPORT IN HUMANS
4-chambered heart, double circulation, arteries, veins, capillaries, lymph; BP 120/80

TRANSPORT IN PLANTS
Xylem: water up (transpiration pull); phloem: food both ways (uses ATP)

EXCRETION ★★★
Kidney → ureter → bladder → urethra; nephron: filtration in glomerulus, reabsorption in tubule; dialysis; plants: leaves, resins, gums, soil

Must-know: digestive juices ★★★

Juice / enzyme	From	Acts on
Salivary amylase	salivary glands	starch → sugar
Pepsin (with HCl)	gastric glands	proteins
Bile (no enzyme)	liver	emulsifies fats
Trypsin	pancreas	proteins
Lipase	pancreas	fats → fatty acids + glycerol

MNEMONIC: 4 LIFE PROCESSES
"Life ki kahani N-R-T-E se suno" = Nutrition, Respiration, Transportation, Excretion.

Chapter 9 ke notes sirf ek baar padh liye

Heart, nephron aur stomata ke diagrams label karke Boss Level test diya

BOSS INTRO
*Notes ek baar padh ke so gaye? Aaj nahi! Aaj **diagrams label** karke Boss harana hai.*

2 SECTION A: MCQS ★★★ MOST ASKED

1. The site of photosynthesis in a plant cell is: ★★★
(a) mitochondria (b) chloroplast
(c) nucleus (d) ribosome

2. Opening and closing of stomata is controlled by: ★★★
(a) guard cells (b) xylem
(c) root hairs (d) phloem

3. The enzyme that digests proteins in the stomach is: ★★★
(a) amylase (b) lipase
(c) pepsin (d) trypsin

4. Bile is secreted by the liver and stored in: ★★☆
(a) pancreas (b) gall bladder
(c) stomach (d) small intestine

5. Anaerobic respiration in yeast gives: ★★★
(a) lactic acid + energy (b) $CO_2 + H_2O$ + energy
(c) ethanol + CO_2 + energy (d) pyruvate only

6. Glucose is broken down into pyruvate in: ★★★
(a) cytoplasm (b) mitochondria
(c) chloroplast (d) nucleus

7. Oxygenated blood from the lungs reaches the heart through: ★★★
(a) pulmonary artery (b) pulmonary vein
(c) aorta (d) vena cava

8. The heart of a fish has: ★★☆
(a) two chambers (b) three chambers
(c) four chambers (d) one chamber

9. In human lungs, the exchange of gases takes place in: ★★★
(a) bronchi (b) trachea
(c) alveoli (d) larynx

How to attack the Boss: 90 minutes

- Section A (MCQs): 15 min; read every A/R option slowly
- Sections B and C: 35 min; draw diagrams in pencil
- Sections D and E: 30 min; label every part you draw
- Last 10 min: check with the key, then score yourself

★★★ *A/R aur diagram-based MCQs Chapter 9 se har paper mein aate hain, Nitesh Bhaiya ki guarantee!*

10. Excess water in plants is removed mainly by: ★★☆

(a) translocation (b) transpiration
(c) photosynthesis (d) root pressure

11. Lymph carries digested and absorbed fat from the: ★★☆

(a) stomach (b) small intestine
(c) liver (d) kidney

12. The correct path of urine is: ★★★

(a) kidney → urethra → bladder → ureter (b) kidney → ureter → bladder → urethra
(c) kidney → bladder → ureter → urethra (d) ureter → kidney → bladder → urethra

13. **Assertion (A):** The ventricles of the human heart have thicker muscular walls than the atria.
Reason (R): Ventricles have to pump blood into various organs of the body. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

14. **Assertion (A):** Aquatic animals breathe much faster than terrestrial animals.
Reason (R): Water holds much less dissolved oxygen than air. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

15. **Assertion (A):** Villi increase the surface area for the absorption of digested food.
Reason (R): Villi are found in the inner lining of the large intestine. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

MNEMONIC: BLOOD PATH IN THE HEART

"Right = Rukha (O_2 kam), Left = Laal (O_2 zyada)."
Body → vena cava → RA → RV → lungs → LA → LV → aorta → body.

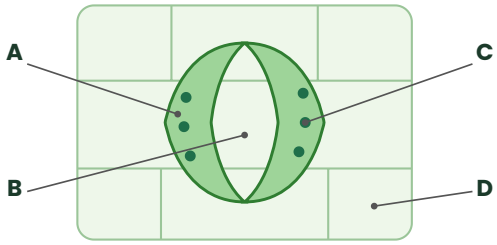
3 SECTIONS B AND C ★★★ MOST ASKED

Section B: Short Answers (2 marks each)

1. Why is diffusion not enough to meet the oxygen needs of multicellular organisms like humans?
2. Name the products of glucose breakdown in (i) yeast, (ii) muscles during heavy exercise. Where does glycolysis occur?
3. State two functions of hydrochloric acid in the stomach.
4. Identify A, B, C and D in the diagram of a stoma below. What makes the pore open?

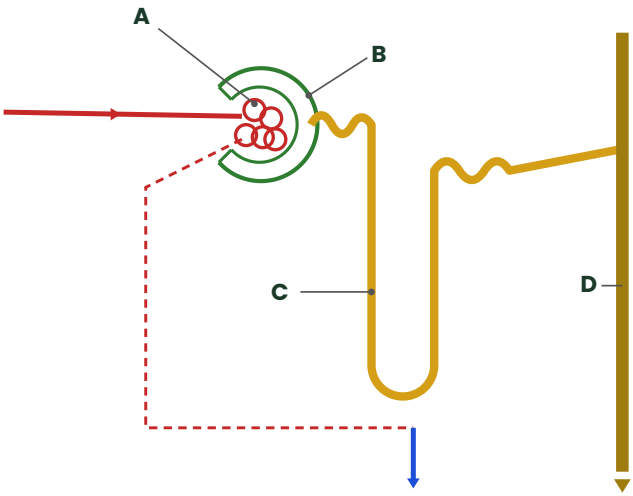
TIP

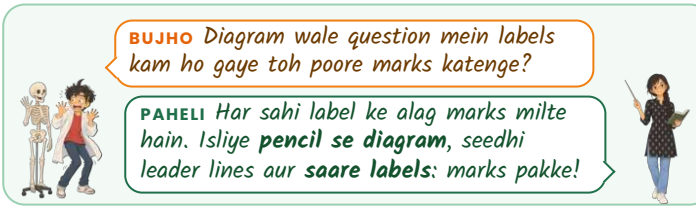
2 marks = **two clear points**; 3 marks = three points or a diagram.



Section C: Short Answers (3 marks each)

1. What is double circulation? Why must oxygenated and deoxygenated blood be kept apart in mammals and birds?
2. What is translocation? How does it take place in plants, and why does it need energy?
3. Write the role of (a) salivary amylase, (b) bile, (c) villi in digestion.
4. Identify parts A, B, C and D of the nephron below and write the function of C.





Bonus rapid-fire round: quick answers ★★☆

Question	Answer
Energy currency of the cell	ATP
Normal blood pressure	120/80 mm Hg
Valves in the heart stop	backflow of blood
Chambers: fish / frog	2 / 3
Lacteals in villi absorb	fat
Filtrate made by kidneys per day	about 180 L
Protein-digesting enzyme of pancreas	trypsin

MCQ TIP

In A/R questions, first decide if **A** is true, then **R**, and only then ask: does R **explain** A?

4 SECTIONS D AND E ★★★ MOST ASKED

Section D, Case 1: the variegated leaf (4 marks)

Riya kept a potted plant with variegated (green and white) leaves in the dark for three days. Then she kept it in sunlight for six hours, plucked a leaf, boiled it in alcohol on a water bath, washed it and added iodine solution.

1. Why was the plant kept in the dark for three days? **(1)**
2. Why was the leaf boiled in alcohol? **(1)**
3. Which part of the leaf turned blue-black, and what does the experiment prove? **(2)**

Case 2: the blood pressure check (4 marks)

During a health camp, Mr Rao's blood pressure was 150/100 mm Hg. The doctor said the normal value is about 120/80 mm Hg and asked him to cut salt and walk daily.

1. Name the instrument used to measure blood pressure. **(1)**
2. What do 120 and 80 stand for? **(1)**
3. Name Mr Rao's condition. What causes it, and why is it harmful? **(2)**

Case 3: the limewater test (4 marks)

Aman took two test tubes of fresh limewater. Into tube X he pumped ordinary air with a syringe; into tube Y he blew his exhaled breath through a straw.

1. Which tube turns milky first, and why? **(1)**
2. Name the gas that turns limewater milky. **(1)**
3. Give two differences between aerobic and anaerobic respiration. **(2)**

MEME BREAK
*Might-ochondria? Nahi bhai, **mitochondria** hi powerhouse hai: aerobic respiration ka ATP wahin banta hai!*

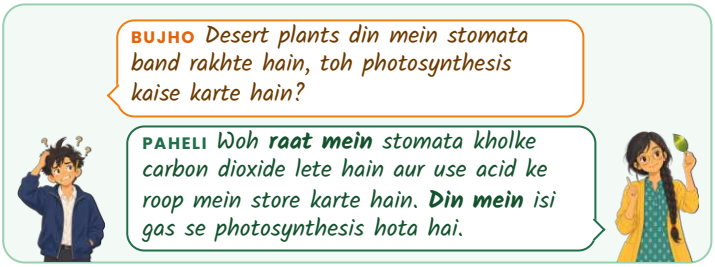
× WRONG	✓ RIGHT
Bile is an enzyme	Bile has no enzyme; it emulsifies fats
Pulmonary vein carries deoxygenated blood	It carries oxygenated blood to the heart
Lactic acid is made in yeast	Yeast makes ethanol; muscles make lactic acid
Phloem moves food only downward	Phloem moves food both ways

Section E: Long Answers (5 marks each)

1. Draw a labelled diagram of the human alimentary canal. Describe the digestion of food in the small intestine.
   OR
   Draw a labelled diagram of the human respiratory system. Explain how air is breathed in and out, and why the lungs always hold some air.
2. Draw a sectional view of the human heart and label: right atrium, left ventricle, aorta, pulmonary artery, vena cava. Trace the path of blood through it.
   OR
   Draw the structure of a nephron. Explain how urine is formed. How does an artificial kidney work?
3. (a) What are the raw materials for photosynthesis? Write the three events of photosynthesis. (b) How do desert plants take up CO_2 without losing much water?

Common mistakes that cost marks

- ✔ Writing "oxygen is a raw material" for photosynthesis; it is a **product**
- ✔ Drawing arrows the wrong way in the heart: blood enters **atria**, leaves from **ventricles**
- ✔ Forgetting **"selective"** in selective reabsorption
- ✔ Leaving labels off a diagram: each label carries marks



★★★ ***Heart, nephron aur digestive system ke diagrams: ek na ek Boss paper mein pakka, Nitesh Bhaiya ki guarantee!***

Most asked in Chapter 9: Eduspark PYQ analysis

Topic	Rating
Human digestive system (diagram), enzymes	★★★
Heart, double circulation (diagram)	★★★
Nephron and urine formation (diagram)	★★★
Breakdown of glucose pathways	★★★
Photosynthesis, stomata	★★☆
Transpiration, translocation	★★☆
Artificial kidney, plant excretion	★☆☆

Diagram drill: draw these today

- ✔ Human alimentary canal with glands
- ✔ Human respiratory system with alveoli
- ✔ Sectional view of the heart and double circulation
- ✔ Human excretory system and a nephron

5 ANSWER KEY ★★★ MOST ASKED

Answer key: Section A (MCQs)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	(b)	2	(a)	3	(c)	4	(b)	5	(c)
6	(a)	7	(b)	8	(a)	9	(c)	10	(b)
11	(b)	12	(b)	13	(a)	14	(a)	15	(c)

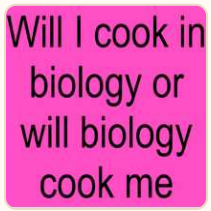
Why: Q10 transpiration removes extra water; Q11 lymph carries fat from the intestine; Q13–14 both true, R explains A; Q15 A true, R false (villi are in the **small** intestine).

Model answers: Sections B and C

- **B1:** large body volume; diffusion is too slow to reach cells deep inside; need a transport system
- **B2:** (i) ethanol + CO_2; (ii) lactic acid; glycolysis in the cytoplasm
- **B3:** makes the medium acidic for pepsin; kills germs in food
- **B4:** A guard cell, B stomatal pore, C chloroplast, D epidermal cell; water entering guard cells makes them swell and open the pore
- **C1:** blood passes twice through the heart in one cycle; separation gives efficient O_2 supply for high energy needs (warm-blooded)
- **C2:** transport of food in phloem; sucrose loaded with ATP, osmotic pressure rises, water enters, food moves to low-pressure areas
- **C3:** (a) starch → sugar; (b) emulsifies fats, makes medium alkaline; (c) increase surface area for absorption
- **C4:** A glomerulus, B Bowman's capsule, C tubule, D collecting duct; C reabsorbs glucose, amino acids, salts, water

Model answers: Sections D and E

- **Case 1:** to remove (destarch) stored starch; alcohol removes chlorophyll; only the **green** part turns blue-black, so chlorophyll is needed for photosynthesis
- **Case 3:** tube Y, exhaled air has more CO_2; carbon dioxide; aerobic uses O_2, gives CO_2 + H_2O and more energy, in mitochondria; anaerobic without O_2, less energy
- **Case 2:** sphygmomanometer; 120 systolic (contraction), 80 diastolic (relaxation); hypertension, caused by constriction of arterioles; may rupture an artery, internal bleeding
- **E1:** small intestine gets bile (emulsifies fats) and pancreatic juice (trypsin, lipase, amylase); intestinal juice completes digestion; villi absorb. OR: diaphragm and rib muscles enlarge the chest, air rushes in; residual volume left for gas exchange
- **E2:** body → vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta → body. OR: filtration, selective reabsorption, urine; dialysis by diffusion, no reabsorption
- **E3:** CO_2, water, minerals, sunlight, chlorophyll; light absorbed, water split into hydrogen and oxygen, CO_2 reduced to carbohydrate; desert plants take CO_2 at night



MEME BREAK
*Will I cook in biology? Agar aaj ka Boss **50+ / 62** hai, toh haan bhai, tum **cook** kar rahe ho!*

Score card: how did you fight the boss?

	A	B	C	D	E	Total
Marks	15	8	12	12	15	**62**
Mine	__	__	__	__	__	____

Day 46 checklist: tick before you sleep

- ☐ I revised the whole Chapter 9 mind map
- ☐ I solved all 15 MCQs and checked them with the key
- ☐ I drew and labelled the stoma and nephron diagrams
- ☐ I wrote all three case-based answers in full
- ☐ I scored myself and marked my weak topics

My score: ____ / 5

Total 62 marks. 50+ = Boss defeated! Kam aaye? Day 41–45 dobara.

Boss Level tracker: write your percentage

Boss Level	Day	My %
Chapter 9: Life Processes	46	_____
Chapter 10: Control and Coordination	50	_____
Chapter 11: Reproduction	54	_____
Chapter 12: Heredity	57	_____
Chapter 13: Our Environment	60	_____

LEVEL 46 COMPLETE +200 XP earned. Next: Level 47

BUJHO *Boss khatam! Nutrition se excretion tak, poora Chapter 9 clear.*

PAHELI *Shabaash! Kal Level 47: **Control and Coordination**.*

Coming up: Day 47

Control and Coordination begins: **neurons, synapse, reflex arc** and the human brain. Keep your nephron and heart diagrams handy for weekly revision.

NITESH SIR SAYS
*Life Processes board mein sabse zyada marks wala chapter hai. Diagrams roz ek baar banao. **Kal milte hai Day 47 pe!***

BIOLOGY | CHAPTER 9 COMPLETE | LEVEL 46 ASTRA SERIES

CHAPTER 9 SWAHAA!

Life Processes | Days 41–46

BUJHO *Photosynthesis, digestion, respiration, heart aur nephron! Chapter 9 ke saare boss khatam, Swahaa!*

PAHELI *Shabash Bujho! Ab body kaise control hoti hai, woh dekhenge. Brain ready rakhna!*

CHAPTER XP EARNED
+900 XP

LEVELS CLEARED
6 levels (Days 41–46)

BOOK PROGRESS
Level 46 / 60 5900 XP total

WHAT YOU HAVE MASTERED

- ✔ Photosynthesis and autotrophic nutrition
- ✔ Aerobic and anaerobic respiration
- ✔ Xylem, phloem and transpiration
- ✔ Human digestive system and enzymes
- ✔ Heart, blood and double circulation
- ✔ Nephron and excretion

NEXT CHAPTER Day 47: Control and Coordination

Chapter 9 cleared!

DAY 47
LEVEL 47

BIOLOGY | CHAPTER 10 | PART 1

NERVOUS SYSTEM & THE HUMAN BRAIN

Goal: neurons, reflexes and the brain.

by Nitesh Prakash

01 Receptors | 02 Neuron | 03 Reflex Arc | 04 Human Brain | 05 Limitations | +125 XP today | Level 47 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHY DO WE NEED COORDINATION? ★★★ MOST ASKED

DEFINITION

- **Stimulus:** any change in the surroundings (light, heat, touch, sound, smell) to which an organism responds
- **Coordination:** the working together of different organs in a controlled, systematic way so that the body gives a **proper response** to a stimulus
- Movement in response to a stimulus is controlled; animals use two systems: the **nervous system** and the **endocrine (hormonal) system**

Receptors: the sensors of the body ★★☆

Receptor	Where	Detects
Gustatory	tongue	taste
Olfactory	nose	smell
Photoreceptors	eye	light
Phonoreceptors	inner ear	sound, balance
Thermoreceptors	skin	heat and cold
Touch receptors	skin	touch, pressure, pain

RECEPTOR KYA HAI?

Receptors are **specialised tips of nerve cells** in our sense organs. They detect information from the environment and start the nerve impulse.

BUJHO *Nervous system aur coordination, ek hi cheez hai kya?*

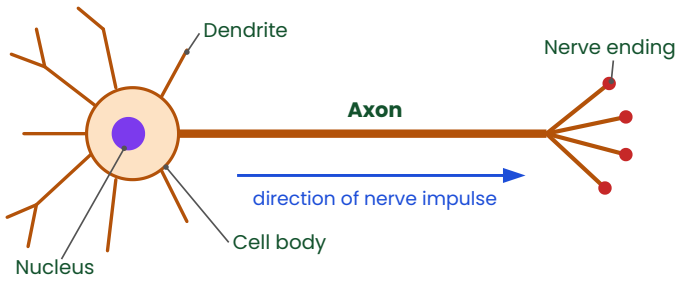
PAHELI *Nahi Bujho! Coordination **kaam** hai, aur nervous system uska ek **tool**. Animals mein doosra tool hai **hormones**, plants mein sirf hormones.*

MEME BREAK

*Ek neuron ne signal pakda aur poori chain **activate**! Dendrite → cell body → axon → nerve ending, yahi order.*

2 NEURON: THE UNIT OF THE NERVOUS SYSTEM ★★★

A **neuron (nerve cell)** is the structural and functional unit of the nervous system. It is the **longest cell** of the human body and carries messages as **electrical impulses**.

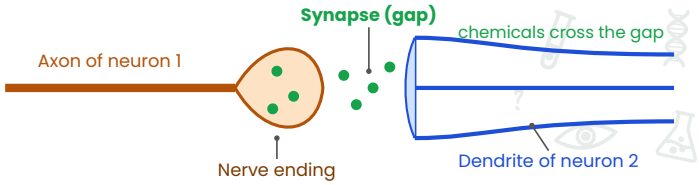


Parts and their jobs ★★★

Part	Job
Dendrite	picks up information from receptors or another neuron
Cell body	has the nucleus; the impulse passes through it
Axon	long fibre that carries the impulse away from the cell body
Nerve ending	releases chemicals that pass the message on

How a message travels (NCERT) ★★★

1. Information is picked up at the **dendrite tip**, which sets off a **chemical reaction** creating an **electrical impulse**.
2. The impulse travels: **dendrite → cell body → axon → nerve ending**.
3. At the nerve ending the electrical impulse makes the neuron release **chemicals**; they cross the gap (**synapse**) and start a new impulse in the dendrite of the next neuron.
4. A similar synapse passes the impulse from a neuron to a **muscle cell or gland**.



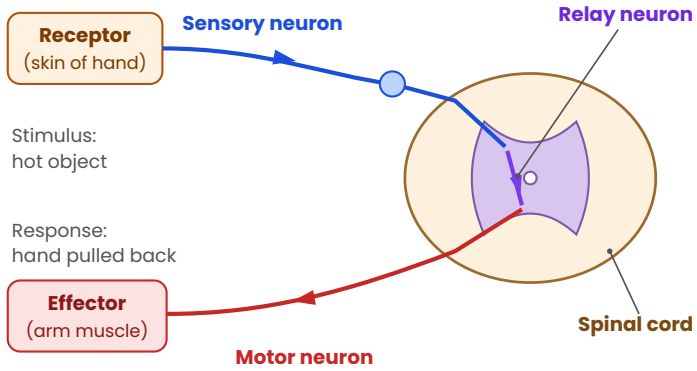
3 REFLEX ACTION AND REFLEX ARC

★★★ MOST ASKED

DEFINITION

- **Reflex action:** a sudden, **automatic** and involuntary response to a stimulus, done **without thinking** (without the conscious control of the brain)
- **Reflex arc:** the pathway along which the nerve impulse travels during a reflex action

Examples: pulling the hand back from a hot object, blinking when an object comes near the eye, sneezing, coughing, the knee-jerk, the pupil shrinking in bright light.



Why reflex arcs are in the spinal cord ★★★

- ✔ Thinking in the brain is **too slow** for an emergency; the spinal cord gives a **quick** response.
- ✔ Input nerves meet output nerves in a bundle in the spinal cord, so the reply is sent straight back.
- ✔ The information **also reaches the brain**, which becomes aware of the event a moment later.
- ✔ Reflex arcs evolved early; they work well even in animals with little or no thinking power.

MNEMONIC: REFLEX ARC IN ORDER

"Ram Se Raavan Mara Ekdum"
Receptor → Sensory neuron → Relay neuron (spinal cord) → Motor neuron → Effector

MEME BREAK
*Reflex ka seedha rasta: receptor se spinal cord, wahan se muscle. Dimaag ko **news baad mein** milti hai.*

★★★ *Neuron ka diagram aur synapse ka kaam: 2–3 marks pakke, Nitesh Bhaiya ki guarantee!*

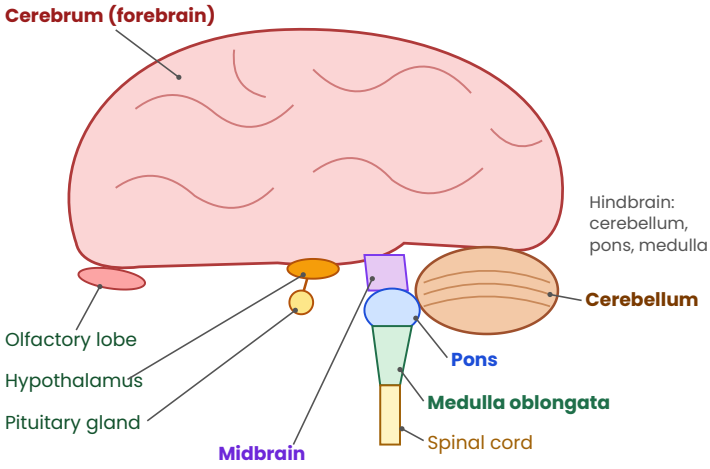
Voluntary vs involuntary vs reflex ★★☆

Action	Control	Examples
Voluntary	by our will; forebrain (cerebrum)	writing, talking, walking
Involuntary	automatic; midbrain and hindbrain (medulla)	heartbeat, breathing, salivation
Reflex	sudden and very fast; mostly the spinal cord	blinking, hand off a hot plate

MEME BREAK
*Haath garam tawe se hata, dimaag ko pata baad mein chala. Ye kaam **spinal cord** ka hai, aliens ka nahi!*

4 HUMAN BRAIN ★★★ MOST ASKED

The **brain** and the **spinal cord** form the **central nervous system (CNS)**. The brain is the main coordinating centre; it has three regions: **forebrain, midbrain** and **hindbrain**.



★★★ *Brain ka labelled diagram aur parts ke kaam, ye Nitesh Bhaiya ki guarantee wala question hai!*

Brain + spinal cord + nerves

- **CNS** = brain + spinal cord: they take decisions and send orders.
- **PNS (peripheral nervous system)** = **cranial nerves** (from the brain) + **spinal nerves** (from the spinal cord); they link the CNS with all body parts.
- **Spinal cord:** carries messages to and from the brain and handles most reflex actions.

★★★ *Reflex arc ka labelled diagram har saal pucha jata hai, Nitesh Bhaiya ki guarantee!*

Parts of the brain and their functions ★★★

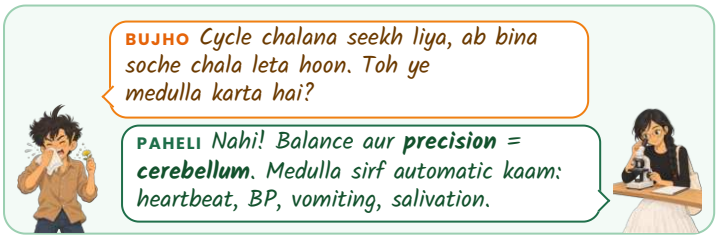
Region	Part	Main functions
Forebrain	Cerebrum	thinking, memory, intelligence; voluntary actions; areas for **hearing, smell, sight**
	Olfactory lobe	sense of smell
	Hypothalamus	hunger, thirst, body temperature, sleep; controls the **pituitary gland**
Midbrain		reflex movements of the head, neck and eyes
Hindbrain	Cerebellum	precision of voluntary actions; **posture and balance** (walking straight, riding a cycle, picking up a pencil)
	Pons	regulates breathing
	Medulla	involuntary actions: **blood pressure, heartbeat, salivation, vomiting**, swallowing

MNEMONIC: HINDBRAIN TEAM

Cycle, Pranayam, Machine
Cerebellum = cycle balance, Pons = breathing,
Medulla = automatic machine (BP, heartbeat, vomiting)

How the brain and spinal cord are protected ★★☆

- ✔ The brain sits inside a bony box, the **cranium (skull)**.
- ✔ Inside the box it floats in a fluid-filled balloon, the **cerebrospinal fluid (CSF)**, which acts as a **shock absorber**.
- ✔ The spinal cord is protected by the **vertebral column** (backbone).



EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Many axons are wrapped in a fatty **myelin sheath** that makes the impulse travel faster. A common chemical released at a synapse is **acetylcholine**.
Humans have **12 pairs** of cranial nerves and **31 pairs** of spinal nerves.

× WRONG	✔ RIGHT
Reflexes are controlled by the brain	Mostly by the spinal cord; the brain is informed later
Cerebellum controls heartbeat	Medulla controls heartbeat and BP; cerebellum gives balance
Impulse flows axon → dendrite in a neuron	Dendrite → cell body → axon → nerve ending

5 HOW NERVOUS TISSUE CAUSES ACTION ★★☆

1. The nerve impulse reaches a **muscle** through a nerve ending (synapse).
2. Muscle cells have special **proteins** that change their **shape and arrangement** in response to the impulse.
3. This makes the muscle cell **shorter (contract)**, and the body part moves.

Voluntary muscles (arms, legs) move by our will; **involuntary muscles** (heart, gut) work on their own.

6 LIMITATIONS OF NERVE IMPULSES ★★☆

- ✔ They reach only those cells that are **connected by nervous tissue**, not every cell of the body.
- ✔ After making an impulse, a cell needs time to **reset**; it cannot send impulses continuously.
- ✔ So the body also uses **chemical communication (hormones)**, which reach every cell through the blood; plants, with no nervous system, depend only on chemicals.

MEME BREAK

*Exam hall mein wo ek **braincell** bhi impulse ke baad **reset** leta hai. Isliye body ko hormones bhi chahiye hote!*

Exam tips

- ✔ In the reflex arc, draw **arrows**: towards the spinal cord on the sensory neuron, away from it on the motor neuron.
- ✔ Write "**relay neuron**" inside the spinal cord; examiners look for it.
- ✔ Brain diagram: label all three regions plus cerebellum, pons, medulla and hypothalamus.

7 PRACTICE MCQS

1. Which part of a neuron picks up information from the environment? ★★★

(a) axon (b) dendrite
(c) nerve ending (d) myelin

2. The gap between the nerve ending of one neuron and the dendrite of the next is called: ★★★

(a) axon (b) synapse
(c) impulse (d) receptor

3. Which part of the brain maintains posture and balance of the body? ★★★

(a) cerebrum (b) medulla
(c) cerebellum (d) pons

4. Involuntary actions like blood pressure, salivation and vomiting are controlled by the: ★★★

(a) medulla in the hindbrain (b) cerebrum
(c) hypothalamus (d) midbrain

5. In a reflex arc, which comes immediately after the sensory neuron? ★★★

(a) receptor
(b) relay neuron in the spinal cord
(c) motor neuron
(d) effector

6. Olfactory receptors detect: ★★☆

(a) taste
(b) light
(c) smell
(d) sound

7. The brain is protected by a fluid-filled balloon that acts as a shock absorber. This fluid is: ★★☆

(a) lymph
(b) cerebrospinal fluid
(c) plasma
(d) bile

8. **Assertion (A):** Pulling the hand away from a hot object is a reflex action.
Reason (R): The response is generated by the spinal cord, without waiting for a decision from the thinking part of the brain. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9. **Assertion (A):** A nerve impulse can reach every cell of the body.
Reason (R): Nerve impulses can reach only those cells that are connected by nervous tissue. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. What is a synapse? How is a message passed across it? **(2 marks)**
2. Name the parts of the brain that control (a) posture and balance, (b) blood pressure, (c) thinking. **(3 marks)**
3. Draw a neat labelled diagram of a neuron. Trace the path of a nerve impulse through it. **(3 marks)**
4. What is a reflex action? Draw the reflex arc for pulling the hand away from a hot object, and explain why such reflexes are controlled by the spinal cord. **(5 marks)**

Model answers

1 Synapse
- Gap between the nerve ending of one neuron and the dendrite of the next (1)
- Electrical impulse at the nerve ending releases chemicals; they cross the gap and start a new impulse (1)

2 Brain parts
- (a) cerebellum (b) medulla (c) cerebrum, 1 mark each

3 Neuron
- Diagram with dendrite, cell body, nucleus, axon, nerve ending (2)
- Dendrite → cell body → axon → nerve ending (1)

4 Reflex arc
- Definition (1); labelled arc: receptor, sensory neuron, spinal cord (relay), motor neuron, effector (2)
- Quick response, no wait for the brain; brain still informed (2)

Answer key: MCQs

Q	Ans	Why
1	(b)	Dendrite tips pick up information.
2	(b)	Synapse: chemicals carry the signal across.
3	(c)	Cerebellum: posture, balance, precision.
4	(a)	Medulla controls involuntary actions.
5	(b)	R S R M E: relay neuron comes next.
6	(c)	Olfactory = smell; gustatory = taste.
7	(b)	CSF cushions the brain in the cranium.
8	(a)	Both true; R explains the speed of A.
9	(d)	A false; R true (only connected cells).

Day 47 checklist: tick before you sleep

- ☐ I can define stimulus, receptor and coordination
- ☐ I can draw and label a neuron and trace the impulse
- ☐ I can draw the reflex arc and explain it
- ☐ I can label the brain and give each part's job
- ☐ I solved all 9 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 47 COMPLETE +125 XP earned. Next: Level 48

BUJHO *Samajh gaya! Neuron ka order, reflex arc ka RSRME, aur hindbrain ka Cycle, Pranayam, Machine.*

PAHELI *Shabaash! Kal Level 48: plants mein coordination aur hormones.*

Day 47 hit-list: Eduspark PYQ analysis

Topic	Rating
Reflex arc: diagram and explanation	★★★
Neuron: labelled diagram	★★★
Brain parts and functions	★★★
Synapse and limitations of nerve impulses	★★☆

NITESH SIR SAYS
*Diagram hi marks dilata hai: **neuron, reflex arc aur brain** roz ek baar bana ke dekho. **Kal milte hai Day 48 pe!***

DAY 48 · LEVEL 48

BIOLOGY | CHAPTER 10 | PART 2 · ASTRA SERIES

COORDINATION IN PLANTS

Goal: plant movements, tropisms and plant hormones.

by Nitesh Prakash

01 Mimosa | 02 Tendrils | 03 Tropisms | 04 Auxin | 05 Hormones

+125 XP today | Level 48 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

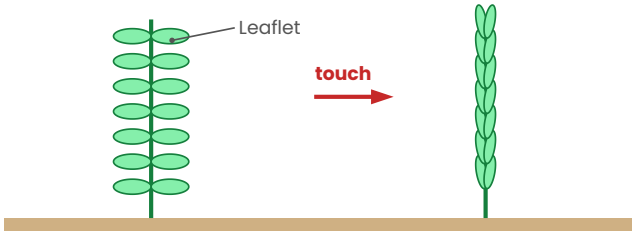
1 COORDINATION IN PLANTS ★★☆

Plants have **no nervous system and no muscles**, yet they respond to light, gravity, water, touch and chemicals. They do it in two ways:

Two kinds of plant movement ★★★

Type	Growth involved?	Example
Immediate response to a stimulus	no, independent of growth	leaves of the touch-me-not (Mimosa) fold
Movement due to growth	yes, slow and directional	tendrils coil; shoots bend to light

2 TOUCH-ME-NOT: QUICK RESPONSE ★★★ MOST ASKED



Before touch: leaflets open — After touch: leaflets fold

How Mimosa folds its leaves ★★★

1. Touch is sensed at one point, but the movement happens **at a different point**, so information must travel.
2. Plant cells pass this information using **electrical-chemical means**, from cell to cell (no special nerve tissue).
3. Cells at the leaf base **change shape** by changing the amount of **water** in them: they swell or shrink.
4. So the leaflets fold quickly. **No growth** is involved.

KITNI SPEED?

Mimosa folds in **seconds** (no growth). Tropisms take **hours to days** because they need growth.

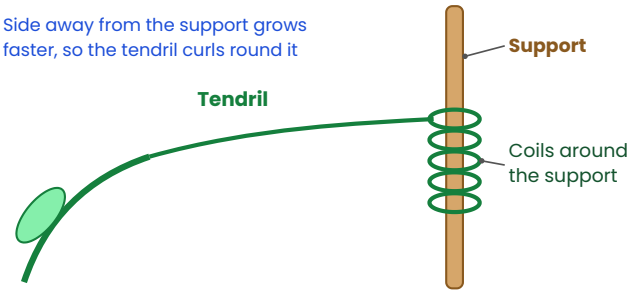
★★★ *Mimosa (touch-me-not) vs tendril ka difference har saal, Nitesh Bhaiya ki guarantee!*

BUJHO *Plant ke paas muscles hi nahi, toh Mimosa ki pattiyan hilti kaise hain?*

PAHELI *Paani ka khel hai Bujho! Cells mein **paani kam-zyada** hota hai, cells ka **shape badalta** hai, aur patti band. Muscle protein wala system animals ka hai.*

3 MOVEMENT DUE TO GROWTH ★★☆

Some plants (pea, sweet pea, grapevine) climb with **tendrils**. Tendrils are sensitive to **touch**; when they touch a support they **grow around it**.



WHY THE TENDRIL CURLS

The part of the tendril **in contact** with the support grows **slowly**; the part **away** from it grows **faster**. This uneven growth makes the tendril coil round the support.

DEFINITION

- **Tropism (tropic movement):** a slow, **directional** growth movement of a plant part in response to a stimulus
- Growth **towards** the stimulus = **positive** tropism; **away** from it = **negative** tropism

MEME BREAK

*Tendril ka scene: jab tak support **touch** nahi hota, bas wait. Touch hua, toh **growth** se lipat gaya.*

4 TROPISMS: DIRECTIONAL GROWTH

★★★ MOST ASKED

The four tropisms of the syllabus ★★★

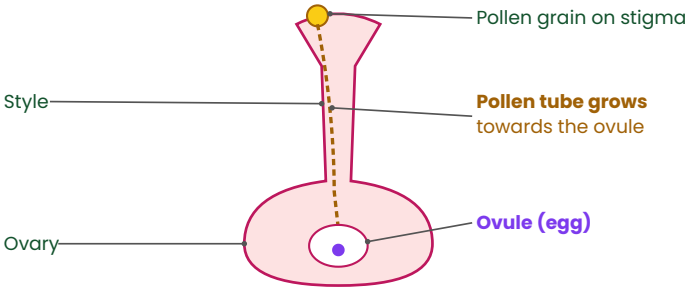
Tropism	Stimulus	Shoot	Root
Phototropism	light	+ bends towards light	− bends away
Geotropism	gravity	− grows upward	+ grows downward
Hydrotropism	water		+ grows towards water
Chemotropism	chemicals	pollen tube grows towards the ovule	

MNEMONIC: FOUR TROPISMS

"Pappu Ghar Hamesha Chalta"
Photo = light, Geo = gravity, Hydro = water, Chemo = chemical

YAAD RAKHNE KI TRICK: SHOOT VS ROOT

Shoot suraj ki taraf, zameen se door.
Root zameen aur paani ki taraf, roshni se door.



Chemotropism: after pollination, the **pollen tube** grows down the style towards the **ovule**, attracted by chemicals; it carries the male gamete to the egg.

Exam tips

- ✔ Always say whether a tropism is **positive or negative**, and for which part (shoot or root).
- ✔ Mimosa is **not** a tropism: it is an immediate response with no growth.
- ✔ Write "auxin is made at the **shoot tip**" in every phototropism answer.

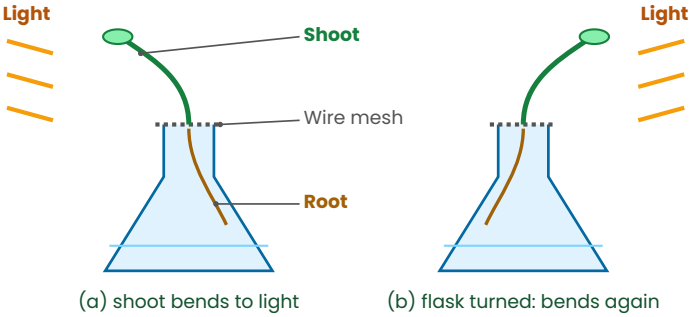
★★★ *Phototropism aur geotropism ke examples har paper mein, Nitesh Bhaiya ki guarantee!*

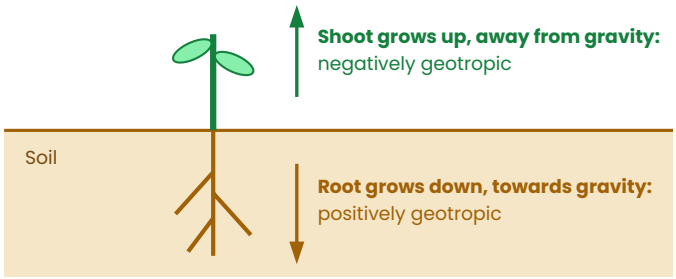
BUJHO *Plant ki toh aankh hi nahi, use pata kaise chalta hai roshni kis taraf hai?*

PAHELI *Shoot ki **tip** light sense karti hai. Wahan bana **auxin** chhaon wali side jaata hai, wo side zyada badhti hai, aur shoot roshni ki taraf jhukta hai.*

NCERT Activity: plant in a flask ★★☆

1. Fill a conical flask with water, cover its mouth with a **wire mesh** and place two or three freshly germinated bean seeds on it.
2. Keep it in a cardboard box **open on one side**, facing light from a window.
3. After 2–3 days: **shoots bend towards the light**, **roots bend away**.
4. Turn the flask so the shoots face away from the light: in a few days the old parts stay put, but the **new parts bend towards light again**.





Shoot when it finally finds the sunlight from the window

MEME BREAK

*Shoot ko roshni mil gayi, ab bas usi taraf jhukna hai **(positively phototropic)**. Root ko roshni pasand hi nahi.*

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Growth in response to touch (as in tendrils) is called **thigmotropism**; the folding of Mimosa leaves is a **nastic** (non-directional) movement. Ethylene is a gaseous hormone that helps fruits ripen.

SOCHO ZARA

A pot kept on its side for a few days: the shoot **turns up** (negative geotropism) and the root **turns down** (positive geotropism). Both are slow growth movements.

5 PLANT HORMONES ★★★ MOST ASKED

DEFINITION

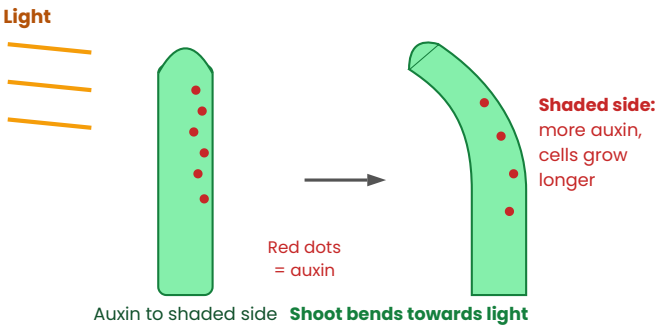
> **Plant hormones (phytohormones)** are chemical substances that coordinate **growth, development** and **responses** to the environment; they are made at one place and **diffuse** to the area of action

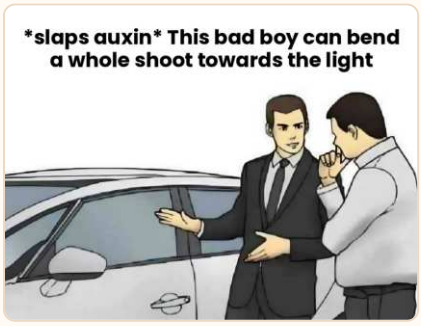
Animals vs plants: passing information

	Animals	Plants
Special tissue	nervous tissue and muscles	none
Signals	electrical impulses + hormones	electrical-chemical signals + hormones
Movement	muscle proteins change shape	cells change water content, or growth

Auxin: how a shoot bends to light ★★★

1. **Auxin** is made at the **tip of the shoot**.
2. When light comes from one side, auxin **diffuses towards the shaded side** of the shoot.
3. Auxin makes cells **grow longer**, so cells on the shaded side elongate more.
4. The shaded side grows faster than the lit side, so the shoot **bends towards light**.





MEME BREAK

*Auxin: shoot tip pe bana, **shady side** pe gaya, cells lambe kiye, aur shoot roshni ki taraf mud gaya.*

AUXIN KE AUR KAAM

Auxin also helps **tendrils** coil (the side away from the support grows faster) and promotes growth at the shoot tip.

★★★ *Auxin se phototropism wala question har saal, Nitesh Bhaiya ki guarantee!*

Plant hormones and their jobs ★★★

Hormone	Main job	Example
Auxin	cell elongation (growth)	shoot bends to light; tendril coiling
Gibberellins	growth of the **stem**	stem elongation
Cytokinins	promote **cell division**	high in fruits and seeds
Abscisic acid	**inhibits** growth	wilting of leaves; stress hormone

MNEMONIC: 3 PROMOTERS, 1 STOPPER

Auxin = Aage badho (lambe cells)
Gibberellin = Girafe jaisa tall stem
Cytokinin = Cell division
Abscisic acid = Absolutely ruko

MEME BREAK

*Teen hormones growth badhate, ek **(abscisic acid)** rokta. Perfectly balanced, jaise plant ko chahiye.*

BUJHO *Hydrotropism aur hydro... matlab root paani dekh ke bhagti hai?*

PAHELI *Bhaagti nahi, **badhti** hai! Tropism = growth movement. Root dheere dheere paani ki taraf **grow** karti hai, isliye ye slow hota hai.*

6 PRACTICE MCQS

1. The folding of leaves of the touch-me-not plant on touching is due to: ★★★

(a) growth of cells
(b) change in the water content of cells
(c) action of muscles
(d) auxin

2. The growth of a pollen tube towards the ovule is an example of: ★★★

(a) hydrotropism
(b) chemotropism
(c) geotropism
(d) phototropism

3. Roots growing downwards into the soil show: ★★★

(a) positive geotropism
(b) negative geotropism
(c) positive phototropism
(d) negative hydrotropism

4. Which plant hormone promotes cell division? ★★★

(a) auxin
(b) gibberellin
(c) cytokinin
(d) abscisic acid

5. Which hormone inhibits growth and causes wilting of leaves? ★★☆

(a) auxin
(b) abscisic acid
(c) cytokinin
(d) gibberellin

6. When light falls on a shoot from one side, auxin: ★★★

(a) moves to the lit side (b) is destroyed
(c) moves to the shaded side (d) moves to the root

7. A tendril coils around a support because: ★★☆

(a) the side touching the support grows faster (b) the side away from the support grows faster
(c) cells lose water (d) of gravity

8. **Assertion (A):** The shoot of a plant bends towards light.
Reason (R): Auxin collects on the shaded side and makes cells there grow longer. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9. **Assertion (A):** Movement of Mimosa leaves is a growth movement.
Reason (R): Mimosa leaflets fold because their cells change shape by changing water content. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. How do the leaves of a sensitive plant move on touch? How is this different from the movement of a shoot towards light? **(3 marks)**
2. Name the plant hormone that (a) promotes cell division, (b) inhibits growth, (c) helps the stem grow. **(3 marks)**
3. Describe an activity to show that the shoot is positively phototropic and the root negatively phototropic. Draw a labelled diagram. **(5 marks)**

Model answers

1 Mimosa vs shoot

- Mimosa: no growth; cells change shape by losing/gaining water; quick (1.5)
- Shoot: growth movement; auxin on shaded side; slow and directional (1.5)

2 Hormones

- (a) cytokinin (b) abscisic acid (c) gibberellin, 1 mark each

3 Flask activity

- Steps: flask, wire mesh, germinated seeds, box open on one side (2)
- Result: shoot to light, root away; diagram (3)

× WRONG	✓ RIGHT
Mimosa leaves move by growth	Immediate response; cells change water content, no growth
Auxin collects on the lit side	On the shaded side, which grows faster
Abscisic acid promotes growth	It inhibits growth; the other three promote it

Answer key: MCQs

Q	Ans	Why
1	(b)	Water moves out, cells change shape.
2	(b)	Chemicals attract the pollen tube.
3	(a)	Root goes towards gravity.
4	(c)	Cytokinins: cell division.
5	(b)	ABA: growth inhibitor, wilting.
6	(c)	Auxin diffuses to the shade.
7	(b)	Away side grows faster: coiling.
8	(a)	Both true; R explains A.
9	(d)	A false (no growth); R true.

Day 48 checklist: tick before you sleep

- [ ] I can explain how Mimosa leaves fold without growth
- [ ] I can explain tendril coiling and define tropism
- [ ] I can name four tropisms with examples
- [ ] I can explain auxin and phototropism with a diagram
- [ ] I solved all 9 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 48 COMPLETE +125 XP earned. Next: Level 49

BUJHO Samajh gaya! Pappu Ghar Hamesha Chalta, aur 3 promoters, 1 stopper.

PAHELI Shabaash! Kal Level 49: animals mein hormones.

Day 48 hit-list: Eduspark PYQ analysis

Topic	Rating
Auxin and phototropism (diagram)	★★★
Tropisms with examples	★★★
Plant hormones and functions	★★★
Mimosa vs tendril movement	★★☆

NITESH SIR SAYS

*Plants chup chaap bhi kitna kuch karte hain! **Tropism ka diagram** aur hormones ki table aaj hi yaad karo. **Kal milte hai Day 49 pe!***

DAY 49 LEVEL 49

BIOLOGY | CHAPTER 10 | PART 3 ASTRA SERIES

HORMONES IN ANIMALS & FEEDBACK

Goal: name each endocrine gland, its hormone and its job.

by Nitesh Prakash

01 Adrenaline | 02 Endocrine Glands | 03 Thyroid, GH | 04 Insulin | 05 Feedback | +125 XP today | Level 49 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 HORMONES: THE CHEMICAL MESSENGERS ★★★

DEFINITION

- **Hormone:** a chemical messenger secreted by an **endocrine gland** directly into the **blood**, which carries it to target cells far away
- **Endocrine glands** are **ductless**: they pour their secretion straight into the blood

Why animals need hormones too ★★★

- Nerve impulses reach only cells **connected by nerves**, and a nerve cell needs time to reset before the next impulse
- Many changes must happen in **all cells at once** and **last long**: growth, puberty, sugar balance

Exocrine vs endocrine glands ★★☆

Point	Exocrine gland	Endocrine gland
Duct	has a duct	**ductless**
Secretion goes to	a surface or cavity	**blood** directly
Examples	salivary, sweat, gastric glands	thyroid, pituitary, adrenal

NCERT Q: How does chemical coordination take place in animals? ★★★

- Endocrine glands release hormones into the **blood** in response to a stimulus
- Blood carries them all over the body; only **target cells** with matching receptors respond

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

The adrenal gland has two parts: an outer **cortex** (hormones like cortisol and aldosterone) and an inner **medulla**, which makes adrenaline and noradrenaline.

BUJHO *Hormone bhi nerve jaisa wire se jaata hai na, seedha target tak?*

PAHELI *Nahi Bujho! Hormone ka koi wire nahi. Gland use seedha **blood** mein chhod deta hai, blood poore body mein le jaata hai, aur sirf target cells react karte hain.*

2 ADRENALINE: FIGHT OR FLIGHT ★★★ MOST ASKED

Emergency hormone of the adrenal gland ★★★

Scary moment (a dog chasing you, a sudden exam surprise): the **adrenal glands** release **adrenaline** straight into the blood. It reaches the heart, muscles and other organs and gets the body ready to **fight or run**.

Organ	What adrenaline does	Why it helps
Heart	beats **faster**	more oxygen to muscles
Breathing	rate **rises** (diaphragm, rib muscles contract)	more oxygen in
Digestive system, skin	blood supply **cut down** (small arteries constrict)	blood diverted to skeletal muscles

Flow: from fear to action

- **Stimulus:** sudden danger is sensed by the senses
- **Adrenal glands** pour adrenaline into the blood
- **Response:** faster heartbeat and breathing, more blood to skeletal muscles

MEME BREAK

*Adrenal = **ad** (near) + **renal** (kidney). Gland kidney ke upar topi jaisa baitha hai, aur emergency mein poori body ko alert kar deta hai!*

MNEMONIC: ADRENALINE KE 3 TEZ

"Dil tez, saans tez, khoon muscles ki taraf tez." Pet aur skin ka khoon kam, kyunki bhaagte waqt khaana digest karna zaroori nahi!

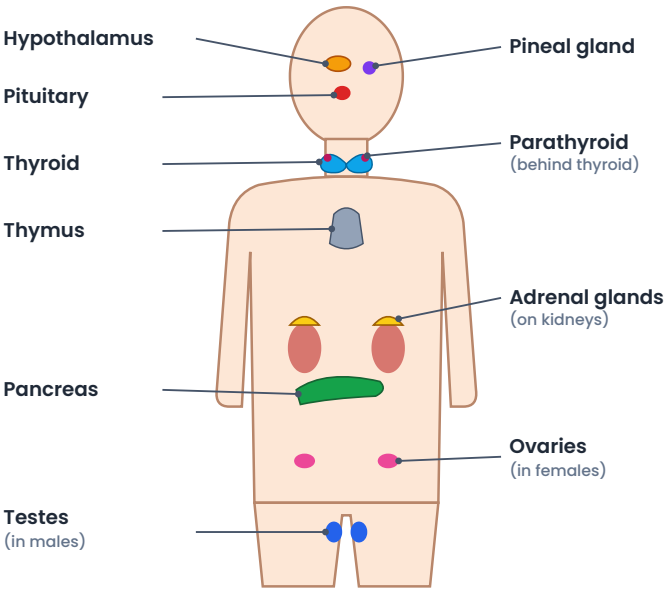
COMMON MISTAKE

Adrenaline does **not** travel along nerves to the heart. It is carried by the **blood**, and that is why it can act on many organs at the same time.

★★★ *Adrenaline ke effects (heart, breathing, blood flow) har saal pooche jaate hain, Nitesh Bhaiya ki guarantee!*

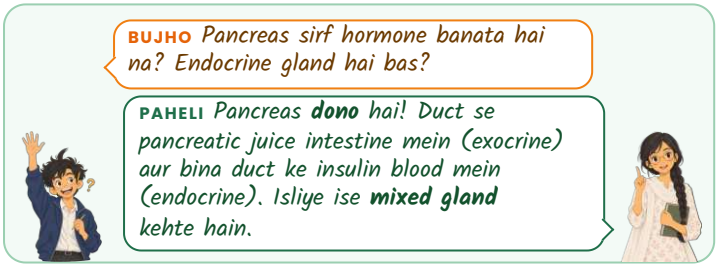
3 ENDOCRINE SYSTEM OF THE HUMAN BODY

★★★ MOST ASKED



MNEMONIC: GLANDS FROM HEAD TO TOE

"Har Pal Pyaar, Thoda Pyaar, Thoda Adjust, Phir Tension Over." Hypothalamus, Pituitary, Pineal, Thyroid, Parathyroid, Thymus, Adrenal, Pancreas, Testes/Ovaries.



Key features of hormones

- ✔ Secreted in **tiny amounts**, yet very powerful
- ✔ Act **slowly** but the effect **lasts longer**
- ✔ Both **too little** and **too much** cause disorders
- ✔ Timing and amount are controlled by **feedback mechanisms**

5 THYROID, IODINE AND GOITRE ★★★ MOST ASKED

Why iodised salt is advised (NCERT) ★★★

- The thyroid gland needs **iodine** to make **thyroxin**
- Thyroxin controls the metabolism of **carbohydrates, proteins and fats** for balanced growth
- Too little iodine → too little thyroxin → **goitre**: the thyroid swells and the neck looks swollen
- Iodised salt supplies enough iodine in daily food

MNEMONIC: KAM HUA TOH KYA HUA

"Iodine kam, gala jam (goitre). GH kam, height kam (dwarfism). Insulin kam, sugar ka gham (diabetes)."

Note: the four small **parathyroid** glands sit on the back of the thyroid; write **thyroxin** (NCERT spelling).

4 GLAND, HORMONE AND FUNCTION

★★★ MOST ASKED

The table you must learn line by line ★★★

Gland	Hormone	Main function
Hypothalamus	releasing hormones	tell the pituitary to release its hormones (e.g. growth hormone releasing factor)
Pituitary	growth hormone	growth and development of the body
Pineal	melatonin	sleep–wake cycle
Thyroid	thyroxin	metabolism of carbohydrates, proteins and fats
Parathyroid	parathormone	calcium level in blood
Thymus	thymosin	development of immunity in children
Adrenal	adrenaline	prepares body for emergency
Pancreas	insulin	lowers blood sugar level
Testes	testosterone	male puberty changes
Ovaries	oestrogen	female puberty changes, menstrual cycle

MEME BREAK

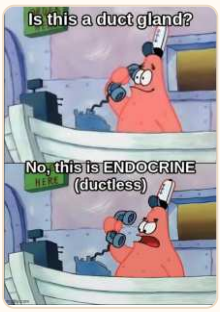
*Yahi lab scientist wala feeling aata hai jab table ki 10 lines ek baar mein yaad ho jaati hain. Gland, hormone, kaam: **teen column, poore marks!***

★★★ *Gland–hormone matching ka MCQ ya table har paper mein milta hai, Nitesh Bhaiya ki guarantee!*

6 GROWTH HORMONE: HEIGHT CONTROL ★★☆

Growth hormone from the pituitary ★★☆

Condition	In childhood	Result
Normal	right amount	normal height
Dwarfism	**too little** growth hormone	very short stature
Gigantism	**too much** growth hormone	very tall stature



MEME BREAK

*Endocrine gland ka koi duct nahi hota, phone seedha **blood** pe lagta hai. Duct wala gland exocrine hai, yaad rakho!*

7 HORMONES OF PUBERTY ★★☆

Testosterone and oestrogen ★★☆

Point	Testosterone	Oestrogen
Made by	**testes** (males)	**ovaries** (females)
Starts at	puberty	puberty
Main role	development of male sex organs and male features	development of female sex organs and the menstrual cycle
Visible changes	deeper voice, facial hair, more muscle	breast development, broader hips

MNEMONIC: T-T AND O-O

"T se Testes, T se Testosterone; O se Ovary, O se Oestrogen."
Gland aur hormone ka pehla akshar same!

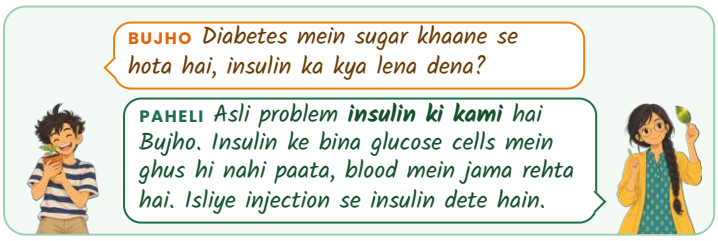
8 INSULIN AND DIABETES ★★★ MOST ASKED

Insulin keeps blood sugar in check ★★★

- **Insulin** is secreted by the **pancreas** (by special groups of cells in it)
- It helps cells take up glucose and **lowers** the blood sugar level
- Too little insulin → sugar level stays **high** → **diabetes**, which harms many organs

NCERT Q: Why are some diabetes patients given insulin injections? ★★★

- Their pancreas does not secrete **enough insulin**, so blood sugar remains high
- Injected insulin does the missing job and brings blood sugar **down to normal**
- It is injected because insulin is a protein and would be **digested** if taken by mouth (extra point)



EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

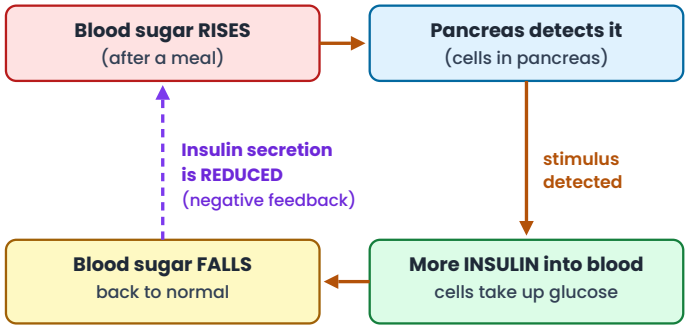
The pancreas also makes **glucagon**, which **raises** blood sugar when it falls too low. Insulin and glucagon work as a pair to keep sugar steady.

Common mistakes that cost marks

- ✔ Writing that **thyroxin** controls blood sugar (it is **insulin**)
- ✔ Calling the pituitary hormone "growth enzyme": it is **growth hormone**
- ✔ Saying hormones travel through **nerves** (they travel in **blood**)
- ✔ Mixing up goitre (iodine) with diabetes (insulin)

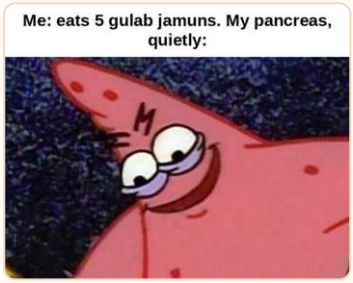
Remember: puberty changes start slowly because they are run by hormones carried in the blood, not by nerves.

9 FEEDBACK MECHANISM: BLOOD SUGAR ★★★ MOST ASKED



How feedback controls the amount ★★★

- Hormones must be released in the **right amount** at the **right time**
- Sugar rises → pancreas detects it → **more insulin** → sugar falls
- Sugar falls → insulin secretion is **reduced**: the result itself switches the gland down
- This self-checking loop is called a **feedback mechanism**



MEME BREAK

*Gulab jamun gaya, sugar badhi, aur pancreas ne chupke se **insulin** chhod diya. Feedback system kabhi chhutti nahi leta!*

10 NERVOUS VS HORMONAL COORDINATION ★★★ MOST ASKED

Compare: NCERT exercise favourite ★★★

Point	Nervous	Hormonal
Message	electrical impulse (chemical at synapse)	chemical (hormone)
Carried by	nerve fibres	blood
Speed	very fast	slow
Effect	short-lived	long-lasting
Reaches	only cells linked by nerves	all cells; target cells respond
Example	reflex action	growth, puberty

MNEMONIC: NERVE VS HORMONE

"Nerve = WhatsApp call (fast, ek bande tak). Hormone = newspaper (slow, sabke ghar, der tak asar)."

Exam tips

- ✔ In a comparison answer, write **at least four points** in a two-column table
- ✔ Draw the feedback loop with arrows: easy marks

11 PRACTICE MCQS

1. Which gland secretes growth hormone? (NCERT) ★★★
(a) adrenal (b) thyroid
(c) pituitary (d) pancreas

2. Iodine, whose lack causes goitre, is needed to make: ★★★
(a) insulin (b) thyroxin
(c) adrenaline (d) testosterone

3. Which of these is NOT an effect of adrenaline? ★★★
(a) faster heartbeat (b) faster breathing
(c) more blood to the digestive system (d) more blood to skeletal muscles

4. Endocrine glands release their secretions: ★★☆
(a) into ducts (b) directly into the blood
(c) into the stomach (d) on the skin

5. Excess growth hormone in childhood causes: ★★☆
(a) dwarfism (b) goitre
(c) gigantism (d) diabetes

6. The gland that acts as both an exocrine and an endocrine gland is: ★★☆
(a) thyroid (b) pancreas
(c) pituitary (d) adrenal

7. Which hormone regulates the calcium level in blood? ★☆☆
(a) thyroxin (b) parathormone
(c) thymosin (d) melatonin

8. Less insulin is secreted once blood sugar falls. This is an example of: ★★☆
(a) reflex action (b) feedback mechanism
(c) tropism (d) osmosis

9. **Assertion (A):** Hormonal coordination is slower than nervous coordination.
Reason (R): Hormones are carried to target cells by the blood. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Iodised salt is advised in our diet.
Reason (R): Iodine is needed by the pancreas to make insulin. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Day 49 checklist: tick before you sleep

- [ ] I can name all ten endocrine glands and show them on a diagram
- [ ] I can match each gland with its hormone and function
- [ ] I can explain adrenaline, goitre, dwarfism and gigantism
- [ ] I can draw the blood sugar feedback loop
- [ ] I can compare nervous and hormonal coordination

My score: ____ / 5

Short and long answer questions

1. How does our body respond when adrenaline is secreted into the blood? **(2 marks)**
2. Why are some patients of diabetes treated by giving injections of insulin? **(2 marks)**
3. Draw a diagram of the human endocrine system and label the thyroid, adrenal gland, pancreas and pituitary. Write one hormone of each. **(5 marks)**
4. Compare nervous and hormonal mechanisms of control and coordination in animals (any four points). **(3 marks)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Pituitary makes growth hormone.
2	(b)	Thyroid needs iodine for thyroxin.
3	(c)	Blood to digestive system is reduced.
4	(b)	Ductless: straight into blood.
5	(c)	Excess GH in childhood: gigantism.
6	(b)	Juice via duct; insulin into blood.
7	(b)	Parathormone controls calcium.
8	(b)	The result switches the gland down.
9	(a)	Blood transport is slow; R explains A.
10	(c)	A true; R false: thyroid makes thyroxin.

Model answers (marking points)

1 Adrenaline response
- Heart beats faster, so more oxygen reaches muscles (1)
- Breathing rate rises; blood to digestive system and skin falls, more to skeletal muscles (1)

2 Insulin injections
- Pancreas secretes too little insulin, blood sugar stays high (1)
- Injected insulin lowers blood sugar to normal (1)

3 Endocrine diagram
- Correct positions and labels (3); pituitary: growth hormone, thyroid: thyroxin, adrenal: adrenaline, pancreas: insulin (2)

LEVEL 49 COMPLETE +125 XP earned. Next: Level 50

BUJHO *Samajh gaya! Gland seedha blood mein hormone chhodta hai, aur feedback amount control karta hai.*

PAHELI *Shabaash! Kal Level 50: Chapter 10 ka Boss Level.*

NITESH SIR SAYS
Gland, hormone, kaam: ye teen column wali table roz ek baar likho. Diagram mein glands ki position pakki karo. ***Kal Boss Level, taiyaar rehna!***

DAY 50 LEVEL 50

BIOLOGY | CHAPTER 10 | BOSS LEVEL

CONTROL AND COORDINATION BOSS

Goal: finish the chapter test and score 80% or more.

by Nitesh Prakash

01 MCQs + A/R | 02 Short Answers | 03 Diagrams | 04 Case-based | 05 Long Answers | +200 XP Boss Level | Level 50 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 10 REVISION MAP ★★★ MOST ASKED

Chapter 10 mind map: revise in 5 minutes ★★★

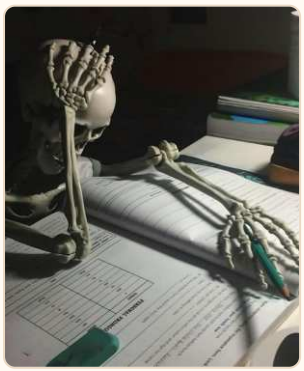
Branch	Key points
Nervous system	receptors → sensory neuron → CNS → motor neuron → effector; neuron: dendrite, cell body, axon, nerve ending; synapse is chemical
Reflex arc	quick, automatic; controlled by **spinal cord**; brain informed later
Brain	forebrain (thinking, senses, hypothalamus), midbrain, hindbrain (cerebellum: balance; pons and medulla: involuntary actions)
Plants	no nerves; touch (Mimosa, water changes), tropisms (photo, geo, hydro, chemo); hormones auxin, gibberellin, cytokinin, abscisic acid
Animal hormones	adrenaline, thyroxin, GH, insulin, testosterone, oestrogen; feedback control

How to attack the Boss: 90 minutes

- Section A (12 MCQs incl. A/R): 15 minutes, no blanks
- Sections B and C: draw neat labelled diagrams, they carry marks
- Case-based: read the passage once, then answer each part
- Keep 10 minutes to check arrows in the reflex arc and labels

MNEMONIC: BRAIN KE TEEN BOSS

"Cerebrum = CEO (soch, yaad, senses). Cerebellum = balance coach. Medulla = autopilot (BP, saans, ulti, laar)."



MEME BREAK

*Boss test se pehle ki raat: skeleton bhi padh raha hai! Tension mat lo, mind map dekho aur **diagrams practise karo**.*

★★★ *Reflex arc ka diagram aur brain ke parts ke functions: Chapter 10 ke sabse pakke sawaal, Nitesh Bhaiya ki guarantee!*

2 SECTION A: MCQS (1 MARK EACH) ★★★ MOST ASKED

1. The gap between two neurons is called a: ★★★
(a) dendrite (b) synapse
(c) axon (d) impulse

2. Posture and balance of the body are maintained by the: ★★★
(a) cerebrum (b) medulla
(c) cerebellum (d) pons

3. Blood pressure, salivation and vomiting are controlled by the: ★★★
(a) medulla (b) cerebellum
(c) hypothalamus (d) cerebrum

4. The plant hormone that promotes cell division is: ★★★
(a) auxin (b) gibberellin
(c) cytokinin (d) abscisic acid

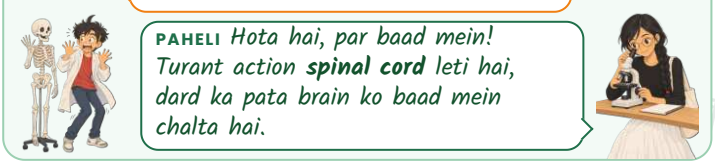
5. Growth of a pollen tube towards the ovule is an example of: ★★☆
(a) phototropism (b) geotropism
(c) hydrotropism (d) chemotropism

6. Leaves of the touch-me-not plant fold on touch because: ★★☆
(a) of growth at the base (b) cells change shape by losing water
(c) auxin is destroyed (d) of gravity

7. The plant hormone that inhibits growth and causes wilting of leaves is: ★★☆
(a) abscisic acid (b) auxin
(c) cytokinin (d) gibberellin

8. Deficiency of which hormone causes diabetes? ★★★
(a) thyroxin (b) adrenaline
(c) insulin (d) growth hormone

9. Which gland is called the master gland? ★★☆
(a) thyroid (b) pituitary
(c) adrenal (d) pancreas



10. In a reflex arc, the correct sequence is: ★★★

(a) receptor → motor neuron → spinal cord → sensory neuron → muscle
(b) receptor → sensory neuron → spinal cord → motor neuron → muscle
(c) muscle → motor neuron → brain → receptor
(d) receptor → brain → sensory neuron → muscle

11. **Assertion (A):** A plant shoot kept near a window bends towards the light.
Reason (R): Auxin diffuses to the shaded side of the shoot and makes cells there grow longer. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

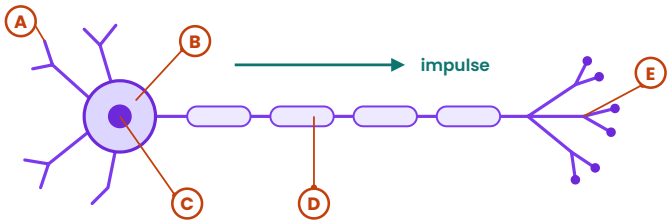
12. **Assertion (A):** The brain is protected by a fluid-filled balloon inside the cranium.
Reason (R): The fluid (cerebrospinal fluid) absorbs shocks. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

3 SECTION B: SHORT ANSWERS (2 MARKS EACH)

★★★ MOST ASKED

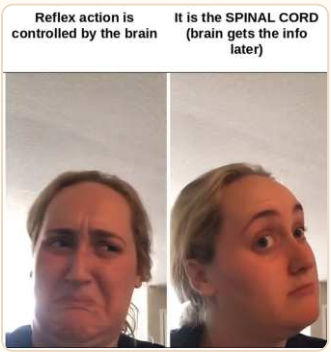
B1. Diagram-based: the neuron ★★★



1. Identify the parts marked A to E.
2. In which direction does the impulse travel: from A to E or from E to A?

B2 to B4 ★★☆

1. What is a synapse? How does an impulse cross it? **(2)**
2. How does the tendril of a pea plant coil around a support? **(2)**
3. Why is the use of iodised salt advised? **(2)**



MEME BREAK
*Reflex action ka control **spinal cord** mein hota hai. Brain ko khabar baad mein milti hai, isliye dard thoda late lagta hai!*

4 SECTION C: SHORT ANSWERS (3 MARKS EACH)

★★★ MOST ASKED

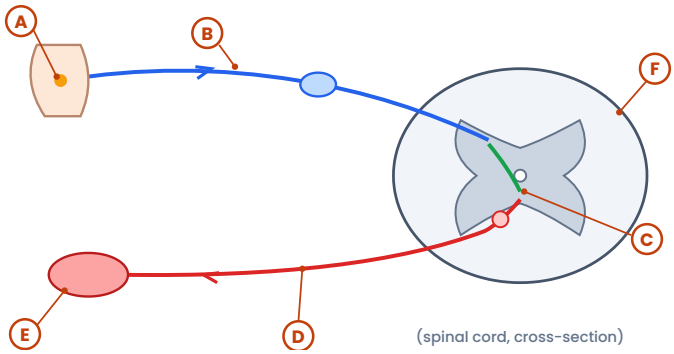
C1 to C3 ★★★

1. What are plant hormones? Name one plant hormone each that (a) promotes cell division, (b) inhibits growth, (c) promotes stem elongation. **(3)**
2. Distinguish between the movement in touch-me-not leaves and the movement of a shoot towards light (any three points). **(3)**
3. How does our body respond when adrenaline is secreted into the blood? Why is it called an emergency hormone? **(3)**

5 SECTION D: CASE-BASED (4 MARKS EACH)

★★★ MOST ASKED

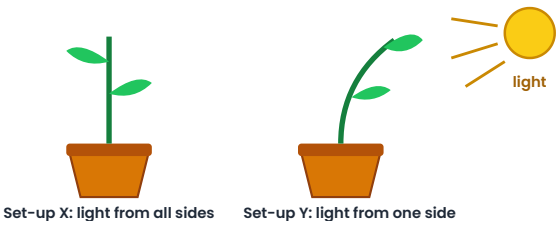
Case 1: the hot pan ★★★



Rohan touched a hot pan and pulled his hand back before he even felt the pain.

1. Identify the parts A, C and E. **(1)**
2. Name the type of neuron marked B and D. **(1)**
3. Where is this action controlled, and why is the brain not involved first? **(2)**

Case 2: the window plant ★★★



Two potted plants were kept for a week: X in a room with light from all sides, Y near a window.

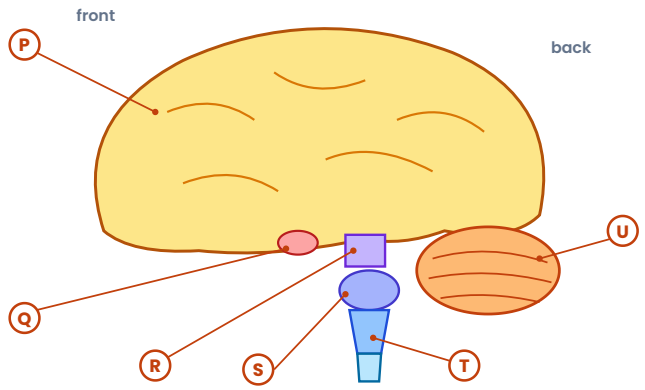
1. Name the response shown by the shoot of Y. **(1)**
2. Which hormone causes it, and where is it made? **(1)**
3. Explain how this hormone makes the shoot bend. How would the roots respond? **(2)**

DIAGRAM MARKS
Leader lines must touch the part; always put **arrows** on a reflex arc.

6 SECTION E: LONG ANSWERS (5 MARKS EACH)

★★★ MOST ASKED

E1. Diagram-based: the human brain ★★★



1. Identify the parts marked P to U. **(3)**
2. Write one function each of P, U and S. **(2)**

E2. Hormones in animals ★★★

1. Name the endocrine glands that secrete (a) thyroxin, (b) insulin, (c) growth hormone, (d) adrenaline. **(2)**
2. What is goitre? How can it be prevented? **(1)**
3. Explain the feedback mechanism with the example of blood sugar. **(2)**

MEME BREAK

*Arrows ki direction sahi = poore marks. Receptor se muscle tak, beech mein **spinal cord**. Gigachad energy!*

Model answers: Section B

B1 Neuron

- A dendrite, B cell body, C nucleus, D axon, E nerve ending (½ each, max 1½)
- Impulse travels from A (dendrite) to E (nerve ending) (½)

B2 Synapse

- Gap between the nerve ending of one neuron and the dendrite of the next (1)
- Electrical impulse releases chemicals at the nerve ending; they cross the gap and start a new impulse (1)

B3 Tendril

- Cells on the side away from the support grow faster than those in contact (1)
- So the tendril bends and coils around the support; this growth is controlled by auxin (1)

B4 Iodised salt

- Iodine is needed to make thyroxin (1); its lack causes goitre (1)

Common mistakes that cost marks

- ✔ Writing that the brain controls a reflex action (it is the **spinal cord**)
- ✔ Drawing reflex arc arrows from muscle to receptor
- ✔ Calling cerebellum the centre of thinking (that is the **cerebrum**)
- ✔ Saying plants have nerves: they use **hormones** and water movement

Answer key: Section A (MCQs)

Q	Ans	Why
1	(b)	Junction between neurons.
2	(c)	Cerebellum: posture and balance.
3	(a)	Medulla controls involuntary actions.
4	(c)	Cytokinins promote cell division.
5	(d)	Pollen tube grows towards a chemical.
6	(b)	Cells lose water and change shape.
7	(a)	ABA inhibits growth, wilting of leaves.
8	(c)	Too little insulin: high blood sugar.
9	(b)	Pituitary controls other glands.
10	(b)	Receptor → sensory → cord → motor → muscle.
11	(a)	Auxin on shaded side; R explains A.
12	(a)	CSF cushions the brain; R explains A.

MEME BREAK

*Thyroxin = metabolism, **insulin** = blood sugar. Ye mix kiya toh fan bhi disappoint ho jayega!*

Model answers: Section C

C1 Plant hormones

- Chemicals that coordinate growth and responses in plants (1)
- (a) cytokinin (b) abscisic acid (c) gibberellin or auxin (⅔ each)

C2 Mimosa vs phototropism

- Mimosa: not growth related, quick, cells change shape by water movement
- Phototropism: growth related, slow, auxin makes shaded side grow more (1 each point)

C3 Adrenaline

- Heart beats faster, breathing faster, blood diverted to skeletal muscles (2)
- Prepares the body to fight or run in danger, so emergency hormone (1)

★★★ Neuron, reflex arc aur brain: teen diagrams pakke, toh Chapter 10 mein poore marks, Nitesh Bhaiya ki guarantee!

Model answers: Section D

1 The hot pan

- A receptor in skin, C relay neuron in spinal cord, E effector muscle (1)
- B sensory neuron, D motor neuron (1)
- Controlled by the spinal cord; going to the brain first would take too long, so the arc gives a quick response that protects the body (2)

2 The window plant

- Positive phototropism (1)
- Auxin, made at the shoot tip (1)
- Auxin moves to the shaded side, cells there grow longer, so the shoot bends towards light; roots bend away from light (2)

Score card: how did you fight the boss?

Score (out of 12 in A)	Verdict
11 to 12	Boss defeated, legend level
8 to 10	Strong, revise the slips
below 8	Re-read Days 47 to 49, then retry

Bonus rapid-fire round: answer in one breath

Question	Answer
Longest part of a neuron	axon
Thinking part of the brain	cerebrum (forebrain)
Centre for hunger and body temperature	hypothalamus
Bony box protecting the brain	cranium
Plant hormone for stem elongation	gibberellin
Root growth towards water	hydrotropism
Hormone needing iodine	thyroxin
Gland on top of kidneys	adrenal

Boss Level tracker: write your percentage

Section A ____ / 12 Sections B to E ____ / 50 Total ____ %

Exam strategy for Chapter 10

- ✔ Start long answers with a neat labelled diagram, then write the points
- ✔ Use the words **receptor, sensory neuron, relay neuron, motor neuron, effector** in every reflex answer
- ✔ For plant answers, always name the hormone: **auxin** for bending, **ABA** for wilting

Day 50 checklist: tick before you sleep

- ☐ I attempted all 12 MCQs and checked them with the key
- ☐ I can label the neuron, reflex arc and brain diagrams
- ☐ I can explain phototropism with the role of auxin
- ☐ I can match glands with hormones and explain feedback
- ☐ I marked my weak topics to revise from Days 47 to 49

My score: ____ / 5

Model answers: Section E

E1 Brain

- P cerebrum, Q hypothalamus, R midbrain, S pons, T medulla, U cerebellum (½ each)
- P: thinking, memory, senses; U: balance and posture; S: involuntary actions such as regulating breathing (with medulla) (2)

E2 Hormones

- (a) thyroid (b) pancreas (c) pituitary (d) adrenal (½ each)
- Goitre: swollen thyroid due to iodine deficiency; prevented by iodised salt (1)
- Sugar rises → pancreas releases more insulin → sugar falls → insulin secretion reduced (2)

Most asked in Chapter 10: Eduspark PYQ analysis

Topic	Rating
Reflex arc (diagram and sequence)	★★★
Parts of the brain and their functions	★★★
Neuron diagram	★★★
Phototropism and auxin	★★★
Gland–hormone–function, goitre, diabetes	★★★
Plant hormones table	★★☆
Nervous vs hormonal comparison	★★☆

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Nerve impulses in some human nerves travel at up to about 100 metres per second, which is why a reflex is over before you even think about it.

LEVEL 50 COMPLETE +200 XP earned. Next: Level 51

BUJHO *Boss khatam! Neuron, reflex arc, brain, plant hormones, animal hormones: sab clear.*

PAHELI *Shabaash! Chapter 10 complete. Kal se Chapter 11: reproduction.*

NITESH SIR SAYS

Is chapter mein diagram hi asli hathiyar hai: neuron, reflex arc, brain, aur phototropism ka set-up. Har din ek baar bina dekhe banao. ***Boss defeated, ab agla level!***

Coming up: Day 51

- Chapter 11 starts: **How do Organisms Reproduce?**
- DNA copying, variation, binary and multiple fission, budding, regeneration

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter.

BIOLOGY | CHAPTER 10 COMPLETE | LEVEL 50

CHAPTER 10 SWAHAA!

Control and Coordination | Days 47–50

BUJHO *Neuron, reflex arc, brain, auxin aur insulin, sab clear! Mera braincell aaj full activate hai!*

PAHELI *Bilkul! Ab dekhte hain organisms apne jaise naye organisms kaise banate hain.*

CHAPTER XP EARNED
+575 XP

LEVELS CLEARED
4 levels (Days 47–50)

BOOK PROGRESS
Level 50 / 60 6475 XP total

WHAT YOU HAVE MASTERED

- ✔ Neuron, synapse and reflex arc
- ✔ Tropic movements in plants
- ✔ Endocrine glands and hormones
- ✔ Human brain and its parts
- ✔ Plant hormones
- ✔ Feedback mechanism

NEXT CHAPTER Day 51: How do Organisms Reproduce?

Chapter 10 cleared!

DAY 51
LEVEL 51

BIOLOGY | CHAPTER 11 | PART 1

ASEXUAL REPRODUCTION & VARIATION

Goal: see how one parent makes many, and why copies vary.

by Nitesh Prakash

01 DNA Copying | 02 Variation | 03 Fission | 04 Regeneration | 05 Budding

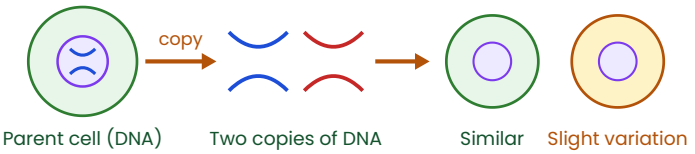
+125 XP today | Level 51 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 DO ORGANISMS CREATE EXACT COPIES? ★★★

DEFINITION

- **Reproduction:** the process by which living organisms produce new individuals of **their own kind**; it is not needed for an individual to stay alive, but it is needed for the **species** to continue
- **DNA** (deoxyribonucleic acid) in the chromosomes of the nucleus carries the **information** to make proteins; different proteins give different body designs
- The basic event in reproduction is the **creation of a DNA copy**, along with extra cellular apparatus, so that one cell becomes two



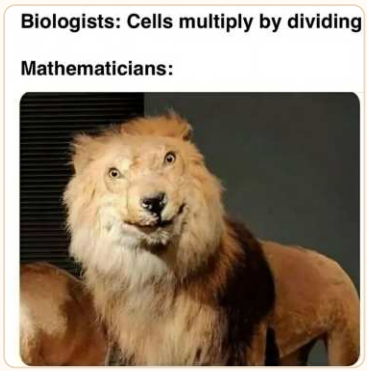
Why copies are never 100% exact ★★☆

- Cells copy DNA using **biochemical reactions**, which are never absolutely accurate
- So each DNA copy is **similar** to the original but may have small changes: this is **variation**
- Too big a change: the new cell may not survive; small changes are the raw material of variation
- Sexual reproduction adds even more variation (Day 52)

Importance of variation ★★★

- **Consistency of DNA copying** keeps the body design suited to the organism's **niche** (its place in the ecosystem)
- **Variation** helps a species survive when the niche changes suddenly
- Example: if water temperature rises, most bacteria die; a few **heat-resistant variants** survive and keep the species alive
- So variation is useful for the **survival of the species** over time, not for the individual

★★★ *"Why is variation useful for the species but not always for the individual?" Nitesh Bhaiya ki guarantee: ye 3 marks ka sawaal pakka aata hai!*



MEME BREAK

*Single cell ka family planning: **divide karo aur multiply ho jao**. Binary fission mein 1 se 2, phir 4, phir 8!*

BUJHO *Toh Amoeba ki beti bilkul photocopy hoti hai na? Xerox machine jaisi?*

PAHELI *Almost! DNA copying ke reactions **100% accurate nahi** hote, isliye har copy mein thoda sa **variation** aa jaata hai. Wahi species ko bachata hai.*

2 MODES OF REPRODUCTION ★★☆

Asexual vs sexual: first look

Feature	Asexual	Sexual
Parents	**one**	usually **two**
Gametes	not formed	formed and fuse
Offspring	nearly identical copies	show more variation
Speed	fast, many offspring	slower

MNEMONIC

"Fatafat Fission, Fir Fragment, Re-bano, Bud nikalo, Veg bhi, Spore bhi"

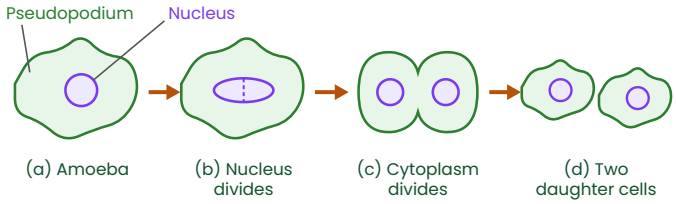
Six asexual modes in order: **F**ission, **F**ragmentation, **R**egeneration, **B**udding, **V**egetative propagation, **S**pore formation (last two on Day 52).

3 FISSION IN UNICELLULAR ORGANISMS ★★★

DEFINITION

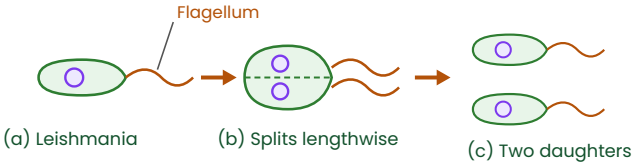
- **Fission:** a single-celled organism splits into two or more **new cells**; the parent cell disappears into its daughters
- **Binary fission:** division into **two** equal halves, e.g. **Amoeba**, bacteria, **Leishmania**, Paramecium

Binary fission in Amoeba ★★★



1. The **nucleus divides** first (DNA copied and shared)
2. The **cytoplasm** then constricts in the middle
3. Two **daughter Amoebae** are formed; Amoeba can split in **any plane**

Binary fission in Leishmania ★★☆

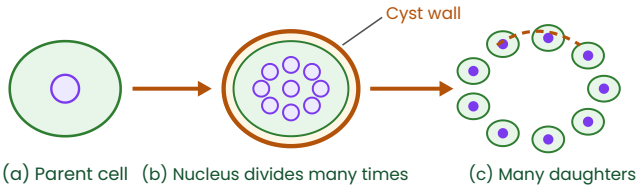


- Leishmania causes **kala-azar**; it has a whip-like **flagellum** at one end
- It splits in a **definite orientation** (lengthwise) with respect to this structure, unlike Amoeba

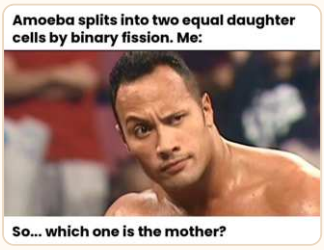
BUJHO *Amoeba aur Leishmania dono binary fission karte hain, toh fark kya hai?*

PAHELI *Amoeba **kisi bhi plane** mein divide ho sakta hai. Leishmania ek **definite plane** (lengthwise, flagellum ke respect mein) mein hi divide hota hai.*

Multiple fission in Plasmodium ★★★



- **Multiple fission:** one cell divides into **many** daughter cells at the same time
- Plasmodium (the **malaria** parasite) forms many daughter cells; the nucleus divides many times first
- In many organisms a **cyst** (protective wall) forms in unfavourable conditions; it bursts later to release the daughters



MEME BREAK *Binary fission mein parent hi gayab ho jaata hai, toh "mother" kaun? **Dono daughters barabar** hain, bhai.*

YAAD RAKHNE KI TRICK

"AL Binary, PM Multiple"

Amoeba and **L**eishmania do **binary** fission; **P**lasmodium does **M**ultiple fission. Like AL and PM on a clock!

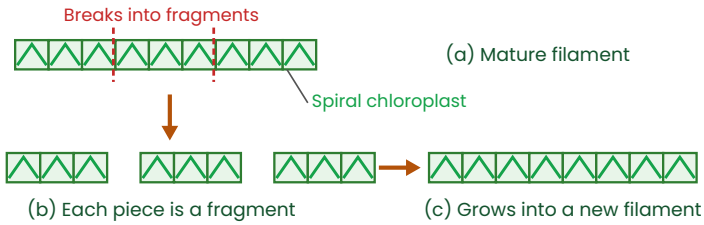
EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Under ideal conditions some bacteria (e.g. *E. coli*) can divide by binary fission about every **20 minutes**, so one cell can become millions in a few hours. This is why food spoils fast in warm weather.

4 FRAGMENTATION ★★☆

DEFINITION

- **Fragmentation:** a simple multicellular organism breaks into smaller pieces (**fragments**) on maturing; each fragment grows into a new individual, e.g. **Spirogyra**



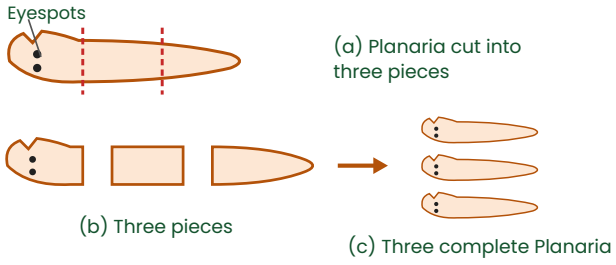
WHY NOT ALL MULTICELLULAR ORGANISMS?

Complex organisms have **specialised** tissues and organs; a random piece cannot rebuild the whole body. They need **specific reproductive cells**, made in reproductive organs.

5 REGENERATION ★★★

DEFINITION

- **Regeneration:** the ability of a fully differentiated organism to give rise to new individuals from its **body parts**, e.g. **Planaria**, **Hydra**



How regeneration happens ★★☆

- It is carried out by **specialised cells**
- These cells **proliferate** (make large numbers of cells)
- The mass of cells then **differentiates** into different cell types and tissues, in an organised sequence (development)

REGENERATION IS NOT REPRODUCTION

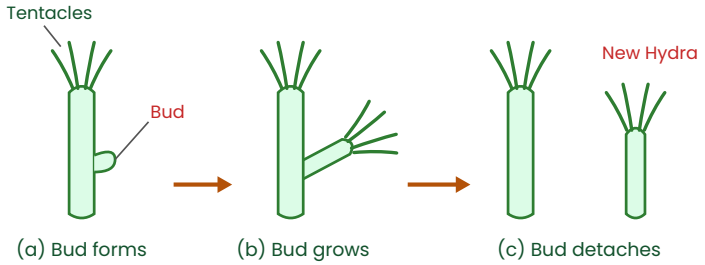
Most organisms do not normally **depend on being cut** into pieces to reproduce, so regeneration is not the same as reproduction.

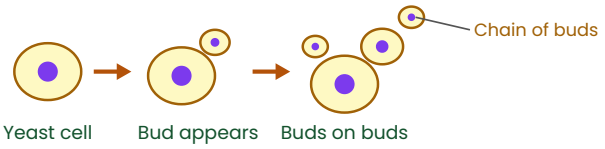
MEME BREAK *Planaria ko kaatoge toh **teen Planaria** mil jayenge. Gussa karna bekaar hai, ye toh regeneration ka master hai!*

6 BUDDING ★★★

DEFINITION

> **Budding:** a **bud** (outgrowth) develops on the parent due to repeated cell division at **one specific site**; it grows into a tiny individual and **detaches** on maturing, e.g. **Hydra**, **yeast**





Budding: key points ★★☆

> Hydra uses **regenerative cells** for budding
> In yeast the buds may stay attached and form a **chain**
> The bud gets a full DNA copy, so it is like the parent

BUJHO *Budding aur regeneration mein confusion hai, dono mein naya Hydra banta hai!*

PAHELI *Budding mein **bud apne aap** ek jagah ugta hai. Regeneration mein **kate hue tukde** se poora organism banta hai.*

MNEMONIC

"HY = Hi Bud! aur PH = Phir se ban jao"

Hydra and **Y**east say **Hi** to their **Bud**; **P**lanaria and **H**ydra **regenerate** (phir se ban jaate). Hydra appears in both!

7 ASEXUAL MODES AT A GLANCE ★★★

Mode	How it happens	Example
Binary fission	one cell splits into two	Amoeba, Leishmania
Multiple fission	one cell splits into many	Plasmodium
Fragmentation	body breaks into pieces	Spirogyra
Regeneration	body part forms a whole organism	Planaria, Hydra
Budding	outgrowth detaches	Hydra, yeast

NCERT in-text questions: model answers

1. **Importance of DNA copying?** It passes the body design to the next generation, and small copying errors give variation.
2. **Binary vs multiple fission?** Binary: one cell gives two (Amoeba). Multiple: one cell gives many at once (Plasmodium).
3. **Why can complex organisms not regenerate into new individuals?** Their bodies are highly specialised; most cells cannot divide and rebuild every organ.
4. **Is Hydra reproducing when it buds?** Yes: budding is asexual reproduction in Hydra.

× WRONG	✔ RIGHT
Amoeba divides in a fixed plane	Amoeba can divide in any plane; Leishmania has a definite plane
Regeneration is the same as reproduction	Organisms do not depend on being cut to reproduce
Hydra reproduces only by regeneration	Hydra shows both budding and regeneration
Variation helps every individual	Variation helps the species survive changes

Exam tips

- ✔ In diagram answers of binary fission, show **all stages** with the nucleus dividing first.
- ✔ Always name an **example** with each mode; examples carry marks.
- ✔ Write "Plasmodium: multiple fission", not binary fission.

8 PRACTICE MCQS

1. The basic event in reproduction is the: ★★★

(a) formation of gametes (b) creation of a DNA copy
(c) division of the cytoplasm (d) formation of a bud

2. Leishmania reproduces by: ★★★

(a) multiple fission (b) budding
(c) binary fission (d) fragmentation

3. Multiple fission is seen in: ★★★

(a) Amoeba (b) Plasmodium
(c) Hydra (d) yeast

4. Spirogyra reproduces asexually by: ★★☆
(a) budding (b) regeneration
(c) fragmentation (d) spore formation

5. Which organism shows both budding and regeneration? ★★☆
(a) Planaria (b) Hydra
(c) Spirogyra (d) Plasmodium

6. Variation is useful for: ★★☆
(a) the individual only (b) the survival of the species
(c) faster cell division (d) making exact copies

7. In yeast, buds that stay attached form a: ★☆☆
(a) cyst (b) filament
(c) chain (d) sporangium

8. Regeneration is carried out by: ★★☆
(a) gametes (b) specialised cells
(c) the cyst wall (d) the flagellum

9. **Assertion (A):** Fragmentation is not seen in complex multicellular organisms.
Reason (R): Complex organisms are made of specialised cells, tissues and organs. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Variation in a population helps the species survive.
Reason (R): DNA copying in reproduction is never absolutely accurate. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. **(2 marks)** Distinguish between binary fission in Amoeba and in Leishmania.
2. **(3 marks)** Why is variation beneficial to the species but not necessarily to the individual?
3. **(3 marks, diagram)** Draw labelled diagrams of budding in Hydra. Can Hydra also regenerate?

Answer key

MCQs: 1 (b), 2 (c), 3 (b), 4 (c), 5 (b), 6 (b), 7 (c), 8 (b), 9 (a), 10 (b)

1 Amoeba vs Leishmania
> Amoeba: splits in **any plane**; Leishmania: **definite** lengthwise plane, with respect to its flagellum (1 + 1)

2 Variation
> Copying errors give variation; when the niche changes (heat), some variants survive and the **species** continues
> An individual with a big change may not survive (1 + 1 + 1)

3 Hydra
> Stages: bud forms, grows tentacles, detaches (2); yes, Hydra regenerates from its body parts (1)

NITESH SIR SAYS
*Har mode ke saath **ek example aur ek diagram** yaad karo, number pakke. **Kal milte hai Day 52 pe!***

Today's checklist: tick before you sleep

- ☐ I can explain why DNA copying leads to variation
- ☐ I can draw binary fission in Amoeba and Leishmania
- ☐ I know multiple fission in Plasmodium
- ☐ I can explain fragmentation and regeneration
- ☐ I can draw budding in Hydra and yeast

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–3 phir se padho.

LEVEL 51 COMPLETE +125 XP earned. Next: Level 52

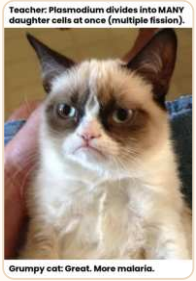
BUJHO *Samajh gaya! Fission, fragmentation, regeneration, budding: sab ek parent se.*

PAHELI *Shabaash! Kal Level 52: plants mein reproduction aur flower ka magic.*

Day 51 hit-list: Eduspark PYQ analysis

Topic	Rating
Binary fission in Amoeba (diagram)	★★★
Budding in Hydra (diagram)	★★★
Importance of variation	★★★
Regeneration in Planaria	★★☆
Multiple fission, fragmentation	★★☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.



MEME BREAK
*Plasmodium ek cell se **bahut saare** bana deta hai. Isliye mosquito net lagao, grumpy cat ki tarah gussa mat karo!*

DAY 52 LEVEL 52

BIOLOGY | CHAPTER 11 | PART 2 ASTRA SERIES

PLANT REPRODUCTION & THE FLOWER

Goal: learn how plants multiply and how a flower makes seeds.

by Nitesh Prakash

01 Vegetative | 02 Tissue Culture | 03 Spores | 04 Flower LS | 05 Fertilisation | +125 XP today | Level 52 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

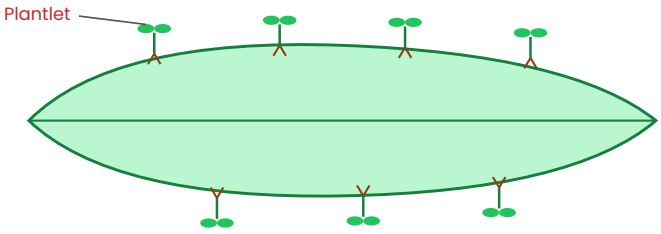
1 VEGETATIVE PROPAGATION ★★★

DEFINITION

- **Vegetative propagation:** new plants develop from **vegetative parts** of the parent (root, stem, leaf) under suitable conditions, without seeds
- It is a mode of **asexual** reproduction; the new plants are genetically like the parent

Natural methods (NCERT examples) ★★★

Plant	Part used	How
Bryophyllum	leaf	buds in the leaf-margin notches fall on soil and grow
Potato	stem (tuber)	"eyes" are buds that sprout into shoots
Sugarcane	stem	stem pieces with nodes are planted
Rose	stem	stem cuttings strike roots
Banana	stem (suckers)	no seeds; grown only vegetatively



Buds in the notches of the leaf margin grow into new plants

Artificial methods ★★☆

- **Cutting:** a piece of stem with buds is planted in moist soil (rose, sugarcane)
- **Layering:** a branch is bent and covered with soil while still on the parent; it roots, then is cut off (jasmine)
- **Grafting:** a cut stem (scion) of one plant is joined to the rooted stem (stock) of another so they grow as one (mango)
- Gardeners use these to copy a plant with **good qualities** exactly

NCERT ACTIVITY

Place a **Bryophyllum leaf** on wet cotton and a **potato** piece with an "eye" in moist soil; in a few days plantlets and shoots appear.

Advantages of vegetative propagation ★★★

1. Plants bear **flowers and fruits earlier** than those grown from seeds
2. Plants that have **lost the capacity to make seeds** (banana, orange, rose, jasmine) can be grown
3. All new plants are **genetically similar** to the parent, so good characters are kept

★★★ *"Write two advantages of vegetative propagation." Nitesh Bhaiya ki guarantee: ye 2 marks har saal ghoom ke aata hai!*

MNEMONIC

"Patti Bryo ki, Tana Aloo ka"

Bryophyllum grows from the **leaf** (patti); potato grows from the **stem** (tana, the tuber). Aloo is a stem, not a root!

MEME BREAK *Aloo **stem** hai, root nahi. Uski "aankhein" actually **buds** hain. Hello there, confusion!*

2 TISSUE CULTURE ★★☆

Steps

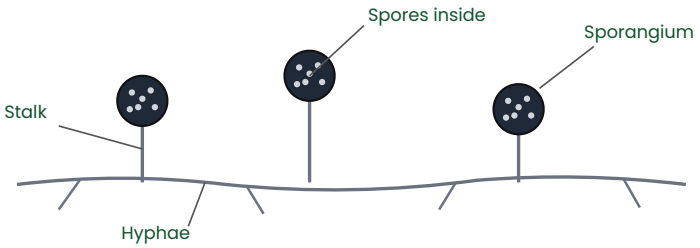
1. Cells are taken from the **growing tip** of a plant
2. They are placed in an **artificial nutrient medium** and divide to form a mass of cells: the **callus**
3. The callus is moved to a medium with **hormones** for growth and differentiation
4. **Plantlets** form; they are shifted to soil to grow into mature plants

Uses: many plants from **one parent**, disease-free, fast; common for **ornamental plants** such as orchids.

3 SPORE FORMATION ★★☆

DEFINITION

- **Spore formation:** many simple multicellular organisms make **spores** in blob-like structures called **sporangia**, e.g. **Rhizopus** (bread mould)



Rhizopus: key points ★★☆

- Thread-like **hyphae** spread on the bread; vertical stalks carry round **sporangia**
- Spores are tiny **asexual reproductive bodies** made in large numbers
- Spores have **thick walls** that protect them till they meet a **moist surface**, then they grow
- NCERT activity: keep a moist slice of bread in a cool, moist, dark place; a white-to-black cottony mould appears

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

A single sporangium can release thousands of spores, which float in air. This is why bread, fruit and even leather goods get mouldy in the rainy season.

4 WHY SEXUAL REPRODUCTION? ★★★

Two big reasons ★★★

- **Variation:** DNA copying errors pile up slowly; combining DNA from **two individuals** creates far more new combinations, which helps the species survive
- **Half chromosomes in gametes:** germ cells carry **half** the chromosomes and half the DNA, so when two fuse, the zygote gets the **normal number** back and DNA does not double every generation
- Germ cells are specialised: one is **large with stored food** (female), one is **small and motile** (male)

BUJHO *Agar gametes mein poore chromosomes hote toh kya problem thi?*

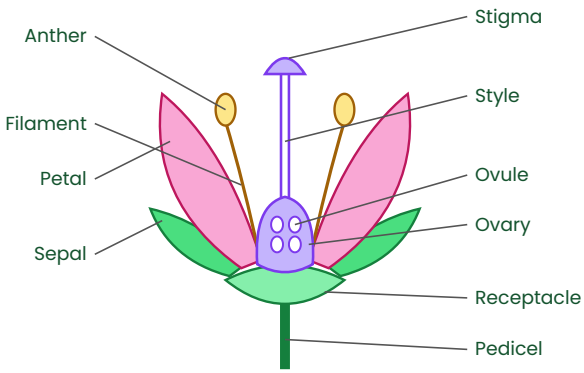
PAHELI *Har generation mein DNA **double** hota jaata! Isliye gametes mein **aadhe** chromosomes hote hain, zygote mein phir se **poore**.*

5 SEXUAL REPRODUCTION IN FLOWERING PLANTS ★★★

DEFINITION

- The **flower** is the reproductive organ; its reproductive parts are the **stamens** (male) and the **carpel or pistil** (female)

Longitudinal section (LS) of a flower ★★★



Stamen = anther + filament (male part, makes pollen) | **Pistil (carpel)** = stigma + style + ovary (female part, has ovules)

Unisexual vs bisexual flowers ★★☆

Type	Has	Examples
Unisexual	either stamens or pistil	papaya, watermelon
Bisexual	both stamens and pistil	Hibiscus, mustard

BUJHO *Hibiscus aur papaya dono phool hain, toh unisexual aur bisexual ka chakkar kya hai?*

PAHELI *Hibiscus **bisexual** hai: ek hi phool mein stamens aur pistil dono. Papaya **unisexual** hai: ek phool mein sirf ek.*

MEME BREAK *Papaya ka phool **unisexual** hota hai, sirf stamens ya sirf pistil. Bisexual flower ke saamne bhi wo kam nahi hai!*

Pollination ★★★

- **Pollination:** transfer of **pollen grains** from the anther to the **stigma**
- **Self-pollination:** pollen reaches the stigma of the **same flower**
- **Cross-pollination:** pollen reaches the stigma of **another flower** of the same kind, carried by wind, water or animals (insects, birds)

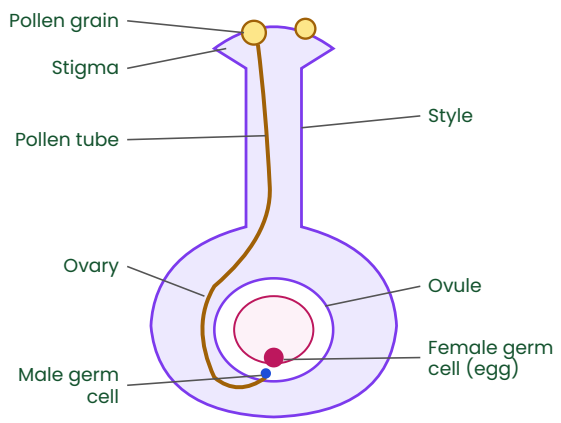
YAAD RAKHNE KI TRICK

"Self = apne ghar ka phool, Cross = padosi ka phool"

Same flower: self-pollination. Another flower of the same kind: cross-pollination. Agents: **Wind, Water, Animals** (WWA).

6 FERTILISATION AND AFTER ★★★

Germination of pollen on stigma ★★★



1. The pollen grain lands on the **stigma** and grows a **pollen tube**
2. The tube grows down through the **style** and reaches the **ovule** in the ovary
3. The **male germ cell** travels down the tube and fuses with the **female germ cell** (egg): this is **fertilisation**, giving a **zygote**

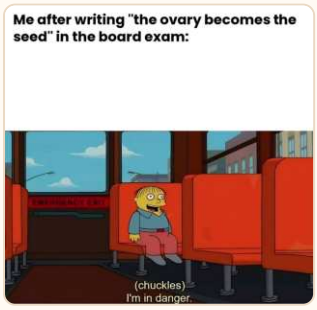
Changes after fertilisation ★★★

This part	becomes
Zygote	**embryo** (divides many times inside the ovule)
Ovule	**seed** (develops a tough coat)
Ovary	**fruit** (grows rapidly and ripens)
Petals, sepals, stamens, style, stigma	usually **shrivel and fall off**

MNEMONIC

"Chhota ovULE se chhota SEED, bada ovARY se bada FRUIT"

The small ovule turns into the seed; the bigger ovary turns into the fruit that holds the seeds. Zygote se embryo, simple!



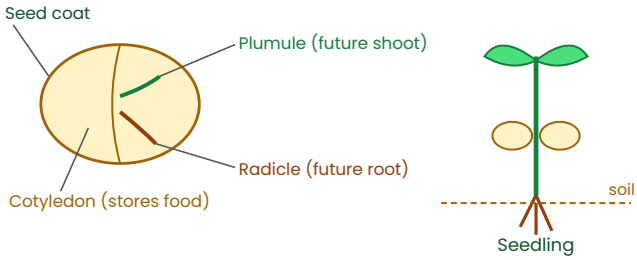
MEME BREAK

*Board exam mein "ovary becomes seed" likh diya? **Ovule → seed** aur **ovary → fruit**. Warna danger pakka!*

BUJHO *Pollination aur fertilisation ek hi cheez hai na?*

PAHELI *Nahi! **Pollination** = pollen ka anther se stigma tak jaana. **Fertilisation** = male aur female germ cell ka **fusion**. Pehle pollination, phir fertilisation.*

Seed germination ★★☆



- The seed holds the future plant, the **embryo**: **plumule** (future shoot), **radicle** (future root) and **cotyledons** (stored food)
- With water, air and a suitable temperature the embryo grows into a **seedling**: this is **germination**

MEME BREAK

*Microscope ke neeche pollen tube ka wait karte karte main bhi skeleton ban gaya. **Patience = fertilisation!***

× WRONG	✔ RIGHT
Potato is a modified root	Potato is a modified stem (tuber)
Ovary becomes the seed	Ovule becomes the seed; ovary becomes the fruit
Pollination and fertilisation are the same	Pollination is transfer of pollen; fertilisation is fusion of gametes
Gametes have the full number of chromosomes	Gametes have half; the zygote restores the full number

Exam tips

- ✔ In the flower diagram, label **all six**: sepal, petal, stamen (anther, filament), stigma, style, ovary with ovule.
- ✔ For pollen germination, draw the **pollen tube** reaching the **ovule** and label male and female germ cells.
- ✔ Give one example with every method of vegetative propagation.

7 PRACTICE MCQS

1. In Bryophyllum, new plants arise from buds on the: ★★★

(a) roots (b) stem
(c) leaf margin (d) flowers

2. Which of these is NOT a reason to use vegetative propagation? ★★☆

(a) earlier flowers and fruits (b) growing seedless plants like banana
(c) producing more variation (d) keeping desired characters

3. Rhizopus reproduces asexually by: ★★☆
(a) budding (b) spore formation
(c) fragmentation (d) binary fission

4. The female reproductive part of a flower is the: ★★★
(a) stamen (b) petal
(c) pistil (d) sepal

5. An example of a unisexual flower is: ★★☆
(a) Hibiscus (b) mustard
(c) papaya (d) rose

6. After fertilisation, the ovary develops into the: ★★★
(a) seed (b) fruit
(c) embryo (d) seed coat

7. The pollen tube carries the male germ cell to the: ★★★
(a) anther (b) ovule
(c) sepal (d) filament

8. Gametes have half the number of chromosomes so that: ★★☆
(a) they divide faster (b) the zygote gets the normal number
(c) they can swim (d) variation stops

9. **Assertion (A):** Tissue culture can give many plants from one parent in a short time.
Reason (R): In tissue culture, cells from the growing tip form a callus that is made to form plantlets. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Cross-pollination needs an agent such as wind, water or animals.
Reason (R): In cross-pollination, pollen of a flower reaches the stigma of the same flower. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. **(2 marks)** List two advantages of vegetative propagation with one example each.
2. **(3 marks, diagram)** Draw a labelled diagram showing germination of pollen on the stigma.
3. **(5 marks)** Draw the LS of a flower. Explain pollination, fertilisation and the changes that follow in the flower.

NITESH SIR SAYS
Flower ka LS aur pollen tube wala diagram ***daily ek baar*** *banao. Board mein diagram = pakke number.* ***Kal milte hai Day 53 pe!***

Answer key

MCQs: 1 (c), 2 (c), 3 (b), 4 (c), 5 (c), 6 (b), 7 (b), 8 (b), 9 (a), 10 (c)

1 Vegetative propagation
> Earlier flowers and fruits (rose); seedless plants grown (banana); 1 + 1

2 Pollen germination
> Pollen grain on stigma, pollen tube through style, ovule with egg; 3 labelled parts = 3 marks

3 Flower LS
> Diagram with six labels (2); pollination (1); fertilisation (1); ovule to seed, ovary to fruit, others fall off (1)

Today's checklist: tick before you sleep

- ☐ I can explain vegetative propagation with examples
- ☐ I know the steps of tissue culture
- ☐ I can draw Rhizopus and explain spores
- ☐ I can draw and label the LS of a flower
- ☐ I can explain fertilisation and ovule to seed, ovary to fruit

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2–3 phir se padho.

LEVEL 52 COMPLETE +125 XP earned. Next: Level 53

BUJHO *Samajh gaya! Ovule se seed, ovary se fruit, aur pollen tube ovule tak.*

PAHELI *Shabaash! Kal Level 53: reproduction in human beings.*

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

In flowering plants one male gamete fuses with the egg (zygote) and the other fuses with two polar nuclei to form food tissue (endosperm). This is called double fertilisation.

Day 52 hit-list: Eduspark PYQ analysis

Topic	Rating
LS of flower (diagram)	★★★
Germination of pollen on stigma (diagram)	★★★
Ovule to seed, ovary to fruit	★★★
Advantages of vegetative propagation	★★★
Spore formation in Rhizopus	★★☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 53 | LEVEL 53

BIOLOGY | CHAPTER 11 | PART 3 | ASTRA SERIES

REPRODUCTION IN HUMAN BEINGS

Goal: master human reproduction and reproductive health.

by Nitesh Prakash

01 Puberty | 02 Male System | 03 Female System | 04 Placenta | 05 Reproductive Health | +125 XP today | Level 53 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 PUBERTY: THE BODY GETS READY ★★☆

DEFINITION

- **Puberty:** the stage of adolescence when the reproductive tissues mature and the body becomes capable of sexual reproduction
- In boys it usually begins around **13–14 years**, in girls a little earlier, around **10–12 years** (varies person to person)
- Body growth slows down only after the reproductive organs mature, because making germ cells needs a **mature body**

Changes at puberty ★★☆

Both	Boys	Girls
Thick hair in armpits and genital area	Facial hair: beard, moustache	Breast size increases
Thinner hair on arms, legs, face	Voice begins to crack and deepen	Skin of nipples darkens
Skin becomes oily, pimples may appear	Testes start making sperms; penis occasionally erect	Menstruation (periods) begins
Growth spurt; awareness of own body	Driven by **testosterone**	Driven by **oestrogen**

BUJHO *Mere dost ki moustache aa gayi, meri nahi. Kya mujhme kuch gadbad hai?*

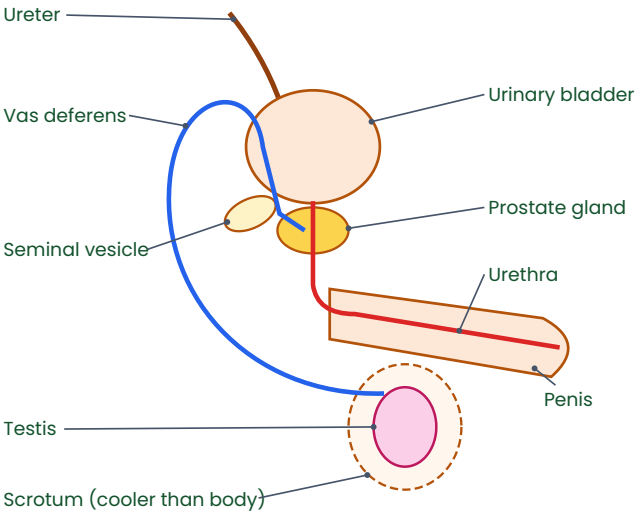
PAHELI *Bilkul nahi Bujho! Puberty ka time har insaan mein alag hota hai. Kuch log 11 mein, kuch 15 mein. Sab normal hai.*

WHY SO MANY CHANGES?

Sexual reproduction needs **germ cells** (sperms and eggs) and, in a woman, a body that can carry a baby. Puberty prepares both. These changes are slow and gradual, and are signals that the body is **sexually maturing**, not that a person is ready to be a parent.

★★★ *Boys, girls aur common changes ki teen list alag likho, 2 marks seedhe. Nitesh Bhaiya ki guarantee!*

2 MALE REPRODUCTIVE SYSTEM ★★★ MOST ASKED



Parts and their jobs ★★★

Part	Function
Testes (2)	Make **sperms** (male germ cells) and the hormone **testosterone**
Scrotum	Pouch that holds the testes **outside** the abdomen, about 2–3 °C cooler
Vas deferens	Carries sperms from the testis; joins the tube from the bladder
Seminal vesicle, prostate	Add fluid: sperms move easily and get **nutrition**
Urethra	Common passage for **sperms and urine**
Penis	Transfers sperms into the female body

★★★ *Testes bahar kyun? Sperm banane ke liye body se kam temperature chahiye. Har saal poochte hai, Nitesh Bhaiya ki guarantee!*

Path of sperms: in order ★★★

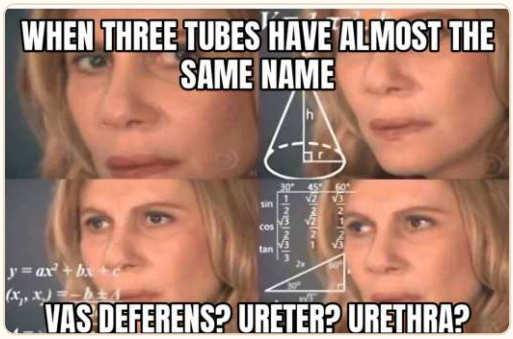
Testis → Vas deferens → (joined by seminal vesicle and prostate fluid) → Urethra → Penis → outside

MNEMONIC: YAAD RAKHNE KI TRICK

"T-V SUP": **T**estis, **V**as deferens, **S**eminal vesicle, **U**rethra, **P**enis. Sperm ka safar ek T-V SUP jaisa: seedha forward!

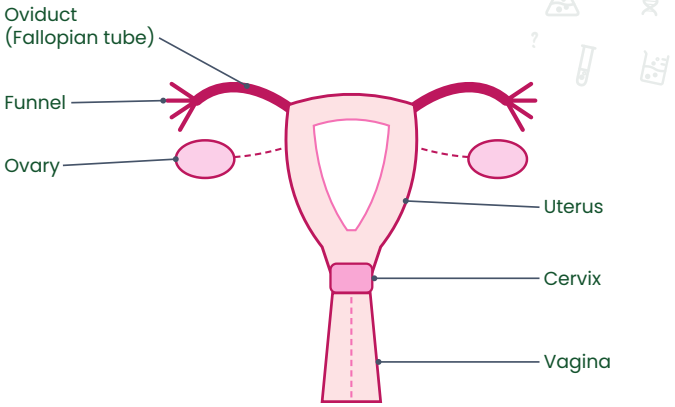
Hormone and germ cell

- **Testosterone** (from testes): regulates sperm formation and brings about the puberty changes in boys
- **Sperm:** tiny cell, mostly genetic material, with a **long tail** that helps it swim towards the female germ cell
- Sperms travel in a fluid (semen) made by the seminal vesicles and prostate gland



MEME BREAK
Vas deferens = sperm, ureter = urine, urethra = dono ka exit!

3 FEMALE REPRODUCTIVE SYSTEM ★★★



Parts and their jobs ★★★

Part	Function
Ovaries (2)	Make **eggs** (ova) and the hormones **oestrogen** and progesterone
Oviduct (fallopian tube)	Carries the egg from the ovary to the uterus; **site of fertilisation**
Uterus (womb)	Elastic bag-like organ where the baby develops
Cervix	Narrow neck by which the uterus opens into the vagina
Vagina	Receives sperms; also the birth canal

- ✔ A girl is born with **thousands of immature eggs** in her ovaries; at puberty some start maturing
- ✔ **One egg** is released every month by one of the two ovaries
- ✔ The two oviducts unite into the uterus

★★★ Oviduct = fertilisation, uterus = baby ka ghar, cervix = gate. Diagram mein ye teen label Nitesh Bhaiya ki guarantee!

4 FERTILISATION TO BIRTH ★★★ MOST ASKED

1. **Fertilisation:** sperms enter through the vagina, travel up and meet the egg in the **oviduct**; their fusion gives a **zygote**
2. **Division:** the zygote divides repeatedly and forms a ball of cells, the early embryo
3. **Implantation:** the embryo gets embedded in the thick, **blood-rich lining** of the uterus
4. **Placenta** forms and nourishes the embryo (see box); the embryo grows into a foetus
5. **Gestation** is about **9 months** (about 40 weeks); birth happens by rhythmic contractions of the uterus muscles

Placenta: the life-line ★★★

- A special **disc-like tissue** embedded in the uterine wall, joining the embryo to the mother
- On the embryo's side it has finger-like **villi**; on the mother's side are **blood spaces** that surround the villi
- Villi give a **large surface area**: glucose and oxygen pass from mother to embryo
- Wastes made by the embryo pass into the mother's blood through the placenta for removal

BUJHO *Baby ko khana mummy ke pet se seedha milta hai kya, jaise hum khate hai?*

PAHELI *Nahi! Baby ko digested food aur oxygen **placenta ke villi** se blood ke through milta hai. Mother aur baby ka blood seedha mix nahi hota.*

5 MENSTRUATION ★★☆

1. Every month the uterus **prepares** for a possible pregnancy: its inner lining becomes **thick and spongy** with a rich blood supply
2. The released egg lives for only about **one day**; if it is not fertilised, the thick lining is not needed
3. The lining **breaks down** and comes out through the vagina as blood and mucus: **menstruation**
4. This cycle repeats roughly every **28 days** and each period usually lasts about **2 to 8 days**

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

The first menstruation at puberty is called **menarche**; the stopping of the cycle at around 45–50 years is called **menopause**.

6 REPRODUCTIVE HEALTH ★★★ MOST ASKED

DEFINITION

- **Reproductive health:** physical, mental, social and emotional well-being in all matters of reproduction, including the freedom to decide when, and how many, children to have

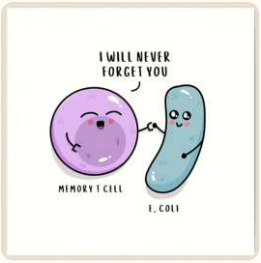
EXAM TIP

Write the **site** with each event: fertilisation in the **oviduct**, implantation in the **uterus** wall, exchange at the **placenta**. Sites alone carry marks in MCQs.

Sexually transmitted diseases (STDs) ★★★

Caused by	Diseases
Bacteria	Gonorrhoea, Syphilis
Viruses	Warts, HIV-AIDS

They spread through sexual contact; HIV can also spread through infected blood and needles. A **condom** (barrier) reduces the risk to a large extent.



MEME BREAK
Memory T cell kabhi germ ko nahi bhoolta. Par HIV inhi immune cells ko target karta hai, isliye AIDS mein body ki defence ki memory bhi kamzor ho jaati!

Contraception: preventing pregnancy ★★★

Method	How it works
Barrier (condom)	Stops sperms reaching the egg; also protects against STDs
Hormonal (oral pills)	Change hormone balance so **eggs are not released**; may have side effects
IUCD (loop, Copper-T)	Placed in the uterus to prevent pregnancy; may cause irritation of the uterus
Surgical	**Vasectomy:** vas deferens blocked (male). **Tubectomy:** oviduct blocked (female)

MNEMONIC: YAAD RAKHNE KI TRICK

"B-H-I-S" = **Bus Hai Itna Simple**: **B**arrier, **H**ormonal, **I**UCD, **S**urgical. Surgical mein: **V**asectomy = **V**as deferens (male), **T**ubectomy = fallopian **T**ube (female).

MEME BREAK
*Copper-T, loop = **IUCD**. Option mein dikhe toh pehchaan lena, ye surgical nahi hai, device hai!*

Female foeticide and sex ratio ★★☆

- Surgery can be misused to remove a pregnancy for wrong reasons, e.g. killing a female foetus after illegal sex determination: **female foeticide**
- Prenatal sex determination is **prohibited by law** in India
- It has lowered the **child sex ratio** sharply in some places; a healthy society needs a balanced ratio of girls and boys

MEME BREAK
*Copper-T ek **device** hai, surgery nahi!*

× WRONG	✓ RIGHT
Fertilisation happens in the uterus	It happens in the **oviduct** (fallopian tube)
Ureter carries sperms	**Vas deferens** carries sperms; ureter carries urine
Oral pills protect against STDs	Only the **condom** (barrier) gives protection from STDs
Placenta mixes mother's and baby's blood	Exchange happens across the **villi**; the bloods do not mix

NITESH SIR SAYS
*Is chapter ko sharmaa ke mat chhodo. Diagrams aur functions **board mein pakke 3–5 marks** laate hai. Labelled diagram roz ek baar banao!*

7 PRACTICE MCQS

1. Sperms are formed in the: ★★★
(a) vas deferens (b) testes
(c) prostate gland (d) seminal vesicle

2. The testes lie in the scrotum outside the abdomen because sperm formation needs: ★★★
(a) a higher temperature (b) a lower temperature
(c) more oxygen (d) more light

3. Fertilisation of the egg in human females normally takes place in the: ★★★
(a) ovary (b) uterus
(c) oviduct (d) vagina

4. The common passage for urine and sperms in human males is the: ★★☆
(a) ureter (b) vas deferens
(c) urethra (d) seminal vesicle

5. Which of these is a viral STD? ★★☆
(a) Gonorrhoea (b) Syphilis
(c) AIDS (d) Tuberculosis

6. Copper-T is an example of: ★★★
(a) barrier method (b) IUCD
(c) surgical method (d) oral pill

7. Which method blocks the vas deferens? ★★☆
(a) tubectomy (b) vasectomy
(c) Copper-T (d) condom

8. In the human embryo, the villi of the placenta mainly: ★★☆

(a) produce eggs
(b) increase surface area for exchange
(c) make testosterone
(d) store urine

9. **Assertion (A):** The lining of the uterus breaks down every month if the egg is not fertilised.
Reason (R): The thick, spongy lining is needed only to nourish an implanted embryo. ★★★

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

10. **Assertion (A):** Oral contraceptive pills protect a person from sexually transmitted diseases.
Reason (R): Oral pills change the hormonal balance so that eggs are not released. ★★☆

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Short and long answer questions

1. Why are the testes located outside the abdominal cavity? Name the hormone they secrete and give one function of it. **(2 marks)**
2. What is placenta? Describe its function in the human body. **(3 marks)**
3. Draw a labelled diagram of the human female reproductive system. What happens to the egg if it is not fertilised? **(5 marks)**

Case-based question (4 marks)

Reena's biology teacher shows a chart of the female reproductive system. An egg is released from the right ovary on day 14 of the cycle and is fertilised a day later.

1. Name the part where fertilisation took place. **(1)**
2. Where will the embryo get implanted? **(1)**
3. Name the tissue that supplies the embryo with food and oxygen, and state how its structure suits this job. **(2)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Testes make sperms and testosterone.
2	(b)	About 2–3 °C below body temperature.
3	(c)	Sperm meets egg in the oviduct.
4	(c)	Urethra carries both.
5	(c)	HIV is a virus; the other two are bacterial.
6	(b)	Copper-T is an intra-uterine device.
7	(b)	Vasectomy = vas deferens blocked.
8	(b)	More area for food, O_2 and wastes.
9	(a)	R explains why the lining is shed.
10	(d)	Pills give no STD protection; R is true.

Model answers

1 Testes outside

- Sperm formation needs a temperature lower than body temperature (1 mark)
- Testosterone: regulates sperm formation and puberty changes in boys (1 mark)

2 Placenta

- Disc-like tissue in the uterine wall joining embryo and mother
- Villi (embryo side) bathed in maternal blood spaces: large surface area
- Glucose and O_2 pass to the embryo; its wastes pass to the mother's blood

3 Female system

- Diagram: ovaries, oviducts with funnels, uterus, cervix, vagina (any five labels, 3 marks)
- Unfertilised egg lives about one day; the uterine lining breaks down and is shed as menstruation (2 marks)

Case-based answers

- (i) Oviduct (fallopian tube)
- (ii) In the thick, blood-rich lining of the uterus
- (iii) Placenta: its villi give a large surface area, bathed in the mother's blood

Day 53 checklist: tick before you sleep

- ☐ I can list the changes at puberty in boys and girls
- ☐ I can draw and label the male reproductive system
- ☐ I can draw and label the female reproductive system
- ☐ I can explain fertilisation, implantation and the placenta
- ☐ I know the STDs and all four contraceptive methods

My score: ____ / 5

LEVEL 53 COMPLETE +125 XP earned. Next: Level 54

Day 53 hit-list: Eduspark PYQ analysis

Topic	Rating
Male / female system diagram and labels	★★★
Placenta: structure and function	★★★
Contraceptive methods; STDs	★★★
Why testes lie in the scrotum	★★☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 54
LEVEL 54

BIOLOGY | CHAPTER 11 | BOSS LEVEL

CHAPTER 11 BOSS LEVEL

Goal: test all of reproduction in 90 minutes.

by Nitesh Prakash

01 Revision Map | 02 MCQs | 03 Short Answers | 04 Case & Long | 05 Answer Key | +200 XP Boss Level | Level 54 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 11 REVISION MAP ★★★ MOST ASKED

Chapter 11 mind map: revise in 5 minutes

ASEXUAL
Binary and multiple fission, fragmentation, regeneration, budding, spores

VEGETATIVE ★★★
Bryophyllum leaf, potato, layering, grafting; tissue culture

HOW DO ORGANISMS REPRODUCE?

FLOWERING PLANTS
Stamen + carpel, pollination, pollen tube, ovule to seed, ovary to fruit

HUMANS ★★★
Testes, ovaries, oviduct, placenta; STDs and contraception

Must-know pairs: write them daily ★★★

Organism / part	Remember
Amoeba, Leishmania	Binary fission
Plasmodium	Multiple fission
Spirogyra	Fragmentation
Planaria, Hydra	Regeneration
Hydra, yeast	Budding
Rhizopus	Spore formation (sporangia)
Bryophyllum	Buds on leaf margins
Anther / ovary	Pollen grains / ovules

BUJHO *Ovary aur ovule same cheez hai na? Dono seed ban jaate hai?*

PAHELI *Nahi Bujho! **Ovule** seed banta hai aur **ovary** fruit. Ek ovary ke andar kai ovules ho sakte hai, isliye ek fruit mein kai seeds.*

MEME BREAK
*Kal Biology test hai aur aaj Boss Level? Perfect! Ye page solve karo, kal ka test **easy level** lagega.*

How to attack the Boss: 90 minutes

1. **Section A** (12 marks): 15 minutes. Mark doubtful ones and come back
2. **Sections B and C** (15 marks): 30 minutes. Draw neat, labelled diagrams in pencil
3. **Sections D and E** (18 marks): 40 minutes. Read every case twice
4. Last 5 minutes: check labels, arrows and units of every diagram
5. Start with the section you are most sure of; never leave an MCQ blank

2 SECTION A: MCQS (1 MARK EACH) ★★★

1. Amoeba reproduces by: ★★★
(a) budding (b) binary fission
(c) fragmentation (d) spore formation

2. Multiple fission is seen in: ★★★
(a) Plasmodium (b) Hydra
(c) yeast (d) Spirogyra

3. Which organism reproduces by budding? ★★☆
(a) Planaria (b) Leishmania
(c) Hydra (d) Rhizopus

4. Spirogyra breaks into pieces and each piece grows into a new individual. This is: ★★☆
(a) regeneration (b) fragmentation
(c) budding (d) fission

5. Buds produced in the notches of the leaf margin grow into new plants in: ★★★
(a) rose (b) potato
(c) Bryophyllum (d) banana

6. Rhizopus reproduces asexually by: ★★☆
(a) binary fission (b) budding
(c) spore formation (d) fragmentation

7. Pollen grains are produced in the: ★★☆
(a) ovary (b) anther
(c) stigma (d) style

8. After fertilisation, the ovule develops into the: ★★★
(a) fruit (b) seed
(c) embryo sac (d) flower

9. Which of these is NOT a part of the carpel? ★★☆

(a) stigma (b) style
(c) ovary (d) anther

10. Which contraceptive method also protects against STDs? ★★★

(a) oral pills (b) Copper-T
(c) condom (d) vasectomy

11. **Assertion (A):** DNA copying during reproduction is not perfectly accurate.
Reason (R): Variations produced during reproduction help a species survive in a changing environment. ★★★

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

12. **Assertion (A):** Human testes are located outside the abdominal cavity in the scrotum.
Reason (R): Sperm formation requires a temperature lower than the normal body temperature. ★★★

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

3 SECTIONS B AND C ★★★ MOST ASKED

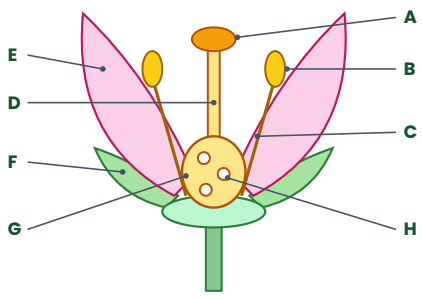
Section B: Short Answers (2 marks each)

B1. What is DNA copying? Why is it essential for reproduction?
B2. Differentiate between self-pollination and cross-pollination.
B3. List two advantages of vegetative propagation. Name one plant grown this way.

MEME BREAK
*Text ratta maar liya, diagram skip? Boss Level mein **flower LS** zaroor aata hai!*

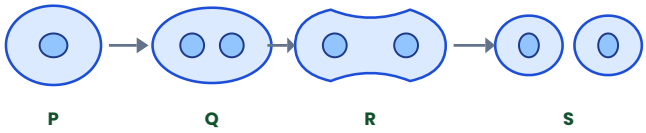
Section C: Short Answers (3 marks each)

C1. The figure shows the longitudinal section of a flower. Name the parts A to H. Which parts form the male and the female reproductive organs?



Section C (continued)

C2. The figure shows stages P to S of a process. Name the process and one organism showing it. How is it different from multiple fission?



C3. Why do the testes lie in the scrotum? Name the hormone they make. What will happen if the vas deferens is blocked?

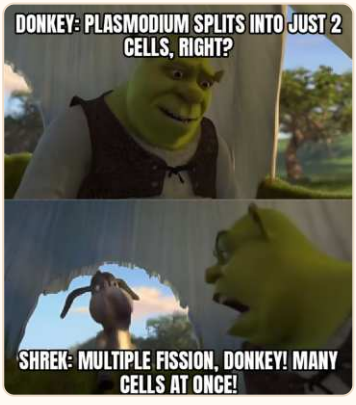
4 SECTIONS D AND E ★★★ MOST ASKED

Section D, Case 1: Meena's nursery (4 marks)

Meena grows roses and sugarcane from stem cuttings, bananas from suckers and Bryophyllum from leaves. She also grafts a good mango variety onto a strong rootstock.
(i) Name this group of methods. **(1)**
(ii) How does a Bryophyllum leaf give new plants? **(1)**
(iii) Give two advantages of these methods over growing from seeds. OR Why is tissue culture useful for nurseries? **(2)**

Case 2: Planning a healthy family (4 marks)

A health worker explains contraceptive methods to a young couple: condoms, oral pills, Copper-T, and surgery for those who want no more children.
(i) Which method also protects against STDs? Name one bacterial STD. **(1)**
(ii) How do oral pills prevent pregnancy? **(1)**
(iii) What is Copper-T? Name the surgical methods for males and females. **(2)**



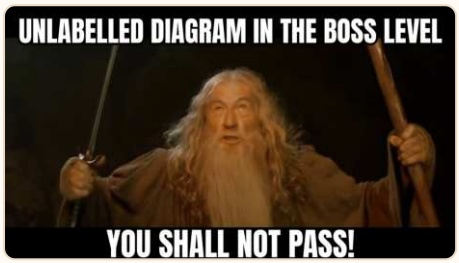
MEME BREAK
*Binary fission = **2** cells. Multiple fission (Plasmodium) = **bahut saare** cells ek saath. Donkey mat bano!*

Section E: Long Answers (5 marks each)

E1. Draw a labelled diagram showing the germination of pollen on the stigma. Describe fertilisation in a flowering plant and state what the zygote, ovule and ovary develop into.
E2. (a) Draw a labelled diagram of the human female reproductive system. (b) How does the developing embryo get nourishment? (c) What happens if the egg is not fertilised?
OR (a) Draw a labelled diagram of the human male reproductive system. (b) Give the functions of the testes, seminal vesicle and prostate gland.

Diagram drill: labels the examiner looks for ★★★

- **Binary fission (Amoeba):** nucleus dividing, cytoplasm dividing, two daughter cells
- **Budding (Hydra):** parent, bud, tentacles of the bud
- **Flower LS:** sepal, petal, stamen (anther, filament), carpel (stigma, style, ovary, ovule)
- **Pollen germination:** pollen grain, pollen tube, style, ovule, female germ cell
- **Human systems:** testis, vas deferens, urethra, prostate; ovary, oviduct, uterus, cervix, vagina



MEME BREAK

*Bina label ka diagram? **Marks nahi milenge!** Har diagram mein 4–6 labels, arrows ruler se.*

× WRONG	✓ RIGHT
Plasmodium divides by binary fission	**Multiple** fission: many daughter cells at once
Ovary becomes the seed	**Ovule** becomes the seed; ovary becomes the fruit
Pollen tube grows through the ovary wall first	It grows down the **style** to reach the ovule
Hydra only regenerates	Hydra shows both **budding** and regeneration

BOSS TIP

Har answer mein **keyword** likho: “multiple fission”, “micropyle”, “villi”, “implantation”. Examiner keywords dhoondhta hai, lambi kahani nahi.

5 ANSWER KEY AND MARKING POINTS ★★★

Answer key: Section A (MCQs)

Q	Ans	Why
1	(b)	Amoeba splits into two.
2	(a)	Plasmodium: many cells at once.
3	(c)	Hydra forms buds.
4	(b)	Pieces regrow: fragmentation.
5	(c)	Bryophyllum leaf-margin buds.
6	(c)	Sporangia release spores.
7	(b)	Anther makes pollen.
8	(b)	Ovule becomes the seed.
9	(d)	Anther belongs to the stamen.
10	(c)	Only a barrier blocks STDs.
11	(b)	Both true; R gives the benefit, not the cause.
12	(a)	R explains A.

Model answers: Sections B and C

B1 DNA copying

- Making a copy of the DNA of a cell before it divides; two copies separate into two cells (1)
- Passes the body design and information to the next generation; small errors give variation (1)

B2 Pollination

- Self: pollen reaches the stigma of the same flower or plant (1)
- Cross: pollen goes to a flower of another plant by wind, water or insects (1)

B3 Vegetative propagation

- Faster flowering and fruiting; all plants are genetically like the parent (1)
- Plants without seeds (banana, orange, rose) can be grown; e.g. potato (1)

C1 Flower LS

- A stigma, B anther, C filament, D style, E petal, F sepal, G ovary, H ovule (2)
- Male: stamen (B + C). Female: carpel or pistil (A + D + G) (1)

C2 Binary fission

- Binary fission in Amoeba; P to S: nucleus divides, then the cytoplasm (2)
- Multiple fission (Plasmodium) gives many daughter cells at once (1)

C3 Testes

- Sperm formation needs a lower temperature (1); testosterone (1)
- Sperms cannot be carried out, so no fertilisation: the idea behind vasectomy (1)

Model answers: Sections D and E

D1 Meena’s nursery

- (i) Vegetative propagation (1)
- (ii) Buds in the notches of the leaf margin fall on soil and grow into plants (1)
- (iii) Faster, identical to parent; seedless plants grown. OR Tissue culture gives many disease-free plants quickly (2)

D2 Healthy family

- (i) Condom; gonorrhoea or syphilis (1)
- (ii) They change the hormone balance so that eggs are not released (1)
- (iii) An IUCD placed in the uterus (1); vasectomy and tubectomy (1)

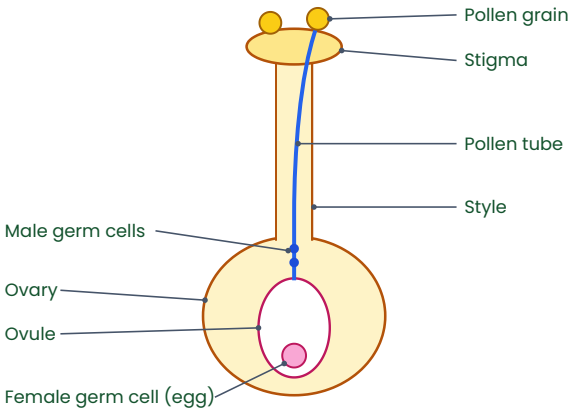
E1 Pollen germination

- Diagram: stigma, pollen grain, pollen tube, style, ovary, ovule, female germ cell (2)
- Pollen tube carries the male germ cell to the egg in the ovule; they fuse to form the zygote (1)
- Zygote → embryo, ovule → seed, ovary → fruit; petals, sepals, stamens wither (2)

E2 Female system

- Diagram: ovaries, oviducts with funnels, uterus, cervix, vagina (2)
- Placenta: villi bathed in mother’s blood supply glucose and oxygen; wastes go back (2)
- Lining breaks and is shed as menstruation (1). OR male: testes make sperms and testosterone; vesicles and prostate add nourishing fluid

Model diagram for E1: pollen on the stigma



★★★ *Pollen tube ka diagram har saal ka favourite hai. Nitesh Bhaiya ki guarantee!*

Bonus rapid-fire round: quick answers

Question	Answer
Process in which DNA makes a copy of itself	DNA replication (copying)
Organism that regenerates from body pieces	Planaria
Sac that holds spores in Rhizopus	Sporangium
Method to grow many plants from tissue	Tissue culture
Male reproductive part of a flower	Stamen
Where the pollen tube enters the ovule	Through the micropyle
Hormone made by the testes	Testosterone
Tissue that nourishes the human embryo	Placenta
Viral STDs	Warts, HIV-AIDS
Approximate human gestation	About 9 months

Score card: how did you fight the boss?

Marks out of 45	Your level
40 and above	Boss defeated: ready for boards
30 to 39	Strong: fix the diagrams once more
Below 30	Re-read Days 51–53, then retry

Day 54 checklist: tick before you sleep

- [ ] I attempted the full Boss test in 90 minutes
- [ ] I can draw the flower LS and pollen germination diagrams
- [ ] I can name the asexual method for each organism
- [ ] I can draw both human reproductive systems with labels
- [ ] I checked every answer with the marking points

My score: ____ / 5

LEVEL 54 COMPLETE

+200 XP earned. Next: Level 55

BUJHO *Boss khatam! Fission se placenta tak, sab yaad ho gaya.*

PAHELI *Shabaash! Kal Level 55: **Heredity** aur Mendel ke experiments.*

NITESH SIR SAYS

*Chapter 11 se board mein lagbhag **har saal ek diagram** aata hai: flower LS, pollen tube ya human system. Roz ek diagram banao, **Kal milte hai Day 55 pe!***

NCERT exercise: quick answers

- **Variation:** helps the species survive a changed environment, though it may not help the individual
- **Spore formation:** spores have thick walls that protect them until they find moist conditions to grow
- **Regeneration** fails in complex animals: their cells are highly specialised and organised into organ systems
- **Vegetative propagation** is used for plants that have lost the capacity to make seeds, e.g. banana, orange, rose, jasmine
- **Fertilised egg:** the zygote divides and forms the embryo, which gets implanted in the uterine wall
- **Contraception** methods: barrier, hormonal, IUCD and surgical (vasectomy, tubectomy)

BUJHO *Fragmentation aur regeneration mein kya farak hai? Dono mein tukde hi toh hote hai!*

PAHELI *Fragmentation mein simple body (Spirogyra) khud tootkar har tukda naya banta hai. Regeneration mein kate hue tukde se **specialised cells** poora organism banate hai (Planaria, Hydra).*

Most asked in Chapter 11: Eduspark PYQ analysis

Topic	Rating
Pollen germination and fertilisation in plants	★★★
Human reproductive systems (diagrams)	★★★
Modes of asexual reproduction	★★★
Vegetative propagation and its advantages	★★☆
Contraception and STDs	★★☆

Coming up: Day 55

Heredity begins: variation, Mendel's pea plants, monohybrid cross and the 3:1 ratio. DNA copying leads straight into genes.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

BIOLOGY | CHAPTER 11 COMPLETE | LEVEL 54 ASTRA SERIES

CHAPTER 11 SWAHAA!

How do Organisms Reproduce? | Days 51–54

BUJHO *Fission, budding, flower ka LS aur pollen tube bhi! Chapter 11 Swahaa!*

PAHELI *Great going! Next stop Heredity: Mendel, pea plants aur Punnett square.*

CHAPTER XP EARNED
+575 XP

LEVELS CLEARED
4 levels (Days 51–54)

BOOK PROGRESS
Level 54 / 60 7050 XP total

WHAT YOU HAVE MASTERED

- ✔ Fission, budding, fragmentation, regeneration
- ✔ Flower parts and pollination
- ✔ Human reproductive systems
- ✔ Vegetative propagation and spores
- ✔ Fertilisation in flowering plants
- ✔ Reproductive health

NEXT CHAPTER Day 55: Heredity

Chapter 11 cleared!

DAY 55 LEVEL 55

BIOLOGY | CHAPTER 12 | PART 1 ASTRA SERIES

MENDEL & THE MONOHYBRID CROSS

Goal: learn how traits pass on and predict ratios.

by Nitesh Prakash

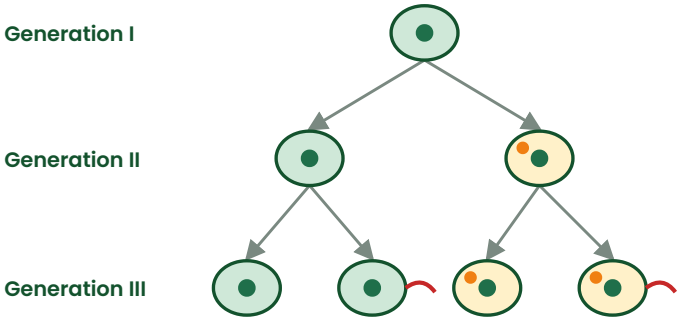
01 Variation | 02 Heredity | 03 Mendel's Pea | 04 Monohybrid | 05 Two Laws | +125 XP today | Level 55 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 VARIATION DURING REPRODUCTION ★★☆

DEFINITION

- **Variation:** the differences in characters (traits) among individuals of the same species, including between parents and their offspring
- It arises because DNA copying during reproduction is **never 100% accurate**, and in sexual reproduction DNA from **two parents** combines



Orange spot and red tail = new variations from DNA copying

Asexual vs sexual: how much variation? ★★☆

Asexual reproduction	Sexual reproduction
one parent, copies of DNA	two parents, DNA of both mixes
very little variation, from small copying errors only	**much greater** variation (new combinations every generation)
offspring almost exactly like the parent	offspring similar, not identical

EXAM POINT

Not every variation is useful; some are harmful and some neutral. The useful ones help a few individuals survive when conditions change, so sexual reproduction (more variation) gives a species a better chance to last.

BUJHO *Variation matlab galti? Toh ye toh bura hua na?*

PAHELI *Nahi Bujho! Garmi badhi toh jin bacteria mein **heat-resistant** variation hai, wahi bachenge. Variation = species ka **survival insurance**.*

Why variation is useful

- ✔ The **environment selects** the variants that suit it; others may die
- ✔ Example: if water gets much hotter, most bacteria die, but a few **heat-resistant** variants survive and multiply
- ✔ So variation helps the **species survive**, even if many individuals die

NCERT in-text questions

1 Trait A in 10%, trait B in 60% of an asexual population: which arose earlier?

> **Trait B.** Variations pile up over generations, so the trait found in more individuals has had more time to spread.

2 How does variation promote survival?

> Some variants suit a changed environment (heat, drought, disease) and survive, so the species does not die out.

2 HEREDITY AND INHERITED TRAITS ★★★

DEFINITION

- **Heredity:** the transmission of characters (traits) from parents to offspring, generation after generation
- **Inherited traits:** features a child gets from the parents through genes, e.g. earlobe type, eye colour, hair type

REMEMBER

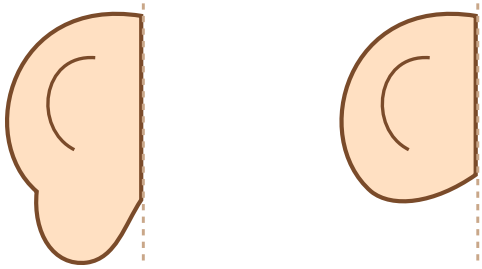
A child shows features of **both** parents because both contribute genetic material, so children never look exactly like either parent.

MEME BREAK

*Hair, earlobe, height: sab ka source ek hi, **DNA**. Mom-dad ka DNA mix hua toh tum bane!*

NCERT activity: earlobe survey ★★☆

1. Look at the lower part of the ear (the earlobe) of everyone in your class.
2. It is either **free** (hangs below the point where it joins the head) or **attached** (joins the side of the head closely).
3. Count both types and find the percentage of each. Both are common, so earlobe type is an **inherited trait** that varies.



Free earlobe
lobe hangs free below the joint

Attached earlobe
lobe joins the side of the head

3 MENDEL AND HIS PEA PLANTS ★★★

Gregor Johann Mendel (1822–1884)

- An Austrian monk who did breeding experiments on the **garden pea** (*Pisum sativum*) and worked out the first rules of heredity
- He studied **one or two traits at a time**, counted the offspring and used maths to find patterns
- Called the **Father of Genetics**; his work was recognised only after his death

Why did Mendel choose the pea plant? ★★★

- ✔ Many clear **contrasting traits** (tall/dwarf, round/wrinkled seeds)
- ✔ Flowers are **bisexual** and normally **self-pollinate**, so pure lines are easy to get
- ✔ Cross-pollination can be done **by hand** when needed
- ✔ **Short life cycle** and many seeds per plant, so many generations and large numbers to count
- ✔ Easy to grow in a garden

Seven pairs of contrasting traits ★★☆

Character	Dominant	Recessive
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green
Flower colour	Violet	White
Pod shape	Inflated (full)	Constricted
Pod colour	Green	Yellow
Flower position	Axial	Terminal
Stem height	Tall	Dwarf

★★★ *Monohybrid Punnett square aur ratios 3:1, 1:2:1 har saal aate hain, Nitesh Bhaiya ki guarantee!*

MNEMONIC: SEVEN DOMINANT TRAITS

"Rohit, Yadav, Virat, Iyer, Gill, Axar, Tilak"

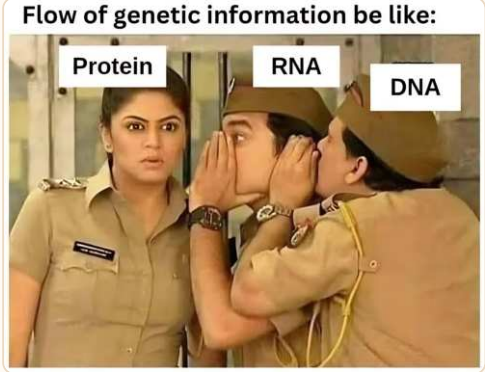
Team India ki **dominant** XI: **R**ound, **Y**ellow, **V**iolet, **I**nflated, **G**reen pod, **A**xial, **T**all. Recessive = inke ulte.

MEME BREAK
*F1 ka tall plant (Tt) apne **dwarf parent** ko yaad kar raha hai. Tension mat lo, F2 mein dwarf **wapas aata hai**!*

4 KEY TERMS OF GENETICS ★★★

Know these words exactly

Term	Meaning
Gene	a section of DNA that carries the information for one trait (one protein)
Alleles	two alternative forms of the same gene, e.g. T (tall) and t (dwarf)
Dominant	the trait that shows up even when only one copy is present (T)
Recessive	the trait that shows up only when both copies are recessive (tt)
Genotype	the gene makeup of an organism, e.g. TT, Tt, tt
Phenotype	the visible character, e.g. tall or dwarf
Homozygous	both alleles same: TT or tt (pure line)
Heterozygous	both alleles different: Tt (hybrid)



MEME BREAK
*Gene = DNA ka chhota hissa jo **protein** ka recipe deta hai. Wahi protein trait banata hai (detail Day 56 mein).*

BUJHO *Tt aur TT dono tall dikhte hain, toh dono same hue na?*

PAHELI ***Phenotype*** *same (dono tall), par* ***genotype*** *alag! TT pure tall hai, Tt hybrid hai jisme dwarf ka* ***t*** *chhupa hai.*

5 MONOHYBRID CROSS ★★★ MOST ASKED

DEFINITION

- **Monohybrid cross:** a cross between two plants that differ in **one pair** of contrasting traits, e.g. tall × dwarf

Parents (P): Tall **TT** × Dwarf **tt**
Gametes: T (from tall) and t (from dwarf)
F1: all **Tt** → all plants **tall** (no medium height)
Selfing F1: Tt × Tt → gametes T, t from each parent
The dwarf trait is hidden in F1 but not lost

F2 Punnett square (Tt × Tt) ★★★

	T	t
T	TT Tall	Tt Tall
t	Tt Tall	tt Dwarf

Ratio	F2 result
Phenotypic	Tall : Dwarf = **3 : 1**
Genotypic	TT : Tt : tt = **1 : 2 : 1**

MNEMONIC: MONOHYBRID RATIOS

"Photo mein 3:1, Gene mein 1:2:1"

Phenotype = jo photo mein dikhe (tall/dwarf) = 3:1. **Genotype** = andar ka gene hisaab (TT, Tt, tt) = 1:2:1.

6 MENDEL'S LAWS ★★★

Law of Dominance

- In a hybrid (Tt), of two contrasting alleles only one shows its effect (**dominant**); the other stays hidden (**recessive**)
- Proof: every F1 plant of TT × tt is tall; none is dwarf or medium

Law of Segregation (purity of gametes)

- The two alleles of a gene **separate** when gametes form, so each gamete carries **only one** allele
- Proof: the hidden dwarf trait (tt) **reappears** in F2 in one-fourth of the plants

MEME BREAK

Bhai, t kahin gaya nahi. Recessive allele F1 mein chhup ke baitha hai, F2 mein tt banke nikalta hai.

COMMON MISTAKES

- Writing F1 as "medium height": wrong, F1 is **all tall** (no blending)
- Mixing up ratios: 3:1 is **phenotypic**, 1:2:1 is **genotypic**
- Writing a gamete as "Tt": a gamete carries **one** allele only, T or t

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Mendel's actual F2 count for stem height was **787 tall : 277 dwarf**, about 2.84 : 1, close to the 3 : 1 he predicted.

Practice MCQs: Eduspark PYQ analysis

1. Which of these is **not** a reason why Mendel chose the pea plant? ★★☆
(a) contrasting traits (b) self-pollination
(c) long life cycle (d) many seeds

2. A cross between TT and tt gives F1 plants that are: ★★★
(a) all dwarf (b) all tall
(c) half tall (d) all medium

3. The genotypic ratio of the F2 generation in a monohybrid cross is: ★★★
(a) 3 : 1 (b) 1 : 2 : 1
(c) 9 : 3 : 3 : 1 (d) 1 : 1

4. In 400 F2 plants of a Tt × Tt cross, the expected number of dwarf plants is: ★★☆
(a) 100 (b) 200
(c) 300 (d) 400

5. An organism with genotype Tt is said to be: ★★☆
(a) homozygous (b) heterozygous
(c) recessive (d) pure line

6. Which pair is a pair of alleles? ★☆☆
(a) T and t (b) T and R
(c) TT and tt (d) R and y

7. Which trait in pea is **recessive**? ★★☆
(a) violet flower (b) axial flower
(c) green seed (d) inflated pod

8. Variation is greatest in organisms produced by: ★★☆
(a) binary fission (b) budding
(c) sexual reproduction (d) fragmentation

9. **Assertion (A):** Dwarf plants reappear in the F2 generation of a cross between pure tall and pure dwarf pea plants.
Reason (R): The two alleles of a gene separate during gamete formation, and each gamete gets only one of them. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** All F1 plants of a TT × tt cross are tall.
Reason (R): Variation does not occur in sexual reproduction. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(c)	Pea has a short life cycle; long is wrong.
2	(b)	T is dominant, so Tt is tall.
3	(b)	TT : Tt : tt = 1 : 2 : 1.
4	(a)	One-fourth of 400 = 100 dwarf.
5	(b)	Two different alleles = heterozygous.
6	(a)	Alternative forms of the same gene.
7	(c)	Yellow seed is dominant; green is recessive.
8	(c)	Two parents mix DNA: most variation.
9	(a)	Segregation explains the return of tt.
10	(c)	A true (dominance); R false.

Short and long answer questions

1. Define variation. Why is variation in a population useful for the survival of a species? Give one example. **(3 marks)**
2. Distinguish between genotype and phenotype, giving one example of each. **(2 marks)**
3. With the help of a labelled Punnett square, show a cross between pure tall (TT) and pure dwarf (tt) pea plants up to F2. State the phenotypic and genotypic ratios and the law it proves. **(5 marks)**

Model answers

1 Variation

- Differences among individuals of a species, caused by inaccurate DNA copying and mixing of parents' DNA
- Helps some individuals survive changed conditions, e.g. heat-resistant bacteria in hot water

2 Genotype vs phenotype

- Genotype = gene makeup (TT, Tt, tt); phenotype = visible trait (tall, dwarf)

3 Monohybrid cross

- P: TT × tt → F1 all Tt (tall) → F1 selfed → F2 Punnett square as on page 62
- Phenotype 3 tall : 1 dwarf; genotype 1 TT : 2 Tt : 1 tt; laws of dominance and segregation

Case-based question (4 marks)

A student crosses pure violet-flowered pea plants (VV) with pure white-flowered plants (vv) and then self-pollinates the F1 plants.

1. What is the flower colour of F1 plants? **(1)**
2. Which colour is dominant? **(1)**
3. Out of 800 F2 plants, how many are white, and how many are pure violet (VV)? **(2)**

EXAM TIPS

- Gametes ko **single letter** (T, t) mein likho; har box mein genotype aur phenotype likho.
- Number wale sawal: dwarf = total ÷ 4, tall = total × 3/4, Tt = total ÷ 2.

Day 55 checklist: tick before you sleep

- [ ] I can explain how variation arises and why it helps survival
- [ ] I know the earlobe example of an inherited trait
- [ ] I can list the seven contrasting traits of pea
- [ ] I can draw the monohybrid Punnett square and give 3:1 and 1:2:1
- [ ] I can state the laws of dominance and segregation

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Section 5 phir se padho.

Case-based answers

- (i) All violet (Vv)
- (ii) Violet is dominant over white
- (iii) White = 800 ÷ 4 = **200**; pure violet VV = 800 ÷ 4 = **200** (Vv = 400)

LEVEL 55 COMPLETE +125 XP earned. Next: Level 56

BUJHO *Samajh gaya! F1 mein dwarf chhupa, F2 mein 1/4 wapas, aur ratio 3:1.*

PAHELI *Shabaash! Kal Level 56: dihybrid cross aur sex determination.*

Day 55 hit-list: Eduspark PYQ analysis

Topic	Rating
Monohybrid cross with Punnett square	★★★
3:1 and 1:2:1 ratios	★★★
Genotype vs phenotype, dominant vs recessive	★★★
Why Mendel chose pea	★★☆
Variation and survival	★★☆
Earlobe example	★☆☆

NITESH SIR SAYS

*Punnett square ko **chart ki tarah** banao: upar ek parent ke gametes, side mein doosre ke. Har box mein dono mila do, ratio khud nikal aayega. **Kal milte hai Day 56 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 56
LEVEL 56

BIOLOGY | CHAPTER 12 | PART 2 ASTRA SERIES

DIHYBRID CROSS & SEX DETERMINATION

Goal: master the 9:3:3:1 cross and how sex is decided.

by Nitesh Prakash

01 Dihybrid | 02 9:3:3:1 | 03 Gene Action | 04 Germ Cells | 05 XX and XY

+125 XP today | Level 56 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 DIHYBRID CROSS ★★★ MOST ASKED

DEFINITION

> **Dihybrid cross:** a cross between two plants that differ in **two pairs** of contrasting traits, e.g. seed shape and seed colour

P: Round yellow **RRYY** × Wrinkled green **rryy**
Gametes: RY and ry
F1: all **RrYy** → all **round yellow**
F1 selfed: RrYy × RrYy; each parent gives **4 kinds** of gametes: RY, Ry, rY, ry
R = round, r = wrinkled; Y = yellow, y = green (R and Y dominant)

F2 Punnett square (RrYy × RrYy) ★★★

Gametes	RY	Ry	rY	ry
RY	**RRYY** RY	**RRYy** RY	**RrYY** RY	**RrYy** RY
Ry	**RRYy** RY	**RRyy** RG	**RrYy** RY	**Rryy** RG
rY	**RrYY** RY	**RrYy** RY	**rrYY** WY	**rrYy** WY
ry	**RrYy** RY	**Rryy** RG	**rrYy** WY	**rryy** WG

RY = round yellow, RG = round green, WY = wrinkled yellow, WG = wrinkled green

F2 result: 16 boxes, 4 phenotypes ★★★ MOST ASKED

Phenotype	Boxes	Type
Round yellow	**9**	parental
Round green	**3**	new combination
Wrinkled yellow	**3**	new combination
Wrinkled green	**1**	parental

Phenotypic ratio **9 : 3 : 3 : 1**. The two **new combinations** prove that seed shape and seed colour are inherited independently.

COUNT IT YOURSELF

Pure **RRYY** and **rryy** fill only 1 box each; the 9 round yellow boxes hold 4 genotypes (RRYY, RRYy, RrYY, RrYy).

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Mendel's actual F2: **315** round yellow, **108** round green, **101** wrinkled yellow, **32** wrinkled green (556 seeds).

2 LAW OF INDEPENDENT ASSORTMENT ★★★

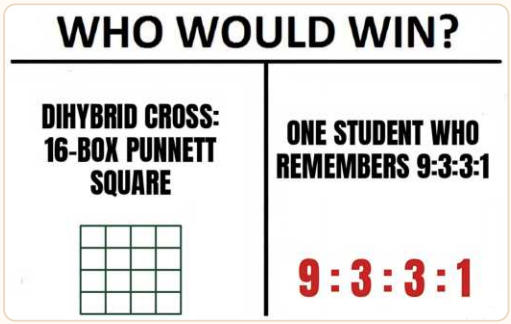
DEFINITION

> **Law of independent assortment:** when two pairs of traits are crossed, the alleles of one pair separate (assort) **independently** of the alleles of the other pair during gamete formation
> So shape and colour do not stay "tied": round can go with green and wrinkled with yellow

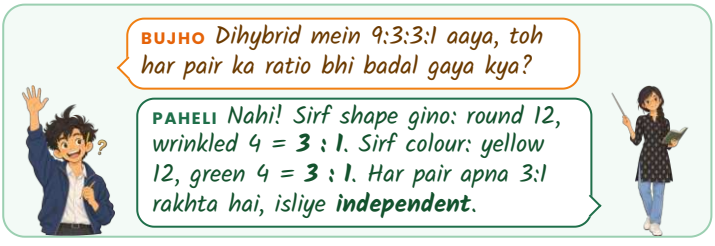
MNEMONIC: 9 : 3 : 3 : 1

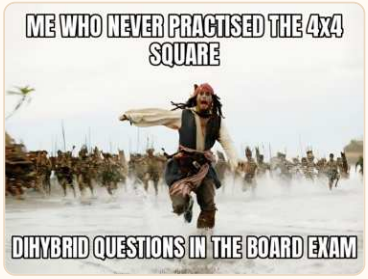
"9 mein dono Dabang, 3-3 mein ek-ek, 1 mein dono Chhupe"

9 = both dominant (round yellow), **3 + 3** = one dominant each (round green, wrinkled yellow), **1** = both recessive (wrinkled green). Total 16.



MEME BREAK
*16 box dekh ke darr gaye? Bas yaad rakho **9 : 3 : 3 : 1**, aur haar Punnett square ki hogi!*





MEME BREAK
*Exam mein bhaagna nahi padega agar aaj ek baar **4 × 4 square** khud bana liya. Practice!*

3 HOW DO TRAITS GET EXPRESSED? ★★★

DEFINITION

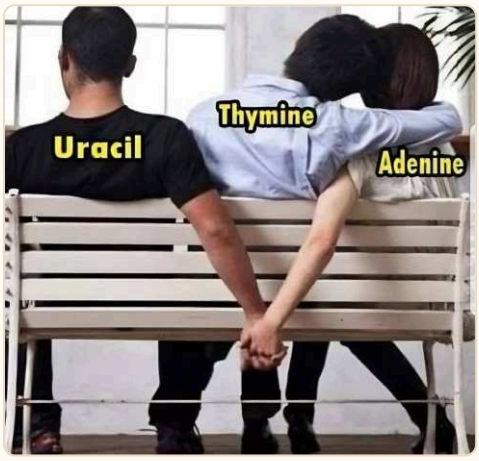
- A **gene** is a section of DNA that carries the information to make **one protein**
- Proteins (often **enzymes**) control the chemical reactions in cells, and these reactions decide the traits

NCERT example: plant height ★★☆

Gene (DNA) → makes a **protein** (enzyme) → enzyme makes a **plant hormone** → hormone controls **height**

Efficient enzyme → more hormone → tall plant; altered gene → less efficient enzyme → less hormone → short plant

- Plant height depends on the amount of a growth **hormone**
- The amount of hormone depends on how **efficient the enzyme** that makes it is
- So the **gene** for that enzyme decides tall or short

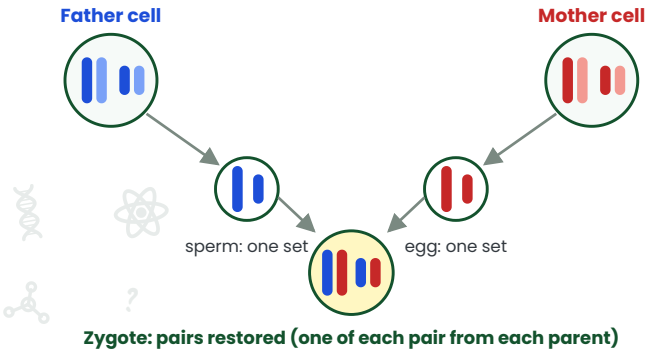


MEME BREAK
*DNA mein **thymine**, RNA mein **uracil**. Gene ka message RNA leke jaata hai protein banane, isliye uracil pe nazar (not for assessment).*

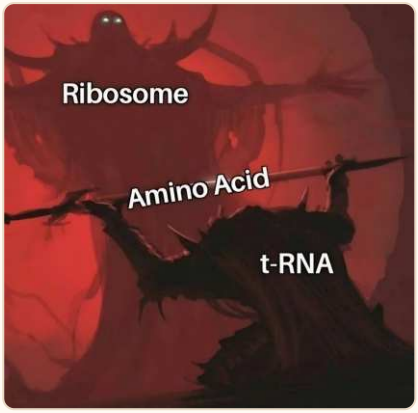
4 BOTH PARENTS CONTRIBUTE EQUALLY ★★☆

How do both parents pass on genes?

- ✔ Genes are carried on **chromosomes**; each body cell has chromosomes in **pairs**, one of each pair from each parent
- ✔ Each **germ cell** (sperm or egg) gets only **one set**: one chromosome from each pair
- ✔ At fertilisation the two sets join, so the **zygote** gets the full number back, half from each parent
- ✔ This keeps the chromosome number **constant** generation after generation and lets both parents contribute equally



Zygote: pairs restored (one of each pair from each parent)



MEME BREAK
*Protein banane ki factory = **ribosome**. Gene ki recipe se amino acids jodkar protein, aur protein se trait!*

5 SEX DETERMINATION ★★★ MOST ASKED

DEFINITION

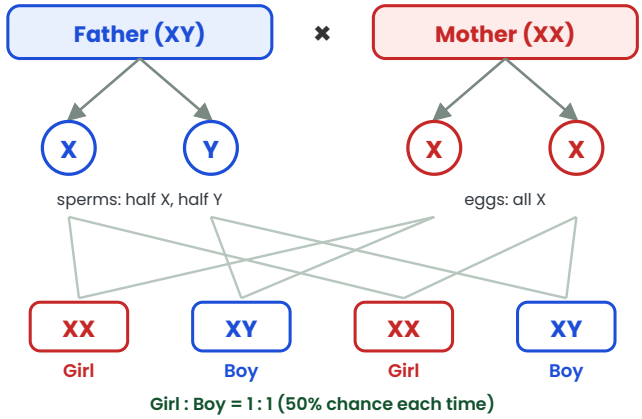
- **Sex determination:** the way the sex (male or female) of a new individual is decided

Different ways in different animals ★★☆

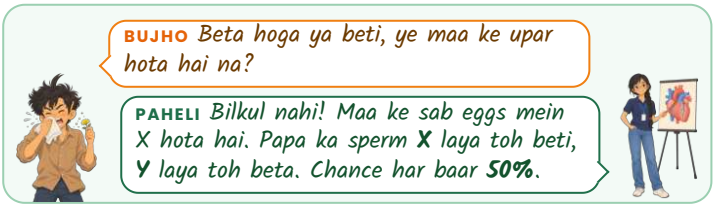
Animal	Sex decided by
Some reptiles, e.g. **turtles**	**temperature** at which the fertilised eggs are kept
Snails	can **change sex**, so sex is not fixed by genes
Humans	**genes**: inherited sex chromosomes

Sex chromosomes in humans ★★★

- ✔ Humans have **23 pairs** of chromosomes: **22 pairs** of autosomes (same in both sexes) and **1 pair** of sex chromosomes
- ✔ Female: **XX** (a perfect pair). Male: **XY** (Y is shorter)
- ✔ All eggs carry **X**; half the sperms carry **X** and half carry **Y**
- ✔ Egg + X sperm → **XX girl**; egg + Y sperm → **XY boy**. So the sex of the child is decided by the **father**



Girl : Boy = 1 : 1 (50% chance each time)



MEME BREAK

*Science clear hai: maa sirf **X** de sakti hai, **Y** sirf papa ke sperm mein. Beti hone pe maa ko blame karna **galat** hai.*

MNEMONIC: SEX DETERMINATION

"Papa ka Y = Ladka, Papa ka X = Ladki"

Maa ka contribution hamesha **X**. Result papa ke sperm pe: 23 pairs = 22 + XX ya 22 + XY.

★★★ *Sex determination ka flow chart (XX × XY) har saal board mein aata hai, Nitesh Bhaiya ki guarantee!*

BUJHO *Toh turtles mein bhi XX aur XY hota hoga?*

PAHELI *Nahi Bujho! Kuch reptiles mein ande kis **temperature** pe rakhe gaye, usse sex decide hota hai. Aur snail toh apna sex **badal** bhi sakta hai.*

NCERT exercise questions

1 Tall violet × short white gives all violet, about half short. Tall parent's genotype?

> **TtWW**: all violet means WW; half short means the tall parent was Tt.

2 How do Mendel's crosses show traits are inherited independently?

> In the dihybrid F2, new combinations (round green, wrinkled yellow) appear.

3 Father A, mother O, daughter O. Can we tell which is dominant?

> **No.** One family is too small, and the father may carry an O allele, so we need many more crosses.

Monohybrid vs dihybrid at a glance ★★☆

Point	Monohybrid	Dihybrid
Trait pairs	one (TT × tt)	two (RRYY × rryy)
F2 boxes	4	16
F2 phenotypes	3 : 1	9 : 3 : 3 : 1
Law shown	dominance, segregation	independent assortment

COMMON MISTAKES

> Writing F1 gametes as only RY and ry: an RrYy plant makes **four** kinds (RY, Ry, rY, ry)

> "Mother decides the sex": wrong, the **father's** sperm (X or Y) decides

> "Humans have 23 chromosomes": no, **23 pairs** = 46 chromosomes

Practice MCQs: Eduspark PYQ analysis

1. The F1 plants of a cross RRYY × rryy are: ★★★
(a) round green (b) wrinkled yellow
(c) round yellow (d) wrinkled green

2. How many kinds of gametes does an RrYy plant form? ★★☆
(a) 1 (b) 2
(c) 3 (d) 4

3. The phenotypic ratio of the F2 generation of a dihybrid cross is: ★★★
(a) 3 : 1 (b) 1 : 2 : 1
(c) 9 : 3 : 3 : 1 (d) 1 : 1 : 1 : 1

4. In a dihybrid F2 of 160 plants, the expected number of wrinkled green plants is: ★★☆
(a) 10 (b) 30
(c) 90 (d) 160

5. A gene controls a trait by making a: ★★☆
(a) carbohydrate (b) protein
(c) fat (d) vitamin

6. The number of autosome pairs in a human cell is: ★★★
(a) 22 (b) 23
(c) 44 (d) 46

7. In some reptiles such as turtles, the sex of the young is decided by: ★★☆
(a) sex chromosomes (b) incubation temperature
(c) the mother (d) diet

8. A child gets an X chromosome from the father. The child will be: ★★★
(a) a boy (b) a girl
(c) either (d) cannot say

9. **Assertion (A):** Round green and wrinkled yellow seeds appear in the F2 of a cross between round yellow and wrinkled green plants.
Reason (R): The alleles for seed shape and seed colour are inherited independently of each other. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The sex of a child is decided by the mother.
Reason (R): The mother has two X chromosomes and all her eggs carry an X chromosome. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

EXAM TIPS

> Flow chart mein teen line zaroor: **Parents → Gametes → Offspring**, aur ratio 1 : 1 likho.

> Dihybrid numbers: wrinkled green = total ÷ 16, round yellow = total × 9/16, round green = total × 3/16.

> "23 pairs" aur "46 chromosomes" dono sahi hain; "23 chromosomes" likha toh number katega.

Answer key: MCQs

Q	Ans	Why
1	(c)	R and Y are dominant: RrYy is round yellow.
2	(d)	RY, Ry, rY, ry.
3	(c)	16 boxes give 9 : 3 : 3 : 1.
4	(a)	1/16 of 160 = 10.
5	(b)	Gene → protein (enzyme) → trait.
6	(a)	22 pairs of autosomes + 1 pair sex chromosomes.
7	(b)	Temperature of the fertilised eggs.
8	(b)	X from father + X from mother = XX.
9	(a)	Independent assortment explains new combinations.
10	(d)	A false (father decides); R true.

Short and long answer questions

1. How do genes control the characteristics of an organism? Explain with the example of plant height. **(3 marks)**
2. "The sex of a child is determined by the father." Justify with a labelled flow chart. **(3 marks)**
3. Mendel crossed pea plants with round yellow seeds (RRYY) and wrinkled green seeds (rryy). Show the F1 and F2 with a Punnett square, give the F2 ratio, and state the law that this cross proves. **(5 marks)**

Model answers

1 Genes and traits

- Gene → protein (enzyme) → hormone → trait; efficient enzyme gives more growth hormone, so a tall plant; altered gene gives less hormone, so a short plant

2 Father decides

- Mother XX gives only X eggs; father XY gives X or Y sperm (flow chart as on page 65)
- X sperm → XX girl; Y sperm → XY boy; so the father's sperm decides

3 Dihybrid cross

- F1 all RrYy (round yellow); F2 from a 4 × 4 square: 9 RY : 3 RG : 3 WY : 1 WG
- Law of independent assortment (new combinations appear)

Case-based question (4 marks)

In a family the first two children are girls. Relatives tell the mother that the third child will surely be a boy, and that she alone is responsible for having girls.

1. How many pairs of chromosomes does a human cell have? **(1)**
2. Which chromosome does the mother always give? **(1)**
3. What is the chance that the third child is a boy? Is the relatives' claim correct? Give a reason. **(2)**

DIAGRAM TIP

Punnett square ki grid ruler se banao, gametes ko circle karo aur ratio neeche likho. Har step ke alag marks milte hain.

Day 56 checklist: tick before you sleep

- [ ] I can draw the 4 × 4 Punnett square of RRYY × rryy
- [ ] I can give 9 : 3 : 3 : 1 and state independent assortment
- [ ] I can explain gene → enzyme → hormone → height
- [ ] I know why both parents contribute equally
- [ ] I can draw the XX × XY flow chart and give the 50% chance

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Punnett square phir se banao.

Case-based answers

- (i) 23 pairs (22 pairs autosomes + 1 pair sex chromosomes)
- (ii) An X chromosome
- (iii) 50% every time, whatever happened before; the claim is wrong, since the father's sperm (X or Y) decides

LEVEL 56 COMPLETE +125 XP earned. Next: Level 57

BUJHO *Samajh gaya! 16 box, 9:3:3:1, aur beta ya beti papa ka sperm decide karta hai.*

PAHELI *Shabaash! Kal Level 57: Heredity ka Boss Level.*

Day 56 hit-list: Eduspark PYQ analysis

Topic	Rating
Sex determination in humans (flow chart)	★★★
Dihybrid cross and 9:3:3:1	★★★
Law of independent assortment	★★★
Genes → proteins → traits (plant height)	★★☆
Both parents contribute equally	★★☆
Sex determination in turtles, snails	★☆☆

NITESH SIR SAYS

*Dihybrid square banate waqt **pehle 4 gametes** likho, phir har box mein R-wale saath aur Y-wale saath. Gadbad kabhi nahi hogi. **Kal Boss Level, taiyaar raho!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 57 · LEVEL 57

BIOLOGY | CHAPTER 12 | BOSS LEVEL

CHAPTER 12 BOSS LEVEL

Goal: test all of Heredity in 90 minutes.

by Nitesh Prakash

01 Revision Map · 02 MCQs · 03 Short Answers · 04 Case & Long · 05 Answer Key · +200 XP Boss Level | Level 57 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 12 REVISION MAP ★★★ MOST ASKED

Chapter 12 mind map: revise in 5 minutes

VARIATION
Arises in reproduction (DNA copying errors, sexual mixing); most in sexual reproduction; helps survival

MONOHYBRID ★★★
TT × tt → F1 all Tt (tall); F2 3 : 1 phenotype, 1 : 2 : 1 genotype

HEREDITY

DIHYBRID ★★★
RRYY × rryy → F1 RrYy; F2 9 : 3 : 3 : 1 (16 boxes)

LAWS
Dominance, segregation (one allele per gamete), independent assortment

TRAITS AND SEX ★★★
DNA (gene) → protein/enzyme → hormone → trait. Humans: 23 pairs, XX girl, XY boy; father decides

Must-know ratios and terms ★★★

Cross	F2 phenotype	F2 genotype
Monohybrid Tt × Tt	3 : 1	1 : 2 : 1
Dihybrid RrYy × RrYy	9 : 3 : 3 : 1	9 types
Gametes of RrYy	RY, Ry, rY, ry	4 types
XY × XX (sex)	1 boy : 1 girl	XY : XX = 1 : 1

MNEMONIC: 3 MENDEL LAWS
"D-S-I: Dil Se Insaan": Dominance (FI mein ek hi dikhta), Segregation (gamete mein ek hi allele), Independent assortment (do traits alag alag chalte).

Monohybrid F2 at a glance: Tt × Tt ★★★

Gametes	T	t
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (dwarf)

3 tall : 1 dwarf (phenotype); 1 TT : 2 Tt : 1 tt (genotype).

MNEMONIC: TWO RATIOS
"Ek gene, teen-ek; do gene, nau-teen-teen-ek": monohybrid 3 : 1, dihybrid 9 : 3 : 3 : 1.

How to attack the Boss: 90 minutes

1. Section A (20 MCQs, 1 mark each): **20 min**
2. Section B (5 × 2 marks): **12 min**
3. Section C (4 × 3 marks): **20 min**
4. Section D (2 cases × 4 marks): **14 min**
5. Section E (2 × 5 marks): **18 min**; check every cross: 6 min

Total 60 marks. In every cross write parents, gametes, Punnett square, then the ratio with labels.

BOSS INTRO
*"I hate biology" bolne wale bhi Heredity mein full marks laate hain: bas **Punnett square** sahi bharna hai. Chalo, Boss se ladein!*

2 SECTION A: MCQS ★★★ MOST ASKED

1. Mendel chose the garden pea because it: ★★☆

(a) has many clear contrasting traits
(b) cannot self-pollinate
(c) has a long life cycle
(d) has no variation

2. A tall plant (TT) is crossed with a dwarf plant (tt). The F1 plants are: NCERT ★★★

(a) all dwarf
(b) all tall (Tt)
(c) 3 tall : 1 dwarf
(d) 1 tall : 1 dwarf

3. The F2 phenotypic ratio of a monohybrid cross is: ★★★

(a) 1 : 2 : 1
(b) 9 : 3 : 3 : 1
(c) 3 : 1
(d) 1 : 1

4. The F2 genotypic ratio of a monohybrid cross is: ★★★

(a) 3 : 1
(b) 1 : 2 : 1
(c) 1 : 1
(d) 9 : 3 : 3 : 1

5. Which pair was NOT among the contrasting traits Mendel studied? ★☆☆

(a) round / wrinkled seed
(b) yellow / green seed
(c) violet / white flower
(d) red / black seed

6. The number of different phenotypes in the F2 of a dihybrid cross is: ★★★

(a) 2
(b) 4
(c) 9
(d) 16

7. In the F2 of RrYy × RrYy, the fraction of round green seeds is: ★★★
(a) 9/16 (b) 3/16
(c) 1/16 (d) 4/16

8. The number of types of gametes formed by a plant of genotype RrYy is: ★★☆
(a) 1 (b) 2
(c) 4 (d) 8

9. The number of pairs of chromosomes in a human body cell is: ★★★
(a) 22 (b) 23
(c) 44 (d) 46

10. A human egg cell normally carries: ★★☆
(a) 22 + X (b) 22 + Y
(c) 44 + XX (d) 22 + XY

11. The sex of a human child is decided by: NCERT ★★★
(a) the mother's egg
(b) the sex chromosome in the father's sperm
(c) the season of birth
(d) the mother's diet

12. In some reptiles such as turtles, the sex of the young is decided by: ★★☆
(a) X and Y chromosomes
(b) the temperature at which the fertilised eggs are kept
(c) the father only
(d) the food of the mother

13. Genes are segments of: ★★☆
(a) protein (b) DNA
(c) the cell membrane (d) starch

14. A pea plant is tall because the gene for tallness makes an enzyme that works efficiently to produce more: ★★☆
(a) plant growth hormone (b) chlorophyll
(c) starch (d) cellulose

15. A pea plant with genotype Tt is: ★★☆
(a) homozygous tall (b) heterozygous tall
(c) homozygous dwarf (d) heterozygous dwarf

3 SECTION B: SHORT ANSWERS (2 MARKS EACH)

★★★ MOST ASKED

Answer in 30–50 words

1. Distinguish between dominant and recessive traits with one example each. **(2)**
2. A man with blood group A marries a woman with blood group O, and their daughter has blood group O. Is this information enough to tell whether blood group A or O is dominant? Why or why not? NCERT **(2)**
3. How do Mendel's experiments show that traits may be dominant or recessive? NCERT **(2)**
4. How is the equal genetic contribution of the male and female parents ensured in the progeny? NCERT **(2)**
5. Why does variation arise during reproduction? Give one way in which it helps a species. **(2)**

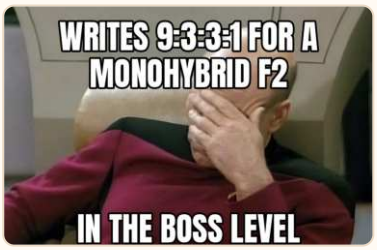
16. In the F2 of the cross TT × tt, the percentage of dwarf plants is: ★★★
(a) 0% (b) 25%
(c) 50% (d) 75%

17. Variation among offspring is greatest in: ★★☆
(a) binary fission (b) budding
(c) sexual reproduction (d) fragmentation

18. **Assertion (A):** In human beings, the sex of the child is determined by the father.
Reason (R): The father produces two types of sperms, carrying X or Y, while the mother produces only X-carrying eggs. ★★★
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

19. **Assertion (A):** Dwarfness reappears in the F2 generation of Mendel's monohybrid cross.
Reason (R): The recessive trait is lost completely in the F1 generation. ★★★
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

20. **Assertion (A):** Both parents contribute equally to the inheritance of traits in the child.
Reason (R): Each germ cell carries only one set of chromosomes, that is one copy of each gene. ★★☆
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true



EXAM FACEPALM
Monohybrid = 3 : 1, dihybrid = 9 : 3 : 3 : 1. Ratio galat likha toh Picard ke saath examiner bhi facepalm karega.

Common mistakes that cost marks

× WRONG	✔ RIGHT
F2 genotype of Tt × Tt = 3 : 1	Genotype 1 : 2 : 1; phenotype 3 : 1
Gametes of Tt written as "Tt"	Each gamete gets one allele: T or t
Mother decides the sex of the baby	Father's sperm (X or Y) decides
Recessive trait is lost in F1	It is hidden in F1 and reappears in F2
Human cell has 23 chromosomes	23 pairs = 46; a gamete has 23

4 SECTION C: SHORT ANSWERS (3 MARKS EACH)

★★★ MOST ASKED

Show your working in every cross

1. Pure tall (TT) pea plants are crossed with pure dwarf (tt) plants. Using a Punnett square, show the F1 and F2 generations and write the phenotypic and genotypic ratios of F2. **(3)**
2. How do Mendel's experiments show that different traits are inherited independently? NCERT **(3)**
3. Explain with the example of plant height how genes control the characteristics of an organism. **(3)**
4. Tall plants with violet flowers are crossed with short plants with white flowers. All progeny have violet flowers, but almost half are short. Work out the genotype of the tall parent and explain your answer. NCERT **(3)**

BUJHO *Do betiyaan hui toh dadi ne mummy ko blame kiya. Kya mummy ki galti?*

PAHELI *Bilkul nahi! Mummy ke saare eggs mein X hi hota hai. Baby ka sex **papa ke sperm** (X ya Y) se tay hota hai, aur har baar chance **50 : 50**.*

Allele terms in one line

- **Gene:** a section of DNA that codes for a protein (trait)
- **Alleles:** two forms of one gene, e.g. T (tall) and t (dwarf)
- **Homozygous:** TT or tt; **heterozygous:** Tt
- **Genotype** = gene make-up (Tt); **phenotype** = what you see (tall)

BUJHO *9 : 3 : 3 : 1 mein kaunsa 9 hai aur kaunsa 1? Har baar confuse ho jaata hoon.*

PAHELI *Simple! **9** = dono dominant (round yellow), **3 + 3** = ek dominant ek recessive (round green, wrinkled yellow), **1** = dono recessive (wrinkled green).*

Any cross in 4 steps (full marks)

1. **Parents:** write both genotypes, e.g. TT × tt
2. **Gametes:** one allele of each gene per gamete (T; t)
3. **Punnett square:** gametes on the top and side, fill every box
4. **Result:** phenotype and genotype ratios, with labels

MNEMONIC: WHO DECIDES BOY OR GIRL

"Y aaya toh Boy aaya": Y sirf papa ke paas hai, isliye baby ka sex papa ka sperm tay karta hai.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Evolution (deleted this year): helpful variations are passed on more often, so a population slowly changes over many generations. **Not for assessment.**

5 SECTION D: CASE-BASED (4 MARKS EACH)

★★★ MOST ASKED

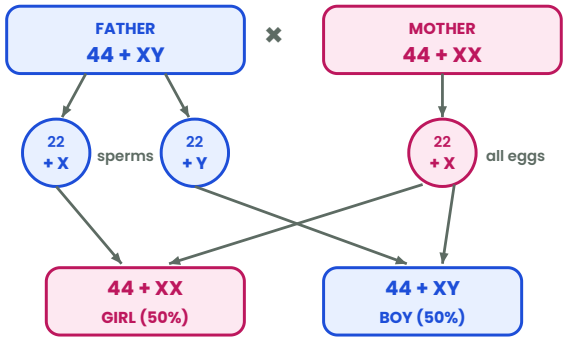
Case 1: Riya's pea garden

Riya crossed pure-breeding pea plants with round yellow seeds (RRYY) with plants having wrinkled green seeds (rryy). She then self-pollinated the F1 plants and collected 1600 F2 seeds.

1. What will be the phenotype and genotype of the F1 plants? **(1)**
2. Write all the types of gametes the F1 plants form. **(1)**
3. Name the four F2 phenotypes with their ratio. How many of the 1600 seeds are expected to be wrinkled and green, and how many round and green? **(2)**

Case 2: Boy or girl?

Study the diagram of sex determination in human beings and answer the questions.



1. Which parent produces two kinds of gametes? Name them. **(1)**
2. What is the sex chromosome make-up of a boy and of a girl? **(1)**
3. A couple already has two daughters. What is the chance that their third child will be a son? Justify it. **(2)**

6 SECTION E: LONG ANSWERS (5 MARKS EACH)

★★☆

Write in points, draw where asked

1. (a) State Mendel's law of dominance and law of segregation. (b) A pure round yellow (RRYY) plant is crossed with a wrinkled green (rryy) plant. Show the F2 with a Punnett square and state the law it proves. **(5)**
2. (a) With a labelled flow diagram, explain how the sex of a child is determined in human beings. (b) How is sex determined in some animals such as turtles and snails? **(5)**

THINK DEEP

*Mummy hamesha X deti hai, toh **X ya Y** papa ka sperm laata hai. Philosoraptor bhi maan gaya!*

Answer key: Section A

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	(a)	2	(b)	3	(c)	4	(b)	5	(d)
6	(b)	7	(b)	8	(c)	9	(b)	10	(a)
11	(b)	12	(b)	13	(b)	14	(a)	15	(b)
16	(b)	17	(c)	18	(a)	19	(c)	20	(a)

- **Q7:** round green = 3 of 16 boxes, so 3/16
- **Q19:** R is false: dwarfness is only hidden in F1, not lost

Marking points: Section B

1 Dominant vs recessive
- Dominant shows even in Tt, e.g. tallness (1); recessive shows only in tt, e.g. dwarfness (1)

2 Blood group A or O
- No (1). One family with one child is too small; O could be dominant or recessive here. Many progeny are needed (1)

3 Mendel and dominance
- TT × tt gave all tall F1: tall is dominant (1); dwarf reappeared in 1/4 of F2, so it was hidden, that is recessive (1)

4 Equal contribution
- Germ cells carry one set of chromosomes (1); fusion restores pairs, one copy from each parent (1)

5 Variation
- Errors in DNA copying and mixing of DNA of two parents (1); variants may survive a change, e.g. heat-tolerant bacteria in a heat wave (1)

Marking points: Section C

1 Monohybrid cross
- TT × tt; gametes T and t; F1 all Tt, tall (1)
- Tt × Tt Punnett square drawn correctly (1)
- F2: 3 tall : 1 dwarf; 1 TT : 2 Tt : 1 tt (1)

2 Independent inheritance
- RRYY × rryy gives F1 RrYy, all round yellow (1)
- F2 shows 4 types in 9 : 3 : 3 : 1 (1)
- New combinations (round green, wrinkled yellow) prove shape and colour are inherited independently (1)

3 Genes control traits
- A gene codes for a protein such as an enzyme (1)
- An efficient enzyme makes more growth hormone, so the plant is tall (1)
- An altered gene gives a weak enzyme, less hormone, a short plant (1)

4 Tall violet parent
- Genotype TtWW (1)
- All progeny violet: parent is WW (1)
- Half short: parent is Tt, as Tt × tt gives 1 tall : 1 short (1)

NITESH SIR SAYS

*Heredity mein marks **Punnett square** se aate hain: parents, gametes, square, ratio. Charon steps likho, full marks pakke. **Kal milte hai Day 58 pe!***

Marking points: Section D

1 Riya's pea garden
- (i) All round yellow, RrYy (1)
- (ii) RY, Ry, rY, ry (1)
- (iii) Round yellow 9 : round green 3 : wrinkled yellow 3 : wrinkled green 1 (1); wrinkled green = 1600 ÷ 16 = 100, round green = 300 (1)

2 Boy or girl?
- (i) Father: X-carrying and Y-carrying sperms (1)
- (ii) Boy 44 + XY, girl 44 + XX (1)
- (iii) 50% (1): every fertilisation is a fresh event; an X or a Y sperm is equally likely to fuse with the egg (1)

Marking points: Section E

1 Laws and dihybrid cross
- (a) Dominance: only the dominant trait shows in F1 (1); segregation: the two alleles separate, each gamete gets one (1)
- (b) F1 RrYy; 4 × 4 square below (1.5); 9 : 3 : 3 : 1 (0.5); law of independent assortment (1)

Gametes	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

2 Sex determination
- (a) Flow diagram as in Case 2 (2); mother gives only X eggs, father gives X or Y sperms; X sperm makes a girl, Y sperm a boy (1)
- (b) Turtles: temperature at which the fertilised eggs are kept (1); snails can change sex, so it is not genetically fixed (1)

Day 57 checklist: tick before you sleep

- [ ] I can draw a monohybrid Punnett square and give both ratios
- [ ] I can work a dihybrid cross and explain 9 : 3 : 3 : 1
- [ ] I can state the laws of dominance, segregation and independent assortment
- [ ] I can draw the sex determination flow diagram
- [ ] I scored at least 45 / 60 in this Boss test

My score: ____ / 5

LEVEL 57 COMPLETE +200 XP earned. Next: Level 58

BUJHO *Boss down! 3 : 1, 9 : 3 : 3 : 1 aur XY sab yaad ho gaya.*

PAHELI *Shabaash! Kal Level 58: Our Environment, food chain aur 10% law.*

SCORE CHECK

45+ = Boss defeated. Below 40? Redo Days 55 and 56.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

BIOLOGY | CHAPTER 12 COMPLETE | LEVEL 57

CHAPTER 12 SWAHAA!

Heredity | Days 55–57

BUJHO 3:1, 1:2:1, 9:3:3:1, sab ratios yaad! Aur XX-XY bhi pakka!

PAHELI Ek last chapter bacha hai, Bujho: Our Environment. Phir poori book complete!

CHAPTER XP EARNED
+450 XP

LEVELS CLEARED
3 levels (Days 55–57)

BOOK PROGRESS
Level 57 / 60 7500 XP total

WHAT YOU HAVE MASTERED

- ✔ Variation and heredity
- ✔ Dihybrid cross (9:3:3:1)
- ✔ How genes control traits
- ✔ Mendel's monohybrid cross (3:1)
- ✔ Dominant and recessive traits
- ✔ Sex determination in humans

NEXT CHAPTER Day 58: Our Environment

Chapter 12 cleared!

DAY 58
LEVEL 58

BIOLOGY | CHAPTER 13 | PART 1 ASTRA SERIES

ECOSYSTEM AND FOOD CHAINS

Goal: ecosystem, food chains, food webs, 10% law.

by Nitesh Prakash

01 Ecosystem | 02 Consumers | 03 Food Chains | 04 Food Web | 05 10% Law | +125 XP today | Level 58 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHAT IS AN ECOSYSTEM? ★★★ MOST ASKED

DEFINITION

- **Ecosystem:** all the living organisms of an area together with the non-living components (air, water, soil, light, temperature) with which they interact
- Living things depend on each other and on their physical surroundings for food, shelter and growth; together they form one functional unit

Components of an ecosystem ★★★

Biotic (living)	Abiotic (non-living)
Plants (producers)	Sunlight, temperature
Animals (consumers)	Water, air, rainfall
Bacteria, fungi (decomposers)	Soil, minerals, humidity

Natural vs artificial ecosystems ★★☆

Natural	Artificial (human-made)
Forest, grassland, desert	Garden, park
Pond, lake, river, sea	Aquarium
Run without human help	Crop field, zoo; humans maintain them

BUJHO *Aquarium mein toh sirf machhliyan hain, woh ecosystem kaise hua?*

PAHELI *Machhli, paudhe, bacteria (biotic) plus paani, oxygen, light, temperature (abiotic) sab hain. Bas insaan banata aur sambhalta hai, isliye* ***artificial ecosystem****.*

MNEMONIC: ABIOTIC FACTORS

"SWAT-S": Sunlight, Water, Air, Temperature, Soil: yeh paanch jaan nahi rakhte, par ecosystem inke bina nahi chalta.

REMEMBER

An ecosystem can be as small as a pond or as big as a forest.

2 PRODUCERS, CONSUMERS, DECOMPOSERS ★★★ MOST ASKED

Producers (autotrophs) ★★★

- Green plants and blue-green algae make food from simple inorganic substances using sunlight and chlorophyll (photosynthesis)
- All other organisms depend on them, directly or indirectly, for food

Consumers (heterotrophs): 4 types ★★★

Type	Eats	Examples
Herbivores	plants only	cow, deer, goat, rabbit
Carnivores	animals only	lion, tiger, frog, snake
Omnivores	plants and animals	human, crow, bear
Parasites	live on a host and feed on it	lice, tapeworm, Cuscuta

Decomposers: nature's cleaners ★★★

- **Bacteria and fungi** break down dead plants, animals and wastes into simple inorganic substances
- These go back into the soil and are used again by plants: **nutrients are recycled**
- Without decomposers, dead bodies would pile up and soil would lose its fertility

FROGS EXIST

Frog bas exist nahi karta, woh ***secondary consumer (T3)*** *hai: keede khaata hai aur saanp ka khana banta hai.*

PRODUCER POWER

Grass sunlight se khana banati hai, deer seedha le leta hai. Yahi hai ***T1 → T2*** *energy transfer.*

3 FOOD CHAINS AND TROPHIC LEVELS

★★★ MOST ASKED

A five-step food chain ★★★

Arrow = direction of energy (food) flow

T1 producer → T2 primary → T3 secondary → T4 tertiary consumer

DEFINITION

- **Food chain:** a series of organisms in which each one eats the one before it; it shows how food and energy pass from one organism to the next
- **Trophic level:** each step or level of a food chain where energy is transferred; producers are always at the first trophic level (T1)

Trophic levels at a glance ★★★

Level	Who	Example
T1	Producers	grass, phytoplankton
T2	Primary consumers (herbivores)	grasshopper, deer
T3	Secondary consumers (small carnivores)	frog, small fish
T4	Tertiary consumers (large carnivores)	snake, lion, eagle

Food chains you can write in the exam ★★★

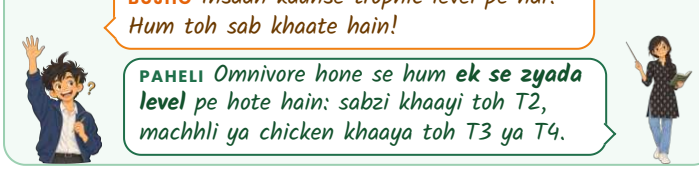
- **Grassland:** grass → deer → lion
- **Pond:** phytoplankton → zooplankton → small fish → large fish
- **Forest:** plants → insects → frog → snake → eagle
- **Field:** grains → mouse → snake → hawk

NITESH BHAIYA KI GUARANTEE

★★★ Food chain likho toh **arrow khane se khane wale ki taraf** (grass → deer), ulta arrow = seedha minus. Board mein yeh har saal aata hai!

NCERT thinker: remove one trophic level? ★★☆

- Kill all herbivores: producers grow unchecked and carnivores starve and die
- No trophic level can be removed without harm; the whole balance of the ecosystem is upset



MNEMONIC: ORDER OF A FOOD CHAIN

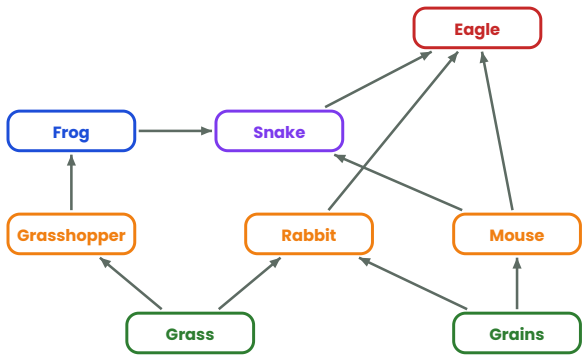
"Pehle Paudha, Phir Pet-bhar": Producer → Primary → Secondary → Tertiary; yaad karo "P-P-S-T: Pani Puri Sabko Tasty".

4 FOOD WEB ★★★ MOST ASKED

DEFINITION

- **Food web:** a network of many interconnected food chains in an ecosystem, because one organism can be eaten by several others
- A food web is more stable: if one food source is lost, consumers can switch to another

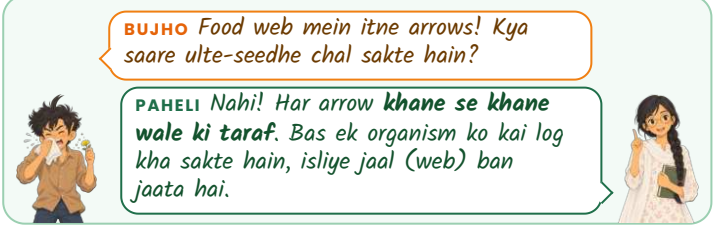
Food web in a grassland ★★★



Arrows show the direction of energy flow, from the food to the eater.

Food chain vs food web ★★☆

Food chain	Food web
A single straight path	Many interlinked chains
Each organism has one food source	An organism may have many food sources
Removal of one link breaks it	More stable; has alternative paths



SUCH ECOSYSTEM

*Sunlight se producer, producer se consumer, aur har step pe sirf **10% aage**. Doge bhi bolta: wow, such loss!*

EXAM TIP

A food web diagram must have at least 5 organisms, labelled, with every arrow from the food to the eater.

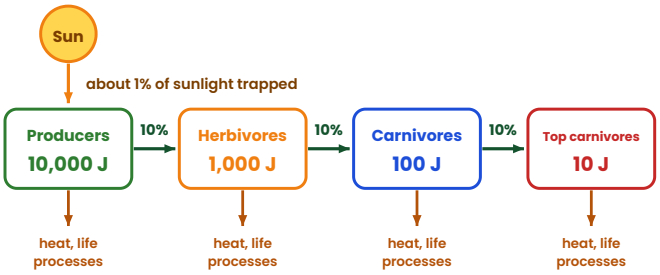
5 ENERGY FLOW AND THE 10% LAW

★★★ MOST ASKED

10% LAW

- **Ten percent law (Lindeman):** only about **10%** of the energy at one trophic level is passed on to the next; the rest is used in life processes and lost as heat
- Green plants trap only about **1%** of the sunlight falling on their leaves and change it into food (chemical energy)

Energy flow through a food chain ★★★



Worked numericals (verified) ★★★

1 Producers trap 10,000 J

- T2 = 10,000 × 10/100 = 1,000 J; T3 = 100 J; T4 = **10 J**

2 Grass → deer → lion, grass has 20,000 J

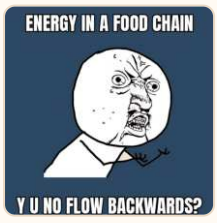
- Deer = 2,000 J; lion = 2,000 × 10/100 = **200 J**

3 A snake at T4 gets 5 J. Energy at T1?

- Go back ×10 per step: 5 → 50 (T3) → 500 (T2) → **5,000 J** (T1)

MNEMONIC: 10% LAW

"Das ka dum, baaki garam": har step pe 10% aage jaata hai, baaki 90% life processes aur heat mein chala jaata.



ONE WAY

*Energy ka rasta **one-way** hai: Sun se producer, producer se consumer. Wapas nahi aata, chahe kitna bhi Y U No bolo.*

6 PRACTICE MCQS ★★★ MOST ASKED

1. Which of these is an artificial ecosystem? NCERT ★★★

(a) forest (b) pond
(c) crop field (d) lake

2. Which of the following is an abiotic component of an ecosystem? ★★☆

(a) bacteria (b) temperature
(c) fungi (d) grass

3. Decomposers in an ecosystem are mainly: ★★★

(a) green plants (b) bacteria and fungi
(c) herbivores (d) parasites

Energy flow is unidirectional ★★★

- Energy moves only one way: Sun → producers → herbivores → carnivores
- Energy passed to the next level never comes back to the previous level; heat lost cannot be reused by plants
- So every ecosystem needs a constant input of energy from the Sun

Why only 3–4 steps in a food chain? ★★★

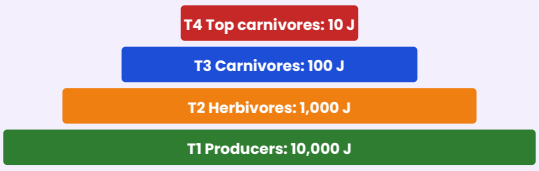
- About 90% of the energy is lost at every step
- After 4 levels very little energy is left (10,000 J → 10 J), not enough to support another level of organisms
- So there are fewer organisms at higher levels, and top carnivores are few

TOP PAR BHI GHAATA

*Top carnivore bano aur mile sirf **10 J out of 10,000 J**. 10% law ne Brian ko phir haraaya.*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

The **pyramid of energy** stacks trophic levels by the energy they hold. Because energy falls ten-fold at each step, it is always upright, never inverted.



Common mistakes that cost marks

× WRONG	✓ RIGHT
Arrow from lion to deer	Arrow from food to eater: deer → lion
Plants trap 10% of sunlight	About 1% of sunlight; 10% is between trophic levels
Decomposers are producers	They break down dead matter; plants are producers
Energy flow is cyclic	Energy is unidirectional; nutrients cycle

Day 58 hit-list: Eduspark PYQ analysis

Topic	Rating
10% law and energy numericals	★★★
Food chain and trophic levels	★★★
Role of decomposers	★★★
Unidirectional flow of energy	★★☆
Food web (diagram)	★★☆
Natural vs artificial ecosystems	★☆☆

4. Which of the following constitute a food chain? NCERT ★★★
(a) grass, wheat and mango (b) grass, goat and human
(c) goat, cow and elephant (d) grass, fish and goat

5. In the chain grass → deer → lion, the lion is at trophic level: ★★☆
(a) T1 (b) T2
(c) T3 (d) T4

6. If producers trap 10,000 J, the energy available to secondary consumers is: ★★★
(a) 1,000 J (b) 100 J
(c) 10 J (d) 1 J

7. The flow of energy in an ecosystem is: ★★★
(a) cyclic (b) unidirectional
(c) bidirectional (d) random

8. Green plants convert about what percentage of the sunlight falling on their leaves into food energy? ★★☆
(a) 1% (b) 10%
(c) 50% (d) 100%

9. **Assertion (A):** Food chains generally have only three or four trophic levels.
Reason (R): Only about 10% of the energy passes to the next level, so very little energy is left after four levels. ★★★
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

10. **Assertion (A):** An aquarium is an artificial ecosystem.
Reason (R): An aquarium has only biotic components. ★★☆
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

Short and long answer questions

1. What are decomposers? What will happen if there are no decomposers in an ecosystem? NCERT **(2 marks)**
2. Why are food chains generally limited to three or four trophic levels? **(2 marks)**
3. What are trophic levels? Give an example of a food chain and state the different trophic levels in it. NCERT **(3 marks)**
4. Draw a food web with at least five organisms. In the chain grass → deer → lion, grass has 20,000 J. Find the energy reaching the lion and explain why energy flow is unidirectional. **(5 marks)**

NITESH SIR SAYS
Food chain ka arrow aur 10% ka hisaab, yeh do cheezein pakki karo toh is topic ke marks pakke. ***Kal milte hai Day 59 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

Answer key: MCQs

Q	Ans	Why
1	(c)	Crop fields are made and kept by humans.
2	(b)	Temperature is non-living.
3	(b)	They break down dead matter.
4	(b)	Producer → herbivore → omnivore.
5	(c)	Grass T1, deer T2, lion T3.
6	(b)	10,000 → 1,000 (T2) → 100 J (T3).
7	(b)	Energy never flows back.
8	(a)	About 1% of the sunlight.
9	(a)	R explains A.
10	(c)	R false: it has abiotic parts too.

Model answers

1 Decomposers
- Bacteria and fungi that break down dead remains into simple substances (1); without them dead matter piles up and nutrients are not returned to the soil (1)

2 3–4 trophic levels
- 90% energy lost at each level (1); after 4 levels too little is left to support another level (1)

3 Trophic levels
- Each step of a food chain where energy is transferred (1)
- Grass → grasshopper → frog → snake (1)
- Grass T1 producer, grasshopper T2, frog T3, snake T4 (1)

4 Food web and energy
- Food web diagram with five or more organisms and correct arrows (2)
- Deer 2,000 J; lion 200 J (1)
- Energy flows Sun → producers → consumers and never returns; heat lost cannot be reused (2)

Day 58 checklist: tick before you sleep

- ☐ I can define an ecosystem and list its biotic and abiotic components
- ☐ I can classify consumers and explain the role of decomposers
- ☐ I can draw a food chain and name its trophic levels
- ☐ I can draw a food web and say why it is more stable
- ☐ I can solve 10% law numericals and explain unidirectional energy flow

My score: ____ / 5

LEVEL 58 COMPLETE +125 XP earned. Next: Level 59

BUJHO *Samajh gaya! Arrow khane wale ki taraf, aur har step pe sirf 10%.*

PAHELI *Shabaash! Kal Level 59: biological magnification, ozone aur garbage.*

DAY 59 LEVEL 59

BIOLOGY | CHAPTER 13 | PART 2 ASTRA SERIES

OUR ENVIRONMENT & BOSS LEVEL

Goal: DDT, ozone and garbage, then the Chapter 13 Boss.

by Nitesh Prakash

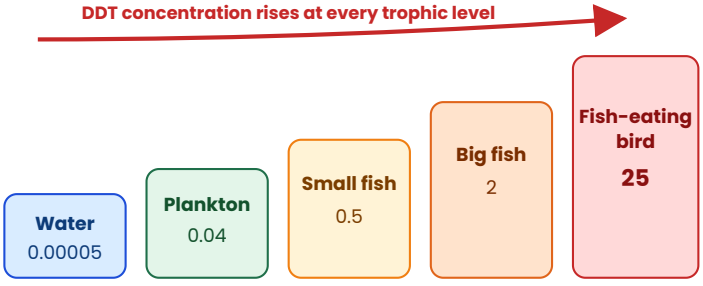
01 Biomagnification | 02 Ozone Layer | 03 CFCs and UNEP | 04 Garbage | 05 Boss Level | +175 XP today | Level 59 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 BIOLOGICAL MAGNIFICATION ★★★ MOST ASKED

DEFINITION

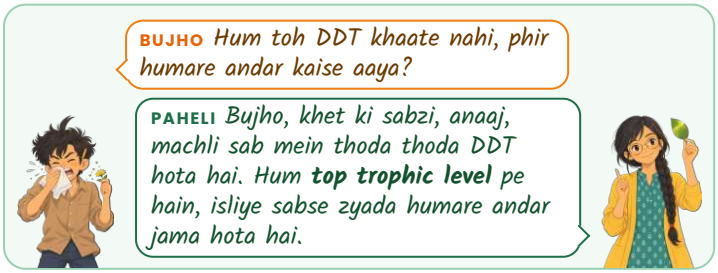
- **Biological magnification:** the progressive **increase in concentration** of harmful, non-biodegradable chemicals (like pesticides) at each higher trophic level of a food chain
- Pesticides sprayed on crops are washed into the **soil and water bodies**, are absorbed by plants with water and minerals, and so **enter the food chain**
- Such chemicals are **not degraded** and not excreted, so they **accumulate** in the body (mostly in fat)



DDT in ppm (parts per million), classic lake study

Why does the amount keep rising? ★★☆

- Each organism eats a **large amount** of the food of the lower level, so it collects the chemical from **many** organisms
- The chemical stays in its body for life, so the dose **adds up** step by step
- Result: the **top consumer** carries the highest concentration



WHY LEAST IN PRODUCERS?

Plants take the pesticide only from soil and water, so their level is the **lowest**. Every consumer above eats many organisms, so its level jumps.

Humans: the top trophic level ★★★

- Humans eat plants, animals and fish, so they occupy the **top level** in many food chains
- Hence the **maximum concentration** of pesticides builds up in humans (found even in food grains, vegetables, fish and milk)
- Common example: **DDT**, a pesticide that is non-biodegradable and fat-soluble

MEME BREAK

Top of the food chain hone ka "reward": sabse zyada DDT! Biological magnification ka matlab yahi hai.

★★★ *"Concentration of pesticides is maximum in humans or the top carnivore" ye one-liner har saal aata hai. Nitesh Bhaiya ki guarantee!*

MNEMONIC: BIOLOGICAL MAGNIFICATION

"Neeche kam, upar dam": neeche wale level mein chemical kam, upar wale level pe sabse zyada jama.

2 OZONE LAYER ★★★ MOST ASKED

DEFINITION

- **Ozone (O_3):** a molecule of **three oxygen atoms**; found in the upper atmosphere (stratosphere) as the **ozone layer**
- It **shields** the surface of the Earth from harmful **ultraviolet (UV)** radiation of the Sun
- At ground level ozone is a **deadly poison**; only the high-altitude layer helps us

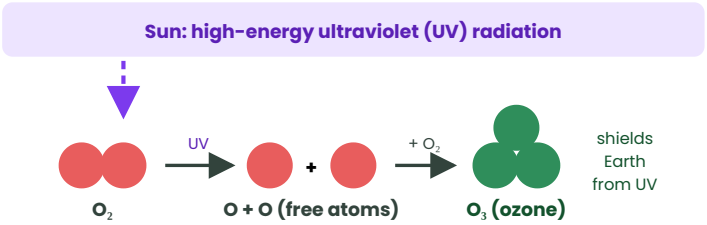
How ozone is formed ★★★

- UV radiation splits an oxygen molecule into free atoms:

$$O_2 \xrightarrow{(UV)} O + O$$

- Each free atom combines with another oxygen molecule:

$$O + O_2 \rightarrow O_3 \text{ (ozone)}$$

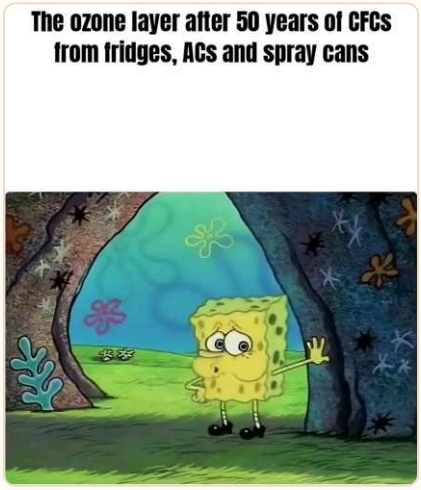


Depletion of the ozone layer ★★★

- The amount of ozone began to **fall sharply in the 1980s** (the "ozone hole")
- Main cause: **CFCs** (chlorofluorocarbons), used as **refrigerants** in fridges and ACs and in **fire extinguishers** and aerosol sprays
- Less ozone means more UV reaches us: **skin cancer**, cataract, weaker immunity and harm to plants and animals
- **1987:** the **UNEP** (United Nations Environment Programme) got an agreement to **freeze CFC production at 1986 levels**; now CFC-free fridges are made

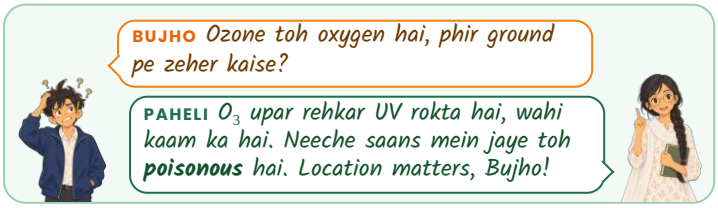
MNEMONIC: OZONE FACTS

"3-O, UV ka shield; CFC villain; 86 pe freeze, 87 ka deal": O_3 teen oxygen, CFC se depletion, UNEP 1987 ne production 1986 level pe freeze kiya.



MEME BREAK

Ozone layer bhi thak gayi hai yaar! Fridge aur AC CFC-free lo, warna UV seedha skin pe.



FACT FILE

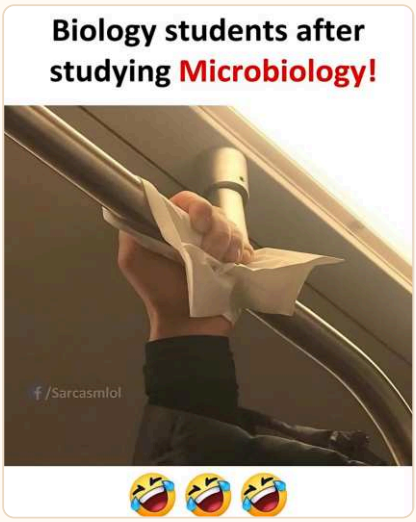
The thinning is worst over **Antarctica**, where an "ozone hole" forms every spring. Thanks to the CFC ban, the layer is now slowly healing.

3 MANAGING THE GARBAGE ★★★ MOST ASKED

Biodegradable vs non-biodegradable ★★★

Basis	Biodegradable	Non-biodegradable
Meaning	broken down by **biological processes** (bacteria, fungi)	not broken down by biological processes
Effect	return to nature, no lasting harm	persist for long, harm the ecosystem
Examples	vegetable and fruit peels, paper, cotton, wood, leaves, cow dung	plastics, polythene, glass, DDT, metal cans, synthetic fibres

Enzymes act on substances in a **specific** way, which is why they cannot break down plastics. Inert substances simply **persist**.



MEME BREAK

Decomposers ka asli kaam: bacteria aur fungi biodegradable kachre ko tod kar mitti mein wapas bhejte hain.

NCERT debate: disposable cups and kulhads ★★☆

- Trains once served tea in **plastic cups** that had to be returned; later **disposable plastic cups** came, creating huge waste
- **Kulhads** (clay cups) were suggested as a substitute, but making them on a large scale would use up **fertile top-soil**
- Now **disposable paper cups** are used: paper is biodegradable, but it still uses trees, so **reuse** is the real answer

Ways to dispose of waste ★★☆

Method	What happens
Segregation	separate wet (green bin) and dry (blue bin) waste at home
Composting	kitchen and garden waste turn into manure; earthworms speed it up (vermicomposting)
Recycling	paper, plastic, glass, metal are made into new products
Landfill	waste is dumped in low-lying areas and covered with soil
Incineration	burning at high temperature; reduces bulk but pollutes air

MEME BREAK

*Green bin mein chips ka packet? Arthur ki tarah gussa control karo aur use **dry waste** wale bin mein daalo.*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

The 3 Rs to save the environment: **Reduce** (use less, e.g. switch off fans and taps), **Reuse** (use things again, e.g. jars, envelopes), **Recycle** (send plastic, paper, glass, metal for remaking).
Reuse is better than recycling because recycling itself uses energy.

Plastics: the big problem

- Plastic is **non-biodegradable**: it stays in the soil for hundreds of years and blocks drains
- Animals like cows eat polythene bags and fall ill; burning plastic releases **toxic gases**
- Carry a **cloth bag**, refuse single-use plastic, and segregate

MNEMONIC: BIODEGRADABLE TEST

"Jo sadta hai, wo bachta hai": jo cheez gal-sad jaati hai (peels, paper, leaves) wo biodegradable; plastic, glass, DDT kabhi nahi sadte.

MEME BREAK

Squid Game ka glass bridge: kulhad wala panel mitti mein mil jaata hai, plastic wala 500 saal tak toot ke bhi nahi jaata. Sahi chuno!

Exam tips and common mistakes

- ✔ Write "concentration **increases** at each trophic level" for biological magnification, not "energy increases"
- ✔ Ozone equations: write **both steps** with UV over the first arrow
- ✔ UNEP agreement: **1987**, freeze at **1986** levels; do not swap the years
- ✔ Kulhad answer must mention **loss of fertile top-soil**

NITESH SIR SAYS

*Environment ke 5 marks pakke hain: DDT, ozone aur garbage ka table yaad karo. Ab Boss Level, poora Chapter 13, ek saath. **Chalo, jalwa dikhao!***

4 BOSS LEVEL: OUR ENVIRONMENT

★★★ MOST ASKED

Boss rules: 60 minutes, 30 marks

1. Section A: 9 MCQs incl. 2 Assertion–Reason (1 mark each) = **9**
2. Section B: 3 × 2 marks = **6**; Section C: 2 × 3 marks = **6**
3. Section D: 1 × 5 marks = **5**; Section E: 1 case-based × 4 marks = **4**
4. Covers the whole chapter: Day 58 (ecosystem, food chains, 10% law) and Day 59

SECTION A (1 mark each)

1. Which of these is an artificial ecosystem? ★★☆

(a) pond (b) forest
(c) crop field (d) lake

2. Which of the following constitutes a food chain? ★★★

(a) grass, wheat, mango (b) grass, goat, human
(c) goat, cow, elephant (d) grass, fish, goat

3. Green plants in a food chain form the: ★★☆

(a) first trophic level (b) second trophic level
(c) top trophic level (d) decomposer level

4. If 20,000 J of energy is trapped by producers, the energy reaching secondary consumers is: ★★★

(a) 2000 J (b) 200 J
(c) 20 J (d) 2 J

5. The flow of energy in an ecosystem is: ★★☆

(a) cyclic (b) bidirectional
(c) unidirectional (d) reversible

6. Which group contains only non-biodegradable items? ★★★

(a) paper, glass, DDT (b) polythene, glass, DDT
(c) peels, wood, plastic (d) leaves, cotton, cans

7. Ozone in the upper atmosphere is formed when: ★★★

(a) UV splits O_2 and O joins O_2 (b) CFCs react with O_2
(c) CO_2 breaks down (d) water vapour splits

8. **Assertion (A):** A food chain usually has only three or four trophic levels.
Reason (R): Only about 10% of the energy of one trophic level is passed on to the next. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9. **Assertion (A):** The concentration of DDT is highest in humans.
Reason (R): DDT is a biodegradable chemical. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B (2 marks each)

10. What are decomposers? What would happen if all the decomposers of an ecosystem were removed? **(2)**

11. What will happen if we kill all the organisms of one trophic level, for example all herbivores? **(2)**

12. Why is damage to the ozone layer a cause of concern? What step was taken in 1987 to limit it? **(2)**

SECTION C (3 marks each)

13. **Diagram-based:** Draw a food chain of four trophic levels in a grassland and label each trophic level. If the producers fix 15,000 J, how much energy reaches the top level? **(3)**

14. What is biological magnification? Will its level be different at different trophic levels? Explain with an example. **(3)**

SECTION D (5 marks)

15. (a) Give two differences between biodegradable and non-biodegradable substances, with two examples of each. (b) Suggest three ways to reduce the garbage produced at home. (c) Why are kulhads not a good large-scale substitute for disposable cups? **(2 + 2 + 1)**

SECTION E (case-based, 4 marks)

Q16. Case: the lake near the cotton fields

A lake beside cotton fields sprayed with a pesticide was tested. Pesticide level (ppm): water 0.02, algae 5, small fish 40, large fish 300, fish-eating bird 1500.

- (i) Name the phenomenon shown by the data. **(1)**
- (ii) Write the food chain and the trophic level of the bird. **(1)**
- (iii) Why does the pesticide not get broken down or excreted? **(1)**
- (iv) Suggest one way farmers can reduce this problem. **(1)**

5 ANSWER KEY ★★★

Answer key: Section A

Q	Ans	Why
1	(c)	Crop fields, gardens, aquaria are made by humans.
2	(b)	Producer → herbivore → omnivore.
3	(a)	Producers (autotrophs) are level 1.
4	(b)	20,000 → 2000 → 200 J (10% law).
5	(c)	Energy does not return to the Sun.
6	(b)	Paper is biodegradable, so (a) is out.
7	(a)	$O_2 \rightarrow O + O$ (UV); $O + O_2 \rightarrow O_3$.
8	(a)	90% loss per step; R explains A.
9	(c)	A true; DDT is non-biodegradable.

BOSS STRATEGY

Section A in 10 min, B and C in 25 min, D and E in 20 min, last 5 min to check units and labels. Food-chain arrows always point **towards the eater**.

6 MARKING POINTS ★★★

Marking points: Sections B to E

10 Decomposers

- Bacteria and fungi that break down dead remains and wastes into simple substances (1)
- Without them, dead matter piles up and nutrients are not returned to the soil (1)

11 Killing one level

- The level above (carnivores) starves and declines (1)
- The level below (plants) grows unchecked; the whole ecosystem is upset (1)

12 Ozone concern

- Less ozone lets UV in: skin cancer, cataract, harm to crops (1)
- UNEP, 1987: CFC production frozen at 1986 levels (1)

13 Food chain diagram

- Grass → grasshopper → frog → snake with arrows (1); levels labelled T1 to T4 (1)
- 15,000 → 1500 → 150 → **15 J** at the top level (1)

14 Biological magnification

- Definition: rise in concentration of harmful chemicals at each higher level (1)
- Yes, highest at the top (1); DDT: water → plankton → fish → birds or humans (1)

15 Garbage (5 marks)

- (a) Breakdown by microbes or not; harmless or persistent (1); examples: peels, paper / plastic, glass (1)
- (b) Segregate wet and dry, compost kitchen waste, carry cloth bags (2)
- (c) Making them on a large scale wastes fertile top-soil (1)

16 Case answers

- (i) Biological magnification (ii) water → algae → small fish → large fish → bird; bird is at the top (fifth) level
- (iii) It is non-biodegradable and stored in fat (iv) Use less pesticide, or biological pest control

Day 59 checklist: tick before you sleep

- [ ] I can explain biological magnification with the DDT example
- [ ] I can write both ozone-formation equations and the CFC story
- [ ] I know the 1987 UNEP step and the 1986 freeze level
- [ ] I can classify waste and argue the kulhad debate
- [ ] I finished the Boss Level and checked it with the key

My score: ____ / 5

LEVEL 59 COMPLETE +175 XP earned. Next: Level 60

BUJHO *Chapter 13 ka Boss bhi gaya! DDT upar jama hota hai, ozone UV rokta hai, kachra alag karo.*

PAHELI *Shabaash! Kal Level 60: Science Grand Mock, poori book.*

DAY 60
LEVEL 60

SCIENCE | ALL 13 CHAPTERS | GRAND MOCK

SCIENCE GRAND MOCK TEST

Goal: write one full 80-mark board paper in 3 hours.

by Nitesh Prakash

01 Instructions | 02 Sections A–E | 03 39 Questions | 04 Answer Key | 05 Marking Hints | +200 XP Boss Level | Level 60 / 60

SCIENCE (086) GRAND MOCK

Class X | Time: 3 hours | Max. Marks: 80

Based on the CBSE 2026-27 syllabus and sample-paper pattern

GENERAL INSTRUCTIONS

1. This question paper has 39 questions in 5 sections, A to E. All questions are compulsory; internal choices are given in some questions.
2. Section A: Q1–20 are multiple choice questions of 1 mark each (Q17–20 are Assertion–Reason).
3. Section B: Q21–26 are very short answer questions of 2 marks each (30 to 50 words).
4. Section C: Q27–33 are short answer questions of 3 marks each (50 to 80 words).
5. Section D: Q34–36 are long answer questions of 5 marks each (80 to 120 words).
6. Section E: Q37–39 are case-based units of 4 marks each, with sub-parts of 1, 1 and 2 marks.

Unit tally: Chemistry 25 | Biology (World of Living) 25 | Light and Eye 12 | Electricity and Magnetism 13 | Our Environment 5 | Total 80

SECTION A — 20 × 1 = 20 marks

1. Which of the following is a decomposition reaction? 1

(a) $CaO + H_2O \rightarrow Ca(OH)_2$

(b) $2AgCl \rightarrow 2Ag + Cl_2$ (in sunlight)

(c) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

(d) $NaOH + HCl \rightarrow NaCl + H_2O$

2. A solution turns red litmus blue. Its pH is likely to be: 1

(a) 1 (b) 4

(c) 5 (d) 10

3. Which metal is kept immersed in kerosene to stop it catching fire? 1

(a) sodium (b) magnesium

(c) aluminium (d) zinc

4. The functional group present in ethanoic acid is: 1

(a) –OH (b) –CHO

(c) –COOH (d) >C=O

5. Pyruvate is broken down completely into CO_2 and water, releasing the most energy, in: 1

(a) cytoplasm (b) mitochondria

(c) chloroplast (d) nucleus

6. The blood vessel that brings oxygenated blood from the lungs to the heart is: 1

(a) pulmonary artery (b) pulmonary vein

(c) aorta (d) vena cava

7. The plant hormone that promotes cell division is: 1

(a) auxin (b) abscisic acid

(c) cytokinin (d) gibberellin

8. Plasmodium reproduces asexually by: 1

(a) budding (b) binary fission

(c) fragmentation (d) multiple fission

9. In the F_2 generation of Mendel's monohybrid cross (TT × tt), the percentage of dwarf plants is: 1

(a) 75% (b) 50%

(c) 25% (d) 0%

10. The power of a lens is −2 D. The lens and its focal length are: 1

(a) convex, +50 cm (b) concave, −50 cm

(c) concave, −2 m (d) convex, +0.5 cm

11. The focal length of the eye lens is adjusted by the: 1

(a) iris (b) pupil

(c) ciliary muscles (d) retina

12. A wire of resistance R is stretched uniformly to double its length. Its new resistance is: 1

(a) R/2 (b) R

(c) 2R (d) 4R

13. One kilowatt hour (1 kWh) is equal to: 1

(a) 3.6×10^3 J (b) 3.6×10^5 J
(c) 3.6×10^6 J (d) 3.6×10^9 J

14. The direction of the force on a current-carrying conductor placed in a magnetic field is given by: 1

(a) right-hand thumb rule (b) Fleming's left-hand rule
(c) Ohm's law (d) Snell's law

15. Which of the following does NOT depend on the length of a wire? 1

(a) resistance (b) resistivity
(c) both (d) neither

16. Which group contains only biodegradable items? 1

(a) grass, flowers, leather
(b) grass, wood, plastic
(c) fruit peels, cake, polythene
(d) cake, wood, glass

Q17–20: ASSERTION–REASON

Two statements are given, Assertion (A) and Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

17. **Assertion (A):** Silver chloride kept in sunlight turns grey. 1
Reason (R): Sunlight decomposes silver chloride into silver and chlorine.

18. **Assertion (A):** Lactic acid is formed in our muscle cells when there is a lack of oxygen. 1
Reason (R): In yeast, pyruvate is converted into lactic acid in the cytoplasm.

19. **Assertion (A):** The image formed by a concave lens is always real. 1
Reason (R): A concave lens always forms a virtual, erect and diminished image.

20. **Assertion (A):** The flow of energy in an ecosystem is unidirectional. 1
Reason (R): The energy that passes to the herbivores does not come back to the autotrophs.

SECTION B — 6 × 2 = 12 marks

21. In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, identify the substance oxidised and the substance reduced. Why is it called a redox reaction? 2

OR

Why should a magnesium ribbon be cleaned before burning it in air? Write the balanced equation for its burning.

22. An electric iron draws a current of 4 A from the 220 V mains. Find its power and the energy it uses in 2 hours, in kWh. 2

23. State the role of (a) salivary amylase and (b) bile juice in the digestion of food. 2

24. What is a reflex arc? Write, in the correct order, the components through which the impulse travels in a reflex action. 2

25. A concave mirror forms a real image three times the size of an object placed 10 cm in front of it. Where is the image located? 2

26. Draw the pattern of magnetic field lines around a straight conductor carrying current. What happens to the field when the current is increased? 2

OR

State Fleming's left-hand rule. What happens to the force on the conductor if the direction of the current is reversed?

SECTION C — 7 × 3 = 21 marks

27. A shiny brown metal X, used in electric wires, turns black when heated in air. Name X and the black compound. Write the balanced equation and name the type of reaction involved. 3

28. Write the chemical name and formula of (a) bleaching powder, (b) baking soda and (c) washing soda. Give one use of each. 3

29. Explain with a labelled diagram how soap molecules form micelles and remove oily dirt from clothes. 3

30. What is double circulation? Why is the separation of oxygenated and deoxygenated blood necessary in mammals and birds? 3

31. (a) What is the role of the placenta? (b) Name one barrier method and one surgical method of contraception. (c) Why is female foeticide a threat to society? 3

32. A 2 cm tall object is placed 15 cm from a convex lens of focal length 10 cm. Find the position, size and nature of the image. 3

33. What is biological magnification? Will its level be the same at every trophic level? Explain with the example of a pesticide such as DDT. 3

SECTION D — 3 × 5 = 15 marks

34. Answer the following about ionic compounds. 5

(a) Show the formation of NaCl and MgO by the transfer of electrons. **(2)**

(b) Why do ionic compounds have high melting points? **(1)**

(c) Why do they conduct electricity in the molten state but not in the solid state? **(2)**

OR

Answer the following about metallurgy.

(a) Distinguish between roasting and calcination with one equation each. **(2)**

(b) Write the thermit reaction and one of its uses. **(2)**

(c) Why are sulphide and carbonate ores changed into oxides before reduction? **(1)**

35. Answer the following about excretion in humans. 5

(a) Draw a neat labelled diagram of a nephron. **(2)**

(b) Explain how urine is formed in it. **(2)**

(c) How is the amount of urine produced regulated? **(1)**

OR

Answer the following about sexual reproduction in plants.

(a) Draw a labelled diagram of the germination of pollen on the stigma. **(2)**

(b) Describe fertilisation in flowering plants. What do the ovule and the ovary develop into? **(3)**

36. Resistors of 2 Ω, 3 Ω and 6 Ω are given. 5

(a) Find their equivalent resistance in series and in parallel. **(2)**

(b) The parallel combination is connected to a 6 V battery. Find the total current and the current through the 2 Ω resistor. **(2)**

(c) Why are household appliances connected in parallel? **(1)**

OR

Answer the following about electric power.

(a) Define electric power and show that $P = I^2R$. **(2)**

(b) A bulb is rated 100 W, 220 V. Find the current it draws and its resistance. **(2)**

(c) Find the energy it uses in 10 hours, in kWh. **(1)**

SECTION E — 3 × 4 = 12 marks

37. Five solutions A to E were tested with universal indicator paper. Their pH values are given below. 4

Solution	A	B	C	D	E
pH	1	4	7	10	13

(i) Which solution is strongly alkaline? **(1)**

(ii) Which solution is neutral? Could it be pure water? **(1)**

(iii) Arrange A to E in increasing order of hydrogen ion concentration. **(2)**

OR

(iii) Name the reaction when A is mixed with E, and give its general word equation. **(2)**

38. Mendel crossed pure pea plants with round yellow seeds (RRYY) with plants having wrinkled green seeds (rryy). All F_1 seeds were round yellow. Selfing F_1 gave 556 seeds: 315 round yellow, 108 round green, 101 wrinkled yellow and 32 wrinkled green. 4

(i) Which traits are dominant? **(1)**

(ii) Write the genotype of the F_1 plants and the gametes they form. **(1)**

(iii) Find the F_2 phenotypic ratio and name the law it proves. **(2)**

OR

(iii) Name the new combinations that appear in F_2. What fraction of the F_2 plants is RRYY? **(2)**

39. Riya, a Class 10 student, can read her book clearly but cannot see the writing on the board clearly from the last bench. The doctor finds that her far point is 2 m. 4

(i) Name the defect of vision she has. **(1)**

(ii) State one cause of this defect. **(1)**

(iii) Find the power of the lens she needs and name the lens. **(2)**

OR

(iii) Draw ray diagrams of her eye before and after correction. **(2)**

BLUEPRINT: MARKS BY UNIT AND SECTION

Unit	A	B	C	D	E	Total
Chemical Substances	5	2	9	5	4	**25**
World of Living	6	4	6	5	4	**25**
Light and Eye	3	2	3	0	4	**12**
Electricity, Magnetism	4	4	0	5	0	**13**
Our Environment	2	0	3	0	0	**5**
Total	**20**	**12**	**21**	**15**	**12**	**80**

END OF QUESTION PAPER

2 ANSWER KEY AND MARKING HINTS

Section A: answers — 20 marks

1 (b)	2 (d)	3 (a)	4 (c)	5 (b)
6 (b)	7 (c)	8 (d)	9 (c)	10 (b)
11 (c)	12 (d)	13 (c)	14 (b)	15 (b)
16 (a)	17 (a)	18 (c)	19 (d)	20 (a)

A–R: 17 (a) sunlight splits AgCl | 18 (c) yeast makes ethanol + CO_2 | 19 (d) concave lens image is virtual | 20 (a) energy never returns

Section B: value points — 2 each

- **21** H_2 is oxidised (gains O), CuO is reduced (loses O); both together = redox. OR: removes the oxide layer; $2Mg + O_2 \rightarrow 2MgO$
- **22** P = VI = 220 × 4 = **880 W**; E = 0.88 kW × 2 h = **1.76 kWh**
- **23** (a) starch → sugar (maltose) in the mouth; (b) emulsifies fats and makes the medium alkaline for lipase
- **24** Nerve pathway of a reflex: receptor → sensory neuron → relay neuron (spinal cord) → motor neuron → effector (muscle/gland)
- **25** m = −v/u = −3, u = −10 cm, so **v = −30 cm**: 30 cm in front of the mirror
- **26** Concentric circles around the wire (diagram); stronger field when the current is increased. OR: forefinger field, middle finger current, thumb force (all at right angles); force reverses

Section C: value points — 3 each

- **27** X = copper, black CuO; $2Cu + O_2 \rightarrow 2CuO$; combination (oxidation) reaction
- **28** Calcium oxychloride $CaOCl_2$ (disinfecting water); sodium hydrogencarbonate $NaHCO_3$ (baking, antacid); sodium carbonate decahydrate $Na_2CO_3.10H_2O$ (removes hardness of water)
- **29** Ionic head in water, carbon tail in oil; tails trap the oil inside a micelle; the micelles are rinsed away with water (diagram 1 mark)
- **30** Blood passes through the heart twice per cycle (body → right side → lungs → left side → body); separation gives the high O_2 supply needed by warm-blooded animals
- **31** (a) supplies food and O_2 to the embryo and removes wastes; (b) condom; vasectomy or tubectomy; (c) lowers the sex ratio and is illegal and unjust
- **32** 1/v = 1/10 − 1/15 = 1/30, **v = +30 cm**; m = v/u = −2; h′ = **−4 cm**: real, inverted, magnified
- **33** Non-biodegradable chemicals build up at each trophic level; highest at the top (humans, birds of prey); e.g. DDT in water → plankton → fish → birds

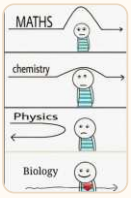
Score tracker

Section	Max.	My marks
A (MCQ, A–R)	20	____
B + C	33	____
D + E	27	____
Total	**80**	____

70+ Topper zone | 60–69 Strong | 45–59 Revise weak chapters | below 45 Redo the chapter Boss Levels

Sections D and E: marking points — 5 and 4 each

- **34** Na(2,8,1) gives $1e^-$ to Cl(2,8,7); Mg(2,8,2) gives $2e^-$ to O(2,6); strong forces of attraction between ions; ions move freely only in the molten state. OR: roasting $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; calcination $ZnCO_3 \rightarrow ZnO + CO_2$; $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ + heat (joining rail tracks); oxides are easier to reduce
- **35** Labels: glomerulus, Bowman's capsule, tubule, collecting duct; filtration in the glomerulus, selective reabsorption of glucose, amino acids, salts and water in the tubule; amount depends on excess water and dissolved wastes. OR: pollen tube grows through the style to the ovule; male gamete fuses with the egg to form the zygote; ovule → seed, ovary → fruit
- **36** Series 11 Ω; parallel 1/R = 1/2 + 1/3 + 1/6, **R = 1 Ω**; I = 6/1 = **6 A**; $I_{2\Omega}$ = 6/2 = **3 A**; same voltage, separate switches, one fault does not stop the rest. OR: P = VI = I^2R; I = 100/220 = **0.45 A**; R = $220^2/100$ = **484 Ω**; E = 0.1 kW × 10 h = **1 kWh**
- **37** (i) E; (ii) C, yes; (iii) E < D < C < B < A. OR: neutralisation, acid + base → salt + water
- **38** (i) round, yellow; (ii) RrYy; gametes RY, Ry, rY, ry; (iii) 9 : 3 : 3 : 1, law of independent assortment. OR: round green, wrinkled yellow; 1/16
- **39** (i) myopia; (ii) excessive curvature of the eye lens or elongated eyeball; (iii) f = −2 m, P = 1/f = **−0.5 D**, concave lens. OR: image in front of the retina; concave lens shifts it on to the retina



BOOK COMPLETE
*Maths, Chemistry, Physics, Biology: graph jaisa bhi ho, **Day 60 tak aa gaye** toh tum already topper ho!*

Day 60 checklist: tick before you sleep

- ☐ I wrote the full paper in 3 hours
- ☐ I attempted every OR choice once
- ☐ I checked my answers with the key
- ☐ I redrew every diagram I got wrong
- ☐ I listed my three weakest chapters

My score: ____ / 5

5/5 = Board ready! Weak chapters ek baar aur.

LEVEL 60 COMPLETE

+200 XP earned. BOOK COMPLETE. Book total: 8000 XP

BUJHO *60 din, 60 levels, 8000 XP! Chemistry, Physics, Biology: sab clear!*

PAHELI *Ab bas revision aur sample papers. Boards mein dikhana hai jalwa!*

NITESH SIR SAYS
*Book khatam, safar nahi. Har hafte ek full paper, time lagake. Diagram saaf, units sahi, steps poore. **All the best, champions!***

BIOLOGY | CHAPTER 13 COMPLETE | BOOK COMPLETE

CHAPTER 13 SWAHAA!

Our Environment | Days 58–60

BUJHO *60 din, 60 levels, 8000 XP! Humne poori Science book khatam kar di, Swahaa!*

PAHELI *Proud of you, Bujho! Ab bas revision aur boards mein full marks. All the best!*

CHAPTER XP EARNED
+500 XP

LEVELS CLEARED
3 levels (Days 58–60)

BOOK PROGRESS
Level 60 / 60 8000 XP total

WHAT YOU HAVE MASTERED

- ✔ Ecosystem and its components
- ✔ Ten percent law of energy flow
- ✔ Ozone layer depletion
- ✔ Food chains, food webs and trophic levels
- ✔ Biological magnification
- ✔ Biodegradable and non-biodegradable waste

BOOK STATUS BOOK COMPLETE: Level 60 / 60, 8000 XP *All 60 levels cleared!*

EDUSPARK

BOOK COMPLETE

LEVEL 60 / 60 CLEARED

SCIENCE SWAHAA!

60 days, 13 chapters, 8000 XP. You did not just read this book, you **finished** it. Revise every topic you marked R or RA, re-attempt the Boss Levels, and walk into the board exam like a topper.

LEVELS CLEARED
60 / 60
Chemistry, Physics, Biology

TOTAL XP EARNED
8000 XP
2500 + 2500 + 3000

NEXT MISSION
BOARD EXAM
You are ready

OTHER BOOKS IN THE ASTRA SERIES

SCIENCE | SOCIAL SCIENCE | MATHEMATICS | ENGLISH | HINDI

EDUSPARK
AN EDUSPARK PUBLICATION

CLASS 10 | 5 BOOKS | 60 DAYS PLAN

YOUR VOICE MATTERS

GIVE YOUR REVIEW: SCAN THE QR

Tell us what helped you most and what we should make better. Every review helps us make the next ASTRA book even better for you and your juniors.

1 Open camera | 2 Scan the code | 3 Rate and write 2 lines

Thank you for learning with Eduspark
Clear. Powerful. Every detail.

ASTRA SERIES
Free e-book by Nitesh Prakash